

Lecture Notes on Data Engineering
and Communications Technologies 275

Andrii Semenov
Iryna Yepifanova
Jana Kajanová *Editors*



Data-Centric Business and Applications

Modern Trends in Financial and
Innovation Data Processes 2025.
Volume 1

Lecture Notes on Data Engineering and Communications Technologies

Volume 275

Series Editors

Fatos Xhafa , Technical University of Catalonia, Barcelona, Spain

Rolando Herrero, Northeastern University, Boston, MA, USA

The aim of the book series is to present cutting edge engineering approaches to data technologies and communications. It will publish latest advances on the engineering task of building and deploying distributed, scalable and reliable data infrastructures and communication systems.

The series will have a prominent applied focus on data technologies and communications with aim to promote the bridging from fundamental research on data science and networking to data engineering and communications that lead to industry products, business knowledge and standardisation.

Indexed by SCOPUS, INSPEC, EI Compendex.

All books published in the series are submitted for consideration in Web of Science.

Andrii Semenov · Iryna Yepifanova · Jana Kajanová
Editors

Data-Centric Business and Applications

Modern Trends in Financial and Innovation
Data Processes 2025. Volume 1

Editors

Andrii Semenov 
Faculty of Information Electronic Systems
Vinnytsia National Technical University
Vinnytsia, Ukraine

Iryna Yepifanova 
Faculty of Management and Information
Security
Vinnytsia National Technical University
Vinnytsia, Ukraine

Jana Kajanová
Faculty of Management
Department of Economics and Finance
Comenius University Bratislava
Bratislava, Slovakia

ISSN 2367-4512 ISSN 2367-4520 (electronic)
Lecture Notes on Data Engineering and Communications Technologies
ISBN 978-3-032-10152-5 ISBN 978-3-032-10153-2 (eBook)
<https://doi.org/10.1007/978-3-032-10153-2>

© The Editor(s) (if applicable) and The Author(s), under exclusive license to Springer Nature Switzerland AG 2026

This work is subject to copyright. All rights are solely and exclusively licensed by the Publisher, whether the whole or part of the material is concerned, specifically the rights of translation, reprinting, reuse of illustrations, recitation, broadcasting, reproduction on microfilms or in any other physical way, and transmission or information storage and retrieval, electronic adaptation, computer software, or by similar or dissimilar methodology now known or hereafter developed.

The use of general descriptive names, registered names, trademarks, service marks, etc. in this publication does not imply, even in the absence of a specific statement, that such names are exempt from the relevant protective laws and regulations and therefore free for general use.

The publisher, the authors and the editors are safe to assume that the advice and information in this book are believed to be true and accurate at the date of publication. Neither the publisher nor the authors or the editors give a warranty, expressed or implied, with respect to the material contained herein or for any errors or omissions that may have been made. The publisher remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

This Springer imprint is published by the registered company Springer Nature Switzerland AG
The registered company address is: Gewerbestrasse 11, 6330 Cham, Switzerland

If disposing of this product, please recycle the paper.

Preface

This volume presents a comprehensive exploration of innovative approaches and data-centric solutions that support sustainable development across diverse economic and technological sectors. Each chapter contributes to the growing body of knowledge on how enterprises, industries, communities, and ecosystems can adapt to the dynamic challenges of the modern world. The authors offer valuable insights grounded in interdisciplinary research and practical applications from strategic development in renewable energy and responsible tourism to digital innovation in agriculture and environmental, social, and governance (ESG) reporting. A particular emphasis is placed on Ukraine's socio-economic context, where the war, decentralisation, and resilience-building efforts highlight the need for adaptive and sustainable strategies. Methodological advances, such as tokenisation of real-world assets, logistic regression for bank stability, and synthetic control methods for investment analysis, enrich the analytical foundation of the book. Several chapters delve into sector-specific innovations—ranging from biomedicine and environmental monitoring to education and scientific research financing—demonstrating the crucial role of data and technology in strategic planning. The inclusion of mathematical modelling and engineering-based studies broadens the monograph's impact, bridging gaps between economics, technology, and applied sciences. Collectively, the research presented here offers both theoretical insights and practical tools for academics, policymakers, and industry practitioners committed to sustainable transformation. We hope this volume will inspire further dialogue and action toward resilient, knowledge-driven development at both local and global levels.

In the first chapter titled “[Innovative Potential in Ensuring Sustainable Development of the Enterprise](#)”, the authors propose to examine the existing approaches in Ukraine to calculating innovation potential and its application to the sustainable development of enterprises. An analysis of approaches used to determine the innovation potential of an enterprise was carried out, and the features of forming a model of the innovative potential for sustainable development were described. The main prerequisites for developing such a model were also identified. As an example of practical application, the chapter developed a model of the innovative potential of

sustainable development of the Private Agricultural Firm «VATAL» and the Agricultural Limited Liability Company «Slavutych». Quantitative indicators for assessing the innovative potential of sustainable development were provided, and based on these, projected indicators of innovation potential were calculated using individual innovations in the production sector of these firms. This chapter may be of particular interest to foreign companies planning to engage in financial and innovative activities in Ukraine.

In the next chapter, “[RWA Tokenization Mechanisms as a Tool for Sustainable Economic Development of Local Hromadas in Ukraine](#)”, the authors argue that the relevance of the study of (RWA—Real World Assets) tokenization mechanisms in the context of local community development is due to the following important factors. Local communities in Ukraine face the problem of limited financial resources for the implementation of infrastructure development and modernization projects. Tokenization of real assets can become a tool for attracting additional investments by creating liquid digital assets secured by real municipal property. Authors develop the mechanisms of the impact of RWA tokenization on sustainable economic development and restoration of local communities in Ukraine. Also, the implementation of RWA tokenization mechanisms provides increased transparency in the management of community assets and reduced transaction costs when carrying out operations with them. The use of this mechanism may become a key factor in the sustainable development of local communities in Ukraine, especially in the process of post-war reconstruction. The methods and mechanisms proposed in the chapter can be useful both for shaping the economic development of communities in developed countries and for managing international investment processes during the post-war development of Ukraine.

In the subsequent chapter, “[Methodology for Assessing the State of Business Processes in Terms of Their Impact on the Economic Security of the Enterprise](#)”, the authors built a mathematical model to optimize decision-making for manufacturing enterprises in an unstable environment. The components of the model are part of the expenses for wages (including taxes), procurement of material resources (raw materials), and payment for electricity used in production. The model is significantly nonlinear and therefore exhibits different behavior for large and small enterprises. The model also takes into account the possibility of self-generation of electricity. As a result, the model’s advantage is the ability to take into account parameters that affect the cost of production, as well as the identification of threats in forecasting the consequences of their implementation. This creates a basis for the development of preventive measures, which is critically important in various conditions in the environment. Thus, the results of the section may be important for the development of effective decision support systems for manufacturing enterprises in modern conditions.

In the next chapter, “[The Strategic Potential of Renewable Energy Microgrids: Global Transformations and Determinants of Sustainable Development](#)” the authors adopt a mixed-methods approach combining conceptual systems analysis and empirical econometric modeling. To overcome the challenges of the renewable energy

source development in Ukraine, a forward-looking solution is to create small electricity generation grids or microgrids that ensure decentralized and stable energy production and distribution at a local level. Structural elements of the microgrid ecosystem and the value system in the context of achieving sustainable development and its sustainable development goals were constructed in the chapter. In support of these conclusions, an empirical regression model was constructed using national-level data from 2015 to 2024. The results reveal that the strongest positive determinant of renewable energy share in electricity capacity is GDP per capita, highlighting the critical role of economic capacity in enabling energy transition. The ICT Development Index also demonstrates a moderate positive correlation with renewable energy penetration, confirming that digital transformation supports decentralized energy infrastructure. Conversely, institutional policy support showed a negative correlation over time, suggesting that the effectiveness of policies may be context-dependent or delayed in impact.

In the next chapter, “[Competency-Based Approach as a Tool for Forming Sustainable Business Ecosystems: Case Studies on the Quality of Education Towards Sustainable Development of Territories](#)”, the authors study the application of the competency-based approach to the problem of forming sustainable business ecosystems in the context of distance education in Ukraine. The quality of education received is studied. For this purpose, a sample of 46 employers (35% female and 65% male aged 35–55 years) and 174 undergraduate and graduate students (55% female and 45% male aged 17–35 years from Lutsk National Technical University, Ukraine) was studied. As a result of the study, 4 main factors were identified that determine the presence of competence in candidates and are important to employers in the context of Ukraine. These are digital competence (F1), language competence (F2), competence in sciences, technologies, engineering, and mathematics (F3), and personal, social, and learning competence (F4). These factors explain 79.4% of the variance of the indicators for level of competence (1st factor—15.5%, 2nd—15.6%, 3rd—32.1%, 4th—12.2%). The results of this chapter will be important for building business processes in creating international companies at the stage of reconstruction of Ukraine after the end of the war.

The next chapter, “[Financial Health Assessment of Ukrainian Banks: A Logistic Regression Model for Bankruptcy Prediction](#)”, explores the use of statistical models to evaluate the financial health of Ukrainian banks. A large number of statistical models developed by Ukrainian scientists were analysed. This part of the paper may be of interest to foreign scientists. The analysis led the authors to conclude that logistic regression is a universal and statistically reliable technique for analysing binary dependent variables. To construct the logistic regression model, 22 indicators were initially calculated and distributed into groups such as liquidity, solvency, financial stability, profitability and business activity (data taken from the official website of the National Bank of Ukraine). Of these, only two relative indicators are critical: (1) administrative and other operating expenses/assets and (2) interest income/assets. This chapter provides a statistically correct mathematical formula for calculating the level of a given bank’s health. The results could be significant for foreign companies planning to engage in financial and innovation activities in Ukraine.

Further chapter “[Development of ESG Reporting Under the Influence of Theoretical Concepts](#)” draws attention to the impact of the theoretical context change on the structure and content of ESG enterprises’ reporting. Within the framework of the sustainable development paradigm, such reporting informs stakeholders about the potential and actual consequences of the enterprise’s activities for the environment and society. The authors analyzed sustainable development reports and identified their universal components. They proposed a set to be considered as central to the concept of sustainable development. The interests of potential stakeholders of the enterprise, which are focused on individual aspects of ESG reports, are described, which may be interesting for participants in the relevant negotiations. The authors make propositions for the life cycle of legitimacy in the context of ESG reporting, characteristics of types of institutional isomorphism, influencing factors on ESG reporting according to stakeholder, legitimacy, and institutional theories, and criteria for evaluating the ESG reporting process. The results of the chapter may be interesting for the formation of databases for relevant reports and developers of analytical tools.

The next chapter “[Strategies for the Development of Responsible Tourism to Support a Sustainable Tourism Ecosystem](#)” aims to identify current trends in the development of responsible tourism and assess the effectiveness of its impact on the preservation of a sustainable tourism ecosystem. The authors draw attention to the fact that the organization of the Ukrainian tourism industry in the post-war period should be reformed using modern methods. Having analyzed the existing systems of tourism organizations in developed countries, the authors developed a business process that takes into account the existing capabilities of Ukraine. The databases involved in the implementation of this business process are highlighted. The results may be important for Ukrainian and international financial investors, start-ups, and developers of information products that support activities in the field of responsible tourism and the sustainable tourism ecosystem.

The subsequent chapter, “[Digital Agro-Innovations as a Strategic Tool for Agricultural Adaptation](#)”, explores directions for forming a practical and conceptual basis for national strategies, digital transformation programs, and agricultural adaptation policies in the context of climate change and post-war economic reconstruction. Particular attention is given to agrodrones due to their high precision, operational flexibility under variable climate conditions, and multifunctionality in monitoring, spraying, and geospatial mapping. The proposed integrated digital agroecosystem framework establishes a foundation for building climate-resilient, resource-efficient, and environmentally balanced agricultural models. Importantly, this study introduces a novel systems-based approach by conceptualizing the synergistic interaction between digital tools as a comprehensive system. The chapter may be of considerable interest to researchers and developers in Europe, the USA, and other developed countries. Here, a projection of the unique experience gained in Ukraine in the use of drones in wartime conditions onto the conditions of agricultural production is presented. Therefore, the results of this chapter will be important for intensifying the formation of a system of innovative start-ups in agriculture in developed countries.

The next chapter, “[The Impact of War on Foreign Direct Investment in Ukraine: An Empirical Assessment Using the Synthetic Control Method](#)”, explores the use of the synthetic control method to estimate the causal impact of the war against Ukraine on foreign direct investment inflows between 2014 and 2021. The period covered was the one in which limited military actions took place in Ukraine (which served in a certain sense as a backdrop), and national political and economic processes also had a significant impact. The paper compares foreign direct investment inflows between Ukraine and 27 countries using 33 predictors (for the placebo test). The comparison of actual and projected trajectories of foreign direct investment inflows indicates heightened sensitivity of foreign investors to the social, political, and economic stability of the country, with investment patterns responding to perceived instability in Ukraine. The results obtained may be important for the formation of databases for foreign direct investment inflows to study the impact of national social and economic processes on them.

The chapter, “[Modelling the Socio-economic Impacts for the Coal Mining: Forecast of Hazardous Properties of Coal Seams](#)”, examines strategic directions for the development of coal mining management technologies that minimize environmental impact and enable a substantial reduction in environmental costs. The chapter proves that the current regulatory framework requires substantial revisions to accommodate the integration of modern coal mining technologies. Particular emphasis is placed on the necessity of enhancing safety standards in mining operations. The introduction of new methods for forecasting the hazardous properties of coal seams and improving the regulatory framework for safe mining operations will increase the socio-economic efficiency of investments in improving the miners’ working conditions. The introduction of an investment support mechanism for the innovative development of coal enterprises, together with open innovation systems and the growth of social capital, will foster partnership relationships between public and private organizations in the innovation sphere, enabling coal industry enterprises to reach a qualitatively higher stage of development.

In the chapter titled “[Managing the Transition Phase Between Stages of the Manufacturing Enterprise Lifecycle as a Key Growth Driver](#)”, the authors developed a model for managing the transition phase between stages of the manufacturing enterprise life cycle. An objective function of the model is an optimization problem based on four key factors of the transition phase, including capital intensity (cost), duration, system ability to change, and external conditions under which the transition is carried out. The model also includes three management parameters. The model is designed to identify optimal conditions for starting the transition phase between stages of the life cycle. The nature of changes in the management system as a response to increasing loads associated with the transition is analyzed in detail. A decision-making model is described to determine the optimal conditions for starting the transition phase based on key influencing factors. The results obtained will be important for the development of analytical information technologies focused on the formation of databases for optimizing business management in a turbulent environment.

In the chapter, “[Method for Competitive Funding Optimization of Scientific Research and Development in Universities and Scientific Institutes \(with Discussion for Ukraine\)](#)”, the authors propose methods for competitively optimization in state funding of higher education institutions and scientific institutions. The optimization for scientific institutions is performed based on indicators of the effectiveness of their scientific activity. The number of citations of scientists actively working at this institution was chosen as an objective indicator for the results of scientific activity. For this, the authors used the Scopus and WoS databases, which have powerful analytical apparatus. Methods have also been developed for cleaning indicators of scientific activity from deception. The optimization of higher education institutions additionally includes indicators of the effectiveness of graduate training, namely the number of graduates who are involved in participating in the economic activity of the state within six months after graduation. The system of conditions for evaluating the optimality of state financing is proposed. The application of the obtained results will allow the Government of Ukraine to make decisions considering a system of objective indicators for the contribution to international science and national economy made by higher education and scientific institutions.

In the context of accelerating digital transformation in agriculture, the chapter “[Mathematical Models of the Milking Processes Duration](#)” presents a significant advancement in modelling the duration of machine milking processes across various technological setups. By analysing real-world data from dairy farms in Ukraine, the authors introduce refined mathematical models that capture the stochastic nature of milking and preparation durations using chi-square and gamma distributions. These models enhance the accuracy of productivity assessment for milking equipment and support the development of intelligent information systems for dairy farm management. Special attention is given to integrating both deterministic and random variables, reflecting the operational realities of manual, semi-automated, and robotic milking systems. The chapter contributes to the broader goals of smart agriculture by offering data-driven tools that optimise farm processes and decision-making. Thus, it provides valuable insights for researchers, engineers, and agrotechnology developers working on the next generation of precision livestock systems.

The chapter titled “[Innovative Magnetosensitive Frequency Microsensors Based on Transistor Structures for Monitoring Systems in Biomedicine](#)” addresses a critical technological challenge at the intersection of microelectronics, sensor design, and biomedical monitoring—developing high-sensitivity magnetic field transducers tailored for miniaturised and data-driven systems. The authors present a novel class of magnetosensitive frequency microsensors based on double-collector transistor structures, capable of accurately converting weak magnetic fields into frequency-modulated signals. By integrating mathematical modelling with experimental validation, the study demonstrates significant improvements in sensor sensitivity, linearity, and thermal stability compared to conventional solutions. The research highlights how semiconductor physics and analogue circuit innovation can enhance data acquisition at the hardware level, crucial for real-time biomedical diagnostics and neurophysiological monitoring. Particularly notable is the compatibility of these sensors with digital signal processing systems, enabling robust performance in compact,

low-power environments such as wearable and implantable medical devices. The chapter contributes to the broader goal of building intelligent, data-centric monitoring infrastructures by ensuring that the signal chain begins with precise and reliable sensing. Through its interdisciplinary approach, this work offers valuable guidance for engineers, researchers, and system designers engaged in the development of next-generation medical and IoT sensor technologies.

The chapter “[Formation of the Concept of Investment Decision-Making in the European Tourism Sector Using Economic Indicators](#)” provides a timely, data-driven analysis of the complexities involved in investment decision-making within the European tourism sector. Through the integration of macroeconomic indicators, regional development metrics, and risk assessment models, the authors present a robust conceptual framework for evaluating tourism investments in various EU contexts. The study emphasises the importance of combining quantitative data with institutional and geopolitical factors in order to inform resilient and adaptive investment strategies. The flexible classification system for decision-making variables is particularly relevant, as it enables scenario modelling under uncertain economic and environmental conditions. This work contributes to the monograph’s broader focus on data-centric approaches, illustrating how structured economic analysis can enhance strategic planning and sustainable development in one of Europe’s most vital service industries.

The chapter “[Optical Measurement Converter “Gas Concentration–Voltage” for Environmental Monitoring of Aggressive Atmospheric Environments](#)” presents an innovative approach to environmental monitoring. This approach involves developing an optical-electronic converter that transforms gas concentration data into voltage signals for real-time analysis. Designed for application in a chemically aggressive or corrosive atmosphere, the proposed system integrates the principles of optical absorption and sensor engineering to enable the accurate, selective and durable detection of gases. Through a combination of physical modelling and experimental validation, the authors demonstrate the converter’s reliability and suitability for deployment in smart environmental infrastructure. Its compact design and long-term stability make it ideal for integration into autonomous monitoring stations, industrial safety systems and smart cities.

In the next chapter “[Formation of Pharmaceutical Enterprises’ Strategy Based on the Development of Operational Capabilities](#)”, the authors present a practical and analytically grounded approach to strategic planning in the pharmaceutical sector. They focus on the development of operational capabilities as a core driver of business resilience. Using EFAS analysis, interrelationship diagrams and ABC analysis, the authors identify external threats, such as supplier instability and competitive pressures, as well as internal inefficiencies relating to low-demand product assortments. The study proposes a structured inventory management model that optimises resource allocation and aligns product offerings with market demand by integrating these insights. The methodological toolkit outlined in this study enables pharmaceutical enterprises to formulate strategies based on objective data, thereby improving

their adaptability and cost-effectiveness. This work exemplifies the value of data-centric decision-making in a sector that is subject to dynamic regulatory, economic and consumer challenges.

In the chapter titled “[Data-Centric Innovations for Psychological Support: Integrating LLMs and Vector Search in Professional Web Applications](#)”, the authors explore the intersection of artificial intelligence and mental health by presenting an innovative digital solution designed to enhance psychological support services. The authors introduce the ‘Smart Psychologist’s Assistant’—a professional web application integrating large language models (LLMs), vector search and modern client-server technologies to support psychologists in clinical decision-making and patient communication. By combining ethical AI deployment with robust system architecture, the project ensures high functionality and strict data protection. Seamless integration of GPT-based models and PostgreSQL vector search enable real-time, context-aware support, preserving the sensitivity required in psychological practice. By aligning data-centric innovations with the needs of mental health professionals, this work offers a forward-looking model for intelligent, secure and ethically grounded digital tools in behavioural health care. It represents a meaningful step towards the development of scalable and responsible AI applications in human-centred domains.

The last chapter “[Cyber Security Manager After NIS2 Directive](#)” discusses the critical role of cybersecurity management in the era of digital transformation, emphasizing the integration of the NIS2 Directive into organizational practices. As organizations increasingly embrace digital transformation, the complexity and frequency of cyber threats have surged, necessitating a proactive and holistic approach to cybersecurity. The authors explored the evolving threat landscape, highlighting the rise of AI-driven attacks, ransomware, and Cybercrime-as-a-Service, which demand adaptive and proactive defense strategies. By focusing on the human factor, the study underscores the importance of structured, mandatory, and role-based training to address persistent gaps in cybersecurity education and awareness. Fostering a proactive security culture, enhancing communication, and embedding cybersecurity into organizational values and leadership accountability are suggested. By exploring real-world case studies, emerging trends, and actionable recommendations, this chapter provides a comprehensive roadmap for organizations to enhance resilience and align with regulatory requirements. Ultimately, this work aims to inspire a shift in organizational priorities, ensuring that cybersecurity becomes a shared responsibility and a cornerstone of sustainable digital growth.

Vinnytsia, Ukraine

Andrii Semenov
semenov.a.o@vntu.edu.ua

Vinnytsia, Ukraine

Iryna Yepifanova
yepifanova@vntu.edu.ua

Bratislava, Slovakia

Jana Kajanová
jana.kajanova@fm.uniba.sk

Contents

Innovative Potential in Ensuring Sustainable Development of the Enterprise	1
Yana Kotko, Ruslana Levkina, Oksana Kulinich, and Dmytro Levkin	
RWA Tokenization Mechanisms as a Tool for Sustainable Economic Development of Local Hromadas in Ukraine	23
Maryna Riabokin, Yevgen Kotukh, Dmytro Nikolaievskyi, and Yuliia Yefimenko	
Methodology for Assessing the State of Business Processes in Terms of Their Impact on the Economic Security of the Enterprise	43
Nataliia Havlovska, Serhii Matiukh, Mykhailo Kryvdyk, and Yevhenii Rudnichenko	
The Strategic Potential of Renewable Energy Microgrids: Global Transformations and Determinants of Sustainable Development	67
Volodymyr Lypov, Natalya Ushenko, Viktoriia Hurochkina, Tomasz Szapiro, Larysa Shaulska, and Viacheslav Dzhedzhula	
Competency-Based Approach as a Tool for Forming Sustainable Business Ecosystems: Case Studies on the Quality of Education Towards Sustainable Development of Territories	99
Oksana Polinkevych, Olena Kuzmak, Vira Shpylova, and Ryszard Kamiński	
Financial Health Assessment of Ukrainian Banks: A Logistic Regression Model for Bankruptcy Prediction	121
Larysa Zomchak and Andriana Seniv	
Development of ESG Reporting Under the Influence of Theoretical Concepts	137
Iryna Zamula, Daria Seletska, Olha Fedorova, and Maryna Tanasiieva	

Strategies for the Development of Responsible Tourism to Support a Sustainable Tourism Ecosystem	167
Iryna Kyryliuk, Viktoriia Krasnomovets, Olena Sushchenko, Oksana Lytvyn, Liudmyla Chvertko, and Liudmyla Neshchadym	
Digital Agro-Innovations as a Strategic Tool for Agricultural Adaptation	187
Tetiana Shabatura, Anush Balian, Margarita Stepanova, Alla Yakovenko, and Olga Petrenko	
The Impact of War on Foreign Direct Investment in Ukraine: An Empirical Assessment Using the Synthetic Control Method	205
Marianna Kichurchak, Iryna Paslavska, Nataliia Datskiv, and Vasyl Antoniv	
Modelling the Socio-economic Impacts for the Coal Mining: Forecast of Hazardous Properties of Coal Seams	233
Yevhen Rudniev, Julia Romanchenko, Kostiantyn Sokolenko, Vadym Tarasov, and Valeriy Sokolenko	
Managing the Transition Phase Between Stages of the Manufacturing Enterprise Lifecycle as a Key Growth Driver	259
Iryna Yepifanova and Anton Kaspirovych	
Method for Competitive Funding Optimization of Scientific Research and Development in Universities and Scientific Institutes (with Discussion for Ukraine)	277
Anatolii Shyian and Liliia Nikiforova	
Mathematical Models of the Milking Processes Duration	295
Pavlo Kulakov, Volodymyr Kucheruk, Tetiana Neskorodieva, Olena Semenova, Roman Lishchuk, Serhii Kontseba, Anna Kulakova, and Wiktorina Mankovska	
Innovative Magnetosensitive Frequency Microsensors Based on Transistor Structures for Monitoring Systems in Biomedicine	309
Volodymyr Martyniuk, Oleksandr Osadchuk, Andrii Semenov, Iaroslav Osadchuk, and Olena Semenova	
Formation of the Concept of Investment Decision-Making in the European Tourism Sector Using Economic Indicators	339
Tetiana But, Daria Mamotenko, Olga Korniienko, Galyna Kozytska, and Nataliia Pavlenchyk	

Optical Measurement Converter “Gas Concentration–Voltage” for Environmental Monitoring of Aggressive Atmospheric Environments	363
Igor Dudatiev, Andrii Semenov, Kostyantyn Ovchynnykov, Maksym Prytula, and Serhii Maltsev	
Formation of Pharmaceutical Enterprises’ Strategy Based on the Development of Operational Capabilities	381
Inna Kuznetsova, Olha Balabash, and Yuliia Karpenko	
Data-Centric Innovations for Psychological Support: Integrating LLMs and Vector Search in Professional Web Applications	397
Inna Baraban, Serhii Baraban, Olha Voitsekhovska, Mariia Baraban, and Nazarii Shportun	
Cyber Security Manager After NIS2 Directive	425
Marián Mikolášik and Daniela Pucicov	

Innovative Potential in Ensuring Sustainable Development of the Enterprise



Yana Kotko , Ruslana Levkina , Oksana Kulinich ,
and Dmytro Levkin

Abstract The study is devoted to defining the main approaches to understanding the essence of the innovation potential of an enterprise. The paper identifies the key components of innovation potential, systematizes the factors of the external and internal environment that influence its formation, and analyzes the indicators for assessing innovation potential. The algorithm for assessing the innovation potential of sustainable development of enterprise should be used to characterize both the efficiency of the enterprise and the implementation of the innovation strategy to substantiate management decisions. When assessing the level of innovation potential of sustainable development, attention should be focused on the selected indicators, which allow not only to determine the innovation potential of an organization, but also to purposefully and timely adjust its activities based on the implementation of the most promising innovations. The modernized model of innovation potential for sustainable development of Private Agricultural Firm VATAL and Agricultural Limited Liability Company Slavutych helps the enterprise to ensure efficiency by promoting innovative approaches, introducing new technologies, and developing products and services based on the principles of sustainable development. The application of this model can help the company conserve resources, reduce environmental impact, increase competitiveness, and ensure sustainable development in the future.

Y. Kotko · O. Kulinich

Department of Economics and Business, State Biotechnological University, Kharkiv, Ukraine

e-mail: kotkoyana@btu.kharkov.ua

O. Kulinich

e-mail: O.Kulinich@btu.kharkov.ua

R. Levkina (✉)

Department of Entrepreneurship and Trade, Odessa National Polytechnic University, Odessa, Ukraine

e-mail: levkina.r.v@op.edu.ua

D. Levkin

Department of Physics and Mathematics, State Biotechnological University, Kharkiv, Ukraine

e-mail: valoi@btu.kharkov.ua

Keywords Innovation potential · Model of innovation potential · Sustainable development · Business entity · Marketing potential · Human resources potential

1 Introduction

Modern global socio-economic processes have gained such speed and intensity that we can talk about the onset of a new era. This era is marked by numerous transformations in the global economy, including changes in the formation of centers of influence on important decision-making, as well as new approaches and tools for managing states and individual companies in an unstable environment.

Unpredictable events and their consequences have led to a decline in economic activity at both the macro and micro levels, triggering negative trends that are of a crisis nature. The existing uncertainty may actually impede the sustainable innovation development of many regions and enterprises. This can hinder the introduction of new ideas and technologies that can improve the performance of companies and contribute to the development of regions. In order to achieve sustainable development, it is necessary to create favorable conditions for innovative thinking and activities, as well as to encourage openness to change and innovation.

Rapid technological transformation and globalization have led to the need for new tools and approaches to business management. Traditional methods often fail to respond effectively to new challenges and may not be sufficient to achieve success. In such circumstances, businesses need to combine traditional approaches with new ones to ensure stability and competitiveness. It is important to develop new skills and use modern technologies to achieve their goals. Understanding market needs, using analytical data and innovations can help businesses successfully cope with modern challenges.

One of the key global challenges that influenced the formation of the model of the innovative potential of sustainable development of an enterprise was the coronavirus pandemic, and today it is a full-scale war. These events have forced all sectors of the economy to maximize the use of innovative technologies to minimize the negative impact on their operations. The escalation of military aggression has increased the signs of aggressiveness and unpredictability in the economic sphere and emphasized the need to study the formation of a model of the innovative potential of sustainable development of an enterprise in the context of global instability. Thus, the study of scientific approaches to the formation of a model of the innovative potential of sustainable development of an enterprise in modern conditions is extremely relevant, and this research has prospects for further development.

2 Formation of the Problem

Innovation capability is a key element for ensuring sustainable development and enhancing a company's competitiveness. The problem of formation and development of innovation potential is to identify strategies and mechanisms that will help create favorable conditions for innovative growth of entrepreneurial activity.

The problem of formation and development of the model of innovation potential for sustainable development is the need to stimulate the creation and implementation of new ideas, technologies and development of new products in the eco-economy. For this purpose, it is important to promote scientific and technological progress, support research and development of innovations in various fields of activity.

To build innovation capacity, it is important to develop the innovation infrastructure, introduce innovative approaches in management, production, and marketing, and attract qualified personnel and investments in science and technology. It is also necessary to participate in incentive programs and support projects from the state to promote innovation development, as well as to establish cooperation and exchange of knowledge and ideas with government agencies, business and research institutions.

3 Analysis of Publications and Research

Determining the essence and components of an enterprise's innovation capacity is an important issue in the research of both foreign and Ukrainian scholars Obidenova T., Chikov [1], Yelnikova G., Kozlovska [2], Dudneva et al. [3]. These scientific works help to understand more deeply what factors contribute to the development of innovation processes and how they can be effectively implemented. It is important to take into account the innovation potential when developing development strategies and planning the activities of companies or organizations.

The issues of formation and development of the innovative economy and formation of the innovative capacity of enterprises are an important topic for scientific research. Economists Moisa [4], Dalyk and Yarmus [5] consider this issue from different points of view, analyzing the impact of innovation on economic development, looking for new approaches to stimulate innovation and assessing the effectiveness of innovation potential.

In particular, the research of Boychuk [6], Gonchar V., Dolobanko et al. [7], Pashchenko V. helps to identify the main reasons that promote or hinder innovative development, determine the strategies and policies necessary to increase innovation activity, and develop new methods for assessing innovation potential. The results of such research are important for solving economic problems and contribute to the creation of sustainable and competitive development of the national economy.

Panchyshyn and Vdovin emphasize the need for balanced development of regions aimed at ensuring proper living conditions for residents, integrated development and increasing the competitiveness of the economy. This may include the allocation of

resources and investments to provide all regions of the country with conditions for sustainable development and improving living standards [8].

According to Petroliuk and Hrebenyuk [9], Makukh et al. [10], the concept of sustainable development involves the balanced use of natural, economic and social resources to meet the needs of the modern generation without jeopardizing the ability of future generations to meet their own needs. Sustainable development of the enterprise contributes to the formation of a balanced environment where there are sufficient resources to meet the needs of the population, and also contributes to the comprehensive development of the enterprise, society and the environment.

We agree with Kifiak [11] and Korobka [12] that sustainable development of an enterprise is a very important aspect in modern business. This means ensuring a balance between economic productivity, social responsibility and environmental protection. The company should work to ensure stable profits, satisfy the needs of employees and customers, and protect the environment. In general, sustainable development contributes to the competitiveness and long-term success of a business.

Theoretical, methodological, and applied issues in the field of these studies are indeed very important. Accordingly, their solution can improve the quality of scientific results and open up new prospects for economic growth. This is an important task that requires attention and study by relevant scientists. In addition, these studies are theoretical in nature.

We have also studied various aspects of sustainable development of business entities, which are related to theoretical and applied issues in the field of international economic relations, agricultural education, strategic management and state support [13]. However, there are still issues that have not yet been studied, such as the development of a model of innovative potential for sustainable development of agrarian business entities.

The study of the model of innovation potential of sustainable development of an enterprise can be an important step in increasing its productivity and competitiveness. This will allow developing strategies that will promote sustainable business development, reducing the negative impact on the environment and society. Ways to implement this model contribute to the introduction of the latest technologies, optimization of business processes and the search for new opportunities for growth. Accordingly, our study has practical value; integration of its results into the strategic planning of enterprises Private agricultural company VATAL and Agricultural Limited Liability Company Slavutych is a key step towards improving their business operations.

4 Objective of the Study

Innovative capacity models play an important role in ensuring sustainable growth and business success. By implementing sustainable practices in their operations, companies can not only reduce their environmental impact but also create long-term value for their stakeholders. The main strategies for building a model of sustainable development innovation potential include investments in renewable energy sources, the

introduction of green technologies, environmentally friendly approaches to design and production, and support for corporate social responsibility initiatives. By prioritizing sustainability in their business strategies, companies can stimulate innovation, increase efficiency, and build a strong reputation as a responsible corporate citizen.

To build a structure of innovation potential for sustainable development, it is necessary to create a framework that brings together new ideas, technologies and approaches to addressing environmental, social and economic challenges. This involves developing a culture of creativity and entrepreneurship, encouraging collaboration between different stakeholders, and investing in research and development. One approach is to create an innovation framework for the exchange of ideas and collaboration on sustainable solutions, which in turn can serve as an incubator for new technologies and business models that promote environmental sustainability and social inclusion. Another important aspect is to stimulate innovation through policies that support research and development in various fields, such as renewable energy, green infrastructure, and sustainable agriculture. This could include providing financing opportunities, tax breaks, and other forms of support to encourage investment in sustainable technologies.

Thus, the purpose of the publication is to form a model of innovation potential for sustainable development of an enterprise, which requires a multifaceted approach that involves creating a favorable environment for innovation, encouraging cooperation between stakeholders and stimulating investment in sustainable solutions.

5 Research Results

The study and clarification of the essential understanding of the concept of innovation potential is based on the resource, performance and systemic-complex approaches, as well as on the definition of aspects of sustainable development in the structure of innovation potential (Fig. 1) [1, 2].

The innovative potential of sustainable development of an enterprise is the ability and capacity of an entity to create, develop and implement new ideas, technologies, processes, products or services that bring value and contribute to growth. It includes the resources (both tangible and intangible) available to an organization, as well as its ability to effectively use these resources to innovate and remain competitive in the market. The sustainability dimension considers the long-term impact of innovation on economic, environmental, and social factors to ensure that progress is made without compromising the ability of future generations to meet their needs. By integrating these perspectives into the analysis of innovation potential, organizations can better assess their readiness to innovate and adapt to changing market conditions, while contributing to sustainable growth and development [3].

In today's environment, it is important to introduce a new economic model focused on investment and innovation. This approach involves policies aimed at increasing investment and stimulating innovation while improving production efficiency. Today, the development of development models with an emphasis on innovation should be a

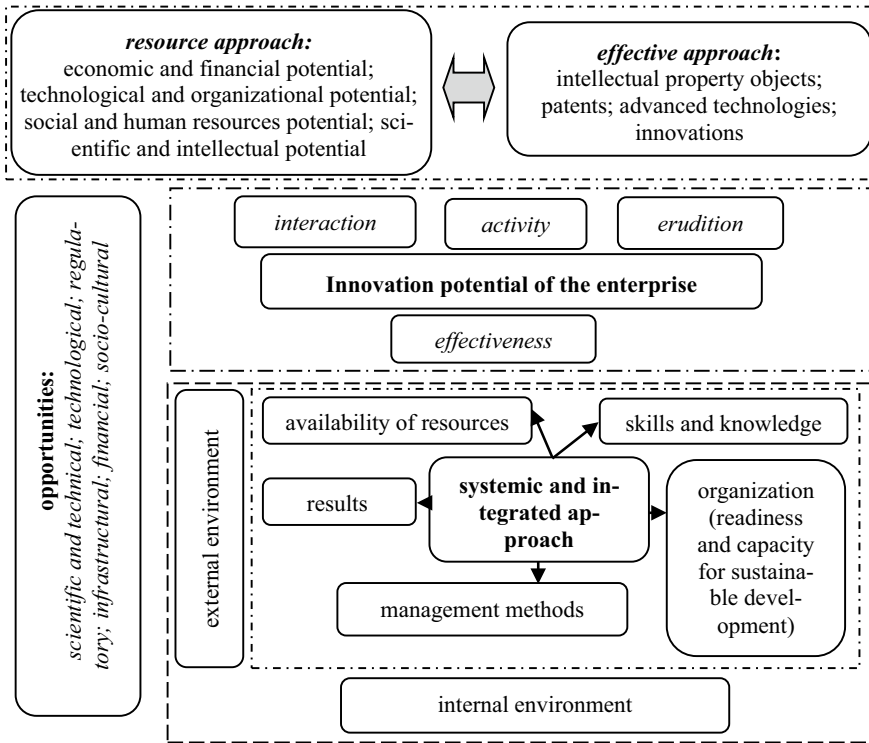


Fig. 1 Analysis of approaches to determining the innovation potential of an enterprise. *Source* Developed by the authors on the basis of [1, 2]

top priority, as it is essential for the modernization of both the economy and society as a whole. This approach requires strategic planning and the desire to promote progress through technological development [4].

Today, the importance of sustainable development goals as an innovative model to be achieved through the interaction of all stakeholders, taking into account relevant modern changes and influences, is significant. In general, it is necessary to ensure economic growth, social security and environmental protection at levels not lower than in developed countries. Sustainable development is crucial for ensuring a balanced and harmonious future for all. Therefore, an innovative model of sustainable development of an enterprise should take into account key components and be implemented in close cooperation with all stakeholders of the process. Sustainable development requires a holistic approach that encompasses economic, social and environmental aspects, and also involves local communities, businesses and government agencies in joint efforts to achieve long-term prosperity and well-being for all stakeholders [5].

Globalization processes and rapid technological changes really require a transition to a model of innovative potential for sustainable development of an enterprise.

Failure to respond to these changes in a timely manner can have a real negative impact on economic development. It is important to introduce new management methods and use modern technologies for the formation and implementation of innovative development models [6].

The concept of the model of innovative potential of sustainable development of an enterprise is a combination of strategies and methods aimed at ensuring the constant growth of the enterprise while preserving natural resources and environmental stability. This model takes into account the needs of the modern market, including the requirements for sustainable development and business efficiency, and also contributes to the competitiveness of the enterprise at various levels [7]. An important element in the formation of the model of innovation potential for sustainable development is the involvement of the social system, in particular, the consumer and user interact with enterprises to promote the successful formation and implementation of innovations and support innovative approaches. Such a partnership between the social system and enterprises can contribute to the successful development and implementation of innovative models in the activities of the enterprise [8]. The formation of a model of the innovation potential of sustainable development of enterprises is influenced by such features as sustainability factors (social, economic, environmental), causes, resources (technological, human, information, financial, material, natural, investment, etc.) and key characteristics (Fig. 2). These factors are crucial in creating the basis for promoting innovation and sustainable development of enterprises. By understanding and utilizing these elements, enterprises can increase their ability to adapt, grow, and succeed in today's rapidly changing business environment [9].

Social, economic, and environmental factors are important prerequisites for the sustainable development of an enterprise. Social factors include ensuring decent working conditions, opportunities for professional and personal growth, maintaining a work-life balance, reducing emissions and environmental impact, using environmentally friendly technologies and materials, participating in social programs, sponsoring cultural and social events, ensuring transparency in interaction with the community, providing opportunities for training and self-improvement of employees, developing leadership and teamwork. These social aspects help an enterprise build a sustainable business that benefits both the enterprise and society as a whole [10].

Economic factors involve the optimal use of resources, such as labor, capital, technology, and natural resources, which is key to the sustainable development of an enterprise. The introduction of new ideas and technologies can help an enterprise maintain its competitiveness in the market and ensure sustainable growth. The ability of the enterprise to adapt to changes in the economic environment and respond quickly to new opportunities or threats is also important. The company must be aware of its responsibility to the community, stakeholders and the environment, as well as be able to manage finances and ensure stable profits for further development. These factors interact with each other and contribute to the sustainable development of the enterprise in the long term.

Environmental factors include the rational use of natural resources, their restoration and integration into the production cycle to reduce waste and efficient use of

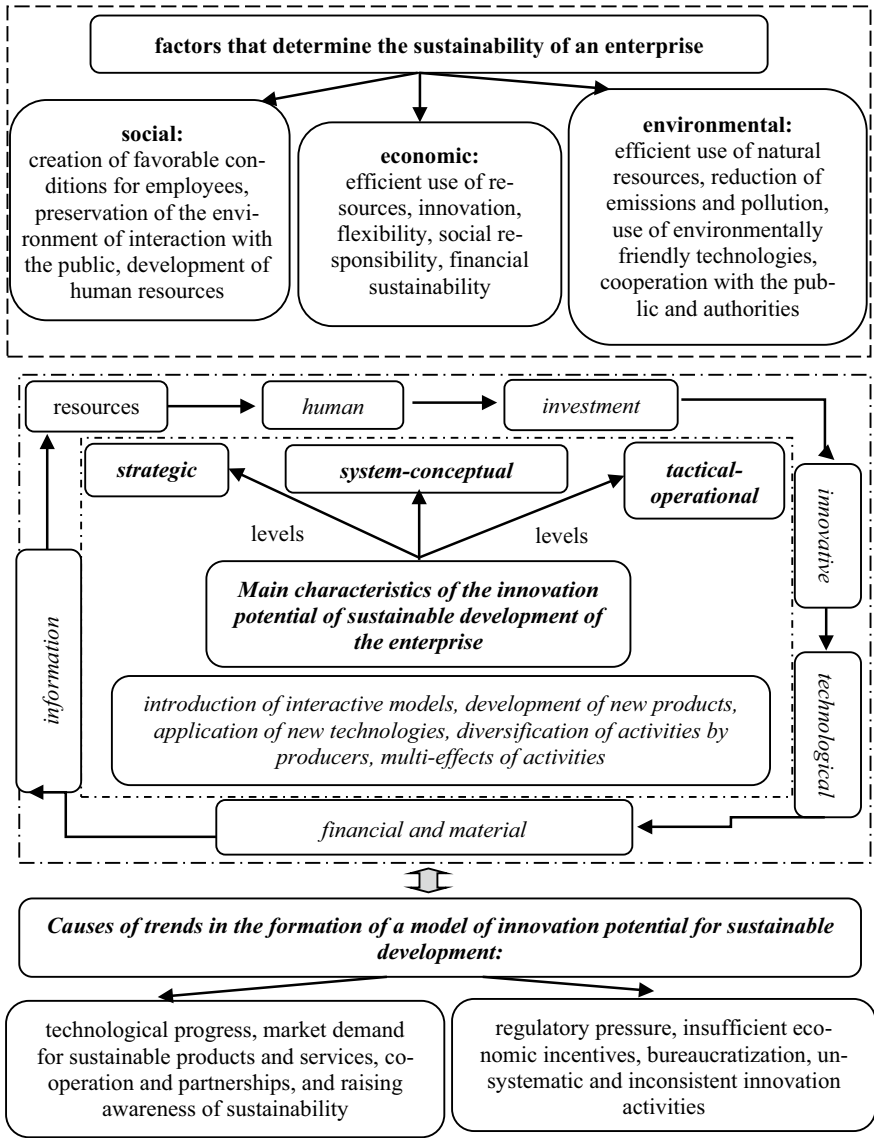


Fig. 2 Peculiarities of forming a model of innovative potential of sustainable development of an enterprise. *Source* Developed by the authors on the basis of [9, 10]

resources. The company must comply with the requirements for limiting emissions of pollutants into the environment and implement wastewater and air treatment technologies. It is important to improve technologies to reduce the negative impact on the environment, including the use of renewable energy sources and waste management. Businesses should be open to communication with the public and the authorities to

jointly find optimal solutions in the field of environmental activities. Environmental issues should be considered as an integral part of sustainable development, balanced with economic and social indicators [11].

In addition, the reasons for the relevance of forming a model of innovation potential for sustainable development may include: growing awareness of the importance of sustainable development for solving global problems such as climate change, resource depletion and social inequality. This awareness is driving the need for innovative solutions that can support sustainable development; consumers are increasingly preferring goods and services that are environmentally friendly and socially responsible. This demand creates opportunities for businesses to innovate and develop sustainable solutions to meet market needs; government regulation and incentives can encourage businesses to adopt more sustainable practices and technologies. This can create a favorable environment for innovation in the field of sustainable development; technological advances, such as renewable energy sources, smart technologies, and sustainable materials, open up new opportunities for innovation in the field of sustainable development, etc.

In addition, the formation of a model of innovation potential for sustainable development may face various challenges (Fig. 3). Some of them include insufficient funding for research and development, insufficient human resources, bureaucratic obstacles, unstable political situation, and lack of interaction between research institutions, industry, and government. To overcome these problems, it is important to create a favorable innovation environment that will stimulate the development of innovative ideas and their implementation.

Various research methods can be used to study the problems of formation and prospects for the development of the model of innovation potential of sustainable development of an enterprise. One of the approaches is to analyze scientific publications and conduct empirical research among the employees of an enterprise.

In addition, it is important to monitor and analyze innovative technologies and trends in the industry, as well as to cooperate with other enterprises and research organizations to share experiences and better understand the potential for innovation. In general, a systematic approach to research, combined with active cooperation and analysis of the current market situation, will ensure the formation and development of a model of the innovation potential of sustainable development of the enterprise [13].

The innovative potential of sustainable development of an enterprise is an important element of the strategy of modern business. The ability of an enterprise to innovate allows not only to ensure competitive advantages but also to guarantee the sustainability of its development in a changing market environment. The practical value of the innovation potential model lies in the ability of an enterprise to introduce new ideas, technologies and processes to improve productivity and meet market needs [14].

In our opinion, it is important to demonstrate the practical value of the model of innovation potential of sustainable development on the example of the enterprises Private Agricultural Firm VATAL and Agricultural Limited Liability Company

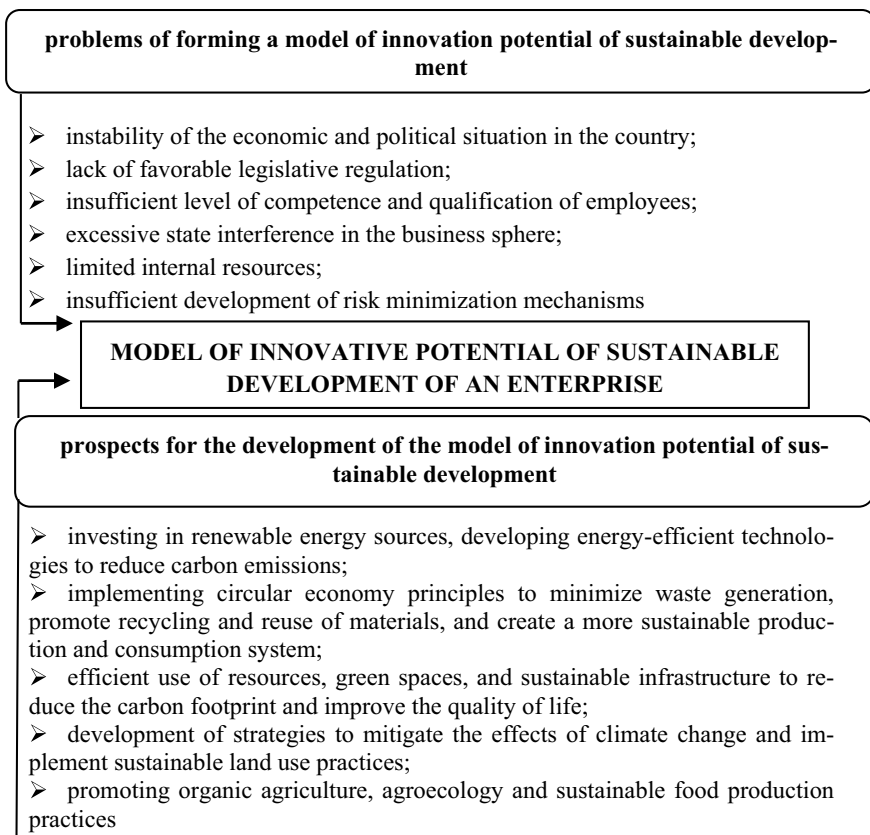


Fig. 3 The main prerequisites for the formation of a model of innovation potential for sustainable development of an enterprise. *Source* Developed by the authors on the basis of [11, 12]

Slavutych (Fig. 4). The concept of sustainable development is important for enterprises in the modern world. This means that companies must take into account the economic, social and environmental impact of their activities. Sustainable development implies that companies should give preference to the production and use of products that have a lower environmental impact, support social programs and set high standards in the area of corporate responsibility. This is important to ensure the long-term success of the enterprise, while also contributing to the protection of the environment and improving the quality of life of employees.

Building a stable foundation for the growth of Private Agricultural Firm VATAL and Agricultural Limited Liability Company Slavutych is essential for their long-term success. To ensure sustainable development, it is important to implement efficient business practices, promote innovation, and adapt to changing market conditions. These enterprises try to plan carefully (develop a detailed business plan with clearly defined goals and strategies); conduct financial control (monitor costs and profits

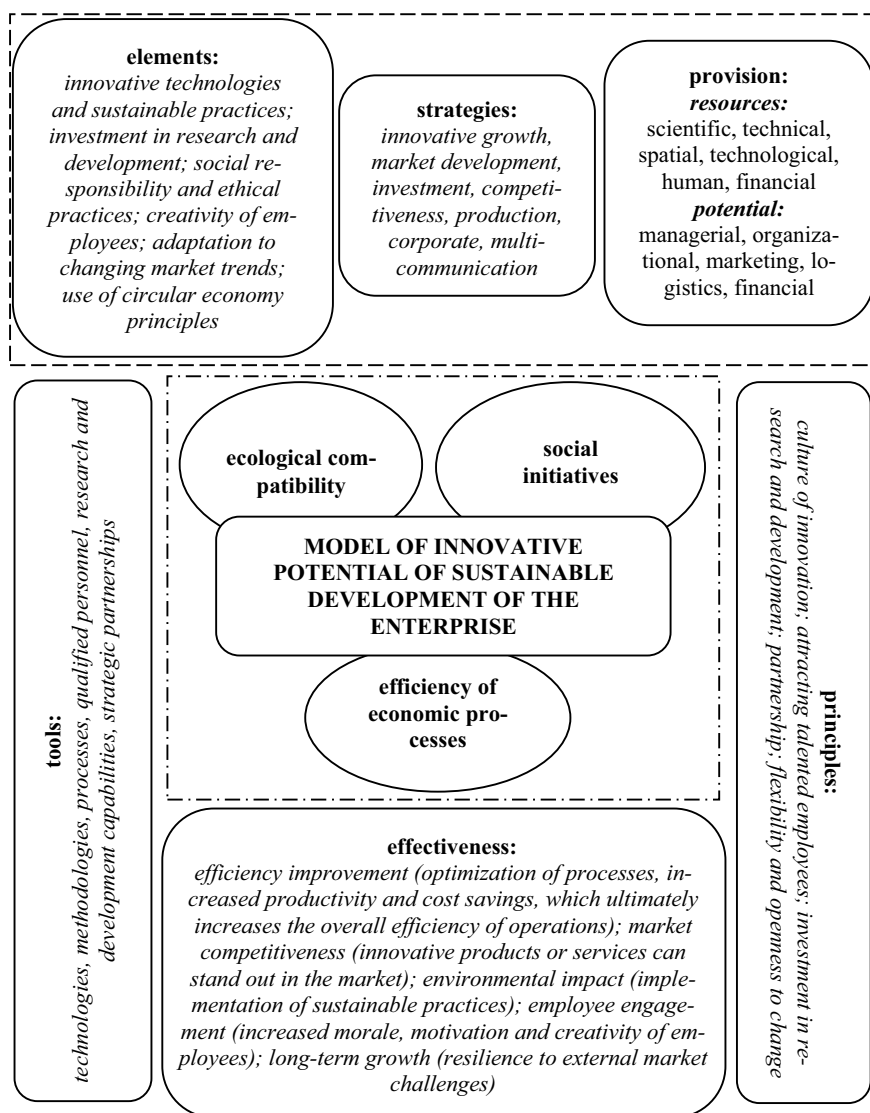


Fig. 4 Modernized model of innovative potential of sustainable development of Private Agricultural Firm VATAL and Agricultural Limited Liability Company Slavutych. *Source* Developed by the authors on the basis of [13, 14]

to prevent financial problems); diversify production (expand the range of products or services to reduce risks and ensure sustainability); and have an effective human resources management system.

Despite the steps taken, they are not enough to make these companies more sustainable. However, it is important to continue working on sustainable development,

looking for new opportunities and strategies. It is important to analyze the results and make the necessary changes to achieve the goals. Therefore, in our opinion, it is advisable to introduce a modernized model of the innovation potential of sustainable development Private Agricultural Firm VATAL and Agricultural Limited Liability Company Slavutych.

To increase the innovative potential of sustainable development of Private Agricultural Firm VATAL and Agricultural Limited Liability Company Slavutych, it is important to introduce innovative technologies, promote cooperation between scientists and business, and ensure the application of innovative practices. This approach will help to promote sustainable development of not only the activities of enterprises but also society and increase the country's competitiveness.

Thanks to the implementation of this modernized model of innovative potential for sustainable development, the enterprises of Private Agricultural Firm VATAL and Agricultural Limited Liability Company Slavutych improve their productivity, reduce the negative impact on the environment and increase their competitiveness in the market. This helps ensure the sustainability and successful operation of the business in the future.

In order to understand how the modernized model of the innovative potential of sustainable development at the enterprises of Private Agricultural Firm VATAL and Agricultural Limited Liability Company Slavutych can stimulate the solution of environmental problems and promote sustainable development, we studied it in the context of modern requirements for the development of society, as well as conducted a comparative analysis between the indicators of enterprises in accordance with the principles of traditional development and the forecast indicators in accordance with the implementation of the model of innovative potential of sustainable development in practice (Table 1).

Table 1 highlights in bold the values of indicators that are the result in each group with the resulting sum of group indicators. Thus, the integral indicator of human resource potential and its values, which represent the total sum of the calculated values: the internal staff mobility coefficient, the employee qualification coefficient, the staff turnover coefficient, the employee education index, the work experience index, and the job satisfaction index at the enterprises PA 'VATAL' and STOV 'Slavutych'. The significance of the impact of each of the coefficients and indices in the sum is equal to 1. The integral indicator of the financial and property potential of enterprises, the integral indicator of marketing potential, the integral indicator of management potential, and the integral indicator of production potential are calculated in the same way. The comprehensive integral indicator of innovation potential is the sum of the integral indicators obtained, multiplied by the integral assessment indicators.

The assessment was carried out using indicators related to the innovation activities of the enterprise: the level of implementation of new technologies (including highly efficient production processes and the use of innovative products); knowledge and innovation management (creating a favorable environment for innovation and effective knowledge management in the organization); cooperation with partners

Table 1 Assessment of the innovative potential of sustainable development of Private Agricultural Firm VATAL and Agricultural Limited Liability Company Slavutych

Indicator	Indicator of integral assessment	PAF VATAL			ALLC Slavutych		
		Value of the indicator	Significance (impact)	Total amount	Value of the indicator	Significance (impact)	Total amount
Integral indicator of human resources potential	0.2	–	1	0.77	–	1	0.61
Coefficient internal staff mobility		0.03	0.1	0.00	0.04	0.05	0.002
Coefficient of employee qualification		0.87	0.2	0.17	0.83	0.15	0.13
Coefficient of staff turnover		0.1	0.05	0.01	0.19	0.1	0.02
Index of education in employees		0.92	0.2	0.18	0.49	0.2	0.1
Index of work experience		0.89	0.2	0.18	0.67	0.25	0.17
Index of job satisfaction		0.9	0.25	0.23	0.81	0.25	0.20
Integral indicator of financial and property potential	0.05	–	1	0.82	–	1	0.61
Coefficient of assets turnover		1.15	0.1	0.12	0.49	0.1	0.05
Coefficient of financial autonomy		0.86	0.2	0.17	0.97	0.15	0.16
Coefficient of equity turnover		1.4	0.15	0.21	0.53	0.1	0.05
Coefficient of financial dependence		1.17	0.2	0.23	1.03	0.25	0.26
Coefficient of financial risk		0.17	0.2	0.03	0.03	0.25	0.01
Coefficient equity maneuverability		0.38	0.15	0.06	0.63	0.15	0.1

(continued)

Table 1 (continued)

Indicator	Indicator of integral assessment	PAF VATAL			ALLC Slavutych		
		Value of the indicator	Significance (impact)	Total amount	Value of the indicator	Significance (impact)	Total amount
Integral indicator of marketing potential	0.25	–	1	0.43	–	1	0.50
Market share		0.01	0.25	0.003	0.01	0.2	0.02
Coefficient of profitability of product sales		0.23	0.25	0.06	0.24	0.25	0.06
Growth rate of product sales		0.98	0.1	0.1	1.11	0.15	0.17
Share of sales expenses		0.18	0.15	0.03	0.39	0.2	0.08
Material efficiency		0.96	0.25	0.24	0.97	0.2	0.19
Integral indicator of managerial potential	0.2	–	1	0.58	–	1	0.55
Coefficient of compliance with the position		0.26	0.3	0.08	0.21	0.4	0.08
Coefficient of efficiency of management decisions		0.71	0.7	0.5	0.77	0.6	0.46
Integral indicator of production potential	0.3		1	0.27		1	0.28
Share of annual product renewal		0.1	0.25	0.025	0.07	0.3	0.02
Level of production capacity utilization		0.88	0.2	0.18	0.82	0.25	0.21

(continued)

Table 1 (continued)

Indicator	Indicator of integral assessment	PAF VATAL			ALLC Slavutych		
		Value of the indicator	Significance (impact)	Total amount	Value of the indicator	Significance (impact)	Total amount
Share of costs for attracting innovations		0.1	0.25	0.03	0.08	0.2	0.02
Coefficient of profitability of innovations		0.15	0.3	0.05	0.13	0.25	0.03
Comprehensive integrated indicator of innovation potential	1	–	–	0.50	–	–	0.47

and stakeholders (active cooperation with other enterprises, government organizations and the public); sustainable development strategy (including environmental protection) [14, 15].

To evaluate the model of innovative development of sustainable development, an easy-to-use integral method was used, for the implementation of which there is enough available information. The possibility of unambiguous interpretation of the results obtained is beyond doubt [15].

The integral indicator of the innovative potential of sustainable development of a company is calculated by the formula:

$$CIIP = I_{hrp} + I_{fpp} + I_{mp} + I_m + I_p,$$

where, CIIP—is a comprehensive indicator of innovation potential for sustainable development;

I_{hrp} —is an integral indicator of human resources potential;

I_{fpp} —is an integral indicator of financial and property potential;

I_{mp} —is an integral indicator of marketing potential;

I_m —is an integral indicator of managerial potential;

I_p —integral indicator of production potential.

To determine the integral indicator of the nth potential, it is necessary to determine a number of indicators, including the coefficient of internal mobility of personnel, indices of education of employees, job satisfaction index, financial dependence coefficient, market share, coefficient of compliance with the industry, efficiency of management decisions, share of annual product renewal, etc.

After determining the indicators, it is necessary to determine the indicator of significance (influence) (α) of their impact on the innovation potential of sustainable development, for each integral indicator of the n th potential it should be 1, and its parts reflect the place of importance.

The significance indicator is determined before processing the enterprise's data, it allows determining the level of influence of indicators on the integral indicator of the n -th potential.

To determine the total amount of the n th potential indicator, use the formula:

$$I_{ta} = i * \alpha,$$

where, I_{ta} —is the total amount of the indicator;

i —indicator of the integral indicator of the n -th potential;

α —is the significance index.

Accordingly, the value of the integral indicator of the n th potential is determined by the formula:

$$I_{ip} = I_{an-th} + I_{an-th} + I_{an-th} \dots + I_{an-th},$$

where, I_{ip} —is the integral indicator of the n -th potential;

I_{ta} —is the total sum of the indicator, taking into account the significance.

This approach corresponds to the specifics of the functioning of agricultural enterprises and the manageability of the management system. In accordance with the structure of innovation potential management, when determining the integral indicator, not only its financial and economic indicators were taken into account, the structure of financial and property status, staffing, marketing policy were studied and analyzed, and the main business processes were diagnosed. Thus, we have selected a number of indicators that affect the innovative potential of these enterprises and are indicators of their problems in terms of innovative development. The numerical values of the indicators allow us to focus on possible risks, to plan a set of preventives and minimizing measures in case of unforeseen events.

The analysis of Table 1 allowed us to conclude that Private Agricultural Firm VATAL uses the innovative potential by 50%, and Agricultural Limited Liability Company Slavutych—only by 47%. The greatest importance for Private Agricultural Firm VATAL in the structure of innovation potential is human resources and management potential. The human resources potential is 0.15 or 30% of the complex integral indicator, and for Agricultural Limited Liability Company Slavutych—0.12 or 26%, which indicates the ability of enterprises to attract, develop and retain highly qualified employees, which is a key resource for achieving sustainable goals, because effective management of human resources potential ensures optimal productivity and competitiveness of the enterprise. The management potential for Private Agricultural Firm VATAL is 0.12 or 24%, and for Agricultural Limited Liability Company Slavutych—0.11 or 23%, which indicates the ability of the management of the enterprises to effectively organize and mobilize resources, manage processes and make strategic

decisions to achieve the set goals, and also covers the ability of managers and staff to effectively cooperate to achieve sustainable goals.

For Agricultural Limited Liability Company Slavutych, the marketing potential plays an important role in the development of the innovative potential of sustainable development and amounts to 28% or 0.13, which indicates the availability of opportunities and resources to promote goods or services in the market, attract customers, establish communication with the audience, develop a sales strategy and increase brand awareness. For Private Agricultural Firm VATAL, the marketing potential is 22% and allows to partially achieve a competitive advantage and increase the company's profit. The financial and property potential, which is 8% (0.04) and 6% (0.03), respectively, and the production potential, which is 0.08 (16%) and 0.08 (17%), are of the lowest importance for Private Agricultural Firm VATAL and Agricultural Limited Liability Company Slavutych, which may indicate possible instability of activity and backwardness of production, insufficient investment activity in innovation, as well as potential problems in financial stability and possible instability in the future. Therefore, to optimize the use of the innovation potential of sustainable development, it is important to make efforts to increase it, possibly by introducing new strategies or approaches to the innovation process.

In order to increase the complex integral indicator of the innovation potential of sustainable development, we believe that more attention should be paid to financial, property and production aspects, which will ensure sustainable development and increase competitiveness. Some opportunities include attracting investment in the production of environmentally friendly products, developing new technologies, and supporting startups in this area. In our opinion, it is necessary to increase the indicators of the production potential, especially for Private Agricultural Firm VATAL: the share of annual product renewal should be increased by 20%, the level of production capacity utilization by 8%, the share of costs for attracting innovations by 2 times and the innovation profitability ratio by 2 times.

For Agricultural Limited Liability Company Slavutych, it is necessary to increase the share of annual product renewal by 50%, the level of production capacity utilization by 15%, the share of costs for attracting innovations by 50% and the return on innovation by 67% (Table 2). The substantiation of such recommendations is based on the results of our studies, which are indirectly related to this publication, but complement it and allow us to move into the plane of designing the enterprise's activities and forecasting indicators [16–18]. For example, Refs. [16, 17] propose biotechnological methods to increase the efficiency of livestock production and, accordingly, the enterprise as a whole. Such innovations, and biotechnological methods are innovative and will remain so for a long time, are aimed at rapid growth of cattle with special productive characteristics (milk yield, live weight, disease resistance, etc.).

Table 2 shows the projected indicators of innovation potential, calculated using the same principle. The first four indicators, highlighted in bold, have not changed and have been transferred from Table 1. The indicators of the integral assessment (their significance) in the formation of a comprehensive integral indicator of innovation potential have also remained unchanged. The integral indicator of production potential has changed, since the innovations proposed for the enterprises under study

Table 2 Projected indicators of innovation potential of Private Agricultural Firm VATAL and Agricultural Limited Liability Company Slavutych based on the use of individual innovations in the production sector

Indicator	Indicator of integral assessment	PAF VATAL			ALLC Slavutych		
		Value of the indicator	Significance (impact)	Total amount	Value of the indicator	Significance (impact)	Total amount
Integral indicator of human resources potential	0.2	–	1	0.77	–	1	0.61
Integral indicator of financial and property potential	0.05	–	1	0.82	–	1	0.61
Integral indicator of marketing potential	0.25	–	1	0.43	–	1	0.50
Integral indicator of management potential	0.2	–	1	0.58	–	1	0.55
Integral indicator of production potential	0.3		1	0.35		1	0.35
Share of annual product renewal		0.1	0.25	0.03	0.1	0.3	0.03
Capacity utilization rate		0.95	0.2	0.19	0.95	0.25	0.24
Share of costs for attracting innovations		0.2	0.25	0.05	0.15	0.2	0.03
Coefficient of innovation profitability		0.25	0.3	0.08	0.2	0.25	0.05
Composite integrated indicator of innovation potential	1	-	-	0.53	-	-	0.50

relate specifically to the production process. It represents the sum of the values of the factors influencing production potential.

Only one measure can increase the efficiency of livestock farming at the enterprise, increase the share of costs for attracting innovations, increase the share of renewed biological assets, load unused livestock facilities and increase the efficiency of production and marketing activities. The introduction of innovations in the marketing sphere, the transition to the production of organic livestock and crop products will contribute to the growth of the integrated indicator of marketing potential [18]. Among the types of crop products for the studied enterprises, we recommend focusing on berry crops (raspberries, blackberries, strawberries) and vegetable crops (crops of the so-called “boron set”), which are already grown and the transition to organic principles or reduction of chemical spraying will be a logical continuation of the chosen strategy.

When determining the forecast indicators of the complex integrated indicator in Table 2, we made changes only to the block of indicators of production potential without the block of marketing potential to make the influence of individual factors more visible and to summarize the impact on the final indicator of innovation potential. Another reason for the unchanged marketing potential is the monopolization of the organic agricultural market, where “new players” have no chance to compete, and thus the marketing policy of enterprises at the first stage remains unchanged. Thus, the reserves for increasing the innovation potential identified by us will contribute to the rapid introduction of innovations and the use of production capacities, which, in turn, will increase competitiveness, create new opportunities for the development of the enterprise on the basis of sustainable development, and promote the development of human resources and management potential.

Thus, the data of Tables 1 and 2 allow us to conclude that the introduction of individual innovations in the activities of Private Agricultural Firm VATAL and Agricultural Limited Liability Company Slavutych allows to increase the complex integrated indicator of innovation potential by 6% and reach the indicator of 0.53 for the first enterprise and 0.50 for the second. According to the accepted evaluation scale, such a potential indicator is considered to be an indicator with an acceptable level (0.40–0.63) [19].

Thus, the modernized model of innovation development at the enterprises of Private Agricultural Firm VATAL and Agricultural Limited Liability Company Slavutych, which we have implemented, ensured the constant readiness and ability of the enterprise to implement the latest technologies, methods and strategies aimed at increasing efficiency, reducing environmental impact and increasing competitiveness. In addition, it is necessary to develop an effective strategy for innovation development, attract qualified specialists and provide them with ongoing support and training. After all, it is important to foster a favorable innovation atmosphere at enterprises where employees feel free to come up with new ideas and solutions to problems.

6 Conclusion

The innovative potential of sustainable development of an enterprise reflects the company's ability to implement innovative ideas, technologies and practices to ensure sustainable development. This includes finding new ways to reduce waste, use renewable energy sources, improve management approaches to minimize environmental impact and increase social responsibility. Thanks to its innovative potential, a company can ensure its long-term success by attracting customers, investors and employees who share its values and goals.

To ensure the successful sustainable development of an enterprise, it is important to develop a model of innovation capability [20, 21]. This may include continuous improvement of business processes, introduction of the latest technologies, and creation of a favorable corporate culture to stimulate creativity and innovation [22]. Such a model will allow the company to be competitive in the market and ensure its stable and profitable development [20–22].

To improve the efficiency of Private Agricultural Firm VATAL and Agricultural Limited Liability Company Slavutych, we have modernized the model of innovation potential for sustainable development, which has allowed the companies to adapt to changes in the economy and society, providing them with a competitive advantage. To succeed in this area, it is important to develop innovative approaches, introduce new technologies and stimulate creativity in the company's team.

Thus, the modernized model of the innovative potential of sustainable development of Private Agricultural Firm VATAL and Agricultural Limited Liability Company Slavutych helps the enterprise to ensure efficiency by promoting innovative approaches, introducing new technologies and developing products and services based on the principles of sustainable development. The application of this model can help a company conserve resources, reduce its environmental impact, increase its competitiveness and ensure sustainable development in the future.

References

1. Chikov I (2024) Innovatsiyni potentsial pidpriemstva: poniatiino-katehorialnyi aparat ta ekonomiko-matematychna model otsinky. *Ekonomika ta suspilstvo*. <https://doi.org/10.32782/2524-0072/2024-68-55>
2. Kozlovska S (2021) Modern mechanisms of management the innovative potential of enterprises in Ukraine. *Eastern Europe Econ Bus Manage*. <https://doi.org/10.32782/easterneurope.33-17>
3. Dudnieva Yu, Obydiennova T, Yelnykova H (2022) Innovatsiyni potentsial pidpriemstv maloho biznesu. *Adap Manage Theory Practice Econ* 13(26). [https://doi.org/10.33296/2707-0654-13\(26\)-06](https://doi.org/10.33296/2707-0654-13(26)-06)
4. Moisa TV (2023) Osoblyvosti formuvannya innovatsiinoho potentsialu pidpriems-tva v suchasnykh umovakh rozvytku. *Problemy suchasnykh transformatsii. Seriya: ekonomika ta upravlinnia* (7). <https://doi.org/10.54929/2786-5738-2023-7-04-05>
5. Dalyk V, Yarmus S (2024) Assessing the innovative potential of an enterprise under risk conditions. *Int Sci J "Internauka" Ser Econ Sci* 2(82). <https://doi.org/10.25313/2520-2294-2024-2-9508>

6. Boichuk N.Ia. Innovatsiyni rozvytok ta potentsial pidpriemstv v Ukraini. Materialy II Mizhnarodnoi naukovo-praktychnoi konferentsii «Biznes, innovatsii, menedzhment: problemy ta perspektyvy». Sektsiia 3. Innovatsii ta pidpriemnytstvo yak osnova ekonomichnoho zrostantia krainy (2021), S 208–209
7. Dolobanko A, Honchar V, Pashchenko V (2023) Innovatsiyni potentsial i metody yoho otsiniuvannia v upravlinni pidpriemnytskoiu diialnistiu. *Dev Serv Indus Manage* 3:207–212. [https://doi.org/10.31891/dsim-2023-3\(31\)](https://doi.org/10.31891/dsim-2023-3(31))
8. Panchyshyn T, Vdovyn M (2023) Komponenty staloho rozvytku terytorialnykh hromad ta rehioniv v umovakh suspilno-politychnykh vyklykiv. *Ekonomika ta suspilstvo* 50. <https://doi.org/10.32782/2524-0072/2023-50-23>
9. Petroliuk Yu, Hrebenuk N (2024) Stalyi rozvytok yak stratehichna perevaha dla pidpriemstva. *Ekonomika ta suspilstvo* (68). <https://doi.org/10.32782/2524-0072/2024-68-148>
10. Makukh T, Korobovych L, Ruban V (2023) Stiikist yak mekhanizm zabezpechennia staloho rozvytku pidpriemstva. *Tavriiskyi naukovi visnyk. Seria: Ekonomika* 18:122–127. <https://doi.org/10.32782/2708-0366/2023.18.13>
11. Kyfiak V (2024) Stratehii innovatsiinoho staloho rozvytku biznesu: model realizatsii. *Ekonomika ta suspilstvo* 59. <https://doi.org/10.32782/2524-0072/2024-59-57>
12. Korobka SV (2024) Stratehiia staloho rozvytku v upravlinni orhanizatsi-yeiu. *Problemy suchasnykh transformatsii. Seria: ekonomika ta upravlinnia* 16. <https://doi.org/10.54929/2786-5738-2024-16-04-06>
13. Levkina R, Kotko Y, Kulinich O (2024) Mechanism of sustainable development of business entities in the system of international economic relations. *Econ J Odessa Polytech Univ* 4(30):57–67. <https://doi.org/10.15276/ej.04.2024.7>
14. Bazyliuk VB, Bazyliuk KF (2021) Metodychni pidkhid do integralnoi otsinky konkurentnoho potentsialu vyrobnychoho pidpriemstva. *Ekonomika ta suspilstvo. Vypusk* 25. <https://doi.org/10.32782/2524-0072/2021-25-85>
15. Yepifanova Iiu, Hladka DO (2022) Metodychni pidkhody do otsiniuvannia innovatsii-noho potentsialu pidpriemstva. *Innov Sustain* 3:152–158. <https://doi.org/10.31649/ins.2022.3.152.158>
16. Levkina R, Levkin A, Petrenko A, Kolomiets N (2020) Current approaches to biotechnology in animal husbandry. *Int J Adv Sci Technol* 29(8):2463–2469. <http://sersc.org/journals/index.php/IJAST/article/view/14746>
17. Levkin A, Petrenko A, Levkina R, Chaliy I (2019). Economic security as a result of modern biotechnology implementation. In: 2019 IEEE International Scientific-Practical Conference Problems of Infocommunications, Science and Technology (PIC S&T). IEEE, pp 139–142. <https://doi.org/10.1109/picst47496.2019.9061533>
18. Levkina R, Petrenko A (2019) Management of innovative marketing techniques as an effective business tool. *Agric Resour Econ Int Sci E-J* 5(1):37–47. <https://doi.org/10.51599/are.2019.05.01.03>
19. Artomova A. V., Artomov I. V. (2016). Protsedura otsiniuvannia resursnoho potentsialu pidpriemstva. *Systemy obrobky informatsii* 7(144):165–170. http://nbuv.gov.ua/UJRN/soi_2016_7_36
20. Yepifanova I, Dzhedzhula V (2021) Modelling of potential level of industrial enterprises. *WSEAS Trans Environ Dev* 17:556–565. <https://doi.org/10.37394/232015.2021.17.53>
21. Yepifanova I, Dzhedzhula V (2020) Methodology of evaluation of innovative potential of enterprises. *Agric Resour Econ Int Sci E-J* 171–190. <https://doi.org/10.51599/are.2020.06.03.10>
22. Engelhardt-Nowitzki C, Kryvinska N, Strauss C (2011) Strategic demands on information services in uncertain businesses: a layer-based framework from a value network perspective. In: 2011 International conference on emerging intelligent data and web technologies. IEEE, pp 131–136. <https://doi.org/10.1109/eidwt.2011.28>

RWA Tokenization Mechanisms as a Tool for Sustainable Economic Development of Local Hromadas in Ukraine



Maryna Riabokin , Yevgen Kotukh , Dmytro Nikolaievskiy ,
and Yuliia Yefimenko

Abstract In the context of digital transformation of the economy, decentralization of management and the need for post-war reconstruction in Ukraine, the search for innovative mechanisms to ensure sustainable economic development of both local hromadas and the national economy is becoming particularly relevant. One of the promising areas is the implementation of the technology of tokenization of real assets (RWA—Real World Assets), which allows creating digital representations of ownership rights to tangible and intangible assets on blockchain platforms. The relevance of the study of RWA tokenization mechanisms in the context of local hromada development is due to the following important factors. Local hromadas in Ukraine face the problem of limited financial resources for the implementation of infrastructure development and modernization projects. Tokenization of real assets can become a tool for attracting additional investments by creating liquid digital assets secured by real municipal property. Also, the implementation of RWA tokenization mechanisms provides increased transparency in the management of hromada assets and reduced transaction costs when carrying out operations with them. This is achieved through the use of smart contracts and a distributed registry, which ensures the immutability and accessibility of information about property rights and transactions with assets.

M. Riabokin (✉)

Kyiv Institute of Business and Technology, Kyiv, Ukraine

e-mail: riabokin@kibit.edu.ua

Y. Kotukh

Dnipro University of Technology, Dnipro, Ukraine

e-mail: yevgenkotukh@ukr.net

D. Nikolaievskiy

V. M. Koretsky Institute of State and Law of National Academy of Sciences of Ukraine, Kyiv, Ukraine

e-mail: nikolaievskiy@thedigital.gov.ua

Y. Yefimenko

Yaroslav Mudryi National Law University, Kharkiv, Ukraine

e-mail: Jilia@i.ua

An important factor is that RWA tokenization creates the prerequisites for the formation of new models of economic interaction between hromada members, local businesses and external investors. Digital tokens can be used as a tool to stimulate local economic activity, implement social projects and ensure the participation of citizens in the management of common resources. In the context of post-war reconstruction of Ukraine, RWA tokenization mechanisms can become an effective tool for attracting international investments to restore damaged infrastructure and implement sustainable development projects at the local level.

Keywords RWA-tokenization · Sustainable development · Local hromadas · Digital transformation · Blockchain

1 Introduction

The intensification of decentralization processes and the strengthening of the role of local government in implementing the principles of sustainable development necessitate the search and implementation of new financial instruments, among which the tokenization of real assets (RWA tokenization) occupies a special place.

Local government, as a key institution in the system of ensuring sustainable development of the national economy, demonstrates a growing influence on strategic decision-making in the areas of economic, social and environmental development of hromadas. The effectiveness of local governments in identifying and meeting the needs of territorial hromadas, as well as their ability to implement locally adapted solutions, create the prerequisites for the effective implementation of RWA tokenization mechanisms in sustainable development processes [1].

In this context, research into the areas of implementation of RWA tokenization mechanisms is of particular importance as a tool for modernizing the local development management system and ensuring the financial autonomy of territorial hromadas.

As of early 2024, the total amount of direct losses to residential and non-residential real estate, infrastructure, vehicles, and inventories exceeded \$157 billion. The largest share of these losses is caused by damage to the housing stock—37.5% or \$58.9 billion, and to infrastructure—23.4% or \$36.8 billion [2].

As a result of the fighting, about 250 thousand housing buildings were damaged or destroyed, of which 222.6 thousand were private houses and 27 thousand were apartment buildings. The total area of damaged buildings is 88.9 million sq.m, which is 8.6% of the total housing stock of Ukraine.

Critical infrastructure facilities suffered significant damage, in particular, as of early February 2024:

- 25.4 thousand km of roads and 344 bridges—\$29.6 billion in damage;
- 507 km of railway track—\$4.3 billion in losses;
- Energy infrastructure—\$9 billion in losses;

- Housing and communal services facilities—\$4.5 billion in losses.

Also, significant losses were recorded in social facilities (schools, hospitals, cultural facilities), corporate assets, and industrial facilities (totaling \$13.1 billion) [2].

Sustainable hromada development in Ukraine is a key factor in the reconstruction of the country, the formation of economically self-sufficient regions, social stability and environmental responsibility. It is based on three main aspects: economic, social and environmental, each of which is critical for the long-term sustainability and prosperity of local territories, especially in the context of post-war reconstruction.

The economic aspect of sustainable hromada development involves creating conditions for local business development, attracting investment and ensuring financial self-sufficiency. One of the key factors is the efficient use of local resources, supporting small and medium-sized businesses, as well as stimulating innovation and digital technologies. Economic decentralization allows hromadas to receive more financial opportunities for the development of their own infrastructure, education, medicine and social services. In addition, the creation of new jobs and increasing the level of income of the population contributes to the strengthening of the local economy and reducing dependence on external financial flows.

The social aspect of hromada development includes improving the quality of life of the population, creating comfortable living conditions, access to education, health-care and social protection. The development of social capital is of great importance, which includes the active participation of citizens in decision-making, supporting public initiatives and establishing effective communication between authorities, business and public organizations. An important component of social development is the formation of a safe environment, preventing social inequality and promoting the integration of different population groups into public life.

The environmental aspect of sustainable hromada development is aimed at conserving natural resources, minimizing the negative impact of human activities on the environment, and implementing environmentally friendly technologies. This includes developing a waste management system, increasing energy efficiency, implementing renewable energy sources, and adapting to climate change. The rational use of natural resources and environmentally responsible behavior contribute not only to environmental conservation, but also to the economic development of hromadas through the introduction of “green” technologies and the creation of new jobs in the field of environmental infrastructure.

2 Background of Digital Transformation in Hromada's Financial Management

In the context of digital transformation and the transition to new financing models, digital assets are becoming increasingly important in the development of local initiatives and support for hromada projects. They open up new opportunities for attracting funding, increasing transparency and efficiency in resource management.

One of the most promising areas is the use of digital assets to finance local projects. Tokenization of assets allows hromadas to raise funding through the issuance of digital tokens that can represent a stake in local infrastructure, real estate, or other strategic assets. This opens up opportunities to attract investment not only from traditional financial institutions, but also from a wide range of citizens and private investors interested in the development of specific territories.

Another important direction is to increase the transparency and accountability of public finances through blockchain technology. The introduction of decentralized registries allows to ensure the openness of transactions, which significantly reduces the risks of corruption and inefficient use of budget funds [3]. This creates conditions for citizens' trust in local authorities and promotes more active participation of the population in financing and implementing projects that directly affect the quality of life in the hromada.

In addition, digital assets can be used to support crowdfunding platforms that help hromadas mobilize financial resources for socially important initiatives. The use of decentralized financial instruments (DeFi) allows hromadas to access financing without involving banks or government agencies, which greatly simplifies the process of raising funds for local needs.

Overall, digital assets are a powerful tool for stimulating the economic development of hromadas, increasing the efficiency of financial flows and improving the quality of life of residents. Their integration into local initiatives contributes to sustainable development, providing hromadas with new opportunities for financing, resource management and implementing innovative solutions. Combined with the principles of sustainable development, digital assets can become a key factor in the formation of modern, efficient and financially independent hromadas.

One of the most promising areas of use for digital assets is tokenization technology, which translates this concept from theoretical considerations to practical application.

Real asset tokenization refers to the process of converting ownership rights to tangible or intangible assets into digital tokens on a blockchain platform. This innovative approach allows a wide range of assets, such as real estate, artwork, commodities, and financial instruments, to be represented in a digital form that can be traded and partially owned [4].

At its core, RWA tokenization involves the creation of digital tokens that represent an ownership of real asset or share, fraction; of ownership of real asset. These tokens are issued on a blockchain (a rule-based, continuous, sequential chain of blocks containing information) that provides a secure and transparent ledger for recording ownership and transferring tokens. Blockchain technology not only facilitates proof

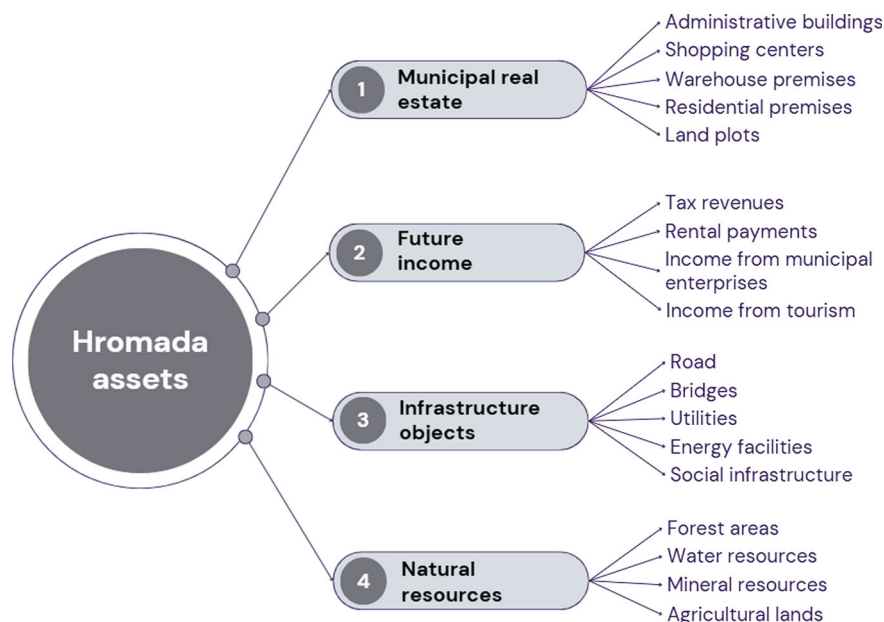


Fig. 1 The range of assets of territorial hromadas that can be tokenized¹

of ownership but also ensures the immutability of transaction records, increasing trust between participants [5].

3 RWA Tokenization as a Tool of Sustainable Economic Development

Territorial hromadas have a wide range of assets that can be tokenized to attract financing, optimize financial management, and implement strategic projects (Fig. 1). One of the most promising assets is municipal real estate, which includes administrative buildings, shopping centers, warehouses, residential buildings, and land plots.

Future hromada revenues, such as tax revenues, rents, utility revenues, or tourism, can also be tokenized, allowing for the raising of funds in the present to cover expenses or implement projects.

Infrastructure assets such as roads, bridges, communications, energy facilities, and social infrastructure can also be tokenized to finance their construction or modernization. For example, future toll revenues from a new road could be the basis for issuing tokens.

¹ Developed by the authors.

Natural resources owned by the hromada, including forests, water resources, minerals, and agricultural land, can also be tokenized. For example, the right to use forest resources can be converted into tokens to raise funds for their restoration.

For local hromadas in Ukraine that have experienced significant damage and need funding for reconstruction, several RWA tokenization methods can be implemented:

Tokenization of municipal real estate—allows for rapid mobilization of financial resources without the need for complete alienation of assets. Hromadas can issue digital tokens representing ownership interest or the right to receive income from municipal real estate.

Tokenization of future tax revenues is an effective tool for closing temporary cash gaps in local budgets. Given the large-scale destruction of infrastructure (25.4 thousand km of roads and 344 bridges with total damage of \$29.6 billion), this method allows hromadas to effectively manage temporary cash gaps to finance urgent reconstruction work.

Creating reserve funds through tokenization of future revenues—allows hromadas to create a financial “safety cushion” to respond to crisis conditions or finance strategic initiatives. Given the losses in energy infrastructure (\$9 billion) and housing and utilities (\$4.5 billion), this approach can ensure stable funding for critical public services.

Tokenization for financing infrastructure projects—allows raising funds without increasing the debt burden for the restoration of destroyed facilities. This is especially relevant for the most affected regions, such as Donetsk region (direct losses \$37.374 billion), Kharkiv region (\$30.224 billion) and Luhansk region (\$17.127 billion).

The implementation of these mechanisms has a number of advantages for the hromada: quick access to funds, reduced debt burden, efficient financial management, improved financial performance, creation of a financial “safety cushion”, ensuring a stable flow of funds, efficient use of assets, attracting a wider range of investors, as well as increasing transparency and trust.

Given the advantages listed above, we propose a basic logical architecture of a platform for tokenizing the assets of territorial hromadas (Fig. 2).

In Fig. 2 the following abbreviations are used:

API (Application Program Interface)—which allows components to integrate and interact with each other;

AML (Anti-Money Laundering) is a set of measures, procedures and regulatory requirements aimed at preventing and detecting money laundering and terrorist financing;

KYC (Know Your Customer) is the process of verifying the identity of customers by financial institutions and other entities to prevent money laundering, terrorist financing, fraud and other illegal activities;

CFT (Combating the Financing of Terrorism) is a set of measures and regulatory requirements aimed at preventing and detecting the financing of terrorist activities;

SPV (Special Purpose Vehicle, or in the context of tokenization, special meta-firm) is a separate legal entity created to achieve a specific financial goal, such

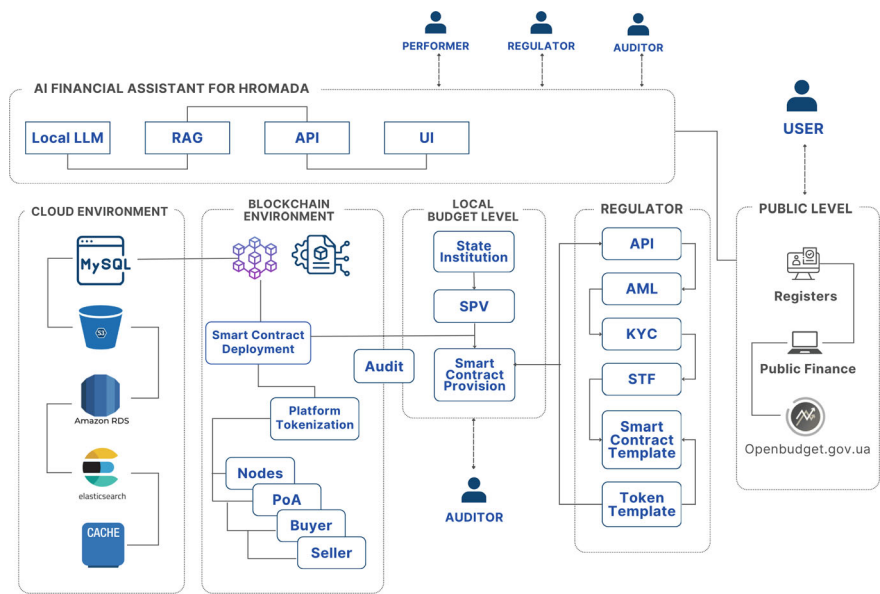


Fig. 2 Basic logical architecture of the platform for tokenization of territorial hromada assets [6]

as isolating financial risks or facilitating the financing of projects by a territorial hromada;

PoA (Power of Attorney, or in the context of tokenization, a digital power of attorney mechanism)—a mechanism of powers of attorney for the rights to represent the interests of investors in a digital system;

Amazon RDS—distributed relational database;

Local LLM (Local Large Language Model) is a set of artificial intelligence algorithms used to understand and generate natural language;

RAG (Retrieval-Augmented Generation) is a method in natural language processing that combines technologies for selecting relevant information and text generation.

Let’s take a step-by-step look at the use case of issuing tokens from a government asset (RWA).

The first step is to identify and value the asset. Experience suggests that this will be implemented in the form of registries, from which information will be available for further actions, such as conducting an independent expert assessment of the asset’s value and analyzing the legal status and potential restrictions on tokenization.

In the second step, it is proposed to implement all necessary economic and legal compliance, namely, the creation of a legal basis for tokenization in accordance with current legislation (creation of a digital identity of a legal entity that owns the rights to the object of tokenization; definition of the rights and obligations of token owners; requirements for AML, KYC and CFT; development of mechanisms for interaction

between the state and investors and other regulated rights and mandatory norms that will become part of the smart contract.

The third step will be to analyze and select a suitable blockchain platform (e.g. Ethereum, Polkadot, Cardano), assess scalability, security, and compatibility with existing financial systems. The state will likely be interested in creating its own blockchain platform, given that digital transformation is recognized as one of the top priority reforms—there are a large number of potential services and economic relations that can be digitized and regulated using blockchain mechanisms.

In the fourth step, a smart contract ready for provisioning will appear from the template. To this end, the main provisions of the smart contract will be developed, defining the functionality of the tokens; revenue distribution, governance, and liquidity mechanisms will be implemented; and an audit will be conducted.

In the fifth step, the total number of tokens, their nominal value and the Initial Token Offering (ITO) procedure will be determined.

The sixth step should involve the registration and settlement of tokens issued for regulatory purposes: obtaining the necessary permits from regulatory authorities for trading, registering tokens as securities (if this is proposed through this mechanism).

Next will be a number of administrative, organizational and technical tasks that embody the life cycle of the overall system, namely: development or integration with existing platforms for secondary token trading; ensuring liquidity and transparency of pricing; implementing a system for token owners to participate in making decisions regarding asset management (it would probably make great sense to implement such management due to the priority of hromada members regarding investment); developing protocols for distributing future income and expenses; implementing multi-level security systems to protect tokens and related transactions; regulatory and legal regulation of conducting independent audits to ensure transparency and trust; creating a regular reporting system for token owners and constant monitoring of the effectiveness of asset use and compliance with regulatory requirements [6].

The implementation of the proposed approach requires close cooperation between government agencies, legal experts, technology specialists and financial consultants. It is important to note that the specific steps may vary depending on the specifics of the asset, the legislative framework and the chosen technological platform. The analysis demonstrates that this approach should be considered as the basis for a future strategy for using tokenization technology as a tool for providing liquidity in local hromadas. It is clear that the strategy should include regular review and updating of smart contracts in accordance with changes in legislation, market needs, scaling the project horizontally (within hromada assets) and vertically (within state assets).

The analysis of challenges provides a complete picture of the complexity of implementing tokenization technology. However, if these difficulties are successfully resolved, tokenization technologies will significantly contribute to increasing interest in investing in local hromada assets. Such challenges include:

The technological infrastructure required for asset tokenization is complex. The proposed principle logic diagram for implementing the tokenization concept provides

an idea of the general approach without revealing the complex details of the implementation. The solution to such difficulties is due to the need to develop a public–private partnership with the involvement of commercial enterprises that have gained sufficient experience in implementing tokenization projects.

The regulatory framework for the tokenization of government assets is not well developed and requires further research. Currently, proactive legislative reviews and regulatory adjustments are needed to fully legalize and protect tokenized assets. Regulating tokenized assets can be complex due to their decentralized nature. It is important to create a clear regulatory framework that ensures transparency, investor protection, and financial stability. In this sense, it would be very important to adopt a sectoral law “On virtual assets”, which is currently being developed, as well as amendments to the Tax Code of Ukraine to introduce a clear concept of taxation of transactions with virtual assets.

The RWA tokenization process, as discussed in more detail above, includes a number of stages from preparatory assessment to the launch of the secondary market. However, the significance of this mechanism goes far beyond purely technical aspects and financial benefits [7]. RWA tokenization is not only an innovative tool for attracting investments, but also a powerful catalyst for achieving more global goals. Modern challenges of sustainable development, recorded in international documents and strategies, require innovative approaches to their solution, and it is in this context that RWA tokenization technology acquires particular importance as a tool for ensuring key sustainable development goals.

In the context of achieving the Sustainable Development Goals (SDGs), RWA tokenization plays a significant role in transforming socio-economic relations and environmental policy (Fig. 3).

In particular, in eradicating poverty (SDG 1) and reducing inequality (SDG 10), it provides access to small investors to assets that were previously only available to large capital, including real estate, infrastructure and natural resources. In addition, tokenization facilitates the financing of small businesses and farms through micro-investment mechanisms, and the decentralization of ownership creates conditions for a more equitable distribution of the benefits of economic growth.

In the area of industrialization, innovation and infrastructure development (SDG 9), tokenization ensures transparency in the financing of infrastructure projects, which increases investor confidence and promotes the implementation of innovative solutions in the field of urban development. The use of smart contracts allows reducing administrative costs and corruption risks within the framework of public–private partnerships.

Tokenization is also an effective tool to support renewable energy and climate initiatives (SDG 7, SDG 13). In particular, tokenization of carbon credits contributes to the formation of an efficient market for trading environmental quotas, incentivizing businesses to reduce CO₂ emissions. In addition, raising financing for renewable energy projects using digital assets facilitates access to capital by retail investors.

In the context of ensuring decent work and economic growth (SDG 8), tokenization contributes to the development of new financial mechanisms that expand access to capital for small and medium-sized enterprises (SMEs), stimulating economic

	1. NO POVERTY	<i>Access for small investors to assets, micro-investing, decentralization of ownership</i>
	7. AFFORDABLE AND CLEAN ENERGY	<i>Financing renewable energy projects through digital assets, stimulating the green economy</i>
	8. DECENT WORK AND ECONOMIC GROWTH	<i>Financing SMEs, lowering business barriers, developing new labor markets</i>
	9. INDUSTRY, INNOVATION AND INFRASTRUCTURE	<i>Transparent infrastructure financing, reduction of administrative costs, stimulation of innovation</i>
	10. REDUCED INEQUALITIES	<i>Fair distribution of assets, accessibility of investments for a wide range of people</i>
	11. SUSTAINABLE CITIES AND COMMUNITIES	<i>Engaging citizens in financing local initiatives, digital asset management</i>
	12. RESPONSIBLE CONSUMPTION AND PRODUCTION	<i>Blockchain for supply chain transparency, sustainable production control</i>
	13. CLIMATE ACTION	<i>Tokenization of carbon credits, financing renewable energy</i>
	16. PEACE, JUSTICE AND STRONG INSTITUTION	<i>Protection of property rights through blockchain, smart contracts for transparency and anti-corruption</i>

Fig. 3 The connection between RWA-tokenization and the Global Sustainable Development Goals²

activity. The introduction of blockchain and digital assets reduces bureaucratic barriers in the financial sector, increasing the efficiency of the labor market and the business environment. Decentralized financial models contribute to the creation of new jobs in the areas of financial technology, asset management and environmental initiatives. In addition, asset tokenization ensures the efficient attraction of financing to innovative and socially responsible businesses, contributing to sustainable economic growth.

Responsible consumption and production (SDG 12) is an important aspect. Using blockchain to track the origin of goods in supply chains ensures transparency, compliance with ethical production standards and environmental regulations. Thanks to distributed ledgers, both consumers and businesses can monitor the sustainability of production processes, which contributes to increasing the level of responsible consumption.

In the area of urban and hromada development (SDG 11), tokenization allows residents to participate in investing in the development of their cities, promoting transparent management of municipal assets. This facilitates the financing of environmental initiatives such as the creation of city parks, the implementation of transportation projects, and the modernization of infrastructure.

Finally, tokenization contributes to strengthening peace, justice and effective institutions (SDG 16). The use of blockchain in the field of property rights registration

² Developed by the authors.

reduces the risks of raider seizures of assets, ensuring their legal protection. Smart contracts contribute to the automatic fulfillment of obligations, which minimizes corruption risks in the field of public administration and business. Thus, tokenization is a powerful tool for the digital transformation of the economy, contributing to sustainable development and increasing the social responsibility of business.

Analysis of the national report “Sustainable Development Goals: Ukraine” reveals significant potential for implementing mechanisms for tokenizing local hromada assets as an innovative tool for financing sustainable development [8]. Let us consider the key areas of implementation of this technology, taking into account current data on the implementation of relevant indicators.

Within Goal 8 “Decent work and economic growth”, in particular target 8.6, aimed at creating institutional and financial opportunities for the self-realization of the economically active part of the population, ambiguous dynamics are observed. Indicator 8.6.1 (number of employees in small and medium-sized enterprises) shows a regressive trend—from 6.5 million people in 2015 to 6.1 million in 2023 [9], with a target of 10.5 million by 2030. In contrast, indicator 8.6.2 (share of value added by small and medium-sized enterprises) shows positive dynamics, reaching 70.5% in 2023, which is significantly closer to the target of 80% by 2030.

Of particular concern is the state of road and transport infrastructure (Goal 9, target 9.3). Indicator 9.3.2 on the share of public roads with hard surface that meet standards does not have current data due to military aggression. Appropriate data collection and compilation were not carried out in 2022–2023, which makes it difficult to assess progress in achieving the target of 70% by 2030 compared to a baseline of 10% in 2015. However, even without official statistics, empirical observations and numerous reports from the regions indicate a critical deterioration in the state of road infrastructure as a result of military actions. Systematic missile strikes, the movement of heavy military equipment, and the reduction of scheduled repair work have led to significant degradation of road surfaces in many regions, especially in the east and south of Ukraine. A significant part of previously repaired roads has been damaged, and some transport arteries have been practically decommissioned due to the destruction of bridges and other critical infrastructure facilities. In this context, tokenization of infrastructure projects becomes particularly important as a mechanism for targeted fundraising for the reconstruction of specific road sections.

In the renewable energy sector (Goal 7, target 7.3), indicator 7.3.1 on the share of energy from renewable sources shows positive dynamics, increasing from 4.9% in 2015 to 9.2% in 2020 and 13.1% in 2024. If the current pace is maintained, the target of 17.1% by 2030 looks quite achievable. Moreover, the introduction of asset tokenization mechanisms in the renewable energy sector creates the prerequisites for a significant overachievement of this indicator.

International experience of EU countries, in particular Germany and Denmark, demonstrates that the use of innovative financial instruments in the renewable energy sector can accelerate the growth of the share of “green” energy in the overall energy balance by 1.5–2 times compared to traditional models. This is facilitated by the mechanism of distributed investment and the formation of energy cooperatives based

on tokenized assets, which allows hromadas not only to invest in energy projects, but also to become their co-owners.

The potential of tokenization is particularly relevant in the context of the post-war reconstruction of Ukraine's energy system, which involves its decentralization and increased resilience to external threats. Tokenization of distributed generation projects, microgrids, and energy storage systems creates financial mechanisms for scaling such solutions at the level of individual hromadas and regions. When applying a systemic approach to tokenizing hromada energy assets, the share of renewable energy can reach 22–25% by 2030, which significantly exceeds the target of 17.1%.

Another important aspect is the possibility of integrating tokenized energy assets with international markets for carbon credits and “green” certificates, which creates additional financial incentives for investors. By providing transparent verification of renewable energy generation volumes and the corresponding reduction in greenhouse gas emissions, energy asset tokens can become an object of international trade, which increases their investment attractiveness and expands the range of potential investors.

The organic sector is developing much more slowly (Goal 15, target 15.3). Indicator 15.3.4 on the area of land under organic production increased only from 410.6 thousand hectares in 2015 to 471.2 thousand hectares in 2023, which is a slight increase compared to the target of 3,000.0 thousand hectares by 2030.

The organic sector is developing much more slowly (Goal 15, target 15.3). Indicator 15.3.4 on the area of land under organic production has increased only from 410.6 thousand ha in 2015 to 471.2 thousand ha in 2023, which is a slight increase compared to the target of 3000.0 thousand ha by 2030. Such dynamics raise reasonable concerns about the possibility of achieving the planned indicator using traditional methods. At the current growth rate (approximately 7.5 thousand ha per year), by 2030 the area of land under organic production will reach only about 530 thousand ha, which is only 17.7% of the target. It is in this context that tokenization of hromada land assets and organic production systems demonstrates significant transformative potential. The introduction of tokenization mechanisms allows to address key barriers to the expansion of organic production, namely: long land conversion periods, limited access to finance and high initial investment costs for farmers. It means that, additionally to the relevant Law “On Virtual Assets,” other legislative acts of Ukraine will require certain changes, such as Land Code of Ukraine, The Law of Ukraine “On the State Land Cadastre,” The Law of Ukraine “On State Registration of Corporeal Rights to Real Estate and Their Encumbrances,” and certain others.

Tokenization of organic land assets involves the creation of digital tokens tied to specific land plots that are in the process of transitioning to organic production or are already certified as organic. Such tokens can serve several functions. First, they serve as a tool for raising capital to finance certification and the transition to organic management methods. Second, they can represent rights to future production from organic lands, forming a system of long-term contracts with processors and consumers. Third, they provide a mechanism for verifying compliance with organic standards through a transparent monitoring system based on blockchain technology.

Tokenization is particularly promising for the formation of hromada-based agroecological cooperatives, where tokens represent a share in shared organic production.

This approach allows for the pooling of land resources of smallholders to achieve economies of scale and reduce transaction costs for certification and marketing [10].

International experience, in particular projects in the United States and the European Union, demonstrates that the implementation of tokenization mechanisms can accelerate the growth of organic land by 5–7 times compared to traditional financial instruments. Extrapolating this experience to the Ukrainian context, it can be predicted that the systematic implementation of tokenization of land assets can ensure an annual increase in organic land at the level of 300–350 thousand hectares, which will allow us to approach the target of 3,000.0 thousand hectares by 2030.

An additional incentive for expanding organic production through tokenization mechanisms is the possibility of integration with carbon credit markets, as organic farming contributes to carbon sequestration in the soil. Tokenization of this ecosystem service creates an additional source of income for organic producers, increasing the economic attractiveness of the transition to sustainable farming methods.

Thus, the introduction of innovative tokenization mechanisms in the organic production sector can become a key catalyst for achieving the ambitious goals set by the National Sustainable Development Strategy and ensure a significant acceleration of the pace of expansion of organic lands in Ukraine by 2030.

The analysis of the above data once again confirms the need to find new financing mechanisms to achieve target indicators, especially in conditions of limited budgetary capabilities of the state. It is in this context that the tokenization of local hromada assets demonstrates significant potential. The use of this technology allows solving a number of problems identified in the analysis of progress in implementing the Sustainable Development Goals.

Firstly, tokenization can ensure the attraction of micro-investments in the development of small and medium-sized enterprises, which will contribute to increasing the number of employees in this sector and increasing its added value. The attraction of the diaspora through investment tokens of their hromadas of origin is especially promising.

Second, in the area of infrastructure development, particularly public roads, tokenization allows for the fragmentation of large infrastructure projects into smaller investment opportunities accessible to a wide range of investors. This approach is especially relevant in the context of limited government funding and the priority of defense spending.

Third, in the renewable energy sector, tokenization enables hromada-based co-investment in solar, wind, and bioenergy projects, which will help diversify the energy mix and decentralize the energy system, which is critical for energy security in the face of military threats.

Fourth, tokenization of hromada land resources can be an effective mechanism for attracting capital for the restoration of degraded lands and the transition to organic production. Given the slow growth rate of organic production areas, this approach will significantly accelerate the process of scaling it up.

In general, the implementation of tokenization mechanisms for local hromada assets creates promising opportunities for bridging the gap between current indicators and target indicators of the Sustainable Development Goals of Ukraine. The technology ensures transparency of asset management, expands access to capital, forms innovative models of cooperation between hromadas, businesses and investors, and also creates additional sources of income for local budgets.

The implementation of asset tokenization requires comprehensive regulatory frameworks, hromada capacity building, and digital infrastructure development. However, given the identified gaps in achieving individual Sustainable Development Goals, these investments are strategically sound and have significant potential for multiplier effects for local and national development.

Analysis of the relationship between the mechanisms of tokenization of real assets and sustainable development goals allows us to state that the potential of this technology significantly exceeds the tasks and indicators defined in the national Strategy [11]. Tokenization of local hromada assets forms fundamentally new approaches to solving complex problems of sustainable development, going beyond existing metrics and creating synergistic effects between different sustainable development goals (Table 1).

Unlike the limited approach of measuring progress through individual indicators, tokenization provides an integrated platform for simultaneously addressing economic, social and environmental challenges at the hromada level. Such integration is particularly valuable given the interdependence of different aspects of sustainable development, which is often not reflected in the Strategy's sectoral approach.

The key advantage of tokenization is to ensure a direct link between investments and results at the local level, bypassing complex bureaucratic procedures. While the Strategy focuses on macroeconomic indicators, tokenization allows hromadas to independently determine development priorities and attract targeted investments according to local needs, which significantly increases the efficiency of resource use.

In the future, Ukraine will need significant resources for large-scale post-war reconstruction, which will require significant financing [12]. For this, it is important to ensure effective resource management, attract both domestic and external investments, as well as develop financial inclusion. In this context, RWA tokenization can become an important tool for stimulating economic growth, restoring infrastructure, and supporting local hromadas.

After the large-scale destruction caused by the war, Ukraine needs huge investments to restore roads, bridges, housing complexes and utility networks. The post-war reconstruction of Ukraine requires effective mechanisms for attracting financing, ensuring transparency in resource management and stimulating the economic development of local hromadas [13]. In this context, RWA tokenization is an innovative tool that allows attracting both domestic and external investments in various sectors of the economy, contributing to the implementation of the Global Sustainable Development Goals (SDGs) [14].

One of the key areas of application of RWA tokenization is the rehabilitation and development of infrastructure (SDGs 9, 11). The tokenization of real estate and land plots contributes to the attraction of private capital in the reconstruction

Table 1 Mechanisms of the impact of RWA tokenization on sustainable economic development and restoration of local hromadas in Ukraine³

Sphere of influence	Mechanisms of influence of RWA tokenization	Relevant SDGs
Infrastructure restoration and development	• Tokenization of real estate and land to attract investment • Financing infrastructure projects through digital assets • Decentralization of ownership in restored facilities	9 “Industrialization” 11 “Sustainable Cities”
Support for small and medium-sized businesses	• Alternative financing for entrepreneurs through tokenized assets • Reducing dependence on bank loans • Increasing transparency for investors	8 “Decent work and economic growth”
Financial inclusion and poverty alleviation	• Access to investment opportunities in democracy • Creation of new microfinance mechanisms • Facilitating access to financial services for the underprivileged	1 “Overcoming poverty” 10 “Reducing inequality”
Restoration of the agricultural sector and land management	• Providing farmers with access to capital through the tokenization of crops and land • Facilitating the purchase, sale or lease of land • Control over the efficiency of resource use	12 “Responsible Consumption and Production”
Digitalization and the development of financial inclusion	• Using blockchain to register property rights • Introduction of local tokenized currencies to support the local economy • Transparent use of charitable and public funds	16 “Peace, Justice and Strong Institutions”
Financing renewable energy	• Facilitating investment in renewable energy sources • Development of energy cooperatives • Transparency of carbon credit trading	7 “Affordable and Clean Energy” 13 “Fighting Climate Change”

(continued)

³ Developed by the authors.

Table 1 (continued)

Sphere of influence	Mechanisms of influence of RWA tokenization	Relevant SDGs
Creating new mechanisms for attracting investments	• Attracting international investments through tokenized assets • Simplification of investment processes • Increasing asset liquidity	8 “Decent work”, 9 “Industrialization”
Increasing transparency and reducing corruption	• Preventing corruption schemes through smart contracts • Creation of open asset registers • Effective control over the distribution of humanitarian aid	16 “Peace, Justice and Strong Institutions”

of residential and commercial facilities, as well as in the modernization of critical infrastructure. This allows local hromadas to effectively manage the restored assets, reducing dependence on public funding and international assistance. In addition, the use of digital assets to finance large infrastructure projects, such as roads, bridges or energy facilities, provides increased liquidity of investments and transparency of transactions.

In the area of supporting small and medium-sized enterprises (SMEs) (SDG 8), RWA tokenization plays an important role in providing alternative sources of financing. The use of tokenized assets allows entrepreneurs to obtain the necessary capital without resorting to traditional banking institutions, which is especially relevant in the context of an unstable financial system [15]. This mechanism also contributes to the involvement of citizens in local entrepreneurship, giving them the opportunity to invest in local businesses through digital assets [15]. This increases the level of investor confidence and ensures greater transparency of financial transactions.

The agricultural sector and land management (SDG 12) have significant potential for tokenization. Tokenization of land and agricultural products can provide farmers with access to the capital needed to restore and develop their economic activities. In addition, tokenization contributes to the efficient use of land resources and reduces corruption in their distribution and sale [16]. The creation of digital assets representing ownership of land or future harvests facilitates the purchase, sale and rental of land plots, reducing bureaucratic barriers [16].

Another important direction is digitalization and the development of financial inclusion (SDG 16). The introduction of blockchain technologies in the registration of property rights to real estate and land allows protecting the property rights of citizens, especially in conditions of risk of loss of documents due to hostilities. It is also possible to introduce local digital currencies, which can stimulate domestic consumption and provide support for the local economy. In addition, tokenization

can be used for the transparent distribution of humanitarian aid and social payments, which increases the effectiveness of state and international support programs.

Of particular note is the potential of tokenization in the field of renewable energy (SDG 13). The reconstruction of Ukraine's energy system should be based on the principles of sustainable energy, and RWA tokenization can play a key role in financing such projects. For example, investors can purchase digital assets representing a share in solar or wind power plants, contributing to the development of energy cooperatives. In addition, tokenization allows for transparency in the trading of carbon credits, which contributes to the reduction of greenhouse gas emissions.

Regarding mechanisms for attracting investment (SDGs 8, 9), RWA tokenization creates an opportunity to attract international capital to the recovery of the Ukrainian economy. The use of blockchain to record assets reduces the risks of fraud and increases the trust of foreign investors. The availability of digital assets simplifies investment processes and makes Ukrainian projects more attractive at the global level. In addition, the increased liquidity of assets in the form of tokens allows you to quickly attract financing for critical industries.

Finally, increasing transparency and reducing corruption (SDG 16) is one of the key effects of implementing RWA tokenization. Thanks to the use of smart contracts, financial transactions become more predictable and independent of human intervention, which minimizes corruption risks. Also, the creation of open asset registries provides the ability to monitor public and private investments, which increases the level of trust in the country's economic system.

4 Conclusion

RWA tokenization can be a key factor in the sustainable development of local hromadas in Ukraine, especially in the post-war reconstruction process. It opens up new opportunities for financing infrastructure, agricultural and energy projects, supports small businesses, increases financial inclusion and contributes to the digital transformation of the economy. The main advantage of this technology is transparency, which is critical for effective resource management and attracting international investors.

The use of RWA tokenization in Ukraine can significantly accelerate regional recovery, ensure a more equitable distribution of resources, and increase trust in public and private initiatives. In the long term, this will help Ukraine not only rebuild the country, but also create a modern, technologically advanced, and resilient economy that meets the UN Global Goals for Sustainable Development.

Tokenization mechanisms create new models of public participation and collective decision-making that go beyond the formal indicators of democratization specified in the Strategy. The use of decentralized autonomous organizations (DAOs) based on tokenized assets allows for the implementation of progressive models of hromada governance with a transparent distribution of responsibilities and benefits.

In the context of climate resilience, tokenization provides tools for monetizing ecosystem services, which is not reflected in existing indicators of the Strategy. Hromadas are given the opportunity to capitalize on their efforts to restore natural capital through mechanisms of carbon credits and biodiversity tokens, creating economic incentives for environmentally responsible behavior.

Of particular importance is the potential of tokenization to build new social protection systems at the hromada level. While the Strategy focuses on traditional social support mechanisms, tokenization allows for the implementation of innovative models of mutual aid, microinsurance, and hromada solidarity funds that increase the resilience of hromadas to economic and social shocks.

The introduction of digital assets at the local level also forms the basis for the development of a circular economy, which is only partially reflected in Goal 12 of the Strategy. Tokenization allows for tracking the life cycle of products, stimulating the reuse of resources, and forming local exchange systems, which significantly reduces the ecological footprint of hromadas.

In the area of cultural heritage, which is practically not reflected in the quantitative indicators of the Strategy, tokenization creates mechanisms for the preservation and monetization of cultural assets of hromadas. This is especially relevant for the preservation of intangible cultural heritage, which is under threat of extinction due to demographic changes and globalization.

Ultimately, tokenization creates new approaches to the development of local innovation ecosystems that go beyond traditional metrics of innovation development. The creation of digital assets based on the intellectual property of hromadas allows us to capitalize on local knowledge and innovation, creating new sources of income and stimulating the creative economy.

Thus, the potential of asset tokenization to ensure sustainable development of local hromadas significantly exceeds the existing tasks and indicators defined in the Strategy. This technology creates fundamentally new models of public development that integrate economic, social and environmental aspects into a single system, increasing the adaptability, inclusiveness and resilience of local hromadas. Expanding the methodology for assessing the impact of tokenization mechanisms on sustainable development and incorporating relevant indicators into national strategic documents is a necessary condition for maximizing the potential of this innovative technology in the context of post-war reconstruction of Ukraine.

References

1. Ryan C, Centieiro H (2023) DigiFT × HashKey deep research: tokenization of real-world assets, leading the next generation of capital markets. ODaily. <https://www.odaily.news/en/post/5191524>
2. Report on direct damage to infrastructure from destruction as a result of Russia's military aggression against Ukraine as of early 2024. https://kse.ua/wp-content/uploads/2024/04/01.01.24_Damages_Report.pdf

3. Fehr K, Roesti Y (2024) Real-world asset tokenization: a game changer for global trade. <https://www.hkdca.com/wp-content/uploads/2024/07/rwa-tokenization-game-changer-global-trade-synpulse.pdf>
4. Chamria R (2021) Tokenization of assets—how Blockchain is rewriting the asset management. <https://www.zeeve.io/blog/tokenization-of-assets-how-blockchain-is-rewriting-asset-management/>
5. Riabokin M, Kotukh Y (2024) RWA-tokenization as a tool for attracting investments and developing post-war Ukraine. *Glob Sci Acad Res J Econ Bus Manage* 3(11):64–77. <https://gsa.rpublishers.com/wp-content/uploads/2024/11/GSARJEBM2192024-Gelary-script.pdf>
6. Guley AI, Kotukh EV, Riabokin MV (2024) RWA-tokenization as an innovative mechanism for attracting investments and increasing local budget revenues. *Econ Soc. Issue No. 64/2024*, <https://doi.org/10.32782/2524-0072/2024-64-22>
7. Lim YQ (2024) US, Indonesia & Türkiye Lead With 33% of RWA Crypto Interest. CoinGecko Research. <https://www.coingecko.com/research/publications/rwa-crypto-interest-countries>
8. Ministry of Economic Development and Trade of Ukraine (2017) National report “Sustainable Development Goals: Ukraine”. <https://me.gov.ua/view/22e86f94-a9dd-421e-adcb-e38748a4b7cb>
9. State Statistics Service of Ukraine (nd) Monitoring the achievement of sustainable development goals in Ukraine. Retrieved from <https://sdg.ukrstat.gov.ua/uk/goals/>
10. Riabokin MV, Kotukh EV (2025) Methods of optimizing financial flows of territorial hromadas using RWA tokenization. *Curr Probl Econ* 1(2025):57–70. <https://doi.org/10.32752/1993-6788-2025-1-283-57-70>
11. Decree of the President of Ukraine No. 722/2019 dated September 30, 2019 “On the Sustainable Development Goals of Ukraine for the period up to 2030”. Verkhovna Rada of Ukraine. <https://zakon.rada.gov.ua/laws/show/722/2019#Text>
12. Holynskiy Yu, Nuriyeva V (2023) Financial capacity of communities during full-scale war. *Econ Soc.* <https://doi.org/10.32782/2524-0072/2023-55-41>
13. Yemets VV (2022) Economic development in the post-war period in Ukraine: regional and local aspects. *Econ State* 5:68–72. <https://doi.org/10.32702/2306-6806.2022.5.68>
14. Cole J (2024) Green assets: tokenizing sustainable real world assets. <https://blockapps.net/blog/green-assets-tokenizing-sustainable-real-world-assets/>
15. Kryvinska N, Auer L, Strauss C (2008) Managing an increased service heterogeneity in a converged enterprise infrastructure with SOA. *Int J Web Grid Serv* 4(4):440. <https://doi.org/10.1504/ijwgs.2008.022546>
16. Sokolova A, Ratoshniuk T, Yepifanova I, Kravchyk Y, Ratoshniuk V, Dzhedzhula V (2024) Assessment of the efficiency of decentralization transformations in the rural areas of Ukrainian Western Polissia: current trends and challenges under the conditions of martial law. In: *Lecture notes on data engineering and communications technologies*. Springer Nature Switzerland, pp 365–385. https://doi.org/10.1007/978-3-031-53984-8_16

Methodology for Assessing the State of Business Processes in Terms of Their Impact on the Economic Security of the Enterprise



Nataliia Havlovska , Serhii Matiukh , Mykhailo Kryvdyk ,
and Yevhenii Rudnichenko

Abstract The paper considers the optimization of enterprise resource provision under critical constraints. The study is based on the analysis of production downtime scenarios and the use of alternative energy generation in the example of enterprises operating in energy-dependent industries. The study summarizes the economic consequences of forced production shutdowns due to power outages. The influence of random factors on the expected financial losses of an enterprise under conditions of energy constraints is considered. The methods of probabilistic analysis have been used. It has been proven that the optimal share of downtime depends on the scale of production—large enterprises are less flexible to sharp reductions in production volumes, while smaller production units can adapt faster to unstable energy supply. The formalized mathematical model presented in the article allows us to assess how different levels of downtime (from partial to full) affect the financial performance of the enterprise. The developed mathematical model allows for estimating the impact of unstable energy supply on production costs, expected profit, and cost variance, which enables the forming of energy consumption management strategies to minimize the financial losses of enterprises. The results of the study confirm that a stable energy supply is a key factor in the economic security of enterprises. The introduction of own electricity generation is economically justified only under the condition of limited use (up to 10%) and an increase in production volumes. Otherwise, a significant increase in electricity costs leads to a decrease in profitability and financial risks.

N. Havlovska · S. Matiukh · M. Kryvdyk · Y. Rudnichenko (✉)
Khmelnitsky National University, Khmelnytskyi, Ukraine
e-mail: rudnichenkoiem@khnmu.edu.ua

N. Havlovska
e-mail: havlovskant@khnmu.edu.ua

S. Matiukh
e-mail: matuh@khnmu.edu.ua

M. Kryvdyk
e-mail: science_new@ukr.net

Keywords Economic security · Volume of resources · Management · Mathematical model · Risks · Threats · Industry enterprise

1 Introduction

Managing the resource support for the activities of an industrial enterprise under certain constraints caused by certain circumstances or processes that do not depend on the business entity itself is a complex engineering and economic task related to the planning, design, and operation of supply chains based on sound management decisions, as well as the management of human capital (personnel, including managerial personnel), technologies, material and technical, information and institutional support [1–3]. Supply chain engineering includes logistics, production and pricing, covering various areas of mathematical modeling and forecasting—general operations research, machine learning and optimization. Management decisions regarding personnel, technology, information and institutional support are made and adopted on the basis of both expert analysis and forecasting of the current situation, and analytical approaches. When using the latter, expert assessment is replaced by introducing uncertainties in the external environment impact into consideration.

Researchers and professional managers generally define supply chains as an integrated network of facilities and transportation options for the supply, production, storage, and distribution of materials and products. They vary considerably in size, complexity, and scope from industry to industry. For construction companies, typical supply chain elements may include suppliers of building materials, manufacturers (processing companies), and distributors of finished products (particularly finished construction materials) [4–6]. An optimally designed supply chain should reflect, using one or more indicators, the most rational (optimal) configuration and operation of all these elements, which is possible only with the effective work of personnel and the effective use of technology and information and institutional support [7]. Therefore, it is in the interests of the industry to optimize its supply chains in a certain way and rationalize the use of non-productive resources in conditions of critical environmental impact [7].

However, the present day is making its adjustments. The supply of electricity has ceased to be uninterrupted, and the influence of the external environment has become threateningly critical in some places [8]. And if on the side of suppliers of construction raw materials and distributors of finished construction products, power outages are not yet extremely critical, then manufacturers, on the contrary, are seriously suffering from this, because the processing of construction raw materials (for example, clay for brick production) requires the starting and preparation of special equipment that is powered by the electric grid [8]. In the context of quasi-periodic power outages, this equipment may be so inefficient that on certain working days, it is not even worth running, since the costs of maintaining such equipment will exceed the potential profits from its use [9]. The fact is that there are a number of institutional restrictions, according to which it is quite difficult to switch to a selective wage payment regime

taking into account forced non-working days (for example, the option with frequent sending of employees on forced leave at their own expense is risky in terms of negative consequences in the future—there may be both civil lawsuits from employees and their outflow to other, more stable enterprises). Re-signing employment contracts with an indication of the possibility of such forced non-working days is not always legally possible.

2 Theoretical Foundations and Methodological Tools for Managing the Resource Supply of Enterprise Activities in the Context of Critical Environmental Impact

Modeling the dilemma regarding energy modernization of production requires formalizing the task of optimizing the profitability of the enterprise's operation under conditions of uncertainty [10]. This task is based on the concept of an optimization model of resource management of the enterprise's activities [10]. This management should be aimed primarily at the controlled (stable, with minor permissible deviations) maximization of net profit [9, 10]. The components of the model here are parts of the expenses for wages (including taxes), procurement of material resources (raw materials), and payment for the use of electricity in production.

Labor costs include the salaries of administrative personnel, the wages of employees directly or indirectly related to production, the wages of carriers (transportation department), the wages of support personnel, as well as taxes. Let's make the following notations. Let N_{Admin} be the number of administrative staff and N_{Pra} be the number of employees, where usually $N_{Pra} > N_{Admin}$, although there are exceptions to this rule. Also, let N_{Transp} be the number of transport personnel and N_{Dop} be the number of support personnel. Support staff should include employees whose activities at a given enterprise do not affect management decisions in any way. That is, they include cleaners, watchmen (guards), and security guards. Support staff should also include computer specialists who do not provide information and analytical support, but only maintain computer equipment and the network, as well as peripheral systems, in proper condition. Analysts should be classified as administrative personnel because they indirectly influence management decision-making (otherwise they should be classified as support personnel, although this is of no fundamental importance, as we will see below). The monthly salary of the k -th member of the administrative staff before tax is denoted by ZP_Admin_k , $k = 1, 2, \dots, N_{Admin}$. Not all of them are necessarily different, but in small enterprises where the number of administrative staff is insignificant, there are practically no identical salaries for the administrative staff. The monthly salary of the l -th employee before tax is denoted by ZP_Pra_l , $l = 1, 2, \dots, N_{Pra}$. Here, the salaries are likely to be the same (even taking into account seniority, which is not always considered, depending on working conditions and the functions performed by the employee), but for the general case, we denote

them as different. The monthly salary of the t -th member of the transportation staff before taxation is denoted by ZP_Transp_t , $t = 1, 2, \dots, N_{Transp}$. Practice shows that small enterprises may have $N_{Transp} = 1$ or $N_{Transp} = 2$. And in general, the number of N_{Transp} is usually less than $\min\{N_{Admin}, N_{Dop}\}$. And, finally, the monthly salary of the d -th member of the support staff before tax will be denoted by ZP_Dop_d , $d = 1, 2, \dots, N_{Dop}$. Thus, the labor costs, including taxes, for the month are:

$$ZP = \sum_{k=1}^{N_{Admin}} ZP_Admin_k + \sum_{l=1}^{N_{Pra}} ZP_Pra_l + \sum_{t=1}^{N_{Transp}} ZP_Transp_t + \sum_{d=1}^{N_{Dop}} ZP_Dop_d \quad (1)$$

for the total number of:

$$N_{Admin} + N_{Pra} + N_{Transp} + N_{Dop} \quad (2)$$

employees of the enterprise.

When calculating the costs of purchasing raw materials, we take into account the volume of raw materials V_C , purchased during the month (measured in kg or in pieces—a quantity unit or QU), the cost C_{var} of a volume unit of raw materials purchased (measured in UAH/QU), as well as the cost T_{var} of transporting a volume unit of raw materials purchased (measured in UAH/QU). The cost of T_{var} includes the cost of moving from the production area to the loading point. Therefore, the average monthly costs for purchasing raw materials are:

$$C_{var} \cdot V_C + T_{var} \cdot V_C = (C_{var} + T_{var}) \cdot V_C \quad (3)$$

The average monthly expenses for paying for electricity consumption in production (including the pre-processing of raw materials, which is usually part of the production process itself) are calculated separately from the expenses for lighting production and other premises (although these expenses are much smaller). Let us denote by E_{var} the cost of one kilowatt-hour for industrial enterprises, where E_{var} is measured in UAH per kilowatt-hour, i.e. UAH/(kW $\times$ h). Let W be the specific power consumed to process a unit volume of raw materials in one hour, where W is measured in UAH/(kW $\times$ h). Then the average monthly expenses for paying for electricity consumption in production are:

$$W \cdot E_{var} \cdot V_C, \quad (4)$$

Expenses for electricity consumption paying to illuminate production and other premises during the month will be denoted by O (unit of measurement—UAH). These expenses are assumed to be constant on average and practically independent of the volume of raw materials V_C , but they still depend on the number of (2) employees of the enterprise. Therefore, taking into account (4) the average monthly payment for electricity consumption is as follows:

$$W \cdot E_{\text{vart}} \cdot V_C + O, \quad (5)$$

It should be noted that, although production costs for electricity (4) and non-production costs for electricity O are not closely related, these costs are planned based on a full working day (possibly with evening and night shifts, if such are provided for in the enterprise's work schedule).

In an idealized formulation, the sum of average monthly expenses for salary payments (1), purchase of raw materials (3) and electricity (5) together with expenses for repair and maintenance needs RE (this also includes depreciation and some other operating expenses) should provide the maximum average monthly net profit P . That is, it is necessary to construct a more or less stable mapping of the sum of average monthly expenses to this maximum. In other words, such an amount of average monthly expenses should maximize the average monthly net profit P :

$$\begin{aligned} V_{yd} &= ZP + (C_{\text{vart}} + T_{\text{vart}}) \cdot V_C + W \cdot E_{\text{vart}} \cdot V_C + O + RE \\ &\rightarrow \text{net profit maximization } P, \end{aligned} \quad (6)$$

where the variables in the mapping (6) are actually all the components of the sum on the left. By personifying the management staff of the enterprise, we can influence components (1), (3), (5) in such a way as to find, in one way or another, the best possible combination of them, at which the average monthly net profit (approximately) reaches its maximum. However, it is important to remember that each component has its own limits of variation. In addition, these components are interconnected. For example, the smaller the purchases of V_C per month, the lower the expected expenses for electricity supply—both for electricity for production needs (4) and for electricity for non-production needs O . Repair and maintenance expenses RE but to a lesser extent. The sum of salary expenses will also decrease slightly if the volume of raw material purchases is reduced. However, such dependencies are not linear. Just as the sum on the left in (6) has a nonlinear effect on net profit: at small volumes of V_C P values can grow only in a narrow range; at medium, rational, volumes of V_C (the limits of these rational volumes have still to be determined) P values will grow more, reaching their possible peak (maximum); with large volumes of V_C , even if production can cope with them, sales of manufactured products will slow down due to restrictions on sales markets, resulting in additional costs for warehousing and temporary storage, which will eventually reduce the average monthly net profit.

It should be noted that the unit cost of production is implicitly included in the sum of V_{yd} , since the latter includes all cost components—purchasing (raw materials), production and non-production costs, and transportation costs, where we will assume that the delivery of finished construction products is carried out by wholesale buyers directly from production sites (in most cases, this is done by mid-level trading companies; small wholesale purchases are also possible). An approximate calculation of the initial cost can be performed by dividing the amount V_{yd} by the number of conditionally indivisible units of finished products. Of course, the cost of production affects the net profit, although it does not determine it by 100%, but its

calculation is inappropriate here due to the uncertainty of the quantitative indicator of finished goods. The sum V_{yd} , formulated in the form of the left-hand side of (6), is a sufficient basis for further considerations of maximizing net profit.

It is important to note that in more specific cases of entrepreneurial activity in the construction business, the cost structure may contain additional components, where sometimes the distinction between fixed and variable costs is blurred and unclear. This is primarily the case for small businesses where production is insignificant in volume and the range of products varies depending on customer requirements. Therefore, the production process is not constant and is modified from time to time, as well as the expenses for it. However, the structure of the sum V_{yd} in (6) contains clearly identifiable fixed costs. The expenses for maintaining the production process consist mainly of purchasing, repairing, and maintaining the equipment, which are classified as fixed costs.

Along with certain uncertainties in transformation (6), which are caused by circumstances beyond the control of the enterprise, such as fluctuations in raw material prices, fluctuations in prices on sales markets, fluctuations in the level of demand (the level of supply here is also “tied” to demand, so the enterprise can adjust the sales of its products), additional uncertainties arise caused by power outages. In the autumn of 2022, such outages were more chaotic, while in 2024, enterprises (outside the list of critical infrastructure) had a clear schedule of outages, although it changed every day. Another problem was outages in waves, when, for example, the mains power supply was lost every three hours. For medium- and large-capacity enterprises, this created additional problems with the starting up of the equipment, which takes a significant period of time (up to several hours) to ‘warm up’ along with the preparation of related components.

Without additional costs to solve the problem of power outages, all these uncertainties are irremovable. Therefore, forced downtime in the production process is inevitable. Taking into account the percentage of hours per day during which electricity was supplied, in a first approximation it can be assumed that the forced downtime is 50%. At the same time, the actual work performed by employees in production and by the personnel of the carriers is approximately half as much. However, administrative and support staff will mostly continue to do the same work (perhaps in some enterprises this will not apply to the staff of cleaners, but this does not mean a 50% reduction in their staff). Taking this into account, the average monthly wage expenditure instead of (1) will be:

$$\begin{aligned}
 ZP_{prost(1/2)} = & \sum_{k=1}^{N_{Admin}} ZP_Admin_k + \frac{1}{2} \cdot \left(\sum_{l=1}^{N_{Pra}} ZP_Pra_l + \sum_{t=1}^{N_{Transp}} ZP_Transp_t \right) \\
 & + \sum_{d=1}^{N_{Dop}} ZP_Dop_d,
 \end{aligned} \tag{7}$$

where it is assumed that the average monthly volume of purchases will be equal to $\frac{V_C}{2}$. Thus, instead of reflecting (6) for the case of 50 per cent forced downtime, the

sum of average monthly costs should maximize the corresponding average monthly net profit $P_{prost}(\frac{1}{2})$:

$$\begin{aligned} Vy d_{prost(1/2)} &= ZP_{prost(1/2)} + \frac{V_C}{2} \cdot (C_{vart} + T_{vart} + W \cdot E_{vart}) + \frac{O}{2} + RE \\ &\rightarrow \text{profit maximization } P_{prost}\left(\frac{1}{2}\right), \end{aligned} \quad (8)$$

where electricity costs for non-production needs are also expected to be half as much as in the normal operating mode of the enterprise, and maintenance and operating costs RE remain practically the same.

However, even at this stage of model development, it becomes clear that the mapping (8) has a significant drawback: it does not take into account either fluctuations in outage schedules or possible shifts in raw material transportation schedules.

In addition, the absence of electricity is not uniform—there may be fewer hours of grid power supply during the day than at night, and vice versa. Even if we assume that there were exactly 12 h of grid power supply around the clock during a given month, the uneven daily distribution of electricity causes certain deviations in the volumes of raw materials purchases. It also affects the illumination of production and other premises throughout the month. Let φ be the value of a standard normally distributed random variable (with zero mathematical expectation and unit variance). Then, a more realistic average monthly volume of (purchased and processed) raw materials is equal to $\frac{V_C}{2} + v\varphi$, where $v > 0$ is a coefficient with a dimension in CV (kg or pieces). Similarly, let us denote by ϑ the value of a standard normally distributed random variable for a realistic representation of electricity expenditures for non-production needs, which is $\frac{O}{2} + u\vartheta$, where $u > 0$ is a coefficient with dimension in UAH. The repair and maintenance costs RE should also be included here. Hence, for the case of a 50 per cent forced downtime, of production it is more correct to use (8) instead of:

$$\begin{aligned} Vy d_{prost(1/2)} &= ZP_{prost(1/2)} + \left(\frac{V_C}{2} + v\varphi\right) \cdot (C_{vart} + T_{vart} + W \cdot E_{vart}) \\ &+ \left(\frac{O}{2} + RE + u\vartheta\right) \rightarrow \text{profit maximization } P_{prost}\left(\frac{1}{2}\right), \end{aligned} \quad (9)$$

where the cost price is also implicitly included, similarly to the basis in the form of the sum $Vy d$ in the left-hand side of mapping (6). It should be noted that mapping (9) assumes the fulfillment of the obvious conditions for the existence of expenses (except for wages): $\frac{V_C}{2} + v\varphi > 0$, $\frac{O}{2} + u\vartheta > 0$.

Thus, the impact of the component $\frac{O}{2} + RE + u\vartheta$ is usually much smaller than the impact of the component $\frac{V_C}{2} + v\varphi$. However, for small enterprises, the gap between the values of these components is not so great, so they should be kept in the form (9). On the other hand, medium and large-capacity enterprises have larger production

areas, for which the average monthly value of O is much higher than for small-capacity enterprises. As for the amplitudes v and u , they are also values of random variables with a distribution law close to the exponential distribution [11] or the Weibull distribution [12]. However, the expected value of average monthly purchase volumes, as well as the expected value of electricity expenditures for non-production needs, corresponds to the mapping (8):

$$E\left(\frac{V_C}{2} + v\varphi\right) = \frac{V_C}{2}, \quad E\left(\frac{O}{2} + RE + u\vartheta\right) = \frac{O}{2} + RE, \quad (10)$$

where $E(x)$ —is the mathematical expectation (expected value) of a random variable, the values of which are denoted by x (although for simplicity x can be considered a random variable). Taking into account (10), it follows that the expected average monthly expenses at the enterprise for the case of a 50 per cent forced production downtime are equal to the left-hand side of (8):

$$\begin{aligned} & E(Vy d_{prost(1/2)}) \\ &= E\left[ZP_{prost(1/2)} + \left(\frac{V_C}{2} + v\varphi\right) \cdot (C_{vart} + T_{vart} + W \cdot E_{vart}) + \left(\frac{O}{2} + RE + u\vartheta\right)\right] \\ &= E(ZP_{prost(1/2)}) + E\left[\left(\frac{V_C}{2} + v\varphi\right) \cdot (C_{vart} + T_{vart} + W \cdot E_{vart})\right] \\ &\quad + E\left(\frac{O}{2} + RE + u\vartheta\right) \\ &= ZP_{prost(1/2)} + \frac{V_C}{2} \cdot (C_{vart} + T_{vart} + W \cdot E_{vart}) \\ &\quad + E[v\varphi \cdot (C_{vart} + T_{vart} + W \cdot E_{vart})] + \frac{O}{2} + RE + E(u\vartheta) \\ &= ZP_{prost(1/2)} + \frac{V_C}{2} \cdot (C_{vart} + T_{vart} + W \cdot E_{vart}) \\ &\quad + v\varphi \cdot (C_{vart} + T_{vart} + W \cdot E_{vart}) \cdot E(\varphi) + \frac{O}{2} + RE + u \cdot E(\vartheta) \\ &= ZP_{prost(1/2)} + \frac{V_C}{2} \cdot (C_{vart} + T_{vart} + W \cdot E_{vart}) + \frac{O}{2} + RE, \end{aligned} \quad (11)$$

where we use the fact that $E(\varphi) = 0$ and $E(\vartheta) = 0$ according to the properties of the random values φ and ϑ given by us. However, with significant fluctuations in procurement volumes, this expected value is not significant (unless such forced downtime lasts for several years, which is extremely unlikely). On the contrary, the fluctuations around such an expected value will be significant, the dispersion (or RMS deviation) of which is determined by the amplitudes v and u . Denoting the dispersion of a random variable with values x by $D(x)$, we have the following dispersion of average monthly expenses (9):

$$\begin{aligned}
& D(Vy d_{prost(1/2)}) \\
&= D\left[ZP_{prost(1/2)} + \left(\frac{V_C}{2} + v\varphi\right) \cdot (C_{vart} + T_{vart} + W \cdot E_{vart}) + \left(\frac{O}{2} + RE + u\vartheta\right)\right] \\
&= D[ZP_{prost(1/2)}] + D\left[\left(\frac{V_C}{2} + v\varphi\right) \cdot (C_{vart} + T_{vart} + W \cdot E_{vart})\right] \\
&\quad + D\left(\frac{O}{2} + RE + u\vartheta\right) \\
&= 0 + (C_{vart} + T_{vart} + W \cdot E_{vart})^2 \cdot D\left(\frac{V_C}{2} + v\varphi\right) + D\left(\frac{O}{2}\right) + D(RE) + D(u\vartheta) \\
&= (C_{vart} + T_{vart} + W \cdot E_{vart})^2 \cdot \left(D\left(\frac{V_C}{2}\right) + D(v\varphi)\right) + 0 + 0 + u^2 \cdot D(\vartheta) \\
&= (C_{vart} + T_{vart} + W \cdot E_{vart})^2 \cdot (0 + v^2 \cdot D(\varphi)) + u^2 \cdot D(\vartheta) \\
&= (C_{vart} + T_{vart} + W \cdot E_{vart})^2 \cdot v^2 + u^2, \tag{12}
\end{aligned}$$

where the properties of dispersion are used, as well as the fact that $D(\varphi) = 1$ and $D(\vartheta) = 1$. From expression (12) for the dispersion of average monthly expenses, it is clear that the main contribution to this dispersion is made by tariffs and the volume of electricity consumption. In contrast, the value of the expenditure dispersion (12) does not depend on the volume of purchases and, even more so, on the level of electricity consumption for non-production needs.

It is necessary to note here one crucial element that is typical for almost any construction company—seasonality. The estimated consideration of seasonality in expected expenditures (11) and their standard (RMS) deviation $\sqrt{D(Vy d_{prost(1/2)})}$ according to (12) is carried out by adjusting the amplitudes of v and u . For the warm season, the amplitude of v will be larger than the amplitude v for the cold season. This follows from some reduction in procurement volumes during the cold season, since at this time the sales markets work more slowly due to lower demand. With the amplitude of u the opposite is true—it will be larger for the cold season, as this is when the need for additional lighting (daylight hours are shorter) and heating increases.

In general, larger values of dispersion (12) will indicate not only high volatility of sales markets [11, 12] but also an even greater negative impact from the critical impact of the external environment. Therefore, along with maximizing net profit, we need to minimize its dispersion (12), otherwise the expected maximum net profit may never occur. This problem, to which we will return below, is a two-criteria optimization problem with two stochastic components based on random values of φ and ϑ . As it is known, even ordinary two-criteria optimization problems usually do not have an exact solution. In our case, the approximate solution will have a stochastic component.

Let us now formalize a less specific, generalized case, leaving aside the instability and heterogeneity of the outage schedule. For now, let's just assume that switching on and off the power supply for two to three hours does not significantly affect

the process of bringing equipment back to normal operation. In the general case of downtime with a proportion h , where h is the average proportion of the day during which production continues, $0 \leq h \leq 1$, instead of mapping (9) we will have

$$\begin{aligned} Vyd_{prost(h)} &= ZP_{prost(h)} + (h \cdot V_C + v\varphi) \cdot (C_{vart} + T_{vart} + W \cdot E_{vart}) \\ &\quad + (h \cdot O + RE + u\vartheta) \\ &\rightarrow \text{profit maximization } P_{prost(h)}, \end{aligned} \quad (13)$$

where the expenses for the average monthly salary during the forced downtime are:

$$\begin{aligned} ZP_{prost(h)} &= \sum_{k=1}^{N_{Admin}} ZP_Admin_k \\ &\quad + h \cdot \left(\sum_{l=1}^{N_{Pra}} ZP_Pra_l + \sum_{t=1}^{N_{Transp}} ZP_Transp_t \right) \\ &\quad + \sum_{d=1}^{N_{Dop}} ZP_Dop_d, \end{aligned} \quad (14)$$

as well as the fulfilled conditions of the existence of costs (except for salaries) for any partial downtime, except for a complete shutdown: $h \cdot V_C > v\varphi$, $h \cdot O > u\vartheta$ for $0 < h \leq 1$.

We did not immediately consider this formulation for two reasons. Firstly, a 50 per cent forced downtime is the most likely of all possible. Secondly, the formulation of the mapping of the amount of expenses for the maximum net profit in the form (13) with the generalization of salary expenses (14) contains two limiting cases, due to which the mapping (13) is conveniently integrated into the formulation (6) without taking into account stochastic components [11]. Thus, the value of $h = 1$ corresponds to the case of a fully functioning enterprise, without any stops or downtime, and the value of $h = 0$ corresponds to a complete shutdown of production.

Similarly to the derivation of formula (11), the expected monthly expenses on average are:

$$E(Vyd_{prost(h)}) = ZP_{prost(h)} + h \cdot V_C \cdot (C_{vart} + T_{vart} + W \cdot E_{vart}) + h \cdot O + RE, \quad (15)$$

The dispersion does not depend on the fraction of downtime $1-h$: for any $0 < h \leq 1$ the identity holds:

$$D(Vyd_{prost(h)}) = (C_{vart} + T_{vart} + W \cdot E_{vart})^2 \cdot v^2 + u^2 = D(Vyd_{prost(1/2)}), \quad (16)$$

However, the amplitudes of v and u are nonlinearly proportional to h . The smaller the proportion of h , the smaller these amplitudes are. In the case of a complete

production shutdown ($h = 0$) the amplitudes of v and u will be close to 0, although they cannot be equal to 0. For example, the lighting of the premises will remain at least for the administrative staff, and therefore an additional component $u|\vartheta|$ should remain in the composition $h \cdot O + u\vartheta$. Even at a complete shutdown, the final transportation processes (in particular, the transportation of raw materials to storage facilities) may still continue, and therefore an additional component $v|\varphi|$ should remain in the composition $h \cdot V_C + v\varphi$. The only thing that disappears is the procurement component and expenses for repair and maintenance. Thus, in the limiting case of a complete shutdown, the sum of the average monthly costs should simply be minimized:

$$Vyd_{prost(0)} = ZP_{prost(0)} + v|\varphi| \cdot (T_{vart} + W \cdot E_{vart}) + u|\vartheta| \rightarrow \text{minimize}, \quad (17)$$

where the expenses for salaries are:

$$ZP_{prost(0)} = \sum_{k=1}^{N_{Admin}} ZP_Admin_k + \sum_{d=1}^{N_{Dop}} ZP_Dop_d. \quad (18)$$

and it is also assumed that the enterprise will not close completely and will eventually at least partially resume production.

From expression (17) it follows that during complete downtime, the expected average monthly expenses are higher than the payment of salaries to administrative and support staff. Indeed:

$$\begin{aligned} E(Vyd_{prost(0)}) &= E[ZP_{prost(0)} + v|\varphi| \cdot (T_{vart} + W \cdot E_{vart}) + u|\vartheta|] \\ &= ZP_{prost(0)} + v \cdot (T_{vart} + W \cdot E_{vart}) \cdot E(|\varphi|) + u \cdot E(|\vartheta|) > ZP_{prost(0)}, \end{aligned} \quad (19)$$

since $E(|\varphi|) > 0$ and $E(|\vartheta|) > 0$.

To accurately determine the expected costs (19), let us recall that the probability density function of the standard normal distribution, which is inherent in the random values of φ and ϑ , is as follows: $f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$ for $-\infty < x < \infty$, where the fulfilled condition of normalization is $\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx = 1$, moreover $\int_0^{\infty} f(x) dx = \int_0^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx = \frac{1}{2}$.

Since there can no longer be negative values through the modules, and the normalization condition, $\int_{-\infty}^{\infty} f(|x|) dx = 1$, must also be satisfied, then $f(|x|) = 0$ for any $x < 0$, whence the probability density function of random values $|\varphi|$ and $|\vartheta|$ is represented as:

$$f(|x|) = 2f(x) = \sqrt{\frac{2}{\pi}} \cdot e^{-\frac{x^2}{2}} \quad \text{for } 0 \leq x < \infty, \quad (20)$$

where the aforementioned mandatory normalization condition has already been met:

$$\int_{-\infty}^{\infty} f(|x|)dx = \int_0^{\infty} f(|x|)dx = \int_0^{\infty} 2f(x)dx = 2 \int_0^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx = 2 \cdot \frac{1}{2} = 1.$$

Hence, the mathematical expectation of the modulus of the values of a standard normally distributed random variable, taking into account (20), is as follows:

$$E(|x|) = \int_0^{\infty} xf(|x|)dx = \int_0^{\infty} \frac{2x}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx = \sqrt{\frac{2}{\pi}} \cdot \int_0^{\infty} xe^{-\frac{x^2}{2}} dx = \sqrt{\frac{2}{\pi}},$$

where it is used the fact that the derivative, $\frac{d}{dx} \left(e^{-\frac{x^2}{2}} \right) = -xe^{-\frac{x^2}{2}}$, that is the integral

$$\int_0^{\infty} xe^{-\frac{x^2}{2}} dx = -e^{-\frac{x^2}{2}} \Big|_0^{\infty} = -e^{-\infty} + e^0 = -0 + 1 = 1.$$

Thus, $E(|\varphi|) = E(|\vartheta|) = \sqrt{\frac{2}{\pi}}$, this is how (19) becomes explicit:

$$E(Vy_{d_{prost}(0)}) = ZP_{prost(0)} + \sqrt{\frac{2}{\pi}} \cdot [v \cdot (T_{vart} + W \cdot E_{vart}) + u]. \quad (21)$$

The dispersion of the modulus of the values of a standard normally distributed random variable, taking into account (20), is as follows:

$$\begin{aligned} D(|x|) &= \int_0^{\infty} x^2 f(|x|)dx - \left(\int_0^{\infty} xf(|x|)dx \right)^2 \\ &= \int_0^{\infty} x^2 f(|x|)dx - [E(|x|)]^2 \\ &= \sqrt{\frac{2}{\pi}} \cdot \int_0^{\infty} x^2 e^{-\frac{x^2}{2}} dx - \frac{2}{\pi} \\ &= \sqrt{\frac{2}{\pi}} \cdot \left(\int_0^{\infty} x^2 e^{-\frac{x^2}{2}} dx - \sqrt{\frac{2}{\pi}} \right), \end{aligned}$$

where, using the previous calculations with integrals and Lopital's rule to get rid of uncertainty of like $\frac{\infty}{\infty}$, we obtain:

$$\begin{aligned} \int_0^{\infty} x^2 e^{-\frac{x^2}{2}} dx &= \left| \begin{array}{l} y = x, \quad dw = x e^{-\frac{x^2}{2}} dx \\ dy = dx, \quad w = -e^{-\frac{x^2}{2}} \end{array} \right| \\ &= -x e^{-\frac{x^2}{2}} \Big|_0^{\infty} + \int_0^{\infty} e^{-\frac{x^2}{2}} dx = 0 + \sqrt{\frac{\pi}{2}} = \sqrt{\frac{\pi}{2}}. \end{aligned}$$

Thus,

$$D(|x|) = \sqrt{\frac{2}{\pi}} \cdot \left(\sqrt{\frac{\pi}{2}} - \sqrt{\frac{2}{\pi}} \right) = 1 - \frac{2}{\pi} = \frac{\pi - 2}{\pi} = D(|\varphi|) = D(|\vartheta|)$$

and, therefore, the dispersion of average monthly expenditures during full downtime is:

$$\begin{aligned} D(Vy d_{prost(0)}) &= D[ZP_{prost(0)} + v|\varphi| \cdot (T_{vart} + W \cdot E_{vart}) + u|\vartheta|] \\ &= D[ZP_{prost(0)}] + D[v|\varphi| \cdot (T_{vart} + W \cdot E_{vart})] + D[u|\vartheta|] \\ &= v^2 \cdot (T_{vart} + W \cdot E_{vart})^2 \cdot D(|\varphi|) + u^2 \cdot D(|\vartheta|) \\ &= \frac{\pi - 2}{\pi} \cdot (v^2 \cdot (T_{vart} + W \cdot E_{vart})^2 + u^2). \end{aligned} \quad (22)$$

Thus, for the general case of production downtime, it is necessary to maximize the net profit as:

$$\max_{Vy d_{prost(h)}} P_{prost}(h) = \max_{V_C, h} P_{prost}(h) = \max_{V_C, h} P(V_C, h) \quad (23)$$

and minimize its dispersion as:

$$\min_{V_C, h} D(Vy d_{prost(h)}) = \min_{u, v} D(Vy d_{prost(h)}) = \min_{u, v} dysp(u, v), \quad (24)$$

where the function under the minimum sign in (24) can only have two variants—(16) and (22), with (16) implicitly depending on V_C through the mediation of the amplitudes v and u , the values of which are also determined by the season during which the net profit is estimated. The variables V_C and h and h are interrelated by a nonlinear proportionality relationship (since it is quite obvious that an increase in the share of production h leads to an inevitable rise in the volume of V_C purchases and vice versa), but its type is generally impossible to determine. The minimum possible value of V_C depends on the size of production—the larger the production, the more powerful the equipment is and, therefore, the more difficult it is to apply it for too small volumes of V_C . Therefore, smaller enterprises will have a lower minimum V_C value than larger enterprises. The maximum value of VC is more clearly dependent on the available production capacities and equipment. The range of volume variation

of V_C will also increase with the increase in potential production capacity, so the solution to problems (23) and (24) will be more difficult to find (estimate) for larger enterprises.

The solution to the problem (23) will be a pair of such V_C^* and h^* , where

$$P(V_C^*, h^*) = \max_{V_C, h} P(V_C, h) \quad (25)$$

The solution to the problem (24) will be a pair of such V_C^{**} and h^{**} , corresponding to the pair (u^{**}, v^{**}) , under which:

$$dysp(u^{**}, v^{**}) = \min_{u, v} dysp(u, v), \quad (26)$$

where $V_C^* \neq V_C^{**}$ and $h^* \neq h^{**}$ are in the general case. Let us consider problem (26) in more detail. Since: $\min_{u, v} dysp(u, v) = \min_{u, v} \{D(Vy d_{prost(1/2)}), D(Vy d_{prost(0)})\}$, then it is necessary to find the difference between the dispersion for general downtime and complete downtime:

$$\begin{aligned} & D(Vy d_{prost(1/2)}) - D(Vy d_{prost(0)}) \\ &= (C_{vart} + T_{vart} + W \cdot E_{vart})^2 \cdot v^2 + u^2 \\ &\quad - \frac{\pi - 2}{\pi} \cdot (v^2 \cdot (T_{vart} + W \cdot E_{vart})^2 + u^2) \\ &= C_{vart}^2 \cdot v^2 + 2 \cdot C_{vart} \cdot (T_{vart} + W \cdot E_{vart}) \cdot v^2 \\ &\quad + (T_{vart} + W \cdot E_{vart})^2 \cdot v^2 + u^2 \\ &\quad - \frac{\pi - 2}{\pi} \cdot v^2 \cdot (T_{vart} + W \cdot E_{vart})^2 - \frac{\pi - 2}{\pi} \cdot u^2 \\ &= C_{vart}^2 \cdot v^2 + 2 \cdot C_{vart} \cdot (T_{vart} + W \cdot E_{vart}) \cdot v^2 \\ &\quad + (T_{vart} + W \cdot E_{vart})^2 \cdot v^2 \cdot \left(1 - \frac{\pi - 2}{\pi}\right) + u^2 \cdot \left(1 - \frac{\pi - 2}{\pi}\right) \\ &= C_{vart}^2 \cdot v^2 + 2 \cdot C_{vart} \cdot (T_{vart} + W \cdot E_{vart}) \cdot v^2 \\ &\quad + (T_{vart} + W \cdot E_{vart})^2 \cdot \frac{2v^2}{\pi} + \frac{2u^2}{\pi} > 0, \end{aligned}$$

i.e., the dispersion at complete downtime is smaller, from which we obtain $V_C^{**} = 0, h^{**} = 0$ and the problem (26) can be rewritten as:

$$\begin{aligned} dysp(u^{**}, v^{**}) &= \min_{u, v} D(Vy d_{prost(0)}) \\ &= \min_{v, u} \frac{\pi - 2}{\pi} \cdot (v^2 \cdot (T_{vart} + W \cdot E_{vart})^2 + u^2), \quad (27) \end{aligned}$$

where, let us recall, the positive amplitudes of v and u in the case of a complete shutdown of production will be the smallest (close to zero).

However, it already becomes obvious here that the two-criteria optimization problem (23), (27) will not have a solution (where $V_C^* = 0$ and $V^* = 0$), since a complete shutdown will not be beneficial to every enterprise, even if its capacity is not taken into account. Moreover, since larger volumes of purchases correspond to larger h , which in turn ‘raises’ the amplitudes of v and u , the dispersion (16) will increase nonlinearly as the share of downtime $1-h$ decreases (i.e. as the share of production h increases). Too high a variance, although corresponding to a sufficiently high value of $P(V_C^*, h^*)$, will reduce the probability of such a net profit to an unacceptably low value.

In the case of incomplete downtime, the expected net profit is calculated as follows:

$$P_{prost}(h) = Vyru_{c_{prost}(h)} - E(Vy d_{prost}(h)) - Pod_{prost}(h), \quad (28)$$

where $Vyru_{c_{prost}(h)}$ is the estimated average monthly revenue for production with a share of h and payment of the relevant taxes $Pod_{prost}(h)$. The value of $Vyru_{c_{prost}(h)}$ directly depends on V_C and average market prices for the wholesale distribution of construction products. To calculate the expected net profit (28) in the case of continuing production with any non-zero share ($0 < h \leq 1$) it is necessary to calculate the average monthly revenue $Vyru_{c_{prost}(h)}$ and expected expenses according to formula (15), substituting into it the expenses $ZP_{prost}(h)$, the monthly purchase volume plan V_C , the tariffs C_{var} , T_{var} , E_{var} , the specific power W (estimated at the enterprise), as well as the average monthly expenses O for illumination of production and other premises (taken as the arithmetic average of such expenses for the period from 2023 to April 2024 inclusive). By changing the values of the variables V_C and h with some steps (it can be from 1 to several percent of the range of possible values), we build the function (28) on the lattice of pairs of values of V_C and h . Each such pair is linked to a pair of values v and u , changing them with the same step, starting with the values $v = 0.01$ and $u = 0.01$ for the pair of minimum values of V_C and h . Each pair of values (v, u) is used to estimate the dispersion (16).

As the values of the pair (V_C, h) increase, the expected net profit (28) will probably increase nonlinearly in some range, while the dispersion of (16) will definitely increase up to the maximum values in the pair (V_C, h) . The choice of a pair of values (V_C^{***}, h^{***}) , at which the expected net profit $P(V_C^{***}, h^{***})$ will be sufficiently high with a moderate dispersion $dysp(V_C^{***}, h^{***})$, depends on the line of intersection of surfaces (28) and (16). And this depends on the particular enterprise and the type of its products.

In the case of complete downtime, the expected net profit will be equal to the value of (21) taken with the opposite sign:

$$P_{prost}(0) = -ZP_{prost(0)} - \sqrt{\frac{2}{\pi}} \cdot [v \cdot (T_{var} + W \cdot E_{var}) + u], \quad (29)$$

Of course, Eq. (29) is essentially the losses of an enterprise that is not yet formally closed, but under conditions of critical impact of the environment, there is no guarantee that (28) will be a positive value, whatever the fraction h . To estimate the losses (29) and the dispersion of losses (22), we also use pairs of values (v, u) .

In conditions of limited or unstable electricity supply, there is another option—switching to auxiliary power generation. If this switch is complete (for a certain period of time, not on a permanent basis), then the transformation of expenses into net profit of P_{gener} looks like this:

$$\begin{aligned} Vy d_{gener} &= ZP + (C_{vart} + T_{vart} + W \cdot G_{vart}) \cdot (V_C + v\varphi) + O_{gen} + RE + u\vartheta + \frac{Gen}{m} \\ &\rightarrow \text{net profit maximization } P_{gener}, \end{aligned} \quad (30)$$

where G_{vart} is the cost of one kilowatt-hour generated at the enterprise [similar to E_{vart} , measured in UAH per kilowatt-hour, i.e. UAH/(kW × h)], and Gen is the cost of generators (petrol or diesel) purchased by the enterprise for its own electricity generation during periods of grid blackouts, m is the expected number of months of alternative generation. In this case, the objective is to minimize the expected costs:

$$E(Vy d_{gener}) = ZP + V_C \cdot (C_{vart} + T_{vart} + W \cdot G_{vart}) + O_{gen} + RE + \frac{Gen}{m}, \quad (31)$$

which is equivalent to maximizing the net profit of P_{gener} , and minimizing the dispersion:

$$D(Vy d_{gener}) = (C_{vart} + T_{vart} + W \cdot G_{vart})^2 \cdot v^2 + u^2, \quad (32)$$

of these expenditures, which differs from the dispersion in (32) only in the cost of G_{vart} . The difference between expected expenditures (15) and (31):

$$\begin{aligned} &E(Vy d_{prost(h)}) - E(Vy d_{gener}) \\ &= ZP_{prost(h)} + h \cdot V_C \cdot (C_{vart} + T_{vart} + W \cdot E_{vart}) + h \cdot O + RE \\ &\quad - \left(ZP + V_C \cdot (C_{vart} + T_{vart} + W \cdot G_{vart}) + O_{gen} + RE + \frac{Gen}{m} \right) \\ &= ZP_{prost(h)} - ZP + (h - 1) \cdot V_C \cdot (C_{vart} + T_{vart}) + V_C \cdot W \cdot (h \cdot E_{vart} - G_{vart}) \\ &\quad + h \cdot O - O_{gen} - \frac{Gen}{m}. \end{aligned} \quad (33)$$

From the difference (33), it follows that in the absence of downtime, it will be as follows:

$$E(Vy d) - E(Vy d_{gener}) = V_C \cdot W \cdot (E_{vart} - G_{vart}) + O - O_{gen} - \frac{Gen}{m}.$$

That is, balancing the costs of alternative generation (assuming that $O \approx O_{\text{gen}}$) is possible only by reducing the cost of G_{vart} , preliminary searching for cheaper generator options, and, if possible, extending the operation of the enterprise (even if the electricity supply interruptions cease, this will not prevent the enterprise from continuing to use its own generation).

But there is also a variant of mixed generation, when production with a share of p is powered by the grid, and the rest of the production with a share of $1-h$ is powered by its own generation. In this case, the expected expenses are equal to the sum of the expenses in (13) and the expenses in (30), taken with a weight of $1-h$, where the objective is to maximize the corresponding net profit of $P_{\text{gener-prost}}(h)$:

$$\begin{aligned}
 & Vy d_{\text{prost}(h)} + Vy d_{\text{gener}(1-h)} \\
 &= ZP + h \cdot (V_C + v\varphi) \cdot (C_{\text{vart}} + T_{\text{vart}} + W \cdot E_{\text{vart}}) \\
 &\quad + h \cdot (O + u\vartheta) + (1-h) \cdot (V_C + v\varphi) \cdot (C_{\text{vart}} + T_{\text{vart}} + W \cdot G_{\text{vart}}) \\
 &\quad + (1-h) \cdot (O_{\text{gen}} + u\vartheta) + RE(h) + \frac{Gen}{m} \cdot (1-h) + Gen_0 \\
 &= ZP + (V_C + v\varphi) \cdot [h \cdot (C_{\text{vart}} + T_{\text{vart}}) + h \cdot W \cdot E_{\text{vart}}] + hO + hu\vartheta \\
 &\quad + (V_C + v\varphi) \cdot [C_{\text{vart}} + T_{\text{vart}} + W \cdot G_{\text{vart}} - h(C_{\text{vart}} + T_{\text{vart}}) - h \cdot W \cdot G_{\text{vart}}] \\
 &\quad + O_{\text{gen}} + u\vartheta - h \cdot (O_{\text{gen}} + u\vartheta) + RE(h) + \frac{Gen}{m} \cdot (1-h) + Gen_0 \\
 &= ZP + (V_C + v\varphi) \cdot h \cdot W \cdot E_{\text{vart}} + hO + (1-h) \cdot O_{\text{gen}} \\
 &\quad + (V_C + v\varphi) \cdot [C_{\text{vart}} + T_{\text{vart}} + W \cdot G_{\text{vart}} \cdot (1-h)] + u\vartheta \\
 &\quad + RE(h) + \frac{Gen}{m} \cdot (1-h) + Gen_0 \\
 &= ZP + (V_C + v\varphi) \cdot [h \cdot W \cdot E_{\text{vart}} + C_{\text{vart}} + T_{\text{vart}} + W \cdot G_{\text{vart}} \cdot (1-h)] \\
 &\quad + hO + (1-h) \cdot O_{\text{gen}} + u\vartheta + RE(h) + \frac{Gen}{m} \cdot (1-h) + Gen_0 \\
 &\rightarrow \text{net profit maximization } P_{\text{gener-prost}}(h), \tag{34}
 \end{aligned}$$

where the expenses for repair and maintenance needs are denoted here by $RE(h)$, since they depend on the share of production $1-h$ from own power generation (generator maintenance expenses will grow with the increase in own power generation). It should be noted that the purchase of own generators by the enterprise here directly depends on the value of h , so the share $\frac{Gen}{m}$ is taken with a weight of $1-h$ for the case $1 < h$. To this share the base cost Gen_0 of the generators is added (they are purchased by the company for any case, including the case $h = 1$, when the power supply will be entirely from the grid). The salary expenses will be the same as in the case of a normal production cycle without downtime. The objective is to minimize the expected expenses:

$$\begin{aligned}
& E(Vyd_{prost(h)} + Vyd_{gener(1-h)}) \\
& = ZP + V_C \cdot [h \cdot W \cdot E_{vart} + C_{vart} + T_{vart} + W \cdot G_{vart} \cdot (1-h)] \\
& \quad + hO + (1-h) \cdot O_{gen} + RE(h) + \frac{Gen}{m} \cdot (1-h) + Gen_0, \tag{35}
\end{aligned}$$

which is equivalent to maximizing the net profit $P_{gener-prost}(h)$, and minimizing the dispersion

$$\begin{aligned}
& D(Vyd_{prost(h)} + Vyd_{gener(1-h)}) \\
& = (h \cdot W \cdot E_{vart} + C_{vart} + T_{vart} + W \cdot G_{vart} \cdot (1-h))^2 \cdot v^2 + u^2. \tag{36}
\end{aligned}$$

It is obvious that problem (34) turns into the problem (30) if we set $h = 0$. It should also be noted that the variance (36) does not depend on the number of months m .

Similar to problems (23) and (24) for the general case of production downtime, for the general case of using alternative generation, it is necessary to maximize the net profit as:

$$\max_{Vyd_{prost(h)} + Vyd_{gener(1-h)}} P_{gener-prost}(h) = \max_{V_C, h, m} P_{gener-prost}(h) = \max_{V_C, h, m} P(V_C, h, m), \tag{37}$$

Similar to problems (23) and (24) for the general case of production downtime, for the general case of using alternative generation, it is necessary to maximize the net profit as:

$$\min_{h, u, v} D(Vyd_{prost(h)} + Vyd_{gener(1-h)}) = \min_{h, u, v} dysp(h, u, v), \tag{38}$$

where in (38) we substitute (36), and under the maximum sign in (37) we substitute the expected net profit:

$$\begin{aligned}
P_{gener-prost}(h) & = Vyruc_{gener-prost(h)} - E(Vyd_{prost(h)} + Vyd_{gener(1-h)}) \\
& \quad - Pod_{gener-prost}(h), \tag{39}
\end{aligned}$$

where $Vyruc_{gener-prost(h)}$ is the estimated average monthly revenue for production with a share of h of power supply from the general grid and a share of $1-h$ of power supply from own generation, and $Pod_{gener-prost}(h)$ —is the taxes paid. The value of $Vyruc_{gener-prost(h)}$ directly depends on V_C and average market prices for the wholesale distribution of construction products.

The solution of problem (37) is a triple of such $V_C^{(*g)}, h^{(*g)}, m^{(*g)}$, where:

$$P(V_C^{(*g)}, h^{(*g)}, m^{(*g)}) = \max_{V_C, h, m} P(V_C, h, m). \tag{40}$$

The solution of problem (38) will be a pair of $(V_C^{(**g)}, h^{(**g)})$, corresponding to the triple $(h^{(**g)}, u^{**}, v^{**})$, for which^

$$\begin{aligned} dysp(h^{(**g)}, u^{**}, v^{**}) &= \min_{h, u, v} dysp(h, u, v) \\ &= \min_{u, v, h} [(h \cdot W \cdot E_{vart} + C_{vart} + T_{vart} + W \cdot G_{vart} \cdot (1 - h))^2 \cdot v^2 + u^2]. \end{aligned} \quad (41)$$

where the pair (u, v) is connected to the volume V_C . As in the pair of problems (23) and (24), in general, these triples are pairwise distinct. Problems (37) and (38) are somewhat more complicated than similar problems (23) and (24) for the general case of downtime, because they contain an additional variable—the number of months m . But this complexity is not excessive. As in the solution of the problem (23) to achieve (25), the values of the pair (V_C, h) in the triple (V_C, h, m) to achieve (40) are gradually increased to a certain volume maximum of V_C and $h = 1$ for $m = 1, 2, 3, \dots$, until the number of months goes beyond the predicted values of market prices and tariffs. The number of months, by the way, can be a fractional (non-integer) number (e.g. half a month, one and a half months, etc.).

3 Testing the Model of Resource Supply Management of Enterprise Activities Under Critical Influence of the External Environment

For the period from April to June 2024, the average monthly expenses of “Khmelnytsky Plant of Construction Materials” ALC (Khmelnytskyi zavod budivelnykh materialiv) (“KhZBM” ALC) were as follows:

- (1) pulverized coal materials and coal—UAH 4,000,000;
- (2) electricity—UAH 1,200,000;
- (3) fuel (transportation)—UAH 750,000;
- (4) salaries and wages—UAH 2,900,000, where the share of administrative and support staff is 18%;
- (5) repair and maintenance needs—UAH 500,000;
- (6) expenses for electricity consumption for illumination of production and other premises—UAH 27,000.

During the warm season, the output at “KhZBM” ALC ranges from 1 million 700 thousand to 1 million 800 thousand bricks per month. During this period all products are sold without storage. The price is UAH 6 excluding VAT per brick. Thus, we have the following parameters for problem (23) at (28), (15) and problem (27) at (39), (35):

$$\begin{aligned} ZP_{prost(h)} &= 0.18 \cdot 2100000 + h(1 - 0.18) \cdot 2100000 = 1680000 \cdot h + 420000, \\ V_C &= 1750000, \quad C_{vart} = 2.28571429, \quad T_{vart} = 0.42857143, \end{aligned}$$

$$W = 0.08568537, E_{\text{var}} = 8.0027, O = 27000, RE = 500000.$$

The expected amount of revenue is as follows:

$$Vyru_{c_{\text{prost}(h)}} = 6 \cdot 1.2 \cdot h \cdot V_c = 7.2 \cdot h \cdot V_c = 12600000 \cdot h,$$

where the corresponding taxes will be proportional to the reduction in production:

$$Pod_{\text{prost}(h)} = Vyru_{c_{\text{prost}(h)}} \cdot 0.2 = 2520000 \cdot h,$$

The minimum dispersion (27) is calculated as follows:

$$dysp(u^{**}, v^{**}) = \min_{v, u} (0.3633802276 \cdot u^2 + 0.4511847561 \cdot v^2),$$

where the minimum possible values of u^{**} and v^{**} are close to zero but not equal to 0. For calculations in the warm season, we take $v = \sqrt{h}$, $u = h^2$.

The “KhZBM” ALC estimated that expenses for repair and maintenance needs grow nonlinearly with the increase in the share of own power generation, so that with 90% of such generation, these expenses will almost triple. Therefore, it is reasonable to present these expenses as follows:

$$RE(h) = \frac{RE}{\sqrt{h}} \text{ for } h \geq 0.1.$$

Taking into account the prices of industrial generator lines, it was estimated that:

$$G_{\text{var}} = 6.32346 \frac{\text{UAH}}{\text{kW} \cdot \text{h}}, \text{ Gen} = 1950000 \text{ UAH}, \text{ Gen}_0 = 900000 \text{ UAH}$$

where the cost of one kilowatt-hour generated at the enterprise was estimated on the basis of the average fuel consumption of a diesel generator with a peak capacity of 200 kW. Thus, we can build the dependencies of the expected net profit (28) and (39) on the production capacity with a fraction of p . Figure 1 shows the net profit (28) for $V_c = 1,750,000$ (star markers) and $V_c = 1,850,000$ (dot markers), as well as the net profit (39) for $V_c = 1,750,000$ (circle markers) for one, two and three months of own generation and for $V_c = 1,850,000$ (square markers) for $m = 1, 2, 3$.

The net profit curve (39) is nonlinear, especially when the share of own generation is large. With an increase in the duration of own generation (the larger m), this curve rises, but still achieving a positive net profit is possible only if the volume of production and sales increases by 100,000 units (bricks) using no more than 10% of own generation for at least two months. At the same time, the maximum expected net profit will be at the level of UAH 136,000, which is significantly less than the expected net profit of UAH 460,000 with a realistic volume of $V_c = 1,750,000$. Figure 1 also shows that the maximum permissible production downtime is about 30%, at which the average monthly expected net profit is equal to UAH 15,400.

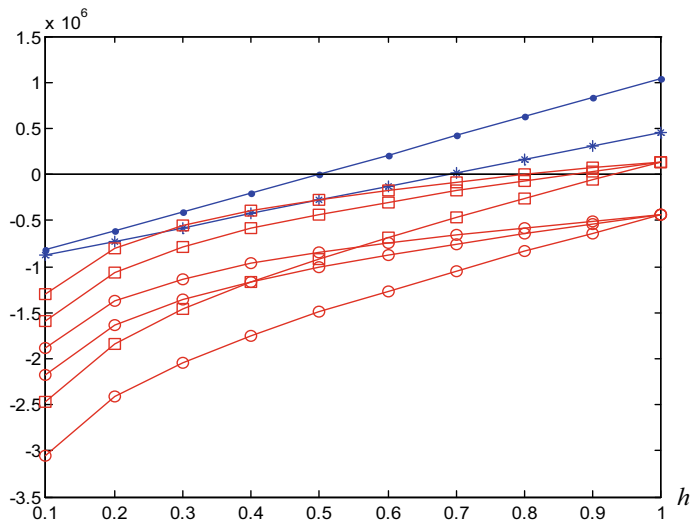
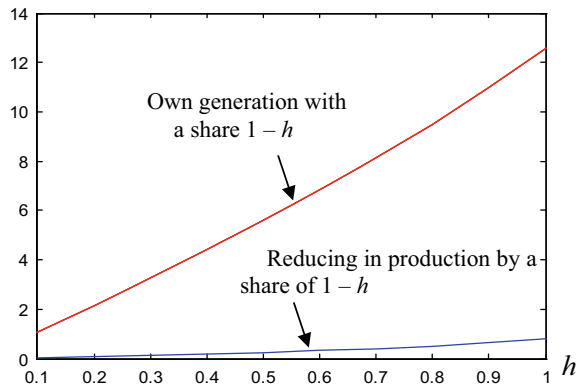


Fig. 1 Dependence of the expected net profit at “KhZBM” ALC on the production capacity with a share h for two basic production volumes

The dispersion practically does not depend on the volume of production and sales V_c , representing a non-linearly increasing curve (Fig. 2).

Own generation is less reliable in terms of net profit stability (39), which is why it is inefficient at “KhZBM” ALC. This is explained by the fact that production consumes significant volumes of electricity, which at current prices for power generators is difficult to compensate for within a month or a quarter.

Fig. 2 Dependence of the dispersions (22) and (36) for “KhZBM” ALC on the share of production h with power supply from the general grid



4 Conclusions

Resource support for an enterprise's operations in the event of a disruption in the continuity of supply of the relevant resource (electricity) affects three key types of resource support: personnel, technology, and logistics. The level of this influence varies depending on the specifics of the enterprise, its size, development dynamics, profitability, and production volumes. Because of this, an important task is to develop effective mechanisms for managing the enterprise's resource supply in the face of an unstable supply of key resources.

The article presents a model that allows enterprise management, based on actual and forecasted indicators, to develop alternative management decisions that will provide the opportunity to manage the resource provision of the enterprise's activities effectively and will positively affect the level of economic security. One of the important advantages of the model is the ability to take into account most of the parameters that affect the cost of production, as well as the clear identification of threats with forecasting the consequences of their implementation. This creates a basis for the development of preventive measures, which is critically important in today's conditions of destabilization of the external environment.

General recommendations for managing the resource support of an enterprise's activities in conditions of critical external influences may contain specific strategic parameters. For example, it is recommended that "KhZBM" ALC increase additional production (up to 7%), which will allow the enterprise to increase the level of economic security and minimize the risks associated with the instability of the energy supply.

References

1. Adedigba SA, Khan F, Yang M (2018) An integrated approach for dynamic economic risk assessment of process systems. *Process Saf Environ Prot* 116:312–323
2. Havlovska N, Rudnichenko Ye, Babiy I, Matiukh S, Liubokhynets L, Lopatovskyi V (2020) The strategy of choosing promising markets for the enterprise—subject of foreign economic activity on the basis of accessibility, safety and profitability criteria. *Qual Access Success J* 21(178):26–34
3. Liubokhynets L, Rudnichenko Ye, Dzhereliuk I, Illiashenko O, Kryvdyk V, Havlovska N (2020) Methodological foundations of flexible management and assessing the flexibility of an enterprise economic security system. *Int J Sci Technol Res* 9(3):4616–4621
4. Bosovska M, Boiko M, Bovsh L, Okhrimenko A (2022) Models of the Industrial Revolution 5.0. In: 2022 IEEE 4th international conference on modern electrical and energy system (MEES). IEEE, pp 1–4. <https://doi.org/10.1109/mees58014.2022.10005761>
5. Gilad-Bachrach R, Navot A, Tishby N (2004) Margin based feature selection: theory and algorithms. In: Greiner R, Schuurmans D (eds) *Proceedings of the twenty-first international conference on machine learning*, Banff, Alberta, Canada. ACM Press, New York, NY, pp 337–344
6. Erić-Nielsen J, Todorović-Spašenić A, Stevanović A (2024) Designing an organizational structure as a prerequisite for achieving business excellence in manufacturing companies. *Ekonomski Horizonti* 26(1):41–61. <https://doi.org/10.5937/ekonhor2401041e>

7. Engelhardt-Nowitzki C, Kryvinska N, Strauss C (2011) Strategic demands on information services in uncertain businesses: a layer-based framework from a value network perspective. In: 2011 international conference on emerging intelligent data and web technologies. IEEE, pp 131–136. <https://doi.org/10.1109/eidwt.2011.28>
8. Belik M, Rubanenko O, Hunko I, Rubanenko O, Baraban S, Semenov A (2025) Tools for researching the parameters of photovoltaic modules. *Electronics* 14(9):1885. <https://doi.org/10.3390/electronics14091885>
9. Dzhedzhula V, Yepifanova I (2021) Optimization of energy saving potential of industrial enterprises. In: 2021 11th international conference on advanced computer information technologies (ACIT). IEEE, pp 433–436. <https://doi.org/10.1109/acit52158.2021.9548428>
10. Dzhedzhula V, Yepifanova I, Kravchik Y (2022) Use of the theory of fuzzy sets in determining the level of enterprise security. In: 2022 12th international conference on advanced computer information technologies (ACIT). IEEE, pp 311–315. <https://doi.org/10.1109/acit54803.2022.9913150>
11. Manly BFJ (2008) *Statistics for environmental science and management*. Chapman & Hall/CRC, Boca Raton, FL, p 295
12. Walpole RE, Myers RH, Myers SL, Ye K (2012) *Probability and statistics for engineers and scientists*. Prentice Hall, Boston, MA, p 791

The Strategic Potential of Renewable Energy Microgrids: Global Transformations and Determinants of Sustainable Development



Volodymyr Lypov , Natalya Ushenko , Viktoriia Hurochkina ,
Tomasz Szapiro , Larysa Shaulska , and Viacheslav Dzhezhula

Abstract The study presents the results of a study aimed at determining the strategic potential of implementing digitization technology in renewable energy production through the creation of renewable energy microgrids and the impact of such technology on the processes of evolutionary globalization. Renewable energy is a key element in ensuring sustainable development. The variety of alternative energy sources and the large number of producers capable of offering electricity highlight the need for small microgrids. The smart grid business model considered in the study can stimulate the restructuring of the production system and electricity distribution markets. The study carried out structural, functional, systemic and institutional analysis. The features of digital platforms as an organizational form are discussed. These

V. Lypov

State Organization “Institute for Economics and Forecasting of the National Academy of Sciences of Ukraine”, Kyiv, Ukraine

e-mail: lypov_vl@ukr.net

N. Ushenko

Borys Grinchenko Kyiv Metropolitan University, Kyiv, Ukraine

e-mail: ushenko_nv@ukr.net

N. Ushenko · V. Hurochkina · T. Szapiro

Warsaw School of Economics, Warsaw, Poland

e-mail: viktoriia.hurochkina@sggw.waw.pl

T. Szapiro

e-mail: tszapiro@sggw.waw.pl

V. Hurochkina

Kyiv Institute of Business and Technology, Kyiv, Ukraine

L. Shaulska

Taras Shevchenko National University of Kyiv, Kyiv, Ukraine

e-mail: shaulska@knu.ua

V. Dzhezhula (✉)

Vinnitsia National Technical University, Vinnitsia, Ukraine

e-mail: djedjula@vntu.edu.ua

platforms provide synergies by connecting small renewable energy producers in microgrids, leading to a two-sector model for the energy industry. The mechanism of building energy systems based on the principles of “Industry 5.0” and aspects of accelerating global sustainable development goals is being analyzed.

Keywords Renewable energy sources · Microgrids · Energy · Sustainable development goals · Model · Ecosystem · Digital platform

1 Introduction

The purpose of the study is to determine the strategic potential for the introduction of technologies for digitalization and renewable energy generation, the formation of microgrids of renewable energy sources (RES) and their impact on the evolutionary processes of globalization. The analysis includes the impact of these processes on the economy, environment and global energy markets, the role of digitalization in optimizing the energy sector, as well as the democratization of relations in this area. The importance of sustainable development, related goals and ways of introducing new energy technologies, especially in the context of challenges and threats, in particular, armed conflicts, is noted. The deepening of globalization processes leads to transformations of civilizational concepts, economic and social needs of stakeholders in national systems. This, in turn, requires changes in the functioning of the energy market, the growth of the role and importance of digital technologies. Ultimately, almost all components of the life of society are transformed. This is especially important when these changes affect the backbone sectors of the economy. The main place among them is occupied by electricity production.

The energy sector plays a paradoxical role in the life of society. The development of technologies based on the use of electric energy ensures an increase in the standard of living of society. At the same time, energy is the largest consumer of non-renewable resources and a source of environmental pollution. Therefore, the sustainable development of individual countries and humanity as a whole largely depends on the ability to solve these contradictory tasks. Increasing the level of satisfaction of human needs should occur simultaneously with limiting the negative consequences of intensifying the use of these resources.

In this context, the digitalization of the energy sector and the introduction of renewable energy generation technologies directly contributes to the achievement of a number of sustainable development goals. Among them, the main ones are: *SDG 7*—“Ensuring access for all people to affordable, reliable, sustainable and modern energy sources”, *SDG 9*—“Creating sustainable infrastructure, promoting inclusive and sustainable industrialization and innovation”, *SDG 11*—“Ensuring the openness, security, resilience and environmental sustainability of cities and human settlements”, *SDG 12* “Ensuring the transition to sustainable consumption and production patterns”. The indirect goals are: *SDG 8*—“Promoting continuous, inclusive and sustainable economic growth, full and productive employment and decent work for

all”, *SDG 13*—“Take urgent action to combat climate change and its consequences”, *SDG 14*—“Conserve and sustainably use the oceans, seas and marine resources for sustainable development”, *SDG 15*—“Protect, restore and promote ecosystem management, rational forest management, combat desertification, halt and reverse land degradation and halt the loss of biological diversity” (“Transformation of Our World”, 2015).

Digitalization and the use of alternative sources of renewable energy open up new prospects for the industry. Increasing the total capacity and increasing the efficiency of renewable energy generation plants is accompanied by a decrease in negative environmental emissions. Optimization of the work of a whole complex of enterprises for generation, accumulation and distribution of energy flows is provided. Opportunities are created to overcome natural monopolies in the markets of production, transmission, distribution, and retail trade in electricity. The potential for deepening the differentiation of wholesale and retail markets, increasing the number of their participants is growing. Prospects for the transition from a hierarchically integrated organization to a two-sector model are opening up. In it, the traditional centralized structure is complemented by local decentralized microgrids. They bring together both ordinary electricity consumers and households, small and medium-sized enterprises that can independently produce and supply surplus electricity to the market. Thus, the prerequisites for the transformation of competitive relations in the local and global markets are formed.

The digitalization of the energy sector is especially important in the face of military aggression by Russia. The events of autumn 2022 demonstrated the importance of reorganizing the entire complex of industry enterprises in the process of prompt response to challenges and improving the country’s energy infrastructure. Under these conditions, the opportunities that ensure the introduction of information technologies and the development of digital platforms are explored, see Hrytsenko and Lipov [1] to overcome the threats of uncertainty and instability of the energy system functioning in wartime, the period of post-war reconstruction of the Ukrainian economy, ensuring sustainable development.

2 Literature Review

Already at the beginning of the new millennium, Rochet and Tirole [2] drew attention to the formation of markets with bilateral characteristics as a result of the use of the business platform model. The recognition of the role of digital platforms as an innovative institutional form and business model is justified by Parker et al. [3]. Moret et al. [4] points to the reduction of transaction costs, which is a key feature of the platform model.

At the same time, significant changes are taking place in the area of electricity production, as evidenced by Rochet et al. [2]. The development of alternative energy based on renewable sources has created the possibility of a transition to the so-called “small energy” within private households, investigated by Fehrenbacher [5]. The next

step in this area is the creation of microgrids for the production and distribution of electricity. Hossain et al. [6] investigate the role of information and communication technologies in the development of grid renewable energy. According to data from Jenkins et al. [7], they enable intelligent evaluation (smart metering) of planning and organizing network options. Olivella-Rosell et al. [8] propose a model for smart platforms of energy service providers. The use of the platform model paves the way for ensuring interaction between suppliers. Kiesling [9] and Tabors [10] focus on the study of the specifics of the organization of this process.

Breuer [11] explores the impact of the integration of digital technologies on the decentralization of energy and the role of microgrids in ensuring the flexibility and stability of energy systems. The author emphasizes that the development of microgrids supported by digital platforms makes it possible not only to increase energy efficiency, but also to democratize access to energy and strengthen the role of local communities in the production and consumption of electricity. In particular, Breuer is looking at the potential of microgrids in terms of sustainability and the reduction of greenhouse gas emissions, which is in line with the global goals of the energy transition [11].

Scientific literature proves the growing understanding of the market potential of the interaction platform model in the area of electricity production and distribution, as explored by the author Srnicek [12]. An important element of ensuring the energy efficiency of microgrids is the possibility of storing and further redistributing temporary surplus generated energy. The economic and political aspects of this process are at the center of research of Timilsina et al. [13], Parag and Sovacool [14] highlight the key role in the ongoing changes in the possibility of connecting households within platform ecosystems, the market positions of electricity producers and consumers, and the entry of this market into the era of pro-consumerism. An important aspect of this process is domestic economic policy in the area of international exchange, taking into account the context of economic security, investigated by Hurochkina et al. [15, 16]. In turn, Morstyn et al. [17] and Matviienko et al. [18] study the impact of the interaction of small microeconomic electricity producers on the transformation of the national energy production system.

Attempts to globally assess the economic development of solar energy in households were made by Lang et al. [19]. Similar results of the US market analysis were presented by the National Renewable Energy Lab (NREL), investigated by Fu et al. [20]. The synergy effects of creating network structures that connect renewable energy sources in the countries of the Global South and Germany are discussed by Kirchhoff et al. [21]. The benefits and problems arising from the development of pro-consumerism in the energy sector in Germany are studied by Schill et al. [22]. Kwan [23] investigates the impact of the specificity of the economic environment on the organization of the territorial distribution network of electricity generated by private photovoltaic installations. Camello et al. [24] focuses on investigating the economic viability of solar energy development in households in Portugal. A comprehensive study of the experience of many countries around the world in the field of renewable energy development and prospective directions for the development of green energy was presented by Korohodova et al. [25].

Scientific literature also notes the issues of stimulating the development of private energy production. Important results in this area were obtained by Abdmouleh et al. [26]. They focus on researching best practices to drive integration in the renewable energy industry. Borenstein [27] on the other hand, examines the impact of tools such as tariffs and tax breaks. Eid et al. [28] explore the use of network cost compensation and cross-subsidisation practices. Hagerman et al. [29] and Lang et al. [19] explore the possibilities of small-grid electricity without state support. The specificity of competition as a stimulus for the development of grid energy is discussed in the works of Comello and Reichelstein [24].

The development of renewable energy sources (RES) around the world has drawn attention to the issues of energy security, resilience and climate change mitigation, which is emphasized by Sevostyanov and Shaforost [30] and Shaulska et al. [31]. Renewables capacity is projected to grow significantly, with a record increase of 320 GW planned for 2022, as considered by Trunina et al. [32]. The transition to renewable energy is possible thanks to factors such as technological progress, cost reduction and increased efficiency, as indicated by Karabegović and Dolecek [33], Karabegović [34]. Karbovska et al. [35] emphasize that the introduction of RES is conducive to reducing dependence on fossil fuels, increasing energy security and supports sustainable development. Countries such as Iceland, Kenya and Latvia are leaders in electricity production from RES, while other countries such as Ukraine and the United Kingdom are showing rapid growth, as indicated by Karbovska et al. [35]. Investments in renewable energy technologies have also led to increased job opportunities around the world. Klepacka [36] notes that the promotion of renewable energy sources is in line with the Sustainable Development Goals and requires constant support and training. Therefore, for the effective implementation and development of renewable energy sources, comprehensive approaches are necessary, including government support, attracting investments, developing educational programs and technical infrastructure. Such a systemic approach will help accelerate the transition to sustainable energy models, reduce greenhouse gas emissions, and achieve energy independence, which is important for countries with high dependence on energy imports, such as Ukraine.

Ukraine has significant potential in this sector and is among the top 20 countries attractive for investments in renewable energy, as evidenced by Voronin and Panasiuk [37]. The country's renewable energy market is showing growth, including solar, wind, and hydropower, according to Pecheniuk et al. [38]. However, such investments also pose challenges, including investment risks caused by political instability and legal uncertainty, as pointed out by Yakubiv et al. [39] and Kharlamova et al. [40]. Demand for renewable energy and readiness for its deployment, as indicated by Kuzior et al. [41]. It is high in Ukraine. Therefore, the literature recommends the introduction of support policies, attracting investments and improving energy management practices in order to trigger further development of this sector in Ukraine, investigated by Klopov [42] and Denysiuk et al. [43]. The ongoing war with Russia has had an impact on the renewable energy sector, which requires a focus on post-war reconstruction and development. However, overall, renewable energy in

Ukraine has significant potential and prospects for further growth, but its implementation requires strategic actions such as creating favourable conditions for investors, ensuring the stability of the legal framework and overcoming existing risks. Despite the challenges of war and political instability, the existing demand for green energy among the public indicates the country's readiness for change. In the post-war period, Ukraine has the opportunity not only to rebuild its infrastructure, but also to transform it in accordance with modern standards of sustainable development, strengthening its energy security and independence.

An analysis of global trends in the growth of renewable energy capacity between 2013 and 2022 shows significant and continuous growth worldwide. The data, as shown in Fig. 1, shows an increase in total renewable energy capacity more than doubled, from 1,566,487 MW in 2013 to 4,448,051 MW in 2024.

Figure 1 shows the trends in the growth rate of renewable energy capacity worldwide. Renewable energy is being actively implemented in many countries by reducing the cost of technology, supporting governments and raising awareness of the importance of green sustainability and security of energy systems. This has become a major factor in the gradual reduction of reliance on traditional energy sources such as coal and oil, and the reorientation towards solar, wind, hydro and other renewable energies.

This increase in the use of renewable energy reflects a strong global drive for cleaner, more sustainable energy solutions, which are driven by factors such as technological advancements, cost reductions, and a global focus on energy security, climate resilience, and greenhouse gas emission reductions. This trend highlights the

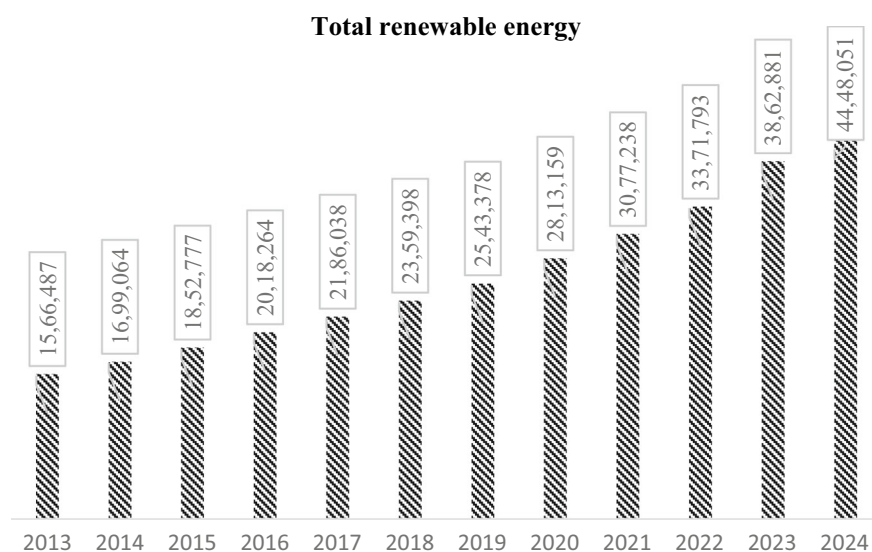


Fig. 1 The dynamics of renewable energy use rates in the world. *Source* Processed by the authors based on statistical data [44, 45]

collective international commitment to renewable energy, emphasizing the importance of continuing to support policy, investment, and technical education in the sector to accelerate the transition to sustainable energy models, which is crucial for countries heavily dependent on energy imports, such as Ukraine.

At the World Economic Forum in Davos on May 24, 2022, a project to increase Ukraine's renewable energy capacity by 2030 from 9 to 30 GW was presented, investigated by Trotsenko [46]. This is in line with global trends in supporting RES for energy security and sustainable development, as noted by the authors of the scientific papers Trunina et al. [32], Zacharia and Kovalchuk [47]. Many authors support this stance with regard to the country's efforts to align with global and European initiatives such as REPowerEU, which aim to diversify energy sources and reduce dependence on fossil fuels, investigated by Denysiuk et al. [43], Hurochkina and Kohut [16]. The legal and organizational basis for the implementation of this project was set out in the Law of Ukraine "On Alternative Energy Sources", adopted by the Parliament of Ukraine in December 2003. The Act sets out the legal, economic, environmental and organisational rules for the use of alternative energy sources and to support their development in the fuel and energy sector (2003).

3 Methodology

3.1 Research Design

This study adopts a mixed-methods approach combining conceptual systems analysis and empirical econometric modeling. The methodology is aimed at identifying and quantifying the key determinants of renewable energy (RE) microgrid growth and their correlation with the achievement of Sustainable Development Goals (SDGs). The methodological framework integrates system-level thinking, policy analysis, and econometric techniques to ensure both depth and rigor in interpreting the transformative potential of RE microgrids at local, national, and global levels.

3.2 Conceptual and Theoretical Framework

The conceptual framework is built on the assumption that RE microgrids represent a catalyst for achieving multiple SDGs, by enabling energy democratization, reducing carbon emissions, fostering local economic development, and enhancing resilience to crises. These microgrids, supported by digital platforms and innovation ecosystems, contribute to both technological and social sustainability. The regression model serves as a tool to quantify these relationships and validate their empirical relevance.

Table 1 Variables, operationalization, and link to SDGs

Variable	Definition	Expected effect	Linked SDGs
RE microgrid growth	Annual growth of decentralized renewable energy generation capacity (e.g., MW/year, number of sites)	Dependent variable	SDG 7, SDG 13
DigitalIndex	Composite index of digital adoption (e.g., DESI, World Bank Digital Index)	Positive (+)	SDG 9 (Industry, Innovation), SDG 11 (Resilient Infrastructure), SDG 17 (Partnerships for Goals)
FDI	Foreign direct investment in energy sector, % of GDP	Positive (+)	SDG 17 (Global Partnerships), SDG 8 (Decent Work and Economic Growth)
PolicySupport	Institutional index of renewable energy policy (e.g., incentives, tax relief, green tariffs)	Positive (+)	SDG 7 (Affordable and Clean Energy), SDG 12 (Responsible Consumption and Production)
GlobalTrade	Trade openness index or RE technology imports/exports (as % of GDP or total trade)	Positive (+)	SDG 17 (Global Trade), SDG 9 (Infrastructure), SDG 13 (Climate Action)
GDPpc	GDP per capita in constant USD. Proxy for investment capacity and affordability	Positive (+)	SDG 1 (No Poverty), SDG 10 (Reduced Inequalities), SDG 11 (Inclusive Cities), SDG 7 (Energy Access)

Source Developed by the authors

We posit that the growth of RE microgrids is a function of the country’s digital transformation, global integration, institutional policy environment, and socio-economic capacity. These determinants, in turn, correlate with specific SDGs as detailed in Table 1.

3.3 Regression Model Specification

The core model is specified as:

$$RE\ Microgrid\ Growth = \beta_0 + \beta_1 \cdot Digital\ Index + \beta_2 \cdot FDI + \beta_3 \cdot Policy\ Support + \beta_4 \cdot Global\ Trade + \beta_5 \cdot GDP_{pc} + \varepsilon, \tag{1}$$

where: i denotes the country, t is the time period, ε is the error term capturing unobserved influences.

3.4 Analytical Strategy

The empirical model is estimated using Ordinary Least Squares (OLS), and in the case of panel data availability, Fixed Effects or Random Effects estimators are used to control for unobserved heterogeneity. Diagnostic procedures include tests for multicollinearity (VIF), heteroskedasticity (White's test), and autocorrelation (Durbin-Watson). Log-transformation of selected variables may be applied to correct for skewness and scale disparity. This approach allows for robust analysis of how macro-level drivers affect microgrid development and how these, in turn, facilitate the progress toward SDG targets.

3.5 Hypotheses Formulation

The following hypotheses are proposed to guide the empirical analysis:

- H1: Higher digitalization levels (DigitalIndex) positively impact the growth of RE microgrids (linked to SDG 9 and SDG 7).
- H2: Strong policy support (PolicySupport) for renewables enhances the deployment of microgrids (linked to SDG 7 and SDG 12).
- H3: Higher economic capacity (GDP per capita) enables broader participation in decentralized energy systems (linked to SDG 10 and SDG 11).

3.6 SDG-Oriented Policy Implications

The model's structure provides a pathway for deriving policy-relevant insights on how targeted interventions in digital infrastructure, investment facilitation, regulatory support, and global cooperation can reinforce sustainable energy transitions. The results are expected to inform multi-level governance strategies by:

- strengthening digital and financial inclusion for clean energy access (SDG 7);
- enhancing innovation systems and infrastructure modernization (SDG 9);
- promoting resilience and environmental sustainability at community levels (SDG 11 and SDG 13);
- facilitating international collaboration in green energy technology and knowledge transfer (SDG 17).

The integrated methodological approach adopted in this study enables both a theoretical and empirical investigation of the drivers influencing RE microgrid development within the framework of the Sustainable Development Goals. By establishing direct correlations between macro-level determinants and SDG outcomes, the model offers a robust foundation for evidence-based energy policy and strategic planning in the context of global energy transitions.

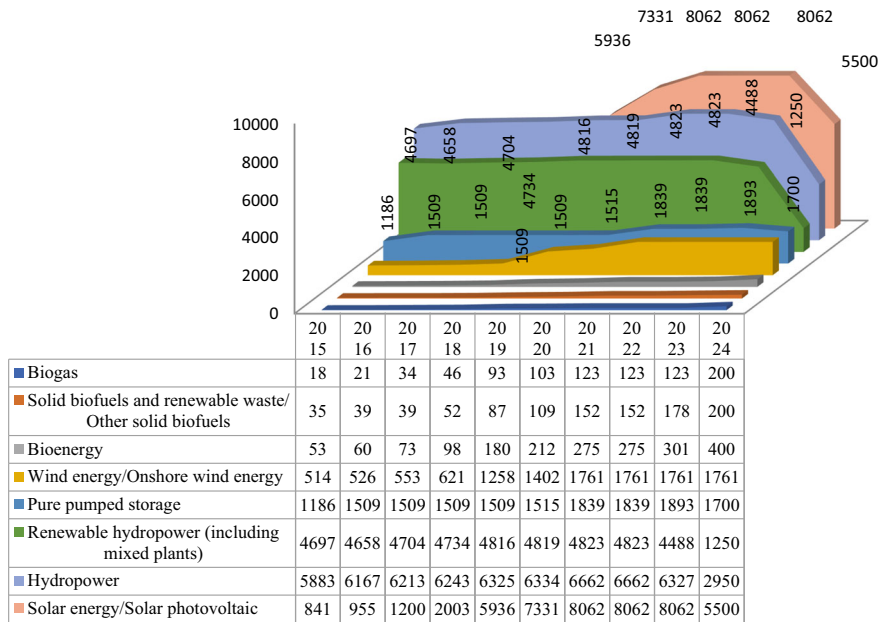


Fig. 2 Development of renewable energy in Ukraine in 2015–2024¹ (megawatts). *Source* Developed by the authors on the basis of Energy (2025)

4 Results

At the World Economic Forum in Davos on May 24, 2022, a project to increase Ukraine’s renewable energy capacity by 2030 from 9 to 30 GW was presented, as evidenced by Trotsenko [46]. This is in line with global trends in supporting RES for energy security and sustainable development, as noted by the authors of the scientific papers Trunina et al. [32], Zacharia and Kovalchuk [47]. Many authors support this stance with regard to the country’s efforts to align with global and European initiatives such as REPowerEU, which aim to diversify energy sources and reduce dependence on fossil fuels, as evidenced by Denysiuk et al. [43]. The legal and organizational basis for the implementation of this project was set out in the Law of Ukraine “On Alternative Energy Sources” [48, 49], adopted by the Parliament of Ukraine in December 2003. The Act sets out the legal, economic, environmental and organisational rules for the use of alternative energy sources and to support their development in the fuel and energy sector (2003). The data presented in Fig. 2 indicate the potential for the development of renewable energy in Ukraine and the successes achieved in this area.

¹ By 2014, including capacities located in the Autonomous Republic of Crimea and the occupied parts of the Luhansk and Donetsk oblasts (a total of 633.7 MW).

Large energy companies, small and medium-sized enterprises and households are involved in the development of energy from renewable sources. The increase in electricity production in households using solar panels in 2020 and 2021 was 40.9% and 54.7%, respectively. Over the same period, energy production using biomass by small farms increased by 62.8% and 67.0%.

The total number of home photovoltaic systems (DSF) at the end of 2021 was 44,888 units, the total installed capacity was 1205 MW, and the value of electricity sold at the “green” tariff was 1094 million kWh [45]. At the same time, the potential for the development of renewable energy sources in Ukraine is not fully exploited. Factors limiting growth include limited financial resources, insufficient return on investment and increased risks associated with high dependence on weather conditions. It is also worth noting that investment risks have increased as a result of Russia’s military aggression. Another effect of this is the emergence of problems with the implementation of “green” tariffs.

The chart presents the installed capacity trends of various renewable energy sources from 2015 to 2024. The data show a significant and continuous increase in solar photovoltaic (PV) energy, which grew from 841 MW in 2015 to a peak of 8062 MW in 2023, with a projected decrease to 5500 MW in 2024. Wind energy and hydropower exhibit more stable values, with hydropower peaking around 6662 MW in 2020 and decreasing thereafter. Renewable hydropower (including mixed plants) and pure pumped storage remain relatively consistent across the period. Moderate growth is observed in bioenergy sources, including biogas and solid biofuels. The figure highlights solar PV as the fastest-growing renewable energy source during the observed period, although a potential decline is projected in 2024.

Figure 2 shows the trends in the dynamics of renewable energy sources in Ukraine. Solar energy and wind power plants showed particularly high growth, where the capacity increased many times, especially after 2018. Other sources, such as biomass, biogas and small hydroelectric power plants, are also showing gradual growth, but at a slower rate. The overall trend is that interest and investment in the sector is increasing, which is in line with the global trend of clean energy transition, although it leaves room for further development of bioenergy and small hydropower plants.

Large energy companies, small and medium-sized enterprises, as well as individual households are involved in this process. For example, the increase in electricity production by households based on the use of solar panels in 2020 and 2021 was 40.9% and 54.7%, respectively. Over the same period, energy production using biomass by small farms increased by 62.8% and 67.0%. The total number of home photovoltaic systems (SES) at the end of 2021 was 44,888 units, the total installed capacity was 1205 MW, and the value of electricity sold at the “green” tariff was 1094 million kWh [45]. Despite this, the potential for the development of renewable energy sources in Ukraine is not fully exploited.

Factors that hinder or slow down the development of renewable energy sources in Ukraine include aspects such as limited financial resources, low return on investment, increased risk associated with high dependence on weather conditions, and limited opportunities to monetize excess energy generated. This complex of factors is shown in Fig. 3.

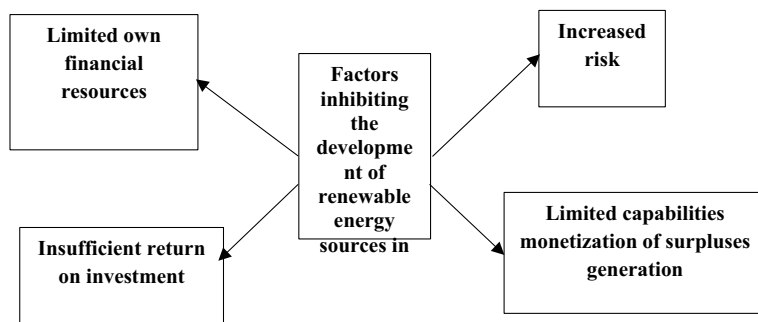


Fig. 3 Factors inhibiting the development of renewable energy sources in Ukraine. *Source* Developed by the authors

The following barriers to the development of processed energy can be identified in Fig. 3:

1. The shortage of public and private funds for investments in renewable energy sources significantly slows down the implementation of new projects, especially small and medium-sized RES facilities. Many Ukrainian companies operating in the energy sector do not have sufficient working capital or access to long-term credit on favourable terms, and public funds are mainly channeled towards urgent defence and reconstruction needs;
2. Investments in renewable energy sources often require significant upfront investment, while their cost-effectiveness is lower compared to traditional energy sources. Due to the uncertainty surrounding green tariffs and potential risk factors, investors tend to rate such projects as less attractive, which negatively affects the level of capital raised;
3. Renewables, including photovoltaics and wind farms, depend heavily on natural conditions that can be unpredictable. Climate change and extreme weather events, which have become more frequent in recent years, can affect the stability of energy production, leading to unpredictable declines in energy production. This forces power system operators to look for ways to balance the grid, which involves additional investments in energy storage systems or integration with other sources;
4. Excess energy production from RES (when the energy produced exceeds the current demand) remains an untapped sphere for monetization. Ukraine lacks modern technologies for large-scale energy storage, such as battery systems or hydrogen installations, that could store excess energy. In addition, the limited integration of the Ukrainian energy system with the European ENTSO-E does not allow for the full use of the opportunities to export excess energy, especially during periods of peak production.

It is worth noting that the investment risk has only been strengthened as a result of Russia's military aggression. Another result is the emergence of problems with the implementation of sources of coverage of "green" tariffs.

In order to overcome these obstacles and ensure the stable development of renewable energy sources, grants and loans and support from international financial organizations such as the EU, the World Bank, and the European Bank for Reconstruction and Development (EBRD) play an important role in financing “green” projects. Investment processes in energy storage systems and creating conditions for the integration of the national energy system with the European ENTSO-E significantly increase the reliability and stability of renewable energy sources. It is also worth highlighting the importance of decentralization processes in the context of providing distributed generation that supports communities by reducing dependence on centralized networks and increasing resilience to crisis situations. Ukraine has made significant progress in the development of renewable energy sources, especially in the solar and wind energy sectors. However, to fully unlock this potential, it is necessary to address financial constraints, increase the return on investment and reduce the risks associated with high weather dependency. These problems are exacerbated by increased investment risks and disruptions caused by Russia’s military aggression, which makes it difficult to finance ‘green tariffs’. Continued support and strategic actions are necessary to ensure and expand renewable energy capacity in Ukraine.

To overcome the challenges of the development of renewable energy sources in Ukraine, a forward-looking solution is to create small electricity generation grids or microgrids that ensure decentralized and stable energy production and distribution at the local level. The next chapter will discuss ways to overcome the limitations of renewable energy by creating and developing digital platform practices for microgrids as a strategic approach to overcoming these challenges and increasing energy independence.

5 Digital Platforms and RES Microgrids

The technological basis and the organizational center for the formation of the RES microgrid ecosystem at the local level are digital platforms, which play an important role in ensuring the efficiency of the value chain between producer and consumer. The specificity of the functioning of RES microgrids is that the producer and the consumer, depending on the situation, can change roles, performing the functions of both producer and consumer, which makes them prosumers, according to the definition of Šajin [50]. RES microgrids, operating on digital platforms, become elements of the energy system of both local communities and the national energy system, combining centralized and decentralized models. The functionality of digital platforms allows to increase the efficiency of the use of RES installations in households or small and medium-sized enterprises (SMEs), shortening the payback period for private owners and reducing costs for consumers, as well as minimizing the risks associated with the continuity of service provision, as noted by Lypov [51].

Modern digital platforms have become a tool for data collection, analysis, coordination and distribution of energy flows, enabling network interaction between

producer and consumer through continuous monitoring of data on the functioning of microgrids, as indicated by Lypov and Ushenko [52].

Digital transformation is a major source of change in the overhaul of local and centralized energy systems, and digital platforms perform the function of modernizing and creating combined energy systems. The concept of “platformization” in the energy sector focuses on data-driven production processes, networking between independent energy producers and the transition to horizontal interaction between them. It is worth noting that the proliferation of digital platforms brings new challenges to the sector, including reindustrialization, P2P interaction, changes in ownership relations and competition policy, as well as expanded opportunities to ensure national security and economic stability [53, 54].

The next step will be to determine the impact of platform uptake on both large energy producers and SMEs and to establish a conceptual basis for understanding the transition to decentralised and combined energy networks. The directions of change in the energy sector are determined by key trends in the development and implementation of new technologies, the growing dynamics of energy markets and the strengthening of the role of state regulations in the context of increased volatility and uncertainty in the business environment.

The U.S. Department of Energy (DOE), in the context of recognizing the national importance of this process, promotes the integration of renewable energy sources through the systemic design, planning, and operation of electric grids, presented DOE [55]. The priority actions include:

- reducing emissions of carbon dioxide and other air pollutants by increasing the use of renewable energy sources and other forms of clean distributed generation;
- Increase asset utilization by integrating distributed systems and customer workloads to reduce peak demand and reduce energy costs.
- supporting the implementation of renewable energy portfolio standards and energy efficiency;
- improving the reliability, security and resilience of microgrid applications in the protection of critical infrastructure and high-confinement areas of power grids;
- reducing fuel consumption by ensuring the operation of electric vehicles.

In conclusion, the current digital transformation in the energy sector will be a gradual process, but it will eventually change the structure of the industry, moving from centralized to decentralized and combined systems. The integration of RES microgrids focuses on the incorporation of renewable, distributed energy production, energy storage and demand response in the electricity distribution and transmission system.

6 Industry 5.0—A Model for the Development of RES Microgrids

Currently, an analysis of spatial mechanisms of change in the energy sector will be presented, taking into account the specificity of the impact of digital platforms and RES microgrids on structural transformations, operational efficiency and the role of state regulations in the context of the integration of new technologies with existing structures. Renda et al. [56], notes that the introduction of digital platforms for small-scale energy production allows the energy sector to implement an Industry 5.0 model that integrates energy production, storage and distribution processes as components of a single system of socially responsible interaction.

Through projects such as Siemens IREN2 and Brooklyn microgrids, this digital transformation uses technologies such as the Internet of Things, blockchain algorithms, and artificial intelligence to create “smart virtual power plants”—self-regulating hybrid (cyber-physical) systems that support decentralized, resilient, and efficient energy grids at both local and global levels.

The use of digital platforms in a microgrid network shows the innovative potential for the development of the global energy market in the context of the implementation of “Industry 5.0”. This includes the full cycle of production and management processes—generation, storage, support, distribution and consumption of energy in one system, and democratization of the involvement of local residents in solving problems related to strengthening energy stability and improving the quality of life in communities.

A characteristic feature of the construction of digital platforms in the area of energy production based on “Industry 5.0” is the full automation of interactions between all participants in the energy production and consumption process. Digital platforms enable rapid response to changes in the internal structure of the system under the influence of the external environment. The core of the platform as a technological component of the “Industry 5.0” model in the energy sector ensures the coordination of the activities of all components. The digital twin of RES microgrids allows physical objects to track the processes taking place in the digital platform ecosystem. The Internet of Things, digital ecosystems, big data, algorithmic management systems and artificial intelligence are all elements of technological support for the operation of RES microgrids. Figure 4 shows the structure of a digital platform in a global energy production system.

Figure 4 illustrates the structural elements of the digital platform of RES microgrids. The model of applying the principles of “Industry 5.0” in the energy sector assumes the integration of owners of individual rooftop photovoltaic systems, wind turbines, biogas and hydropower installations, as well as energy storage stations, into a uniform local ecosystem of a digital platform. This provides an incentive for owners to monetize their surplus energy generation, allowing them to join the retail electricity market. At the same time, the low transaction costs of exchanging energy flows remain minimal.

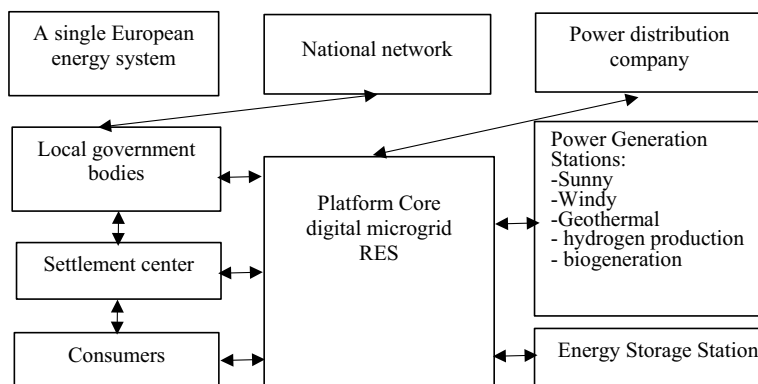


Fig. 4 Structure of the Digital Platform of RES microgrids in the Global Energy Production System.
Source Developed by the authors

The core of the digital platform is the main technological element of the “Industry 5.0” model in the RES microgrid, which ensures the coordination of the activities of all components. Their use allows the maximum number of participants or technological components to be combined in a single production process, regardless of their spatial distribution (local/global) and the nature of the interaction (hierarchical subordination, horizontal interaction). This chapter considers the importance of integrating RES microgrids into Industry 5.0 in the context of strengthening the social dimension of local community interaction to increase energy resilience. The structural element of the RES microgrid model includes components ensuring the generation, storage, distribution of energy flows, collection and analysis of relevant data, rapid response and management of energy flows in RES microgrids, as well as integration with and through the national energy system with the Single European Energy Grid.

The realism of Industry 5.0 in the socio-political perspective of a globalizing world is determined by the following two aspects. Firstly, it is full support for the implementation of the global sustainable development goals with a focus on the synergies of digital and green transformations. Secondly, it is influenced by strengthening the resilience of value chains to crisis phenomena and processes (warfare, epidemics, etc.). As a whole, this contributes to the tendency of socio-economic systems, and in particular energy systems, to global integration on the basis of the transformation of conditions and components of the system for ensuring people’s lives.

7 Microgrid Ecosystem and Accelerating Sustainability

The concept of Smart Grid is the implementation of innovative technologies, including information and telecommunications, into power systems in order to increase their efficiency, profitability and reliability. This makes the power grid more flexible, automated, and self-supervisable. Expanding the microgrid ecosystem to accelerate sustainability fosters the integration of renewable energy sources and minimizes environmental impact, reducing dependence on traditional energy sources, as well as ensuring local energy independence of communities, increasing resilience to external crises. Andriushchenko et al. [57] consider the implementation of Smart Grid technology as a natural evolution of an energy system that improves energy efficiency and provides accurate calculations. RES microgrids connect small electricity producers (farms, small businesses, households, municipal companies) at the local level. Kwaśiński [58] describe them as “...a group of associated loads and distributed energy resources within clearly defined electrical boundaries that act as a controlled unit with respect to the grid....”.

The main features of smart microgrids include: management automation and monitoring, two-way exchange of information and energy, integration of renewable energy sources, reduction of grid losses and resource savings, development of decentralized systems [59].

1. Management automation and monitoring. With smart sensors and monitoring systems, microgrids automatically tracks the status of the grid and responds quickly to changes to avoid power outages.
2. Bilateral exchange of information and energy. Microgrids technology allows energy companies and consumers to exchange data in real time, which enables energy consumption to be optimized, reducing costs for users and helping to maintain system stability.
3. Integration of renewable energy sources (RES). Microgrids facilitates the operation of solar and wind power plants, which are typically characterized by generation instability, and helps stabilize their impact on the overall power grid.
4. Reduce grid losses and save resources. Optimized management and energy-saving technologies reduce energy losses during transportation and distribution, making the system greener.
5. Development of decentralized systems. Supporting microgrids and distributed generation allows for autonomous power systems at the community or enterprise level.

The Microgrids concept aims to create a more stable, reliable and adaptive power system that will be able to respond to contemporary challenges, such as the growing demand for electricity, the integration of renewable energy sources and the need to improve energy efficiency.

Microgrids contribute to achieving sustainable energy development by ensuring the integration of renewable energy sources and fostering energy autonomy at the local level. Building a microgrid ecosystem involves the interplay of several

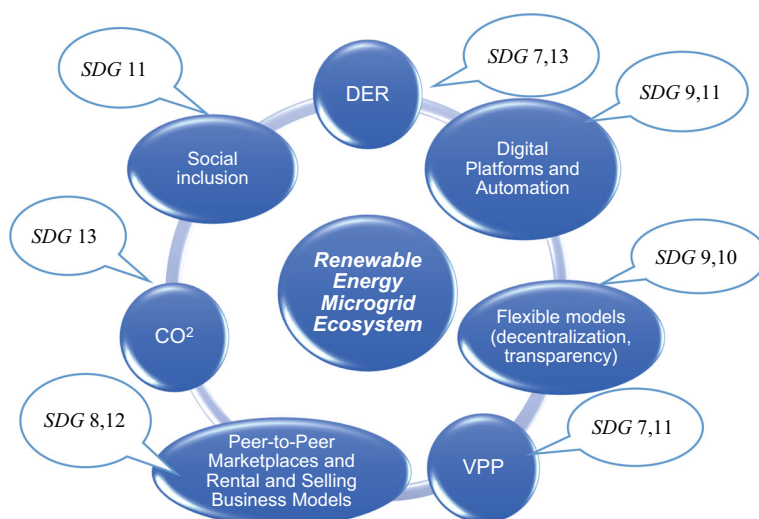


Fig. 5 Microgrid ecosystem through the lens of sustainability. *Source* Developed by the authors

key components, including local power generation, *Distributed Energy Resources (DERs)*, digital platforms for automation and efficient management of energy flows, and modern business models for renting and selling excess energy. Such systems not only contribute to the reduction of greenhouse gas emissions, but also improve energy security and the flexibility of power grids, reducing dependence on central energy supply.

Given the numerous ecological and socio-economic benefits, microgrids are becoming a key element in the development of a sustainable energy future, contributing to the achievement of the global sustainable development goals. Figure 5 presents the structural elements of the microgrid ecosystem and presents the value system in the context of achieving sustainable development and its SDGs.

In order to accelerate sustainable development, in order to build a microgrid ecosystem, it is worth distinguishing the following key structural elements:

1. Local energy production and DER.
2. Integration with digital platforms and automation.
3. Operation of models of rental and sale of excess energy
4. Modular and Virtual Power Plants (VPPs).
5. Decentralized energy exchange through the digital energy market.
6. Ecological and socio-economic benefits of microgrids.

Local energy production and DER. Local energy producers—small businesses, farms, households and municipal utilities that produce electricity from renewable sources such as solar panels, small wind farms, bioenergy systems. Distributed energy resources (DERs) are small autonomous resources that can both produce and store energy. These are, m.in, photovoltaic panels on roofs, home batteries, small wind

turbines that provide flexibility in the use of energy at the local level. Local renewable energy sources, such as solar panels, support SDG 7 (Accessible and Clean Energy) because they expand access to stable and green energy, especially in isolated areas. They also support SDG 13 (Combating climate change) by reducing greenhouse gas emissions that arise during traditional fossil fuel energy production.

Integration with digital platforms and automation is firstly the use of digital platforms to automate the processes of energy exchange and sale within local microgrid networks. These platforms allow you to reduce transaction costs and improve control over the distribution of energy resources. Second, the implementation of DER management systems that coordinate the operation of different energy resources within the microgrid ensures the stability of energy supply. Automation ensures that devices such as photovoltaic panels and batteries are integrated into virtual power plants and other larger modular networks.

The development of digital platforms for business and *peer-to-peer* markets is closely related to the implementation of distributed energy resource (DER) technologies. DER assets, such as photovoltaic panels or small batteries, enable both self-consumption of energy and its sale within the microgrid. This creates a new level of energy autonomy, allowing resources to be combined into virtual power plants and reducing the load on central power grids. The use of digital platforms simplifies transactions in a DER environment by automating energy exchange and the integration of devices such as distributed photovoltaic panels. This fosters the development of digital energy markets, where participants can buy or sell excess energy through information platforms. Combined with smart grid technologies, digital automation allows you to optimize energy production and consumption in decentralized systems, increasing the resilience and efficiency of energy consumption. Such changes stimulate the development of the home energy management market, support the increase in the share of renewable energy sources and encourage consumers to actively manage their own energy.

The use of digital platforms to manage DER supports SDG 9 (Industry, Innovation and Infrastructure) by implementing innovative solutions in the energy sector. Process automation helps ensure a stable energy supply, supporting SDG 11 (Sustainable development of cities and communities).

Peer-to-peer markets and business models for renting and selling excess energy, such as local markets where DER owners allow the sale of excess generated energy to other microgrid participants who need it. This increases the efficiency of the energy generated and reduces the load on the main grid. Innovative business models for renting and sharing (e.g. pay-per-use models or renting devices (solar panels, batteries) for households that cannot afford full installation).

Local markets foster the development of sustainable consumption and production models (SDG 12) by stimulating a more efficient use of energy resources and minimizing the excessive load on central grids. The ability to sell excess energy also helps to achieve SDG 8 (Decent Work and Economic Growth) by creating new business models and markets.

Combining local DERs into a modular structure is conducive to functioning in one virtual power plant. VPP allows you to centrally control a large number of distributed

resources, maintain the energy balance in the network and provide backup capacity. In addition, modularity allows for flexible integration, allowing for rapid adaptation and integration of new energy resources, adapting to the needs of local consumers and responding to changes in demand.

Virtual power plants support SDG 7 by ensuring a stable and clean energy supply through the coordination of distributed resources. The modularity of VPPs supports flexibility and rapid adaptation to changes in demand, which is important to ensure the reliability of energy supply in cities and rural areas (SDG 11).

The decentralised exchange of energy through the digital energy market and the information platform provides increased transparency and fosters flexible energy management models. It supports the operation of a decentralized market by allowing microgrid participants to exchange energy in real time and interact with the main energy system. The use of digital technologies allows for the automation of energy exchange processes, reducing the costs of doing business and increasing the competitiveness of the local energy sector. This component supports SDG 9, stimulating the development of digital solutions for energy infrastructure and increasing the competitiveness of local energy markets. At the same time, it supports SDG 10 (Reducing Inequalities), ensuring access to cheap and clean energy for more people.

The ecological and socio-economic benefits of microgrids using RES significantly reduce dependence on fossil fuels, which helps reduce greenhouse gas emissions and contributes to a better ecological situation. In general, microgrids ensure inclusion in the energy market, stable energy supply, especially in crisis conditions or interruptions in the operation of the main grid. The use of microgrids from RES contributes to increasing energy security and sustainable development of local communities, reducing dependence on fossil fuels, supporting SDG 13. In addition, ensuring a stable energy supply even during a crisis increases the energy security of communities, which corresponds to SDG 11.

The structural elements of the microgrid ecosystem increase its efficiency and flexibility, support economic growth through the development of new market relationships and business models, and ensure sustainability through the integration of renewables and digital technologies.

The modern approach to the decentralization and diversification of energy supply, as well as energy management technologies, improves economic and environmental performance, enabling more efficient resource management and fostering innovation in the energy sector.

8 Globalization Context of the Development of RES Microgrids

The formation of RES microgrids is related to the processes of globalization. It is worth taking into account the two-sided nature of this process, the mutual impact of globalization and the development of RES microgrids. Paradoxically, the intensification of localization in the energy sector leads to significant changes at the global level. There is a profound transformation of the energy generation sector, global energy markets, and the systems of interconnection and interdependence between countries and regions participating in global interaction. Researchers also deserve attention in the fact that the spread of the effects of this transformation goes far beyond economic relations. The democratization of the relationship between energy producers and consumers is ensured, the liquidation of monopoly in centralized energy production systems, presented Moret and Pinson [4]. The democratization of the interaction of RES microgrid participants at the local level, thanks to the reduction of dependence on fossil fuels, is an impulse for the process of democratization of international relations in the energy sector. The development of RES microgrids is conducive to reducing the dependence of fossil fuel importers on supplier countries. It opens up the possibility of transforming the centers of global influence, changing the role and place of individual countries and even regions, and democratizing interactions at the global level. The development of cooperation between RES power plant owners at the local level is of particular importance in the context of activities to support sustainable development, stated by the authors Bauwens and Eyre [60]. Let's look at some aspects of the interaction of RES microgrid development and globalization in more detail.

The authors of the study "The impact of economic globalization on renewable energy in OECD countries" propose treating economic globalization as a mechanism supporting the spread of renewable energy sources as an element supporting sustainable development, stated by the authors Gozgor et al. [61]. Several components of this process can be distinguished. It is highly expected that the intensification of international interaction supported by globalisation will contribute to the development of renewable energy production. The more intense the interaction of a country with its environment, the faster innovative technologies, which include renewable energy technologies, reach the internal markets of other countries, facilitating the borrowing of relevant knowledge and skills to work with them, stated by the authors Zeren and Akku [62]. At the same time, Doytch and Narayan [63] note that the international flow of investments aimed at supporting the development of renewable energy sources in various regions of the world is facilitated. The level of development of renewable energy production in the country depends on indicators such as the level of development of the financial sector, the availability of foreign direct investment, the level of income of the population and electricity prices, presented Anton and Afloarei [64]. Manufacturers of relevant devices, financial institutions of the countries they represent, are interested in supporting the promotion of their products on global markets. Ultimately, globalisation has made it much easier for renewable

energy technologies to spread and scale up rapidly. On the other hand, connecting individual RES power plants in a microgrid, thanks to the possibility of monetizing excess energy production, ensures increased efficiency in the use of equipment and accelerates its return. The key factors influencing the intensification of this process and its specific manifestations in individual countries are income and the level of carbon dioxide emissions per capita, the degree of development and organization of the centralized energy production system, the proportions and level of prices of fossil fuels and renewable energy sources, as well as the level and nature of support for the implementation of the latter by the state.

In the context of the research topic, the relationship between globalization, the implementation of digital technologies and the development of renewable energy production is particularly noteworthy. The digitalisation of economic relations opens up ample opportunities for the trade of digital products and the use of digital technologies to facilitate international trade operations and international direct investment as key components of economic globalisation. Combined with stimulating the further development of digital technologies, it contributes to accelerating technological progress and increasing the potential for green development. In the energy sector, these processes are manifested by improving energy storage and distribution technologies, reducing costs, reducing related expenses, maximizing efficiency and increasing the efficiency of RES cycles. The result is an optimization of the energy structure (see Fig. 6), presented Zhang et al. [65].

At the same time, it is worth noting that the development of RES microgrids has the opposite impact on globalization processes. The creation of RES microgrids contributes to increasing the efficiency of using the potential of renewable

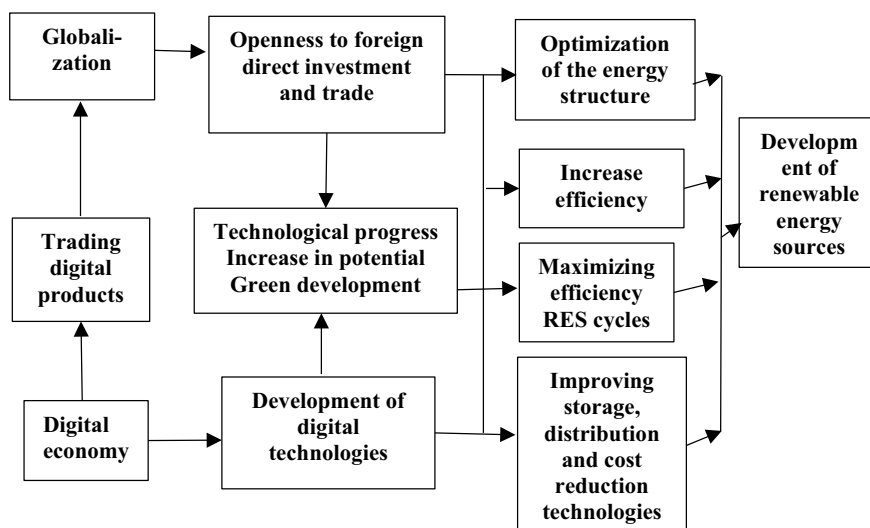


Fig. 6 The mechanism of the impact of globalization and digitization of the economy on the development of renewable energy production. *Source* Developed by the authors

energy sources, which leads to a reduction in energy costs. A situation is already emerging in which the traditional, centralized energy supply system is becoming uncompetitive compared to distributed energy generation systems. The growing share of RES in the structure of energy consumption leads to significant changes in global energy markets. Experts from McKinsey predict that by 2050 the share of renewable sources in global energy consumption will reach between 65% and 80%, presented McKinsey & Company [66].

Figure 6 presents a structural model of the mechanism of the impact of globalization and digitization of the economy on the development of RES. The model shows how openness to foreign direct investment, trade and the development of digital technologies contribute to the optimization of the energy structure, increasing efficiency and maximizing the efficiency of renewable energy sources. In addition, the importance of improving energy storage and distribution technologies to reduce costs was emphasized. Such systematization of data allows to identify key processes affecting the energy transition, supports the planning of RES development strategies and indicates the links between globalization, digitization and sustainable development of the energy sector.

The growing importance of distributed energy production systems creates the possibility of demonopolization of energy markets. First of all, developed countries, located in climatic zones conducive to solar energy generation, are actively striving to develop distributed generation systems. Centralized systems lose their competitive advantage. Already today, some cities in the US meet 80% of their energy needs with RES. Microgrids are gaining widespread popularity as a key component of the energy supply system. The combination of the development of renewable energy technologies and technologies increasing the efficiency of energy resources, based on digitization, creates opportunities to significantly reduce the consumption of fossil fuels. Countries exporting oil and gas face the threat of rapid loss of sources of financing for the development of their own economy, and also lose tools of influence in a globalized world. The prospect of another redistribution of role and place in the global economy is opening up, not only for individual countries, but also for their groups and regions of the world.

The development of RES microgrids is both a product of globalization and a tool for its further development, even at a time when the general trend of intensive development of globalization is changing in the opposite direction. The paradox of this process is underlined by the fact that the idea of creating RES microgrids is based on the desire to solve local problems of a given community without relying on external energy sources. However, even in this case, the result obtained has an impact far beyond this community.

9 Empirical Estimation of the RE Microgrid Growth Model

This section presents the empirical analysis aimed at quantifying the influence of selected macroeconomic, institutional, and digital development factors on the growth of renewable energy microgrids in Ukraine. Drawing on annual data for the period 2015–2024, the analysis integrates key indicators such as digitalization level (ICT Development Index), economic openness and institutional quality (proxied by the Index of Economic Freedom), GDP per capita, and Ukraine’s ranking in global digital competitiveness. The regression model is estimated using Ordinary Least Squares (OLS), based on the assumption of a linear relationship between RE microgrid expansion and the explanatory variables. This approach enables the identification of the most statistically significant factors driving renewable energy transformation in the country. Additionally, this analysis supports the validation of earlier conceptual assumptions regarding the role of digitalization and institutional policy support in accelerating the energy transition aligned with Sustainable Development Goals (SDGs). The dataset for the econometric analysis of renewable energy microgrid development in Ukraine from 2015 to 2024, which was systematised by the authors based on Refs. [67–69], is presented in Table 2.

The strongest positive correlation is observed with GDP per capita ($r = +0.91$), indicating that higher economic capacity substantially enhances the deployment of renewable energy technologies. A moderate positive correlation is found with the ICT Development Index ($r = +0.60$), suggesting that digital infrastructure and technological adoption support the energy transition. In contrast, the Policy Support index demonstrates a moderate negative correlation ($r = -0.57$), potentially reflecting a

Table 2 Dataset for econometric analysis of renewable energy microgrid development in Ukraine 2015–2024

Year	RE microgrid/RES growth (renewable energy share of electricity capacity)	ICT Development Index/World Digital Competitiveness Index		GDP per capita, million UAH	Policy support/ Index of economic freedom
2015	10.8	79	59	46,210.2	46.9
2016	10.7	72	59	55,853.5	47
2017	10.7	79	60	70,224.3	51
2018	12.2	77	58	84,192	51.9
2019	14.2	78	60	94,589.8	52.3
2020	21.7	–	58	100,432.5	54.9
2021	23.8	–	54	131,907.2	56.2
2022	25.2	–	–	126,152.3	54.1
2023	24.8	80	–	–	30
2024	35.4	81	–	–	21.5

Source Systematized by the authors [67–69]

Table 3 Correlation matrix (2015–2024)

	Year	RE microgrid/ RES growth (renewable energy share of electricity capacity)	ICT Development Index/World Digital Competitiveness Index		GDP per capita, million UAH	Policy support/ Index of Economic Freedom
Year	1.0	0.9421	0.5930	−0.6371	0.9819	−0.5024
RE microgrid/RES growth (renewable energy share of electricity capacity)	0.9421	1.0	0.5976	−0.7814	0.9141	−0.5657
ICT Development Index	0.5930	0.5976	1.0	0.3074	0.2084	−0.5176
World Digital Competitiveness Index	−0.6371	−0.7814	0.3074	1.0	−0.7303	−0.6186
GDP per capita, ml. grn	0.9819	0.9141	0.2084	−0.7303	1.0	0.9383
Policy support/ Index of Economic Freedom	−0.5024	−0.5657	−0.5176	−0.6186	0.9383	1.0

Source Developed by the authors

time-lag effect or structural decline in policy incentives in the later years of the period. Similarly, the World Digital Competitiveness Index shows a strong inverse relationship ($r = -0.78$), likely influenced by incomplete data in the later years or divergent measurement frameworks. These findings support the hypothesis that economic and digital development are primary drivers of renewable energy growth, while the institutional environment may present more complex or delayed effects.

The correlation analysis reveals several significant relationships between macro-level variables and the share of renewable energy in electricity capacity (RES Share) is presented in Table 3.

10 Discussion of Hypotheses Based on Correlation Findings

The empirical correlation results offer preliminary insights into the validity of the proposed hypotheses:

H1: Higher digitalization levels (DigitalIndex) positively impact the growth of RE microgrids (linked to SDG 9 and SDG 7).

The moderate positive correlation between the ICT Development Index and RES Share ($r = +0.60$) lends support to this hypothesis. It confirms that digital advancement contributes to the energy transition by enabling smart grid infrastructure, efficient energy management, and prosumer participation. However, data limitations in later years necessitate cautious interpretation.

H2: Strong policy support (PolicySupport) for renewables enhances the deployment of microgrids (linked to SDG 7 and SDG 12).

Interestingly, the negative correlation ($r = -0.57$) contradicts the expected direction of this hypothesis. This may be attributed to recent declines in Ukraine's policy support indicators (e.g., Index of Economic Freedom), despite continued RES expansion. It suggests a decoupling effect where past policies continue to influence growth, or that RES development is increasingly driven by market forces rather than direct policy incentives.

H3: Higher economic capacity (GDP per capita) enables broader participation in decentralized energy systems (linked to SDG 10 and SDG 11).

The very strong positive correlation ($r = +0.91$) robustly supports this hypothesis. It confirms that countries with higher economic capacity are more likely to invest in renewable energy infrastructure, indicating that GDP per capita is a powerful enabler of decentralized energy systems.

Among the five hypotheses, H1 and H3 are strongly supported, H2 is contradicted. These findings emphasize the dominant role of economic and technological factors in accelerating renewable energy deployment, while suggesting the need for further research into the timing and effectiveness of institutional support mechanisms.

11 Conclusions

The current geopolitical landscape, characterized by intensified actions by aggressive countries to threaten Ukraine's energy security, highlights the urgent need for innovative solutions to maintain the stability of the country's energy system. In this context, the development of small-scale renewable energy sources is particularly promising. Ukraine has achieved significant successes in increasing solar and biomass capacity, especially on small farms. Despite these achievements, the full potential of renewable energy generation remains largely untapped. Factors influencing this include the limited financial resources available to firms and households, insufficient state support, and increased investment risks related to the ongoing armed conflict, which exacerbates uncertainty about the cost-effectiveness of green tariff commitments.

A viable strategy to overcome these challenges is to integrate small energy producers—those who use renewable sources such as solar, wind, hydro and biomass—into local microgrids [70]. The creation of these microgrids facilitates the development of digital platforms that provide peer-to-peer interaction in energy markets and the exchange of surplus energy production. These platforms enable the exchange of electricity between participants, increasing the overall efficiency

of energy systems and reducing payback periods for energy generation and storage equipment through optimised use [71]. This integrated model not only supports efficient use of equipment, but also increases the reliability of local energy supply, which in turn reduces electricity costs and allows surpluses that might otherwise go unused to be sold.

Ukraine continues its path of integration with Europe, and at the same time there is significant potential for further cooperation at the cross-border level, in particular through initiatives such as Euroregion projects. However, the transition to digital and platform energy production and distribution comes with its own challenges. These include the need to accurately collect data from platform participants, the growing importance of virtual components in energy production, and the creation of network structures that act as value-added chains in the sector. This integration process fosters the connection of numerous small power producers into a single virtual production network, which fundamentally changes ownership relationships in the industry.

The development of RES microgrids at the local level is both the result of global trends and a factor of global change. The emergence and rapid spread of microgrids in many countries on all continents is the result of the increasing openness of national economies at the turn of the millennium, the facilitation of the exchange of material, financial and intellectual assets between countries, and the increased attention to problems related to ensuring sustainable development. On the other hand, the success of the development of renewable energy sources, largely supported by RES microgrids, is leading to significant global changes that go beyond the economic sphere. The demonopolization of energy markets goes hand in hand with the democratization of relations between their participants. This process is moving from the local to the global level. The increasing share of renewable energy in global energy consumption is leading to changes in fossil fuel markets, changing the role and position of exporting countries in the global economy.

To fully reap the benefits offered by small renewables grids, a comprehensive approach is needed, covering a wide range of state policies, including structural investment, scientific and technological development, regional initiatives, pricing regulation and support for entrepreneurial activities. Specific measures could include grants and preferential loans, targeted financial strategies, tax and customs breaks, as well as the creation of extensive training and counselling schemes for operators of small energy grids. In addition, creating a favourable legal environment and simplifying documentation processes will be key to promoting microgrid initiatives.

The emergence of small renewable energy networks could significantly change the dynamics of competition in energy markets. Increasing competition between many small producers is complemented by their rivalry with centralized energy production systems, potentially prompting them to look for opportunities in international markets. This shift marks a shift away from the classic natural monopoly model toward a more competitive, four-tier market structure that is open to global competition.

In support of these conclusions, an empirical regression model was constructed using national-level data from 2015 to 2024. The results reveal that the strongest positive determinant of renewable energy share in electricity capacity is GDP per

capita, highlighting the critical role of economic capacity in enabling energy transition. The ICT Development Index also demonstrates a moderate positive correlation with renewable energy penetration, confirming that digital transformation supports decentralised energy infrastructure. Conversely, institutional policy support showed a negative correlation over time, suggesting that the effectiveness of policies may be context-dependent or delayed in impact. These findings underscore the importance of combining economic investment with targeted digital and institutional reforms to fully unlock the potential of local renewable microgrids.

In the face of the ongoing threats to Ukraine's energy system, effective coordination and adaptation of all system components is extremely important. The potential for energy generation, distribution and storage needs to be maximised, and small-scale renewables offer significant flexibility and adaptability to changing conditions. The creation of microgrids brings the expected synergistic benefits that increase the efficiency of the use of energy production stations, support the stability of the energy market and, consequently, strengthen national energy security. This study examines the prospects and challenges of the development of small-scale renewables networks in Ukraine, focusing on the role of digital platforms and microgrids in increasing energy efficiency, stability and security. Integrating data from the existing literature, the analysis harmonizes global trends with Ukraine's current energy landscape, highlighting the urgent need for strategic investments in small renewables. This approach is important to ensure Ukraine's energy future in the face of ongoing geopolitical tensions, which strengthens the country's resilience to external threats.

Acknowledgement The article was prepared within the framework of the project implemented by the Centre of Ukrainian Research Studies at the Warsaw School of Economics (SGH).

References

1. Hrytsenko A, Lypov V (2020) Information platforms as a network institutional transformation. *J Inst Stud* 12(2):132–148 (in Russian). <https://doi.org/10.17835/2076-6297.2020.12.2.132-148>
2. Rochet J, Tirole J (2003) Platform competition in two-sided markets. *J Eur Econ Assoc* 1(4):990–1029
3. Parker G, Van Alstyne M, Choudary S (2016) Platform revolution: how networked markets are transforming the economy and how to make them work for you. W W Norton & Company, 352 pp
4. Moret F, Pinson P (2018) Energy collectives: a community and fairness based approach to future electricity markets. *IEEE Trans Power Syst* 34:3994–4004. <https://doi.org/10.1109/TPWRS.2018.2808961>
5. Fehrenbacher K (2017) Five trends for the smart energy home of the future. Greentech Media, EIA. <https://www.greentechmedia.com/articles/read/5-trends-for-the-smart-energy-home-of-the-future>. Accessed 2 Nov 2024
6. Hossain MS, Madloul NA, Rahim NA, Selvaraj J, Pandey AK, Khan AF (2016) Role of smart grid in renewable energy: an overview. *Renew Sustain Energy Rev* 60:1168–1184
7. Jenkins N, Long C, Wu J (2015) An overview of the smart grid in Great Britain. *Engineering* 1(4):413–421

8. Olivella-Rosell P, Bullich-Massagué E, Aragüés-Peñalba M, Sumper A, Ottesen SØ, Vidal-Clos J-A, Villafafila-Robles R (2018) Optimization problem for meeting distribution system operator requests in local flexibility markets with distributed energy resources. *Appl Energy* 210(C):881–895
9. Kiesling L (2016) Implications of smart grid innovation for organizational models in electricity distribution. *Smart Grid Handbook*. <https://ssrn.com/abstract=2571251>
10. Tabors R, Caramanis M, Ntakou E, Parker G, Van Alstyne M, Centolella P, Hornby R (2017) Distributed energy resources: new markets and new products. In: *Proceedings of the 50th Hawaii international conference on system sciences*, pp 38–45
11. Breuer H (2017) A microgrid grows in Brooklyn. <https://www.siemens.com/global/en/company/stories/research-technologies/energytransition/a-microgrid-grows-in-brooklyn.html>
12. Srnicek N (2017) *Platform capitalism*. Polity Press, Cambridge, p 92
13. Timilsina G, Kurdgelashvili L, Narbel P (2012) Solar energy: markets, economics and policies. *Renew Sustain Energy Rev* 16(1):449–465
14. Parag Y, Sovacool B (2016) Electricity market design for the prosumer era. *Nat Energy* 1(4):16–32
15. Hurochkina V, Nahaieva O, Rymar O, Chornyi O, Makhanets L, Shyshpanova N (2021) Foreign state economic policy in the context of economic security. *J Manage Inform Decis Sci* 24(3):1–14. <https://www.abacademies.org/journals/month-march-year-2021-vol-24-issue-3-journal-jmids-past-issue.html>
16. Hurochkina V, Kohut S (2024) Formation of the energy balance of Ukraine using renewable energy sources. *Econ Finances Manage Topical Issues Sci Pract Activity* 4(66):109–133. <https://doi.org/10.37128/2411-4413-2023-4-8>
17. Morstyn T, Farrell N, Darby Sarah J, McCulloch M (2018) Using peer-to-peer energy-trading platforms to incentivize prosumers to form federated power plants. *Nat Energy* 3(2):94–101
18. Matvienko H, Kucherkova S, Yanovska V, Ternovsky V, Hurochkina V, Kęsy M (2023) Governmental management and regulatory measures for advancing AI in the Ukrainian energy sector as a basis for rapid and sustainable development of the Ukrainian Economy. In: *Proceedings of the 13th international conference on advanced computer information technologies (ACIT)*, pp 303–307, 21–23.09.2023. http://acit.wunu.edu.ua/Download/ACIT%2023_Conference%20Program.pdf
19. Lang T, Gloerfeld E, Girattemptedmpt B (2015) Don't just follow the sun—a global assessment of economic performance for residential building photovoltaics. *Renew Sustain Energy Rev* 42(Feb):932–951
20. Fu R, Feldman D, Margolis R, Woodhouse M, Ardani K (2017) US solar photovoltaic system cost benchmark: Q1 2017. Technical report National Renewable Energy Lab. (NREL), Golden, CO. <https://www.nrel.gov/docs/fy17osti/68580.pdf>
21. Kirchhoff H, Kebir N, Neumann K et al (2016) Developing mutual success factors and their application to swarm electrification: microgrids with 100% renewable energies in the Global South and Germany. *J Clean Prod* 128:190–200
22. Schill W-P, Zerrahn A, Kunz F (2017) Prosumage of solar electricity: Pros, cons, and the system perspective. *Econ Energy Environ* 6(1):7–32
23. Kwan C (2012) Influence of local environmental, social, economic and political variables on the spatial distribution of residential solar PV arrays across the United States. *Energy Policy* 47(Aug):332–344
24. Comello S, Reichelstein S (2017) Cost competitiveness of residential solar PV: the impact of net metering restrictions. *Renew Sustain Energy Rev* 75(Aug):46–57
25. Korohodova OO, Chernenko NO, Moiseienko TY, Hlushchenko YI (2022) Analysis of the attractiveness of renewable energy in individual countries. *J Econ Reforms* 3(47):74–82
26. Abdmouleh Z, Alammari R, Gastli A (2015) Review of policies encouraging renewable energy integration and best practices. *Renew Sustain Energy Rev* 45:249–262
27. Borenstein S (2017) Private net benefits of residential solar PV: the role of electricity tariffs, tax incentives, and rebates. *J Assoc Environ Resour Econ* 4(S1):85–122

28. Eid C, Reneses G, Javier F, Marín P, Hakvoort R (2014) The economic effect of electricity net-metering with solar PV: consequences for network cost recovery, cross subsidies and policy objectives. *Energy Policy* 75(Dec):244–254
29. Hagerman S, Jaramillo P, Morgan M (2016) Is rooftop solar PV at socket parity without subsidies? *Energy Policy* 89(Feb):84–94
30. Sevostyanov NA, Shaforost AO (2024) Core directions of renewable energy sources development. In: 2024 6th international youth conference on radio electronics, electrical and power engineering (REEPE), pp 1–5
31. Shaulska L, Kovalenko S, Allayarov S, Sydorenko O, Sukhanova A (2021) Strategic enterprise competitiveness management under global challenges. *Acad Strat Manage J* 20(4):1–7
32. Trunina IM, Pryakhina KA, Yakymets S (2022) Research on the development of renewable energy sources in the world due to the war in Ukraine. In: 2022 IEEE 4th international conference on modern electrical and energy system (MEES), pp 1–5
33. Karabegović I, Dolecek V (2015) Development and implementation of renewable energy sources in the world and European Union. *Contemp Mater VI-2*:130–148
34. Karabegović I (2021) Tendency of global capacity development of renewable energy sources in the world in the last ten years. *Contemp Mater XII-2*:166–183. <https://doi.org/10.7251/COMEN2102166K>
35. Karbovska L, Yakushik I, Feshchenko EY, Kalina I, Kozlova A (2021) Sustainable development of the economy and increasing energy security based on the use of res: problems and prospects. In: Financial and credit activity: problems of theory and practice, vol 2, pp 438–446
36. Klepacka A (2019) Significance of renewable energy sources in sustainable development. *Annals PAAAE XXI(1)*:55–64. <https://doi.org/10.5604/01.3001.0013.0852>
37. Voronin A, Panasiuk P (2022) State of renewable energy of Ukraine and prospective directions of development. *Sci View Econ Manage* 1(77):70–75
38. Pecheniuk A, Garasymchuk I, Potapyskyi P, Vusatyi M, Dubik V, Pukas V (2022) Renewable energy of Ukraine in global energy transformations. *Grassroots J Nat Resour* 5(4):19–33. <https://doi.org/10.33002/nr2581.6853.050402>
39. Yakubiv VM, Maksymiv Y, Hryhoruk II, Popadynets NN, Piatnychuk I (2019) Development of renewable energy sources in the context of energy management. *J Vasyl Stefanyk Precarpathian Natl Univ* 6(3–4):77–87
40. Kharlamova G, Nate S, Chernyak O (2016) Renewable energy and security for Ukraine: challenge or smart way? *J Int Stud* 9:88–115
41. Kuzior A, Lobanova A, Kalashnikova L (2021) Green energy in Ukraine: state, public demands, and trends. *Energies* 14:7745. <https://www.mdpi.com/1996-1073/14/22/7745>
42. Klopov I (2015) World trends in the development of renewable energy sources. Collection of scientific papers of the Cherkasy State University of Technology. *Ser Econ Sci* 40(1):59–64. http://nbuv.gov.ua/UJRN/Znpchdtu_2015_40%281%29__11 (in Ukrainian)
43. Denysiuk S, Bielokha H, Cherneshchuk I, Lysyi V (2023) Global trends in implementation of renewable energy sources and features of their implementation during the recovery of Ukraine's economy. *Power Eng Econ Tech Ecol* 2022(4):7–23
44. A unique object. How the completion of the Dniester hydro-accumulating power plant will affect the reduction of the deficit of maneuvering capacities (2021). https://uhe.gov.ua/media_tsentr/novyny/unikalniy-obekt-yak-dobudova-dnistrovskoi-gaes-poznachitsya-na-sko-rochenni (in Ukrainian)
45. Energy Efficiency (2022) State agency on energy efficiency and energy saving of Ukraine. <https://sae.gov.ua/uk/content/energy-efficiency>. Accessed 26 Nov 2022
46. Trotsenko L (2022) DTEK initiates an increase in the capacity of green energy in Ukraine. <https://ua.korrespondent.net/business/companies/4480792-dtek-initsiuie-zbilshennia-potuzhnosti-zelenoi-enerhetyky-v-ukraini>. Accessed 24 May 2022 (in Ukrainian)
47. Zaharii VK, Kovalchuk T (2021) Renewable energy: world development trends and Ukraine. In: Herald UNU. International Economic Relations and World Economy, pp 70–76. <http://www.visnyk-econom.uzhnu.uz.ua/>

48. Transforming our world: the 2030 Agenda for Sustainable Development. Resolution of the UN General Assembly dated September 25, 2015 (2015). <https://sdgs.un.org/2030agenda>. Accessed 03 Jan 2023
49. Przekształcamy nasz świat: Agenda na rzecz zrównoważonego rozwoju 2030. Rezolucja przyjęta przez Zgromadzenie Ogólne w dniu 25 września 2015 r. https://www.unic.un.org.pl/files/164/Agenda%202030_pl_2016_ostateczna.pdf
50. Šajin N (2016) Electricity 'Prosumers'. [https://www.europarl.europa.eu/RegData/etudes/BRIE/2016/593518/EPRS_BRI\(2016\)593518_EN.pdf](https://www.europarl.europa.eu/RegData/etudes/BRIE/2016/593518/EPRS_BRI(2016)593518_EN.pdf). Accessed 26 Nov 2022
51. Lypov VV (2023) "Mosquito flotilla" on the energy front: small energy networks based on digital platforms. *Econ Theory* 2:53–70. <https://doi.org/10.15407/et2023.02.053> (in Ukrainian)
52. Lypov V, Ushenko N (2023) The impact of platformization on the development of the renewable energy market in Ukraine: risks and prospects for development. In: *Modeling the development of the economic systems*, vol 4, pp 244–251. <https://doi.org/10.31891/mdes/2023-10-33> (in Ukrainian)
53. Elblox Platform (2021) The regional P2P energy market. <https://www.fingrid.fi/globalassets/dokumentit/fi/palvelut/alkuperatakuun-sertifikaatti/tapahtumamatskut/case-elblox.pdf>
54. Kryvinska N, Auer L, Strauss C (2008) Managing an increased service heterogeneity in a converged enterprise infrastructure with SOA. *Int J Web Grid Serv* 4(4):440. <https://doi.org/10.1504/ijwgs.2008.022546>
55. U.S. Department of Energy (DOE) (2022) 2020 Smart grid system report. https://www.energy.gov/sites/default/files/2022-05/2020%20Smart%20Grid%20System%20Report_0.pdf
56. European Commission: Directorate-General for Research and Innovation, Renda A, Schwaag Serger S, Tataj D, Morlet A, Isaksson D, Martins F, Mir Roca M, Hidalgo C, Huang A, Dixon-Declève S, Balland P, Bria F, Charveriat C, Dunlop K, Giovannini E (2021) *Industry 5.0, a transformative vision for Europe: governing systemic transformations towards a sustainable industry*. Publications Office of the European Union. <https://doi.org/10.2777/17322>
57. Andriushchenko K, Tepluk M, Boniar S, Ushenko N, Liezina A (2019) Influence of cost drivers on value-oriented management of investment activity of companies. *Invest Manage Financ Innov* 16(3):353–364
58. Kwasinski A, Weaver W, Balog RS (2016) *Microgrids and other local area power and energy systems*. Cambridge University Press, Cambridge. <https://doi.org/10.1017/CBO9781139002998>. Accessed 26 Nov 2022
59. Kwasinski Alexis (2016) Quantitative model and metrics of electrical grids' resilience evaluated at a power distribution level. *Energies* 9(2):93
60. Bauwens T, Eyre N (2017) Exploring the links between community-based governance and sustainable energy use: quantitative evidence from Flanders. *Ecol Econ* 137:163–172. <https://doi.org/10.1016/j.ecolecon.2017.03.006>
61. UNCTAD (2019) *The role of science, technology, and innovation in promoting renewable energy by 2030*. UNCTAD, Geneva, 51 p
62. Zeren F, Akkus HT (2020) The relationship between renewable energy consumption and trade openness: new evidence from emerging economies. *Renew Energy* 2020(147):322–329. <https://doi.org/10.1016/j.renene.2019.09.006>
63. Doytch N, Narayan S (2016) Does FDI influence renewable energy consumption? An analysis of sectoral FDI impact on renewable and non-renewable industrial energy consumption. *Energy Econ* 54:291–301. <https://doi.org/10.1016/j.eneco.2015.12.010>
64. Anton SG, Afloarei Nucu AE (2020) The effect of financial development on renewable energy consumption. A panel data approach. *Renew Energy* 147:330–338. <https://doi.org/10.1016/j.renene.2019.09.005>
65. Zhang Z, Su L, Jin W, Yang Y (2022) The impact of globalization on renewable energy development in the countries along the belt and road based on the moderating effect of the digital economy. *Sustainability* 14(10):6031. <https://doi.org/10.3390/su14106031>
66. McKinsey&Company (2024) *Global energy perspective 2024*. <https://www.mckinsey.com/industries/energy-and-materials/our-insights/global-energy-perspective>

67. International Renewable Energy Agency (IRENA). <https://www.irena.org/>
68. Change in electricity demand by region, 2021–2027. <https://www.iea.org/data-and-statistics/charts/change-in-electricity-demand-by-region-2021-2027>
69. REN21 reports. <https://www.ren21.net/reports/ren21-reports/>
70. Yepifanova I, Dzhezdhula V (2020) Methodology of evaluation of innovative potential of enterprises. *Agric Resour Econ Int Sci E-J* 171–190. <https://doi.org/10.51599/are.2020.06.03.10>
71. Yepifanova I, Dzhezdhula V (2021) Modelling of potential level of industrial enterprises. *WSEAS Trans Environ Dev* 17:556–565. <https://doi.org/10.37394/232015.2021.17.53>

Competency-Based Approach as a Tool for Forming Sustainable Business Ecosystems: Case Studies on the Quality of Education Towards Sustainable Development of Territories



Oksana Polinkevych , Olena Kuzmak , Vira Shpylova ,
and Ryszard Kamiński

Abstract The article considers the competency-based approach as a tool for forming sustainable business ecosystems. It is noted that in the digital economy, the importance of distance learning for students is increasing, where quality is the main indicator of the sustainable development of territories. A sustainable business ecosystem combines participants, a platform, and an enterprise. Effective interaction can be built based on the main indicator of development—the quality of distance education throughout life. The ecosystem as an enterprise is self-developing, using innovative approaches to management. To achieve success, the ecosystem must be closely connected to the main business, education, and territorial community. Its participants must adhere to dynamic experimentation and a constant search for innovative solutions. The study was conducted on the example of the quality of education at Lutsk National Technical University in the context of sustainable development. It was determined that the main skill of the twenty-first century is digital literacy, particularly its role in forming sustainable business ecosystems. The thesis that students in wartime Ukraine require significant coordination of employer requirements with the offers of educational institutions is confirmed. Business and administration education is identified as a cornerstone for building sustainable business ecosystems in Ukraine given the ongoing military conflict. The study describes the theoretical, methodological,

O. Polinkevych (✉) · O. Kuzmak
Lutsk National Technical University, Lutsk, Ukraine
e-mail: kravomp@gmail.com

O. Kuzmak
e-mail: o.kuzmak@lutsk-ntu.com.ua

V. Shpylova
Bohdan Khmelnytskyi Cherkasy National University, Cherkasy, Ukraine
e-mail: vera_bsh.21@vu.cdu.edu.ua

R. Kamiński
Adam Mickiewicz University Poznań, Poznań, Polska
e-mail: ryszard.kaminski@amu.edu.pl

and analytical aspects of lifelong distance education in the context of sustainable development of territories. The main approaches to the competency-based approach in distance education in Ukraine are summarized and the features of obtaining education in business and administration from the point of view of quality and sustainability are highlighted. The key competencies necessary for distance education in the field of business and administration from the point of view of students and employers are identified. The article recommends that educational institutions that train specialists in business and administration pay attention to the balance of hard and soft competencies.

Keywords Competencies · Sustainable development · Lifelong learning · Employers' expectations · Quality of education · Ecological business system · Sustainable Development of Territories · Social Welfare

1 The Role of Distance Education and Lifelong Learning in Shaping Sustainable Business Ecosystems

1.1 Trends in the Development of Distance Education in Ukraine

The characteristics of the development of distance education are marked by its relevance during the global pandemic, hence its popularity is increasing, especially in the context of ensuring lifelong education. The main advantages of distance learning are the opportunities it provides, namely, the ability to receive education without leaving work, saving resources and time, learning in a convenient format, and improving professional skills or acquiring new knowledge and skills [1].

Distance learning is a flexible combination of independent cognitive activities of students with various sources of information, educational materials specifically developed for the given course; prompt and systematic interaction with the leading course instructor; group work similar to collaborative learning with the participants of the course; exchange of opinions and information with course participants as well as any other partners, including international ones, via the Internet. The goal of distance learning through the competency-based approach is not only to acquire knowledge, skills, and abilities but also to obtain and deepen professional competence, which is a more complex socio-personal structure. It is based on values, orientation, knowledge, and experience acquired by an individual both in the process of learning and outside of it.

The war in Ukraine and the COVID-19 pandemic have forced us to change approaches to learning and advanced training through the introduction of distance forms of education. Global trends in the development of distance education are implemented in such areas as: adaptive or personalized learning; augmented, virtual,

and mixed reality AR/VR/MR; smart assistants/chatbots; gamification and game-based learning; microlearning; selection of content materials for learning; interactive video-based learning; social learning, artificial intelligence and helping students learn. The development of the global distance education market has received a huge boost with the development of information technologies, and, first of all, computer networks, which have made it possible to reach a huge audience of students almost anywhere in the world. The main goal of the development of distance technologies is to attract foreign students. The leader in distance education is currently the USA, which together with Canada occupies more than 50% of the entire e-education market [2]. According to scientists' forecasts, by 2025 the number of those studying remotely could reach 650 million people [2]. Supporters of distance learning highlight the following features and advantages: continuity, accessibility, cost-effectiveness, individualization, differentiation of learning, innovation, and mobility. Critics of distance education claim that its quality cannot be compared even with the correspondence form, and such educational technologies only allow you to get a diploma with minimal investment of effort and resources without acquiring actual knowledge. The low quality of distance education, according to several experts, can be explained by the lack of high-quality distance teaching methods, the average quality of electronic courses, as well as the electronic educational environments themselves. In addition to the low quality of the knowledge received, opponents of distance education express doubts that students who study in distance programs receive the same qualifications and have the same knowledge and skills as those who study in full-time education. The lack of direct communication between the teacher and the student, in their opinion, reduces the emotional coloring of knowledge transmitted remotely and indicates that such education is not suitable for everyone, linking this to the need for constant self-control and self-discipline on the part of the higher education seeker himself.

The dynamics of distance education in educational institutions of Ukraine are shown in Fig. 1.

From Fig. 1 it can be seen that distance education in 2023–2024 was about 27%, and its highest level was 82% in March 2022. However, in Ukraine, there is still a low level of full-time education, which is 23.8%. Educational services in Ukraine are provided mainly in a blended format and its level is 48.9%.

The purpose of this study is to summarize the approaches to the competency-based approach in distance learning for students and to identify the factors that affect the quality of education through adherence to the competency-based approach in the context of sustainable development. Based on the aim of the work, the following main tasks will be performed:

To summarize the existing theoretical approaches to understanding the competency-based approach in distance education in the context of sustainable development of society;

Develop a factor model of key competencies for students of business and administration Education using SPSS Statistics, taking into account the integration of opinions from students and employers.

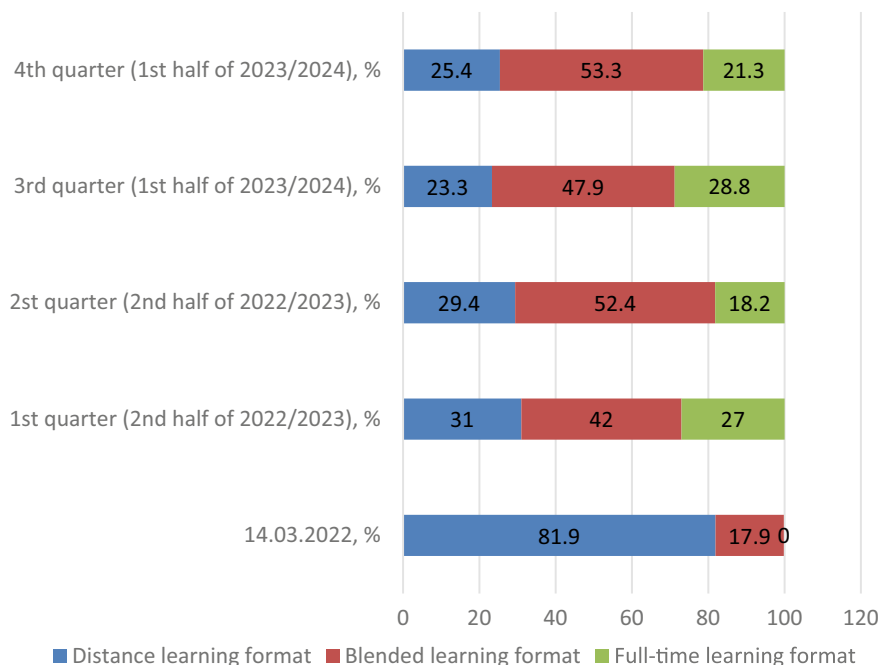


Fig. 1 Format of education in Ukraine, % [3]

Research Methods: (1) Content analysis of the competence approach in distance education; (2) online survey questionnaire (Google Forms) for experts in the field of business and administration (sample size: 55, response rate: 84.0%, 46 responses received; survey date: June 2024); (3) online survey questionnaire (Google Forms) for business and administration Education students of LNTU 1–4 undergraduate courses, 1 master’s course (sample size: 210, response rate: 83.0%, 174 responses received; survey date: May 2024).

1.2 Bridging Skills Gaps Through Lifelong Learning to Build Sustainable Business Ecosystems

Ukraine’s integration into the EU member states should involve the development of an effective lifelong learning strategy based on the principles and procedures for the recognition of prior learning. Modern education is gradually becoming one of the most important components of both the social and economic development of humanity and also acts as the main means of solving a set of global problems. The idea of lifelong learning is one of the key competencies defined in the Law “On Education”. Lifelong education can be considered as learning, the duration of which

is equal to the duration of a person's life. It provides that people need to continue learning and renew their learning throughout their entire life path. Lifelong education is both a need for every person and a general requirement of time for the whole society. Such education contributes to the adaptation of each person to social and economic conditions, which in the modern world are changing quite quickly. This is a kind of way to protect the individual from unforeseen events in the world and from general uncertainty. That is why lifelong education should be a need for every person. In parallel, the implementation of such activities requires the presence of all necessary conditions, updating the methodological principles of formal education, and changing the attitude towards non-formal education. Lifelong education is considered the most important prerequisite for achieving economic, social, environmental, and other goals. Therefore, every person should learn throughout his life, and the education system, in turn, should help him and provide all opportunities. In Ukraine, due to war, migration processes and rapid technological development, personnel imbalances arise. Even experienced specialists may find that their skills are outdated and no longer meet market requirements. At the same time, mastering new, relevant competencies may be difficult or undesirable for them. Therefore, it is important to develop fast, effective and inclusive educational opportunities for employees. In 2024, employers most value such employee qualities as sociability, responsibility, friendliness, knowledge of foreign languages, stress resistance, adaptability and flexibility, willingness to learn, ability to work with new technologies, critical thinking and change management skills [4]. There is a new trend in the labor market: employers are increasingly moving away from searching for ready-made specialists and prefer candidates who are motivated for professional development, seek to learn, improve their skills, and adapt to modern requirements. The European Business Association, based on research, has identified a list of skills that will be relevant in 2024–2025: “stress management, multitasking, adaptability and flexibility, emotional intelligence, willingness to learn, knowledge of foreign languages, working with new technologies, critical thinking, work-life balance and understanding the first signals of burnout, effective communication, change management” [5].

One of the key ways to overcome the skills gap is through continuous learning and retraining, in line with the concept of lifelong learning. This is necessitated by rapid technological change, the evolution of market conditions, and the transition to the sixth technological order. The future workforce is expected to have the following competencies:

1. Information literacy—the ability to effectively process large amounts of data, work synchronously with parallel information flows, think critically and isolate key information.
2. Technical literacy—the ability to work with various technical devices, manage automated systems, and master information and communication technologies (ICT).
3. Inter-sectoral communication—understanding of technological processes and their dynamics in different areas, even those that are not directly related.

4. Project and process management—the ability to organize and control work processes using digital technologies.
5. Multiculturalism and language skills—fluency in English and other foreign languages, understanding of the cultural and national context of partner countries, familiarity with the specifics of working in an international environment.
6. Communication and leadership skills—the ability to work effectively in a team, communicate with people, organize the work process, and demonstrate leadership qualities.
7. Flexibility and stress resistance—the ability to work in conditions of instability, assess risks, make decisions quickly, adapt to changes, effectively allocate resources, and manage one's own time.
8. Analytical and logical thinking—the ability to work with complex systems, and understand their structure and functionality.
9. Creativity and initiative—the ability to generate new ideas, show a creative approach, and develop aesthetic thinking.
10. Environmental awareness and social responsibility—taking into account the principles of rational use of nature, energy conservation, and general trends in one's activities.

Human capital in the national wealth of world co-authorship is 64%, natural capital is 20%, and physical capital is only 16%. The share of human capital in developed countries (Finland, Switzerland, Germany, Japan, and the USA) reaches 80% of their national wealth. The main increase in national wealth is determined today primarily by the conditions created for the disclosure of human potential. Currently, about 5% of theoretical and 20% of professional knowledge are updated annually. The unit of measurement of the aging of a specialist's knowledge, adopted in the USA, is the "half-life" period of competence, i.e. its decrease by 50% due to the emergence of new information, shows that for many professions this period occurs in less than five years, i.e. about the domestic higher education system, often earlier than the completion of training [6]. According to official data, the share of adults studying in the formal and non-formal education system at the beginning of the twenty-first century in the EU countries exceeded 37%, and in Sweden, 73% of the population was involved in the adult education system at that time, in Norway and Finland—55% each, in Ukraine—less than 15% [7].

In Ukraine in 2022, after a full-scale invasion, the number of students aged 25 to 60 who are pursuing second and third higher education has tripled [8]. In Ukraine, almost 80% of school graduates enter higher education institutions, and the share of people with higher education among young people aged 25–34 exceeds 50% [9]. Currently, the participation rate of the population under 70 in formal and non-formal education and training is about 9%. The participation rate in non-formal education of citizens of OECD member countries aged 25–64 ranges from 17% in Turkey to 64% in New Zealand. The Global Competitiveness Index in 2014–2015, 2016–2017, and 2018 includes indicators such as the level of coverage of higher education, the quality of the education system, the quality of management schools, the availability of

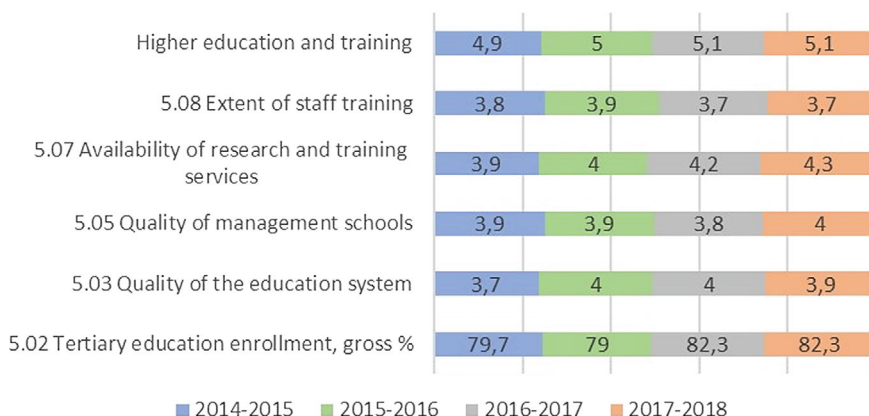


Fig. 2 Score of the Higher Education and Vocational Training indicator according to the Global Competitiveness Index in Ukraine in 2014–2018. *Source* Built by The Global Competitiveness Report [10]

specialized research and training services at the local level, and the level of personnel training (Fig. 2).

From the data in Fig. 2, we can conclude that the indicator of higher education and vocational training is estimated in 2014–2018 at 4.9–5.1 points (rating scale from 0 to 7 points). The higher education coverage rate in Ukraine is quite high. The quality of the education system has decreased and averaged 3.9 points. The indicator of availability of specialized educational services at the local level increased in 2018 compared to 2014, the average value is 4.1 points. Thus, for all indicators, higher education and vocational training have an average score of 4.1. Let us consider the place of Ukraine in the global competitiveness index for these indicators in 2014–2018 (Fig. 3).

From Fig. 3, we can conclude that according to the indicator of higher education and vocational training, Ukraine ranks 40–35 out of 144–137 countries in 2014–2018. The higher education coverage ratio in Ukraine ranks quite low—13–16 in 2014–2018. The quality of the education system decreased from 72nd place in 2014–2015 to 56th place in 2017–2018. The indicator of availability of specialized educational services at the local level also tended to decrease from 84 to 68th place among countries in 2014–2018. Thus, according to all indicators, higher education and vocational training lost their place in the ranking, except for the quality of management.

In 2019, the indicators of the Global Competitiveness Index changed. They include the indicator of the level of personnel training, the quality of professional training, the skills of graduates, and the skills of the future workforce. These indicators did not differ significantly in terms of points from the similar period of the study in 2014–2018 and amounted to 4–4.3 points (rating scale from 0 to 7 points). According to the indicator of the level of personnel training and the quality of professional training, Ukraine in 2019 ranks 65th, and according to the skills of graduates in 2019, Ukraine ranks 54th, and according to the skills of the future workforce—26th. Thus, if we

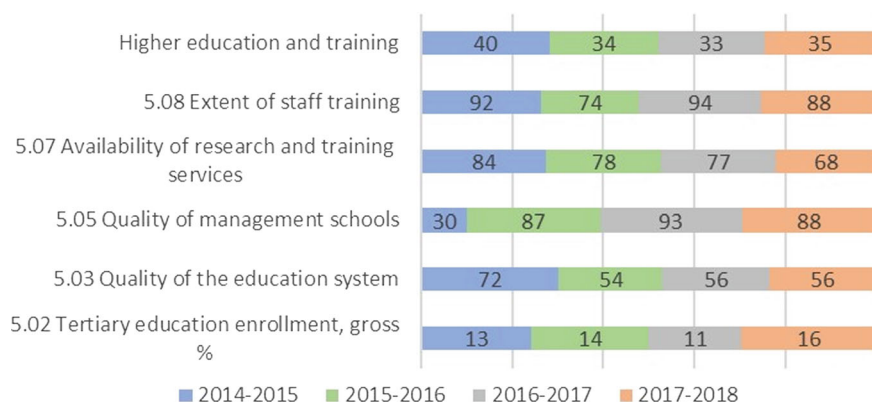


Fig. 3 Place of indicators of Higher Education and Vocational Training according to the Global Competitiveness Index in Ukraine in 2014–2018. *Source* Built by The Global Competitiveness Report [10]

compare them with the indicators of 2014–2018, the skills indicators in the Global Competitiveness Index of Ukraine in 2019 rank 44th. This is a significantly lower position than in 2014–2018.

Since 2020, the indicators of the Global Competitiveness Index of Ukraine have not been calculated [10, 11] (Fig. 4).

Thus, in terms of the quality of higher education and vocational training, Ukraine, according to the Global Competitiveness Index, is placed at the 30th percentile, i.e. among all the countries studied. Such a result requires the development of a strategy for improving the quality of education in Ukraine, which can be represented at three

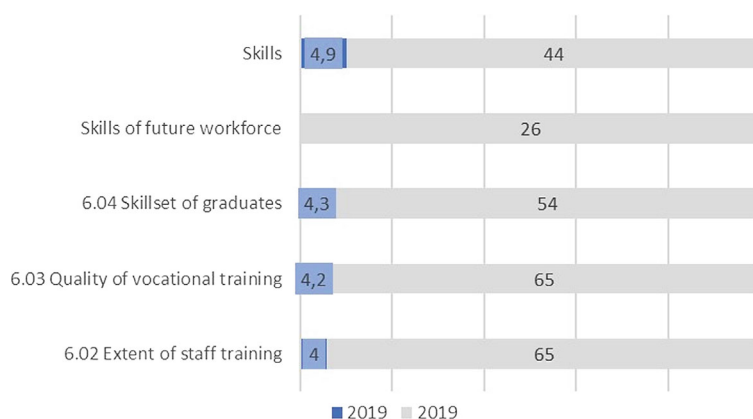


Fig. 4 Ranking of Higher Education and Vocational Training indicators according to the Global Competitiveness Index in Ukraine in 2019. *Source* Built by The Global Competitiveness Report [10]

levels: individual, enterprise or organization level, and society level. The combination of actions at all these levels forms a sustainable business ecosystem that can ensure the improvement of the quality of education, especially in the digital economy.

1.3 Building Sustainability Business Ecosystems in Ukraine

An ecosystem can be viewed as a combination of participants, a platform, and an enterprise. The ecosystem as a collection of participants includes all entities interacting with the business and contributing, either directly or indirectly, to the “value chain” (educational institutions, product and service distributors, communities), as well as customers. The ecosystem as a platform for goods and services (marketplace) represents a space where diverse, integrated products and services are offered to meet the broadest range of client needs within a specific segment. The ecosystem as a self-developing enterprise involves an organization utilizing innovative management approaches and perceiving itself as a “living organism”.

The shift toward an ecosystem model is a response to evolving market demands and technological advancements. Businesses must undergo more frequent and fundamental transformations. Companies face intense competition, rapid change, and increasing uncertainty. However, large-scale and rapid transformations come with significant risks. To succeed, an ecosystem must be closely aligned with the core business, and its participants need to move away from rigid strategies and directive models toward dynamic experimentation and continuous exploration of innovative solutions. For instance, connecting major companies and users through networks enables the direct exchange of diverse values among them.

A business ecosystem can be defined as a network of market participants, including educational institutions, businesses, and government entities, collectively engaged in achieving sustainable development goals and fostering societal well-being.

There is no universal approach to understanding ecosystems. They can be viewed from the perspectives of structure and affiliation. The first approach sees ecosystems as intentionally organized entities designed to achieve a shared goal among participants, driven by a deliberate business strategy. The second approach views ecosystems as emerging naturally within the economy due to the blurring of traditional sector boundaries, increased interdependence, and symbiosis among participants in the process of value creation.

The emergence of ecosystems is driven by several key processes:

Firstly, the boundaries between goods and services are becoming increasingly blurred due to regulatory changes and digital transformation.

Secondly, legal protections granting exclusive privileges to certain firms for meeting specific customer needs have been eliminated.

Thirdly, modern technologies have revolutionized the ways firms can serve their customers. The growing reliance on mobile devices and the influence of the Internet on consumer behavior have significantly enhanced the ability to connect

previously unrelated goods and services, amplifying the effects of the first two factors.

Given this, businesses alone cannot create a competitive and secure ecosystem; they must collaborate with educational institutions and authorities. Accordingly, three components of a business ecosystem can be identified:

1. Developer community (a space for value creation),
2. Digital platform (a shared resource),
3. Business ecosystem (a space for commercializing innovative proposals).

The developer community focuses on designing and promoting ideas for creating business ecosystems. Digital platforms serve as multi-sided marketplaces that connect various participant groups through digital technologies. These platforms facilitate interaction among all participants, identify challenges, and work collaboratively on solutions. The business ecosystem involves seeking resources (human, informational, and financial) to implement the activities and initiatives of the business ecosystem.

The key element in the formation and functioning of a business ecosystem is business and administration education, given the ongoing military actions and the pressing issues within this field. Neither businesses nor government bodies alone can resolve these challenges. A symbiosis of education, business, and authority is required to address these problems. Business and administration education serves as a vital resource through which sustainable development goals can be achieved and societal well-being enhanced.

2 Competency-Based Approach in Distance Education as the Basis for the Formation of Competitive Business Ecosystems

2.1 Theoretical Basis for Understanding the Competence Approach in Distance Education

To prevent the occurrence of discussions, we propose to interpret the concept of “competence” as awareness, mastery of the appropriate set of competencies capable of being immediately implemented in practical activities, and “competence” as each element of a certain range of subjects and processes realized at different levels, i.e., including various mental operations, practical skills, common sense, and endowed with their hierarchy and classification. Accordingly, we understand the competence-based approach as the orientation of the educational process towards the formation and development of key and subject-specific competencies of an individual, which they will use during professional activities [12]. Experts of the Council of Europe hold the opinion that competencies imply the ability of an individual to meet individual

and social needs; a set of attitudes, values, knowledge, skills, and abilities [13]. Competence should be formed through the mobility of knowledge, flexibility, and critical thinking. Distance education provides all these characteristics.

Many countries were forced to replace traditional in-person education with distance learning in response to the COVID-19 crisis, often without being fully prepared or having sufficient experience. Research into the Arab education system has shown that the quality of the learning process can be affected by low preparedness and incompetence [14]. Many countries were forced to replace traditional in-person education with distance learning in response to the COVID-19 crisis, often without being fully prepared or having sufficient experience. Research into the Arab education system has shown that the quality of the learning process can be affected by low preparedness and incompetence [15]. Lytras et al. investigated the effectiveness of distance learning methods in Mexico using a regression model. The model for instructor satisfaction with e-learning included factors such as educational quality, human resources, job satisfaction, lack of financial resources, and organizational policies. The model for student satisfaction with e-learning included human resources, job satisfaction, low sense of community, student satisfaction, fairness, lack of financial resources, technology access, and organizational policies [16].

Jaekel et al., in their research, found that opportunities for social connection and interaction are fundamental to students' perceptions of teaching quality and their learning outcomes. The quality of teaching was assessed from the perspectives of students and parents, using six parameters: monitoring, structure, support in learning, feedback, challenging tasks, and practicality. The study established that teachers' communication with their colleagues was most strongly associated with students' perceptions of teaching quality. Regarding learning objectives, the goal of maintaining good relations with students determines the quality of teaching as perceived by students and parents. Teachers spent a significant amount of time on preparation and actual teaching. The interconnection between all elements depends on students' social engagement [17]. The authors' research was based on data obtained from "Teaching and Learning in Distance Education" (CUNITAS) in June/July 2020 in the German federal state of Baden-Württemberg [18].

According to an OECD report, "the readiness of countries to take advantage of the benefits of digital transformation largely depends on the skills of their population..." [19]. Today, such skills are even more critical for students in the context of war.

Workers need to be actively engaged in lifelong learning to acquire new competencies necessary for adapting well to the rapidly changing work environment [19–21].

The quality of distance education must become a priority in the preparation of specialists who meet the labor market's demands. This thesis is supported by the works of Polinkevych et al. [22, 23]. The quality of education is a priority for sustainable development, which is important for the formation of urban lifelong learning centers.

Holubnychy et al. explored the competency-based approach to teaching and learning within the framework of the modern educational paradigm. In the contemporary education system, the competency-based approach is one of the central elements,

as the existence of a competency framework facilitates the description of academic and professional profiles, levels, and outcomes of higher education at the international level [24].

Pontrandolfo and Scozzi note that quality lifelong education should be the foundation of forming smart cities [25].

Arnold et al. note that over 60% of surveyed employees mentioned that they constantly need to upgrade their skills at work due to the rapid emergence of technological innovations [26].

Today, the primary goal of master's programs is to prepare students for a globalized world [27–29]. Consequently, to remain competitive in the modern world, higher education must support students in successfully acquiring many relevant competencies [30].

Currently, higher education faces a particular challenge in enhancing students' competencies, including creativity, collaboration, communication, critical thinking, innovation, problem-solving, and self-management [31], which are recognized as “21st-century skills” or 21st-century competencies. The competency-based approach, adopted worldwide a few years ago in teaching/learning activities, stems from social changes that have occurred over the past decades. However, this approach concludes with the rapid emergence and expansion of globalization. Some examples of this digital revolution include the continuous renewal of ICT, the exponentially growing volume of data, the implementation of various data resources and digital media, and their extensive use [32], the call for learning stages that can provide remote teaching/learning, and the initiation of emergency remote teaching (ERT) [33], accelerated by the health emergency caused by COVID-19.

Mei et al. concluded that today, companies and governments increasingly understand the need to bridge the gap between existing and required labor resources. They note a significant gap between academia and practice. The study used a hybrid method with Pythagorean fuzzy perspective and the PF-entropy-RS-MARCOS method. The most important future competencies include data and information processing, the ability to interact with modern interfaces, social intelligence, conflict self-resolution, time management, and a value-oriented management style. These are the most important competencies for the future workforce to implement digital technologies in higher education [34].

Stepura and Kuzmak research the features of digital competencies using factor analysis by the principal component method. They identified the main factors influencing the digital-inclusive components of human potential, which include the profitability of virtual activities, the ability to conduct such activities, and the readiness of the population to engage in the digital space due to the absence of language barriers [35].

Guzina analyzed perspectives on the essence of the competency-based approach and noted the insufficient number of studies regarding its significance in ensuring effective strategic management at various levels. The competency-based approach is considered not only from the standpoint of setting strategic goals while accounting for the dynamics of changes in the external environment but also emphasizes the necessity of involving personnel in the strategic management process. These personnel

should possess the so-called required model of professional behavior, which not only facilitates the development and implementation of the enterprise's business strategy but also encompasses independent knowledge acquisition, skill development, and enhancement of individual abilities [36].

The novelty of the study lies in ranking the competencies that are important for employers in the training of students and for students during their education in the field of business and administration Education, by the recommendations of the European Parliament and the European Council. Additionally, the importance of competencies for women and men has been identified.

2.2 Research Methodology of the Competence Approach in Distance Learning

Let's explore the key skills that are important for employers when preparing students in business and administration Education using a competency-based approach. According to the recommendations of the European Parliament and the European Council dated December 18, 2006, and the ANNEX to the Proposal for a Council Recommendation on Key Competencies for Lifelong Learning, eight key competencies for lifelong learning were identified in 2006 and 2008 [37, 38]. These were proposed to employers in business and administration for assessment regarding their importance on a Likert scale from 1 (strongly disagree that the competence is important for the company) to 5 (strongly agree that the competence is important for the company). These competencies include Literacy competency; Language competency; Mathematical competence and competence in science, technology, and engineering; Digital competency; Personal, social, and learning competency; Civic competency; Entrepreneurship competency; and Cultural awareness and expression competency. The sample of employers consisted of 46 individuals. It was randomly selected from a list of email addresses of enterprises requiring business and administration specialists. The selection was done using Excel. The sample included 35% women and 65% men aged 35–55 years. A similar study was conducted for business and administration students at LNTU from the 1st to the 4th year of bachelor's studies and the 1st year of master's studies. The student sample consisted of 174 individuals. The sample included 55% women and 45% men aged 17–35 years.

3 Case Study of the Competence-Based Approach in Shaping the Quality of Distance Education in the Business Ecosystem Toward Sustainable Development

3.1 *Research of Competencies by Employers and Acquirers Throughout Life*

We conducted a study of the importance of the competencies of business and administration students on a sample of 174 people. The respondents were students of LNTU in the following specialties: D3 Management; D5 Marketing; D2 Finance, banking, insurance and stock market.

As a result of testing students of LNTU using a questionnaire, the results presented in Table 1 were obtained.

Both male and female students among those studying business and administration consider key competencies that students should acquire to be digital, personal, learning, and social skills. Ranked second in importance are entrepreneurship, civic competence, language and mathematical competence, and competence in sciences, technology, and engineering. The least important is literacy and competence in cultural awareness and expression. It is worth noting that male students consider mathematical competence and competence in sciences, technology, and engineering

Table 1 Answers of respondents of business and administration students to questionnaire questions regarding key competencies

Key competence	Generalized assessment average score	Generalized assessment men	Generalized assessment women
Literacy competence	2	1	3
Languages competence	3	3	3
Mathematical competence and competence in science, technology, and engineering	3	4	2
Digital competence	5	5	4
Personal, social, and learning competence	4	3	5
Civic competence	3	3	2
Entrepreneurship competence	3	4	2
Cultural awareness and Expression competence	2	2	2

Source Own research

to be more important compared to personal, social, and learning competence. However, female students rank it as one of the most important. Thus, soft skills are more important for female students than for male students. Overall, soft skills are preferred by business and administration students.

As a result of testing employers who require specialists in business and administration using a questionnaire, the results are presented in Table 2.

Both male and female employers seeking specialists in business and administration consider key competencies that their employees should possess to be digital and mathematical competencies, competencies in science, technology, and engineering, cultural awareness and expression competency, and language competency. The second most important competencies are entrepreneurship, personal, learning, social, and civic competencies. The least important is literacy. It is worth noting that male and female employers almost agree in their assessments. The exception is civic competence, which men prefer, while women rank it third. This also applies to entrepreneurship, which men rate highest, while women rate it only 3 out of 5. However, female employers consider personal, learning, and social competence to be important, while men give it an average rating. Thus, both soft and hard competencies are equally important for employers seeking specialists in business and administration.

The averaged overall scores by the degree of importance of competencies for students and employers who need specialists in business and administration are presented in Fig. 5.

Table 2 Answers of employer respondents to questions of the questionnaire regarding key competencies

Key competence	Generalized assessment average score	Generalized assessment men	Generalized assessment women
Literacy competence	2	2	2
Languages competence	5	4	5
Mathematical competence and competence in science, technology, and engineering	5	5	4
Digital competence	5	5	5
Personal, social, and learning competence	4	3	5
Civic competence	3	4	2
Entrepreneurship competence	4	5	3
Cultural awareness and Expression competence	5	4	5

Source Own research

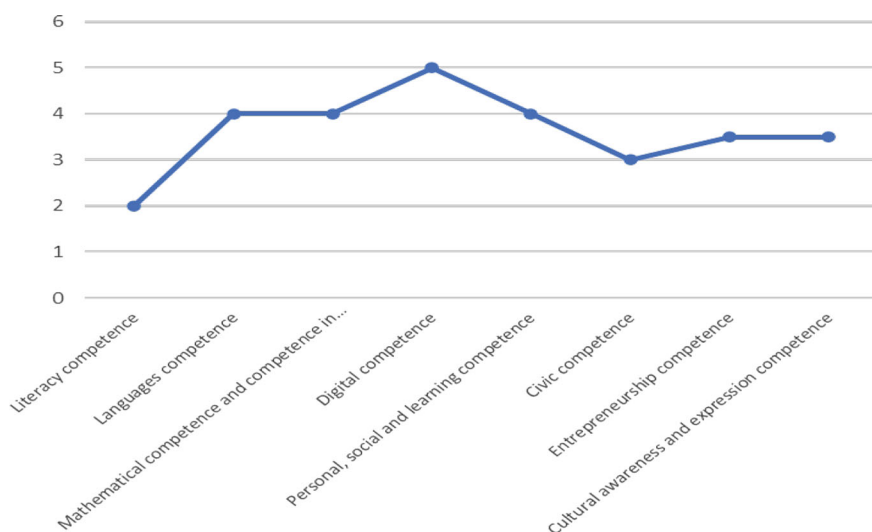


Fig. 5 Graphical representation of generalized assessments of key competencies of business and administration students and employers. *Source* Own research

Based on the summarization of opinions from employers and business and administration students, it has been established that the main factors determining the presence of competence in candidates and important to employers are: digital competence (F1)—continuous learning in the field of digitalization, proficiency in specialized software for performing production tasks; language competence (F2)—possession of a diploma of fluent foreign language proficiency or an international certificate; mathematical competence and competence in sciences, technologies, and engineering (F3)—thorough knowledge of the subject area, productivity, quality, and accuracy in performing assigned tasks; personal, social, and learning competence (F4)—possession of certificates in soft skills.

3.2 Factorial Model of Distance Learning Competencies of Business and Administration Students

Using the method of factor analysis using the method of main components, we will distinguish the key competencies of employers and students. For this, we will use the SPSS Statistics program. According to the “stone scree” criterion, 5 factors are determined that are balanced for employers and students of business and administration (Fig. 6). The research was based on the methodology of the researchers [35, 36].

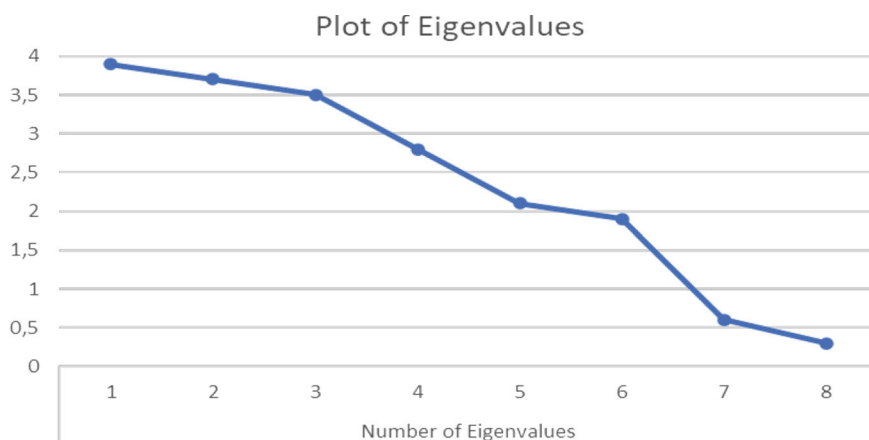


Fig. 6 The “stone scree” graph as a criterion for deciding on the number of factors in the factor model of competencies. *Source* Calculated by the author in the SPSS Statistica environment

For the study, four factors were chosen: digital competence; language competence; mathematical competence and competence in science, technology, and engineering; and personal, social, and academic competence.

In Table 3 we present the main factor loadings on four factors.

Thus, the factor model explains 79.4% of the variance of the indicators included in the model (1st factor—15.5%, 2nd—15.6%, 3rd—32.1%, 4th—12.2%). Based on the content of the indicators, the 1st factor describes digital and information literacy, communication and collaboration, digital content creation (including programming), cyber security, and problem-solving. The second factor reflects the ability to appreciate cultural diversity, interest in different languages, and intercultural communication, and implies respect for the individual language characteristics of each person. The third factor—involves understanding the changes caused by human activity and his responsibility for the consequences of such changes, expertise, and professionalism. The fourth factor includes resilience, the ability to cope with stress, cooperate in a team, and be able to negotiate, knowledge of perception styles and learning strategies, one’s abilities, and development needs (Table 4).

Based on the data in Table 4, it can be concluded that the key competency in the distance learning of business and administration students is professional literacy. This competency is fundamental and significant from the perspective of all participants in the educational process. However, digital literacy is equally important, considering the current level of economic development. It should manifest in computer and information literacy, which includes the ability to interact with gadgets, work with large volumes of data, navigate search engines, and identify misinformation in the information space. Additionally, students should be able to create digital content, communicate in the digital world, ensure a safe digital environment, and engage in lifelong learning.

Table 3 System-creating factors of competencies for business and administration education

Indicator	Factor loadings F1	Factor loadings F2	Factor loadings F3	Factor loadings F4
Possession of a foreign language fluency diploma or international certificate		0.99		
Continuous education in the field of digitalization	0.79			
Basic knowledge of the subject area of knowledge			0.75	
Possession of specialized programs for the performance of production tasks	0.93			
Labor productivity			0.89	
Quality and accuracy of tasks			0.94	
Availability of soft skills certificates				0.99
Contribution of the main components to the variance (%)	19.5	15.6	32.1	12.2

Source Own research

Table 4 Interpretation of results of factor analysis regarding key competencies in distance education of business and administration students

Order of factors	Factorial loading	Content of the factor
F1	1.72	Digital skills
F2	0.99	Language skills
F3	2.58	Professional skills
F4	0.99	Soft skills for learning and team building

Source Summarized by the author based on own research

Linguistic literacy and soft skills related to learning and team-building are considered less important by participants in the educational process [39]. However, these skills play a crucial role in enhancing professional and digital literacy [40, 41]. Accordingly, a model for measuring instructors' satisfaction with e-learning should include components such as education quality, human resources, professional and

digital literacy, lack of financial resources, and organizational policies. The model for assessing students' satisfaction with e-learning should encompass elements such as human resources, professional and digital literacy, personal satisfaction with work and the educational environment, fairness, access to informational and educational technologies, and organizational policies [39].

4 Conclusions

So, summarizing the statistical data, it can be determined that at the current stage of development, a decisive role in the formation of a quality approach to distance education throughout life is played by a competence approach and a view of this process both from the point of view of students of business and administration education and from the point of view of employers who need these specialists.

It has been established that the competency-based approach has gained popularity in teaching and learning. Its emergence is driven by the social changes of recent decades. These changes include a decreasing lag between financial and economic crises, pandemics, military conflicts, the openness and nonlinearity of systems, and the prolonged state of systems in pre-bifurcation (high-entropy) conditions. However, the competency-based approach is also evolving under the influence of globalization. This evolution is facilitated by the digital revolution, which drives the constant development of ICT, the exponential growth of data volume, and the implementation and active use of various information sources and digital media. Additionally, distance learning and the implementation of emergency remote learning have become widespread due to the public health crisis caused by COVID-19 and the rise of infectious diseases driven by FLiRT strains.

Under these conditions, it is essential to maintain the quality of teaching at an appropriate level. The quality of teaching can be evaluated from the perspective of active service consumers using parameters such as monitoring of classes and course content, mentoring support in learning, feedback and consultation, and the practicality and target orientation of the proposed competencies. Active service consumers include students and academic staff, while passive service consumers are parents and employers. Passive consumers assess the quality of educational services through parameters such as the practicality of acquired knowledge, the presence of soft skills (foreign language proficiency, stress resistance, diligence, readiness for self-learning, time management, conflict management, and team-building skills), and the level of software proficiency. It should be noted that communication among different stakeholder groups can lead to a bifurcation point, enabling a high-level assessment of the quality of distance learning. Regarding the objectives of distance learning, communication emerges as the primary competency that should be emphasized by all participants in the educational process. However, a challenging aspect is the irrationality of behavior, which makes it difficult to predict equilibrium between all elements in the student education system.

To establish the integration of competencies that modern students of business and administration education need, a questionnaire survey was used, which made it possible to identify the following key competencies: digital, language, mathematical competencies, competencies in science, technology, and engineering, and personal, social and educational competencies.

The results of scientific research indicate that distance education is becoming more and more common in the modern space. It is recognized both among employers and students. At the same time, there is a need to provide quality education that will meet the goals of sustainable development, guarantee the acquisition of sufficient competences that meet the demands of the labor market.

Recommendations include increasing the focus of educational institutions on providing opportunities for business and administration education students to acquire soft competencies on par with hard ones. Increasing attention to the digital and language literacy of business and administration education students, developing educational distance programs that integrate the requirements of all stakeholders. Accordingly, distance education can be of high quality and ensure the acquisition of exactly those competencies defined by the labor market and the EU as key. Thus, high-quality business education will contribute to the improvement of social welfare in society and the sustainable development of local communities.

The prospects for further research include identifying mechanisms for conveying information to learners about the importance of competencies defined by employers and approaches to their popularization.

References

1. Miastkovska M, Kobylanska I (2019) Prospects for the development of distance education in the context of the competence approach. *Secur Pedagogy* 1:72–77
2. Distance education in Ukraine: the realities of today. *Educ Discourse* 24(6):56–67 (2020). [https://doi.org/10.33930/ed.2019.5007.24\(6\)-4](https://doi.org/10.33930/ed.2019.5007.24(6)-4)
3. Information and analytical note on the organisation of the educational process in professional pre-higher and higher education institutions of Ukraine under martial law (based on the results of a quarterly online survey, IV quarter of 2023), Main—State Service of Education Quality of Ukraine. https://sqe.gov.ua/wp-content/uploads/2024/08/IAD_IV_kvartal_2023.pdf
4. Kucherenko IA, Yudenkova OP (2024) Qualification gap in the ecosystem of education and labour market of Ukraine. *Bull Postgrad Educ Ser Pedagog Sci* 28:103–122
5. Melnyk SV (2023) Research of the labour market in Ukraine (2023). <https://eba.com.ua/doslidzhennya-rynku-pratsi-ukrayiny-2/>
6. Approaches and directions of formation of the adult education system in Ukraine, adapted to the needs of the economy in wartime and post-war (2024). https://iea.gov.ua/wp-content/uploads/2024/05/az_osvita-doroslyh_2024.pdf
7. Maslyanikova IV (2020) Modern approaches to adult learning: theoretical and empirical aspects. *Scientific notes of Vernadsky TNU. Psychology* 31(70):1, 28–36. https://psych.vernadskyjournals.in.ua/journal/1_2020/6.pdf
8. The number of students receiving higher education from 25 to 60 years old has tripled—Venislavsky. *Ukrinform* 04.09 (2023). <https://www.ukrinform.ua/rubric-society/3756769-kil-kist-studentiv-aki-zdobuvaut-visu-osvitu-vid-25-do-60-rokiv-zrosla-vtrici-venislavskij.html>

9. Akimova L, Semygina T, Fedyuk V, Markevych N, Kolodko I (2022) 41% of young people have higher education in the EU. <https://nqa.gov.ua/news/visu-osvitu-v-es-maut-41-molodih-ludej/>
10. Global Competitiveness Report. <https://www.weforum.org/search/?query=Global+Competitiveness+Report>
11. Indicator 16.7.1. <https://sdg.ukrstat.gov.ua/uk/16-7-1/>
12. Miloradova NE, Shevchenko VV (2020) Competency approach as a methodological basis for the study of the professional competence of an individual. *Habitus* 16:233–237. <https://doi.org/10.32843/2663-5208.2020.16.38>
13. Hutmacher W (1997) Key competencies for Europe. In: Report of the Symposium Berne, Switzerland 27–30 March, 1996. Council for Cultural Co-operation (CDCC). Secondary Education for Europe, Strasbourg
14. Lily AE, Al Ismail AF, Abunasser FM, Alhajhoj Alqahtani RH (2020) Distance education as a response to pandemics: coronavirus and Arab culture. *Technol Soc*. <https://doi.org/10.1016/j.techsoc.2020.101317>
15. Roman M, Ploeanu A-P (2021) The effectiveness of the emergency e-learning during COVID-19 pandemic. The case of higher education in economics in Romania. *Int Rev Econ Educ*. <https://doi.org/10.1016/j.iree.2021.100218>
16. Lytras MD, Serban AC, Ruiz MJT, Ntanos S, Sarirete A (2022) Translating knowledge into innovation capability: an exploratory study investigating the perceptions on distance learning in higher education during the COVID-19 pandemic—the case of Mexico. *J Innov Knowl* 7(4):100258. <https://doi.org/10.1016/j.jik.2022.100258>
17. Jaekel A-K, Fütterer T, Göllner R (2023) Teaching characteristics in distance education—associations with teaching quality and students' learning experiences. *Teach Teach Educ* 132:104174. <https://doi.org/10.1016/j.tate.2023.104174>
18. OECD Homepage. <https://www.oecd-ilibrary.org/sites/df80bc12-en/index.html?itemId=/content/publication/df80bc12-en>
19. Carnevale A, Smith N (2013) Workplace basics: the skills employees need and employers want. *Hum Resour Dev Int* 16:491–501. <https://doi.org/10.1080/13678868.2013.821267>
20. Cascio WF, Montealegre R (2016) How technology is changing work and organizations. *Annu Rev Organ Psych Organ Behav* 3:349–375. <https://doi.org/10.1146/annurev-orgpsych-041015-062352>
21. Beinicke A, Bipp T (2020) Digital competencies: a review of the literature and applications in the workplace. *Comput Educ* 146:103752. <https://doi.org/10.1016/j.compedu.2019.103752>
22. Polinkevych O, Khovrak I, Trynchuk V (2023) Transformation of lifelong digital education in the context of sustainable development of society. In: 2023 IEEE 5th international conference on modern electrical and energy system, MEES, pp 1–6. Kremenchuk, Ukraine. <https://doi.org/10.1109/MEES61502.2023.10402428>
23. Polinkevych O, Kuzmak O, Kuzmak O (2024) The strategy of sustainable development of digital business in the conditions of the variability of the business environment and European integration. In: Semenov A, Yepifanova I, Kajanová J (eds) *Data-centric business and applications*. Lecture notes on data engineering and communications technologies, vol 194. Springer, Cham. https://doi.org/10.1007/978-3-031-53984-8_2
24. Holubnychy L, Shchokina T, Soroka N, Besarab T (2022) Development of Competency-Based Approach to Education. *Educational Challenges* 27(2):54–65. <https://doi.org/10.34142/2709-7986.2022.27.2.04>
25. Pontrandolfo P, Polinkevych O, Scozzi B, Kuzmak O (2023) Smart cities for the sustainable development of local communities on the example of Volyn and the city of Lublin, Central and Eastern Europe. *Comp Econ Res* 26(3):53–85. <https://doi.org/10.18778/1508-2008.26.21>
26. Arnold D, Butschek S, Steffes S, Müller D (2016) Digitalisierung am Arbeitsplatz: Bericht (Forschungsbericht/Bundesministerium für Arbeit und Soziales, FB468). In: Nürnberg: Bundesministerium für Arbeit und Soziales. GmbH; Universität Köln. <https://nbn-resolving.org/urn:nbn:de:0168-ssoar-47712-7>

27. Di Pietro, L, Mugion RG, Musella F, Renzi MF, Vicard P (2015) Reconciling internal and external performance in a holistic approach: a Bayesian network model in higher education. *Expert Syst Appl* 42:2691–2702. <https://doi.org/10.1016/j.eswa.2014.11.019>
28. Islam MdSh, Stamp K (2020) A reflection on future directions: global international and intercultural competencies in higher education. *Res Comp Int Educ* 15:69–75. <https://doi.org/10.1177/1745499920901951>
29. Zevallos Z (2012) Context and outcomes of intercultural education amongst international students in Australia. *Intercult Educ* 23:41–49. <https://doi.org/10.1080/14675986.2012.664753>
30. Martha ASD, Junus K, Santoso HB, Suhartanto H (2021) Assessing undergraduate students' e-learning competencies: a case study of higher education context in Indonesia. *Educ Sci* 11:189. <https://doi.org/10.3390/educsci11040189>
31. Ferreras-Garcia R, Sales-Zaguirre J, Serradell-López E (2021) Developing entrepreneurial competencies in higher education: a structural model approach. *Educ Train* 63:720–743. <https://doi.org/10.1108/ET-09-2020-0257>
32. Basilotta-Gómez-Pablos V, Matarranz M, Casado-Aranda L-A, Otto A (2022) Teachers' digital competencies in higher education: a systematic literature review. *Int J Educ Technol High Educ* 19:8. <https://doi.org/10.1186/s41239-021-00312-8>
33. Trust T, Whalen J (2024) Should teachers be trained in emergency remote teaching? Lessons learned from the COVID-19 pandemic. *J Technol Teacher Educ* 28(2):189–199. Society for Information Technology & Teacher Education, Waynesville, NC. <https://www.learntechlib.org/primary/p/215995/>. Retrieved 27 June 2024
34. Mei L, Feng X, Cavallaro F (2023) Evaluate and identify the competencies of the future workforce for digital technologies implementation in higher education. *J Innov Knowl*. <https://doi.org/10.1016/j.jik.2023.100445>
35. Stepura TM (2024) Analytical methods of researching the quality of the labor market of Ukraine. *Bus Inform* 1:208–217. <https://doi.org/10.32983/2222-4459-2024-1-208-217>
36. Guzina I (2020) The essence and features of the competency-based approach to strategic enterprise management. *Dev Manage* 18(4):30–40. [https://doi.org/10.21511/dm.18\(4\).2020.04](https://doi.org/10.21511/dm.18(4).2020.04)
37. Recommendation 2006/962/EU of the European Parliament and of the Council (EU) “On core competences for lifelong learning” of December 18 (2006). https://zakon.rada.gov.ua/laws/show/994_975#Text
38. Council recommendation of 22 May 2018 on key competences for lifelong learning (Text with EEA relevance) (2018/C 189/01). Official Journal of the European Union. [https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32018H0604\(01\)](https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32018H0604(01))
39. Rauer JN, Kroiss M, Kryvinska N, Engelhardt-Nowitzki C, Aburaia M (2021) Cross-university virtual teamwork as a means of internationalization at home. *Int J Manage Educ* 19(3):100512. <https://doi.org/10.1016/j.ijme.2021.100512>
40. Kirzhetska M, Novakivskiy I, Musiiiovskia O, Kirzhetskyy Y, Yepifanova I (2024) Transformation of a regional IT cluster into a cross-border IT cluster as a direction of it business development under the conditions of negative influence of external factors. In: *Lecture notes on data engineering and communications technologies*. Springer Nature Switzerland, pp 233–249. https://doi.org/10.1007/978-3-031-54012-7_10
41. Dzhezdzhula V, Hurochkina V, Yepifanova I, Telnov A (2022) Fuzzy technologies for modeling social capital in the emergent economy. *WSEAS Trans Bus Econ* 19:915–923. <https://doi.org/10.37394/23207.2022.19.80>

Financial Health Assessment of Ukrainian Banks: A Logistic Regression Model for Bankruptcy Prediction



Larysa Zomchak and Andriana Seniv

Abstract This research explores the prediction of bank bankruptcy in Ukraine using logistic regression, a statistical method specifically designed for binary outcome. A comprehensive set of 22 financial indicators, encompassing liquidity, solvency, profitability, and business activity ratios, were extracted from the National Bank of Ukraine to construct the predictive model. Rigorous analysis revealed that among the multitude of indicators, only two emerged as significant predictors of bank bankruptcy: administrative and other operating expenses relative to assets, and interest income relative to assets. These findings underscore the importance of operational efficiency and interest income generation in maintaining bank stability. The logistic regression model demonstrated robust predictive capabilities, accurately classifying 88.9% of banks as either bankrupt or stable. The results of this study highlight the critical need for early warning systems to identify banks at risk of failure. By employing advanced statistical modeling techniques, financial regulators can proactively implement measures to mitigate systemic risks and safeguard the stability of the banking sector, particularly in economies like Ukraine that are susceptible to economic shocks.

Keywords Bank · Bankruptcy · Financial health · Financial distress · Prediction · Logistic regression · Financial risk · Risk management · Bank stability · Finance · Predictive modelling · Machine learning · Model · Bank system · Crisis

L. Zomchak (✉) · A. Seniv
Ivan Franko National University of Lviv, Lviv, Ukraine
e-mail: larysa.zomchak@lnu.edu.ua

A. Seniv
e-mail: andriana.seniv@lnu.edu.ua

L. Zomchak
Slovak University of Technology in Bratislava, Bratislava 1, Slovak Republic

1 Introduction

1.1 Bank System of Ukraine

Analyzing the modern banking system in Ukraine, it is evident that political and military instability, significant inflation, and general economic destabilization have negatively impacted the performance indicators of banks.

Currently, the banking sector in Ukraine is in a challenging position. The reform program for the banking system, proposed by the National Bank of Ukraine, has led to a significant reduction in the number of banks and radical changes in the operations of major banks. Ukrainian banks have transitioned to international accounting and reporting standards, though full implementation is ongoing. Adopting modern accounting norms and regularly publishing balance sheets and profit and loss reports will enhance transparency and gradually increase customer trust in banks. The reform of the banking system has yielded tangible results for the national economy and societal development.

The low stability of the Ukrainian banking system and the lack of effective regulation mechanisms led to numerous bankruptcies during the recent economic crisis. In recent years, the National Bank of Ukraine (NBU) [1] has been reducing the number of banks and increasing capital requirements. Many restrictive measures are now in place to stabilize the banking system and the economy as a whole. Due to the instability and stringent monetary control in Ukraine, public trust in banks remains low. When choosing a bank, it is important to verify its financial condition, consider the branch network, internet banking options, fees, and the level of bureaucracy. A wise decision would be to choose a bank that is part of an authoritative European banking group. Among European banks, leaders in Ukraine include Raiffeisen Bank Aval, Ukrsibbank, Kredite Agricole Bank, OTP Bank, Citibank, Kredobank, ING Bank, and others.

Between 2022 and 2024, the Ukrainian banking system faced significant challenges due to Russia's war in Ukraine and ongoing combat. Overall, the banking system remains stable, with banks generating profits from their core activities, allowing them to operate without major disruptions. However, in the first half of 2022, the banking system experienced a general loss of 4.6 billion UAH, marking its first unprofitable period in the last five years. Despite this, the loss of 4.6 billion UAH is relatively small compared to the reserves. The substantial difference between reserves and losses indicates that banks were still profitable: in six months, they earned about 53 billion UAH. In the second half of 2022, out of 68 solvent banks, 47 were profitable with a net profit of 10.0 billion UAH, while 21 banks were unprofitable with losses totaling 14.6 billion UAH [1]. More information about number of commercial banks in Ukraine and key profitability indicators in Table 1.

The provided data in Table 1 elucidates key trends in the structural composition and profitability of the Ukrainian banking system over the period spanning 2017 to 2023. A consistent consolidation is evident in the number of operating commercial banks, declining from 96 institutions in 2017 to 67 by the end of 2023, indicating a market

Table 1 Number of commercial banks in Ukraine and key profitability indicators

	2017	2018	2019	2020	2021	2022	2023
Number of operating banks (units)	96	82	77	75	74	71	67
With foreign capital including	38	38	37	35	33	33	30
With 100% foreign capital (units)	17	18	23	23	23	23	22
Return on assets (%)	– 12.6	– 1.94	1.65	4.35	2.54	4.09	1.04
Return on capital (%)	– 116.74	– 15.9	10.73	34.18	19.97	35.15	9.68

Source Created by authors

restructuring potentially driven by regulatory pressures, mergers, acquisitions, or market exits. The subset of banks with foreign capital also experienced a gradual decrease, from 38 in 2017 to 30 in 2023, while those with 100% foreign capital exhibited relative stability, fluctuating within a narrow range and ending at 22 in 2023. In terms of profitability, the banking sector demonstrated a recovery from significant negative returns on assets (– 12.6%) and capital (– 116.74%) in 2017. Positive profitability, as indicated by both return on assets and return on capital, was achieved from 2019 onwards, with notable peaks in 2020 and 2022.

The problems that arose in the banking sector during the war led to significant changes in its operating conditions. However, the fundamental principles of the banking system must remain resilient, even in times of crisis. In order to maintain customer trust, the National Bank of Ukraine (NBU) and all structural units of the banking system had to quickly develop and implement a series of measures to stabilize the banking market. Despite the state of war, the banking system managed to avoid any collapses, primarily due to the fact that commercial banks were well-capitalized.

In February 2022, operations of Russian banks PJSC “Prominvestbank” and JSC “International Reserve Bank” were suspended in Ukraine. At the beginning of the year, these banks held 2% of net assets in the banking sector. As a result of their liquidation, the number of operational banks in Ukraine decreased to 69 by the end of March. The sector’s assets shrank due to a reduction in the activities of private and foreign financial institutions. The stability of state-owned banks led to an increase in their share of net assets and individuals’ deposits by 1.1 percentage points and 0.4 percentage points, respectively, reaching 47.8% and 55.9%. At the same time, sector concentration increased.

Overall, the assets of Ukrainian banks increased from 2019 to 2023, with the most significant growth seen in state-owned banks and PrivatBank. In contrast, foreign and private banks experienced slower asset growth during this period.

Several factors influenced the asset dynamics of Ukrainian banks during these years, including the growth of the Ukrainian economy, changes in monetary policy, the COVID-19 pandemic, and the ongoing war in Ukraine, among others.

Following Russia’s full-scale war in Ukraine on February 24, 2022, the assets of the banking sector began to decline sharply. This decline was largely due to an increase in defaults, deposit outflows, reduced liquidity, and fluctuations in exchange rates. However, by 2023, a certain stabilization in the dynamics of bank assets was

observed. Despite this stabilization, bank assets remain lower than they were before the war.

The most significant losses were experienced by banks with foreign capital. Their share of net assets dropped from 31.1% in 2021 to 27% in 2023. This decline was largely due to many foreign banks either exiting the Ukrainian market or reducing their presence because of the war.

Meanwhile, the share of net assets held by state-owned banks increased slightly from 26.9% in 2021 to 27.7% in 2023. This increase can be attributed to substantial financial support from the government aimed at maintaining the stability of the financial system. Notably, PrivatBank also saw a rise in its net assets, growing by UAH 12.2 billion during 2022–2023.

Private banks saw the largest increase in the share of individual deposits. As the population became more cautious with their assets, many moved their deposits from state-owned banks to private banks, especially those with foreign capital. The share of private banks' individual deposits grew from 17.5% in 2022 to 18.6% in 2023, while the share of deposits in state-owned banks slightly decreased from 24.8% in 2022 to 24.5% in 2023.

Thanks to external support and balanced economic policies, the government and the NBU managed to maintain economic control under extremely difficult conditions. Critical budget expenditures (defense, social protection, wages, etc.) were fully financed; inflation remained controlled with signs of stabilization; the state budget's financing shortfall was covered through a mix of sources, not solely through emission; and the banking system, including payment systems, continued to function uninterrupted even in the early days of the full-scale war.

The growth prospects for 2024 are uncertain due to the ongoing full-scale war. Persistent Russian attacks on energy infrastructure have led to a gradual deterioration of growth forecasts for 2022 and the following years. This highlights the need for successful completion of the Monitoring Programs by the Ukrainian government, which were finally approved by the IMF on December 19. Successful completion of these programs will provide a reliable macroeconomic framework for securing necessary financing from other donors and facilitate the transition to post-war recovery.

One year after the beginning of the full-scale war, the Ukrainian banking system demonstrated significant increases in reserves and profitability. In 2022, the banking sector accumulated 119 billion UAH in reserves, a phenomenal increase of 35 times compared to 3.4 billion UAH in 2021. Only 10% of banks reported losses in January 2023, while the remaining financial institutions earned a total of 14.7 billion UAH over the months of the war. January 2023 was very profitable for the Ukrainian banking system, with banks earning 14.7 billion UAH, more than double the profit of 7.1 billion UAH in January 2022 [1]. Despite the challenges, the Ukrainian banking system has managed to endure, with the official exchange rate of the hryvnia fixed and interest rates increased to curb inflation.

In conclusion, the Ukrainian banking system has navigated a period of profound turbulence, marked by economic instability, regulatory reforms, and the unprecedented challenges posed by the ongoing war. Despite a significant contraction in

the number of operating institutions and initial financial losses following the full-scale war in 2022, the sector has demonstrated remarkable resilience. Supported by robust capitalization, timely interventions from the National Bank of Ukraine, and international financial assistance, the banking system has largely maintained operational stability and returned to profitability. Shifts in asset distribution, favoring state-owned banks, and changes in deposit patterns reflect the public's adaptive behavior in response to the prevailing uncertainties. While the long-term growth trajectory remains contingent on the resolution of the war and successful implementation of economic recovery programs, the banking sector's ability to withstand such severe shocks underscores its fundamental importance to the Ukrainian economy and the effectiveness of the measures implemented to safeguard its stability.

1.2 Literature Review

The Ukrainian banking system has experienced profound changes driven by economic volatility, geopolitical tensions, and regulatory reforms. This literature review synthesizes key research on the structure, performance, and evolution of the Ukrainian banking sector.

Frolov and Shukairi [2] underscore the bank-centric nature of Ukraine's financial system, which signifies that banks play a predominant role in financial intermediation. This central role can both fortify and expose the system to various vulnerabilities. Khalatur et al. [3] address the diversification of bank lending, an essential factor for maintaining banking stability. Diversification in lending portfolios can help mitigate risks associated with economic downturns and sector-specific shocks. Chaikovskiy et al. [4] examine the efficiency of Ukrainian banking institutions, providing insights into their operational performance, profitability, and asset quality. Complementarily, Bukhtiarova et al. [5] focus on the efficiency of financial monitoring mechanisms, which are crucial for preventing financial crimes and ensuring the integrity of the banking system. Blikhar et al. [6] analyze the evolution of the National Bank of Ukraine's roles and functions. Understanding the central bank's evolving responsibilities is vital for evaluating its impact on monetary policy, financial stability, and the overall banking sector. Vasylytsiv et al. [7] contribute to this discourse by exploring the interaction between monetary policy and financial policy in Ukraine. Druhov and Druhova [8] offer a critical assessment of the Ukrainian banking system in the context of pre-war conditions and the ongoing war. Their research likely highlights severe challenges faced by the sector, such as liquidity risks, asset quality deterioration, and operational disruptions. Yakubiv et al. [9] investigate the growth of electronic banking in Ukraine. Digitalization is a significant area of development, with potential benefits including enhanced financial inclusion, increased efficiency, and greater competition within the banking sector. Kachula et al. [10] explore the relationship between economic growth and the development of the banking sector in Ukraine. A robust banking system is often viewed as a catalyst for economic growth, facilitating credit provision, trade, and financial risk management.

A wide range of economic and mathematical methods and models are used to study the banking system, in particular, as a critically important component of the financial system. Examples of some of them are given in Table 2.

Logistic regression method is well-suited for modeling banking stability and predicting bank failures due to its effectiveness in analyzing binary outcomes, such as bankruptcy probabilities. Logistic regression accommodates both numerical and categorical variables, enabling the integration of diverse financial indicators and economic factors. The method offers interpretable results through coefficients that quantify the impact of explanatory variables on the likelihood of financial distress. Additionally, it can model non-linear relationships between variables and provide accurate risk forecasts, which is crucial for assessing the financial stability of banks, particularly in the context of economic instability in Ukraine.

In conclusion, the reviewed literature provides a comprehensive understanding of the multifaceted nature of the Ukrainian banking system, highlighting its structural characteristics, operational dynamics, and the significant challenges it has faced. The research underscores the system's bank-centricity, the importance of lending diversification and operational efficiency, and the critical role of regulatory oversight and monetary policy in maintaining stability. Furthermore, the growing influence of digitalization and the intrinsic link between banking sector development and broader economic growth are evident themes. The application of a diverse

Table 2 Mathematical methods and models in bank systems research

Publication	Method
Mishchenko et al. [11]	Regression analysis
Adamyk et al. [12]	Machine learning
Blahun et al. [13]	Fuzzy logic
Yemets et al. [14]	Neural network
Chukhray et al. [15]	Stacking machine learning
Davydenko et al. [16]	Quantitative analysis
Zomchak and Lapinkova [17]	ARIMA and VAR models
Izonin et al. [18]	Ensemble methods
Zomchak and Hakava [19]	Multidimensional ranking
Guryanova et al. [20]	Methods of business analytics of multidimensional processes
Wheat and Oliskevych (2018) [21]	Systems dynamics
Bielinskyi et al. [22]	Network analysis
Velykoivanenko and Korchynskyi [23]	Clustering in the dimensionality reduction algorithms
Yemets et al. [24]	FFT-LSTM time-series forecasting model
Kyshakevych and Klymkovych [25]	Z-score

Source Created by authors

array of economic and mathematical modeling techniques, including regression analysis, machine learning, fuzzy logic, and neural networks, reflects the complexity of analyzing and predicting the behavior of this vital sector.

2 Methods and Materials

In the realm of statistical modeling, understanding the relationship between various factors and a specific outcome is paramount across numerous scientific disciplines. When the outcome of interest is binary, representing the occurrence or non-occurrence of an event, specialized regression techniques are required. This section is devoted to logistic regression, a powerful analytical tool specifically designed for examining the association between one or more independent variables and a dichotomous dependent variable.

By estimating the probability of a particular event occurring, logistic regression provides a framework for both understanding the underlying drivers of binary outcomes and for classifying observations based on their predicted likelihood.

This method stands as a cornerstone in analyzing phenomena where the response variable is inherently categorical with two levels, offering significant advantages over traditional linear regression approaches in such contexts.

Logistic regression is a type of regression analysis that examines the relationship between several independent variables (known as regressors or predictors) and a dependent variable. When the dependent variable has only two possible values, binary logistic regression is used. Logistic regression allows for the estimation of the probability that a certain event will occur for a given set of predictors.

In general, a regression model can be expressed with the formula:

$$y = F(x_1, x_2, \dots, x_n) \quad (1)$$

where y is the dependent variable;

$x_1, x_2, \dots, x_n$ are the independent variables.

In multiple linear regression, it is assumed that the dependent variable reflects a linear dependence on the independent variables:

$$\ln\left(\frac{p}{1-p}\right) = b_0 + b_1x_1 + b_2x_2 + \dots + b_nx_n \quad (2)$$

where p is the dependent variable;

$x_1, x_2, \dots, x_n$ are the independent variables;

$b_0, b_1, \dots, b_n$ are the coefficients.

To model probabilities of a binary dependent variable, a special function that monotonically increases within the range from 0 to 1 is used. Common functions employed in binary choice models include:

- The logistic function
- The standard normal distribution function

Models based on the logistic function for binary outcomes are known as logistic regression or logit models. Models using the standard normal distribution function are referred to as probit models.

Logistic regression differs from multiple linear regression by accounting for the binary nature of the dependent variable and predicting values in the range from 0 to 1, avoiding the limitations of predicting values outside this range. Logistic regression estimates probabilities within the interval $[0, 1]$ for any values of the independent variables, using the following regression equation (logit transformation):

$$P_i = \frac{1}{1 + e^{-Z_i}} \quad (3)$$

where P_i is the probability of the event occurring,

e is the base of the natural logarithm,

Z_i is the linear combination of the predictor variables.

To estimate the coefficients of a logistic regression model, the maximum likelihood method is used. This statistical method evaluates parameters based on sample data by maximizing the likelihood function, which accounts for the significance of each coefficient simultaneously.

Logistic regression is used to predict the probability of the dependent variable based on the independent variables included in the model. By predicting probabilities, it is possible to classify observations into two groups.

To assess the classification efficiency of logistic regression models, ROC (Receiver Operating Characteristic) analysis is employed. This analysis is based on the ROC curve, which illustrates the relationship between true positive observations and false negative observations. The model's performance is evaluated using a cut-off point, which affects classification accuracy and determines the values of type I and type II errors.

In practice, logistic regression is used for two main purposes: modeling the relationship between variables and classifying observations.

In conclusion, logistic regression stands as a versatile and statistically robust technique for analyzing binary dependent variables. Its ability to model the probability of an event's occurrence based on a set of predictor variables, coupled with the interpretability of its coefficients and the capacity for classification, renders it invaluable across diverse fields of research. The utilization of the logit transformation ensures that predicted probabilities remain within the meaningful range of 0–1, addressing a key limitation of linear regression when applied to binary outcomes. Furthermore, the application of maximum likelihood estimation provides a rigorous framework for determining the model's parameters, while ROC analysis offers a comprehensive method for evaluating its classification performance and understanding the trade-offs between different types of classification errors. Whether the primary goal is to elucidate the relationships between predictors and a binary response or to effectively

classify observations into distinct categories, logistic regression provides a powerful and well-established methodological approach.

3 Results

In today's world, where the banking system is crucial for overall economic stability, early detection of financial difficulties in banks enables timely actions to prevent systemic crises and maintain trust in the banking system, especially during global crises such as wars, pandemics, and other external shocks. Hence, developing models to predict the financial health of Ukrainian banks has become essential.

At the beginning of the year, the National Bank published information indicating that 10 active banks were on the brink of bankruptcy. To address this issue, machine learning models were employed to predict bankruptcy effectively, with logistic regression being selected for this purpose.

To build these models, 22 indicators were calculated, categorized into groups such as liquidity, solvency, financial stability, profitability, and business activities. The data for these indicators were sourced from the official website of the National Bank. Data for the study were taken from the official website of the National Bank of Ukraine, which help to assess the financial condition of the bank [1]: the calculated indicators are listed in Table 3.

Furthermore, the study's reliance on publicly available data from the National Bank of Ukraine enhances its transparency and replicability. The detailed list of 22 initial indicators, spanning various facets of banking activity, provides a comprehensive foundation for understanding the multifaceted nature of financial stability. The categorization of these indicators into liquidity, solvency, financial stability, profitability, and business activities offers a structured framework for analyzing the underlying factors contributing to bank vulnerability. The subsequent analysis and results derived from this methodology will provide valuable insights for regulators, banking institutions, and researchers seeking to understand and predict financial distress within the Ukrainian banking sector, particularly in the context of ongoing economic and geopolitical uncertainties.

The dependent variable used in the model was binary (0 = stable bank, 1 = bankrupt). Based on the information that 10 banks were bankrupt, each bank was classified accordingly.

Before constructing the models, an initial check of the incoming data was performed. A correlation matrix was built to examine the indicators for any strong correlations. Indicators with a correlation value greater than 0.7 were excluded to avoid multicollinearity.

Subsequently, using a stepwise elimination method, indicators that were not statistically significant were removed from the model. Building upon the initial data

Table 3 Indicators of banking activity in Ukraine

No.	Indicator	No.	Indicator
x1	Administrative and other operating expenses/Assets	x12	Credits and arrears from customers/ Liabilities
x2	Interest income/Assets	x13	Customer deposits/Liabilities
x3	Equity/Assets	x14	Bank funds/Liabilities
x4	Cash and cash equivalents/Assets	x15	Debt securities issued by the bank/ Liabilities
x5	Investments in associated and subsidiary companies/Assets	x16	Subordinated debt/Liabilities
x6	Profit (loss) after taxation/Bank liabilities	x17	Reserves/Liabilities
x7	Profit (loss) after taxation/Assets	x18	Statutory capital/Equity
x8	Profit (loss) after taxation/Equity	x19	Fixed assets and intangible assets/ Equity
x9	Net interest income (Net interest expenses)	x20	Undivided profit (uncovered loss)/ Equity
x10	Net commission income (Net commission expenses)	x21	Interest income/Interest expenses
x11	Credits and arrears from clients/Assets	x22	Commission income/Commission expenses

Source Created by authors

preprocessing, the subsequent stage of model development involved rigorous evaluation and refinement of the logistic regression model. The selection of logistic regression was predicated on its suitability for binary classification problems, offering interpretability in assessing the impact of various financial indicators on the likelihood of bank bankruptcy. The process of stepwise elimination of statistically insignificant variables aimed to identify the most parsimonious and impactful set of predictors for financial distress in Ukrainian banks. This iterative approach not only enhances the model's predictive power by focusing on key determinants but also mitigates the risk of overfitting, ensuring the model's generalizability to unseen data.

4 Discussion

As a result of the analysis, only 2 out of the 22 initially selected indicators were deemed suitable for building the models: x1—Administrative and other operating expenses/Assets, and x2—Interest income/Assets.

The reduction from 22 initial indicators to a final set of only two (x1: Administrative and other operating expenses/Assets, and x2: Interest income/Assets) highlights the critical importance of these specific financial metrics in discerning the financial

Table 4 Omnibus test results of the logistic regression model for bank bankruptcy prediction

	Chi-square	df	Sig
Step	55.132	2	< 0.001
Block	55.132	2	< 0.001
Model	55.132	2	< 0.001

Source Calculated by authors

health of Ukrainian banks during the study period. The fact that these two indicators emerged as statistically significant predictors after rigorous correlation analysis and stepwise elimination suggests that the efficiency of operational management (as reflected by administrative expenses relative to assets) and the core revenue generation capacity from lending activities (indicated by interest income relative to assets) are particularly influential factors in determining the likelihood of bank bankruptcy in the Ukrainian context. This parsimonious model, while utilizing a limited number of variables, demonstrates a strong overall statistical significance, as confirmed by the omnibus test results, implying that these two key ratios collectively provide a robust basis for predicting bank financial distress.

First, Table 4 presents the results of the omnibus tests for the coefficients of the logistic regression models. The omnibus test is a general test for the statistical significance of the model coefficients. It compares the model to a baseline model (a model without any independent variables) to determine whether there are significant differences between the two models.

Table 4 shows that the p -value is < 0.01, indicating a high statistical significance of the model. The chi-square values are substantial, suggesting that the logistic regression model constructed is statistically significant, adequate, and suitable for predicting binary outcomes based on the independent variables. The presented omnibus test results provide strong statistical evidence supporting the overall validity and usefulness of the developed logistic regression model. The highly significant chi-square statistic (55.132 with 2 degrees of freedom) and the corresponding p -value (< 0.001) unequivocally demonstrate that the model, incorporating the selected independent variables (x_1 and x_2), significantly improves upon a null model that includes no predictors. This indicates that the inclusion of administrative expenses relative to assets and interest income relative to assets contributes meaningfully to the model's ability to discriminate between stable and potentially bankrupt banks within the Ukrainian financial system during the studied period.

Table 5 displays the classification results and the relationship between the number of correctly and incorrectly classified observations according to the model.

As shown in Table 5, 49 banks were correctly classified as stable, representing 92.5% of the sample. Among the bankrupt banks, 7 were correctly classified, which is 70% of that sample. Overall, the percentage of correctly classified observations is 88.9%, indicating high quality of the built model. The classification table provides a more granular assessment of the model's predictive accuracy at the individual bank level. The high percentage of correctly classified stable banks (92.5%) suggests a strong ability of the model to identify institutions that are not facing imminent

Table 5 Classification table for the logistic regression model on Ukrainian banks

	Bankrupt/Non-bankrupt			Percentage correct
	0		1	
Bankrupt/Non-bankrupt	0	49	4	92.5
	1	3	7	70.0
Overall Percentage				88.9

Source Calculated by authors

financial distress. While the classification accuracy for bankrupt banks (70.0%) is slightly lower, it still represents a significant proportion of correctly identified cases of financial difficulty. The overall classification accuracy of 88.9% underscores the model's strong discriminatory power in distinguishing between stable and bankrupt banks based on the selected financial indicators. This level of accuracy suggests that the logistic regression model, utilizing administrative expenses to assets and interest income to assets, can be a valuable tool for early warning systems and regulatory oversight within the Ukrainian banking sector.

The statistical significance of the model indicators was also evaluated. Table 6 displays the model parameters, standard errors, and p-values.

Table 6 provides a detailed examination of the individual predictors within the logistic regression model, outlining their estimated coefficients (B), standard errors (S.E.), Wald chi-square statistics, degrees of freedom (df), significance levels (Sig.), and the exponentiated coefficients (Exp(B), also known as the odds ratio). For the indicator "Administrative and other operating expenses/Assets," the positive coefficient ($B = 1.702$) suggests that higher levels of administrative and operating expenses relative to assets are associated with an increased likelihood of a bank being classified as bankrupt. This relationship is statistically significant ($\text{Sig.} = 0.002$), as indicated by the low p-value. The odds ratio of 5.487 implies that for every one-unit increase in this ratio, the odds of a bank being bankrupt increase by approximately 448.7% ($5.487 - 1$).

Conversely, the indicator "Interest income/Assets" exhibits a negative coefficient ($B = -2.579$), indicating that higher levels of interest income relative to assets are associated with a decreased likelihood of bank bankruptcy. This relationship is also statistically significant ($\text{Sig.} = 0.018$). The odds ratio of 0.076 suggests that

Table 6 Logistic model of the bank bankruptcy of Ukraine parameters and their significance

	B	S.E	Wald	df	Sig	Exp(B)
Administrative and other operating expenses/Assets	1.702	0.542	9.88	1	0.002	5.487
Interest income/Assets	- 2.579	1.094	5.56	1	0.018	0.076
Constant	0.544	1.761	0.095	1	0.0758	1.723

Source Calculated by authors

for every one-unit increase in this ratio, the odds of a bank being bankrupt decrease significantly, by approximately 92.4% ($1 - 0.076$). Finally, the constant term has a positive coefficient ($B = 0.544$), but its significance level ($\text{Sig.} = 0.758$) is above the conventional threshold of 0.05, suggesting that the intercept does not significantly contribute to the model's predictive power on its own. Overall, this table confirms the individual and statistically significant contributions of both selected financial indicators to the logistic regression model's ability to predict bank bankruptcy in Ukraine.

The table 5 shows that the standard errors of the parameters are relatively small, and the p-values are < 0.05 , demonstrating the statistical significance of the selected indicators.

Based on the calculations, the logistic model for predicting bank bankruptcy is represented by the following equation:

$$L = 0.544 + 1.702x_1 - 2.579x_2,$$

where x_1 are administrative and other operating expenses,
 x_2 is interest income.

The constant term (intercept), 0.544, reflects the estimated log-odds of bankruptcy when the predictor variables are zero. While a literal interpretation of this value in the context of financial ratios is limited due to the practical improbability of zero values for x_1 and x_2 , the constant serves a crucial role in positioning the regression hyperplane within the model's parameter space.

The coefficient associated with the first predictor, x_1 (Administrative and other operating expenses/Assets), is 1.702. This positive coefficient indicates a direct relationship between the relative magnitude of administrative expenses and the log-odds of bankruptcy. Specifically, an increase in the ratio of administrative and operating expenses to total assets is associated with an elevated likelihood of a bank being classified as bankrupt. The magnitude of the coefficient suggests the extent to which this indicator influences the predicted outcome.

Conversely, the coefficient for the second predictor, x_2 (Interest income/Assets), is -2.579. The negative sign of this coefficient denotes an inverse relationship, whereby higher levels of interest income relative to total assets are associated with a decreased log-odds of bankruptcy. The larger absolute value of this coefficient, compared to that of x_1 , implies a comparatively stronger influence of interest income in mitigating the predicted probability of bank failure within the developed model.

Using this model, the probability of bankruptcy can be calculated with the formula provided in Eq. (3). This yields a value between 0 and 1, where 0 indicates a stable financial condition and 1 represents a high probability of bankruptcy. By plugging the calculated $\text{Logit}(P(\text{Bankruptcy}))$ value into this formula, the model provides a predicted probability of bankruptcy for a given bank based on its administrative expenses/assets (x_1) and interest income/assets (x_2) ratios. A probability closer to 1 suggests a higher likelihood of bankruptcy, while a probability closer to 0 indicates a more stable financial condition according to the model's predictions.

5 Conclusions

The present study aimed to develop a predictive model for assessing the financial health of Ukrainian banks and identifying those at risk of bankruptcy. By employing logistic regression and a comprehensive set of financial indicators, we sought to contribute to the enhancement of early warning systems within the Ukrainian banking sector.

Our findings indicate that the logistic regression model exhibits significant predictive power in differentiating between stable and bankrupt banks. The model's accuracy of 88.9% in classifying banks highlights its potential as a valuable tool for financial regulators and risk managers. Moreover, the identification of administrative and other operating expenses to assets ratio, and interest income to assets ratio as the primary determinants of bank bankruptcy provides crucial insights into the factors influencing bank stability.

It is essential to acknowledge that the model's predictive capabilities are contingent upon the quality and availability of data. As the banking landscape evolves and new financial instruments emerge, the model may require periodic recalibration to maintain its effectiveness. Furthermore, while the model offers valuable insights, it should be considered as a complementary tool to human expertise and judgment in assessing bank risk.

In conclusion, this study demonstrates the applicability of logistic regression in predicting bank bankruptcy within the Ukrainian context. The findings underscore the importance of robust financial monitoring and early intervention strategies to safeguard the stability of the banking system. Future research could explore the incorporation of additional macroeconomic variables and advanced machine learning techniques to enhance the predictive power of the model.

References

1. National Bank of Ukraine Homepage. <https://bank.gov.ua/en/>. Accessed 01 Mar 2025
2. Frolov S, Shukairi F (2020) Bank-centric nature of the financial system of Ukraine: analysis of the current situation. *Banks Bank Syst* 15(3):184
3. Khalatur S, Zhylenko K, Masiuk Y, Velychko L, Kravchenko M (2018) Assessment of bank lending diversification in Ukraine. *Banks Bank Syst* 13(3):141–150
4. Chaikovskiy Y, Chaikovska I, Chaikovskiy Y (2023) Efficiency of the activities of banking institutions in Ukraine. *World Finance* 3(72):127–140
5. Bukhtiarova AH, Semenoh AY, Razinkova M, Nebaba NO, Józef AH (2020) Assessment of financial monitoring efficiency in the banking system of Ukraine. *Banks Bank Syst* 1
6. Blikhar M, Natyrkach R, Chystokletov L, Shyshko V, Hula I (2024) Evolution of functions of the national bank of Ukraine: economic and legal study. *Financ Credit Activ Probl Theory Pract* 3(56)
7. Vasylytsiv TH, Klipkova OI, Lupak RL, Mitsenko NG, Mishchuk IP (2019) Monetary and financial policy of Ukraine: theoretical-empirical connections and priorities of state regulation. *Financ Credit Activ Probl Theory Pract* 4(31):320–330
8. Druhov O, Druhova V (2022) Banking system in Ukraine 2022: before and in the Wartime. *Bezpіeczny Bank* 86(1):33–48

9. Yakubiv V, Sodoma R, Hrytsyna O, Pavlikha N, Shmatkovska T, Tsymbaliuk I, Brodska I (2019) Development of electronic banking: a case study of Ukraine. *Entrepreneursh Sustain Issues* 7(1):219
10. Kachula S, Zhytar M, Sidelnykova L, Perchuk O, Novosolova O (2022) The relationship between economic growth and banking sector development in Ukraine. *WSEAS Trans Bus Econ* 19:222–230
11. Mishchenko V, Naumenkova S, Mishchenko S (2021) Assessing the efficiency of the monetary transmission mechanism channels in Ukraine. *Banks Bank Syst* 16(3):48–62
12. Adamyk B, Skirka A, Snihur K, Adamyk O (2019) Analysis of Trust in Ukrainian banks based on machine learning algorithms. In: 2019 9th international conference on advanced computer information technologies (ACIT). IEEE, pp 234–239
13. Blahun IS, Blahun II, Blahun SI (2020) Assessing the stability of the banking system based on fuzzy logic methods. *Banks Bank Syst* 15(3):171
14. Yemets K, Izonin I, Dronyuk I (2025) Time series forecasting model based on the adapted transformer neural network and FFT-based features extraction. *Sensors* 25(3):652
15. Chukhray N, Shakhovska N, Mrykhina O, Lisovska L, Izonin I (2022) Stacking machine learning model for the assessment of R&D product's readiness and method for its cost estimation. *Mathematics* 10(9):1466
16. Davydenko N, Boiko S, Cherniavska O, Nehrey M (2023) Analysis of the impact of state-owned banks on the sustainability of public finances. *Economies* 11(9):229
17. Zomchak L, Lapinkova A (2022) Key interest rate as a central banks tool of the monetary policy influence on inflation: the case of Ukraine. In: *The international symposium on computer science, digital economy and intelligent systems*. Springer Nature Switzerland, Cham, pp 369–379. https://doi.org/10.1007/978-3-031-24475-9_32
18. Izonin I, Tkachenko R, Verhun V, Zub K (2021) An approach towards missing data management using improved GRNN-SGTM ensemble method. *Eng Sci Technol Int J* 24(3):749–759
19. Zomchak L, Hakava S (2025) Unveiling disparities and resilience in Ukrainian regional labor markets: multidimensional ranking approach. In: Štarchoň P, Fedushko S, Gubíniava K (eds) *Developments in information and knowledge management systems for business applications. Studies in systems, decision and control*, vol 578. Springer, Cham
20. Guryanova L, Bolotova O, Gvozdytskyi V, Olena S (2020) Long-term financial sustainability: An evaluation methodology with threats considerations. *Rivista di Studi Sulla Sostenibilita* (2020/1)
21. Wheat ID, Oliskevych, M (2018) The canonical New Keynesian monetary policy model: a system dynamics translation. In: *Proceedings of the 36th International System Dynamics Conference*, Reykjavik, Iceland
22. Bielinskyi AO, Soloviev VN, Hushko SV, Kiv AE, Matviychuk AV (2022) High-order network analysis for financial crash identification. In: *M3E2-MLPEED*, pp 132–149
23. Velykoivanenko H, Korchynskyi V (2022) Application of clustering in the dimensionality reduction algorithms for separation of financial status of commercial banks in Ukraine. *Univ J Account Finance* 10(1):148–160
24. Yemets K, Izonin I, Dronyuk I (2025) Enhancing the FFT-LSTM time-series forecasting model via a novel FFT-based feature extraction-extension scheme. *Big Data Cogn Comput* 9(2):35
25. Kyshakevych B, Klymkovych I (2018) Estimation of Z-score for Ukrainian banking system. *Sci J Polonia Univ* 30(5):43–51

Development of ESG Reporting Under the Influence of Theoretical Concepts



Iryna Zamula , Daria Seletska , Olha Fedorova ,
and Maryna Tanasiieva 

Abstract Business entities can compile ESG reports, a specialized type of sustainability reporting, to inform stakeholders about potential and actual environmental and social impacts. ESG reporting focuses more on specific, measurable indicators and can be utilized for internal assessment of non-financial risks and external disclosure regarding their management. This study aims to characterize the evolution of ESG reporting under the influence of the concept of sustainable development and other theoretical frameworks. ESG factors play a central role in sustainable development as a tool for demonstrating the contribution to achieving the SDGs to stakeholders. Moreover, some differences have been identified between classical sustainability reporting and ESG reporting, which allowed the formulation of the definition of “ESG reporting” as a type of non-financial reporting within the field of sustainable development, which is a subset of the sustainability report. It is created to provide stakeholders with information regarding the non-financial aspects of the company’s operations by highlighting clearly defined indicators related to environmental risks, social responsibility, and corporate governance, primarily for the present time. The main provisions of three theories—stakeholder theory, legitimacy theory, and institutional theory—were analyzed within the research. A criteria system for assessing the ESG reporting process has been developed and proposed. It describes the criteria for evaluation, their characteristics, possible indicators, and evaluation procedures

I. Zamula (✉) · D. Seletska · O. Fedorova
Zhytomyr Polytechnic State University, Zhytomyr, Ukraine
e-mail: zamula@ztu.edu.ua

D. Seletska
e-mail: koa_sdo@ztu.edu.ua

O. Fedorova
e-mail: fedorova.olga@ztu.edu.ua

M. Tanasiieva
Yuriy Fedkovych Chernivtsi National University, Chernivtsi, Ukraine
e-mail: m.tanasiieva@chnu.edu.ua

within each criterion. It is believed that applying these criteria will ensure a systematic approach to identifying areas for improvement in the ESG strategy and ESG reports of businesses.

Keywords ESG reporting · Sustainable development · Sustainability reporting · Reporting development

1 Introduction

1.1 Relevance

In September 2015, 193 countries joined forces to address global challenges in social, economic, and environmental spheres by adopting 17 Sustainable Development Goals (SDGs), which are planned to be achieved by 2030. Ukraine has committed to this process as a member of the United Nations. Currently, representatives of the global business community, guided by a series of directives and internationally recognized regulatory frameworks, actively compile sustainability reports to inform stakeholders and society about their compliance with the SDGs in their operations. However, despite global trends, sustainability reporting in Ukraine remains underdeveloped—only a small fraction of businesses prepare such reports, and mostly at the initiative of foreign partners.

On June 23, 2022, Ukraine obtained the status of a candidate for EU membership, which necessitated the alignment of national legislation with European requirements, including those related to sustainability reporting. Consequently, on October 18, 2024, the Cabinet of Ministers of Ukraine approved a Strategy to establish the regulatory foundations for mandatory sustainability reporting by domestic business entities by 2030 [1]. According to the Strategy, businesses will begin compiling and submitting sustainability reports in 2026. However, we believe that to attract foreign investment and facilitate access to international capital markets, Ukrainian enterprises should already be implementing and disclosing such reports.

Business entities can compile ESG (Environmental, Social, and Governance) reports, a specialized type of sustainability reporting, to inform stakeholders about potential and actual environmental and social impacts. ESG reporting focuses more on specific, measurable indicators and can be utilized for internal assessment of non-financial risks and external disclosure regarding their management. Moreover, ESG reporting holds greater practical value as investors and financial institutions increasingly consider ESG ratings when making investment decisions.

Many domestic scholars have confirmed in their research the necessity of submitting such reporting to stakeholders to demonstrate compliance with the principles of social responsibility and the Sustainable Development Goals (SDGs). However, the methodological approaches to information disclosure and the role of accounting support in preparing sustainability and ESG reporting remain insufficiently explored.

Ukraine lacks a unified regulatory framework for forming data used in ESG reporting. Therefore, developing methodological proposals will facilitate the standardization of the data collection, recording, and aggregation processes for ESG reporting purposes, ultimately ensuring the possibility of generating transparent and reliable reports.

Prospects for legislative regulation further reinforce the relevance of research for ESG reporting. On November 7, 2024, the Ministry of Finance of Ukraine published a Draft Law of Ukraine “On Amendments to the Law of Ukraine ‘On Accounting and Financial Reporting in Ukraine’ Regarding the Introduction of Sustainability Reporting” [2]. According to the draft law’s provisions, the role of the chief accountant (or another responsible individual) will change upon the law’s enactment. This individual will be required to ensure compliance with the methodological principles for the preparation and submission of sustainability reports and, consequently, will bear responsibility for the accuracy of the accounting information reflected in such reports.

1.2 Analysis of Research and Publications

Theoretical, methodological, and practical aspects of non-financial and integrated reporting as types of sustainability reporting have been explored by T. Baranovska, K. Bezverkhyi, O. Budko, I. Vasylichuk, V. Hyk, D. Hrytsyshen, I. Derun, N. Yershova, I. Zhygley, V. Zhuk, V. Kostiuchenko, P. Kutsyk, S. Levytska, S. Lehenchuk, N. Lokhanova, O. Nesterenko, O. Oliinyk, M. Prodanchuk, and O. Sokil. These scholars have also studied the representation of information in accounting records that can be utilized in reporting to confirm enterprises’ environmental and social responsibility. However, in this context, it is also important to mention researchers such as O. Baryshnikova, N. Brazilii, I. Demko, V. Derii, V. Yevdokymov, P. Kaminskyi, V. Len, Yu. Serpeninova, L. Shvets, and K. Shymanska. Scholars who have contributed to the study of sustainability reporting (including ESG reporting), notable authors include A. Amran, N. Bellantuono, N. Brown, P. Gocht, S. Gocht, K. Deegan, M. Clarkson, H. Christensen, C. Kuzey, K. Leuz, Y. Ozerhan, P. Pontrandolfo, R. Roberts, B. Scozzi, A. Uyar, R. Haniffa, L. Hale, J. Chen, and others.

Despite the significant contributions of the aforementioned scholars, the issues surrounding the development of ESG reporting as a distinct form of sustainability reporting under the influence of various theoretical concepts remain insufficiently researched.

1.3 The Study Aims and Research Methods

This study aims to characterize the evolution of ESG reporting under the influence of the concept of sustainable development and other theoretical frameworks.

The methodological foundation of this study is based on general philosophical and scientific methods for understanding phenomena and processes within the economic sciences. The examination of ESG reporting within the framework of the sustainable development concept was conducted using analysis and synthesis, as well as historical and comparative analysis. Abstraction, generalization, logical, and comparative analysis methods were employed to assess the influence of theoretical concepts on the formation of ESG reporting. Tabular and graphical methods were applied to visually represent research findings, along with the bibliometric mapping technique.

2 Development of ESG Reporting in the Context of the Sustainable Development Concept

The concept of sustainable development has permeated all spheres of life, including corporate economics. Businesses integrating sustainable development principles into their activities have started disclosing relevant information through various reporting formats.

An analysis of published articles in the Scopus scientometric database related to sustainability reporting (Fig. 1) indicates that this concept is predominantly explored in the following academic fields: business, management, and accounting; economics, econometrics, and finance; social sciences; and environmental studies. The study of sustainability reporting within these disciplines confirms that sustainability (and its associated reporting) is a crucial component of strategic business management and performance assessment.

Disclosing information on the impact of enterprises on the environment, society, and the economy enables stakeholders to evaluate corporate activities in the context of

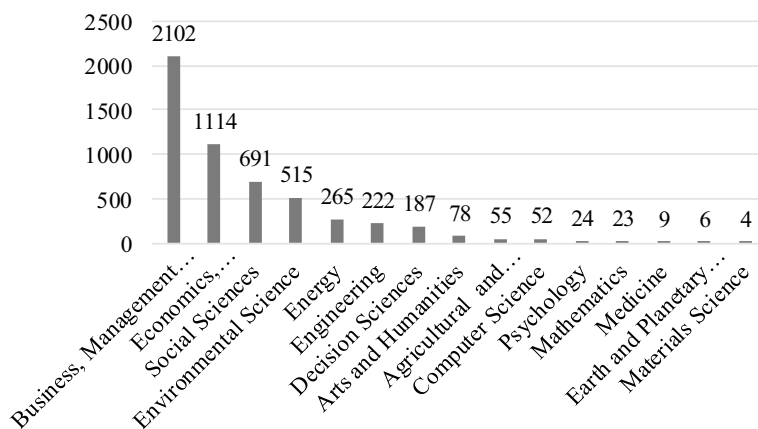


Fig. 1 Number of scientific studies in the Scopus database across different fields with the keyword “Sustainability reporting” (2018–2023). *Source* Compiled by the authors

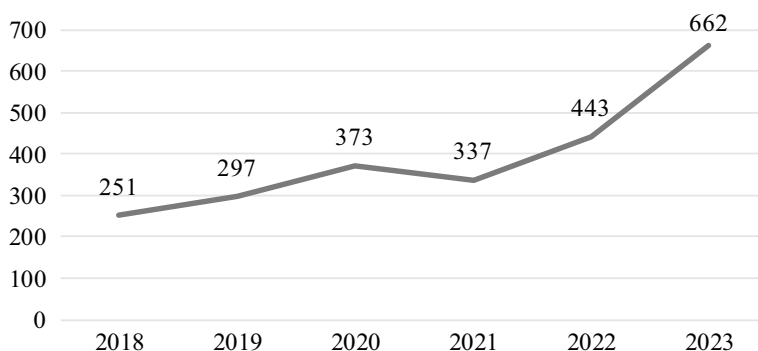


Fig. 2 Number of scientific studies in the Scopus database with the keyword “Sustainability Reporting” (2018–2023). *Source* Compiled by the authors

sustainable development. Additionally, it allows businesses to establish a constructive dialogue with various stakeholder groups. Moreover, such reporting helps companies identify risks associated with sustainable development, such as environmental impact, labor relations issues, and corruption risks, and implement appropriate mitigation measures.

Furthermore, data from the Scopus scientometric database reveal a positive trend: from 2018 to 2023, the number of scientific publications related to sustainability reporting research increased significantly (Fig. 2). This indicates both the growing interest of contemporary scholars in the subject and the relevance of ongoing research in this area. A slight decline in publication activity was observed in 2021 (36 fewer articles than in 2020), likely due to the COVID-19 pandemic. However, the overall trend confirms sustained scientific interest in sustainability reporting research.

The multidimensional nature of the concept of “sustainability reporting,” illustrated in Fig. 3, highlights that the strongest direct relationships (marked in red) are observed with the terms “corporate social responsibility” (the second-largest red circle, labeled “CSR”), “stakeholder theory,” “legitimacy theory,” “non-financial reporting,” and “GRI”. Indirect associations are found with concepts such as “sustainability,” “sustainable development,” and “integrated reporting.” Based on this, we conclude that existing research on sustainability reporting is primarily focused on:

1. defining its role within the sustainable development framework,
2. examining corporate adherence to social responsibility practices,
3. assessing its influence on theoretical concept formation and
4. exploring its application within the Global Reporting Initiative (GRI) framework.

Non-financial reporting has become a key element of corporate sustainability strategies as businesses seek to demonstrate their commitment to the environmental, social, and economic principles outlined in the United Nations General Assembly Resolution adopted on September 25, 2015—“Transforming Our World: The 2030 Agenda for Sustainable Development” [3].

the International Sustainability Standards Board (ISSB), and the adoption of IFRS S1 and IFRS S2, marking the beginning of a unified global approach to high-quality and comparable sustainability disclosures.

An analysis of sustainability reports [5] reveals that the reporting process integrates three primary approaches to information disclosure. These include the mechanism for presenting financial and non-financial indicators, supplemented by a narrative component, as illustrated in Fig. 5.

The balance between financial and non-financial information and the narrative component ensures transparency and completeness of the disclosed information. Specifically, the narrative section of the report establishes a link between financial and non-financial indicators, providing explanations regarding their achievement and highlighting the influencing factors that contributed to changes. Using narrative enhances the impact of sustainable practices on the company’s long-term financial stability and growth while also serving as an informational basis for assessing risks and opportunities that arise during the company’s operations.

The term “ESG” first gained prominence in the 2004 report *Who Cares Wins*, a joint initiative of financial institutions invited by the United Nations [6]. By 2023,

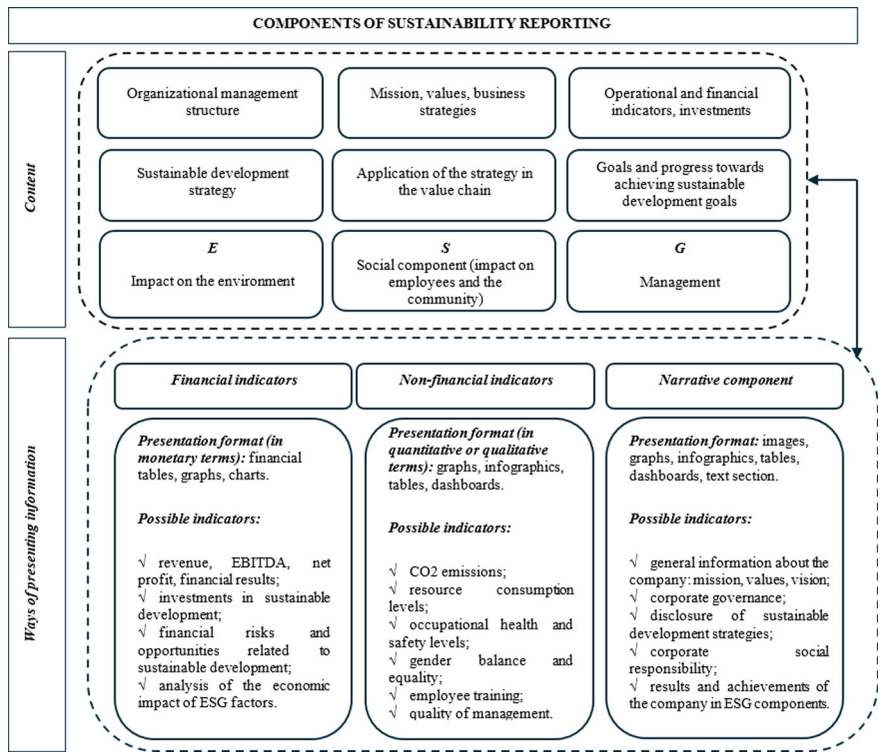


Fig. 5 Components of sustainability reporting. *Source* Compiled by the authors based on sustainability reports

ESG had evolved from a UN corporate social responsibility initiative into a global phenomenon.

ESG (Environmental, Social, and Governance) refers to qualitative and quantitative factors that measure an entity's impact on the environment, society, and corporate governance. These aspects are central to the sustainable development concept (Table 1), as they enable corporate leadership to integrate sustainability principles into management strategies, achieving both short-term financial objectives and long-term business resilience.

The systematization of Sustainable Development Goals (SDGs) by ESG aspects, developed by the author (Table 1), demonstrates their interdependence. Accordingly, a company's achievement of SDGs 6, 7, 12, 13, and 14 can be assessed through its implemented mechanisms to minimize environmental impacts (including greenhouse

Table 1 The role of ESG in the concept of sustainable development

ESG aspects	Sustainable development goals (SDGs)
Environmental (E)	SDG 6: Clean Water and Sanitation —Managing water resource usage and reducing pollution levels. SDG 7: Affordable and Clean Energy —Increasing energy efficiency and promoting renewable energy sources. SDG 12: Responsible Consumption and Production —Developing toxic and pollutant disposal mechanisms, optimizing resource utilization. SDG 13: Climate Action —Reducing greenhouse gas emissions and implementing climate-resilient practices. SDG 14: Life Below Water —Reducing ocean and marine pollution, protecting marine ecosystems. SDG 15: Life on Land —Conserving and restoring biodiversity, combating land degradation, and halting deforestation
Social (S)	SDG 1: No Poverty —Creating job opportunities, supporting low-income populations, and expanding access to education. SDG 3: Good Health and Well-being —Conducting immunization campaigns, implementing disease prevention measures, and promoting health education. SDG 4: Quality Education —Investing in primary, secondary, and higher education, as well as employee professional development. SDG 5: Gender Equality —Preventing sexual violence and exploitation, eliminating discrimination, and ensuring equal rights for men and women. SDG 8: Decent Work and Economic Growth —Establishing fair working conditions, improving labor productivity, and eradicating forced labor, slavery, and human trafficking. SDG 10: Reduced Inequalities —Promoting safe migration, enhancing financial market regulation, and developing policies that empower individuals to achieve equitable income
Governance (G)	SDG 16: Peace, Justice, and Strong Institutions —Ensuring the rule of law, transparent, accountable, ethical governance, combating corruption, and strengthening human rights. SDG 17: Partnerships for the Goals —Encouraging global partnerships and collaboration with businesses, governments, and civil society organizations to support sustainable development, including assistance to developing nations

Source Compiled by the authors

gas emissions, wastewater discharge, resource consumption, and waste management). This also involves innovations aimed at improving and/or restoring environmental conditions, as well as measures for controlling the use of non-renewable resources.

For SDGs 1, 3, 4, 5, 8, and 10, the focus should be on issues related to human rights, gender and social equality, working conditions, health, and safety. In achieving these goals, business interaction with communities is crucial—this includes monitoring the impact on local populations, implementing social justice practices, and adapting business strategies to stakeholder expectations.

The governance aspect is directly aligned with SDGs 16 and 17, as companies demonstrate compliance with these goals through transparency in reporting, adherence to ethical norms, anti-corruption measures, continuous engagement with internal and external stakeholders, and balancing the interests of all concerned parties with corporate objectives.

All SDGs may indirectly correspond to multiple ESG aspects since they are complementary. For instance, goals classified under the social aspect may be achieved through initiatives originally developed within the environmental domain—such as emissions reduction measures, which positively impact both the environment and public health, aligning with SDGs 13 and 3.

Having examined the evolution and essence of sustainable development and the relationship between ESG aspects and SDGs, it is possible to determine the role of *ESG reporting* within the sustainable development framework. The term ESG reporting is not explicitly defined in any regulatory documents. In Ukrainian academic and periodical literature, it is often equated with *sustainability reporting*, leading to inconsistencies in interpretation. To clarify this distinction, let us analyze their differences.

Sustainability reporting is a form of non-financial reporting that enables companies to inform stakeholders and society about their progress in achieving SDGs across various parameters and the risks and impacts they face now or may encounter in the future.

A sustainability report discloses information in three key dimensions, although companies may report solely on their environmental impacts at their discretion [7]. The three main reporting areas include:

1. Environmental—addressing environmental needs;
2. Social—meeting the needs of stakeholders, including communities, employees, consumers, and other interested parties;
3. Economic—fulfilling business needs, including analyzing the business model and assessing its resilience within the current economic context.

A distinctive feature of sustainability reporting is its time focus, which spans the past, present, and future. At the same time, its target audience includes a wide range of stakeholders such as employees, clients, and investors [7].

At the same time, ESG reporting measures and discloses a company's environmental, social, and governance performance, along with associated risks and policies

(e.g., emissions, labor relations, antitrust behavior) [8]. It communicates with stakeholders regarding progress toward sustainable development goals within the ESG framework. Unlike sustainability reporting, ESG reporting is primarily focused on the present (with some indicators extending into future projections), and its primary audience consists of external stakeholders.

Sustainability reports are broader in scope, covering comprehensive metrics that reflect a company's impact on society and the environment (e.g., effects on public health and well-being, contributions to reducing social inequalities, and alignment with SDGs). In contrast, ESG reports contain specific and narrowly focused quantitative indicators that measure a company's environmental, social, and governance impacts (e.g., water consumption, waste levels, gender diversity, pay equity, board composition, and anti-corruption programs).

ESG reporting metrics are more measurable, comparable, and tailored to specific stakeholder concerns. In contrast, sustainability reporting encompasses broader strategic issues and assesses a company's long-term societal, environmental, and economic impact. The latter emphasizes long-term corporate strategy and its influence on society.

The ESG reporting and sustainability reporting are significantly aligned; however, ESG reports are characterized by the presentation of more specific, measurable indicators of interest to stakeholders, while sustainability reports have a broader temporal and thematic focus, including a description of the impact of the company's activities, including on economic aspects.

Based on the analysis of ESG reporting and its place within the sustainable development concept, we propose the following interpretation of the term "ESG reporting": it is a type of non-financial reporting in the field of sustainable development, which is a subset of sustainability reporting, and is formulated to provide stakeholders with information regarding the non-financial aspects of the company's activities through the presentation of clearly defined indicators related to environmental risks, social responsibility, and corporate governance, predominantly focused on the present time.

We believe that the proposed definition highlights the specific role of ESG reporting within the concept of sustainable development—it serves as a more specific tool for informing interested parties through clearly defined indicators regarding the current impact of the company's activities on the environment and society. Moreover, such reporting allows for demonstrating management practices that can contribute to achieving sustainable development goals, even if they were not directly designed for this purpose.

3 Theoretical Concepts Influencing the Development of ESG Reporting

Sustainability reporting (including ESG reporting) is one of the key components of modern business activity, as it responds to societal and market demands for transparency and accountability in its operations. Therefore, various theoretical approaches are consistently used by scholars to analyze the influences on the formation of such reports and the motivation behind their development.

In the scientific paper “Comparing Research on Sustainability Reporting with the UN Sustainable Development Goals” by R. Raman et al., one of the findings is identifying a list of theories applied in research on the motivations for developing sustainability reporting (Table 2). According to the agency theory, a company’s management should act as an agent for its shareholders (principals). By using sustainability reporting as a communication tool with stakeholders, the knowledge gap between shareholders and management can be bridged, transparency can be increased, risks associated with information asymmetry among stakeholders can be reduced, and as a result, transaction costs can be minimized.

The application of this theory can contribute to accountability and effectiveness in the field of sustainable development, aligning with Sustainable Development Goals (SDGs) 12 “Responsible Consumption and Production” and 16 “Peace, Justice, and Strong Institutions” [9].

In general, signaling theory explains how one party in a transaction (e.g., company management) can convey information to another (e.g., investors) through deliberate signaling via certain actions. Key elements of this theory include: (1) information asymmetry—one party has better information than the other; (2) signals—actions or attributes that convey reliable information; (3) cost of the signal; (4) interpretation; (5) equilibrium—achieved when the receiving party correctly interprets the signals and makes optimal decisions based on that information. For example, Lehenchuk et al. [7], in their paper “Institutional Signaling through ESG Reporting,” determined that ESG reporting also serves as an institutional signal sent by companies to inform about the presence and extent of social and environmental impacts, which, in turn, facilitates communication with stakeholders. The authors concluded: “...the more reliable the signal provided by the company, the stronger the feedback from stakeholders, which results from reducing the information asymmetry between them” [10].

Signaling Theory can contribute to the implementation of sustainability reporting practices by companies, which aligns with SDG 8, “Decent Work and Economic Growth,” and SDG 12, “Responsible Consumption and Production” [9].

Although the theories mentioned can be used to analyze ESG reporting practices, it is considered appropriate to focus on the most common theories in sustainability reporting research: Stakeholder Theory, Legitimacy Theory, and Institutional Theory.

Stakeholder Theory: The founder of Stakeholder Theory is rightfully considered to be R. Freeman—his book “*Strategic Management: A Stakeholder Approach*”

Table 2 Theories used in research on the influences on the formation of sustainability reporting and their connection to the sustainable development goals

Theories	Formation of the report from the perspective of	Guiding points	Action logic	Connection to sustainable development goals
Stakeholder theory	Internal and external stakeholders	How does the company meet the demands of various stakeholders?	Obtain approval from influential stakeholders	Goals 8 and 12
Legitimacy theory	Social value system	Does the organization’s value system align with society’s value system?	Fulfill social obligations and gain social approval	Goals 8 and 12
Institutional theory	Institutionalized social structures	How to fit into norms already established in similar organizations?	Meet the standards and expectations of society	Goals 8, 9, and 12
Agency theory	Principal-agent relationships	How to address the problem of agency relations?	Reduce agency costs and maximize organizational value	Goals 12 and 16
Signaling theory	Signaler-recipient relationships	How to leverage company activities to its benefit in the market?	Reduce information asymmetry and increase organizational value	Goals 8 and 12

Source Compiled using [9]

formed the basis for classifying stakeholder groups and proposed methods by which management can consider the interests of these groups (Fig. 6).

Freeman [11] defines a stakeholder as “any group or individual who can affect or is affected by the achievement of a company’s objectives” while noting that the categories of stakeholders presented are merely examples. The list of stakeholders is thus formed based on the individuals or groups that cooperate with the business entity.

Additionally, in the paper “Stakeholder Theory: Current Status” by Bihdan et al. (Parmar et al. 2023), the essence of stakeholder theory is clearly defined. When considering the relationships between a company and the individuals (or groups) who can influence its activities or are impacted by them, three key issues are highlighted that can be addressed:

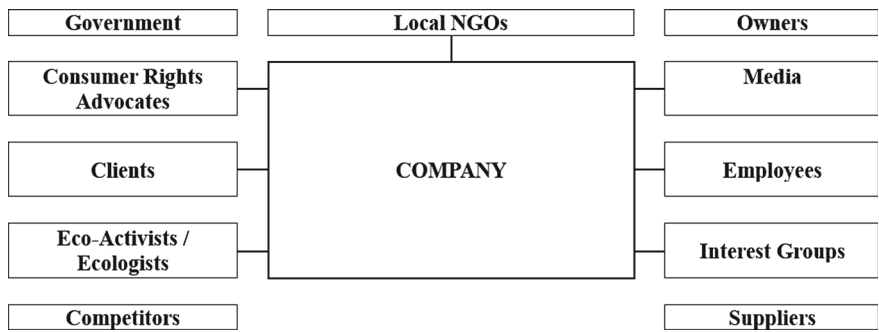


Fig. 6 Stakeholder groups in a company, according to [11]

1. The problem of value creation and trade (in a fast-changing and global business context, how is value created and exchanged?);
2. The problem of capitalism ethics (what links exist between capitalism and ethics?);
3. The problem of managerial thinking (how should managers think about management to better create value and clearly link business and ethics?) [12].

At this point, the relevance of using stakeholder theory to explain the motivation behind the formation of ESG reporting becomes clear. One of the issues that R. Freeman believed could be resolved by applying stakeholder theory is the problem of developing managerial thinking, which involves creating a strategy to combine effective business management with ethical norms. The creation of ESG reporting directly aligns with these requirements, as it allows companies to demonstrate their social responsibility to a wide range of stakeholders—from shareholders and employees to the public and environmental activists. ESG reporting serves as a tool that enhances business transparency and contributes to building sustainable relationships with stakeholders by informing them about the company’s responsible approach to environmental, social, and governance aspects of its operations, as per stakeholder demands.

To better understand the essence of stakeholder theory, it is useful to refer to the scientific study by T. Donaldson and L. Preston, “*The Stakeholder Theory of the Corporation: Concepts, Evidence, and Implications*”—the authors formulated four key propositions explaining the importance of this theory:

1. Proposition 1: The stakeholder theory is undoubtedly descriptive;
2. Proposition 2: The stakeholder theory is also instrumental;
3. Proposition 3: While propositions 1 and 2 are important aspects of stakeholder theory, its core foundation is normative and involves adopting the following ideas: (a) Stakeholders are individuals or groups who have legitimate interests in procedural and/or substantive aspects of corporate activity; (b) The interests of all stakeholders have intrinsic value;
4. Proposition 4: The stakeholder theory is managerial in the term’s broadest sense [13].

The first proposition emphasizes that stakeholder theory describes how a company works by showing it as a system where different stakeholders interact with each other and have varying interests, and the company takes these into account. The key idea is to compare this description with reality, allowing an analysis of whether the company is truly a place where different interests are considered and whether various stakeholders and management perceive the company this way. In practice, one can simultaneously analyze the interests of suppliers (who want stable orders), customers (who seek affordable prices and quality products), and employees (who desire bonuses). Still, working with all these groups is necessary, even if their demands conflict.

The second proposition argues that stakeholder theory can be a tool for understanding whether managing relationships with various stakeholder groups helps achieve better business outcomes. Through stakeholder theory, it is possible to explore whether a company that actively considers the interests of different stakeholders functions more effectively (higher profits, more investors, faster expansion) compared to a business that does not. For example, if a company actively collaborates with its consumers and improves its product based on customer feedback, such collaboration leads to higher product quality and improved customer loyalty, boosting profitability and business stability.

The third proposition focuses on the ethical foundation of stakeholder theory. The interests of each stakeholder deserve attention not only because they may be economically beneficial for the company now or in the future but also because all interests have intrinsic value. For instance, if factory workers request a change in workwear (from thinner fabric), the company's management decides to approve it, knowing that it will increase current costs and reduce profits (since production levels, sales, and prices will not change). From the stakeholder theory perspective, this decision is ethical because the interests of the workers have intrinsic value and are not considered solely for economic benefit to the company.

The fourth proposition summarizes that stakeholder theory is also a management approach. It not only describes how things work but also advises on how to act. It recommends considering the interests of all stakeholders when making decisions and developing policies. This applies to management and other individuals who influence the company's activities, including the government, shareholders, and investors. However, the theory does not suggest that all stakeholders should participate equally in company management; it simply points out that a balance between the interests of different groups needs to be found, and achieving this balance may be challenging. For example, if a company decides to switch to environmentally friendly packaging, it must consider the interests of customers who desire products in safe packaging, as well as government regulators who set environmental standards. Meanwhile, management needs to assess how this change will impact profitability and logistics while also considering the interests of shareholders and suppliers to find a balance between stakeholder interests when making the final decision.

Bashatweh [14] notes that for a long period, stakeholder theory was considered the dominant and most effective framework for explaining the practice of sustainable development reporting. Therefore, based on the above characteristics of stakeholder theory, we can highlight two key aspects: ethical and managerial. The ethical aspect

implies the equal right to fair consideration of the interests of all stakeholders, regardless of their level of influence on the enterprise. The organization's management should consider the opinions and desires not only of specific privileged individuals who influence the management of critical resources of the enterprise [15].

In contrast, the managerial aspect of stakeholder theory asserts that the management of the enterprise must satisfy the interests of those stakeholders who have the greatest impact on the organization's ability to continue its activities [15]. As Deegan [16] notes in his book *Theory of Financial Accounting*, the managerial aspect can often be tested and empirically researched in contrast to the ethical aspect of stakeholder theory. Lawrence and Fernando [15] stated that there is more evidence supporting the managerial aspect of stakeholder theory with regard to sustainable development reporting practices.

Chen and Roberts [17] also mention that stakeholder theory is frequently used in research on sustainable development reporting, providing a foundation for understanding the factors influencing such reporting. Furthermore, in the publication *Comparing Sustainable Development Reporting with the United Nations Sustainable Development Goals* by Raman et al. [9], it is indicated that many scholars have also used stakeholder theory to explain the motivations behind disclosing sustainable development reports. The authors note that stakeholder theory is connected to several SDGs, particularly SDG 8, "Decent Work and Economic Growth," and SDG 12, "Responsible Consumption and Production," as prioritizing stakeholder interests enables companies to improve economic performance, reduce negative environmental impact, and contribute to the achievement of SDGs.

Vaz et al. [18] emphasize that by publishing information about the economic, social, and environmental impact of their activities in the form of sustainable development reporting, companies can demonstrate how they provide certain benefits or advantages to different stakeholder groups, thereby securing their continued support.

Thus, the application of stakeholder theory in the research on ESG reporting formation defines the interests of different stakeholder groups that influence or are influenced by the enterprise as a key factor in developing the company's ESG strategy and forming information in the context of management, environmental, and social aspects of the business entity. An analysis of scientific studies has shown that in practical activities, enterprises primarily focus on the managerial aspect (considering the interests of privileged stakeholders), which does not meet the requirements for adhering to social responsibility norms and achieving sustainable development goals. The consideration of the interests of all stakeholders (the ethical aspect of stakeholder theory) should be mandatory for entities that form ESG reports, as it contributes to creating a more complete picture of the enterprise's interaction with society and the environment, ensuring trust in business activities from all stakeholder groups. Building sustainable relationships with all stakeholders will lead to reduced risks, enhanced reputation, and additional value for the enterprise for all stakeholders.

As summarized in their research "*Stakeholder Theory in the System of Strategic Management*" by Hevko and Rebyrina [19], "the practical value of stakeholder theory can only be felt when the composition of stakeholders is correctly identified, the

balance of interests is determined, and a system for measuring and assessing the mutual impact of various parties is developed”.

The developed systematization of potential enterprise stakeholders (Table 3) demonstrates that different interests can be revealed through environmental, social, and managerial aspects. Further, it allows for structuring ESG reporting according to stakeholder expectations. Establishing a balance between considering all demands, reflecting in the report how these were addressed, and whether they align with the achievement of the company's goals and SDGs significantly influences the decisions of stakeholders regarding cooperation with the enterprise. First of all, such a strategy creates additional value by emphasizing the reliability of the business. For example, investors are more likely to focus on businesses with a positive reputation, as it reduces the risk of negative publicity. Trust from customers results in support or increased demand, ensuring operational activities' stability.

Legitimacy Theory. As early as 1995, Suchman [20] defined legitimacy as “the general perception or assumption that the actions of an entity are desirable, proper, or appropriate within a socially constructed system of norms, values, beliefs, and definitions”.

The legitimacy theory suggests that a company's actions in sustainable development reporting are driven by a desire to fulfill a social contract—to demonstrate that its activities align with moral and societal values. A business must justify its existence through substantiated economic, environmental, and social actions that do not endanger society or the natural environment in which it operates.

As Spence et al. [21] correctly point out, a company's legitimacy and continued existence are at risk when operating against societal values. This is confirmed by the research of Brown and Diggan [22], which suggests that legitimacy theory can reveal the motivation behind creating sustainable development reporting (and ESG reporting in particular). Companies operate within the norms of society, which can evolve over time, requiring businesses to respond to these changes. At the same time, Misak and Derun [23] note that legitimacy theory is often applied in social and environmental accounting studies. It suggests that business disclosures should be made in such a way as to align with the social and environmental expectations of stakeholders who exert political and economic influence on the business.

Schiopoiu Burlea and Popa [24] argue that legitimacy theory is based on stakeholder and institutional theory, “combining traditional norms and values with contemporary ethics”. The authors emphasize that in this theory, stakeholders who exert internal and external pressure create the “legitimate” status for a company operating within the institutional environment. When discussing sustainable development reporting, it is important to note that legitimacy in this context is not simply a formal disclosure of information: the company must not only declare but also truly adhere to moral and social principles in its activities. At the same time, stakeholders must deeply accept the actions of the business.

As Kagan et al. [25] highlight, the power of a business lies in its legitimacy, which, although not a legally binding norm, demonstrates tacit respect for voluntary moral

Table 3 Interests of potential stakeholders of the enterprise, considering which ESG aspects can be disclosed

Stakeholder category	Stakeholder subcategory	Main interests	ESG aspect
Internal stakeholders	Management	Effective management, stable development, strategic planning, rewards	G (Governance)
	Employees	Occupational safety, fair compensation, working conditions, social guarantees, inclusivity	S (Social)
	Shareholders	Profit maximization, dividends, stock value growth, financial transparency	G (Governance)
	Owners, founders	Business idea realization, financial stability, influence on corporate strategy	G (Governance)
External stakeholders	Customers, consumers	Product quality and sustainability, pricing, service level, innovations, regulatory compliance	E (Environmental), S (Social)
	Suppliers and contractors	Order stability, timely payments, long-term cooperation, interdependence	E (Environmental), G (Governance)
	Investors	Investment risks, interest rates, financial stability, revenue growth, return on investment	G (Governance)
	Competitors	Market position, innovations, competition strategy	G (Governance)
	Government and regulatory authorities	Compliance with laws, tax payments, environmental and social standards, regulatory compliance	G (Governance)
	Local communities and social groups	Social responsibility, environmental impact, natural resource protection, local development	E (Environmental), S (Social)

(continued)

Table 3 (continued)

Stakeholder category	Stakeholder subcategory	Main interests	ESG aspect
	Media and public	Transparency, reputation, corporate news, social and environmental initiatives	G (Governance)
	Non-governmental and charitable organizations	Social initiatives, environmental projects, charitable contributions, cooperation on social projects	E (Environmental), S (Social)
	Political parties and movements	Legislative support, socio-economic policy, influence on regulations	G (Governance)
Financial sector stakeholders	Banks and financial institutions	Loans, interest rates, solvency, financial stability	G (Governance)
	Stock exchanges and investment analysts	Reporting transparency, financial indicators, stock stability	G (Governance)
Environmental sector stakeholders	Environmental organizations and activists	Environmental sustainability, impact on the environment, adherence to environmental standards	E (Environmental)
	Environmental regulators and bodies	Compliance with environmental regulations, emission reduction, environmental responsibility	E (Environmental)
Educational sector stakeholders	Universities and research institutions	Joint research, innovations, educational projects, scientific research collaboration	S (Social), G (Governance)
	Educational and professional organizations	Professional development, certification, corporate training	S (Social)
	Students, faculty, and researchers	Collaboration, participation in research projects, innovations	S (Social), G (Governance)

Source Developed by the authors

standards. In particular, this opens the way for regulatory bodies to increase their influence on the business through “empowering various civil society institutions”.

In the process of stakeholders accepting a business as legitimate (one that carries out its activities under the “social contract” and has the right to exist), the company must pass through three levels of legitimacy: cognitive, moral, and pragmatic, which were developed based on the research of M. Suchman [20].

Cognitive legitimacy is the fundamental level characterized by the interest of stakeholders in the company and the recognition of its activities. As the authors point out, reflecting the company’s image in the external environment helps strengthen or restore cognitive legitimacy. This means the company must constantly highlight its values to society and the environment.

Moral legitimacy, like cognitive legitimacy, is formed in the company’s internal environment, but it is related to its actions’ social acceptability and morality. A business demonstrates compliance with the “social contract” and ethical norms, making it important and useful in the eyes of stakeholders by working on projects that strengthen its moral legitimacy.

Pragmatic legitimacy shows the benefit the company brings to its stakeholders. It develops in the external environment where customers, suppliers, investors, the media, etc., evaluate the real benefit the business provides.

In summary, maintaining legitimacy requires developing and implementing short-term projects in response to stakeholder requests, laying the foundation for implementing more long-term sustainable development initiatives.

For example, if a company manufactures electronic devices and, due to a roundtable discussion with representatives of an environmental protection organization and local communities, a request is made to reduce electronic waste in the area where the company operates, the company may propose a short-term project. This project could involve offering significant discounts on its products in exchange for old devices that consumers would recycle. The initiative quickly gains public support since consumers benefit economically (through discounts), and the company demonstrates environmental responsibility by responding to stakeholder requests.

The long-term initiative might involve the company transitioning to a circular economy. Instead of simply disposing of old devices, it would recycle them to reuse materials in new products. As a result, the need for new resources would decrease, waste levels would be reduced, and production costs would fall.

Thus, the short-term project, which solves a specific problem, becomes the basis for a larger, more long-term initiative aimed at environmental preservation. This, in turn, creates a positive image for the company in the eyes of its stakeholders.

The results of such a project will be reflected in the ESG reporting through specific indicators, such as:

- Amount of collected electronic waste (In 2024, the company collected 500 tons of old devices for recycling, which is 20% more than in 2023);

- Reduction in the consumption of new resources (The company reduced spending on new materials by 350,000 UAH by using recycled components in new products);
- Number of customers participating in the program (50,000 consumers participated in the old device exchange program, 30% more than the previous year);
- Number of educational forums for consumers (The company held 20 educational events on social media and 5 seminars for the public on the importance of recycling used electronics);
- Investments in recycling and sustainable development technologies (The company invested 5 million UAH in developing e-waste recycling technologies).

These results showcase the practical impact of the company's actions and how it aligns with sustainability goals in the ESG framework.

Lehencyuk and Serpeninova [8], in describing types of legitimacy, concluded that an accounting system can be used to ensure them as an instrument for informing stakeholders:

- Pragmatic legitimacy—confirmed by determining the economic efficiency of the business entity through financial reporting;
- Moral legitimacy—established by determining the role of the enterprise from the perspective of sustainable development through the publication of integrated, non-financial, and ESG reports;
- Cognitive legitimacy is established through the communication and information policy of the company, which is supported by the overall reporting system.

Dagiļēnē and Nedžinskienē [26] argue that the theory of legitimacy is one of the most widely used theories to explain the disclosure of information on social responsibility and CSR. Additionally, Kuzey and Uyar [27] emphasize that an enterprise can achieve legitimacy through sustainability reports, demonstrating that the business operates within the acceptable boundaries of the “social contract”. Therefore, in the context of the legitimacy theory, the practice of sustainability reporting can be viewed as a strategic attempt by the company to gain an important resource—legitimacy—and ensure the continuity of its operations.

Returning to the scientific work of Raman et al. [9], it is important to note that the legitimacy theory is also relevant to SDG 8, “Decent Work and Economic Growth,” and SDG 12, “Responsible Consumption and Production,” as it highlights the importance of organizations demonstrating their commitment to sustainable development and social responsibility to stakeholders.

In the context of ESG reporting, the legitimacy of a company is maintained through transparent disclosure of its environmental, social, and governance practices, which meet society's expectations (Table 4). It is important to note that a key factor in maintaining the status of a legitimate business entity is the real demonstration of compliance with its claims through the disclosure of relevant information in ESG reports.

Institutional Theory. According to the book “New Institutionalism in Organizational Analysis” by Powell and DiMaggio [28], businesses were initially identified

Table 4 Life cycle of legitimacy in the context of ESG reporting

Type of legitimacy	Description	Application in ESG reporting
Cognitive	Recognition of the enterprise's activities by society	Disclosing ESG aspects helps stakeholders gain information about the business's activities and response to societal challenges
Moral	Acceptance of social and ethical norms by the enterprise	ESG reporting demonstrates compliance with moral standards, ethical norms, and social responsibility practices
Pragmatic	Assessment of the benefits of the enterprise by stakeholders	ESG reporting highlights the real benefits for stakeholders: projects to reduce environmental impact, implemented social support measures, and transparency in corporate governance

Source *Developed by the authors*

as closed systems with no ties to the institutional environment, relying solely on themselves. However, in the 1960s–1970s, the importance of the institutional environment for business entities was recognized, and institutional theory became an important tool for studying phenomena occurring during business operations [29].

The development of institutional theory (in its modern form) is linked to the research of Meyer and Rowan [30], Powell and DiMaggio [28], which emphasized the homogeneity of organizational practices: practices that arise and develop in specific social or institutional spheres begin to resemble each other over time because businesses strive to gain legitimacy, i.e., to meet the expectations of society or the environment in which they operate. In this way, businesses achieve high stability by employing acceptable, effective practices approved by the institutional and social environment.

As Arman and Haniffa [31] note, the essence of institutional theory is that businesses will adjust their operational strategy to achieve legitimacy in the context of the broader environment in which they function. Therefore, legitimacy is crucial for understanding and applying institutional theory, as positioned by legitimacy theory.

The desire for such homogenization led researchers to the concept of institutional isomorphism. Meyer and Rowan [30], in their work “Institutionalized Organizations: Formal Structure as Myth and Ceremony,” explored and established that businesses create formal structures that align with institutional norms and expectations but do not always enhance the economic efficiency of the business entity. This phenomenon is compared to M. Weber’s theory of the “iron cage”—businesses are forced to comply with a unified set of norms and standards, even if this does not promote their own goals and leads to increased bureaucracy.

In continuation of the development of institutional theory, a study titled “The Iron Cage Revisited: Institutional Isomorphism and Collective Rationality in Organizational Fields” [32] was published, which for the first time described the concept of institutional isomorphism and identified three types: coercive (pressure from the state and other regulators), normative (influence of professional standards), and

mimetic (imitation of other organizations). Based on the scholarly work of Powell and DiMaggio [32], a brief description of the types of institutional isomorphism was formulated (Table 5), establishing that coercive isomorphism manifests through business compliance with current legislative requirements, mimetic through the adoption of best practices from successful competitors and/or partners, and normative through adherence to professional standards. Therefore, business entities adapt their strategies (including those regarding disclosing ESG aspects) according to external pressures to maintain legitimacy.

The analysis of sustainable development reporting practices through institutional theory suggests explaining the influences on such reporting by assuming that businesses (to some extent) possess at least one form of institutional isomorphism. For example, Martínez-Ferrero and García-Sánchez [33] provides an example where mimetic isomorphism regarding reporting is evident when leading companies in the industry implement the practice of producing integrated, non-financial, or ESG reports, and other business entities in the industry generally follow their example. The intention to join best practices can explain the desire for such imitation. Still, the problem is that for high-quality ESG or other sustainable development reporting, businesses must first develop and implement strategies to achieve goals in addressing social and environmental issues. Otherwise, mere “copying” of reporting practices is meaningless. It can negatively affect stakeholders’ trust in these companies once discrepancies between the report (which describes the “beautiful picture” of achieving sustainable development goals) and reality (where no actual projects or tasks related to such achievement are undertaken) are revealed.

Table 5 Characteristics of types of institutional isomorphism

Type of isomorphism	Description	Example
Coercive	Arises under the influence of formal or informal pressure from external institutions, often with a mandatory nature due to legal or legislative requirements	The general data protection regulation (GDPR) in the EU requires all businesses to comply with personal data protection standards, regardless of the industry in which the business operates
Normative	Arises under the influence of professional requirements or standards imposed by external forces, expecting “proper” organizational behavior	Auditing firms must adhere to international auditing standards as defined by professional organizations
Mimetic	Occurs when businesses adopt the structure or practices of other successful business entities to solve their own problems or gain legitimacy; these situations often occur when companies face an unpredictable environment without clear instructions or visible solutions	If one business successfully implements a new production technology, other businesses may begin to copy it to achieve similar success, even if it is not ideally suited to their operations

Source Developed by the authors

If we talk about an example of normative isomorphism in sustainable development reporting, attention should be drawn to companies reporting according to GRI standards. These business representatives are, therefore, obliged to follow the normative requirements of reporting on social, environmental, and governance impact as prescribed in GRI standards. Since most large companies use these standards, other businesses begin to do the same to comply with professional norms and maintain legitimacy.

A good example of coercive isomorphism is provided in the study by Vaz et al. [18]. In South Africa, the publication of an integrated report is mandatory for listing on the Johannesburg Stock Exchange. In general, the mechanism of coercive isomorphism closely intersects with stakeholder theory, as businesses can use sustainable development reports to meet the expectations of their most influential stakeholders [31].

The service “Twenty Now,” which highlights the latest ESG trends and sustainable development initiatives, published a study on isomorphism in ESG and sustainable development reports (Table 6) [34].

It has been established that most companies follow common practices without attempting to stand out or introduce anything new, which is reflected in the standard approaches to reporting. This behavior is best explained by mimetic isomorphism, where companies copy each other to achieve legitimacy, avoiding risks and not seeking innovation or a competitive advantage.

Institutional theory is consistently applied in studies examining businesses’ implementation of sustainable development reporting practices [35]. In summary, this theory focuses on how institutionalized behavior becomes an inherent part of a business’s operations. Institutional theory emphasizes the role of external factors in shaping business reporting practices. It is significant for SDG 8, “Decent Work and Economic Growth,” SDG 9, “Industry, Innovation, and Infrastructure,” and SDG 12, “Responsible Consumption and Production” [9].

Therefore, according to each theory, we can outline a list of influences on ESG reporting. According to the theories studied (Table 7), it can be established that “ESG reporting” is one of the means of achieving and maintaining business legitimacy, which includes highlighting the consideration of stakeholder requests and the pressure from the institutional environment in the reports. Furthermore, the continuous improvement of such reporting practices occurs at the request of society and stakeholders, considering the experience of ESG reporting by leading industry players.

In practice, the principles of stakeholder theory, legitimacy theory, and institutional theory can be integrated into a system of criteria (Table 8), which a working group can apply to assess how well the ESG reporting process is formed and conducted from the perspective of meeting stakeholder demands, maintaining legitimacy, and addressing institutional pressure.

This evaluation can be conducted in five stages:

Table 6 Results of the search for signs of isomorphism in ESG reports and sustainable development reports [34]

Element of the report	Analysis results
Title	The majority of reports contained the term “Sustainable development” in the title (31%) or “Environmental, Social, Governance”/“ESG” (33%); among other less common titles, “Corporate responsibility” (14%) and “Corporate social responsibility” (7%) were noted
Size	The reports ranged from 11 to 273 pages, with an average length of 77 pages
Data table	Over 70% of the reports contained a table with key ESG data
Press release	62% of surveyed companies issued a press release announcing their report
Time	Most companies published their reports within 4–6 months after the end of the financial year, with April being the most popular month (39%), followed by March (18%) and June (17%)
Format	93% of the reports were available in PDF format, and about 27% of companies included interactive landing pages about ESG on their websites
Letter from the CEO or board	89% of the reports included a letter from the CEO, with 10% also including a letter from the board
internal oversight	about 72% of the reports specified that the responsibility for sustainable development was assigned to senior management
Materiality assessment	In 50% of the reports, it was stated that an official materiality assessment had been conducted to help prioritize ESG issues; 40.5% also included an actual materiality matrix, and 34% included disclaimers or qualifiers regarding the use of the term “materiality”
Diversity, equity, and inclusion	47% of the reports published goals for diversity
Employee demographics	91% of the reports disclosed employee demographic data, with 85% of the diversity data focused on the management/senior level
Gender pay gap analysis	About 30% of the reports included information on gender pay gaps among employees
Use of external assurance	54% of companies stated that the data in their reports were externally assured, with 51% providing only environmental data, nearly 43% assuring both environmental and social data, 5% assuring only social data, and less than 4% not specifying which ESG data were assured

Source Developed by the authors

1. Identifying the business stakeholders, determining their interests and demands (including which ESG aspects are most important to them), and assessing the justification for their inclusion or exclusion.
2. Assessing the level of legitimacy of the business, specifically whether the actions of the business align with societal expectations in terms of ethical and social norms and how effectively this is reflected in the reporting.

Table 7 Influencing factors on ESG reporting according to stakeholder, legitimacy, and institutional theories

Theory	Essence of the theory	Role of ESG reporting according to the theory	Identified influencing factors on ESG reporting
Stakeholder theory	The functioning of a business is adjusted according to the demands and interests of its stakeholders	ESG reporting is a communication tool with stakeholders, which should reflect the company's response to their requests or information regarding the consideration of their interests in business operations	Interests of internal and external stakeholders
Legitimacy theory	To maintain legitimacy, a business adapts its activities in line with the fulfillment of a "social contract"	ESG reporting becomes part of fulfilling the "social contract," which involves demonstrating adherence to moral and social norms	Expectations and demands of society, social and ethical norms
Institutional theory	Through isomorphism, businesses adapt their practices to meet the requirements of external institutions and other businesses	The formation of ESG reporting may be accompanied by adopting reporting practices from market leaders, and it also complies with professional and legal standards	External pressure (legislative and legal), best practices of competitors, professional requirements

Source Developed by the authors

- Investigating how regulatory and institutional factors influence the business's ESG reporting, including whether a high degree of mimetic isomorphism is observed in the reports and whether this is justified; what standards or frameworks the business uses for ESG reporting, and whether their provisions are considered; whether current legislation that directly or indirectly pertains to ESG aspects is considered and disclosed in the reports.
- Evaluating the impact of ESG reporting on key performance indicators: reputation, risk resilience, and investment attractiveness.
- Formulating recommendations for further improvement.

The criteria system outlined in Table 8 can also be applied by external auditors, industry experts, or specially hired specialists in sustainable development reporting. Based on the evaluation results, they can provide recommendations regarding:

- modernizing the ESG strategy (e.g., if the business's legitimacy is declining or if stakeholder demands are not being met without substantial justification);
- improving the ESG report (for example, if the "social contract" is being adhered to, stakeholders' views are considered, and there is a healthy isomorphism, but this is not adequately reflected in the report).

Table 8 Criteria for evaluating the ESG reporting process

Criterion	Description	Procedures within the criterion	Possible indicators for evaluation
Interests and requests of stakeholders	Based on stakeholder theory, this involves analyzing how the company addresses the interests of internal and external stakeholders in ESG reporting	Based on past and current stakeholder requests to the company:—Identify key stakeholders;—Evaluate the level of satisfaction with how their requests are addressed;—Highlight ecological, social, or governance aspects most relevant to each stakeholder group	Is there a stakeholder survey form? Does the ESG report provide information on how stakeholders' requests are considered or rejected? Is the information important for each stakeholder group transparently disclosed? Is there a feedback mechanism for stakeholders?
Legitimacy and the fulfillment of the “Social contract”	Based on legitimacy theory, this aims to assess how the company uses ESG reporting to maintain legitimacy with society	Through online conferences, communication on social media, and analysis of public sources:—Analyze the fulfillment of the “social contract,” i.e., how well the company meets societal expectations;—Study changes in the external environment regarding emerging social and environmental trends;—Assess the achievement of all levels of legitimacy (e.g., is society informed about the company's activities, does it benefit society, and does it meet ethical norms)	The number and nature of environmental, social, and governance initiatives; presence in reports of information demonstrating compliance with ethical standards (e.g., anti-corruption, inclusivity); presence of analysis regarding changes in the company's reputation (based on surveys or media analysis)
Institutional pressure and isomorphism	Based on institutional theory, this involves evaluating the company's adaptation to the institutional environment through ESG reporting	Based on monitoring updates in legislative and regulatory acts and reports of industry leaders:—Determine the pressure level from professional organizations, standards, and regulatory bodies;—Assess which best practices in reporting are used by companies and whether they should be adopted	Degree of compliance with standards (e.g., GRI, CDP, SASB); presence of internal and external ESG report audits; frequency of using best practices from competitors

Source Developed by the authors

4 Conclusion

The concept of sustainable development gradually developed in the mid-twentieth century under the influence of social movements and the devastating environmental damage caused by human activities. In response to these new challenges, numerous scholars in sociology, economics, philosophy, and ecology—both domestic and international—examined the prospects for the survival and development of humanity until these issues began to be addressed at the international level. In 1987, in the United Nations report *Our Common Future*, the concept of “sustainable development” was first formulated as the ability to meet the current needs of society without compromising the ability of future generations to meet their own needs. Over time, the global community concluded that there was an objective need to define the general goals of sustainable development—key development directions for countries, which were adopted at the UN Sustainable Development Summit. Since then, businesses have been required to publicly inform about their efforts towards achieving the SDGs, the level of impact their activities have on society, the economy, and the environment, as well as their policies to mitigate these impacts.

ESG factors play a central role in sustainable development as a tool for demonstrating the contribution to achieving the SDGs to stakeholders. Moreover, some differences have been identified between classical sustainability reporting and ESG reporting, which allowed the formulation of the definition of “ESG reporting” as a type of non-financial reporting within the field of sustainable development, which is a subset of the sustainability report. It is created to provide stakeholders with information regarding the non-financial aspects of the company’s operations by highlighting clearly defined indicators related to environmental risks, social responsibility, and corporate governance, primarily for the present time.

Preparing ESG reports should be oriented not only toward legal requirements but also toward existing theoretical concepts that allow for identifying key guidelines for sustainability reporting. The main provisions of three theories—stakeholder theory, legitimacy theory, and institutional theory—were analyzed within the research. By studying the works of scholars from the twentieth and twenty-first centuries, it was established that the key factors influencing the ESG reporting process are the requests of internal and external stakeholders, societal expectations and demands, social and ethical norms, legislative and legal pressures, best practices in report preparation by competitors, and professional requirements. Since the concept of “ESG reporting” is not legally enshrined and has not been extensively researched in the works of foreign and Ukrainian scholars, it is proposed to define it as one of the means of achieving and maintaining the legitimacy of an enterprise based on the presentation of information in reports, taking into account the requests of stakeholders and the pressure of the institutional environment.

In this regard, a system of criteria for assessing the ESG reporting process has been developed and proposed for use, describing the criteria for evaluation, their characteristics, possible indicators, and evaluation procedures within each criterion. It

is believed that applying these criteria will ensure a systematic approach to identifying areas for improvement in the ESG strategy and ESG reports of businesses.

Acknowledgements The research was conducted with the support of the Ministry of Education and Science of Ukraine within the framework of the project “Development of a Mechanism for the Sustainable Development of Economic Systems in the Conditions of Military Actions and Post-War Economic Recovery” (registration number 0124U000463).

References

1. On the approval of the strategy for implementing sustainability reporting by enterprises: resolution of the cabinet of ministers of Ukraine No. 1015-r, dated October 18, 2024. <https://zakon.rada.gov.ua/laws/show/1015-2024-p#Text>
2. Draft regulatory legal acts for 2025. Ministry of Finance of Ukraine. https://mof.gov.ua/uk/draft_normative_legal_acts_in_2025-821
3. Transforming our world: the 2030 agenda for sustainable development. UNDP. <https://www.undp.org/uk/ukraine/publications/peretvorennnya-nashoho-svitu-poryadok-dennyy-u-sferi-staloho-rozvytku-do-2030-roku>
4. Lai A, Stacchezzini R (2021) Organizational and professional challenges amid the evolution of sustainability reporting: a theoretical framework and an agenda for future research. *Meditari Accountancy Res* 29(3):405–429. <https://doi.org/10.1108/MEDAR-02-2021-1199>
5. ESG, CSR, sustainability reporting—what’s the difference? Connect Earth. <https://connect.earth/esg-csr-sustainability-reporting-whats-the-difference/>
6. Who cares wins 2005 conference report: investing for long-term value. IFC. <https://www.ifc.org/en/insights-reports/2000/publications-report-whocareswins2005--wci--1319576590784>
7. Lehenchuk SF, Oliinyk OV, Zavalii TO (2024) Institutional signaling with the help of ESG reporting. *Probl Theory Methodol Account, Control Anal* 2(58):24–31. [https://doi.org/10.26642/pbo-2024-2\(58\)-24-31](https://doi.org/10.26642/pbo-2024-2(58)-24-31)
8. Lehenchuk SF, Serpeninova YS (2024) The legitimacy theory of additional disclosure of accounting information on corporate social and environmental activities. *Econ, Manage, Adm* 1(107):103–110. [https://doi.org/10.26642/ema-2024-1\(107\)-103-110](https://doi.org/10.26642/ema-2024-1(107)-103-110)
9. Raman R, Kumar NV, Shivdas A, Bhukya R, Viswanathan PK, Subramaniam N, Nedung P (2023) Mapping sustainability reporting research with the UN’s sustainable development goal. *Heliyon* 9(8):1–24. <https://doi.org/10.1016/j.heliyon.2023.e18510>
10. Christensen HB, Hail L, Leuz C (2021) Mandatory CSR and sustainability reporting: economic analysis and literature review. *Rev Account Stud* 6:1176–1248. <https://doi.org/10.1007/s11142-021-09609-5>
11. Freeman RE (2010) *Strategic management: a stakeholder approach* (illustrated, reprint). Cambridge University Press. <https://doi.org/10.1017/CBO9781139192675>
12. Parmar BL, Freeman RE, Harrison JH, Wicks AC, Purnell L, De Colle S (2010) Stakeholder theory: the state of the art. *Acad Manage Ann* 4(1):403–445. <https://doi.org/10.1080/19416520.2010.495581>
13. Donaldson T, Preston LE (1995) The stakeholder theory of the corporation: concepts, evidence, and implications. *Acad Manage Rev* 20(1):65. <https://doi.org/10.2307/258887>
14. Bashatweh ADM (2018) Accounting theory and its impact on adopting sustainability reporting dimensions—a field study. *Int J Account Financ Rep* 8(4):82–99. <https://doi.org/10.5296/ijافر.v8i4.13736>
15. Fernando S, Lawrence S (2014) A theoretical framework for CSR practices: integrating legitimacy theory, stakeholder theory, and institutional theory. *J Theor Account* 10:149–178.

- https://www.researchgate.net/publication/290485216_A_theoretical_framework_for_CSR_practices_Integrating_legitimacy_theory_stakeholder_theory_and_institutional_theory
16. Deegan C (2014) Financial accounting theory. McGraw-Hill education Australia. https://books.google.com.ua/books/about/Financial_Accounting_Theory.html?id=zH-_BQAAQBAJ&redir_esc=y
 17. Chen JC, Roberts RW (2010) Toward a more coherent understanding of the organization–society relationship: a theoretical consideration for social and environmental accounting research. *J Bus Ethics* 97(4):651–665. <https://doi.org/10.1007/s10551-010-0531-0>
 18. Vaz N, Fernandez-Feijoo B, Ruiz S (2016) Integrated reporting: an international overview. *Bus Ethics: Eur Rev* 25(4):577–591. <https://doi.org/10.1111/beer.12125>
 19. Hevko V, Rebrina N (2012) The stakeholder theory in the strategic management system. In: Proceedings of the all-Ukrainian scientific and practical conference in memory of the honored professor of TNTU, Academician of the national academy of sciences of Ukraine Mykola Hryhorovych Chumachenko, TNTU named after I. Pulyuy. https://elartu.tntu.edu.ua/bitstream/123456789/7789/2/Conf_2012_Hevko_V-The_stakeholder_theory_in_the_59-60.pdf
 20. Suchman MC (1995) Managing legitimacy: Strategic and institutional approaches. *Acad Manage Rev* 20(3):571–610. <https://doi.org/10.5465/amr.1995.9508080331>
 21. Spence C, Husillos J, Correa-Ruiz C (2010) Cargo cult science and the death of politics: a critical review of social and environmental accounting research. *Crit Perspect Account* 21(1):76–89. <https://doi.org/10.1016/j.cpa.2008.09.008>
 22. Brown N, Deegan C (1998) The public disclosure of environmental performance information—a dual test of media agenda setting theory and legitimacy theory. *Account Bus Res* 29(1):21–41. <https://doi.org/10.1080/00014788.1998.9729564>
 23. Mysaka H, Derun I (2021) Legitimacy theory in the formation of the institutional and legal basis of the companies non-financial reporting system. *Soc Econ* 61:60–71. <https://doi.org/10.26565/2524-2547-2021-61-06>
 24. Schiopoiu Burlea A, Popa I (2013) Legitimacy theory. *Encycl Corp Soc Responsib* 1579–1584. https://doi.org/10.1007/978-3-642-28036-8_471
 25. Gunningham N, Kagan RA, Thornton D (2004) Social license and environmental protection: why businesses go beyond compliance. *Law Soc Inquiry* 29(2):307–341. <https://doi.org/10.1111/j.1747-4469.2004.tb00338.x>
 26. Dagilienė L, Nedzinskienė R (2018) An institutional theory perspective on non-financial reporting. *J Fin Rep Account* 16(4):490–521. <https://doi.org/10.1108/jfra-06-2016-0054>
 27. Kuzey C, Uyar A (2017) Determinants of sustainability reporting and its impact on firm value: evidence from the emerging market of Turkey. *J Cleaner Prod* 143:27–39. <https://doi.org/10.1016/j.jclepro.2016.12.153>
 28. Powell WW, DiMaggio PJ (eds) (1991) The new institutionalism in organizational analysis. University of Chicago Press, p 478
 29. Major MJ, Ribeiro J (2009) A teoria institucional na investigação em contabilidade. In: Major MJ, Vieira R (eds) *Contabilidade e Controlo de Gestão. Teoria, Metodologia e Prática*, pp 37–59
 30. Meyer JW, Rowan B (1977) Institutionalized organizations: formal structure as myth and ceremony. *Am J Sociol* 83(2):340–363. <https://doi.org/10.1086/226550>
 31. Amran A, Haniffa R (2011) Evidence in development of sustainability reporting: a case of a developing country. *Bus Strategy Environ* 20(3):141–156. <https://doi.org/10.1002/bse.672>
 32. DiMaggio P, Powell WW (2010) The iron cage revisited: institutional isomorphism and collective rationality in organizational fields (transl by Yudin G). *J Econ Sociol* 11(1):34–56. <https://doi.org/10.17323/1726-3247-2010-1-34-56>
 33. Martínez-Ferrero J, García-Sánchez I-M (2017) Coercive, normative and mimetic isomorphism as determinants of the voluntary assurance of sustainability reports. *Int Bus Rev* 26(1):102–118. <https://doi.org/10.1016/j.ibusrev.2016.05.009>
 34. Isomorphism in ESG and sustainability reporting—twenty now. *Twenty Now*. (2022) <https://www.twentynow.com/sustainability-initiatives/governance/isomorphism-in-esg-and-sustainability-reporting/>

35. Katsikas E, Manes Rossi F, Orelli RL (2017) Towards integrated reporting. Springer International Publishing. <https://doi.org/10.1007/978-3-319-47235-5>

Strategies for the Development of Responsible Tourism to Support a Sustainable Tourism Ecosystem



Iryna Kyryliuk , Viktoriia Krasnomovets , Olena Sushchenko ,
Oksana Lytvyn , Liudmyla Chvertko , and Liudmyla Neshchadym

Abstract In the context of global environmental change, there is a growing concern about the impact of traditional tourism on ecosystems, so the search for responsible tourism practices aimed at minimizing environmental damage, supporting local communities, and promoting social responsibility are key aspects of maintaining a sustainable tourism ecosystem. The purpose of the study is to identify current trends in the development of responsible tourism and assess the effectiveness of its impact on the preservation of a sustainable tourism ecosystem. The object of the study is the process of identifying the features of responsible tourism and determining its impact on the balanced economic, social and environmental development of tourism activities. The study outlines the conceptual framework of responsible tourism, analyzes its key principles and the relationship with ecological and regenerative tourism. The

I. Kyryliuk (✉) · O. Lytvyn

Department of Hotel, Restaurant, and Tourism Business, Kherson State Agrarian and Economic University, Kropyvnytskyi, Ukraine
e-mail: i.kyryluk@udpu.edu.ua

O. Lytvyn

e-mail: okslit79@ukr.net

V. Krasnomovets

Department of Tourism and Hotel and Restaurant Business, Bohdan Khmelnytsky Cherkasy National University, Cherkasy, Ukraine
e-mail: krasnomovets_v_a@ukr.net

O. Sushchenko

Department of Tourism and Hotel and Restaurant Business, Kyiv National University of Technologies and Design, Kyiv, Ukraine

L. Chvertko

Department of Finance, Accounting and Economic Security, Pavlo Tychyna Uman State Pedagogical University, Uman, Ukraine
e-mail: luda_um@ukr.net

L. Neshchadym

Department Technologies and Organization of Tourism and Hotel and Restaurant Business, Pavlo Tychyna Uman State Pedagogical University, Uman, Ukraine
e-mail: n_lydmila@ukr.net

main trends in the development of responsible tourism and its impact on environmental conservation and support for local communities, the role of environmental awareness in the development of ecotourism are identified. It is proved that the introduction of responsible practices in tourism activities contributes not only to minimizing the negative impact on the environment, but also to the active restoration of ecosystems, support for local communities, responsible use of resources, and the formation of knowledge for a conscious attitude towards the environment. The effectiveness of the regenerative tourism concept depends on systemic changes in the functioning and management of tourism and cooperation with all stakeholders. The implementation of European Union initiatives and financing mechanisms as part of a comprehensive strategy to promote responsible tourism practices for the preservation and restoration of the natural environment is considered. A mechanism for implementing a responsible tourism development strategy has been developed, the application of which contributes to the formation of environmental and social awareness of tourists to form a sustainable environmentally balanced and socially oriented ecosystem. Effective management and a strategic approach will make tourism an important tool not only for economic growth, but also for preserving the planet and improving human well-being.

Keywords Responsible tourism · Sustainable tourism · Sustainable development · Regenerative tourism · Ecological tourism · Ecosystem

1 Introduction

1.1 *Relevance*

In today's environment, globalization processes expand access to international markets, stimulate economic growth, and increase the level of international cooperation and competitiveness of countries. At the same time, globalization creates new challenges, such as increased social inequality, growing environmental problems, and the need for ethical approaches to address global issues. Unsustainable consumption and production patterns have led to a triple planetary crisis: climate change, biodiversity loss, and environmental pollution. It is clear that the transformation of three interdependent systems—energy, food and waste—is critical. Activities aimed at achieving sustainable development goals are a prerequisite for ensuring rational use of natural resources and balancing economic needs with environmental opportunities.

Tourism is one of the most globalized and dynamic sectors of the economy, so it has significant potential for sustainable development through the development of environmental initiatives, preservation of cultural heritage and support for innovative practices in the tourism industry. However, tourism activities often put additional pressure on natural resources, leading to ecosystem degradation, environmental pollution, and social imbalances. A report by the World Tourism Organization states

that by 2030, carbon dioxide emissions from transport used in the tourism industry will account for 5.3% of total anthropogenic emissions [1]. This situation necessitates cooperation of the tourism industry with the transport industry and government associations to support commitments to accelerate the decarbonization process and implement environmental sustainability. There is a growing awareness of the negative impact of tourism on the environment, which encourages all tourism market participants to develop a responsible attitude to the tourism process and implement approaches that reduce the negative impact on the environment and contribute to the preservation of natural and cultural resources. Effective management and a strategic approach will make tourism an important tool not only for economic growth, but also for preserving the planet and improving human well-being.

Numerous publications by scholars are devoted to the problems of transforming the tourism industry into an effective factor of socio-economic development of a country or region through modernization with due regard for environmental requirements and in accordance with the principles of sustainable development. In particular, N. Karpenko considers sustainable tourism development as meeting the current needs of tourists and the regions that host them, while ensuring the economic growth of the tourism business, protection and enhancement of natural, recreational, historical and cultural resources and preservation of such opportunities for the future [2]. According to M. Osypchuk, sustainable tourism development is focused on the long-term period, during which it is possible to achieve a balance in the realization of economic, environmental, social and cultural development goals, taking into account the interests of all stakeholders (tourists, host and generating destinations, local population), based on the rational use of tourism potential and multilateral partnerships [3]. In a study by foreign scholars, sustainable tourism is defined as a form of tourism that promotes long-term resource management to meet economic and social needs, maintain cultural integrity and characteristic ecological processes [4]. Sustainable tourism is tourism that respects both the local population and the traveler, cultural heritage and the environment [5]. The concept of tourism sustainability, from the point of view of C. Bizzarri, G. Querini, has a double meaning: on the one hand, ecological, in terms of maintaining a balance between the components of the natural environment; and on the other—anthropological, in terms of the experience of using the environment [6].

Ecotourism is an important tool for minimizing the negative impact on ecosystems and developing responsible and sustainable practices. The International Ecotourism Society (TIES) and the Global Ecotourism Network (GEN) define ecotourism as “responsible travel to natural areas that preserves the environment, supports the well-being of local people, and provides interpretation and education for all parties involved” [7, 8]. According to O. Savitska, N. Savitska, and L. Pohrebniak, ecotourism, the main idea of which is to harmonize human relations with the environment, care for the preservation of the environment used for tourism purposes, and increase the environmental awareness of society, is a popular and promising area of the tourism industry [9]. C. Bardash and D. Chernysh note that a type of tourism with minimal interference in the natural environment, with optimal use of its exhaustive resources, which contributes to the sustainable development of the region, can

be identified as ecological. The ecological component, which is understood as the desire not to harm the environment where the recreational and cognitive needs of tourists are met, should be present in any type of tourism [10]. The development of environmental initiatives, preservation of cultural heritage and support for innovative practices in the tourism industry contributes to the achievement of sustainable development goals.

The analysis of research by foreign scholars allows us to better understand the multidimensional nature of the concept of responsible tourism. In particular, H. Goodwin, one of the leading researchers in this area, notes that responsible tourism is tourism that minimizes negative impacts on the environment, economy and culture, while enhancing the well-being of local communities and providing a more authentic experience for tourists. In his view, responsible tourism differs from sustainable tourism in that it emphasizes concrete actions by people, businesses and governments to achieve sustainable tourism development by maximizing positive economic, social and environmental impacts [11].

These approaches are reflected in the Cape Town Declaration, according to which responsible tourism:

- minimizes negative economic, environmental and social impacts;
- creates economic benefits for local people and improves the well-being of host communities;
- involves local people in decision-making processes;
- contributes to the preservation of natural and cultural heritage;
- provides a deeper cultural experience for tourists;
- takes into account the needs of people with disabilities;
- promotes respect, understanding and strengthening of local identity [12].

Although Emil Durkheim did not directly study tourism, his ideas of social responsibility and community cohesion have been adapted by modern scholars to the concept of responsible tourism as a tool for social enrichment based on mutual respect between tourists and locals [13].

Modern scholars, in particular R. Fletcher, I. Murray Mas, A. Dianco-Romero, M. Blazquez-Salom, emphasize the need for a systematic approach. In their opinion, responsible tourism should not be a separate area, but an integrated part of the overall strategy for sustainable development of territories where tourists are not only consumers, but also active participants in environmental, social and cultural transformations [14].

1.2 Goals and Objectives

Despite the significant attention of scholars and practitioners to this problem and the large volume of publications and research on this topic, the issue of responsible tourism development remains relevant and requires further study and consideration of modern practices in responsible tourism management to ensure the sustainable

development of the industry. The purpose of the study is to analyze current trends in the development of responsible tourism and assess the effectiveness of its impact on the preservation of a sustainable tourism ecosystem. The object of the study is the process of identifying the features of responsible tourism and determining its impact on the balanced economic, social and environmental development of tourism activities. The analysis of the role of responsible tourism in maintaining a balance between the economic, social and environmental aspects of tourism activities is aimed at assessing how responsible tourism can mitigate environmental damage, support local communities and promote social responsibility.

Focusing on systemic approaches to responsible tourism development can help both tourism researchers and managers and other tourism market participants to gain a better understanding of specific opportunities for consciously adopting responsible practices to maintain a sustainable tourism ecosystem.

This section is organized as follows. First, we provide an overview of the conceptual framework of responsible tourism, analyzing its key principles and the relationship with ecological and regenerative tourism. Next, we look at the main trends in the development of responsible tourism and its impact on environmental conservation and support for local communities, explore the role of environmental awareness in the development of ecotourism, analyze European policies and practices of responsible tourism aimed at preserving natural and cultural resources, and review European Union initiatives and funding mechanisms as part of a comprehensive strategy to promote responsible tourism practices.

As a further contribution, a mechanism for implementing a strategy for the development of responsible tourism has been developed, the application of which contributes to the formation of environmental and social awareness of tourists to create a sustainable environmentally balanced and socially oriented ecosystem.

The study used dialectical and abstract logical methods, methods of systematic approach, comparison, observation, analysis, synthesis, induction, deduction, etc.

2 Theoretical and Conceptual Background

2.1 Tourism Ecosystem Structure

The tourism ecosystem encompasses globalized and interconnected value chains that include information and service providers (tourist offices, digital platforms, travel technology providers), travel agents and tour operators, accommodation providers, destination management organizations, attractions and passenger transportation, research and government agencies [15].

The heterogeneity of the tourism ecosystem requires diversified and tailored support at the global level and a shared vision that promotes concerted action by all stakeholders. Ensuring the effective functioning of the tourism ecosystem requires a balanced combination of the interests and needs of visitors, local residents and

the environment. The transformation required for the green and digital transition requires taking into account trends and changes in both tourism demand and supply. It is important to take into account the needs and characteristics of each region and develop strategic approaches that will effectively integrate the principles of sustainable development into tourism activities. The active participation of governments, businesses, local communities and tourists is key to creating responsible tourism practices that meet today's challenges and help preserve natural and cultural resources for future generations.

In the context of global environmental change, there is growing concern about the impact of traditional tourism on ecosystems, and tourists are increasingly looking for more sustainable and responsible alternatives for their travels. Responsible tourism is able to strike a balance between the economic, social and environmental aspects of tourism activities. Aiming to minimize environmental damage, support local communities, and promote social responsibility are key aspects of responsible tourism development.

2.2 Comparative Analysis: Sustainable, Responsible, Ecological, and Regenerative Tourism

Today, a responsible approach to tourism is one of the key strategies for ensuring the sustainable development of the industry. Responsible tourism involves taking into account environmental, economic and social aspects, which helps to reduce the negative impact on the environment and preserve resources for future generations. The active participation of tourists in environmental conservation through educational initiatives, promotion and motivation to consciously consume resources while traveling helps to minimize pollution, preserve biodiversity, support local communities and develop cultural heritage.

In the modern discourse of sustainable tourism development, the differences between four related concepts are increasingly being considered: responsible, ecological, sustainable, and regenerative tourism. Responsible tourism focuses on minimizing the negative impact on the environment and society, as well as on the active involvement of local communities in tourism activities. It involves conscious decision-making by all participants in the tourism process: tourists, businesses, and state authorities.

Sustainable tourism is a basic concept that encompasses the balanced development of the tourism industry, taking into account economic, socio-cultural and environmental aspects. Its goal is to meet the needs of current generations of tourists and host communities without compromising future generations. Sustainable tourism involves the rational use of resources, respect for cultural heritage, and preservation of ecosystems.

Ecotourism, in turn, is a subtype of responsible tourism and is aimed primarily at protecting nature, preserving biodiversity and ecosystems. The main characteristics

of this type of tourism are visiting natural areas with minimal interference in the environment and raising environmental awareness among tourists.

Regenerative tourism is an emerging concept that goes beyond mere damage minimization. It aims not only to preserve but also to restore the natural and social environment in which tourism activities take place. It is an approach where travel contributes to the improvement of the environment, economy, and culture, leaving the space in a better condition than before the arrival of the tourist.

Thus, while sustainable tourism is a strategic framework for balancing the economy, nature, and society, responsible tourism focuses on ethical behavior, ecological tourism on nature conservation, and regenerative tourism on restoring and transforming the environment for the better. All four approaches play an important role in shaping the sustainable future of tourism, but the regenerative model is increasingly seen as the next evolutionary stage of sustainable development in the industry [16].

The Venn diagram in Fig. 1 clearly shows the relationship between the concepts of responsible, ecological and regenerative tourism and demonstrates that they are components of the broader concept of sustainable tourism.

Awareness of responsibility for the future of the planet is leading to an increase in demand in the ecotourism market. Tourists are attracted to destinations that prioritize environmental preservation, traveling in line with their values, and minimizing their ecological footprint. This trend emphasizes the importance of implementing environmental initiatives in accommodation and catering facilities, using carbon-neutral transportation, and environmentally conscious activities. Compared to other types of tourism, ecotourism has a more pronounced socio-economic focus, and proper organization ensures a balance between the commercial activities of the tourism industry and environmental safety and provides new economic opportunities for both protected areas and host communities [17].

Promoting responsible tourism helps to plan and organize trips aimed at preserving the environment. Today, the global ecotourism market is developing due to the growth of environmental awareness among tourists (Fig. 2).

In 2024, the global ecotourism market grew by 12.5% compared to 2023, indicating an increase in demand for travel that meets the environmental values of tourists, minimizes their ecological footprint and negative impact on nature, and contributes to the sustainable development of local communities. According to experts, by 2034, the volume of ecotourism will grow to \$ 945.34 billion. The average annual growth rate will be 14.31% due to the increasing availability of ecological routes, the development of responsible tourism practices, and the growing interest in authentic natural landscapes and ecosystem conservation.

According to the “Sustainable Travel 2024” study conducted by Booking.com among 31 thousand travelers from 34 countries, tourists’ perception of sustainable travel practices has recently changed. They are becoming more attentive to the choice of destinations and services and make a conscious choice in favor of those that support the principles of sustainable development. According to 83% of tourists, eco-travel aimed at minimizing environmental impact, supporting local communities and using resource-saving practices are important criteria for making their choice. Significantly,

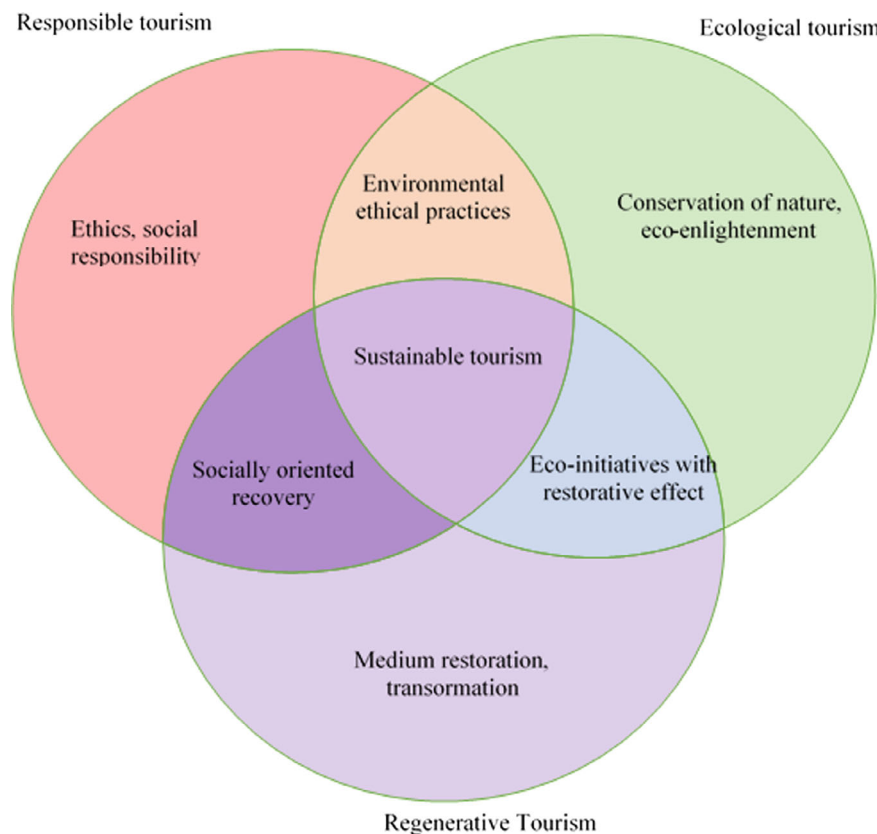


Fig. 1 The relationship between responsible, ecological and regenerative tourism within the conceptual framework of sustainable tourism



Fig. 2 Dynamics of the global eco-tourism market volume, bln. dollars USA [18]

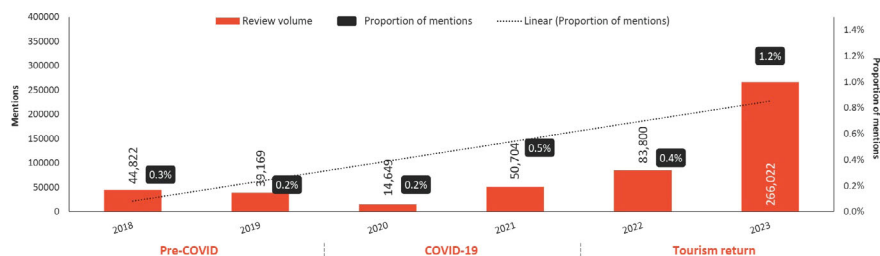


Fig. 3 The volume and share of tourist reviews indicating the use of sustainable development practices in accommodation facilities and tourist attractions in Europe [20]

75% of travelers express a desire to travel more environmentally friendly in the next 12 months; 57% intend to reduce energy consumption during future trips; 54% intend to use more environmentally friendly modes of transportation when traveling; 43% feel guilty when making less environmentally friendly choices when traveling. Among those who intend to travel responsibly, 32% believe it is the right thing to do, 16% say that sustainable travel will improve their experience, 11% are convinced that they will get a more authentic experience of local culture, and 7% believe it will make their trip more interesting. Sustainable practices not only help to reduce the ecological footprint, but also strengthen the sense of responsibility and personal contribution to the preservation of the planet, and promote environmental awareness: 62% of tourists recognize that they are the best version of themselves when they travel sustainably; 67% believe that observing sustainable practices while traveling inspires them to apply sustainable habits in their daily lives [19].

The fact that tourists are increasingly choosing destinations and companies that implement the principles of sustainable development is confirmed by a study conducted by the European Travel Commission and MMGY TCI Research (Fig. 3).

The fourfold increase in the share of reviews in which tourists pay attention to sustainable aspects during their travels in the analyzed period indicates that, in addition to the standard criteria of comfort and cost, responsible attitude to the environment is becoming more important to them. Companies' adherence to the principles of sustainable tourism is becoming an important factor for many tourists when choosing travel services. This should encourage travel service providers to pay more attention to environmental initiatives and responsible business strategies to meet the expectations of tourists who prefer sustainable travel approaches. The growing demand for sustainable tourism from tourists encourages tourism enterprises and the industry as a whole to adapt to new market requirements by implementing appropriate sustainability strategies.

The growth trend is largely due to practical experience, expert knowledge, promotion of dynamic programs and a promising strategy for the development of responsible tourism. An important role in the development of ecotourism is played by tourism policy aimed at ensuring balanced tourism development and protecting the environment from excessive anthropogenic impact. Promoting the development

of responsible tourism through the formation of effective and replicable environmental policies and practices is in the focus of attention of many international organizations, including: World Tourism Organization; United Nations Environment Program, UNEP; United Nations Educational, Scientific and Cultural Organization; International Union for Conservation of Nature, IUCN; Global Environment Facility; Green Climate Fund; World Wide Fund for Nature; The International Ecotourism Society; Global Ecolabelling Network, etc.

Cooperation between international organizations, governments, the scientific community, the private sector, civil society and other entities is an effective tool for promoting environmental initiatives, effectively addressing global challenges, and implementing innovative solutions for sustainable development of the industry. Such multilateral cooperation allows mobilizing resources and knowledge to minimize the negative impact of tourism on ecosystems and ensure the balanced development of the industry. Synergies expand opportunities for improving infrastructure, raising environmental awareness among tourists, and attracting investment in projects aimed at supporting responsible tourism.

2.3 Global and European Trends in Responsible Tourism

The development of responsible tourism is an important area of European tourism policy aimed at preserving natural resources, cultural heritage, improving the quality of life of local communities and supporting sustainable economic development of regions. European countries are increasingly integrating environmental and social standards into the tourism industry to ensure its sustainability and focus on long-term results by integrating innovative technologies, supporting responsible tourism, and promoting social inclusion. In 2023, the volume of European ecotourism amounted to \$ 84.14 billion. This amounted to 38.87% of the total global ecotourism market [21].

The strategic direction for the development of competitive and sustainable tourism in Europe is to focus on an approach that takes into account changes in tourism demand, the need to respect natural resources and harness the innovative potential of digital technologies, while protecting the long-term well-being of local communities, which are key stakeholders in creating attractive tourist destinations and sustainable tourism experiences. The publication of the Transition pathway for tourism by the European Commission in February 2022, created in accordance with the update of the EU Industrial Strategy, was an important step towards promoting the sustainable development of the tourism industry. The initiatives developed are aimed at ensuring the competitiveness and sustainability of tourism through the introduction of environmental, digital and innovative solutions that meet modern challenges and changes in global travel demand. The involvement of industry representatives, public authorities, social partners and other stakeholders allowed us to identify 27 areas for measures aimed at the transition to green and digital technologies and increasing the sustainability of EU tourism [22].

To implement the developed measures, the EU Council adopted the European Agenda for Tourism 2030 on December 1, 2022 [23]. This multi-year program identifies five priority areas for joint work on a greener, more digital and sustainable tourism ecosystem, which will be implemented by EU countries, the European Commission and other stakeholders interested in the development of sustainable tourism (Fig. 4).

The EU Tourism Dashboard [24], developed by the Directorate-General for Internal Market, Industry, Entrepreneurship and SMEs (DG GROW) and the Joint Research Centre (JRC) with the support of the European Commission, together with EU countries and international organizations, is an effective tool for a comprehensive assessment of the results of initiatives aimed at creating a more sustainable tourism ecosystem. The platform visualizes 19 indicators grouped into the following priority areas: green transition, digital transition, and socioeconomic indicators. The use of statistical data from Eurostat, Eurocontrol, and the European Commission Joint Research Center enables the calculation of indicators with the highest possible territorial and thematic detail. The presented indicators help to analyze changes and assess the impact of tourism activities at the regional and national levels, identify problems and support policy initiatives aimed at strengthening a sustainable tourism ecosystem.

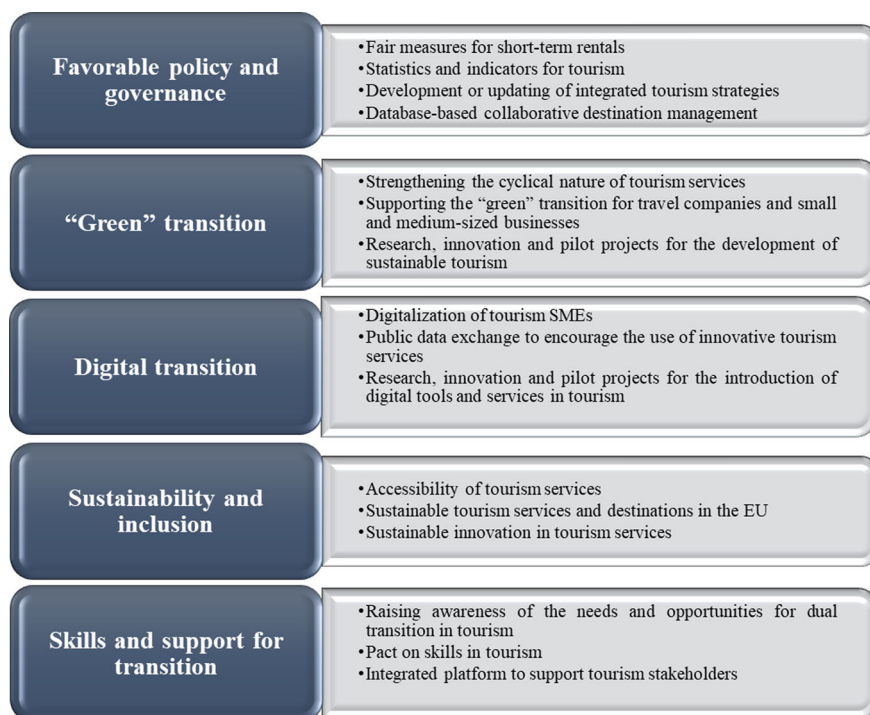


Fig. 4 Priority areas for the implementation of the European agenda for tourism until 2030

Obviously, financing plays an important role in the green and digital transition and increasing the sustainability of EU tourism, which promotes the implementation of environmental initiatives, support for innovative projects, and infrastructure development in accordance with the principles of sustainable development. The European Union offers various financial instruments: private investments and loans, technical support for the transformation of the tourism ecosystem, regional and national programs to support sustainable tourism development. In particular, in 2022, the European Commission plans to allocate €723.8 billion to stimulate the sustainable development of the tourism industry and the implementation of environmentally friendly projects, digital innovations, and socially responsible initiatives.

To inform interested individuals, small and medium-sized enterprises, corporations, academic institutions and public authorities working in the tourism sector, DG GROW has developed a Guide on EU funding for tourism [25]. It covers a wide range of funding programs financed from both the budget and EU funds, in particular: Recovery and Resilience Facility (RRF), European Regional Development Fund (ERDF), Cohesion Fund, European Social Fund Plus (ESF+), European Agricultural Fund for Rural Development (EAFRD), European Maritime, Fisheries and Aquaculture Fund (EMFAF), LIFE Program, Horizon Europe, Creative Europe Program, Erasmus+, Just Transition Fund (JTF), Digital Europe Program, Single Market Program (SMP), InvestEU, REACT-EU, Support by the European Bank for Reconstruction and Development (EBRD). Attracting grant funds helps empower local communities and ensures their participation in tourism initiatives that support long-term economic and environmental sustainability.

Through loans and grants provided to Member States through the RRF, the NextGenerationEU program is actively investing in a number of important areas: green transition, digital transformation, social and territorial cohesion, policies for the next generation, smart, sustainable and inclusive growth, health and economic, social and institutional resilience. In line with Member States' requests for funding under the Recovery and Resilience Facility and the funding needs of other programs supported by NextGenerationEU, the EU expects to raise €712 billion by 2026 (out of a maximum program package of €806.9 billion) [26].

2.4 Best Practices and Case Studies

Many European organizations are focused on implementing initiatives aimed at supporting responsible tourism and achieving sustainable development goals, fostering a responsible attitude to environmental protection, and preserving cultural diversity. For example, the European Ramblers Association (ERA) [27], founded in 1969 in Germany, is a recognized expert in the field of hiking. The Association unites about 65 organizations from 35 European countries, with a total membership of over 3 million people. The ERA's mission is to develop network and cross-border cooperation with organizations that promote responsible attitudes towards nature, share experiences, and promote the values of sustainable tourism for both hosts and

tourists. Since October 11, 2022, ERA has been a member of the Council of Europe, which allows it to strengthen its position as an international organization and to actively influence the development of sustainable tourism in Europe.

The main objectives of the association are:

- promoting the development of hiking and mountaineering, fostering a sense of responsibility and care for the countryside and environmental protection;
- creation, marking and protection of European long-distance tourist routes;
- protecting the rights and traditions of border crossing in European countries and preserving European cultural heritage;
- promoting mutual understanding between people through walks and communication with nature;
- representing the interests of member associations in international organizations such as the European Union and the Council of Europe;
- raising awareness of the importance of physical activity as a basis for a healthy lifestyle.

ERA plays an important role in promoting hiking, facilitating cooperation between organizations and preserving Europe's diverse natural and cultural heritage. By creating a variety of routes, providing expert advice and promoting environmentally responsible practices, ERA inspires tourists from different social groups to embark on unforgettable journeys, connect with nature, experience cultural diversity and create lasting memories. Each trip recommended by the association aims to provide a transformative experience that promotes a deeper understanding of Europe's natural beauty and cultural heritage.

An important area of ERA's activities is the development and implementation of initiatives aimed at achieving sustainable development goals, fostering a responsible attitude to environmental protection, preserving cultural diversity, and supporting local communities. In particular, the Green Trails initiative, developed in 2024, aims to define criteria for routes that promote the development of ecological hiking and responsible visits to natural areas. This system of criteria sets a certain standard for ecological hiking trails that provides tourists with reliable information about trails, promotes environmental practices, and encourages responsible behavior.

The introduction of areas in ERA's activities that ensure the formation of a responsible attitude of tourists to the environment during their travels, aimed at achieving sustainable development goals (Fig. 5).

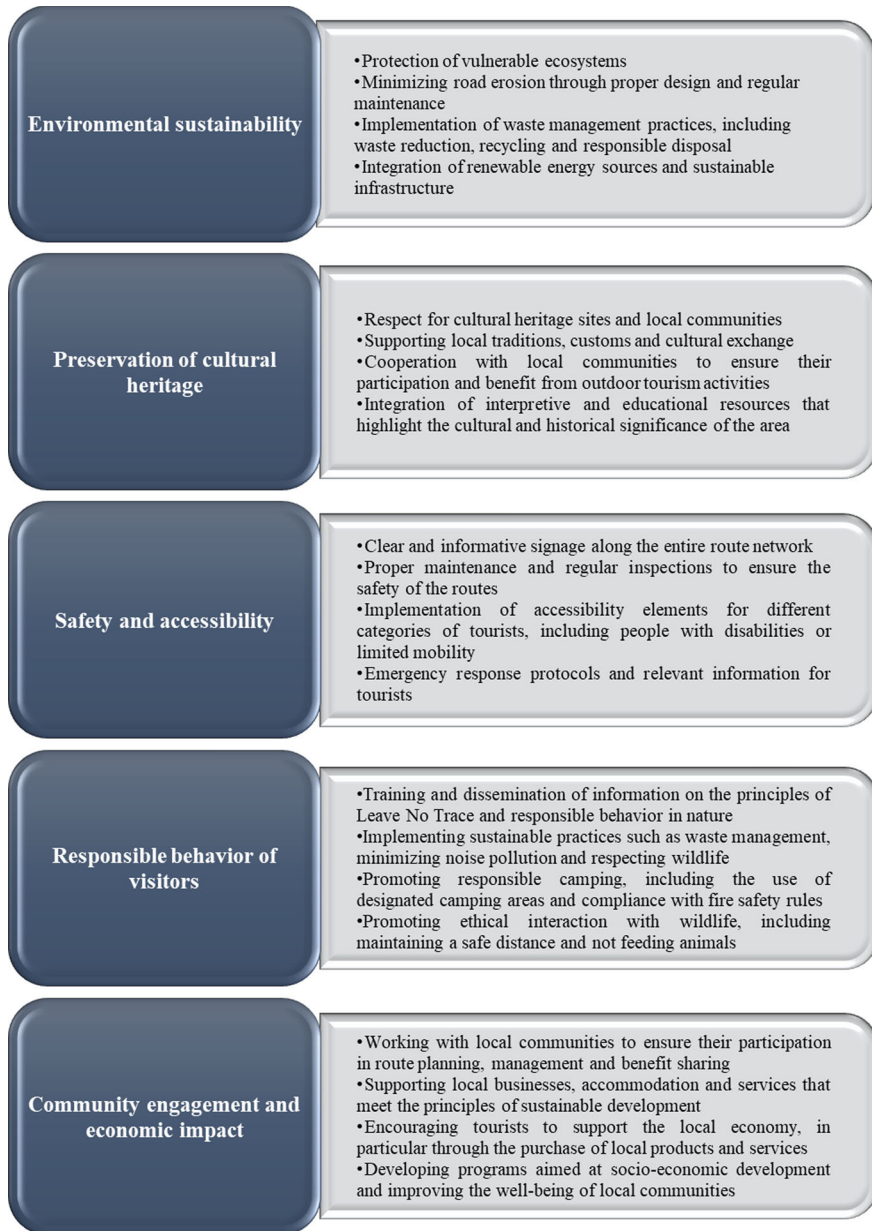


Fig. 5 ERA activities aimed at achieving sustainable development goals

3 Strategies for the Development of Responsible Tourism

3.1 Education and Skills for Sustainable Tourism Development

For effective management of responsible tourism, it is important not only to apply environmental and socially responsible approaches, but also to ensure that tourists and industry employees are aware of the principles of sustainable development. The use of special educational programs and initiatives aimed at raising awareness of the environmental and social aspects of tourism is a key element in ensuring the sustainable development of the industry. This will not only contribute to the formation of responsible tourism practices, but also increase the level of professional training of specialists capable of implementing sustainable tourism strategies in practice. For example, the European Commission's Pact for Skills program focuses on skills development in the tourism sector by supporting public and private organizations in upskilling and retraining for efficient operations and the transition to green and digital technologies. It brings together more than 80 partners, including tourism companies, government agencies and educational institutions, to create training programs that help tourism workers adapt to the challenges of sustainable development, ensure the sustainability of the tourism industry, build environmental awareness and knowledge of sustainable resource management and the preservation of cultural and natural sites. Such training initiatives, implemented through local and regional partnerships, play an important role in building a resilient tourism ecosystem that can withstand crises and maintain long-term economic, social and environmental sustainability.

Various educational projects are being implemented to implement green growth strategies. In particular, the Global Green Growth Initiative, in partnership with the Youth Climate Lab (YCL), has developed the Greenpreneurs project. This is a free 12-week virtual incubator that supports young people in developing social enterprise solutions that address sustainability and climate change in their communities. The incubator program targets existing innovative early-stage medium, small and micro enterprises, including women-owned firms with demonstrated growth potential that are still looking for a scalable and repeatable business model, as well as individuals with relevant experience and entrepreneurial ambitions [28]. Such initiatives help build knowledge and skills to implement responsible and sustainable practices in tourism activities, reducing negative impacts and supporting local communities and the natural environment.

An important tool for promoting responsible tourism is digital technology, which helps not only to improve the tourist experience but also to promote responsible decision-making regarding the impact on the environment and local communities [29]. Mobile apps and websites allow tourists to get information about the environmental and social impacts of their travels, help them find eco-friendly hotels, restaurants, or tourist routes, and choose travel options with a minimal environmental footprint.

Through VR and AR technologies, tourists can visit cultural or natural places virtually, reducing the physical impact on these destinations. This is especially important for protected areas or historical sites that are negatively impacted by an excessive number of visitors. Virtual tours also help to preserve the authenticity of places and reduce the environmental impact on sensitive ecosystems.

Digital tools, such as data collection and analysis platforms, help travel companies to be more transparent about their environmental and social practices. For example, hotels can use the platforms to showcase their responsible tourism certifications or to provide reports on resource use. This data helps tourists make responsible decisions when choosing accommodation or tour operators.

AI technologies help to manage resources in tourist facilities. For example, energy, water, and waste management systems can be optimized with the help of AI, which not only reduces the environmental impact but also ensures the rational use of resources, which is a key aspect of responsible tourism.

Smart technologies are a powerful tool for developing responsible tourism. For example, smart thermostats and lighting systems, energy management systems, automatic sensors on water taps and showers, smart water catchers and water reuse, eco-transportation and smart parking systems, electric vehicles and bike rental, smart parking systems, smart waste management systems, sensors for analyzing tourist flows, e-tickets with environmental analysis help reduce the environmental impact of tourism, conserve resources, protect the environment and improve the interaction of tourists with the local community.

Digital educational resources, such as online courses, webinars, trainings, and educational videos, allow tourists to learn about the importance of responsible tourism and sustainable development before they start their trip.

3.2 Mechanism for Implementing the Responsible Tourism Development Strategy

The effectiveness of the concept of responsible tourism depends on systemic changes in the functioning and management of tourism, as well as synergies between government agencies, tourism operators, local communities, and tourists. An important role in promoting this concept is played by the implementation of policies that promote sustainable development, the integration of environmental, social and economic principles into the activities of tourism enterprises, and cooperation between all stakeholders to create a sustainable tourism ecosystem. The mechanism for implementing a responsible tourism development strategy (Fig. 6) requires an integrated approach aimed at optimizing resource management, minimizing negative environmental impact, supporting sustainable community development, and building environmental awareness among all participants in the tourism process.

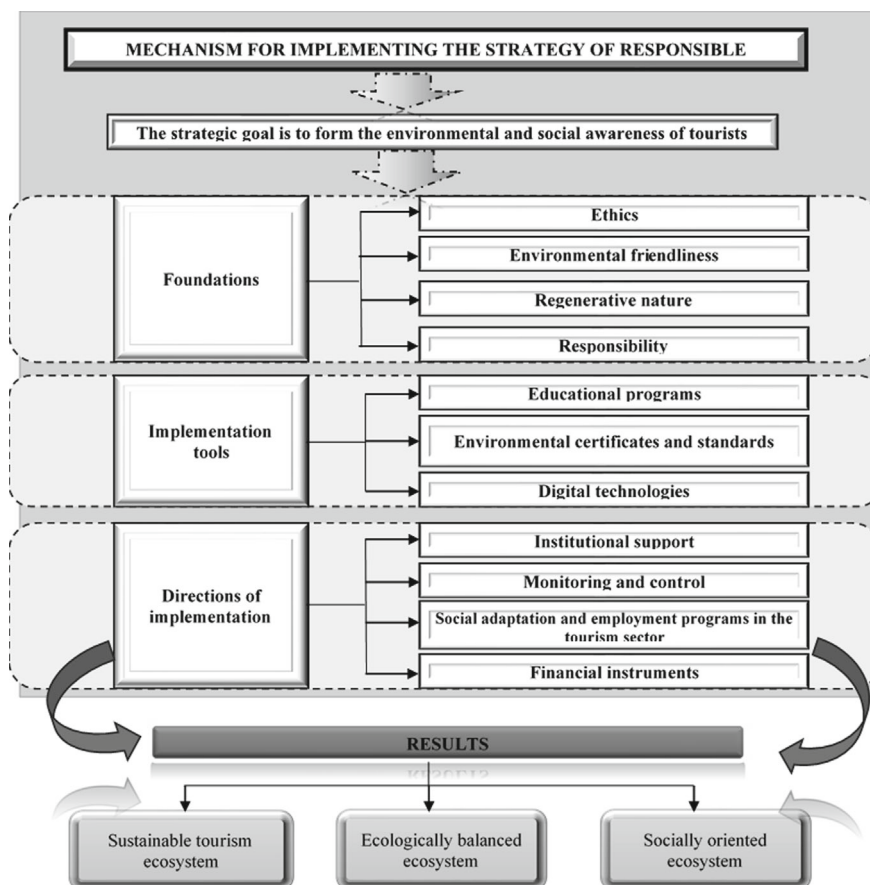


Fig. 6 Mechanism for implementing a responsible tourism development strategy

Ensuring sustainable development, preservation of natural and cultural resources, improving the quality of life of local communities and increasing the tourist attractiveness of territories is possible through integrated destination management in compliance with the principles of holistic management and responsible tourism. The implementation of these principles will allow:

- minimize the negative economic, environmental and social impacts of tourism through the preservation and rational use of resource potential through the use of innovative technologies, minimizing risks, and improving security [30];
- to obtain financial and social benefits for tourism business entities operating on the principles of sustainability;
- empower local communities and their participation in destination planning and management;

- popularize the values of the destination at the local, national and international levels;
- develop strategies for the integrated development of sustainable tourism infrastructure;
- provide economic benefits for local residents and improve their well-being through involvement in the tourism system;
- contribute to the preservation of the natural environment and cultural heritage through environmental education and the formation of environmental awareness among tourists, understanding of local cultural, social and environmental issues and obtaining a responsible tourist experience;
- building mutual respect between tourists and local residents [31].

4 Conclusion

4.1 Synopsis

The study contributes to the understanding of how responsible tourism, together with other approaches such as sustainable, ecological and regenerative tourism, can lead to the evolution of the tourism industry towards a more sustainable and responsible future. The analysis of the main approaches of sustainable, responsible, ecological and regenerative tourism shows the need to integrate their principles into the development strategies of tourist destinations. Global and European practices demonstrate the effectiveness of integrated solutions that combine digital tools, institutional support, environmental education, and community engagement. Of particular importance are educational initiatives aimed at creating a new culture of tourism consumption and developing environmental awareness. Implementation of innovative strategies aimed at minimizing the negative impact of tourism activities on the environment and supporting local communities will contribute to the development of responsible tourism in order to maintain a sustainable tourism ecosystem.

The mechanism for implementing the responsible tourism development strategy proposed in this study creates the basis for the transition from a declarative approach to practical actions aimed at achieving long-term results. This approach allows not only to adapt the tourism sector to modern challenges, but also to ensure its active participation in shaping a sustainable future.

4.2 Further Research

The study confirms that in modern conditions, responsible tourism is becoming a key factor in ensuring the sustainable development of the tourism industry. Therefore, further research in the area of responsible tourism development should be aimed at expanding the empirical base for the implementation of environmental, social, ethical

and economically sustainable practices in tourist destinations of various sizes. In order to implement the best responsible practices in specific destinations, a comparative analysis of the effectiveness of implementing responsible tourism strategies in the European Union should be conducted. The development of indicators for assessing the level of responsibility of tourism entities, as well as the formation of a system for monitoring the impact of tourism activities on local ecosystems, will increase control over the environmental sustainability of destinations and help minimize the negative impact on the environment.

Further areas of research aimed at studying the role of digital technologies, in particular smart tourism, big data and blockchain, in shaping responsible tourism practices will contribute to the effective management of sustainable tourism processes. Particular attention should be paid to tools for engaging young people, educational institutions, and civil society in the implementation of responsible tourism policies to create sustainable behavioral models among all tourism market participants to ensure a balance between the economic, social, and environmental aspects of tourism activities.

References

1. World Tourism Organization (2019) Tourism's carbon emissions measured in landmark report launched at COP25. <https://www.unwto.org/news/tourisms-carbon-emissions-measured-in-landmark-report-launched-at-cop25>. Accessed 4 Dec 2019
2. Karpenko N (2015) Tourism in the context of sustainable development and ecological modernization [Turyzm u konteksti staloho rozvytku ta ekolohichnoi modernizatsii]. In: Skliar HP (ed) Modernization of tourism studies: theory and practice of partnership. PUET, Poltava, pp 113–117
3. Osypchuk M (2014) International tourism in the context of foreign economic activity modernization of Ukraine on the basis of sustainable development [Mizhnarodnyi turizm u konteksti modernizatsii zovnishnoekonomichnoi diialnosti Ukrainy na zasadakh staloho rozvytku]. Sci Bull Kherson State Univ 6(1):107–111. http://www.ej.kherson.ua/journal/economic_06/26.pdf
4. Barlet S, Colombon M (2004) Approaches to some definitions [Approches de quelques définitions]. In: Solidarity tourism and sustainable development [Tourisme solidaire et développement durable]. Gret Publications, Paris, 120 p. https://gret.org/wp-content/uploads/2021/11/AGR-12_Tourisme-solidaire-et-developpement-durable.pdf
5. UN Tourism (2017) Economic cooperation for sustainable tourism development. Policy paper on tourism. <http://www.tourism4development2017.org>
6. Bizzarri C, Querini C (2006) Economics of sustainable tourism. Theoretical analysis and case studies [Economia del turismo sostenibile: analisi teorica e casi studio. Analisi teorica e casi studio]. Milano, Franco Angeli, 224 p
7. The International Ecotourism Society. <https://ecotourism.org/what-is-ecotourism/>
8. The Global Ecotourism Network. <https://www.globalecotourismnetwork.org/definition-and-key-concepts/>
9. Savytska O, Savytska N, Pohrebniak L (2017) Ecology tourism an important component of sustainable development strategies Ukraine [Ekoturizm yak vazhlyva skladova stratehii staloho rozvytku Ukrainy]. Glob Natl Probl Econ 15:122–128. <http://global-national.in.ua/archive/15-2017/24.pdf>
10. Bardash S, Chernysh D (2023) The essence of ecotourism and its transformation in the context of sustainable development [Sutnist ekolohichnoho turyzmu ta yoho zminy v umovakh staloho

- rozvytku]. Sci Coll «InterConf» 152:482–488. <https://archive.interconf.center/index.php/conference-proceeding/article/view/3204/3235>
11. Goodwin H (2016) Responsible tourism, 2nd edn. Goodfellow Publishers, Oxford. <https://doi.org/10.23912/978-1-910158-84-5-3101>
 12. Cape Town Declaration on Responsible Tourism. <https://responsibletourismpartnership.org/cape-town-declaration-on-responsible-tourism/>
 13. Durkheim E (1982) Introduction. In: Lukes S (ed) The rules of sociological method. Contemporary social theory. Palgrave, London. https://doi.org/10.1007/978-1-349-16939-9_1
 14. Fletcher R, Murray Mas I, Dlanco-Romero A, Blazquez-Salom M (2019) Tourism and degrowth: an emerging agenda for research and praxis. J Sustain Tour 27(12):1745–1763. <https://doi.org/10.4324/9781003017257-1>
 15. EU Tourism Dashboard. <https://tourism-dashboard.ec.europa.eu/background-methodology?lng=en&ctx=tourism>
 16. Krasnomovets V (2023) Regenerative tourism: general characteristics and specifics of interpretation. Bus Inform 12:83–89. <https://doi.org/10.32983/2222-4459-2023-12-83-89>
 17. Kyryliuk I (2023) Ecological tourism in the context of sustainable development [Ekolohichniy turizm u konteksti staloho rozvytku]. In: Povorozniuk IM (ed) Ecotrends in the hospitality industry: realities and prospects [Ekotrendy v industrii hostynnosti: realii ta perspektyvy]. Tvory, Vinnytsia, pp 204–228
 18. Ecotourism Market Size, Share, and Trends 2024 to 2034 (2024). <https://www.precedenceresearch.com/ecotourism-market>
 19. Booking.com (2024) Sustainable travel 2024. <https://news.booking.com/download/904910bb-db77-4886-9ead-accbf87ad891/sustainabletravelreport2024.pdf>
 20. European Travel Commission (2024) Climate change and tourism in Europe: exploring the impact of climate change on future tourism in Europe. MMGY TCI research. https://etc-corporate.org/uploads/2024/06/2024-ETC_Travelsat_Special-Report-on-Climate-Change.pdf
 21. Ecotourism market size, share & industry analysis. Fortune business insights (2024). <https://www.fortunebusinessinsights.com/ecotourism-market-108700>
 22. European Commission (2022) Transition pathway for tourism. Publications Office of the European Union. <https://data.europa.eu/doi/10.2873/344425>
 23. European Agenda for tourism 2030 (2022). <https://data.consilium.europa.eu/doc/document/ST-15441-2022-INIT/en/pdf>
 24. European Commission EU. Tourism dashboard. <https://tourism-dashboard.ec.europa.eu/background-methodology?lng=en&ctx=tourism>
 25. European Commission (2021) Guide on EU funding for tourism. https://single-market-economy.ec.europa.eu/news/new-online-guide-eu-funding-tourism-2021-05-12_en
 26. European Commission. NextGenerationEU. https://commission.europa.eu/strategy-and-policy/eu-budget/eu-borrower-investor-relations/nextgenerationeu_en
 27. European Ramblers Association. <https://www.era-ewv-ferp.org/interal/era/>
 28. Greenpreneurs (2021) <https://gggi.org/globalprogram/greenpreneurs/>
 29. Kyryliuk I, Neshchadym L, Blahopoluchna A (2022) Digitalization as the main factor of hospitality industry development in the conditions of modern challenges [Didzhytalizatsiia yak osnovnyi faktor rozvytku industrii hostynnosti v umovakh suchasnykh vyklykiv]. Actual Probl Reg Econ Dev 18(2):217–231. <https://doi.org/10.15330/apred.2.18.217-231>
 30. Dzhezdzhula V, Hurochkina V, Yepifanova I, Telnov A (2022) Fuzzy technologies for modeling social capital in the emergent economy. WSEAS Trans Bus Econ 19:915–923. <https://doi.org/10.37394/23207.2022.19.80>
 31. Redko V (2021) Responsible tourism in the holistic destination model [Vidpovidalnyi turizm v kholistychnii modeli destynatsii]. In: XV International scientific conference geography, economics and tourism: national and international experience, Lviv, 8 October 2021, pp 235–238. <https://geography.lnu.edu.ua/wp-content/uploads/2021/10/materialy-15-Heohrafiia-ekonomika-i-turizm-2021.pdf>

Digital Agro-Innovations as a Strategic Tool for Agricultural Adaptation



Tetiana Shabatura , Anush Balian , Margarita Stepanova ,
Alla Yakovenko , and Olga Petrenko 

Abstract This study identifies key climate risks affecting Ukraine’s agricultural sector and examines the strategic role of digital agro-innovations in addressing these challenges. It analyzes the functionality of core technologies—including agrodrones, satellite monitoring, the Internet of Things (IoT), and artificial intelligence (AI)—and proposes an integrated digital agroecosystem model. Unlike prior studies that focus on individual tools, this research introduces a systems-based approach that enables the synergistic interaction of digital solutions for building climate-resilient, resource-efficient, and environmentally sustainable agriculture. The proposed framework offers a practical and conceptual basis for national strategies, digital transformation programs, and climate adaptation policies. The findings are particularly relevant for Ukraine’s war-affected agricultural sector and contribute to advancing innovation, sustainability, and adaptive capacity in agri-food systems.

Keywords Climate change adaptation · Digital agriculture · Smart farming · Agrodrones · Artificial Intelligence (AI) · Internet of Things (IoT) · Satellite monitoring

T. Shabatura (✉) · M. Stepanova · A. Yakovenko
Institute of Climate-Oriented Agriculture of NAAS, Odesa Region, Ukraine
e-mail: ta_sha77@ukr.net

A. Yakovenko
e-mail: alla_capital@ukr.net

A. Balian
The National Academy of Agrarian Sciences of Ukraine, Kyiv, Ukraine

O. Petrenko
Odessa State Agrarian University, Odesa, Ukraine
e-mail: leka_m@ukr.net

1 Introduction

1.1 Relevance

The global climate is undergoing rapid transformation, characterized by rising average annual temperatures, increased frequency of extreme weather events, and the destabilization of hydrothermal conditions. These shifts exert significant pressure on agricultural systems, particularly in vulnerable economies such as Ukraine, where agro-industrial resilience is limited. Simultaneously, agricultural practices themselves contribute to climate change through greenhouse gas (GHG) emissions, resource inefficiencies, and environmental degradation. This bidirectional relationship between agriculture and climate variability highlights the urgent need for adaptive, data-driven, and technologically enabled management frameworks.

Emerging digital technologies—specifically artificial intelligence (AI), satellite-based remote sensing, the Internet of Things (IoT), and unmanned aerial vehicles (UAVs)—are reshaping the landscape of agricultural production by enabling higher levels of precision, responsiveness, and automation. These tools collectively facilitate the transition from reactive to proactive agricultural practices, and from linear to adaptive decision-making models. However, current research efforts often treat these technologies in isolation, failing to explore their synergistic potential or their integration into a unified agro-digital ecosystem. Furthermore, prevailing frameworks seldom consider the unique structural and geopolitical constraints facing Ukraine's agricultural sector, including ongoing military conflict, infrastructure deterioration, and limited access to digital and financial resources [1–3].

In light of these challenges, there is a critical need to develop a comprehensive, systems-level vision for the digital transformation of agriculture as a core strategy for climate change adaptation. Such an approach must leverage interdisciplinary methodologies and promote the convergence of heterogeneous digital technologies. This chapter addresses that need by proposing a conceptual and methodological foundation for sustainable agricultural production, specifically tailored to the evolving climate and operational realities of Ukraine.

1.2 Goals and Objectives

Recent advances in digital technologies have led to the accumulation of substantial empirical evidence regarding their impact on agricultural productivity, resource efficiency, and operational decision-making. Nevertheless, current research remains fragmented. Most studies isolate individual technologies—such as UAVs, environmental sensors, or artificial intelligence systems—or apply foreign use cases that fail to account for the structural and contextual peculiarities of Ukraine's agri-industrial systems.

Moreover, the synergistic integration of digital tools to enhance climate resilience, operational scalability, and environmental sustainability in agricultural production remains underexplored. This chapter addresses this gap by presenting a digital transformation framework that integrates AI, IoT, satellite monitoring, and UAV-based solutions into a unified agro-digital ecosystem specifically adapted to the Ukrainian context under climate stressors.

In contrast to earlier studies focused on discrete technological components—robotics [3], artificial intelligence [4], drone applications [5], or energy optimization [6]—this work introduces a systems-level digital methodology aimed at implementing an integrated, adaptive agricultural model. The proposed framework leverages cross-technology interoperability and real-time data analytics to support decision-making under uncertainty and climate variability.

This chapter contributes to the formation of a holistic digital agroecosystem architecture incorporating satellite data processing, edge and cloud-based IoT networks, machine learning algorithms, and drone-enabled precision agriculture. The methodological approach facilitates a shift from isolated digital deployments to a comprehensive digital agriculture stack capable of real-time monitoring, adaptive control, and strategic planning.

Structurally, the chapter proceeds as follows:

- It outlines the spectrum of climate-induced stressors impacting Ukrainian agriculture and motivating digital intervention;
- It reviews key enabling technologies—including satellite imaging, AI, IoT-based field monitoring, and UAVs—with a focus on their integration capabilities;
- It emphasizes UAVs (agrodrones) as a critical interface for high-precision sensing, targeted spraying, and variable-rate fertilization, contributing to both resource optimization and ecological preservation;
- It introduces the conceptual design of an integrated agro-digital platform, emphasizing interoperability, task automation, and distributed analytics.

This model supports the transition to a next-generation agricultural management paradigm—one that is climate-adaptive, resource-optimized, and aligned with sustainable development goals.

2 Digital Agro-Innovations in the Context of Climate Change

2.1 Major Climate Challenges for the Agricultural Sector

Climate change presents significant and escalating challenges to agriculture, particularly in the form of increasing average annual temperatures, a higher frequency of extreme weather events, and declining soil moisture levels. One of the most prominent indicators of recent climatic shifts is global warming, with recorded increases in

average air temperature reaching 2–3 °C over the past decade [7]. This thermal rise adversely impacts both crop yields and livestock productivity, leading to reductions in the overall volume of agricultural output. As a result, agricultural producers are increasingly compelled to adopt innovative practices to maintain yield stability and ensure production sustainability.

Scientific projections suggest that even a 1 °C increase in average temperature may result in a 10% decline in agricultural productivity [7]. Research conducted by the Food and Agriculture Organization (FAO) [8] and the European Environment Agency (EEA) [9] identifies several key climate-related threats to agriculture:

- Alteration of crop vegetation cycles due to elevated average temperatures;
- Proliferation of drought- and heat-tolerant weed species, which suppress crop productivity;
- Heightened risk of soil degradation, particularly wind and water erosion, notably in the southern and eastern regions of Ukraine;
- Rainfall irregularity, which impairs crop planning and induces yield losses through water stress.

In Ukraine, the agricultural sector is particularly susceptible to climate variability. The expansion of arable land, combined with the increasing prevalence of monocultures, necessitates substantial investment in advanced agricultural technologies. Climate-induced disruptions are diminishing the efficacy of traditional crop protection practices, thus necessitating a reevaluation of agronomic strategies.

In addition to being affected by climate change, agriculture also contributes to its intensification. The sector generates a significant volume of greenhouse gas (GHG) emissions. In 2021, total GHG emissions from Ukraine's agriculture sector rose by 12.8% compared to the previous year. This increase is largely attributed to emissions from land cultivation and the expanded use of inorganic nitrogen fertilizers [10].

Given this bidirectional relationship, it is imperative to assess both the environmental impact of agricultural activities and their adaptive capacity. Intensive farming and excessive reliance on chemical inputs intensify environmental pressures, underscoring the urgent need for the adoption of more efficient and environmentally sustainable technological solutions. Emerging digital innovations offer significant potential to enhance the resilience of agriculture to climate change while simultaneously reducing its ecological footprint.

2.2 Modern Digital Technologies in Agricultural Production

Conventional agricultural practices face increasing technological and ecological constraints, including:

- High input consumption (e.g., water, fuel, agrochemicals);
- Significant environmental burdens caused by excessive chemical use;

- Limited precision in application techniques, resulting in inefficient crop coverage and reduced effectiveness of plant protection strategies;
- Strong dependency on soil and weather conditions, restricting the use of heavy machinery, particularly in terrains with irregular topography or excessive moisture [11].

The urgent need to reduce resource inputs and mitigate environmental externalities positions the adoption of digital technologies as a critical vector for agricultural modernization. Digital agriculture enables the optimization of agrochemical applications, reduction of chemical load, and enhancement of the overall efficiency of agronomic interventions.

Recent studies emphasize the transformative role of robotic platforms and UAVs (Unmanned Aerial Vehicles) in advancing sustainable farming practices. Precision agriculture drones have demonstrated the ability to reduce agrochemical use by up to 40%, while autonomous ground robots can significantly improve fertilizer efficiency by minimizing nutrient loss [6].

In the European Union and the United States, digital agriculture technologies are widely adopted to increase climate resilience. Integration of satellite remote sensing, artificial intelligence (AI), and UAVs enables the optimization of decision-making in resource management.

Satellite Monitoring and IoT Integration. Programs such as Copernicus (EU) and Landsat (USA) provide high-resolution imagery for soil and crop health assessment. Coupling this data with IoT-based in-field sensors facilitates real-time environmental monitoring. For example, the Sen4CAP system is deployed by French viticulture operations in Burgundy for microclimatic analysis, while NASA's Harvest program in Iowa demonstrated a 15–20% reduction in water usage through forecast-based irrigation planning. OpenET, used in California, supports water resource forecasting during droughts.

AI and Agricultural Data Analytics. Artificial intelligence offers advanced capabilities in processing large-scale agricultural datasets. AgroScout (Israel/USA/EU) employs UAVs and computer vision to detect early-stage crop diseases, reducing pesticide usage by up to 40%. Similarly, Microsoft's AI for Earth initiative supports yield forecasting in France and nitrogen optimization in Indiana (USA) [12].

Agrodrone in Precision Agriculture. AI-powered UAVs enhance operational precision and environmental safety. John Deere's See & Spray system achieves up to 77% reduction in herbicide usage via computer vision targeting. XAG drones are employed in Germany and Spain to apply biological agents in cereal and vineyard systems, reducing chemical input by 50% [13]. The Hungarian Agrodrone system applies microbiological preparations in maize fields under high-temperature stress conditions.

These technologies have significantly improved input efficiency and yield optimization, key metrics for sustainable development. In Ukraine, similar systems are implemented via EOS Data Analytics, NASA Harvest, and CropX. These platforms

support field condition assessment, real-time data collection, and dynamic production optimization.

AI-driven platforms deployed by Ukrainian firms such as AgriLab and Smart-Farming leverage machine learning to reduce crop losses by 15–20% through early disease and pest detection [4]. However, broader adoption requires infrastructure upgrades and improved access to agronomic data.

Towards an Integrated Agro-Digital Ecosystem. Digital farming is evolving from isolated tool usage to a cohesive, intelligent decision-support infrastructure. The convergence of agrodrones, satellite monitoring, AI, and IoT enables full-stack agricultural automation:

- Satellite monitoring provides strategic field-level insights;
- IoT sensors deliver granular, real-time environmental data (soil moisture, weather, nutrients);
- AI algorithms perform predictive analytics, anomaly detection, and optimization;
- Agrodrones execute targeted interventions such as site-specific spraying, irrigation, or nutrient delivery.

This integrated approach reduces chemical and water consumption, optimizes machinery and labor utilization, minimizes crop losses, and lowers ecological footprints.

Digital agro-innovations are thus positioned as the core instruments for addressing climate-related agricultural challenges. Their deployment ensures proactive climate risk management, efficient resource use, and increased systemic resilience.

The synergistic application of satellite data, AI, IoT, and UAV technologies represents a foundational component of climate adaptation strategies. As such, the digital transformation of agriculture is no longer optional but essential for building a competitive, innovation-oriented, and climate-smart agri-food economy.

3 Agrodrone and Digital Agricultural Technologies: Functionality and Application Prospects

3.1 The Role of Agrodrones in the Transformation of Agricultural Production

One of the promising solutions in precision farming is the use of agrodrones for automated and high-precision application of crop protection products, crop condition monitoring, field mapping, and agronomic performance analysis. This is especially true in the face of increasing droughts, uneven rainfall and soil degradation, which makes traditional treatment methods less effective. Agrodrones allow you to reduce the impact of negative factors, ensuring the optimization of the use of resources.

The use of agrodrones provides:

- increasing the efficiency of monitoring the condition of crops, timely detection of foci of disease or pest damage;
- reducing the use of agrochemicals due to spot application, which reduces environmental pollution;
- expanding the possibilities of working in extreme conditions, in particular in waterlogged or difficult terrain areas, where the use of ground equipment is limited;
- field mapping and creation of digital terrain models for precision farming;
- monitoring soil moisture and plant condition, which contributes to effective irrigation management;
- ensuring uniformity of spraying, which improves the effectiveness of crop protection;
- reduction of operating costs, including savings in fuel, labor resources and time [6, 12, 13].

The growing interest in this technology among farmers is confirmed by statistics. In particular, in 2023, the area of agricultural land cultivated by agrodrones increased by 3.5% compared to the previous season. The increase in the use of agrodrones indicates their active integration into agricultural production and an increase in the level of adaptation of technology to the needs of farmers.

In Ukraine, the use of agrodrones in agriculture is due to a number of technological advantages, primarily the prompt application of plant protection products, monitoring the condition of crops and mapping of agricultural landscapes in situations where traditional ground equipment cannot be used due to adverse weather factors or soil conditions. This is especially important in conditions of soil degradation, where minimizing the impact of agrochemicals on the environment is critical.

Scientists from different countries emphasize the technological advantages of agrodrones in precision farming, their role in distributed spraying, monitoring soil moisture and chemical composition, as well as field mapping [3, 5, 14].

3.2 Comparative Advantages of Agrodrones Over Traditional Agricultural Machinery

Compared to conventional ground-based agricultural technologies, agrodrones offer several critical performance advantages that enhance the efficiency, precision, and sustainability of field operations.

High Operational Productivity and Speed. Due to their aerial mobility and independence from terrain-induced obstacles, agrodrones can rapidly process large land areas without delays related to soil condition or physical obstructions. This capability enables timely interventions in response to emerging biotic or abiotic stressors.

Precision in Application. Utilizing GPS-guided flight paths and automated route optimization, agrodrones ensure uniform distribution of agrochemicals with minimal

overlaps or omissions. This level of spatial accuracy significantly reduces the risks associated with uneven pesticide or fertilizer application, which are common in mechanized ground spraying—especially in irregular or elongated field geometries.

Adaptability to Challenging Field Conditions. Agrodrones can operate efficiently on waterlogged or uneven terrains where heavy machinery may become immobilized or cause crop damage. Their deployment ensures timely agronomic operations, independent of unfavorable soil conditions.

Rapid Deployment Within Narrow Weather Windows. Agrodrones require minimal setup time and can be deployed immediately during short intervals of favorable weather between rain or high winds. This is particularly important for the timely application of fungicides, herbicides, and insecticides, where delays may lead to considerable yield losses.

Reduced Soil Compaction and Environmental Impact. As unmanned aerial platforms, agrodrones avoid mechanical contact with the soil surface, thereby eliminating compaction issues commonly associated with traditional ground equipment. Soil compaction impairs water infiltration, aeration, and root development. The use of agrodrones contributes to maintaining natural soil structure, an essential principle in sustainable and regenerative agriculture [3, 5, 12–14].

Based on recent research and operational benchmarks, Table 1 summarizes key performance differentials between agrodrones and traditional ground-based machinery across selected indicators.

Agrodrones represent one of the most advanced and multifunctional technologies in modern agriculture, integrating capabilities for aerial monitoring, variable-rate fertilization, pest control, and yield estimation. Their deployment enables significant reductions in input resource consumption while enhancing the operational efficiency of field processes. However, the performance of agrodrones must be evaluated in comparison with other contemporary digital technologies—such as autonomous ground-based systems and satellite remote sensing platforms. Such a comparative framework facilitates an objective assessment of the strengths and limitations of agrodrones under varying agro-environmental conditions and use cases.

Table 1 Comparison of the effectiveness of agrodrones and traditional machinery (based on [3, 5, 12–14])

Parameter	Agrodrones	Traditional technique
Processing speed	Up to 100 ha/hour	20–30 ha/hour
Working solution consumption	5–10 l/ha	200–300 l/ha
Operation on wet soils	Possible	Limited
Soil compaction	Missing	Significant
Application accuracy	High	Moderate
Access to hard-to-reach areas	Enabled	Limited

3.3 Comprehensive Approach to the Use of Digital Agro-Innovations in the Agricultural Sector

In contemporary agriculture, technological efficiency is significantly enhanced through the integration of multiple digital solutions into unified systems. Agrodrones—serving as high-performance tools for spraying, aerial monitoring, and field mapping—demonstrate optimal performance when deployed in conjunction with satellite-based remote sensing, artificial intelligence (AI), and Internet of Things (IoT) technologies. This section presents an integrated framework for the combined use of these technologies, analyzing their comparative advantages, operational limitations, and development prospects within the agricultural sector of Ukraine.

Agrodrones perform a range of critical functions, including real-time crop monitoring via multispectral imaging, precision application of fertilizers and pesticides, and high-resolution mapping to identify areas of crop stress or non-uniform growth. However, the standalone deployment of agrodrones offers only localized insights and lacks broader environmental context. Their effectiveness is significantly amplified when integrated with satellite imagery, AI-powered analytics, and ground-based IoT sensor networks [13].

The study by Zamlynskyi et al. [12] presents the architecture of a digital agricultural platform that unifies artificial intelligence, sensor data streams, and cloud-based analytics to support decision-making across multiple levels of the agricultural value chain. This supports the transition toward a unified digital agroecosystem capable of managing agricultural operations adaptively under climate variability [12].

Satellite remote sensing enables macro-level assessment of vegetation health, soil moisture, thermal conditions, and crop phenology. It provides temporal and spatial insights that inform strategic decisions related to fertilization, irrigation, and crop risk management. For example, when satellite data indicates a zone of poor vegetation performance, an agdrone can be deployed for high-resolution diagnostics and site-specific intervention.

The synergistic interaction between satellite platforms, AI, IoT, and UAVs ensures an optimized, data-driven approach to agricultural management, increasing productivity while reducing environmental externalities.

To better understand the functional complementarities and trade-offs among these technologies, Table 2 provides a comparative analysis of agrodrones, autonomous ground-based robotic systems, and satellite monitoring based on selected performance indicators.

The comparative analysis demonstrates that each of the assessed technologies—agrodrones, robotic ground systems, satellite monitoring, artificial intelligence (AI), and the Internet of Things (IoT)—offers distinct advantages, limitations, and implementation prospects within the context of Ukraine's agricultural sector.

Agrodrones represent the most mobile and flexible solution, particularly suitable for operating in complex field topographies and hard-to-access areas. Their high precision in agrochemical application significantly reduces resource use and

Table 2 Comparison of agrodrones, ground robots, and satellite monitoring (based on [3, 5, 12–14])

Parameters	Agrodrones	Robotic ground systems	Satellite/sensor monitoring
Mobility	High	Moderate	Low
Cargo/payload capacity	Limited	High	–
Application accuracy	High	Very high	High
Impact on soil surface	None (no contact)	Negative (causes compaction)	None
Weather resistance	Limited (sensitive to wind/rain)	High	Independent of local weather
Initial investment costs	High	Very high	Very high
Personnel requirements	Trained operator required	Operator with technical training	Data scientists and analysts required
Representative technologies	DJI Agras T40, XAG P100, kray technologies	Naio Oz, AgBot II, FarmDroid FD20	Planets labs, SkySat, sentinel-2, John deer field analyzer

environmental impact. However, their performance may be constrained by adverse weather conditions and current regulatory frameworks governing UAV operation.

Robotic ground systems offer robust performance in uniform terrain, delivering superior dosing accuracy. While these systems are promising for large-scale agricultural enterprises, their operation may lead to soil compaction and they require substantial capital investment and technical maintenance.

Satellite monitoring provides macro-scale insights into vegetation dynamics, crop health, and spatial variability. It enables early identification of problem zones and supports long-term planning. However, it lacks the capacity for direct operational response and thus requires integration with active technologies such as UAVs or ground robots.

Artificial intelligence plays a critical role in processing large-scale agronomic datasets. Applications include yield forecasting through the fusion of satellite imagery, meteorological data, and historical productivity records; anomaly detection in spectral images for early disease or nutrient deficiency identification; and optimization of irrigation and fertilization schedules via predictive modeling. Integration of AI with UAVs enables fully automated decision loops—for instance, targeted spraying based on AI-identified disease hotspots or moisture stress zones.

IoT technologies serve as the foundational layer for real-time environmental data acquisition. Agricultural IoT systems comprise in-field sensors measuring soil moisture, temperature, and nutrient levels; localized weather stations; and smart irrigation systems. These systems generate continuous data streams used by AI engines to produce dynamic field recommendations. For example, in the event of detected low

soil moisture, AI algorithms can schedule immediate localized intervention using UAVs to apply water or foliar nutrients.

In Ukraine, agrodrones demonstrate strong potential due to their adaptability, precision, and operational efficiency. Their applications extend beyond spraying to include remote sensing, health diagnostics, mapping, automated seeding, bioproduct treatment, and data acquisition. The convergence of agrodrones with satellite technologies, AI, and IoT fosters the emergence of an integrated agro-digital ecosystem capable of real-time monitoring, precision intervention, and resource-efficient production.

This integrated approach significantly enhances the accuracy of agrotechnical operations, facilitates rapid detection of field-level anomalies, and reduces environmental externalities. While each technology offers unique capabilities, their combined deployment forms a synergistic framework for next-generation agricultural management. For Ukraine, this digital transformation unlocks substantial opportunities to boost productivity, reinforce climate resilience, and advance toward a sustainable, innovation-driven agricultural model.

4 Digital Agro-Innovation as a Strategic Response to Climate Challenges

4.1 International Guidelines for Digital Transformation of the Agricultural Sector

The digital transformation of agriculture necessitates a holistic, systems-oriented methodology that integrates technical, organizational, and socio-economic components. In this context, a system-based approach serves as a foundational principle for the efficient integration of digital solutions across all levels of the agricultural production chain—from primary data acquisition to high-level strategic decision-making enabled by large-scale analytics.

At the core of this transformation lies the concept of smart farming, which promotes adaptive management based on multilevel monitoring (e.g., satellite imagery, IoT-enabled sensors), autonomous and semi-autonomous operations (e.g., UAVs, robotic platforms), and intelligent analytics (e.g., artificial intelligence and machine learning).

The smart farming model is functionally structured into three interconnected domains:

- Information and analytical domain: responsible for the acquisition, processing, analysis, and visualization of agricultural data;
- Operational domain: focused on executing agrotechnical interventions through robotic and autonomous systems;

- Managerial and strategic domain: supporting decision-making through predictive analytics, scenario-based risk assessment, and system optimization.

International practice confirms the viability of this approach. For instance, the SmartAgriHubs project (EU) operates as a decentralized network of digital innovation hubs, promoting cross-sectoral collaboration between farmers, research institutions, and agritech startups. The OECD Digital Agriculture Framework provides strategic guidance for policy development by addressing key pillars such as infrastructure, data governance, digital competencies, and regulatory frameworks.

In the Ukrainian context, these models are being adapted and applied within several national initiatives:

EOS Data Analytics: combines satellite imagery and AI-driven analytics for crop monitoring, yield forecasting, and anomaly detection;

AgriLab: implements precision agriculture via agrochemical diagnostics, geospatial analytics, and sensor networks;

SmartFarming: offers automated field management tools, vegetation index analysis, and remote operation control;

Agrouprade (UCAB): enhances digital competencies among farmers through training on CRM tools, agri-platforms, and precision farming technologies;

Academic institutions (NUBiP, KhNAU, Odesa State Agrarian University): introduce educational programs and establish demo sites equipped with drones, IoT systems, and AI solutions.

According to Linde et al. [15], AI-based technologies in Ukrainian agriculture are already integrated into digital management platforms, computer vision systems, and agronomic data pipelines—highlighting the feasibility of building a national model for smart agriculture [15].

Importantly, the adaptation of international frameworks requires not only the implementation of ready-made technological solutions, but also the localization of digital strategies in alignment with Ukraine's agricultural specificity. This includes considerations of climate vulnerability, infrastructural constraints, and disruptions caused by military conflict. Consequently, the creation of a national smart farming model must be inclusive, scalable, and capable of serving both large agribusinesses and smallholder farms [16, 17].

A review of international digitalization strategies further reveals distinct yet effective transformation models across countries facing climate-related agricultural stress. The European Union emphasizes platform decentralization and precision farming; the United States promotes AI, big data, and autonomous machinery; while Israel focuses on IoT-intensive models for resource optimization. These strategic orientations, when contextualized for Ukraine, offer valuable blueprints for national adaptation (see Table 3).

Ukraine possesses significant potential to localize and effectively adapt international digital agriculture models by leveraging its unique climatic, economic, and socio-political context. The synthesis of Israeli precision in resource use, American

Table 3 International comparison of digital strategies in the agricultural sector (based on [12–14, 18])

Country/ region	Digitalization strategies	Efficiency in the context of climate change	Adaptable tools for Ukraine
European Union	SmartAgriHubs, CAP support schemes, open agri-data platforms	Reduction of agrochemical use by up to 40%, yield increases up to 20%, reduction in GHG emissions	Open data platforms, sentinel-based satellite monitoring, regional digital innovation hubs
United States	USDA digital Ag programs, AI for earth, public–private partnerships	Forecast accuracy up to 90%, water usage reduced by 25%	AI-based weather analytics, cloud-based agro-analytical platforms
Israel	High-density IoT deployment, automation for arid-zone farming	Increased yields, reduced pest-related losses, irrigation efficiency improved by 30–40%	IoT-enabled drip irrigation systems, soil moisture sensors, autonomous weather stations

analytical depth in data-driven decision-making, and European systemic integration frameworks creates a strong foundation for developing a competitive and resilient national model of agro-digital transformation.

Building on both global best practices and domestic innovation initiatives, Ukraine is well-positioned to deploy digital agro-innovations as strategic tools for climate adaptation and sustainable productivity growth. These frameworks form the basis for scalable and interoperable digital systems tailored to the needs of both large-scale agricultural enterprises and smallholder farms.

The following subsection explores the practical application of these integrated approaches, offering actionable recommendations for their implementation in Ukraine’s agricultural sector under current and projected climate challenges.

4.2 Digital Agricultural Innovations in the Context of Climate Transformations: Practical Recommendations

A comprehensive assessment of climate-induced challenges, coupled with the evaluation of emerging digital technologies, clearly indicates the need for a systemic technological transformation of the agricultural sector. Global and regional climate shifts—including rising average temperatures, more frequent extreme weather events, soil degradation, and water regime instability—are intensifying the vulnerability of agricultural production systems, undermining productivity, economic viability, and ecological resilience.

In this context, digital agricultural innovations are becoming strategic tools for climate adaptation. This transformation is not limited to technological upgrades

but involves the development of a fundamentally new data-driven and automation-enabled management paradigm based on the principles of precision agriculture.

The integration of key digital technologies—satellite monitoring, Internet of Things (IoT), artificial intelligence (AI), and unmanned aerial vehicles (UAVs/agrodrone)s—enables the operation of an intelligent agricultural ecosystem. This integration supports the following capabilities:

Real-time field monitoring using high-resolution multispectral data and sensor networks;

Detection of plant stress, nutrient deficiencies, and disease outbreaks via AI-powered image and sensor data analysis;

Optimization of fertilizer, water, and pesticide application volumes to reduce overuse and environmental impact;

Reduction in resource consumption, including agrochemicals, fuel, and labor costs;

Enhanced environmental safety through lower chemical input loads and reduced mechanical disturbance of soil.

The growing role of artificial intelligence also extends to agricultural waste management, enabling the integration of precision farming with circular economy principles. As demonstrated by Linde et al. [15], AI algorithms can predict, classify, and manage agricultural by-products, reducing environmental impact and improving resource efficiency [19].

Our findings also highlight the value of reusing grain production residues—such as straw, husks, and stem biomass—as renewable energy sources. This approach reduces the environmental footprint of production processes and contributes to the energy independence and sustainability of agri-enterprises [20].

Digital transformation holds particular relevance for Ukraine amid the current wartime and post-war recovery context. It addresses not only the need to restore productivity but also to enhance long-term sustainability, climate resilience, and ecological balance in agricultural systems. Within this framework, the deployment of agrodrone)s—known for their high precision, mobility, and adaptability under complex field and weather conditions—plays a pivotal role in implementing the smart farming concept.

Based on the individual advantages of digital tools and the current national context, the following practical recommendations are proposed for Ukrainian farms:

- Phase-wise investment in digital infrastructure, starting with high-impact tools such as agrodrone)s, soil moisture sensors, and cloud-based agro-analytical platforms;
- Establishment of local digital monitoring systems that combine satellite imagery, IoT networks, and UAV data streams to support operational decisions;
- Development of public–private partnerships through agritech clusters and pilot demonstration projects to promote technology transfer and localization;
- Engagement of qualified personnel in precision agriculture and data analytics, or access to targeted training programs on digital platforms and AI-assisted tools;

- Utilization of national and international funding mechanisms, including technical assistance programs focused on sustainable agriculture, climate adaptation, and green transformation.

These recommendations reflect a scalable approach for accelerating digital transformation in Ukraine's agri-food system, particularly under the constraints and opportunities created by the ongoing recovery process.

For clarity and operationalization, the core recommendations are summarized in Table 4.

Digital agro-innovations should be conceptualized as a strategic framework for achieving climate resilience, economic efficiency, and environmental sustainability in the agricultural sector. In the context of Ukraine's transformational development and in response to global food security imperatives, the digitalization of agriculture is no longer optional—it is essential. The deployment of satellite monitoring, artificial intelligence (AI), Internet of Things (IoT), and agricultural drones enables

Table 4 Practical recommendations for implementing digital agro-innovations in Ukraine (based on [12, 18, 19])

Direction	Recommended practices	Expected impact
Agrodrones	• Utilize drones for spot spraying and variable-rate fertilization; • Deploy multispectral cameras to detect plant stress zones; • Generate digital field maps for zone-based management	• Reduction in agrochemical usage by 20–30%; • Improved resource allocation; • Yield enhancement
Satellite monitoring	• Integrate remote sensing services (Sentinel, Landsat, EOSDA); • Use zonal imagery for sowing and irrigation planning; • Apply early warning systems for climate risk detection	• Timely reaction to abiotic stressors; • Improved yield forecasting; • Lower crop losses
IoT Sensors & Automation	• Install soil and microclimate sensors (humidity, pH, temperature); • Deploy local weather stations for short-term forecasting; • Integrate smart irrigation systems responsive to sensor input	• Water-use optimization; • Increased input efficiency; • Reduced energy consumption
Artificial Intelligence (AI)	• Apply AI-driven platforms (e.g., AgroScout, CropX, Microsoft AI for Earth); • Integrate AI with UAVs for targeted field operations; • Upskill staff in agri-data interpretation and digital tools	• Automated, data-informed decision-making; • Lower dependence on manual intervention; • Enhanced risk forecasting

the construction of flexible, adaptive, and sustainable agri-production systems. This integrated approach contributes not only to efficient resource utilization, but also to cultivating a culture of innovation and laying the foundation for a new agricultural economy—digital, sustainable, and climate-adapted.

5 Conclusions

5.1 Synopsis

This study identifies a broad spectrum of climate risks that directly impact the Ukrainian agricultural sector, including thermal stress, water regime instability, and extreme weather events. In response, it substantiates the potential of digital agro-innovations as strategic instruments for climate adaptation. The work provides a functional overview of key technologies—satellite remote sensing, IoT infrastructures, AI-driven analytics, and UAV systems—and emphasizes the value of their integration into a unified, intelligent agroecosystem.

Particular attention is given to agrodrones due to their high precision, operational flexibility under variable climate conditions, and multifunctionality in monitoring, spraying, and geospatial mapping. The proposed integrated digital agroecosystem framework establishes a foundation for building climate-resilient, resource-efficient, and environmentally balanced agricultural models.

Importantly, unlike previous studies that explored individual technologies in isolation, this research introduces a novel systems-based approach by conceptualizing the synergistic interaction between digital tools as a comprehensive, adaptive response to climate risks. This constitutes the primary scientific novelty of the work.

This systemic vision of digital agricultural transformation can serve as the basis for the formulation of national strategies for sustainability, digital policy roadmaps, and climate adaptation programs. The findings contribute to the analytical and methodological groundwork for future research in smart farming, the prototyping of digital platforms, and the enhancement of innovation support mechanisms in Ukraine's agri-sector. The practical significance of these results lies in their potential to increase productivity, improve environmental performance, and strengthen the resilience of Ukrainian agriculture under conditions of war and recovery.

5.2 Further Research

The study opens several pathways for continued scientific and applied investigation. A critical next step involves empirical evaluation of the effectiveness of digital technologies in specific regions of Ukraine, characterized by diverse climatic zones, infrastructure levels, and technology access.

One key research priority is the modeling of adaptive agro-digital ecosystems under various climate change scenarios. This will inform the development of long-term strategic models to enhance the resilience and adaptive capacity of agricultural systems.

A central direction for further work is the development and calibration of AI algorithms tailored to localized agricultural datasets. These tools must ensure high predictive accuracy, adaptability to regional conditions, and alignment with site-specific agronomic features.

Additionally, there is a need to explore the economic feasibility of digitalization for small and medium-sized farms, which represent the most vulnerable segment of the agri-food sector. This includes conducting cost–benefit analyses, designing scalable solutions, and assessing their socio-economic implications.

Equally important is the institutional and regulatory alignment necessary to support the integration of digital agricultural technologies into national agricultural policy. This includes legislative frameworks, financial incentives, public–private partnerships, and education initiatives aimed at capacity building.

In summary, this work contributes not only to the systematization of current digital solutions but also to the conceptual and methodological foundation for future research. By clearly articulating new scientific directions and practical development pathways, the study reinforces the transformative potential of digital innovations in Ukraine’s agricultural sector and supports the emergence of an adaptive, climate-smart, and innovation-driven agricultural model.

References

1. Sharma V, Tripathi AK, Mittal H (2022) Technological revolutions in smart farming: current trends, challenges & future directions. *Comput Electron Agric.* <https://doi.org/10.1016/j.compag.2022.107217>
2. Karunathilake EM, Le AT, Heo S, Chung YS, Mansoor S (2023) The path to smart farming: Innovations and opportunities in precision agriculture. *Agriculture.* <https://doi.org/10.3390/agriculture13081593>
3. Das S (2024) Transforming agriculture: Harnessing robotics and drones for sustainable farming solution. *J Exp Agric Int* 46(7):219–231. <https://doi.org/10.9734/jeai/2024/v46i72577>
4. Kuchmiiova TS, Moroz TO, Sheshunova AV (2023) Use of artificial intelligence in agriculture. *Mod Econ* 39:69–74. [https://doi.org/10.31521/modecon.V39\(2023\)-10](https://doi.org/10.31521/modecon.V39(2023)-10)
5. Vasylykovska K, Andriienko O, Malakhovska V (2023) Analysis of the energy efficiency of agricultural drones in the system of precision agriculture. *Des, Prod Exploit Agric Mach* 53:131–138. <https://doi.org/10.32515/2414-3820.2023.53.131-138>
6. Yan X, Zhou Y, Liu X, Yang D, Yuan H (2021) Minimizing occupational exposure to pesticide and increasing control efficacy of pests by UAV application on cowpea. *Appl Sci* 11:9579. <https://doi.org/10.3390/app11209579>
7. EOS (2024) Zmina klimatu ta silske hospodarstvo. <https://eos.com/uk/blog/zmina-klimatu-ta-silske-hospodarstvo/>. Accessed 19 Jan 2025
8. FAO (2022) Ukraine: impact of the war on agriculture and rural livelihoods—findings of a nation-wide rural household survey, December 2022. Rome. <https://doi.org/10.4060/cc3311en>
9. European Environment Agency (n.d.) Environmental glossary. <https://www.eionet.europa.eu/gemet/uk/concept/2997>. Accessed 19 Mar 2025

10. Rusan VM, Zhurakovska LA, Zhalilo YA et al (2024) Perspectives for the development of Ukraine's agricultural sector under climate change: analytical report. NISS, Kyiv. <https://doi.org/10.53679/NISS-analytrep.2024.09>
11. Kucher A (2017) Adaptation of agricultural land use to climate change. *Agric Resour Econ: Int Sci E-J* 3(1):119–138. www.are-journal.com. Accessed 19 Jan 2025
12. Zamlynskyi V, Shabatura T, Zamlynska O, Borysevych E (2024) Perspective chapter: exploring the possibilities and technologies of the digital agricultural platform. *IntechOpen*. <https://doi.org/10.5772/intechopen.112358>
13. Agravery (n.d.) Drony u silskomu hospodarstvi abo yak pochynalosia tochno zemlerobstvo. <https://agravery.com/uk/posts/show/droni-u-silskomu-gospodarstviabo-ak-pocinalosa-tocne-zemlerobstvo>. Accessed 22 Jan 2025
14. Ivezić A, Trudić B, Stamenković Z et al (2023) Drone-related agrotechnologies for precise plant protection in Western Balkans: applications, possibilities, and legal framework limitations. *Agronomy* 13(10):2615. <https://doi.org/10.3390/agronomy13102615>
15. Linde N, Balian A, Shabatura T, Gryshova I, Yakovenko A, Hnatieva T (2024) Artificial intelligence in waste management in the context of implementing circular economy. *Grassroots J Nat Resour* 7(3):s149–s172. <https://doi.org/10.33002/nr2581.6853.0703ukr08>
16. Voynarenko M., Dzhezdzhula V., Yepifanova, I. (2016). Modelling the process of making decisions on sources of financing of innovation activity. *Econ Ann-XXI* 160(7–8):126–129. <https://doi.org/10.21003/ea.v160-25>
17. Yaremko SA, Kuzmina EM, Savina NB, Yepifanova IY, Gordiichuk HB, Mussayeva D (2022) Forecasting business processes in the management system of the corporation. *Informatyka, Automatyka, Pomiary w Gospodarce i Ochronie Środowiska* 12(4):53–59. <https://doi.org/10.35784/iapgos.3249>
18. Lelechenko L (2024) Vilnyi dostup ukrainskym fermeram do EOSDA crop monitoring. <https://eos.com/uk/blog/vilnyi-dostup-ukrainskym-fermeram-do-eosda-crop-monitoring/>. Accessed 19 Mar 2025
19. Linde N, Balian A, Shabatura T, Gryshova I, Petrenko O, Shevchenko A (2024) Agricultural technologies as a tool for integrating AI into the agricultural infrastructure of Ukraine. *Grassroots J Nat Resour* 7(3):s30–s51. <https://doi.org/10.33002/nr2581.6853.0703ukr02>
20. Shabatura T, Shevchenko A, Petrenko O, Halytskyi O (2024) The by-products of grain production as an ecological and economic solution to the energy crisis. *BIO Web Conf* 114:01019. <https://doi.org/10.1051/bioconf/202411401019>

The Impact of War on Foreign Direct Investment in Ukraine: An Empirical Assessment Using the Synthetic Control Method



Marianna Kichurchak , Iryna Paslavska , Nataliia Datskiv ,
and Vasyl Antoniv

Abstract The ongoing war presents a significant challenge to investment activity and the attraction of foreign capital into Ukraine's economy. This study employs the synthetic control method (SCM) to estimate the causal impact of the unprovoked war against Ukraine on foreign direct investment (FDI) inflows between 2014 and 2021. Net FDI inflows as a percentage of gross domestic product (GDP) serve as the outcome variable, while the predictor set includes thirty-three key macroeconomic, demographic, social, institutional, security, and technological indicators. The control group comprises twenty-seven carefully selected countries. The analysis period is divided into two phases: the pre-war period (2000–2013) and the active phase of the military conflict, including the year of the war's outbreak (2014–2021). Based on this framework, empirical and counterfactual (synthetic) trajectories of net FDI inflows to Ukraine were constructed. The placebo test confirms that the observed divergence between the actual and synthetic FDI flows is statistically significant. The Root Mean Square Prediction Error Ratio (RMSPE Ratio) reveals a substantial discrepancy between the actual and predicted values, underscoring the profound impact of the war on FDI inflows to Ukraine from 2014 to 2021. Additionally, the analysis identifies considerable volatility in both empirical and counterfactual FDI flows to Ukraine. This volatility is driven by the influence of political events such as election campaigns of 2004–2005, 2010, 2014, and 2018–2019, as well as global economic shocks, including the 2008–2009 financial crisis and the COVID-19 pandemic. The findings highlight the responsiveness of foreign investors to adverse

M. Kichurchak (✉) · I. Paslavska · N. Datskiv · V. Antoniv
Ivan Franko National University of Lviv, Lviv, Ukraine
e-mail: marianna.kichurchak@lnu.edu.ua

I. Paslavska
e-mail: iryna.paslavska@lnu.edu.ua

N. Datskiv
e-mail: nataliya.datskiv@lnu.edu.ua

V. Antoniv
e-mail: vasyl.antoniv@lnu.edu.ua

shocks and the effectiveness of investment policy measures implemented during the studied period. These insights shed light on the formulation of evidence-based policy initiatives aimed at economic recovery and revitalizing FDI inflows into Ukraine's economy in the post-war period.

Keywords Foreign direct investment · War · Synthetic control method · Treated unit · Control units · Counterfactual scenario · Placebo test · Investment policy · Economic recovery

1 Introduction

Foreign direct investment (FDI) serves as a crucial mechanism channel for attracting additional financial resources essential for structural economic transformation, technology transfer, enhanced competitiveness, and deeper integration into the global economic environment. The dynamics of FDI reflects investor responses to geopolitical developments, macroeconomic stability, and the institutional framework's effectiveness. Amid the ongoing war, investment flows to Ukraine may experience significant fluctuations. While increased risks could lead to capital outflows, adaptive strategies of businesses, international financial support, and government measures aimed at strengthening investment security may foster economic growth.

At the same time, the war significantly heightens risks related to the destruction of capital assets, disruptions in logistics and energy infrastructure, destabilisation of the business environment, the severance of economic ties between enterprises. These factors may discourage foreign capital inflows by exacerbating uncertainty and elevating the cost of business operations. Nevertheless, beyond these immediate challenges, the war also serves as a critical test of the resilience of state institutions and the capacity of businesses to adapt to an exceptionally volatile economic and social environment. Consequently, despite the adverse conditions, foreign investors may perceive Ukraine as a strategic market for post-war recovery, potentially sustaining or even intensifying investment activity. The confidence of international capital may be driven by assessments of future economic growth potential, access to resources, and favourable conditions for investment in strategically significant sectors. Given this evolving landscape, exploring trends in FDI inflows during the war is particularly pertinent for understanding investment dynamics and long-term economic recovery.

The war initiated by the Russian Federation against Ukraine in 2014, following the annexation of the Crimean Peninsula and the occupation of parts of the Donetsk and Luhansk regions, resulted in a significant decline in the country's investment attractiveness. In 2014, FDI stocks fell by 17.2 billion USD, reaching 49.83 billion USD—a 25.7% decline compared to 2013 [1]. In response to these economic challenges, between 2014 and 2021, the Ukrainian government implemented a series of initiatives aimed at improving the investment climate and attracting foreign capital into the national economy.

In December 2024, the National Investment Council was established as an advisory body to the President of Ukraine [2]. Earlier, in October 2016, the Office for Attracting and Supporting Investments, initially a temporary advisory body under the Cabinet of Ministers of Ukraine, was created. This institution was subsequently restructured into the State Institution Office for Attracting and Supporting Investments (UkraineInvest) in 2021 [3].

The primary objective of these institutions was to enhance foreign investors' confidence in Ukraine by promoting transparency in investor-government interactions, assisting international companies throughout the investment process, and highlighting the country's most promising sectors. Complementing these efforts, in 2017, Ukraine simplified procedures for employing foreign specialists to improve the investment climate [4]. Additionally, in 2021, new state policy measures were introduced to support large-scale investment projects [3]. As a result of these initiatives, between 2014 and 2021, Ukraine actively developed an institutional framework designed to attract and facilitate foreign investment in its national economy.

Policy initiatives aimed at improving the investment climate and facilitating foreign investment have been key drivers of Ukraine's economic development and integration into the European and global economy, even amid wartime challenges. Despite Russia's full-scale invasion of Ukraine in February 2022, the country has continued to strengthen its economic framework for investment and cooperation. As part of these efforts, the Advantage Ukraine digital platform was launched in September 2022, presenting approximately 500 investment project proposals across ten sectors of the national economy to attract potential foreign investors [5]. Against this backdrop, a quantitative assessment of the war's impact on FDI in Ukraine from 2014 to 2021 is crucial for understanding investment dynamics under conditions of external shocks, heightened uncertainty, and data limitations. Such an analysis also serves as a foundation for evidence-based policy decisions. The synthetic control method (SCM) provides a robust approach for estimating the causal effect of the war on FDI. Its application can enhance understanding of the challenges, opportunities, and long-term consequences of the war for foreign investment in Ukraine. Moreover, it offers valuable insights for policymakers seeking to stimulate investment both during the ongoing war and in the post-war recovery phase.

2 The Purpose of the Study

The purpose of the study is to explore the causal effects of the war on FDI inflows by comparing actual investment volumes in Ukraine with a counterfactual scenario that estimates their likely trajectory in the absence of military conflict.

This research is guided by three key hypotheses. Firstly, the dynamics of FDI inflows into the Ukrainian economy, measured as net FDI inflows as a percentage of gross domestic product (GDP), are expected to differ significantly between wartime and peacetime conditions. Secondly, FDI inflows may exhibit extreme volatility

following the onset of war due to shifts in factors such as the macroeconomic conditions, financial sector stability, technological development, export dynamics, political expectations, and election cycles. Thirdly, the counterfactual trajectory of FDI inflows to Ukraine's economy in the absence of war can be estimated using data from countries that have not experienced military conflicts. This approach enables a quantitative assessment of the potential investment losses attributable to war.

3 Study of Existing Solutions of the Problem

Scientific discussions on investment determinants, FDI attraction, and the creation of a favourable investment climate remain central to economic research, particularly in the context of fostering economic development, driving social change, and addressing urgent environmental challenges. Notably, foreign investments are highly sensitive to shifts in the socio-economic landscape, institutional frameworks, and sustainability requirements. Empirical studies employing the structural gravity approach provide valuable insights into these dynamics. For instance, Kox and Rojas-Romagosa [6] found that preferential trade agreements and bilateral investment agreements significantly increased bilateral FDI, while partnership and cooperation agreements led to an average of 30% rise in FDI flows. Similarly, Bruno et al. [7] applied a structural gravity model to analyse annual bilateral FDI data for nearly all countries from 1985 to 2018. Their findings indicate that EU accession boosts FDI inflows by approximately 60% from non-EU sources and 50% from EU member states.

Analysing a 25-year sample of developed and emerging economies prior to COVID-19 pandemic), Jardet et al. [8] employed a push–pull model, controlling for global and local factors, and found that political uncertainty had a significant and robust impact on FDI inflows. Examining the effects of crises on FDI, Saleh [9] identified two key impacts: indirect (macrolevel) and direct (microlevel). Furthermore, multinational enterprises respond to crises through three strategic approaches: flight, defence, and risky behaviour. Similarly, based on data from 43 countries spanning 1985–2022, Kapopoulos et al. [10] investigated the relationship between FDI inflows and geopolitical risks. Their findings indicate that foreign investors perceive high energy intensity of the economy and water shortages as significant deterrents to capital attraction. Accordingly, attracting FDI requires a comprehensive policy approach that accounts for environmental factors and the broader international landscape.

Beyond these determinants, a country's investment attractiveness is closely linked to the level of human capital development, as confirmed by empirical research. Brintseva [11], through a comparative analysis of human capital investment in Poland and Ukraine, found that Ukraine's human capital efficiency remains low due to negative economic dynamics and limited public expenditures on education and healthcare. Conversely, employing pooled regression models, Kichurchak [12] examined the role of investment factors in human capital accumulation across EU countries. The

study identified public expenditures on higher education, household income, employment prospects, and career opportunities as key determinants of this process. Further analysis by Kichurchak [13], using a system of simultaneous equations, confirmed the positive impact of public investment in education on human capital development in the EU countries.

An analysis of OECD country data from 1960 to 2010 by Pantelopoulos [14] highlighted the crucial role of human capital in shaping the policies and strategies of both host governments and multinational corporations. Specifically, secondary and higher levels of education facilitate the attraction of FDI, while advanced education influences the volume of foreign portfolio investment. Similarly, Dankyi et al. [15], using panel data from 1990 to 2017 for the ECOWAS region, examined the relationships between economic growth, human capital, FDI, CO₂ emissions, and urbanization. Their findings underscore the significance of human capital investment in enhancing the region's investment attractiveness. Overall, investments, especially FDI, can contribute positively to social development within a country, reinforcing the broader economic progress.

At the same time, a country's investment attractiveness is shaped by its institutional environment, human and natural resources, and the level of technological advancement. Harkavenko and Yershova [16] found that between 2000 and 2019, both direct and portfolio foreign investment in Ukraine did not facilitate structural economic transformations and the development of high-tech industries. Instead, these investments reinforced the raw material orientation of the national economy and exacerbated internal economic imbalances due to the significant inflow of speculative capital. Similar conclusions were drawn by Getzner and Moroz [17], who, using panel time-series data for 2003–2016, demonstrated that the impact of FDI on the economic performance of Ukrainian regions was limited. This was attributed to political instability, weak governance, the ongoing military conflict in the east of the country, and incomplete implementation of economic reforms. Furthermore, Cieřlik and Gurshev [18], employing the knowledge-capital model and bilateral panel data on direct investment from 140 partner countries for the period 2013–2017, identified access to cheap labour as the primary factor driving FDI inflows to Ukraine.

Given the critical role of investment, Bashynska et al. [19] developed a methodology for ranking the business activity of innovation-driven and investment-attractive enterprises in Ukraine, considering their territorial and sectoral characteristics. This approach facilitates a more precise assessment of regional and industry-specific investment attractiveness, enabling potential investors to make more informed capital allocation decisions. Paslavskaya et al. [20] examined FDI dynamics, sectoral distribution, and its contribution to Ukraine's GDP, emphasizing the necessity of increasing both overall FDI and investment in high-tech industries as key drivers of Ukraine's post-war economic recovery. Furthermore, based on econometric modelling, Kovtoniuk et al. [21] found that net FDI inflows to Ukraine increase by 0.075 USD and 0.004 USD with rising imports and gross capital formation in the key investor countries, primarily EU member states. However, during wartime, these inflows remained at the lower bound of the forecast corridor, reflecting the realisation of a pessimistic scenario for Ukraine's economy. Consequently, foreign investment in Ukraine has

largely been resource-oriented, with investors maintaining a cautious stance due to substantial geopolitical and macroeconomic risks.

Moreover, wars act as significant disruptive forces, complicating efforts to attract FDI into national economies. Examining FDI inflows in 73 countries affected by civil wars, Joshi, and Quinn [22] found that governments aiming to maximise FDI for post-war economic recovery should prioritize negotiating and implementing comprehensive peace agreements. Similarly, Klosek [23], utilizing random-effects logistic models and UCDP data on interventions from 2001 to 2009, confirmed that protecting existing FDI and upholding prior arms trade agreements significantly increases the likelihood of state military intervention in civil wars. Employing a survival analysis to examine the FDI recovery in a global post-war sample (1970–2008), Moore [24] found that foreign aid can serve as an indirect indicator of persistent challenges in the private investment climate. Given the constraints on international financial resources and the unfavourable environment for foreign investment, Farzanegan [25] applied the SCM, demonstrating that international banking and energy sanctions (2012–2015) led to an average annual decrease of 117 USD in Iran's military spending per capita. Similarly, Nguyen et al. [26], using econometric modelling, established that a 1% increase in areas contaminated by unexploded ordnance reduces the density of FDI firms in Vietnam by 0.69% at the district level. These findings underscore the complexities of attracting foreign investment to war-affected economies, emphasizing the need to account for both economic and political factors, as well as the long-term consequences of conflict-related destruction.

In this regard, academic research on investment processes, their determinants and their effects on economic and social development has significantly advanced the understanding of the strategies for attracting foreign investors and enhancing investment policies. These insights are crucial for strengthening national economic resilience and mitigating the consequences of military conflicts. Quantitative methods have been extensively applied to examine the role of foreign investment in human capital development, economic growth, and post-war recovery across various countries, including Ukraine. In this context, the present study employs the SCM to identify key trends and quantitatively assess the impact of the war on FDI inflows to Ukraine from 2014 to 2021.

4 Methods of Study

The SCM is a powerful tool for assessing the impact of exogenous shocks on macroeconomic indicators and trends. It enables the construction of a synthetic trajectory that simulates the behaviour of a treated unit in the absence of intervention. Comparing this trajectory with actual empirical dynamics allows for identifying and quantifying the causal effect. The synthetic trajectory is built using a control group of units that closely resemble the treated unit in the pre-intervention period but have not been exposed to the intervention. An optimal linear combination of these control units recreates the counterfactual scenario, making the SCM particularly

suitable for analysing unpredictable events. Its algorithmic transparency in determining weighting coefficients ensures high interpretability and reliability, making it a valuable tool for exploring complex political, economic, and social processes.

The principles of applying the SCM are detailed in the works of Abadie [27], Abadie and Vives-i-Bastida [28], and Abadie and Zhao [29]. A review of academic publications from 2015 to the present confirms the high effectiveness of the SCM in quantifying the macroeconomic impact of political and economic shocks, structural changes, public policy reforms, and global crises. For instance, Pinotti [30] employed this approach to analyse the economic consequences of terrorist attacks, demonstrating their significant impact on macroeconomic stability. Similarly, Cavallo et al. [31] utilized the SCM to measure the long-term effects of natural disasters on regional GDP. Moreover, the SCM is widely utilised to assess the effectiveness of monetary and fiscal policies, the impact of trade sanctions, changes in employment, and the socioeconomic effects of pandemics—further underscoring its versatility and reliability in macroeconomic analysis.

Additionally, academic research has examined the impact of Russia's war against Ukraine on macroeconomic dynamics in both Ukraine and European countries. Audretsch et al. [32] analysed the war's effects on entrepreneurship in Ukraine and Russia, employing the SCM to construct counterfactual scenarios of business development in the absence of military conflict. Their findings reveal a significant negative impact on self-employment and a decline in the number of small and medium-sized enterprises. By comparing actual economic outcomes with a synthetic control group, the study highlights substantial deviations caused by the war. Similarly, Kichurchak et al. [33] assess the war's impact on higher education in Ukraine, utilizing the SCM to model a counterfactual trajectory based on countries with comparable educational systems and macroeconomic conditions. Their results indicate a sharp divergence between actual and synthetic trends, underscoring the war's detrimental effects on the accessibility and quality of higher education.

Consequently, the effectiveness of the SCM is reinforced by numerous empirical studies demonstrating its ability to assess the impact of various exogenous shocks on macroeconomic indicators. In particular, research on the impact of Russia's war against Ukraine on business and higher education highlights its relevance for analysing the consequences of armed conflicts. Given that FDI is highly sensitive to political and economic stability, applying the SCM will enable an objective evaluation of the impact of war and revolution on investment flows in Ukraine. By comparing actual FDI dynamics with a synthetic trajectory, this approach facilitates a precise quantification of the causal relationship between exogenous shocks and fluctuations in FDI volumes.

The applicability of the SCM for analysing FDI during periods of war and revolution stems from the abrupt and unpredictable changes these events introduce into the economic environment, making traditional econometric methods less effective. By constructing a counterfactual scenario, the SCM enables an accurate estimation of the net effect of such shocks. Moreover, the method offers a high degree of interpretability, as it allows for the identification of control group countries that

most significantly influence the formation of the synthetic control. This is particularly valuable for FDI analysis, as it enables the identification of key factors shaping foreign investors' decisions in an uncertain environment.

Another advantage of the SCM is its intuitive graphical representation of the counterfactual scenario, which is crucial for interpreting results in political and economic analysis. The effectiveness of the SCM largely depends on the quality of the data used to construct the synthetic scenario. The key data requirements include:

1. Sufficient number of observations. To ensure statistical reliability, there must be an adequate number of observations for both the treated country and the countries in the donor pool.
2. Relevance of variables. The variables used to construct the synthetic control should be significant in explaining FDI dynamics. These may include macroeconomic indicators, political factors, and institutional characteristics.
3. High-quality data. The data should be accurate, reliable, and comparable across countries.
4. No direct exposure of donor pool countries to the shock. The countries in the donor pool should not be directly affected by the shock under study to ensure a valid counterfactual.
5. Relevance of predictors. The predictors should have a significant impact on the treated unit.

The mathematical foundation of the SCM relies on several key assumptions that ensure the accuracy and validity of the causal impact assessment. The first assumption is the exogeneity of the intervention, which means that the studied event (intervention) should not correlate with the prior values of the studied indicator [34]. Formally, this implies that the impact of the intervention at the moment of its occurrence is independent of the variable's values in the pre-intervention period. A violation of this assumption may lead to endogeneity, distorting the estimated effect. For instance, if the studied event was anticipated, its effects might have been partially reflected in the indicator's prior values, making it difficult to isolate the true causal impact of the intervention.

The second assumption is the absence of between-unit interaction [29]. It means that countries in the control group must remain unaffected by the intervention through direct or indirect links with the treated country. A violation of this assumption may result in spillover effects, leading to contamination of the control group and biasing the estimated impact. For instance, in assessing the economic consequences of war, it is essential to exclude countries with strong economic ties to the treated unit or those directly affected by the conflict to ensure the validity of the counterfactual scenario.

A critical challenge in the SCM implementation is the selection of an appropriate donor pool. The validity of the counterfactual scenario, and consequently the reliability of the results, depends on the quality of the donor pool. Its selection may rely on expert judgment or statistical criteria to ensure methodological rigor. It is essential to balance the number and quality of donor units, ensuring that they exhibit structural similarity to the treated unit both in the pre-intervention period and consistently over the entire analysis horizon. Furthermore, addressing donor pool heterogeneity

is crucial, requiring methods that account for structural differences among donor units.

The synthetic control model is constructed using a set of units indexed by $i = 1, \dots, J + 1$, where $J + 1$ represents the total number of units in the study. Among these, one unit (the treated unit) has been exposed to the intervention, while the remaining J units serve as the control group. The analysis covers a time period indexed by $t = 1, \dots, T$.

The intervention period is denoted by $T_0 + 1$, where $1 \leq T_0 < T$, and T_0 represents the number of pre-intervention periods. Let Y_{it}^N denote the value of a given indicator for unit i in period t before the intervention (N —pre-intervention), and let Y_{it}^I represent its value after the intervention (I —post-intervention). We assume that for all $t = 1, \dots, T_0$, the relationship $Y_{it}^I = Y_{it}^N$ holds. This implies that before the intervention, the treatment does not influence the indicator values for the treated unit, nor does the analysed factor affect the control group units.

The effect of exposure is denoted by α_{it} . Since the treatment effect is observed only for the first unit ($i = 1$) for $t > T_0$, the objective of the SCM is to estimate α_{1t} , defined as $Y_{1t}^I - Y_{1t}^N$, where Y_{1t}^N represented the counterfactual outcome that would have occurred in the absence of the intervention. This counterfactual outcome can be modelled using a factor model:

$$Y_{it}^N = \delta_t + \theta_t \cdot Z_i + \lambda_t \cdot \mu_i + \varepsilon_{it}, \quad (1)$$

where δ_t represents the time-specific effect that influences all units in period t , Z_i is a vector of observed covariates characterising unit i , and θ_t is a vector of coefficients that capture the impact of these covariates on the outcome variable at time t . The term λ_t denotes a vector of unobserved latent factors that account for heterogeneity across units, while μ_i is a vector of factor loadings for unit i , indicating the extent to which it is influenced by these latent factors. Finally, ε_{it} is an error term that captures random fluctuations not explained by the model.

Given the assigned weights, the model (1) can be reformulated as follows:

$$\sum_{j=2}^{J+1} w_j Y_{jt}^N = \delta_t + \theta_t \sum_{j=2}^{J+1} w_j Z_j + \lambda_t \sum_{j=2}^{J+1} w_j \mu_j + \sum_{j=2}^{J+1} w_j \varepsilon_{jt}, \quad (2)$$

where w_j is the j -th value of the weight vector $W = (W_2, \dots, W_{J+1})$, such that for $\forall j: w_j > 0 \wedge \sum w_j = 1$.

The task of the SCM is to find the optimal set of weights ($w_2', \dots, w_{J+1}'$) such that the linear combination of the investigated indicator's value for donor units in the pre-intervention period closely approximates the actual values for the treated unit. The weights are chosen to ensure that the synthetic control accurately reproduces the covariate values of the treated unit, thereby minimizing the discrepancy between the synthetic and actual pre-intervention trajectories.

Doudchenko et al. [35] demonstrate that if model errors are small, the difference between the predicted value of the indicator for the synthetic control and

the actual value of the studied indicator in the pre-intervention period will be negligible. Accordingly, the effect of the intervention can be estimated as follows ($\alpha_{it} = Y_{it}^I - Y_{it}^N$):

$$\hat{\alpha}_{it} = Y_{it} - \sum_{j=2}^{J+1} \hat{w}_j Y_{jt}. \quad (3)$$

The calculation of the optimal weights involves minimising the norm with respect to the weight vector W , where X_I is the vector of covariate values for the treated unit up to time $T_0 + I$, and X is the matrix of covariate values for the control units. Regardless of the researcher's choice of a positively defined matrix V , the optimized norm is expressed as follows:

$$\|X_I - X_0 W\|_V = \sqrt{(X_I - X_0 W) \hat{V} (X_I - X_0 W)}. \quad (4)$$

To calculate the final value of V , an external optimisation of the parameter V is performed using a discount factor β that increases the weight of recent observations. This optimisation is formulated as follows:

$$\sum_{t=1}^T \beta^{T-t} \left(Y_{It} - \sum_{j=2}^{J+1} \hat{w}_j(V) Y_{jt} \right)^2 \rightarrow \min, \quad (5)$$

where $\hat{w}$ is the vector of optimal weights obtained in the previous step.

The parameters of the vector V indicate the relative importance of each factor in replicating the dynamics of the target variable. Higher values signify a greater contribution of the respective factor to the formation of a weighted combination of donor units. The sign of the parameter determines the direction of influence: a positive value implies that an increase in the relevant factor enhances the similarity between the treated unit and the synthetic control, whereas a negative value suggests an inverse relationship between the factor and the treated unit. Negative weights may indicate that the treated unit exhibits atypical characteristics compared to other units in the donor pool, potentially signalling the presence of unaccounted-for factors influencing the outcome under study. Additionally, negative weights can serve to offset the effects of other predictors that positively impact the outcome, thereby improving the accuracy of the synthetic control's pre-intervention trajectory. However, their presence complicates the interpretation of the SCM results, as they may contradict intuitive expectations. In such cases, a careful reassessment of the covariate selection and a robustness check of the results are necessary to ensure the validity of the findings.

The weights of the vector W determine the contribution of each donor unit to the synthetic control. Higher weight values indicate a greater similarity between the

corresponding control unit and the treated unit, as well as a stronger influence on the synthetic combination.

Assessing the validity of results within the SCM is a crucial step in verifying the reliability of the conclusions and identifying potential sources of error. Common validation approaches include graphical analysis—comparing the trajectories of the actual and synthetic units before and after the intervention—and residual analysis, such as examining the gap between empirical and predicted values. The accuracy of the approximation is assessed using statistical measures like root mean square error (RMSE) and mean absolute error (MAE). Additionally, structural break tests help identify significant changes in the trend of the treated unit post-intervention, while the p -value of the estimated effect assesses its statistical significance.

To assess the robustness of the results, a sensitivity analysis is conducted to account for variations in model specification. Specifically, the stability of the estimates is examined under different configurations of the control group (including or excluding specific donor units), alternative time periods, modifications of weight coefficients, and adjustments in the modelling of unobserved effects. Additionally, analogous calculations are performed for control units unaffected by the intervention to ensure the validity of effect identification.

To ensure a comprehensive validation of the results, robustness checks are complemented by statistical significance testing. One of the most widely used approaches in the SCM is the placebo test. This methodology involves estimating the intervention effect for units that were not exposed to the treatment, thereby verifying the validity of effect identification. The core idea behind placebo tests is to assess the distribution of estimated effects among control units known to have experienced no intervention. To achieve this, a separate synthetic control is constructed for each donor unit within the control group, and the difference between actual and predicted values is computed. This process generates an empirical distribution of effects for the control units, which serves as a benchmark for comparison with the observed effect in the treated unit. If the magnitude of the intervention effect substantially deviates from this distribution, it indicates statistical significance, reinforcing confidence in the findings and minimizing the likelihood that the results are driven by random fluctuations.

Moreover, one of the key advantages of placebo tests is their non-parametric nature, which does not rely on strict assumptions regarding the distribution of residuals or the dependency structure between variables. This flexibility enhances the robustness of the test against deviations from normality, which are common in real-world empirical data, and provides additional verification of the reliability of the estimates. As a result, placebo tests serve as an additional validation tool, reinforcing the reliability of the obtained estimates and increasing confidence in the causal interpretation of the intervention effect.

Despite its advantages, the SCM also has several limitations and challenges. One of the primary difficulties lies in selecting an appropriate control group [29]. Inaccuracies in this selection process can lead to significant issues that compromise the reliability and interpretation of the study results.

Firstly, if the control group does not adequately represent the counterfactual scenario—i.e., the situation that would have occurred in the absence of the intervention—the estimated effect may be biased. For example, if the donor pool includes countries that have experienced similar shocks or differ substantially in key characteristics, the synthetic control may fail to accurately reproduce the trajectory of the treated unit. This can result in either an overestimation or underestimation of the intervention effect. Secondly, an improperly chosen control group can lead to increased variance in estimates, reducing the precision of the analysis and making it difficult to detect statistically significant effects. This is particularly problematic when studying events with relatively small impacts, as a lack of statistical power may obscure real effects. Finally, an inaccurate control group selection may lead to incorrect conclusions regarding causality. If the synthetic control does not sufficiently capture the underlying dynamics of the indicator under study, observed changes might be mistakenly attributed to the intervention rather than to other external factors.

To enhance the accuracy of SCM results, various advanced approaches to constructing a counterfactual scenario have been proposed in the academic literature. These include the use of negative weights [36], which allow for greater flexibility in matching the treated unit's pre-intervention trajectory, and the construction of prediction intervals [37], which provide a measure of uncertainty around the estimated effect.

A methodologically rigorous approach with carefully selected covariates is employed to empirically assess the statistically significant negative impact of Russia's prolonged military aggression against Ukraine on FDI dynamics from 2014 to 2021. Given the macroeconomic volatility and political instability characterising this period, the careful selection of covariates is crucial to ensuring the validity of the synthetic control method.

The application of the SCM is justified by the exogeneity of military aggression, which aligns with the method's key assumption. A major advantage of the SCM lies in its ability to construct a counterfactual scenario of FDI dynamics in the absence of war—an outcome unattainable through traditional econometric approaches. Additionally, the method accounts for the heterogeneity of donor pool countries by weighting their characteristics to generate a synthetic control that closely approximates Ukraine in the pre-intervention period. The graphical representation of SCM results highlights the divergence between observed FDI dynamics and the counterfactual scenario, facilitating a more precise interpretation of the war's impact.

Moreover, the SCM allows for the assessment of the impact of other macroeconomic shocks, such as revolutions and the COVID-19 pandemic, on FDI in Ukraine, enabling the identification of their relative and cumulative effects on the country's investment attractiveness. This approach enhances the understanding of the transformation of Ukraine's economic structure and investment climate under the influence of armed conflict, which is crucial for developing recovery strategies and attracting capital in the post-war period. The findings may hold practical implications for shaping public policies aimed at restoring Ukraine's investment appeal and attracting FDI for economic development, as well as for providing recommendations to international institutions supporting Ukraine.

5 Study Results

The unprovoked armed aggression of the Russian Federation against Ukraine, which began in 2014, has been a key driver of transformation of the country's investment climate, significantly influencing FDI dynamics. To conduct a comprehensive analysis of the macroeconomic, demographic, social, institutional, security, and technological determinants of FDI and to construct a robust synthetic control, 33 predictor variables were selected, encompassing a broad spectrum of economic and institutional factors. The appropriateness of these predictors is substantiated by their extensive coverage, ensuring a detailed representation of the key forces shaping FDI inflows. Specifically, the predictor framework includes indicators of macroeconomic stability, economic structure and competitiveness, the financial sector and regulatory environment, political risks and security, technological development and human capital, infrastructure and energy efficiency, as well as demographic and social factors. This multidimensional approach minimizes the risk of omitting critical variables and enables a comprehensive assessment of the determinants of FDI.

The selected predictors are highly relevant for explaining FDI dynamics. Macroeconomic indicators, such as GDP, inflation and the trade balance, serve as key determinants of investment decisions, while institutional factors, including political stability, regulatory quality and corruption, are critical for assessing investment risks. The use of verified World Bank data further enhances the reliability and validity of the empirical results. The dataset, obtained from the World Bank website [38] is as comprehensive as possible, allowing for the construction of a high quality synthetic control. As the result, the architecture of 33 predictors provides an accurate representation of the key determinants of FDI, ensuring the correct application of the SCM. Table 1 presents a classification of all selected predictors according to their respective areas of influence.

The period from 2000 to 2013 serves as the pre-intervention phase for constructing the synthetic model, while 2014 marks the intervention year. The post-intervention period, spanning from 2014 to 2021, is used to monitor deviations and assess the impact.

The selected outcome variable, Y_{it} , denotes net inflows of FDI expressed as a share of gross domestic product. This indicator is essential for assessing the country's ability to attract foreign capital and reflects the amount of investment resources flowing into the economy. Given the impact of exogenous factors such as armed conflicts or economic recessions, this measure is highly sensitive to geopolitical, economic, and social shifts, providing insight into how these transformations influence investment attractiveness.

The relevance of this variable lies in its ability to capture not only the volume of investment flows but also the level of international investor confidence, the competitiveness of the national economy, and its resilience to external shocks. In the case of Ukraine, this indicator facilitates an analysis of the impact of Russian aggression on foreign capital inflows while also assessing the effectiveness of government policies aimed at enhancing the investment climate.

Table 1 Classification of factors by their impact on foreign direct investment

Groups	Indicators
Macroeconomic stability and financial balance (Represents the overall economic environment, financial stability and balance of foreign economic flows)	Current account balance, (% of GDP)
	Inflation, GDP deflator (annual %)
	Consumer price index (2010 = 100)
	Gross capital formation (% of GDP)
	Gross domestic savings (% of GDP)
	Interest payments (% of revenue)
	Net foreign assets (current LCU)
	Total reserves (months of imports)
Economic structure and competitiveness (Reflects economic growth dynamics, structural composition, and investment attractiveness)	GDP growth (annual %)
	GDP per capita growth (annual %)
	GDP per person employed (constant 2021 PPP \$)
	Industry (including construction), value added (% of GDP)
	Trade (% of GDP)
	Exports of goods and services (% of GDP)
	Imports of goods and services (% of GDP)
Financial sector and regulatory support for business (Reflects access to financial resources and the business environment)	Domestic credit to private sector by banks, % of GDP
	Expense (% of GDP)
	Tax revenue (% of GDP)
	Cost of business start-up procedures (% of GNI per capita)
Political risks, security, and military expenditures (Reflects political stability, corruption risks, and military threats)	Political stability and absence of violence/terrorism
	Regulatory quality
	Control of corruption: percentile rank
	Military expenditure (% of GDP)
Technological development and human capital (Reflects research potential, innovation development, education level, and healthcare)	Research and development expenditure (% of GDP)
	Individuals using the Internet (% of population)
	ICT goods exports (% of total goods exports)
	High-technology exports, % of manufactured exports
	Government expenditure on education, total (% of GDP)
	Current health expenditure (% of GDP)
Energy efficiency	Electric power consumption (kWh per capita)

(continued)

Table 1 (continued)

Groups	Indicators
Demography and social stability (<i>Reflects the level of urbanisation and social inequality</i>)	Urban population (% of total population)
	Gini index
	Life expectancy at birth, total (years)

Source Classified by the authors based on the World Bank [38]

To ensure the validity and reliability of the results, the donor pool for analysing the impact of Russian aggression on FDI in Ukraine was selected in two stages.

At the first stage, a broad donor pool was constructed, comprising 32 countries from various regions worldwide: Albania, Bulgaria, the Slovak Republic, Serbia, Slovenia, Greece, Mongolia, Morocco, Poland, Romania, Estonia, Latvia, Lithuania, Tajikistan, Jordan, North Macedonia, Kazakhstan, the Kyrgyz Republic, Chile, Cyprus, Croatia, the Czech Republic, France, Germany, Hungary, Italy, Ireland, Malaysia, the Netherlands, Norway, Spain, and the United Kingdom. This wide selection minimized the risk of overlooking potentially relevant countries and ensured a diverse and heterogeneous control group. Crucially, none of the countries in the donor pool experienced large-scale revolutions or wars between 2014 and 2021 that could have distorted the analysis results. The only exception is Ukraine, which serves as the focus of this study.

The SCM was applied using the R statistical software package. The synthetic control analysis, conducted in the first stage using the donor pool of 32 countries and 33 predictors, yielded the results presented in Tables 2 and 3, as well as Figs. 1a and b. The mean squared prediction errors of the synthetic model are $MSPE(\text{loss } v) = 2.003$ and $MSPE(\text{loss } w) = 5.2172$. These results indicate that the model demonstrates limited adequacy, as the majority of countries in the control group have minimal impact, and the $MSPE(\text{loss } w)$ value of 5.2172 remains relatively high.

The analysis of the graphs (Fig. 1) reveals a marked positive deviation of actual FDI dynamics from the synthetic scenario following the Orange Revolution of 2004–2005. This pattern suggests a notable surge in investor confidence, likely driven by the announcement of a political trajectory toward European integration, democratic reforms, and economic liberalization initiatives. Specifically, the easing access to EU markets may have stimulated economic activity, thereby exerting positive influence on FDI inflows. The volatility of investment flows during the 2004–2005 period necessitated a reduction of the optimization window to ensure model accuracy.

Conversely, after 2014, a pronounced negative deviation of actual FDI from its synthetic control emerges, signaling a significant decline in investment potential. This downturn is attributable to the exogenous shock induced by the onset of military conflict.

At the second stage, the quality of the synthetic control was evaluated (Table 3) and the donor pool was adjusted accordingly. The selection criterion sought to reduce the

Table 2 Synthetic control results for the 32-country control group using 33 predictors

Predictor identifiers	Predictors	Solution. v	Control Group	Solution. w
Factor1	Net foreign assets (current LCU)	0.0240	Albania	$1.521 \cdot 10^{-7}$
Factor2	Current account balance, % of GDP	0.0487	Bulgaria	$9.278 \cdot 10^{-7}$
Factor3	Control of corruption: percentile rank	0.0258	Chile	$1.164 \cdot 10^{-7}$
Factor4	Domestic credit to private sector by banks (% of GDP)	0.0175	Croatia	$5.741 \cdot 10^{-7}$
Factor5	Expense (% of GDP)	0.0491	Cyprus	$1.921 \cdot 10^{-5}$
Factor6	Exports of goods and services (% of GDP)	0.0279	Czechia	$1.511 \cdot 10^{-6}$
Factor7	Electric power consumption (kWh per capita)	0.0287	Estonia	$4.664 \cdot 10^{-7}$
Factor8	GDP growth (annual %)	0.0500	France	$8.049 \cdot 10^{-6}$
Factor9	GDP per capita growth (annual %)	0.0354	Germany	$8.455 \cdot 10^{-6}$
Factor10	GDP per person employed (constant 2021 PPP \$)	0.0261	Greece	$2.574 \cdot 10^{-2}$
Factor11	Gross capital formation (% of GDP)	0.0246	Hungary	$9.362 \cdot 10^{-6}$
Factor12	Gross domestic savings (% of GDP)	0.0362	Ireland	$9.677 \cdot 10^{-7}$
Factor13	Imports of goods and services (% of GDP)	0.0373	Italy	$1.502 \cdot 10^{-6}$
Factor14	Industry (including construction), value added (% of GDP)	0.0255	Jordan	$8.525 \cdot 10^{-2}$
Factor15	Inflation, GDP deflator (annual %)	0.0136	Kazakhstan	$1.562 \cdot 10^{-6}$
Factor16	Trade (% of GDP)	0.0286	Kyrgyz Republic	$2.987 \cdot 10^{-1}$
Factor17	Consumer price index (2010 = 100)	0.0325	Latvia	$1.977 \cdot 10^{-7}$
Factor18	Gini index	0.0279	Lithuania	$3.921 \cdot 10^{-6}$
Factor19	Urban population (% of total population)	0.0412	Malaysia	$1.249 \cdot 10^{-7}$
Factor20	Research and development expenditure (% of GDP)	0.0370	Mongolia	$1.248 \cdot 10^{-7}$

(continued)

Table 2 (continued)

Predictor identifiers	Predictors	Solution. v	Control Group	Solution. w
Factor21	Interest payments (% of revenue)	0.0282	Morocco	$9.399 \cdot 10^{-8}$
Factor22	Individuals using the Internet (% of population)	0.0279	Netherlands	$1.045 \cdot 10^{-3}$
Factor23	Current health expenditure (% of GDP)	0.0136	North Macedonia	$3.044 \cdot 10^{-7}$
Factor24	Life expectancy at birth, total (years)	0.0217	Norway	$3.478 \cdot 10^{-1}$
Factor25	Government expenditure on education, total (% of GDP)	0.0270	Poland	$1.111 \cdot 10^{-6}$
Factor26	ICT goods exports (% of total goods exports)	0.0319	Romania	$2.667 \cdot 10^{-7}$
Factor27	High-technology exports (% of manufactured exports)	0.0278	Serbia	7.83710^{-2}
Factor28	Political stability and absence of violence/terrorism: estimate	0.0521	Slovak Republic	$8.877 \cdot 10^{-6}$
Factor29	Regulatory quality: estimate	0.0183	Slovenia	$1.539 \cdot 10^{-1}$
Factor30	Cost of business start-up procedures (% of GNI per capita)	0.0342	Spain	$9.221 \cdot 10^{-7}$
Factor31	Tax revenue (current LCU)	0.0391	Tajikistan	$8.301 \cdot 10^{-3}$
Factor32	Total reserves in months of imports	0.0244	United Kingdom	$8.080 \cdot 10^{-4}$
Factor33	Military expenditure (% of GDP)	0.0159		

Source Calculated based on the World Bank [38]

imbalance between Ukraine's pre-intervention characteristics and those of the potential control group countries, as reflected in the MSPE_pre value (Mean Square Prediction Error in the pre-intervention period). Additionally, MSPE_post and RMSPE_post (Mean Squared Percentage Error and Root Mean Squared Percentage Error in the post-intervention period, respectively) were considered to assess the accuracy of the synthetic control model after 2014.

Based on the analysis of the statistical data, the final donor pool was composed of countries with the lowest MSPE_pre values, indicating a high degree of similarity to

Table 3 Placebo test results for the synthetic control method (32 countries, 33 predictors)

Number	Country	P_value	MSPE_pre	MSPE_post	RMSPE_pre	RMSPE_post
	Ukraine	0.000	2.6585	9.6951	1.6305	3.1137
1	Albania	0.875	6.9195	$4.1056 \cdot 10^1$	2.6305	6.4075
2	Bulgaria	0.500	$3.8773 \cdot 10^1$	2.6794	6.2268	1.6369
3	Chile	0.750	1.9057	$2.3229 \cdot 10^1$	1.3805	4.8197
4	Croatia	0.375	1.0071	4.9777	1.0036	2.2311
5	Cyprus	1.000	$1.2267 \cdot 10^4$	$1.4518 \cdot 10^4$	110.7585	120.4889
6	Czechia	0.125	2.4901	2.1197	1.5780	1.4559
7	Estonia	1.000	$1.6995 \cdot 10^1$	$1.2024 \cdot 10^1$	4.1224	10.9654
8	France	0.250	$3.1631 \cdot 10^1$	1.2058	0.5624	1.0981
9	Germany	0.375	7.5847	2.7173	2.7540	1.6484
10	Greece	0.375	1.4378	1.4793	1.1991	1.2163
11	Hungary	0.875	$1.9491 \cdot 10^2$	$2.6201 \cdot 10^3$	13.9611	51.1868
12	Ireland	1.000	$3.7959 \cdot 10^1$	$5.4882 \cdot 10^2$	6.1611	23.4269
13	Italy	0.375	$6.4158 \cdot 10^{-1}$	1.8444	0.8010	1.3581
14	Jordan	0.375	$1.2770 \cdot 10^1$	4.6143	3.5735	2.1481
15	Kazakhstan	0.625	$1.5991 \cdot 10^1$	$2.3499 \cdot 10^1$	3.9989	4.8476
16	Kyrgyz Republic	0.875	6.8038	$5.9885 \cdot 10^1$	2.6084	7.7385
17	Latvia	0.750	$6.9598 \cdot 10^1$	$3.2999 \cdot 10^1$	8.3425	5.7445
18	Lithuania	0.500	1.5676	6.0924	1.2520	2.4683
19	Malaysia	0.250	2.1713	4.9335	1.4735	2.2211
20	Mongolia	1.000	$1.6996 \cdot 10^2$	$3.0838 \cdot 10^2$	13.0367	17.5607
21	Morocco	0.375	1.4151	1.4562	1.1896	1.2067
22	Netherlands	0.875	$6.3521 \cdot 10^2$	$5.6672 \cdot 10^2$	25.2033	23.8059
23	North Macedonia	0.500	4.1569	3.6915	2.0389	1.9213
24	Norway	0.500	3.3392	$1.2363 \cdot 10^1$	1.8274	3.5162
25	Poland	0.500	$8.6018 \cdot 10^{-1}$	2.6729	0.9275	1.6349
26	Romania	0.125	3.1971	$5.1134 \cdot 10^{-1}$	1.7880	0.7151
27	Serbia	0.625	$1.0644 \cdot 10^1$	$1.1033 \cdot 10^1$	3.2625	3.3216
28	Slovak Republic	0.500	3.6490	5.49630	1.9102	2.3444
29	Slovenia	0.125	3.6556	1.6754	1.9120	1.2944
30	Spain	0.125	1.1266	$6.6022 \cdot 10^{-1}$	1.0614	0.8125
31	Tajikistan	0.250	7.9008	2.4342	2.8108	1.5602
32	United Kingdom	0.625	5.8268	$1.5686 \cdot 10^1$	2.4139	3.9606

Source Calculated based on the World Bank [38]

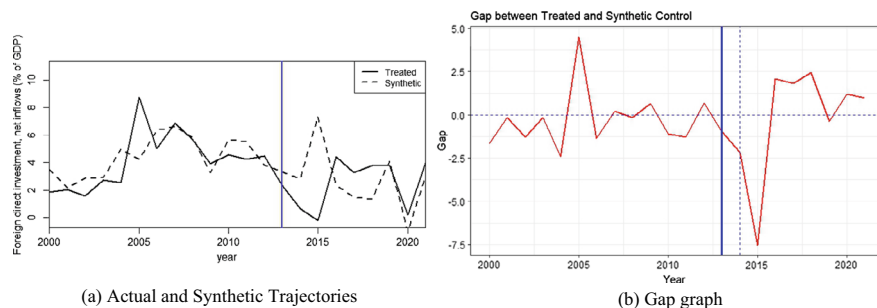


Fig. 1 Results of synthetic control for the control group of 32 countries using 33 predictors *Source* Calculated based on the World Bank [38]

Ukraine's pre-intervention characteristics. Specifically, countries with low MSPE_{pre} values, along with those from Central and Eastern Europe that share comparable historical and economic trajectories were selected to form the control group. The final donor pool includes: Albania, Bulgaria, Chile, Croatia, the Czech Republic, Estonia, France, Germany, Greece, Italy, Jordan, Kazakhstan, the Kyrgyz Republic, Latvia, Lithuania, Malaysia, Morocco, North Macedonia, Norway, Poland, Romania, Serbia, the Slovak Republic, Slovenia, Spain, Tajikistan, and the United Kingdom.

To ensure the robustness of the findings, sensitivity analyses were repeatedly conducted to assess the impact of variations in the donor pool composition. This process resulted in a synthetic model based on a control group of 27 countries and 33 predictors for the period 2006–2021. The results are presented in Table 4 and Fig. 2.

With the given input data, MSPE(loss.v) is 0.0478, and MSPE(loss.w) is 3.1506. The low mean square error of the forecast for the outcome variable (MSPE(loss.v)) indicates the high degree of accuracy and reliability in the synthetic model, demonstrating its effectiveness in replicating the trajectory of the outcome variable during the pre-intervention period. While the MSPE(loss.w) value for the donor pool weights is relatively higher, suggesting some sensitivity to the selection of donor countries and predictors, its magnitude (3.1506) remains within acceptable range. Given the inherent volatility of the input data, this level of error does not undermine the validity of the results.

The analysis of the predictor weights (solution.v) revealed that the key determinants in constructing the synthetic Ukraine were GDP growth (annual %), gross capital formation (% of GDP), interest payments (% of revenue), research and development expenditure (% of GDP), the Gini index, expenses (% of GDP), and the cost of business start-up procedures (% of GNI per capita). These predictors encompass a broad spectrum of macroeconomic and institutional characteristics, highlighting the complex impact of the intervention on the Ukrainian economy.

Regarding the weights assigned to the control group countries (solution.w), Norway, the Kyrgyz Republic, Kazakhstan, and Greece had the greatest influence in constructing the synthetic Ukraine. This suggests that these countries exhibited the most comparable economic and institutional characteristics to Ukraine during

Table 4 Synthetic control results for the 27-country control group using 33 predictors

Predictor identifiers	Predictors	Solution. v	Control Group	Solution. w
Factor1	Net foreign assets (current LCU)	0.0153	Albania	$6.153 \cdot 10^{-7}$
Factor2	Current account balance, % of GDP	0.0007	Bulgaria	$3.306 \cdot 10^{-2}$
Factor3	Control of corruption: percentile rank	0.00003	Chile	$1.061 \cdot 10^{-6}$
Factor4	Domestic credit to private sector by banks (% of GDP)	0.00001	Croatia	$4.672 \cdot 10^{-6}$
Factor5	Expense (% of GDP)	0.0761	Czechia	$2.806 \cdot 10^{-6}$
Factor6	Exports of goods and services, (% of GDP)	0.0095	Estonia	$2.726 \cdot 10^{-6}$
Factor7	Electric power consumption (kWh per capita)	0.0014	France	$1.350 \cdot 10^{-6}$
Factor8	GDP growth (annual %)	0.0878	Germany	$6.172 \cdot 10^{-7}$
Factor9	GDP per capita growth (annual %)	0.0227	Greece	$2.008 \cdot 10^{-1}$
Factor10	GDP per person employed (constant 2021 PPP \$)	0.00005	Italy	$1.082 \cdot 10^{-6}$
Factor11	Gross capital formation (% of GDP)	0.0854	Jordan	$3.188 \cdot 10^{-5}$
Factor12	Gross domestic savings (% of GDP)	0.0002	Kazakhstan	$1.475 \cdot 10^{-1}$
Factor13	Imports of goods and services (% of GDP)	0.0003	Kyrgyz Republic	$1.626 \cdot 10^{-1}$
Factor14	Industry (including construction), value added (% of GDP)	0.00002	Latvia	$5.894 \cdot 10^{-4}$
Factor15	Inflation, GDP deflator (annual %)	0.0007	Lithuania	$1.168 \cdot 10^{-2}$
Factor16	Trade (% of GDP)	0.00003	Malaysia	$1.053 \cdot 10^{-6}$
Factor17	Consumer price index (2010 = 100)	0.0177	Morocco	$1.974 \cdot 10^{-6}$
Factor18	Gini index	0.0807	North Macedonia	$1.222 \cdot 10^{-6}$
Factor19	Urban population (% of total population)	0.1845	Norway	$4.436 \cdot 10^{-1}$
Factor20	Research and development expenditure (% of GDP)	0.0867	Poland	$1.219 \cdot 10^{-4}$

(continued)

Table 4 (continued)

Predictor identifiers	Predictors	Solution. v	Control Group	Solution. w
Factor21	Interest payments (% of revenue)	0.0854	Romania	$9.596 \cdot 10^{-6}$
Factor22	Individuals using the Internet (% of population)	0.00007	Serbia	$5.371 \cdot 10^{-7}$
Factor23	Current health expenditure (% of GDP)	0.0435	Slovak Republic	$2.026 \cdot 10^{-6}$
Factor24	Life expectancy at birth, total (years)	0.0025	Slovenia	$2.515 \cdot 10^{-7}$
Factor25	Government expenditure on education, total (% of GDP)	0.0177	Spain	$8.130 \cdot 10^{-7}$
Factor26	ICT goods exports (% of total goods exports)	0.0068	Tajikistan	$1.133 \cdot 10^{-9}$
Factor27	High-technology exports (% of manufactured exports)	0.0157	United Kingdom	$5.051 \cdot 10^{-6}$
Factor28	Political stability and absence of violence/terrorism: estimate	0.0003		
Factor29	Regulatory quality: estimate	0.00009		
Factor30	Cost of business start-up procedures (% of GNI per capita)	0.0714		
Factor31	Tax revenue (current LCU)	0.0035		
Factor32	Total reserves in months of imports	0.0619		
Factor33	Military expenditure (% of GDP)	0.0245		

Source Calculated based on the World Bank [38]

the pre-intervention period. The most unexpected outcome is the predominant role of Norway in shaping the synthetic Ukraine. Despite the evident disparities between the two countries—particularly in terms of economic development and overall welfare—several factors may account for this finding.

Firstly, both Norway and Ukraine have economies significantly reliant on natural resource exports. While Norway's economy is primarily driven by oil and gas exports, Ukraine depends on the export of raw materials, including grains, oilseeds, and metallurgical products. This structural commonality could potentially explain the

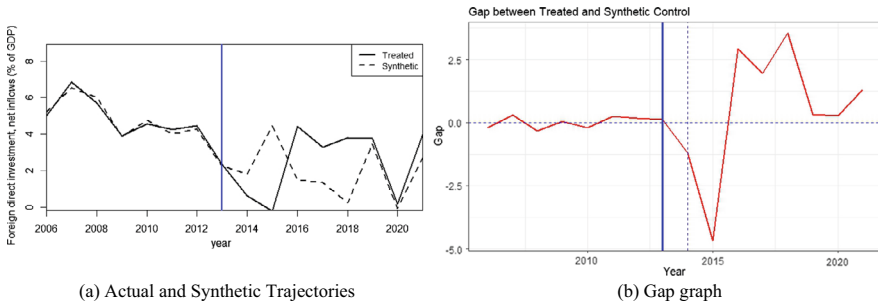


Fig. 2 Results of synthetic control for the control group of 27 countries using 33 predictors *Source* Calculated based on the World Bank [38]

similarity in their economic trajectories. Secondly, the key macroeconomic indicators influencing the weight of the control group countries—such as GDP growth, gross capital formation, government expenditures, and financial indicators—may have exhibited similar dynamics in both economies during specific periods.

Finally, another possible explanation is related to the role of state intervention in the economy and management of state assets. Norway, through its State Pension Fund and strategic oversight of oil resources, maintains a high level of government involvement in key economic sectors. Similarly, Ukraine has historically implemented active regulation of strategic industries, particularly during periods of heightened economic intervention. These shared characteristics may have contributed to the observed weighting of Norway in the synthetic control model.

Kyrgyzstan and Kazakhstan were assigned high weightings due to their shared structural characteristics as transition economies, their substantial reliance on the commodity sector, and the significant role of the state in shaping economic policy. Greece, on the other hand, exhibits similarities with Ukraine due to a range of macroeconomic challenges, including the economic crisis that affected its development—similar to Ukraine’s situation in the pre-intervention period.

According to the placebo test results (Table 5), the p -value for Ukraine is 0.000, indicating a statistically significant gap between the actual and synthetic indicators following the intervention. The low MSPE_pre value (0.0478) confirms that the synthetic Ukraine closely mirrored the trajectory of actual FDI in the pre-intervention period. In contrast, the substantial increase in MSPE_post (6.2534) and RMSPE_post (2.5007), compared to RMSPE_pre (0.2187), underscores the significant impact of the intervention.

To further validate these findings, placebo test was conducted on control countries. In most cases, the MSPE_post/MSPE_pre ratio for Ukraine is significantly higher than that of the other countries, suggesting that the observed gap resulting from the intervention is exceptional.

An analysis of the control group countries reveals that, in most cases, the p -value exceeds 0.1, indicating no statistically significant gap in the post-intervention period.

Table 5 Placebo test results for the synthetic control method (27 countries, 33 predictors)

Country	P_value	MSPE_pre	MSPE_post	RMSPE_pre	RMSPE_post
Ukraine	0.000	$4.7828 \cdot 10^2$	6.2534	0.2187	2.5007
Albania	0.750	9.9352	$3.3055 \cdot 10^1$	3.1520	5.7494
Bulgaria	0.875	$1.8201 \cdot 10^1$	$1.8266 \cdot 10^2$	4.2663	13.5150
Chile	0.625	1.6146	$1.0708 \cdot 10^1$	1.2707	3.2723
Croatia	0.625	$1.4452 \cdot 10^1$	7.0268	3.8015	2.6508
Czechia	0.750	1.5813	7.2637	1.2575	2.6951
Estonia	0.625	$1.3721 \cdot 10^1$	$5.9146 \cdot 10^1$	3.7042	7.6907
France	0.250	$8.8502 \cdot 10^3$	1.0134	0.0941	1.0067
Germany	0.125	1.1196	2.2536	1.0581	1.5012
Greece	0.250	1.3737	$8.2942 \cdot 10^1$	1.1721	0.9107
Italy	0.375	1.0071	1.7667	1.0036	1.3292
Jordan	0.500	$1.5953 \cdot 10^1$	6.0336	3.9941	2.4563
Kazakhstan	0.625	9.5327	$2.3292 \cdot 10^1$	3.0875	4.8262
Kyrgyz Republic	1.000	5.5330	$6.0167 \cdot 10^1$	2.3522	7.7568
Latvia	0.500	1.3643	7.6013	1.1680	2.7570
Lithuania	0.500	1.1634	5.3638	1.0786	2.3160
Malaysia	0.250	2.1205	1.7992	1.4562	1.3413
Morocco	0.375	2.9619	2.5593	1.7210	1.5998
North Macedonia	0.375	$2.1275 \cdot 10^1$	2.9129	0.4613	1.7067
Norway	0.500	1.8556	$1.3753 \cdot 10^1$	1.3622	3.7085
Poland	0.375	$6.0169 \cdot 10^1$	4.1432	0.7757	2.0355
Romania	0.375	2.0869	1.0806	1.4446	1.0395
Serbia	0.625	2.0919	$1.0393 \cdot 10^1$	1.4463	3.2238
Slovak Republic	0.625	$4.9739 \cdot 10^1$	8.7119	0.7053	2.9516
Slovenia	0.250	2.3453	1.9989	1.5315	1.4138
Spain	0.125	$7.6022 \cdot 10^1$	1.4164	0.8719	1.1901
Tajikistan	0.250	3.5359	1.8192	1.8804	1.3488
United Kingdom	0.625	4.7349	$1.7534 \cdot 10^1$	2.1760	4.1874

Source Calculated based on the World Bank [38]

For instance, countries such as Bulgaria (p -value = 0.875), the Czech Republic (p -value = 0.750), Norway (p -value = 0.500), Kazakhstan (p -value = 0.625), and Latvia (p -value = 0.500) exhibit an increase in standard error following the intervention. However, this increase does not reach a threshold that would signify a meaningful deviation from the synthetic counterpart.

The highest level of RMSPE_post/RMSPE_pre ratios after Ukraine is observed in the Kyrgyz Republic (7.76), Estonia (7.69), and Bulgaria (13.52). However, their

relatively high p -values suggest that these deviations are likely random rather than indicative of a significant intervention effect.

Moreover, the high deviations in $\text{RMSPE}_{\text{post}}/\text{RMSPE}_{\text{pre}}$ for some countries can, in part, be attributed to economies of scale. Compared to more developed economies, Ukraine has relatively low absolute volumes of FDI. Under such conditions, even minor fluctuations in FDI flows—whether driven by geopolitical events or economic crises—can result in relatively large percentage deviations. In contrast, for countries with higher FDI volumes, such as Germany, France or Italy, absolute changes in investment are less pronounced in relative terms, explaining their lower $\text{RMSPE}_{\text{post}}/\text{RMSPE}_{\text{pre}}$ and $\text{MSPE}_{\text{post}}/\text{MSPE}_{\text{pre}}$ values. For example, in France, $\text{MSPE}_{\text{post}}$ is 1.01, while MSPE_{pre} is 0.00885, indicating the stability of the prediction error before and after the intervention. For countries with moderate economic development, such as the Kyrgyz Republic, Kazakhstan, or Bulgaria, the economies of scale may be more comparable to those of Ukraine, which partially accounts for their elevated $\text{RMSPE}_{\text{post}}$ values. Notably, in the Kyrgyz Republic, $\text{MSPE}_{\text{post}}$ is 60.16, while MSPE_{pre} is 5.53, indicating a significant gap; however, its p -value (1.000) suggests that this effect is not statistically significant.

Accordingly, the results of the placebo test confirm that the change in FDI in Ukraine after 2014 was atypical among the control countries. This suggests a significant negative impact of the intervention—the outbreak of the Russian Federation’s war against Ukraine—on foreign investment, an effect not observed in other countries with comparable macroeconomic characteristics.

The analysis of Ukraine’s real FDI trajectory from 2015 to 2021 reflects foreign investors’ responses to economic, social, political, and security developments in the country. According to official statistics, 2015 saw the net FDI outflow of – 331 mln USD [1], driven by heightened uncertainty due to ongoing military operations, macroeconomic instability, and the urgent need for structural reforms. However, from 2016 to 2018, investment flows turned positive, signalling a gradual recovery in foreign investor confidence. By 2019, FDI inflows had risen to 6017 mln USD [1], likely reflecting the economy’s adaptation to previous shocks, partial macroeconomic stabilisation, and the implementation of policies aimed at enhancing the investment climate.

At the same time, Ukraine experienced another FDI outflow in 2020 due to the impact of the COVID-19 pandemic, driven by global economic instability, heightened uncertainty, and a decline in business activity. In 2021, however, FDI surged 7320 mln USD [1], likely reflecting a revival in business activity as pandemic-related restrictions were gradually lifted. The analysis of FDI dynamics in 2022–2023 reveals similar fluctuations, primarily driven by foreign investors’ responses to the full-scale war in Ukraine. Synthetic control analysis of FDI trends suggests that such volatility is not exclusive to periods of military conflict but also occurs during peacetime when investors assess risks related to election cycles, pandemic threats, and economic reforms. For Ukraine’s post-war economic recovery, this underscores the critical need for predictable economic policies to attract long-term foreign investment.

Meanwhile, FDI in Ukraine, both in the pre-war period and after the military invasion, was shaped by the phenomenon of “round tripping”—a practice in which

the ultimate controlling investor is a resident of the country. Between 2010 and 2013, this type of investment accounted for an average of 32.7% of total FDI, with 89% of these funds directed toward real-sector enterprises [39]. The notable feature of this phenomenon is the dominance of investments from countries such as Cyprus, Austria, the Netherlands, and Switzerland, which have traditionally been the key sources of foreign investment in Ukraine. A particularly concerning aspect, however, is that the ultimate controlling investors often included companies from the Russian Federation. Their responses to the events in Ukraine between 2016 and 2021 was a gradual increase in net inflows from “round tripping” operations, rising from 3.8% in 2016 to 68.5% of total FDI inflows in 2021 [39]. Considering these dynamics, ensuring national security and facilitating an effective post-war economic recovery necessitates stricter oversight of the origins of ultimate controlling foreign investors. Strengthening such controls would help reduce the presence of companies linked to the aggressor state while fostering genuine partnerships with international investors committed to Ukraine’s long-term economic development.

6 Conclusions

This study has identified key trends in FDI inflows into Ukraine’s economy in the context of the war initiated by the Russian Federation in 2014. By utilizing the Synthetic Control Method (SCM), both empirical and counterfactual (synthetic) trajectories of net FDI inflows to Ukraine from 2014 to 2021 were constructed. The results highlight the significant disruption caused by the war, with a sharp decline in net FDI inflows during 2014–2015, effectively leading to an outflow of foreign capital. This decline stands in stark contrast to the counterfactual scenario, where FDI inflows would have followed a positive trajectory in the absence of the conflict.

Additionally, the study reveals significant volatility in both the empirical and synthetic trajectories of FDI inflows over the observed period. This volatility corresponds to key political and economic events, such as the Orange Revolution and election cycles in 2004–2005, 2010, and 2018–2019, as well as the global financial crisis of 2008–2009. These fluctuations underscore the heightened sensitivity of foreign investors to the social, political, and economic stability of the country, with investment patterns responding to perceived instability in Ukraine.

The findings also suggest that reforms aimed at improving the investment climate in Ukraine could generate positive signals for foreign investors, even in the context of ongoing conflict. Particularly, these reforms contribute to mitigating institutional risks, fostering a more favourable regulatory environment, and facilitating business operations, which, in turn, may encourage foreign capital inflows even amid wartime conditions. These results also emphasize that FDI could serve as a crucial driver of Ukraine’s post-war recovery, supporting the reconstruction of critical infrastructure, fostering technological modernization, and integrating the country into global value chains.

The analysis of the predictor weights further underscores the pivotal role demographic, macroeconomic and innovation activity indicators in shaping FDI dynamics. The findings highlight that, in the post-war recovery phase, Ukraine's foreign investment strategy should prioritize creating an enabling environment for research and development (R&D), particularly within the technology, defence, and energy sectors. Additionally, urban infrastructure development also emerges as a crucial priority, as the significant weighting of urbanization indicators highlights their substantial influence on economic processes and the role of cities as hubs of business activity [40]. Furthermore, improving the overall business climate remains imperative, given the pronounced impact of debt burdens and income inequality on macroeconomic stability.

The study also highlights the need for a more effective regulatory framework. The low weights attributed to political stability and regulatory quality indicate inefficiencies in public regulation amidst uncertainty. To mitigate these issues, Ukraine should focus on enhancing its regulatory framework, ensuring transparency and reducing risks for foreign investors. Additionally, the revision of investment strategies should be based on the key factors identified by the SCM, enabling the effective allocation of resources to stimulate economic growth and structural transformation.

In conclusion, the SCM findings provide a foundation for further analysis of the factors influencing FDI inflows to Ukraine. Given the ongoing war, addressing its economic consequences alongside long-term recovery prospects is crucial, with FDI playing a central role in this process.

References

1. National Bank of Ukraine (n.d.) Direct investments. Available via National Bank of Ukraine. <https://bank.gov.ua/en/statistic/sector-external#5>. Accessed 26 Feb 2025
2. President of Ukraine (2014) Decree No. 950/2014 on the National Investment Council. Available via Verkhovna Rada of Ukraine. <https://zakon.rada.gov.ua/laws/show/950/2014>. Accessed 26 Feb 2025
3. Verkhovna Rada of Ukraine (2020) Law of Ukraine No. 1116-IX on state support for investment projects with significant investments in Ukraine. Available via Verkhovna Rada of Ukraine. <https://zakon.rada.gov.ua/laws/show/1116-20>. Accessed 26 Feb 2025
4. Verkhovna Rada of Ukraine (2017) Law of Ukraine No. 2058-19 on amendments to certain legislative acts of Ukraine to eliminate barriers to attracting foreign investments. Available via Verkhovna Rada of Ukraine. <https://zakon.rada.gov.ua/laws/show/2058-19>. Accessed 26 Feb 2025
5. Advantage Ukraine (n.d.) Available via advantage Ukraine. <https://advantageukraine.com/ua/>. Accessed 26 Feb 2025
6. Kox HLM, Rojas-Romagosa H (2020) How trade and investment agreements affect bilateral foreign direct investment: results from a structural gravity model. *World Econ* 43:3203–3242. <https://doi.org/10.1111/twec.13002>
7. Bruno RL, Campos NF, Estrin S (2021) The effect on foreign direct investment of membership in the European Union. *J Common Mark Stud* 59:802–821. <https://doi.org/10.1111/jcms.13131>
8. JarDET C, Jude C, Chinn M (2023) Foreign direct investment under uncertainty: evidence from a large panel of countries. *Rev Int Econ* 31(3):854–885. <https://doi.org/10.1111/roie.12646>

9. Saleh EA (2023) The effects of economic and financial crises on FDI: a literature review. *J Bus Res* 161:113830. <https://doi.org/10.1016/j.jbusres.2023.113830>
10. Kapopoulos T, Sakkas A, Drakos K (2024) Geopolitical risk and foreign direct investment inflows: the moderating role of water and energy risks. *Int J Finance Econ*. <https://doi.org/10.1002/ijfe.3106>
11. Brintseva O (2023) The effectiveness of investment in human capital in Poland and Ukraine: directions for growth based on comparative analysis. *Krakow Rev Econ Manag Zeszyty Naukowe Uniwersytetu Ekonomicznego W Krakowie* 1(999):31–46. <https://doi.org/10.15678/ZNUEK.2023.0999.0102>
12. Kichurchak M (2021) Development of the higher education market in the EU countries as a factor of human capital accumulation: experience for Ukraine. *Econ Ann-XXI* 192(7–8(2)):52–62. <https://doi.org/10.21003/ea.V192-05>
13. Kichurchak M (2024) Interconnected roles of human capital, employment, and sustainable development in the EU countries. *IOP Conf Ser: Earth Environ Sci* 1415(1):012062. <https://doi.org/10.1088/1755-1315/1415/1/012062>
14. Pantelopoulous G (2022) Foreign direct investment versus foreign portfolio investment with respect to human capital: the impact of education on the host OECD countries. *Appl Econ Lett* 30(15):2060–2068. <https://doi.org/10.1080/13504851.2022.2091104>
15. Dankyi AB, Abban O, Yusheng K, Coulibaly TP (2022) Human capital, foreign direct investment, and economic growth: evidence from ECOWAS in a decomposed income level panel. *Environ Challenges* 9:100602. <https://doi.org/10.1016/j.envc.2022.100602>
16. Harkavenko V, Yershova H (2020) Impact of foreign capital on the economic development of Ukraine. *Econ Forecast* 1:43–56. <https://doi.org/10.15407/eip2020.01.047>
17. Getzner M, Moroz S (2020) Regional development and foreign direct investment in transition countries: a case study for regions in Ukraine. *Post-Communist Econ* 32(6):813–832. <https://doi.org/10.1080/14631377.2020.1745564>
18. Ciešlik A, Gurshev O (2020) Determinants of inward FDI in Ukraine: does political stability matter? *Int J Manag Econ* 56(3):243–254. <https://doi.org/10.2478/ijme-2020-0021>
19. Bashynska I, Smokvina G, Yarenko L, Lemko Y, Ovcharenko T, Zhang S (2022) Assessment of investment and innovation image of the regions of Ukraine in terms of sustainable transformations. *Acta Innov* 43:63–77. <https://doi.org/10.62441/actainnovations.v43i.287>
20. Paslavskaya I, Kvasnii P, Lysa V (2020) Analysis and modelling of the foreign direct investment impact on the economy of Ukraine. In: *Zarządzanie przedsiębiorstwem w dobie globalnych problemów: monografia*. Wydawnictwo SIZ, pp 249–264. <https://www.wydawnictwo-siz.pl/wp-content/uploads/2021/05/Całość-pop1.pdf>. Accessed 20 Feb 2025
21. Kovtoniuk K, Molchanova E, Savych O, Demkiv Y, Piankova O, Kudina A (2023) Estimation of the impact of macroeconomic factors on investment activity in Ukraine. Paper presented at the 13th international conference on advanced computer information technologies (ACIT), West Ukrainian National University, Ukraine, 21–23 Sept 2023
22. Joshi M, Quinn JM (2020) Civil war termination and foreign direct investment, 1989–2012. *Confl Manag Peace Sci* 37(4):451–470. <https://doi.org/10.1177/0738894218778260>
23. Moore RJ (2021) Emerging from war: public policy and patterns of foreign direct investment recovery in postwar environments. *J Int Bus Policy* 4:455–475. <https://doi.org/10.1057/s4214-020-00084-4>
24. Klosek KC (2020) Military interventions in civil wars: protecting foreign direct investments and the defence industry. *Civ Wars* 22(1):87–113. <https://doi.org/10.1080/13698249.2020.1724726>
25. Farzanegan MR (2021) The effects of international sanctions on Iran's military spending: a synthetic control analysis. *Def Peace Econ* 33(7):767–778. <https://doi.org/10.1080/10242694.2021.1941548>
26. Nguyen C, Tran T, Van Vu H (2024) The long-term effects of war on foreign direct investment and economic development: evidence from Vietnam. *J Urban Econ* 143:103680. <https://doi.org/10.1016/j.jue.2024.103680>

27. Abadie A (2021) Using synthetic controls: feasibility, data requirements, and methodological aspects. *J Econ Lit* 59(2):391–425. <https://doi.org/10.1257/jel.20191450>
28. Abadie A, Vives-i-Bastida J (2022) Synthetic controls in action. arXiv: preprint. <https://arxiv.org/abs/2203.06279>. Accessed 18 Feb 2025
29. Abadie A, Zhao J (2024) Synthetic controls for experimental design. arXiv: methodology. <https://arxiv.org/pdf/2108.02196.pdf>. Accessed 20 Feb 2025
30. Pinotti P (2022) The economic consequences of terrorism: evidence from the Italian experience. *J Public Econ* 211:104–630. <https://doi.org/10.1016/j.jpubeco.2022.104630>. Accessed 20 Feb 2025
31. Cavallo E, Galiani S, Noy I, Pantano J (2021) Catastrophic natural disasters and economic growth. *Rev Econ Stat* 103(5):817–832. https://doi.org/10.1162/rest_a_00952
32. Audretsch DB, Momtaz PP, Motuzenko H, Vismara S (2023) The economic costs of the Russia–Ukraine war: a synthetic control study of (lost) entrepreneurship. SSRN Working Paper No. 4378827. <https://doi.org/10.2139/ssrn.4378827>
33. Kichurchak M, Paslavskaya I, Hrynkevych O, Mishchuk H, Bilan Y (2024) Examining tertiary education amid the war in Ukraine: a synthetic control approach. *Eur J Interdiscip Stud* 16(2):95–115. <https://doi.org/10.24818/ejis.2024.13>
34. Pang X, Liu L, Xu Y (2022) A bayesian alternative to synthetic control for comparative case studies. *Polit Anal* 30(2):269–288. <https://doi.org/10.1017/pan.2021.22>
35. Doudchenko N, Khosravi K, Pouget-Abadie J, Lahaie S, Lubin M, Mirrokni VS, Spiess J, Imbens G (2021) Synthetic design: an optimization approach to experimental design with synthetic controls. *Neural Inf Proc Systems*. arXiv: methodology. <https://doi.org/10.48550/arXiv.2112.00278>
36. Wang Y, Zubizarreta JR (2020) Minimal dispersion approximately balancing weights: asymptotic properties and practical considerations. *Biometrika* 107(1):93–105. <https://doi.org/10.1093/biomet/asz050>
37. Cattaneo MD, Feng Y, Titiunik R (2021) Prediction intervals for synthetic control methods. *J Am Stat Assoc* 116(536):1865–1880. <https://doi.org/10.1080/01621459.2021.1979561>
38. World Bank (2025) World development indicators. World bank open data. <https://databank.worldbank.org/source/world-development-indicators>. Accessed 22 Feb 2025
39. National Bank of Ukraine (2024) Assessment of the volume of foreign direct investment, where the ultimate controlling investor is a resident (round tripping) for 2010–2023. National Bank of Ukraine, department of statistics and reporting. https://bank.gov.ua/admin_uploads/article/FDI_round_trippling_ICL-DFS_pr.pdf?v=12. Accessed 26 Feb 2025
40. Dzhedzhula V, Hurochkina V, Yepifanova I, Telnov A (2022) Fuzzy technologies for modeling social capital in the emergent economy. *WSEAS Trans Bus Econ* 19:915–923. <https://doi.org/10.37394/23207.2022.19.80>

Modelling the Socio-economic Impacts for the Coal Mining: Forecast of Hazardous Properties of Coal Seams



Yevhen Rudniev , Julia Romanchenko , Kostiantyn Sokolenko ,
Vadym Tarasov , and Valeriy Sokolenko

Abstract The social effect of investments (increasing working capacity, reducing injuries and morbidity, increasing the prestige of the profession, the degree of job satisfaction, and the like) should always be positive. The economic results of investment are expressed in the form of the economic effect of the implementation of occupational safety measures and efficiency indicators. Improvement of normative documents for the safe conduct of mining operations in terms of the formation of hazardous properties of coal seams during geological processes, their reliable forecast, and the development of preventive measures are very relevant for all coal-mining countries of the world. The regularities of changes in the elemental content of the main components (carbon, oxygen, hydrogen, nitrogen and sulfur) of organic mass and mineral impurities (ash yield) for different states of coal samples under the influence of geological processes of coalification were established. Three characteristic stages of metamorphic transformations of coal seams have been established based on the change in elemental content per conventional organic mass of carbon and hydrogen, as well as moisture content of analytical samples. The introduction of new methods for forecasting the hazardous properties of coal seams and improving the regulatory framework for safe mining operations will increase the socio-economic effectiveness of investments in improving the working conditions of miners.

Y. Rudniev · J. Romanchenko · K. Sokolenko · V. Tarasov (✉) · V. Sokolenko
Volodymyr Dahl East Ukrainian National University, Kyiv, Ukraine
e-mail: tarasov@snu.edu.ua

Y. Rudniev
e-mail: rudnev_es@snu.edu.ua

J. Romanchenko
e-mail: romanchenko_ja@snu.edu.ua

K. Sokolenko
e-mail: sokolenko@snu.edu.ua

V. Sokolenko
e-mail: sokolenko_1@snu.edu.ua

Keywords Social effect · Investment efficiency · Coal industry · Assessment · Optimization · Labor protection · Coal seams · Metamorphism · Coalification · Hazardous properties · Safety · Mining engineering · Regulatory framework

1 Introduction

The coal industry of Ukraine has remained one of the key sectors of the Ukrainian economy for many years. The products of coal mining and coal processing enterprises are a stable basis for the production of electricity, fuel, chemical products, metal melting, etc. Ensuring the energy security of the state, which is the basis of the entire economic complex, is a priority task for Ukraine. The fuel and energy complex is the basis for the activities of other industries, therefore, its innovative development will become an accelerator of the innovative development of the Ukrainian economy as a whole. The coal industry is one of the priority industries from the point of view of the innovative development of the Ukrainian economy.

In this regard, issues of innovative development of the industry are relevant, including the modernization of coal enterprises, improvement of the regulatory framework for the safety of mining operations and for scientific and researches. To form innovative programs of the coal industry at all management levels, in addition to the starting conditions, trends in their development in the future are determined [1]. These trends are established using various forecasting methods, one of the objects is the selection of scientific problems priority in field environmental protection.

Analytical research of the coal industry condition allows us to identify a number of systemic problems, in particular:

- fixed assets modernization in order to increase the cost of production [2];
- increasing the labor protection level at the enterprises of the industry [3];
- increasing funding institutions engaged in research of the coal industry, since having an extensive network of scientific institutions, the funding reduction of which led to the outflow of scientists and the closure of relevant research [4];
- privatization of other enterprises of the industry in order to attract investment [5];
- high capital intensity and a rather long development period for new coal deposits [6].

Further in the work (chapter of the monograph) one of the areas of innovative activity of the coal industry in Ukraine is considered, which is the increasing of production safety.

Underground mining of minerals provides to take into account a large number of natural and man-made factors that lead to increased danger for people and production facilities.

Analysis of accidents that have occurred in the mines of coal-mining countries over the past few decades shows that the main cause of their occurrence is the joint influence of factors from three blocks. These include factors of geological processes

of the starting material transformation in natural conditions, mining and geological conditions of the deposits, technological parameters of the coal enterprise operation.

The most reliable parameters are factors that related to the mining and geological conditions of the coal deposit. They are determined at the exploration stage and then specified during the construction and operation of the coal enterprise in accordance with its technological characteristics.

The most difficult to establish are the factors and their quantitative values that determine the formation conditions of dangerous properties of mine deposits in the process of geological transformations.

Information about accidents that have occurred over the last thirty years in the world's major coal-mining countries is conventionally divided into two categories. One category includes accidents that occurred under the influence of the natural properties of mine deposits (gas content, tendency to sudden emissions of coal and gas, occurrence of spontaneous combustion, explosivity of coal dust, and some other features). The second category includes the remaining accidents to which the hazardous properties of mine deposits are not related. In all major coal-mining countries, the number of victims of the hazardous properties of mine deposits is several times higher than the number of victims in other accidents.

Periodically recurring accidents due to the hazardous properties of mine deposits with serious consequences indicate existing of unsolved problems regarding their timely prevention. To assess the hazardous properties of mine deposits the production safety regulatory documents use only a few classification indicators (mass and volume yield of volatile substances, thickness of the plastic layer, logarithm of the electrical resistivity of anthracites and coal grade). These indicators do not allow us to investigate the patterns of changes in the coal elemental composition in the metamorphic transformations process, which determine the properties of mine deposits, according to the modern interpretation of metamorphism. Each of these indicators characterizes only one property out of several dozen that could arise during the carbonization process. All of them characterize the state of dry ashless organic (combustible) mass, and the composition of fossil coal also includes mineral impurities and moisture. It is not possible to establish changes in the organic matter components of mineral impurities and moisture during metamorphic transformations using these indicators.

Establishing the patterns of composition changes, properties of fossil coal and improving on this basis the predicting methods of dangerous properties manifestation of mine deposits and the early development of preventive measures in accordance with the requirements of current regulatory documents of safety is a relevant scientific and applied problem in the field of labor protection, which is of significant importance for reducing the level of accidents with serious consequences, industrial injuries at coal mining enterprises and occupational morbidity of miners. Scientific research aimed at improving the regulatory framework of mining operations safety is relevant for all coal-mining countries of the world.

The research results allow us to develop proposals for improving the regulatory framework for safety of mining operations regarding the prediction of the manifestation of hazardous properties of mine deposits.

2 Regulatory Framework for Safe Mining Operations

In the modern regulatory framework for safe mining operations ([7–11] and some other documents), the indicator of the release of volatile substances during the thermal decomposition of coal without access to air (V^{daf}) is adopted as the main criterion for predicting the hazardous properties of coal seams.

An analysis of information on dustiness and gas content of production processes in coal mines in Ukraine shows that mines with high dust and methane hazards and high emissions constitute a significant part of the total number of Ukrainian mines, the coal seams of which with a volatile matter yield of 15% or more are particularly hazardous in terms of dust explosions [7, 11]. The presence of dust in the methane-air mixture during an explosion contributes to a significant increase in the force of the explosion and the formation of a large volume of highly concentrated toxic gases.

To predict the gas content of mine workings based on the methane content of coal seams, the volatile matter yield indicator is used to calculate the value of the residual methane content of broken coal [8].

Underground fires from spontaneous combustion of coal are among the natural hazards that accompany mining operations in coal mines. Endogenous fires in development mine workings occur in zones of geological disturbances, in voids behind the support, and also after sudden outbursts. The endogenous fire hazard of coal accumulations during sudden outbursts is assessed by determining the incubation period of spontaneous combustion of coal (τ_{inc}) using the formula in which the residual gas content of coal (x_r) is determined depending on the yield of volatile substances [9].

The yield of volatile substances is used to establish the outburst hazard of coal seams [10]. The outburst hazard of coal seams is established separately by the V^{daf} factor in the ranges of its change of 9–29% and more than 29%.

For coal, the yield of volatile substances (V^{daf}) was determined using methods similar to those outlined in standard [12]. For this reason, the experimental values of the yield of volatile substances for coal seams given in reference and regulatory documents [13–15] are quite reliable, since they were determined by standard methods with a predetermined accuracy. Despite these circumstances, the yield of volatile substances is not the main criterion for the transformation of organic matter during metamorphic processes [16]. According to the officially accepted definition, metamorphism is understood as the transformation of brown coal successively into hard coal and anthracite as a result of changes in the chemical composition, structure and physical properties in the depths of the Earth, mainly under the influence of increased pressure and temperature. In a more general case, such transformations apply not only to coal seams, but to all metamorphic rocks. They consist of a change in the mineral composition or the size and structure of grain aggregates without a significant change in the chemical composition (except for the content of H_2O , CO_2) under the influence of fluids, temperature and pressure [17].

In accordance with generally accepted definitions the degree of metamorphic transformations must be judged by the change in the elemental composition of coal and its physical and mechanical properties. For this reason, the use of volatile matter

yield indicators as a criterion for the degree of metamorphic transformations of coal seams to predict the manifestation of their hazardous properties during mining operations according to [7–11] is insufficiently substantiated.

Thanks to a detailed study of the consumer properties of coals of coal seams of different coal basins, a significant amount of experimental data has been accumulated ([13–15] and some others) characterizing changes in the composition and properties of coal seams during their metamorphic transformations. The use of these data to predict the hazardous properties of coal seams is difficult, since these indicators were determined for different states of the fuel. For example, the elemental content of carbon, hydrogen, nitrogen and total sulfur were determined by classical standard methods in the analysis of analytical samples brought to an air-dry state. The mass fraction of oxygen in solid mineral fuel was calculated by the difference between 100% and the sum of the mass fractions of moisture, carbon, hydrogen, nitrogen, sulfur, and ash content, expressed as a percentage. Additional indicators determined by standard methods are ash content, analytical moisture and total moisture.

It has been established that the most variable quality indicator is ash content. For this reason, all calculations of sampling rates are usually based on ash content calculated for dry coal. The error in determining ash content serves to assess the error of sampling as a whole (sample selection, processing and analysis). This error value is established in advance in the standard or in the fuel supply agreement and is the basis for calculating sampling rates. The number of point samples taken in a combined sample depends on the degree of heterogeneity of the fuel, i.e. on its ash content, and the mass of each point sample depends on the size of the coal pieces. The higher the ash content of the fuel, the more point samples must be taken to obtain a representative sample. Point samples must be taken evenly from the entire mass being tested. In accordance with this, 32 point samples are taken from 1000 tons of run-of-mine coal, and 16 from enriched and sorted coal. The standards for taking point samples are calculated in each specific case in accordance with current standards. Processing of samples or their cutting is the second stage of testing the consumer qualities of coals for the preparation of laboratory and analytical samples.

The moisture of coal in a seam in its natural occurrence is called seam moisture. Usually coal seams contain water in a much larger quantity than is typical for the nature and age of the coal, therefore the content of seam moisture is not always a characteristic indicator of the consumer qualities of coal. Coal extracted from a seam loses moisture in the air, the so-called wetting moisture or free moisture. After removing this moisture, a piece of freshly mined coal, which is completely saturated with water, contains moisture typical of this coal, its chemical nature, petrographic composition and degree of coalification. The content of this moisture in coal (moisture content) can be approximately estimated using the maximum moisture capacity index [18].

In reference and regulatory documents of previous years, different types of moisture were established for coal seams when determining the consumer qualities of coals. The moisture of analytical samples was determined when developing a geological and coal-chemical map [13]. The values of hygroscopic moisture are given in the

reference book [14], and the maximum moisture capacity for hard coals in the document [15] on the quality of fuel of the Donetsk and Lviv-Volyn basins. In the cases under consideration, the types of moisture were determined using different methods based on different physical and chemical processes. Their combined use to establish the dust-forming capacity of coal seams in the catalog [7] led to additional errors in predicting the hazardous properties of coal seams. Hard coals and anthracites have different molecular structures, which determines different ratios between the content of free and bound moisture.

The content of mineral impurities and hydrated moisture at different stages of metamorphic transformations for a separate coal seam are random. These features of the presence of different types of moisture in fossil coals, as well as the methods of selection and preparation of laboratory and analytical samples and the results of their tests must be taken into account when developing scientific foundations for methods of predicting the hazardous properties of coal seams.

When predicting the manifestation of hazardous properties of coal seams, it is necessary to take into account the specified features of establishing the parameters of consumer qualities of coals. They, for the most part, do not coincide with the indicators that directly characterize the hazardous properties of coal seams by the elemental composition of the organic matter and mineral impurities.

For example, the yield of volatile substances (V^{daf}), determined according to analytical samples (a) in laboratory conditions for dry (d) and ash-free (f) state, is used as the main criterion for the degree of metamorphic transformations of coal seams in documents [7–11]. This method of determining V^{daf} excludes the possibility of analyzing the influence of the presence of mineral impurities on the formation of hazardous properties of coal seams during metamorphic transformations. In particular, such impurities include sulfate and pyrite sulfur as well as external moisture, which is removed during the preparation of analytical samples of coal. The presence of these components of mineral impurities has a significant impact on the manifestation of hazardous properties of coal seams which must be taken into account when improving the regulatory framework for the safe conduct of mining operations.

3 Main Components of Fossil Coals, Their Consumer Qualities and Hazardous Properties of Coal Seams

Fossil coals contain organic compounds and mineral impurities. The organic mass, which usually forms the main substance of coal, consists of carbon (C_o), hydrogen (H_o), oxygen (O_o), nitrogen (N_o) and organic sulfur (S_o). Carbon, hydrogen and oxygen are simultaneously part of both organic matter and mineral impurities. In mineral impurities, carbon is found in the form of carbonates, hydrogen in moisture and hydrated water of silicates, oxygen in compounds such as silicates (aluminum and silicon oxides), iron oxides, carbonates, sulfates and others.

Nitrogen (N_o) and organic sulfur (S_o) are elements associated exclusively with organic matter in coal [19, 20]. Sulfur is present in all types of solid fuel in the form of various compounds that are part of the organic and mineral parts of the fuel. In the mineral mass of coal, sulfur is found mainly in the form of sulfates (mainly calcium and iron) and iron disulfates FeS_2 (pyrite and marcasite), and in the organic mass in the form of organic compounds. The total sulfur content in the organic (S_o) and mineral (S_{mm}) masses is an indicator of total sulfur (S_t). The total sulfur content in coals ranges from 0.2 to 10% [19, 20].

The variety of organic sulfur (S_o) has not been studied sufficiently, since there are no reliable methods for isolating organic compounds from coals. Conventionally, the content of organic sulfur is taken to be the difference between the content of total sulfur (S_t) and the sum of sulfate (S_{SO_4}) and pyrite (S_p) [19, 20]:

$$S_o = S_t - (S_{SO_4} + S_p). \quad (1)$$

The elemental form of sulfur (the fourth) is not determined due to its insignificant content, and sulfur in ash is usually calculated as sulfur trioxide (SO_3).

The idea is to use, for assessing the degree of metamorphic transformations of coal seams and predicting their hazardous properties during mining operations, instead of one indicator of the yield of volatile substances during thermal decomposition of coal without air access (V_{daf}), to consider the change in the ratios between the main components of organic matter and mineral impurities under the influence of geological processes of coal formation.

The goal is to establish correlational dependencies or trends in the variation of elemental composition of the main components of organic matter and mineral impurities under different coal states during metamorphic transformations of coal seams. This aims to achieve a deeper understanding of the essence of geological coalification processes and improve the regulatory framework for the safe conduct of mining operations.

The research methodology is based on the scientific foundations of developing a method for predicting hazardous properties of coal seams during mining operations. They consist of examining three blocks that influence the manifestation of hazardous properties of coal seams, factors independent of each other. The factors of the first block are associated with metamorphic transformations of coal seams. They characterize the elemental change in the organic matter and mineral impurities of coals, as well as their physical, mechanical and other properties.

The factors of the second block include the mining and geological conditions of the occurrence of coal seams: the depth of mining operations, the thickness of the seams being developed and their structure, dip angles, the presence of geological disturbances, the proximity of satellite seams, the properties of the host rocks, the values of the geothermal gradient, etc.

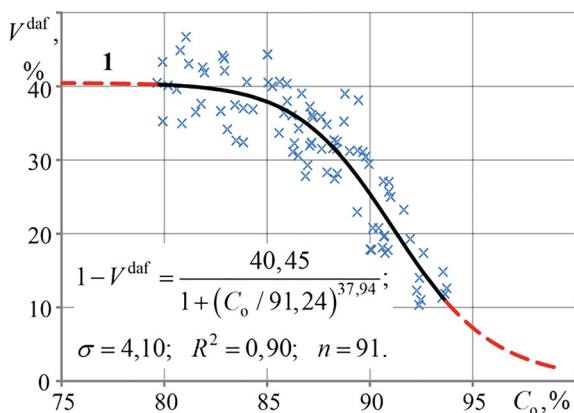
Based on the information about the parameters of the factors of the first two blocks, the mining and technical indicators of the factors of the third block are laid down at the stages of design and operation of a coal enterprise. They are associated with the schemes of opening and preparation of the mine field, development systems,

methods of controlling rock pressure, schemes of ventilation of the mine, extraction areas and individual workings, technology of coal breaking in working faces, method of destruction of rocks during construction of main workings and other technical solutions.

To achieve the set goals, instead of the indicator of the yield of volatile substances during thermal decomposition of coals without air access (V^{daf}), it is necessary to consider the elemental content of carbon (C_o) as a criterion for the metamorphic transformation of organic matter. These indicators are closely correlated with each other (Fig. 1) [19]. The determination coefficient (R^2) is 0.90, but C_o has advantages in predicting the hazardous properties of coal seams. They consist in reliable determination of C_o values in the entire range of coal metamorphic transformations and practically functional control of the content of the sum of the remaining components of the organic mass (O_o, H_o, N_o, S_o). Reliable experimental determination of V^{daf} is possible only at C_o values approximately in the range of 80–93%. At $C_o > 93\%$, the coal seams are classified mainly as anthracite, and the evaluation criterion for them is the volumetric yield of volatile substances V_V^{daf} [13]. In both cases, when experimentally determining V^{daf} and V_V^{daf} , only the sum of the released gases CO , CO_2 , CH_4 and H_2 is established. Their ratios are not constant during metamorphic transformations, which makes it impossible to analyze changes in the elemental content of organic matter when predicting the hazardous properties of coal seams.

When compiling the Geological and coal chemical map of the Donetsk basin [13], the yield of volatile substances (V^{daf} and V_V^{daf}) was adopted as the only and uncontested criterion for the metamorphic transformations of the coal seams. The calculated determination of the oxygen content was carried out without taking into account the content of organic sulfur in the organic mass by the difference between 100% and the sum of only three main components (C_o, H_o, N_o). This approach to assessing the degree of metamorphic transformations of coal seams and calculating the oxygen content without taking into account the content of organic sulfur led to a discrepancy, in one case with Hilt's rule, and in the second case to an insufficiently

Fig. 1 Relationship between the yield of volatile substances (V^{daf}) and the carbon content per conditional organic mass (C_o) [19]: 1—averaging curve; × —experimental data [13]; σ —standard deviations; R^2 —coefficients of determination; n —is the number of processed data pairs



specific determination of the types of coal seams according to their reduction at individual stages of coal formation [19, 20]. This casts doubt on the existence of metamorphic stages associated with recovery processes due to their irreversibility. The above-mentioned methodological shortcomings in assessing the degree of metamorphic transformations of mine beds and calculating the elemental oxygen content in the organic mass without taking into account organic sulfur do not reduce the practical and scientific value of the experimental data established over many decades by standard methods [13]. This is confirmed by the graphs of the dependence of some indicators on the elemental carbon content C_o in the organic mass (Fig. 2) [19].

They made it possible to eliminate methodological shortcomings in the assessment of metamorphic transformations of mine beds and the calculation of the elemental oxygen content in organic matter.

Calculation of the elemental oxygen content without taking into account organic sulfur had little effect on the tightness of the correlation dependence of the averaging curve 1 and the calculated oxygen values (Fig. 2a). This is confirmed by the practically functional dependence (determination coefficient $R^2 = 0.98$) and an insignificant standard deviation ($\sigma = 0.58\%$). The average content of organic sulfur (S_o) for the developed coal seams of Donbass during the period of compiling the geological and coal chemical map [13] was in the range of 0.74–2.88% [19].

The values of organic sulfur in this range of its content change were reflected only in the mutual arrangement of averaging curves 1 and 7. The difference between them in the arrangement corresponds approximately to the range of organic sulfur content change. This indicates the reliability of the experimental determination by standard methods of the remaining main components of organic matter (C_o , H_o , N_o) used to calculate the organic oxygen content.

With an increase in the influence of metamorphic processes on the transformation of coal seams (an increase in the C_o content to 93%), the location of the averaging curve of the change in hydrogen content 2 differs slightly from curve 7 (Fig. 2b). A similar location of the averaging curves of nitrogen content (3 and 7) is observed with an increase in carbon content to 93% (Fig. 2c). Some differences in the location, respectively, between averaging curves 2, 7 (Fig. 2b) and 3, 7 (Fig. 2c), are due to the lack of experimental data [19] for anthracite seams (at $C_o > 93\%$).

The content of hydrogen (Fig. 2b) in the organic matter and oxygen, is practically functionally related to the change in C_o ($R^2 = 0.94$; $\sigma = 0.48\%$).

The nitrogen content remains more stable and independent of the influence of metamorphic processes (Fig. 2c). For the overwhelming majority of mine beds, the elemental nitrogen content was in the range of 1.0–2.0%. As a result, there is no correlation dependence of N_o on C_o ($R^2 = 0.32$; $\sigma = 0.36\%$). In parallel with the presence of such a narrow range of elemental nitrogen content for individual coal seams, there are anomalous values. For example, for seam l_5 of main No 3 “Kochegarka”, the nitrogen content was [13] more than four percent ($N_o = 4.23\%$). The increased nitrogen content had a significant effect on the redistribution of the ratio between the remaining main components of the organic mass, which was reflected in the change in the hazardous properties of the coal seam [19, 20]. The case under consideration indicates possible significant deviations from the general pattern of

Fig. 2 Qualitative indicators of consumer properties of coals based on the results of processing laboratory and analytical samples according to [13] for Donbass coal seams and their dependence on the conditional organic mass from the carbon content (C_o) [19]: (a) oxygen (O_o); (b) hydrogen (H_o); (c) nitrogen (N_o); (d) total sulfur (S_t^d); (e) ash yield (A^c); 1, 2, 3, 4, 5—averaging curves of the dependencies of oxygen (1), hydrogen (2), nitrogen (3), total sulfur (4) and ash yield (5), respectively; 2'—average hydrogen content ($\bar{H}_o = 5.09\%$) for hard coal seams with $C_o < 93\%$; 6—dependence of $\Sigma O_o, H_o, N_o$ on C_o , adopted in the calculation of O_o [13]; 7—averaging dependencies of the average coal composition, including the sulfur content of organic matter S_o (d), for coal seams of the Donetsk Basin according to [19]; I, II, III—gradation of coal seam stages by hydrogen content per conventional organic mass, respectively, with C_o changing in the ranges of 79–89% (I), 89–93% (II) and more than 93% (III); $\times$, $\circ$ —experimental data [13] for hard coals and anthracites, respectively; σ —standard deviations; R^2 —coefficients of determination, r —coefficients of correlation; n —number of processed data pairs according to [13]

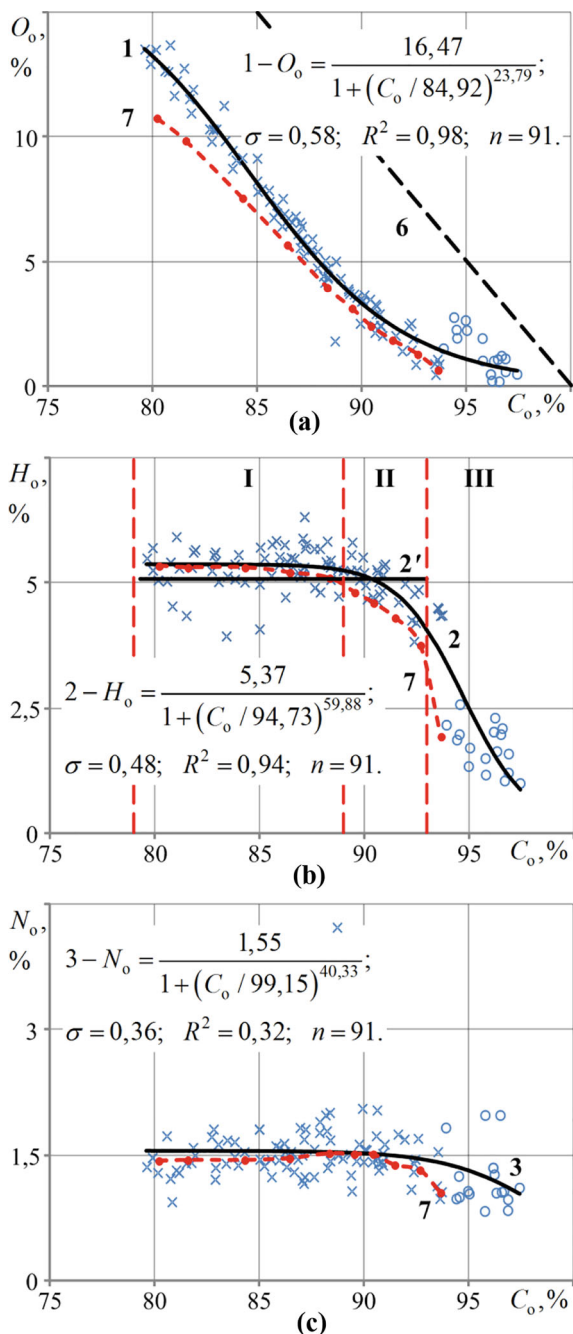
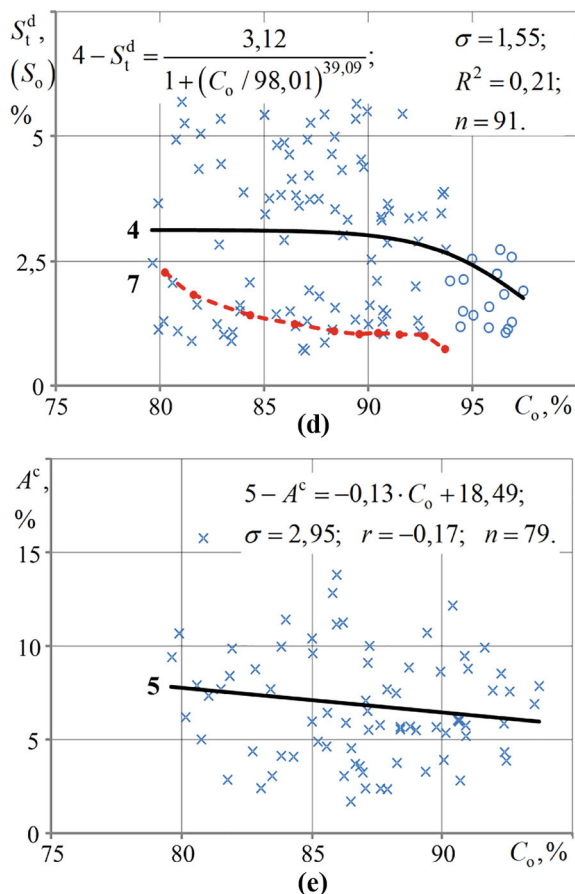


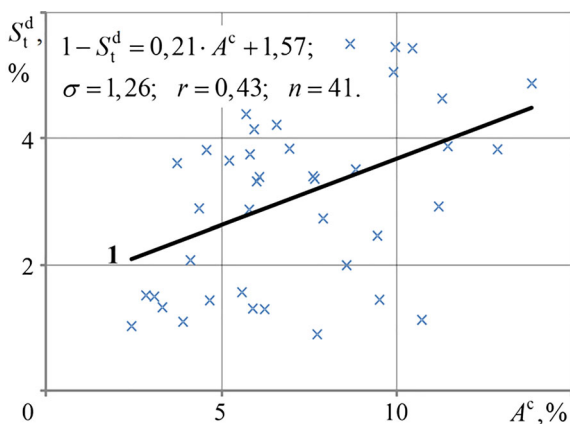
Fig. 2 (continued)



formation of ratios for individual coal seams between the main components of the organic mass at earlier stages of coal formation. Such circumstances must be taken into account when predicting the manifestation of hazardous properties of coal seams based on the factor of their metamorphic transformations.

At earlier stages of coal formation, compared to metamorphism (change in C_o in the range of 80–97%), accumulation and formation of the final content of total sulfur S_t^d occurred (Fig. 2d) [19, 20]. During metamorphic processes, its content changed little. This is evidenced by the practical parallelism of the averaging curve 4 to the abscissa axis for the Carboniferous stage (at $C_o < 93\%$) and the absence of a correlation with the carbon content indicators ($R^2 = 0.21$; $\sigma = 1.55\%$). Judging by the location of the averaging curve 7, the sulfur content of the organic matter also changed insignificantly during metamorphic processes. It follows from the graphs provided (Fig. 2d) that the elemental content of total and organic sulfur for a separately considered coal seam at the stage of metamorphic transformations are practically unpredictable parameters. Along with this, given their stable value for a specific

Fig. 3 Dependence of total sulfur content (S_t^d) on ash yield (A^c) [19]: 1—averaging line; × —experimental data [13]; σ —standard deviations; R^2 —coefficients of determination; n —is the number of processed data pairs



coal seam, they have a significant individual effect on the ratios between the main components of the organic matter and mineral impurities. Such indirect or direct effect of the elemental content of total and (or) organic sulfur must be considered when predicting the hazardous properties of a specific coal seam. The individual need to study the ratios between the main components of mineral impurities for each coal seam under consideration also follows from the absence of a close correlation ($r = 0.43$; $\sigma = 1.26\%$) between the content of total sulfur S_t^d and the yield of ash A^c (Fig. 3) [19].

The ash yield characterizes, to a large extent, the mass content of mineral impurities. Considering the absence of a correlation between the ash yield and the carbon content (Fig. 2e, straight line 5, $r = -0.17$; $\sigma = 2.95\%$), it can be assumed that there is no correlation with the content of mineral impurities.

The ambiguous relationships between the main components of organic matter and mineral impurities for each mine seam are taken into account when calculating the oxygen content according to [21] by the difference:

$$O_d = 100 - (C + H + N + S + A), \% \quad (2)$$

where O_d is the oxygen content by the difference in components in the analyzed sample, %; C , H , N , S , A —respectively, the mass fractions of carbon, hydrogen, nitrogen, sulfur and ash content in the analyzed analytical sample without corrections for the content of the mass fraction of moisture in the analytical sample (W^a), %.

Corrections for moisture content in the analytical sample are introduced for hydrogen

$$H^a = H - 0.1119 \cdot W^a, \% \quad (3)$$

and oxygen

$$O_d^a = O_d - 0.8881 \cdot W^a, \% \quad (4)$$

Calculation according to [ДСТУ ISO 17247:2010, 2012] is also possible from the ratio of components of the analytical sample:

$$O_d^a = 100 - (C^a + H^a + N^a + S_t^a + A^a), \% \quad (5)$$

It is necessary to take into account that hydrogen is contained in moisture and hydrated water of silicates. Oxygen, in addition to water, can be found in silicates (aluminum and silicon oxides), iron oxides, carbonates and sulfates.

The accuracy of forecasting hazardous properties of coal seams is largely determined by the state of the coal samples under consideration [19, 20]. The developed methods [21] allow recalculation from the analytical state to the working (r), dry (d) and (or) working (r, w) states, including hydrogen and oxygen of moisture. The analytical state of the samples (a) is the initial version for recalculation to all other states. Using experimental data on the mass fraction of moisture (W^a) of the analytical samples [19] and the method [21], the authors clarified the elemental content of hydrogen H^a (according to Eq. 3) and oxygen O_d (according to Eq. 4) in the analytical samples was clarified, the results are presented in [19].

Calculation of the elemental oxygen content according to Eq. 4 made it possible to eliminate the drawback of the method [13], associated with ignoring the influence of organic sulfur. In addition, it allows oxygen to be distributed between its content directly in the dry part of the analytical sample O_d^a and in its moisture ($0.8881 \cdot W^a$).

Equations 4 and 5 for determining the calculated values are equivalent [21]. The values O_d^a calculated using Eq. 4 using W^a are equal to the values O_d^a determined according to Eq. 5. This made it possible, having previously calculated O_d^a according to Eq. 4 [19], to determine, using dependence 5, the total values of S_t^a and A^a from the expression:

$$S_t^a + A^a = 100 - (C^a + H^a + N^a + O_d^a), \% \quad (6)$$

The use of the values of the mass fraction of moisture in analytical samples W^a (Eq. 4) significantly affected the reduction in the tightness of the correlation dependencies between the calculated oxygen content O_d^a and carbon C_o (Fig. 4a). When calculating oxygen according to the method [13], the determination coefficient (R^2) was 0.98 with a standard deviation of 0.58% (Fig. 2a). Bringing (recalculating) the samples to their analytical state according to the standard [21] caused a decrease in R^2 to 0.82 and an increase in the standard deviation to 1.41% (Fig. 4a) [19]. This indicates a significantly smaller dependence of the sulfur content in the analytical samples S_t^a and analytical moisture W^a on the metamorphic processes of transformation of coal seams compared to the change in the elemental composition of oxygen.

This assumption is also confirmed by a fairly close correlation ($r = 0.84$) between the oxygen content O_d , determined from the difference in experimental data [13], and its calculated content in analytical samples according to [21] (Fig. 5a) [19].

Fig. 4 Results of recalculation of experimental data [13] of the elemental content of the main components of fossil coals to their analytical state according to the standard [21]: 1, 2, 3, 4—averaging lines of dependences of oxygen (O_d^a), hydrogen (H^a), nitrogen (N^a), respectively; total ash yield and total sulfur content ($A^a + S_t^a$); 5—averaging curves of the content of oxygen (a), hydrogen (b), nitrogen (c); organic sulfur (d) for the initial analytical sample for coal seams of the Donetsk Basin according to [19]; $\times$ —experimental data [13] for hard coals; σ —standard deviations; R^2 —coefficients of determination

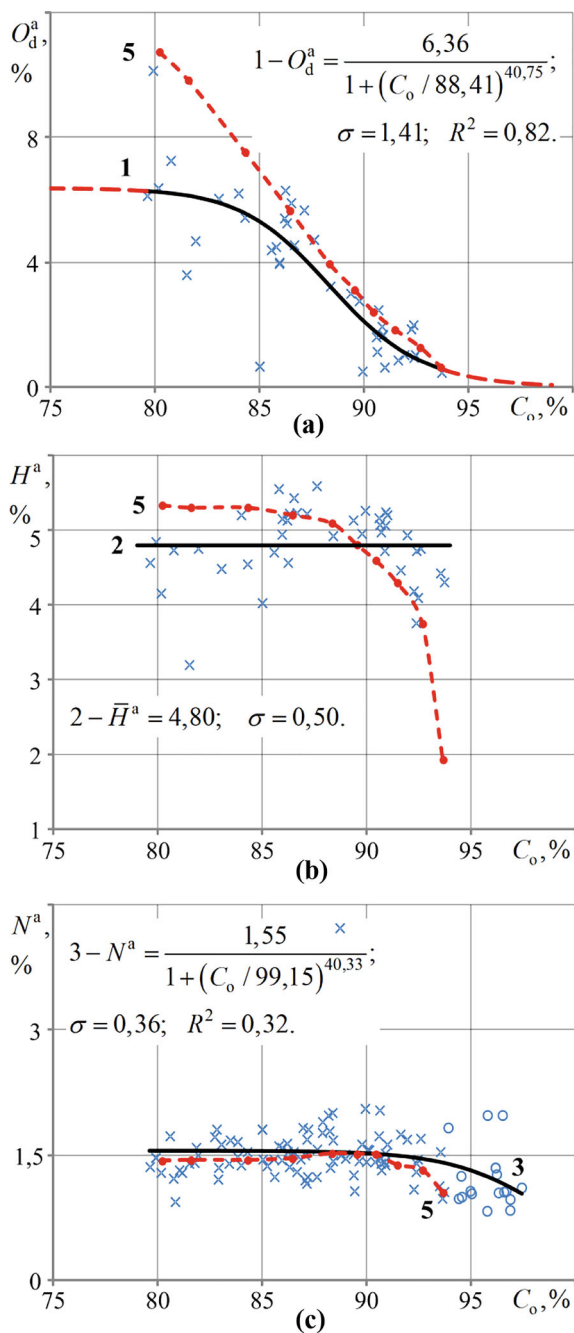
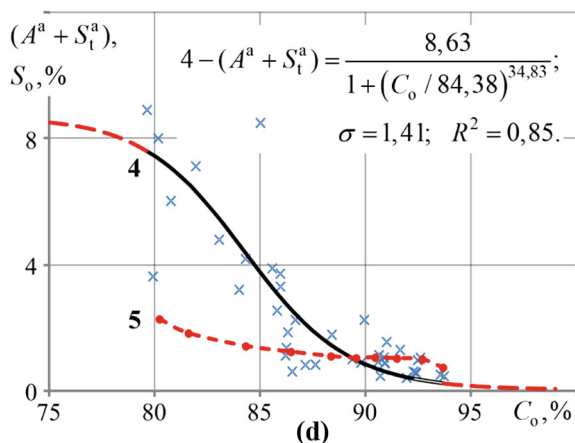


Fig. 4 (continued)

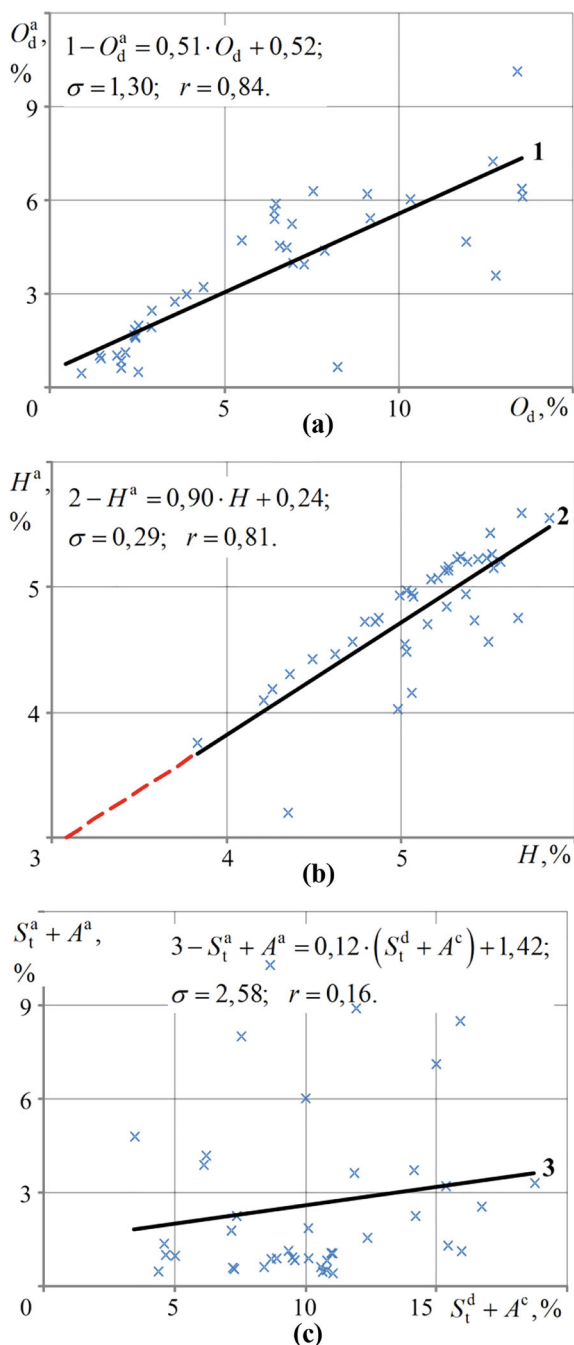


Along with such correlationally confirmed ratio of oxygen content according to the results of experiments (O_d) and their recalculations for the analytical state of samples (O_d^a), the interdependence of the total yield of ash and total sulfur for different states of samples has not been established (Fig. 5c). In the case under consideration, it is possible to state the complete absence of correlation ($r = 0.16$; $\sigma = 2.58\%$), which was one of the reasons for the decrease in the tightness of the correlation dependence of oxygen content when recalculated according to Eq. 4.

The reduction in oxygen content with the intensification of metamorphic transformation of coal seams (increase in C_o) was not always accompanied by a reduction in analytical moisture in the samples. This is evidenced by the parallelism (at $C_o < 89\%$) of one of the sections of curve 2 of the H_o dependence of the abscissa axis (Fig. 2b). Along with the unchanged average hydrogen content at this stage of metamorphic transformations ($C_o < 89\%$), a reduction in oxygen occurred (Fig. 2a). Obviously, these processes of reducing oxygen content at the considered stage of metamorphism (coalification) are due to its removal from other components with an unchanged content in the moisture of analytical samples.

In the whole range of metamorphic transformations of coal seams (with growth of C_o from approximately 80–97%), a nonlinear, practically functional dependence of hydrogen (H_o) on the conditional organic mass was established based on the results of processing analytical samples (Fig. 2b, curve 2, $R^2 = 0.94$, $\sigma = 0.48\%$). Based on this dependence, the value of the carbon content, equal to 93%, is not an unambiguous boundary between hard coals and anthracites according to the hydrogen content criterion. After an increase in the C_o content, approximately to 89%, the straight section of curve 2 smoothly passes into its nonlinear dependence of a decrease in the H_o content as the influence of metamorphic processes further increases. A parallel decrease in the oxygen (Fig. 2a) and hydrogen (Fig. 2b) content indicates the removal of a certain proportion of moisture from the analytical samples at the stage of metamorphic transformations at $C_o > 89\%$.

Fig. 5 Interdependence of the results of the experimental determination of the mass content of components according to [13] with their values after recalculation for the analytical state of samples according to the method [21]: 1, 2, 3—averaging lines of oxygen content (a), hydrogen (b) and total ash yield and total sulfur content (c); $\times$ —ratio of experimental and calculated values of components for coal seams; σ —standard deviations; r —correlation coefficients



A similar curve of the dependence of the hydrogen content on the analytical state of the samples according to Eq. 3 could not be established due to the lack of experimental data on moisture for anthracite coal seams. The Geological and coal chemical map of the Donetsk basin [13] provides information on the moisture of analytical samples (W^a) presumably only for hard coal seams ($C_o < 93\%$). This made it possible to determine the hydrogen content (H^a) in the analytical samples only for these coal seams according to Eq. 3 (Fig. 4b). In the considered range of C_o change, the H^a values changed insignificantly within 3.20–5.60% [19], and its average value (line segment 2 in Fig. 4b) was 4.80% with a standard deviation of 0.50%.

The average value $\overline{H}^a = 4.80\%$ was obtained by recalculating the experimental data on the hydrogen content (H_o) to the conventionally organic mass for 41 coal seams [19]. For this set of coal seams, the average initial values $\overline{H}_o$ of and analytical moisture $\overline{W}^a$, used for recalculation according to Eq. 3 to the average analytical state of hydrogen ($\overline{H}^a$), were 5.09 and 2.60%, respectively. The average value (line segment 2', Fig. 2b) differs significantly from the general averaging curve 2 in the range of C_o change within 89–93%. According to [13], the coals of these coal seams are classified as hard coals. In terms of hydrogen content per conditional organic mass, they are transitional from hard coals to anthracites. This follows from the deviation of the general averaging curve 2 from the straight line segment 2' within the limits of increasing C_o content from 89 to 93% (Fig. 2b). This provides grounds for a more detailed examination of the change in hydrogen content per conditional organic mass in three ranges of C_o change (Fig. 2b). As a first approximation, the approximate boundaries of these ranges are C_o values within 79–89%, 89–93% and more than 93%, respectively. Adhering to this gradation, we examined the available experimental data on the relationship between the indicators used to calculate the average hydrogen values in analytical samples ($\overline{H}^a$) according to Eq. 3 (Table 1) [19].

In the ranges of change of C_o of analytical samples within 79–89% and 89–93% (Fig. 2b, stages I and II) the average values of hydrogen content practically did not differ from each other. Their calculated values were 4.87 and 4.86, respectively (Table 2). Despite the obtained practically identical values $\overline{H}^a$, there is no reason to classify the sets of these coal seams according to the factor of hydrogen content in analytical samples to the same degree of metamorphic transformations (coalification). The reason for this is the different moisture content (W^a) in analytical samples (Table 1). It is also one of the criteria of metamorphic transformations of coal seams [19, 20]. The most unpredictable W^a significantly changed in the range of $C_o = 79$ –89%. With the average value $\overline{W}^a = 4.04\%$, the range of change of moisture of analytical samples fluctuated within 0.62–10.32% and the standard deviation from the average value of 2.95%. The hydrogen content per conventional organic mass was not subject to such significant changes. At $\overline{H}_o = 5.32\%$ its values were within 3.94–6.33, and the standard deviation was 0.43% (Table 1). The values of these indicators and the parallelism of the section of curve 2 to the abscissa axis at stage I (Fig. 2b), indicate the independence of the hydrogen content per conventional organic mass from the metamorphic processes at the considered stage of the transformation of the

Table 1 Information on the experimental values of the indicators [13, 16] used to calculate the average hydrogen content ($\bar{H}^a$) in different ranges of carbon changes per conditional organic mass to Eq. 3 [21]

	Stage I	Stage II	Stage III
C_o , %	79–89%	89–93%	more 93%
	Hydrogen information on conventional organic mass, H_o [13]		
Number of coal seams, pcs	61	26	22
Range of change, %	3.94–6.33	3.83–5.82	1.00–4.51
Standard deviations, %	0.43	0.45	1.14
Average value, $\bar{H}_o$, %	5.32	4.97	2.23
	Information on moisture content of analytical samples. W^a [13]		
Number of coal seams, pcs	22	17	2
Range of change, %	0.62–10.32	0.42–2.26	0.47–0.52
Standard deviations, %	2.95	0.45	0.04
Average value, $\bar{H}_o$, %	4.04	0.98	0.50
$\bar{H}^a$, %	4.87	4.86	2.17

mine beds (at $C_o = 79$ –89%). The transformations at this stage are almost entirely associated with the reduction of oxygen content not in the organic mass, but in other components of mineral impurities. Mineral impurities also include a certain proportion of moisture in analytical samples. The stable hydrogen content per conditional organic mass ($\bar{H}_o = 5.32\%$, $\sigma = 0.43\%$) and unpredictable general fluctuations in the moisture content of analytical samples (the range of change is 0.62–10.32% at $\bar{W}^a = 4.04\%$ and $\sigma = 2.95\%$) indicate individual differences between the mine seams of the considered stage of their metamorphic transformations only in terms of the moisture content factor in analytical samples.

The second distinguishing criterion between coal seams may be the amount of external moisture removed during the preparation of analytical samples. In most

Table 2 Key performance indicators of coal mining enterprises

Indicators of market performance	Indicators of social efficiency
1. Increase in coal production	1. Increase in tax revenues to budgets
2. Increase in coal product sales	2. Preservation or growth of employment
3. Decrease in coal cost	3. Increase in wages
4. Increase in coal product quality	4. Improvement of professional qualifications of employees
5. Increase in production profitability	5. Development of social infrastructure
6. Increase in profits	

cases, when establishing the consumer qualities of coals, the presence of this type of moisture is not determined due to the lack of practical need. For a reliable forecast of the hazardous properties of coal seams, it is necessary to take into account the ratio between these types of moisture, since the presence of total moisture affects the physical and mechanical state of coal seams. Confirmation of this need is the gradation of brown coals according to the standard [16] by the values of moisture presence. Additionally, it was also established [19, 20] that at the early and middle stages of coal transformation, the properties of coal seams are associated mainly with the mechanical removal of moisture from the system. Considering the invariance of the average hydrogen content per conditional organic mass at the first stage of metamorphic transformations of coal seams at ($C_o = 79\text{--}89\%$), mainly only external moisture was mechanically removed from the system.

In general, changes in moisture and oxygen content determine the type of fuel formed during geological processes. When moving from peat to brown coal, the main distinguishing features are only moisture and oxygen content. The moisture content in peat is about 75%, in brown coal—60% or less. The oxygen content is 31–40% and 18–26%, respectively [19, 20].

Mechanical removal of external moisture continued at the first stage of transformation of coal seams ($C_o = 79\text{--}89\%$). This follows from the parallelism of the part of the averaging curve 2 of the abscissa axis of the dependence of hydrogen content on the conditional organic mass and the insignificant (0.43%) standard deviation of the experimental data (Fig. 2b, Table 1).

The decrease in the average hydrogen content ($\overline{H}_o$) to 4.97% at the second stage ($C_o = 89\text{--}93\%$) indicates the removal of some share of the system, together with external moisture and, the moisture of the analytical samples. The decrease in the average moisture value of the analytical samples at the second stage, compared to the first, occurred from 4.04% to 0.98%. At the same time, the standard deviation decreased more than six times from 2.95 to 0.45% (Table 1), which indicates the stability of the processes of reducing the moisture of the analytical samples for coals at the second stage of the transformation of coal seams.

At the third stage of metamorphic transformations of the coal seams ($C_o > 93\%$), together with external moisture, based on the minimum values of $\overline{H}^a = 2.17\%$ and $\overline{W}^a = 0.50\%$, there was a further reduction in the moisture content of the analytical samples to their minimum values (Table 1).

On the basis of the conducted research, the peculiarities of the selection of indicators of metamorphic transformations of coal are established, which are used in parallel, respectively, to establish the quality of fuel and forecast the dangerous properties of coal seams. Inconsistencies in the indicators of the degree of metamorphism of the state of fuel during mining operations in underground conditions, which are used in the current regulatory framework for the safe conduct of mining operations, were revealed. In order to improve normative documents on the safe conduct of mining operations, it is necessary to consider indicators that characterize not only the organic (combustible) part of the fuel, but also the presence of mineral impurities and moisture in coal in the area of mining operations [22].

4 Economic Assessment and Optimization

The direct contribution of the coal mining industry to Ukraine's GDP in 2022 amounted to 5.69%. The invasion of Russian troops into the territory of Ukraine had a major impact on the activities of Ukrainian enterprises, which led to a drop in Ukraine's nominal GDP by 259.82 billion UAH (4.77%) compared to the previous period. Thus, in 2022, there was a negative growth rate of GDP in the coal mining industries by 15.83%, with a corresponding decrease of GDP by 55.6 billion UAH. Accounting this negative trend, the role and importance of innovation as an effective type of economic activity where added value is created is becoming more relevant.

The implementation of innovative development programs for the enterprises, requires special attention to the management process, which takes into account the cyclical nature of economic processes both at the enterprise level, at the industry level, at the state level and at the global level. Very relevant are the issues of the methodological nature of entrepreneurial activity in the conditions of cyclical economic processes, where the variability of the business environment, rather rapid changes in production and commercial activities require scientific and methodological tools, forecasting of trends, strategic planning and foresight in development.

Investment support of an enterprise is the process of implementing measures to create a set of organizational and economic conditions for carrying out investment activities, attracting and using investment resources and assessing the level of investment attractiveness necessary for effective investment support in order to justify the feasibility of implementing an investment decision.

Investment support includes the following main components: financial, technical and technological, informational, social and other components that stimulate the income formation or social effect [23].

One of the most important indicators of investments effectiveness in the improvement of working conditions is their social effect, which for any investment option must be positive. The economic efficiency of investments in the improvement of working conditions is the economic expression of the social effect, its inevitable consequence. The working conditions of coal mine workers that involved in underground mining operations are characterized by a significant number of dangerous and harmful factors, investments in specific occupational safety measures often do not ensure a reduction in the share of workers that work in dangerous or harmful conditions, therefore the social effect of such investments must be determined through integral indicators calculated according to the data of the "Working Conditions Map", by comparing the actual values of harmful factors with their predicted values.

The costs of measures by improvement of working conditions in coal mines in Ukraine are usually several times higher than at the other enterprises. However, in conditions of shortage of financial resources, they are often insufficient, and the measures often turn out to be ineffective, and sometimes economically inexpedient. One of the reasons for this is the imperfection of known methods for assessing the effectiveness of costs aimed at the improvement of working conditions and increasing the safety.

From the point of view of economics, expenses for the improvement of working conditions are investments [24], since they are made with the aim of obtaining long-term benefits due to increased labor productivity, improved product quality, preservation of labor potential, etc. Investments play a crucial role in restoring and increasing production resources, and, accordingly, ensuring sustainable economic growth rates. They constitute the material basis of its development. According to the economic dictionary, investments are long-term capital investments in one's own country or abroad in enterprises of various industries, business projects, socio-economic programs, innovative projects. These can be both monetary and property or intellectual values. The main difference between investments and other expenses is that they yield returns after a significant period of time. Investments can be made in fixed (buildings, machinery, equipment, etc.) and current (stocks of goods and materials, etc.) funds, in intangible resources and assets (securities, patents, licenses, etc.). Investments also include expenses for education, scientific research and training of specialists which represent long-term investments in «human capital», which is currently gaining increasing importance, because the main factor of modern economic development is the intellectual product, which determines the economic position of the country in the world hierarchy of states.

The effectiveness of investments in the improvement of working conditions can be expressed in social, socio-economic and economic effects. It is recommended to calculate the social and economic effectiveness of investments both at operating enterprises and during the design, construction and reconstruction of enterprises.

The social effect of investments (increasing of the working capacity, reducing of injuries and illnesses, reducing the staff turnover, increasing the profession prestige of the profession, the degree of job satisfaction, etc.) should always be positive. It usually leads to increased labor productivity, improved product quality, preservation of labor resources, and other positive consequences.

In cases where the social effect can be expressed in monetary form (for example, when increasing labor productivity), the socio-economic efficiency of investments is considered. That is, the economic efficiency of investments in improving working conditions is nothing more than the economic expression of the social effect, its inevitable consequence.

The economic results of investment are expressed in the form of the economic effect of the implementation of occupational safety measures and efficiency indicators or comparative efficiency, if its necessary, a comparative assessment of several areas of investment in improvement of working conditions. The assessment of both social and economic effects, which are closely related, can be carried out independently using individual indicators, as well as on the basis of their combination.

In the context of innovations implementation, in addition to indicators of market (economic) efficiency, ensuring social efficiency is a priority (Table 2). Foreign and domestic practice of economic transformation of coal mining regions demonstrates the necessity and effectiveness of application for them of special investment activity regimes in order to improve working conditions, create new jobs, and overcome social and other problems of unemployment [25].

The occurrence of accidents and emergency situations, as well as accidents that cause injuries, is probable [26]. Therefore, for the determination of predicted damage amount from accidents, it is necessary to take into account both the possible expected economic losses in the event of a certain type of accident and the probable nature of their occurrence [27]. The forecast assessment of material damage due to accidents in coal mines should be differentiated by the type of accident, the place and time of its occurrence, the condition and operation time of the equipment, the preparedness of workers to eliminate accidents, presence and conditions of the equipment intended for its elimination.

As a basic predicted value of the damage amount of a coal enterprise from a certain type of accident the average statistical value of the damage during the time of the accident liquidation determined according to the results of the accident analysis performed by State Paramilitary Mining Rescue Service can be taken.

The estimated value of the predicted damage of a coal enterprise takes into account its base value, adjusted depending on the mining-geological and mining-technical conditions. The correction is carried out in the case of establishing a relationship between the amount of actual damage and factors characterizing the mining-geological and mining-technical conditions at the mine.

Therefore, the implementation of the mechanism for investment support of the innovative development of coal enterprises, as well as the openness of innovation systems and the growth of social capital, will contribute to the establishment of partnerships between enterprises and organizations (both public and private) in the innovation sphere, which is capable of bringing coal industry enterprises to a qualitatively new level of development [26, 27].

5 Conclusions

Currently, there is a negative growth rate of GDP in the coal mining industry, which highlights the role and importance of innovative activity as an effective type of economic activity that creates added value, including the modernization of coal enterprises, improving the regulatory framework for safe mining operations, and conducting scientific and research developments.

The conducted studies allowed to establish and eliminate methodological inconsistencies in determining the calculated oxygen content per conditional organic mass and the degree of metamorphic transformations of coal seams when determining the consumer qualities of coals. The inconsistencies consisted in ignoring the presence of organic sulfur and assessing the degree of metamorphic transformations of coal seams by one indicator (V^{daf}) without considering the change in the ratios between the main components of the organic mass and mineral impurities. In addition, the reliability of experimental data obtained by standard methods in compiling a Geological and coal chemical map [13] and some other reference and regulatory documents [14, 15] determining the consumer qualities of coals in developed mine seams has

been proven. Thanks to the development of these and some other reference and regulatory documents on the consumer qualities of coals over several decades, a solid information base on developed mine seams has been accumulated. These reliable experimental data have not been properly reflected in modern regulatory documents on the safe conduct of mining operations.

Their use will undoubtedly contribute to improving the quality of forecasting the manifestation of hazardous properties of coal seams based on the factor of their metamorphic transformations. At the same time, it is necessary to take into account some features of the methods of experimental determination of the content of components of organic matter and mineral impurities based on the analysis of analytical and laboratory samples. First of all, this is caused by a change in the content of moisture and mineral impurities during their preparation for analysis, which causes a difference in the indicators of consumer qualities of coals from the state of coal seams in the mining zone. Based on the studies conducted, in addition to such differences, it is necessary to take into account the features of the change in the actual ratios between the main components of organic matter and mineral impurities.

For scientifically substantiated application of indicators of consumer qualities of coals in forecasting hazardous properties of coal seams, it is necessary to take into account some features of the elemental composition and changes in the ratio between components during metamorphic transformations. In the first approximation, they are reduced to the following provisions:

- mineral impurities together with organic matter during mining operations determine the hazardous properties of mine seams. Their content in some cases, based on the ash yield, can significantly exceed 40%;
- when preparing laboratory and analytical coal samples for analysis after their preliminary enrichment, the ash yield, as a rule, does not exceed 10%;
- crushing and drying of laboratory and analytical samples is accompanied by the removal of external moisture, and during analysis, a more stable type of analytical moisture is considered;
- the basis of the organic mass (about 100%) is made up of five main components: carbon, hydrogen, oxygen, nitrogen and organic sulfur. Carbon, hydrogen and oxygen are simultaneously part of both the organic mass and mineral impurities. Nitrogen and organic sulfur are elements associated exclusively with the organic matter of coal. The total sulfur content in the organic and mineral masses refers to the total sulfur content;
- the carbon content in the organic mass practically functionally controls the sum of its other main components in the process of metamorphic transformations of the mine seams. Close correlation nonlinear dependencies in the form of sigmoid (logistic function) of changes in the oxygen and hydrogen content with an increase in the carbon content were established. The content of nitrogen and organic sulfur changes insignificantly in the process of metamorphic transformations. The formation of mineral impurities occurred independently of the change in the ratio between the main components of the organic mass, this is evidenced by

the absence of a correlation between the ash yield and the carbon content in the organic mass;

- the change in the ratio between the oxygen and hydrogen content in the analytical samples has established the transition stage from hard coal to anthracite and proven the mechanical removal of external moisture at the previous stages of the transformation of hard coal seams;
- in order to improve normative documents on the safe conduct of mining operations, it is necessary to consider indicators that characterize not only the organic (combustible) part of the fuel, but also the presence of mineral impurities and moisture in coal in the area of mining operations;
- all indicators of consumer properties of coals determined by standard methods and given in reference and regulatory documents on quality can be reduced to their analytical state. To do this, it is necessary to introduce adjustments for the hydrogen and oxygen content, taking into account the moisture of the analyzed samples. This makes it possible to recalculate the elemental analysis for various states of coals, including in the mining zone.

References

1. Endl A, Tost M, Hitch M, Moser P, Feiel S (2021) Europe's mining innovation trends and their contribution to the sustainable development goals: blind spots and strong points. *Resour Policy* 74:101440. <https://doi.org/10.1016/j.resourpol.2019.101440>
2. Lay J, Wiewiora A, Kashan AJ, Bradley L (2022) Operationalising a process model of innovation for the mining industry. *Resour Policy* 79:102988. <https://doi.org/10.1016/j.resourpol.2022.102988>
3. Kryshchanovych M, Akimova L, Akimov O, Kubiniy N, Marhitich V (2021) Modeling the process of forming the safety potential of engineering enterprises. *Int J Saf Secur Eng* 11(3):223–230. <https://doi.org/10.18280/ijssse.110302>
4. Brodny J, Tutak M (2022) Challenges of the polish coal mining industry on its way to innovative and sustainable development. *J Clean Prod* 375:134061. <https://doi.org/10.1016/j.jclepro.2022.134061>
5. Tian T, Nie B, Zhang X, Li X, Hong X (2024) Analyzing the socially sustainable impacts of private investments in the mining sector in rural areas. *Resour Policy* 98:105347. <https://doi.org/10.1016/j.resourpol.2024.105347>
6. Kashan AJ, Lay J, Wiewiora A, Bradley L (2022) The innovation process in mining: integrating insights from innovation and change management. *Resour Policy* 76:102575. <https://doi.org/10.1016/j.resourpol.2022.102575>
7. Rukovodstvo po bor'be s pyl'ju v ugol'nyh shahtah (1979) Nedra, Moskva, p 319
8. Rukovodstvo po proektirovaniju ventiljacii ugol'nyh shaht (1994) Osnova, Kiev, p 311
9. Rukovodstvo po preduprezhdeniju i tusheniju jendogennyh pozharov na ugol'nyh shahtah Ukrainy: KD 12.01.402-2000 (2000) NIIGD, Doneck, p 216
10. Pravyla vedennia hirnychkyh robit na plastakh, skhylnykh do hazodynamichnykh yavysshch: SOU 10.1.00174088.011-2005 (2005) Minvuhleprom Ukrainy, Kyiv, p 224
11. Medvedev JN, Martovickij VD, Kashuba OI, Krutenko SA, Topchij SE (2006) Teorija i praktika ohrany truda na ugol'nyh shahtah. MakNII Promdruk, Makeevka, p 600
12. DSTU 9220:2023 (2024) Solid mineral fuel. Methods for determination of volatile matter. UkrNDNTs

13. Donetsk State Coal Research Institute (1954) Geological and coal-chemical map of the Donetsk Basin, vol 8. Ugletekhizdat, Moscow, p 430
14. Spravochnik po kachestvu i obogatimosti kamennyh uglej i antracitov Ukrainskoj SSR (Donbass v granicah USSR, L'vovsko-Volynskij bassejn). Harakteristika kachestva kamennyh uglej i antracitov Ukrainskoj SSR (1965) Nedra, Moskva, p 204
15. Spravochnik po kachestvu kamennyh uglej i antracitov Doneckogo i L'vovsko-Volynskogo bassejnov. Doneckij nauchno-issledovatel'skij ugol'nyj institut (1972) Nedra, Moskva, p 168
16. Antoshchenko MI, Tarasov VY, Rudniev YS, Zakharova OI (2022) Using indices of the current industrial coal classification to forecast hazardous characteristics of coal seams. *Min Miner Deposits* 16(2):7–13. <https://doi.org/10.33271/mining16.02.007>
17. Pal D, Ghosh S, Varma AK (2021) Metamorphism in coal. Dept. of Applied Geology Indian Institute of Technology (Indian School of Mines, Dhanbad. <https://doi.org/10.13140/RG.2.2.17190.24642>
18. Rudniev Y, Galchenko A, Tarasov V, Antoshchenko M (2022) Moisture as an indicator of the manifestation of hazardous properties of coal seams. *Geofizicheskiy Zhurnal* 44(3):66–79. <https://doi.org/10.24028/gj.v44i3.261969>
19. Ye R, Filatieva E, Antoshchenko M, Brozhko R (2023) Main components of fossil coals, their consumer qualities and hazardous properties of coal seams. *J Donetsk Min Inst* 53(2):82–100. <https://doi.org/10.31474/1999-981X-2023-2-82-100>
20. Rudniev Y, Filatieva E, Antoshchenko M, Popovich V (2024) Material balance of metamorphic transformations of coal seams and their dangerous properties. *Geofizichnyi Zhurnal* 46(3):85–110. <https://doi.org/10.24028/gj.v46i3.298964>
21. DSTU ISO 17247:201 Coal. Ultimate analysis (ISO 17247:2006, IDT). UkrNDNTs
22. Rudniev Y, Popovych V, Brozhko R, Tarasov V (2024) Innovative method of forecasting the manifestation of dangerous properties of coal seams. In: Semenov A, Yepifanova I, Kajanová J (eds) *Data-centric business and applications. Lecture notes on data engineering and communications technologies*, vol 194. Springer, Cham, pp 337–364. https://doi.org/10.1007/978-3-031-53984-8_15. ISSN: 2367-4512
23. Riepina IM, Shvydanenko HO, Kukoba VP (2016) *Upravlinnia efektyvnistiu funktsionuvannia i rozvytku pidpriemstv: monohrafiia*. KNEU, Kyiv, p 399
24. Novikova OF et al (2022) *Transformatsiia sotsialno-trudovoi sfery v umovakh tsyfrovizatsii ekonomiky: monohrafiia*. IIE of NAS of Ukraine, Kyiv, p 385
25. Amosha O, Cherevatskyi D, Lyakh O, Soldak M, Zaloznova Y (2021) Canvas model of the mining regions' industrial ecosystem based on a circular economy. *E3S Web Conf* 255. <https://doi.org/10.1051/e3sconf/202125501001>
26. Voynarenko M, Dzhezdzhula V, Yepifanova I (2016) Modelling the process of making decisions on sources of financing of innovation activity. *Econ Ann XXI* 160(7–8):126–129. <https://doi.org/10.21003/ea.v160-25>
27. Yaremko SA, Kuzmina EM, Savina NB, Yepifanova IYu, Gordiichuk HB, Mussayeva D (2022) Forecasting business processes in the management system of the corporation. *Informatyka, Avtomatyka, Pomiary w Gospodarce i Ochronie Środowiska* 12(4):53–59. <https://doi.org/10.35784/iapgos.3249>

Managing the Transition Phase Between Stages of the Manufacturing Enterprise Lifecycle as a Key Growth Driver



Iryna Yepifanova and Anton Kaspirovych

Abstract The study explores the nature and operational specifics of manufacturing enterprises under conditions of atypical lifecycle stress. It distinguishes and analyses the stage and transition phases within the enterprise lifecycle. The components of the transition phase are described, and key attributes and characteristics are identified. The paper highlights critical influencing factors that necessitate the adaptation of management systems to ensure the long-term efficiency and sustainability of manufacturing enterprises, even under forced acceleration through the early lifecycle stages. It establishes the relationships determining how the transition phase management system should evolve in response to increasing pressure on the transition phase. Within the broader research, the authors propose a system of models as tools for managing the transition phase. This study introduces an optimization model to minimise the complexity of navigating the transition phase, serving as a foundation for identifying optimal baseline conditions for initiating the transition to the next lifecycle stage. Through this model, the study compares how increased transition phase pressure and reduced durations of initial lifecycle stages affect the transition's difficulty, and how the transition management system should respond to these new requirements.

Keywords Lifecycle management · Manufacturing enterprises · Transition phase · Enterprise growth · Model · Management systems · Sustainability

I. Yepifanova (✉) · A. Kaspirovych
Vinnitsia National Technical University, Vinnitsia, Ukraine
e-mail: yepifanova@vntu.edu.ua

A. Kaspirovych
e-mail: anton.kaspirovych@pa.ua

1 Introduction

1.1 Relevance

The concept of an enterprise lifecycle (ELC) is a model that outlines the sequential stages of organizational development from inception to termination, each characterized by specific challenges, opportunities, and management approaches.

The theoretical foundations of the enterprise lifecycle concept were laid in the seminal works of I. Adizes [1], L. Greiner [2], D. Miller [3], and other scholars [4–8]. These studies provided a basic understanding of organisations' developmental stages and the distinct features of each stage.

Within the enterprise lifecycle concept, two main phases can be distinguished as differentiating factors for describing various models: the stage phase and the transition phase between stages. Most studies focus on managing the stage phase (its duration, efficiency, etc.), while management of the transition phase remains insufficiently explored.

For manufacturing enterprises, the effectiveness of the transition phase is particularly important, as it is the most capital-intensive across all sectors [9]. This is indirectly confirmed by the extended overall lifecycle typical for manufacturing businesses. The war has created non-standard, unstable, and unpredictable business conditions in Ukraine. Combined with the growing demand for new manufacturing enterprises, especially those with high production localization [10], this context fuels interest in enhancing transition phase management.

Among international studies, a significant contribution to the theory of the manufacturing enterprise lifecycle was made by R. Hayes and S. Wheelwright [11], who proposed the product-process matrix. This framework enables analysis of the interdependence between operational and market changes during enterprise growth.

1.2 Goals and Objectives

Under current conditions, one of the key requirements for manufacturing enterprises is the speed at which they move from conception to the realization of production volumes. This, in turn, necessitates reducing stage durations and raising the standards for the quality and effectiveness of the transition phase. The shortening of stages primarily produces an increase in risks [12], including:

- Incomplete realization of stage potential, thus limiting the full potential of the enterprise.
- Reduced organizational stability, possibly leading to premature decline.
- Investment overload, requiring accelerated returns and a shift in focus from long-term strategy to short-term results.

This study first presents a detailed description of the transition phase between lifecycle stages of manufacturing enterprises as a distinct object of management. Special attention is given to managing the transition phase itself. The paper proposes a model of transition phase governance, analyses the nature of changes within the management system as a response to increasing transition-related burdens, and introduces a decision-making model for determining optimal conditions for launching a transition phase based on key influencing factors.

2 Theoretical and Conceptual Foundations of the Transition Phase

2.1 Increasing Requirements for the Transition Phase

One of the key challenges facing Ukraine today is the need to quickly launch and scale up production, which is a response to current market demands, particularly in defence, critical infrastructure and import substitution [13]. As a result, there is a tendency to reduce the duration of the first stages of the enterprise's life cycle, in particular the creation, technological development and initial growth phases. Whereas in traditional conditions these stages could last from several years to decades, modern practice involves their compression to months or weeks.

The reduction in the duration of the first phases of the enterprise life cycle has several characteristic features [14]:

- Insufficient accumulation of resource potential (human, production, intellectual).
- Adjustments to business models before their full implementation.
- Parallel execution of R&D and commercial production without completed technological cycles.
- Premature scaling without validated technological and organizational solutions.
- Fragmented or reactive application of mature management practices.

These signs indicate that the enterprise's operational and organisational architectures are incomplete, which increases the risks to the organisation's existence as a whole and the share of prematurely bankrupt businesses.

Reducing the first stages of the life cycle significantly increases the burden on the transition phase between stages. In the classical approach, transitions are a logical continuation of an enterprise's stable evolution, but in modern conditions, they become forced, abrupt, risky, and often a reaction to external shocks or internal lack of stability [15].

In summary, the transition phase becomes the main phase of risk, cost, and management decision concentration. It serves as an 'adaptive gateway' between the underdeveloped stages, within which the enterprise must restructure high-speed. The nature of the rising pressure on the transition phase is described in detail in Table 1 [16].

Table 1 Characteristics of transition phase pressure factors (based on [16])

Risk driver	Description of pressure characteristics
Intensity of change	Modern transitions involve not one, but a sequence of parallel transformations—technological, organizational, product, infrastructure, and managerial. This creates high systemic pressure requiring synchronized management across domains
Decreasing of stabilization time	Transitions are often launched before the previous stage has stabilized—production lines are not fully operational, staff not fully trained, and processes not standardized. Transitions are initiated from a state of instability
Lack of managerial capacity	Classical management systems are designed for routine operation, not change. Enterprises face deficits in • project management, • change implementation, crisis management, • rapid business model adaptation
Uncertainty and risk in the environment	Economic and geopolitical turbulence prevent reliable forecasting. Most decisions are made under high uncertainty, where transitions are forced rather than planned. The focus shifts from development to survival
Capital intensity	Transitions require infrastructure restructuring: purchasing or upgrading equipment, reorganizing facilities, implementing new technologies, and sometimes complete architectural shifts. This makes transition phases the most investment-heavy
Pressure on personnel and culture	Frequent and rapid transitions lead to: • fatigue from constant change, • resistance to innovation, • staff turnover due to burnout, • erosion of corporate unity due to lack of stability

In summary, in the context of shorter life cycle stages, the transition phase is no longer just an intermediate element between development stages. It is transforming into a central area of management attention that requires separate strategic, project, financial, and personnel support. Its successful completion determines the effectiveness of the next stage and the enterprise’s survival in the face of turbulence.

2.2 Nature and Characteristics of the Transition Phase

The transition phase represents an intermediate stage between two enterprise life-cycle phases. It is characterized by deep transformations in one or more core organizational elements, such as the management structure, production technologies, business models, or market strategies. While not a full-fledged lifecycle stage, the transition phase requires specific managerial approaches and heightened attention.

The internal structure of the transition phase can be divided into four key stages:

- **Initiation:** identification of transition drivers and type, decision-making, assessment of feasibility.

- Preparation: planning resources, team formation, risk assessment, scenario design.
- Implementation: execution of changes, control and adjustment of processes, response to unforeseen factors.
- Stabilization: institutionalization of new processes, organisational architecture adaptation, and results evaluation.

The key characteristics of the transition phase are summarized in Table 2 [15].

Determining the transition phase's key attributes (Table 3) [17] is appropriate to formalise it as a fully fledged management object.

In summary, the transition phase becomes an independent object of management, which requires appropriate adaptation of the enterprise management system to the new requirements and challenges of doing business.

Table 2 Characteristics of the transition phase (based on [15])

Characteristic	Description
Temporality	Time-bounded and significantly shorter in duration compared to full lifecycle stages
Transformativity	Involves substantial organizational change, often requiring structural or cultural adaptation
Systemicity	Affects multiple or all subsystems of the enterprise, with broad operational impact
Intensity	Demands concentrated financial, technological, and managerial efforts within a short timeframe
Riskiness	High uncertainty and potential variability of outcomes; mistakes are amplified in impact
Resource intensity	Typically requires intensive capital investment, often of an infrastructure or technological nature
Complexity	Requires approaches distinct from routine management, tailored to conditions of rapid change

Table 3 Key attributes of the transition phase (based on [17])

Attribute	Description
Transition driver	Internal (e.g., end of stage or restructuring), external (e.g., market or regulatory factors), or mixed (a combination of both)
Transition type	Organizational, technological, financial, or complex (involving two or more dimensions simultaneously)
Timeliness	Premature (the organization is not yet ready), optimal (the transition occurs under favourable conditions), delayed (the optimal moment was missed)
Significance	Local (affecting a single unit or subsystem), systemic (impacting several subsystems), and architectural (affecting the entire enterprise structure)
Duration	Short-term (up to 3 months), medium-term (3–12 months), and long-term (over 12 months)
Cost	Budgetary (up to 3 months of revenue), tactical (3–6 months of revenue), strategic (over 6 months of revenue)
Uncertainty level	Low (variance in expected financial outcomes $\leq 25\%$), medium (25–50%), high ($> 50\%$)
Management capacity	Incompetent (the system lacks the competence to implement the transition), competent (sufficient capacity), highly competent (competence exceeds task complexity)

3 Transition Management System as a Subsystem of an Integrated Enterprise Management System

3.1 *Impact of Increased Requirements to the Transition Phase on the Complexity of the Transition Management System*

The enterprise management system is complex in nature and is designed to maximise the efficiency of the enterprise as a whole as an object of management. In the context of the organisation’s life cycle management concept, an integrated management system is formed from two key subsystems: stage management and transition management [1], as shown in Fig. 1.

At the same time, the current conditions of the business environment encourage a significant and rapid modernisation of the architecture and tools of the management system, precisely in the transition phase. Referring to the fundamental law of system management, effective management is possible only when the management entity has sufficient complexity, flexibility or adaptability to respond to all possible changes or variations that occur in the managed object or environment [18]. Thus, significant changes are made to the basic system by changing one of the subsystems.

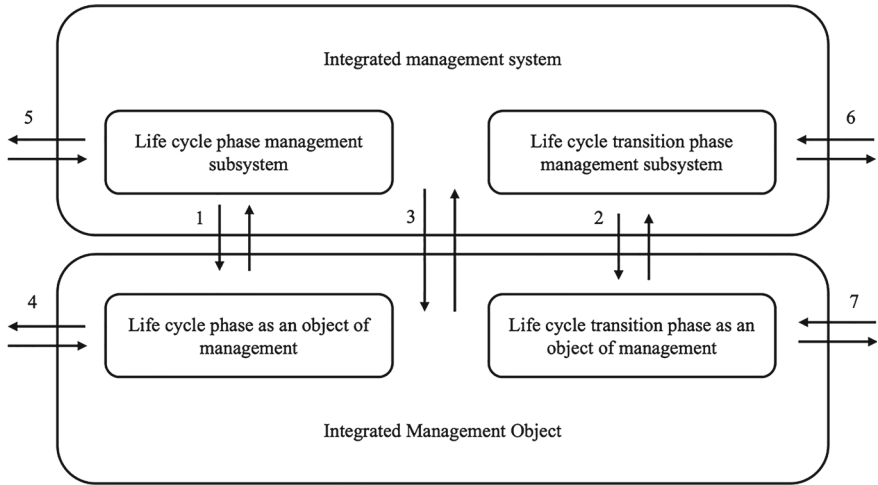


Fig. 1 Basic architecture of the enterprise management system in terms of the concept of organisational life cycle management (based on [1])

3.2 The Impact of Changing the Management System of the Transition Phase on the Integrated Enterprise Management System

Changes in the operating environment of enterprises require increasingly polar management tools for the stage and transition phases. Figure 2 shows how the top-level architecture of the management system of a manufacturing enterprise in Ukraine is changing in 2023–2025. A certain manifestation of these changes is the polarisation of management approaches [19], with lean production [20], TOC [20], creation of process offices [21] and digitalisation of business processes on ERP systems [22] on the one hand, and Agile [23], Scale [24], creation of project offices and/or transformation offices [25] on the other. Accordingly, two separate management subsystems are formed, which are increasingly moving away from each other, further increasing the difficulty of effectively combining the two components of one integrated system.

In summary, a logical chain is formed: the increased requirements for the transition phase (7) further increases the requirements for the transition management system (2) and leads to the emergence of a new subsystem for managing growing diversity (8), as shown in Fig. 2.

The appearance of a new subsystem allows increasing the variety of the integrated management system sufficiently to maintain the quality of the relationships (3) from Fig. 2. As we can see, changes in the transition phase management system significantly affect the restructuring of the architecture of the entire enterprise management system.

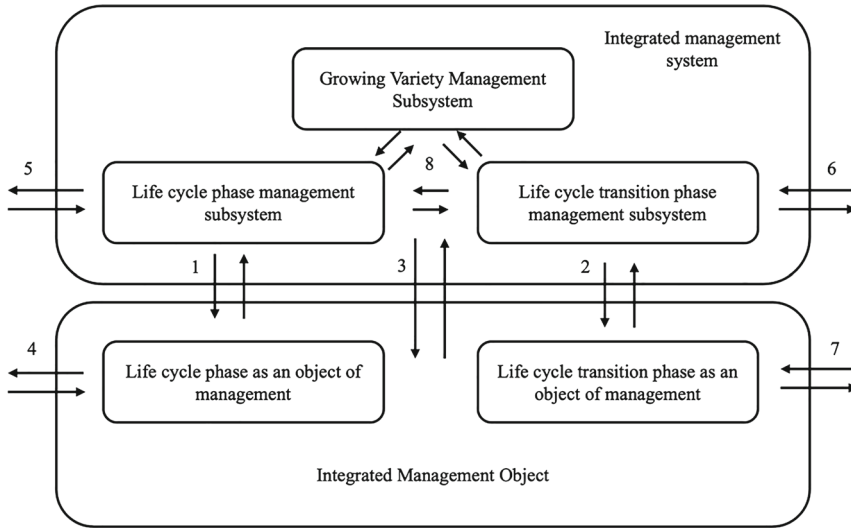


Fig. 2 Model of change of the management system architecture under the influence of changes in requirements to the transition phase management subsystem (adopted from [1])

3.3 *Characteristics of the Transition Phase Management System*

In detailing the transition management system, it is necessary to identify internal subsystems, among which it is advisable to distinguish the following: goal setting, information and analytical, organisational and structural, functional, regulatory and controlling.

Subsystems can be described in terms of function (what they are for) and tools (what they use to produce results). Table 4 [26] provides more details.

In summary, the transition phase and the transition phase management system have been described sufficiently to formulate the objective function and the transition phase management model.

Table 4 Characteristics of subsystems of the transition phase management system

Subsystem	Functions	Core tools/methods
Goal-setting	Establishing strategic transition objectives, defining expected outcomes, KPIs	Goal-based management, performance indicator systems, balanced scorecards
Information-analytical	Monitoring the environment, scenario analysis, generating decision-making insights	SWOT, PESTLE, scenario planning, risk maps, digitalization tools
Organizational-structural	Defining the governing body, roles, responsibilities, and decision-making hierarchy	Organizational structures, RACI matrices, business processes, stakeholder management
Functional	Implementing changes in processes, structures, and technologies	Project management, Agile, change management tools
Regulatory-control	Overseeing the implementation of the transition, deviation analysis, feedback, post-transition audits	KPI monitoring, control points, adaptive regulation

4 Transition Phase Management Model Between the Stages of the Manufacturing Enterprise Life Cycle

4.1 Description of the Objective Function of the Transition Management Model

The objective function of the proposed model is formulated as an optimization problem based on four core factors and three adjustment coefficients. The foundation is built on key attributes of the transition phase, including capital intensity (cost), duration, system capacity for change, and the external conditions under which the transition is executed. The model is intended to identify optimal conditions for launching the transition phase between lifecycle stages.

The objective function of the model is described as follows:

$$\min_t F(t) = \alpha \cdot \sum_i^n (1 - w_i d_i) + \beta \cdot T(t) + \delta \cdot C(t) + \gamma$$

where

- $F(t)$ —integral complexity of executing the transition phase when the decision is made (to be minimised).
- α —transition significance coefficient (ranging from 0.5 to 1.5).
- $\sum_i^n (1 - w_i d_i)$ —inverse function of the system's capacity for transition, calculated as the inverse sum of all available weighted management decisions.
- $T(t)$ —assessment of the transition duration at decision time.

- β —timeliness coefficient (from 0 to 2).
- $C(t)$ —assessment of the transition cost at the decision moment.
- δ —investment risk coefficient (from 0 to 2).
- γ —level of uncertainty in the external environment (from -1 to 1).

The constraint system for this function is defined as:

- $\alpha \in \{0.5, 1.0, 1.5\}$ —the basic lifecycle model assumes three transitions, each subsequent transition is more complex than the previous one. Discrete proportional values are used in the model.
- $\beta, \delta \in [0, 2]$ —coefficients may range from 0 to 2, depending on the circumstances under which the transition decision is made.
- $T(t)$ —assessed as the ratio of the planned transition phase duration to the duration of the previous stage.
- $C(t)$ —calculated as the ratio of the planned transition budget to the total revenue from the last three years (or approximated if there are less than three).

The goal of the model is to maintain the integral complexity indicator below 1. The lower the function value, the more justified it is to initiate the transition phase, the more prepared the management system, and the higher the likelihood of successfully progressing to the next lifecycle stage under planned conditions.

This model enables a comprehensive evaluation of transition phase management scenarios, accounting for financial, temporal, organizational, and contextual uncertainty factors. It serves as a strategic and operational decision-making tool in unstable environments.

4.2 Parameters and Coefficients

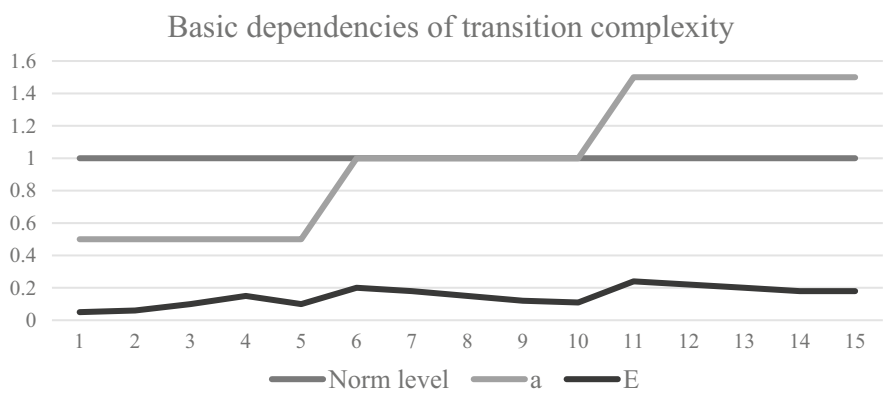
The set of management decisions d_i currently available to the enterprise's governance system reflects its overall capacity to carry out a transition. Each decision is associated with a specific competency that determines its feasibility [26]. Moreover, every decision is weighted by a coefficient w_i that represents its impact on the system's total capacity for change. The weighted sum of these decisions $w_i d_i$ lies within a range from 0 to 1. Consequently, the inverse function $1 - w_i d_i$ demonstrates the inability to make a specific decision and its impact on the decision-making system as a whole. And the sum of all inverse functions $\sum_{i=1}^n (1 - w_i d_i)$ summarises the inability of the system to make the transition. The lower the value of this inverse function, the higher the actual capacity of the system to execute the transition.

The transition significance parameter α reflects the complexity of the transition. The more mature and larger the organization, the more complex the transition process becomes and the higher the demands on the governance system [26]. This study proposes the use of discrete coefficient values: 0.5, 1.0, and 1.5, representing

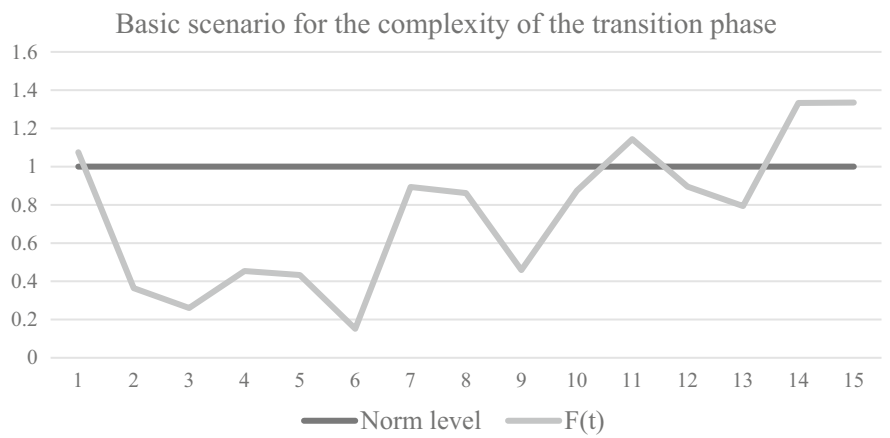
increasing levels of transition difficulty. Accordingly, the general curve of the function $F(t)$, representing transition complexity over time, trends upward, as illustrated in Graph 1.

The graph shows that the basic period under study is 15 years, divided into three segments within which the relevant transition should occur [10]. As the transition complexity threshold increases, the management system’s capacity decreases. This dependence forms the basic graph of the function $F(t)$. Other indicators complement the basic graph, increasing the volatility and more accurately determining when and under what conditions it is advisable to launch the transition phase (Graph 2).

According to the model, a transition between lifecycle stages is considered feasible when the integral function value remains below 1. Over time, complexity increases,



Graph 1 Basic dependencies of transition complexity (adopted from [10, 26])



Graph 2 Basic scenario for the complexity of the transition phase (adapted from [10, 26])

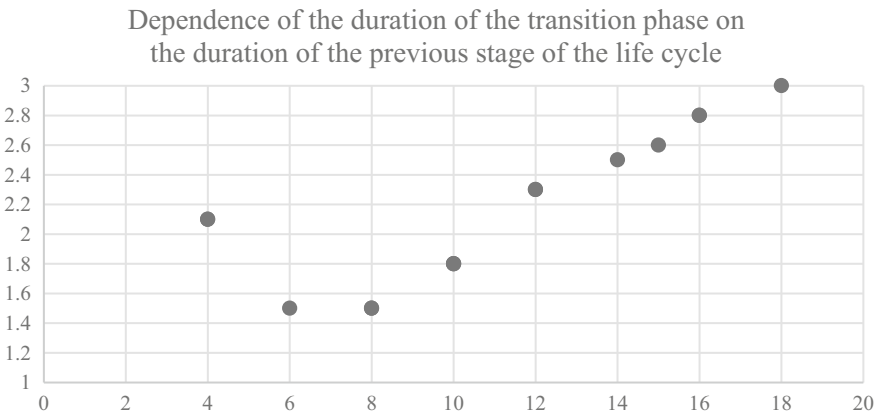
making transitions more difficult. However, depending on the system’s competency, the timeliness of transition initiation, and the investment risk level, temporary “windows of opportunity” may arise [26]. In the basic scenario, these windows appear around years 2, 6, 9, and 13. This is not a universal pattern for all manufacturing enterprises but a demonstration of how systemic factors influence transition complexity.

The timeliness coefficient β identifies optimal “windows of opportunity.” Starting the transition earlier, when the management system’s capacity is still insufficient and the risks of investment are higher, significantly increases the transition’s complexity. Similarly, if the optimal transition window is missed, the transition phase’s complexity immediately increases [26]. This dependence further narrows the window of opportunity.

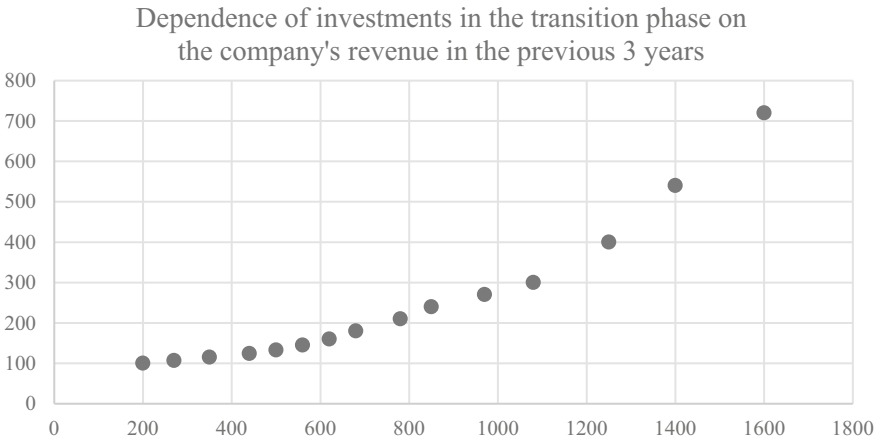
The duration assessment $T(t)$ is defined as the ratio between the planned duration of the transition and the duration of the preceding lifecycle stage. Analysis of this relationship yields a parabolic curve, where very short or very long stages equally require a long transition period, while the average (balanced) stage duration allows for a shorter transition duration, as shown in Graph 3 [26].

The investment risk parameter δ reflects the riskiness of investments, particularly the risk of investing in launching the transition. This parameter varies by industry and geography and is influenced by historical investment effectiveness at a given stage of the enterprise lifecycle [26]. As a result, the model becomes more adaptable and individual to various industries and enterprise profiles.

The transition cost assessment $C(t)$ is calculated by comparing the planned transition budget to the enterprise’s revenue from the past three years [26]. If the decision to transition is made earlier than three full years of revenue history, an extrapolated estimate based on prior performance is used. As can be seen from Graph 4, this



Graph 3 Dependence of the duration of the transition phase on the duration of the previous stage of the life cycle (based on [26])



Graph 4 Dependence of investments in the transition phase on the company's revenue in the previous 3 years (based on [26])

dependence has an exponential form, i.e. the higher the company's income, the more money the business needs for effective restructuring [26].

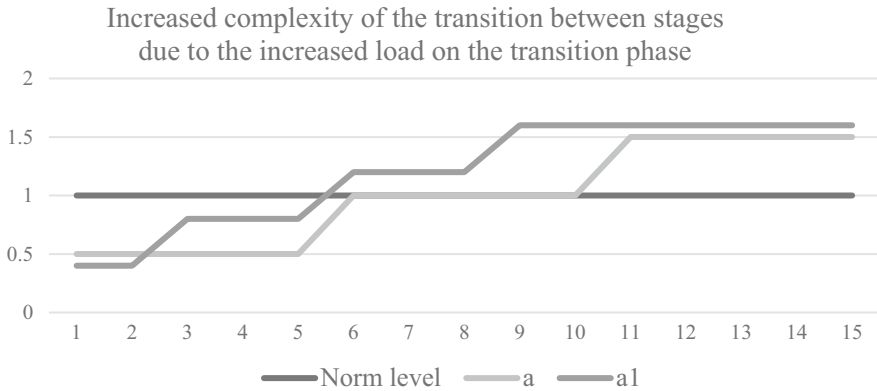
The level of uncertainty in the external environment γ is the only indicator that can have a negative value. Transformation experience shows that a favourable environment can significantly correct and improve internal errors and overall system unpreparedness. Likewise, unfavourable conditions can make it very difficult for even a well-prepared management system to implement changes. The assessment is based on the PESTLE analysis, and accordingly, based on these six factors, an integral score from -1 to $+1$ is assigned [27].

In the current context in Ukraine, for example, different manufacturing enterprises face different conditions: for example, military production aimed at supplying the armed forces has all possible favourable conditions for starting up and scaling up quickly, while traditional civilian output, on the contrary, has rather difficult and unattractive external conditions for scaling up.

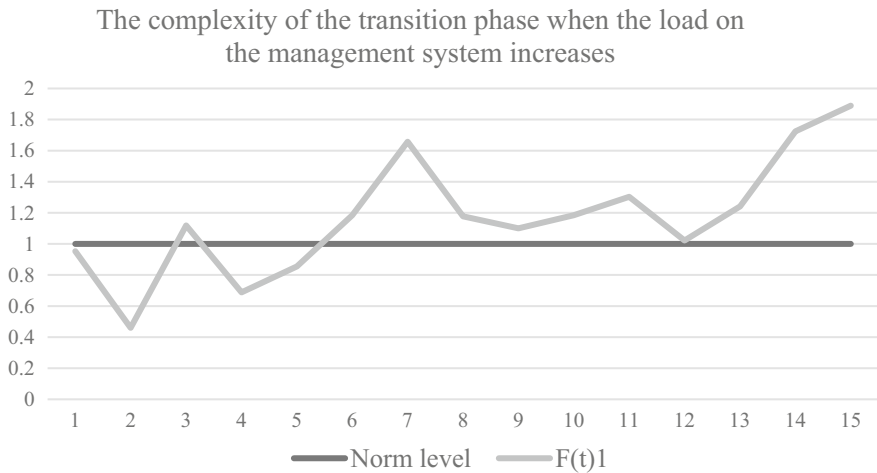
4.3 Effect of Increased Load on the Transition Phase

In Sect. 3.1, the increase in the load on the transition phase, i.e., the increase in requirements for how and when to implement it, was analysed. First, by imposing appropriate restrictions on the graph, the transition complexity graph shifts along the α parameter, as shown in Graph 5.

As a result, the fourth phase of the transition appears. All 4 phases are discretely distributed in the intervals 0.4, 0.8, 1.2, 1.6. This accelerates the growth of integral complexity and provokes the rise of the graph line of the function $F(t)$ 1 on average above the line of the normal level of 1, as shown in Graph 6.



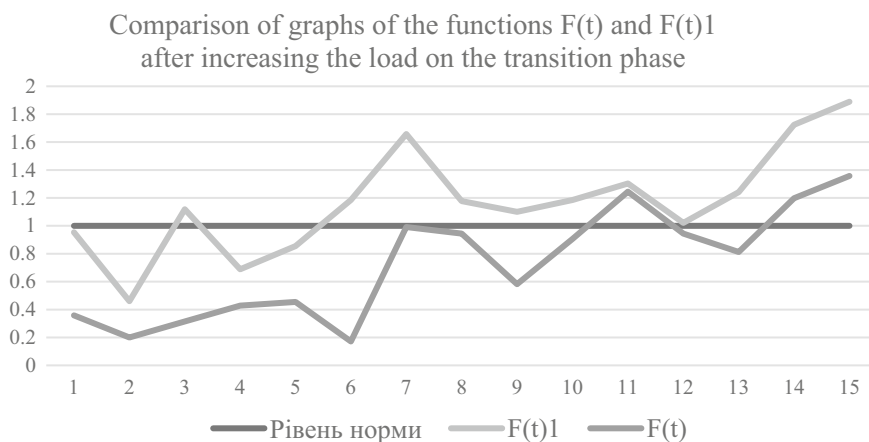
Graph 5 Increased complexity of the transition between life cycle stages due to the increased load on the transition phase (adopted from [26])



Graph 6 The integral complexity of the transition phase when the load on the management system increases (adopted from [26])

The difference between the graphs of the functions $F(t)1$ and $F(t)$ is demonstrated in Graph 7.

In summary, it has been clearly demonstrated that the biggest difference is in the early years, when there is a requirement to make at least 2 phase transitions in the first 3–4 years compared to 1 in the first 5–6 years before. This significantly overloads the management system and provokes premature exit of manufacturing startups from the market [28]. After two phase transitions, the difference decreases, as management systems are balanced over time in accordance with market requirements, but in such conditions, due to early overload, the next windows of opportunity for transition



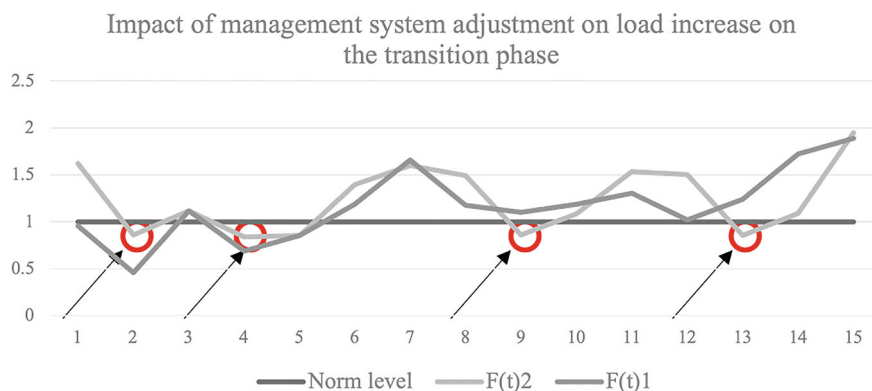
Graph 7 Comparison of graphs of the functions $F(t)$ and $F(t)1$ after increasing the load on the transition phase (based on Graphs 2 and 6)

appear above the normal level, which is shown in the graph as the lowest points in years 9 and 12, which are above line 1 [29].

4.4 Compensation by the Management System for Deviations in Transition Complexity

Sections 3.1 and 3.2 describe the mechanism of the transition management system's corrective impact on the overall system in response to the increased load on the transition phase. This influence can be seen graphically in Graph 8, where the main task of such a corrective influence is to ensure the availability of windows of opportunity and the ability to use them.

First and foremost, the burden on the management system causes the need to increase competencies, i.e. to increase the diversity of the management system [18]. This, in turn, requires an increase in investment from the first years of the enterprise's existence [26]. And this leads to the fact that the area of windows of opportunity in the first years of the enterprise's operation decreases, the total area of the objective function graph above the level of the norm increases, but this allows creating the necessary windows of opportunity in the following years, which prolongs the life and increases the efficiency of the enterprise in the long term [30, 31].



Graph 8 Impact of management system adjustment on load increase on the transition phase

5 Conclusion

5.1 Synopsis

The paper raises the issues faced by manufacturing enterprises in Ukraine in the 2020s in the context of the need to go through the first stages of the life cycle much faster. The essence of the problem can be summarised as follows: shortening the life cycle stages and early transition to the next stage impose additional burdens and increase the requirements for the transition phase between stages. The authors examine the characteristics and attributes of the transition phase to reveal the complexity of transition management in more detail. The peculiarities of the manufacturing business are shown in the higher capital intensity, especially during the transition phase, and in the specific management competencies that the management system must have for effective and sustainable development.

The paper proposes a mathematical model for minimising the integral complexity of transition management. This means that the proposed model's task is to find conditions that make the transition between stages in complicated conditions possible, most efficient, and in compliance with the possibility of long-term sustainable development of the enterprise.

The study reveals the key factors and the nature of their influence on the complexity of the transition phase. Using the proposed model, it compares how increased pressure on the transition phase and the reduced duration of the initial lifecycle stages affect the overall difficulty of executing the transition, and how the transition phase management system should respond to meet these new requirements.

5.2 Further Research

The authors' next steps will be aimed at a broader study of improving the efficiency of transition management in the context of the overall sustainability of the entire life cycle of manufacturing enterprises. As part of testing the current model, an analysis will be carried out on a sample of Ukrainian manufacturing enterprises in different conditions (geography, production direction, etc.) to test the model's effectiveness at different life cycle stages. It is also planned to analyse a relevant sample of Ukrainian enterprises in the pre-war period to compare the behaviour of the management system under different loads.

The model for minimising the complexity of the transition phase between life cycle stages should become a tool of the management system. The authors plan to extend their research with other models, such as a model for managing the transition phase, a model for assessing the level of risk of investing in the transition phase, a model for assessing the timeliness of launching the transition phase, etc.

In summary, this research aims to develop a model system as a set of tools for managing the transition phase between lifecycle stages in manufacturing enterprises that face increased pressure—specifically, the need to accelerate through several early lifecycle stages. Such research is intended to support the design of new policy initiatives for accelerating economic recovery in countries affected by war or other systemic disruptions. Moreover, the proposed toolkit can be applied in developing countries as a catalyst for industrialisation and as a strategic foundation for future economic breakthroughs.

References

1. Adizes I (1989) Corporate life cycles: how and why corporations grow and die and what to do about it. Prentice Hall, Englewood-Cliffs
2. Greiner LE (1972) Evolution and revolution as organizations grow. Harvard Bus Rev 37–46
3. Miller D, Friesen PH (1984) A longitudinal study of the corporate life cycle. Manag Sci 30(10):1161–1183
4. Haire M (1959) Modern organization theory. Wiley, 324 p
5. Lippitt GL, Schmidt WH (1967) Crises in a developing organization. Harvard Bus Rev. <https://hbr.org/1967/11/crises-in-a-developing-organization>
6. Churchill NC, Lewis VL (1983) The five stages of small-business growth. Harvard Bus Rev. Retrieved from: <https://hbr.org/1983/05/the-five-stages-of-small-business-growth>
7. Hanks SH (1990) The organization life cycle: integrating content and process. J Small Bus Strategy 1–12
8. Flamholtz EG, Randle Y (2007) Successful organizational development and growing pains. University of California Los Angeles. Retrieved from: https://web.archive.org/web/20220419142852/https://www.anderson.ucla.edu/documents/areas/fac/hrob/org_dev_growing_pains.pdf
9. Quinn RE, Cameron K (2021) Organizational life cycles and shifting criteria of effectiveness. Manag Sci 29(1):33–51
10. Porter ME (2019) Competitive strategy: techniques for analyzing industries and competitors. Free Press, New York, USA

11. Hayes RH, Wheelwright SC (2019) Link manufacturing process and product life cycles. *Harvard Bus Rev* 1:133–140
12. Beuren IM, Rengel S, Rodrigues MJ (2015) Relation of management accounting attributes and the organizational life cycle stages. *Revista Innovar* (57):63 p
13. The impact of the war on industrial sectors in Ukraine. Inclusive and Sustainable Industrial Development. United Nations Industrial Development Organization, Vienna, Working Paper Series. WP 3 (2024)
14. Lihonenko LO (2021) Crisis management of enterprise: theoretical and methodological principles and practical tools [Antykryzove upravlinnia pidpriemstvom: teoretyko-metodolohichni zasady ta praktychnyi instrumentarii]. KNTEU, Kyiv, Ukraine (in Ukrainian)
15. Milakovsky B, Vlsiuk V (2024) Industrial policy for Ukraine's survival. Reversing 30 years of deindustrialization. Friedrich Ebert Stiftung, 21 p
16. New Monthly Enterprises Survey (2024) Issue 32. Ukrainian Business in Wartime. Retrieved from: <https://www.slideshare.net/slideshow/new-monthly-enterprises-survey-issue-32-12-2024-ukrainian-business-in-wartime/276690486>
17. Mosca L, Gianecchini M, Campagnolo D (2021) Organizational life cycle models: a design perspective. *J Organ Des* 10(1):3–18
18. Ashby WR (1958) Requisite variety and its implications for the control of complex systems. *Cybernetica* (Namur) 1(2)
19. Arefieva OV, Prokhorova VV (2020) Management of economic systems development: theory, mechanisms of regulation and management [Upravlinnia rozvytkom ekonomichnykh system: teoriia, mekhanizmy rehuliuвання ta upravlinnia], UkrDAZT, Kharkiv, Ukraine (in Ukrainian)
20. McLean T (2024) Theory of constraints vs LEAN—which makes sense for your business? Retrieved from: <https://txm.com/theory-of-constraints-vs-lean-which-makes-sense-for-your-business/>
21. Brocke J, Rosemann M (2015) Handbook on business process management. Introduction, methods and information systems. Springer, Heidelberg
22. Le H (2022) ERP implementation at a production company. JAMK University of Applied Sciences, Jyväskylä, 102 p
23. Brinks HH, Johnson PC (2019) The agile way of working within the manufacturing industry. Jönköping International Business School, Jönköping University, 113 p
24. Taponen I (2022) Lean production scale-up. How a startup can scale the manufacturing of physical product innovation. Haaga-Helia, University of Applied Sciences Ltd, 72p
25. McKinsey & Company (2019) Transformation practice. The transformation office: key success factors
26. Transition management (2014) The National Association of Pension Funds Limited, London, 24 p
27. Samoiluk M (2025) Ukraine war economy tracker. Centre for Economic Strategy. Retrieved from: <https://ces.org.ua/en/tracker-economy-during-the-war/>
28. Phan TN, Baird K, Blair B (2016) The use and success of activity-based management practices at different organisational life cycle stages. *Int J Prod Res* 52(3):787–803. Retrieved from: from EBSCO Online Database Business Source Ultimate. Retrieved from: <http://search.ebscohost.com/login.aspx?direct=true&db=bsu&AN=94003962&site=ehost-live>
29. Fang S, Chang E, Ou C, Chou C (2014) Internal market orientation, market capabilities and learning orientation. *Eur J Mark* 48(1/2):170–192. Retrieved 23 Oct 2016, from EBSCO Online Database Business Source Ultimate. Retrieved from: <http://search.ebscohost.com/login.aspx?direct=true&db=bsu&AN=95390955&site=ehost-live>
30. Denton F, Halsnæs K, Akimoto K, Burch S, Diaz Morejon C, Farias F, Jupesta J, Shareef A, Schweizer-Ries P, Teng F, Zusman E (2022) Accelerating the transition in the context of sustainable development. In: IPCC. Cambridge University. Press, Cambridge, UK and New York, NY, USA
31. Kryvinska N, Kaczor S, Strauss C (2020) Enterprises' servitization in the first decade—retrospective analysis of back-end and front-end challenges. *Appl Sci* 10(8):2957. <https://doi.org/10.3390/app10082957>

Method for Competitive Funding Optimization of Scientific Research and Development in Universities and Scientific Institutes (with Discussion for Ukraine)



Anatolii Shyian  and Liliia Nikiforova 

Abstract We propose methods for competitively optimizing state funding of higher education institutions and scientific institutions based on the use of information technologies. These organizations are financed, not the results of their activities. Optimization for scientific institutions is carried out based on indicators of the effectiveness of their scientific activity. The optimization of higher education institutions additionally includes indicators of the effectiveness of graduate training. Methods have also been developed for cleaning indicators of scientific activity from deception. Application of the obtained results will allow the Government of Ukraine to make decisions based on a system of objective indicators for the contribution to the international science and the country's economy made by higher education and scientific institutions. The specific choice of motivational conditions is a decision of the Government of Ukraine, which will have the opportunity to rely on objective indicators and characteristics.

Keywords Funding · Competition · Optimization · Development · University · Scientific institution

1 Introduction

World experience has proven that research and development (R&D) funding is most effectively implemented in conditions of competition for allocated funds. In this case, it is not individual scientists and their teams that compete, but the structures of the organization of support for scientific research and development that are created in these universities and scientific institutes.

A. Shyian · L. Nikiforova (✉)
Vinnitsia National Technical University, Vinnitsia, Ukraine
e-mail: nikiforova@vntu.edu.ua

In developed countries, the results of competitive funding are necessarily analyzed. First of all, the effectiveness of funding of organizations where scientific R&D is created is analyzed, that is, the “economic return” that the obtained scientific results bring both to the financing organization (fund) and to society and/or Humanity as a whole.

Let us give several examples of the results of such an analysis.

In [1], an analysis of the activities of the University of Birmingham, Great Britain, on the economic results of its activities for 2021–2022 is given. We will focus only on indicators that characterize the effectiveness of this university in the field of scientific R&D. The authors write that “our analysis implies a spillover multiplier such that every £1 invested in the University’s research activities generates additional annual economic output of £5.30 across the UK economy in terms of positive productivity spillovers to the UK private sector, totaling £858 million.”

In [2], the authors note that Toledo, Ohio, USA, creates \$10 of economic impact on the local economy for every dollar spent by the state of Ohio on the university.

In [3], data are provided that «Over \$4.79 billion in total business revenue was generated within Tennessee by the added value of MTSU degrees earned».

In Argentina, the impact of competitive funding of scientific research and the level of communication between scientists was analyzed [4]. It was found that grant funding of scientists increases both the number of their contacts with other scientists and the number of co-authors in their publications. The role of the influence of grant funding on the functioning of scientific activity is emphasized in the analytical material of the World Bank [5] using the example of the University of Wisconsin-Madison, USA: “Overall, across the disciplines, Madison figures that it spends an average of \$1.2 million in start-up costs for each new professor. It typically takes eight years for a professor to bring in enough research money to cover that cost. A professor who stays at Madison for 25 years after earning tenure brings in an average of about \$13 million in research money.”

Unfortunately, the reports of many universities and scientific institutes do not provide objective indicators that could indicate the effectiveness of their funding. For example, consider the report on the activities of the Institute of Science and Technology of Austria (ISTA) in 2023 [6]. It provides detailed information on both the development of ISTA’s material base and scientific infrastructure (including scientific equipment that is unique in Europe and the world). The Report provides a detailed description of the best scientific results obtained in the current year, introduces new professors and indicates how they will increase ISTA’s capabilities. Information is provided on all 78 scientific groups available as of 2023, with a description of the scope of their scientific activities and brief information about their leaders. Infographics are widely presented, for example, to illustrate ISTA’s international relations. The average age of ISTA researchers is given, which is 34.3 years. Information on the funding of ISTA’s scientific activities (grants) is provided: the total amount is rounded to 39,822,000 euros (the sources of funding and the amounts for them are also provided). The sources and amounts of funding for each scientific group are provided. The list of researchers who have received international and domestic

scientific awards and distinctions is provided. The list of scientific events (conferences, public lectures, scientific days, etc.) at ISTA that took place during the year is presented. Information on the ISTA board and donors who finance the Institute's activities is provided.

In Ukraine, analytical activities on the effectiveness of competitive state funding of scientific research and development (R&D) are still insufficient. There are only databases on the overall productivity of scientific activity in Ukraine, for example [7–10] for 2020–2024. They use quantitative indicators that relate mainly to activities within Ukraine, which makes it practically impossible to compare with indicators in other countries of the world. Therefore, both methods for analyzing the effectiveness of scientific activity in the R&D sector for individual funded projects and methods for aggregating the results of completed projects are still being developed.

The Ministry of Education and Science of Ukraine (MESU), along with the National Research Foundation of Ukraine, is one of the powerful sources of state competitive funding for scientific R&D. These channels cover the majority of Ukrainian scientists, so their impact on the development of science is decisive.

The MESU competitions for state funding for scientific research and development are carried out in several formats. The first format is focused on scientists and scientific teams who have the opportunity to submit their scientific projects to the competition [11]. There are competitions that are specifically focused on young scientists [12]. In [11, 12] all the information necessary for the participant-scientist is provided: deadlines, application requirements, etc.

In 2024, the MESU proposed a new format for competitive state funding for scientific projects [12]. Their executors are higher education institutions (HEIs) and scientific institutions (NUs) that fall under the scope of the MESU. For this format, new forms (document templates) for submitting applications were introduced, which are different from those used in [11, 12].

In Ukraine these organizations are financed, not the results of their activities [13]. However, today the financing of expenses in Ukraine requires serious optimization due to the war. At the same time, funding higher education and science was and remains one of the priorities of the state. Therefore, in the conditions of both ongoing war and post-war reconstruction, state funding for higher education and science should be optimized. The state finances the system of higher education institutions and scientific institutions (hereinafter referred to as HEI/SI respectively) so that they simultaneously fulfill the following two tasks for building a developed society in Ukraine:

- (1) carry out scientific activities in the field of world-class research and development (for HEI/SI);
- (2) trained a new generation for effective participation in socio-economic processes at the level of development and decision-making based on the scientific results obtained (only for HEI).

These two tasks are closely related. It should be emphasized that it is the scientific activity that is decisive for the higher education institutions to be able to fulfill these two tasks.

The problem of optimization of state financing of HEI/SI has a long history. Thus, as early as the mid-1990s in [14], attention was drawn to the fact that “Universities and industry, up to now relatively separate and distinct institutional spheres, are assuming tasks that were formerly largely the province of the other. The role of state concerning these two spheres is changing in apparently contradictory directions. States are offering incentives, on the one hand, and pressing academic institutions, on the other, to go beyond performing the traditional functions of cultural memory, education, and research, and make a more direct contribution to wealth creation”. In [15] it is emphasized that “developed countries began to appear through the results of their efforts and scientific and research achievements”. Therefore, on the way to a developed socio-economic state, Ukraine cannot ignore the need to optimize state funding of HEI/SI. This is especially important because practically all scientific activity and most of the activities of the higher school function precisely at the expense of state funding. And increasing the efficiency of the use of public funds is a problem that must be solved as soon as possible.

Optimal funding should be sufficiently adaptive to respond in time to changes in the field of science development, research and development. For example, in [16], the activities of 12,720 scientists over 15 years were analyzed and it was found that the funding of the “elite” among scientists is ineffective, because “most funded researchers do not stand out in terms of output and scientific impact”. In other words, over time, the “scientific elite” time loses its position both in terms of obtaining scientific results and creating an impact on the development of the economy and science. These results support those of [17], which analyzed the concentration of funding, articles, and citations at the level of individual researchers for 13,439 university professors. The article found that “only 3.2% of the researchers are in the top 10% for all indicators, while about 20% are in the top 10% for at least one of the indicators”. That is, it is also confirmed that there is a fairly high level of rotation among the “scientific elite”.

Research results indicate that the development of science in higher education institutions leads to the rapid economic growth of both the respective regions and the economy of the state as a whole. For example, in the work [18] the contribution of research structures of the USA to the development of the quantitative and qualitative level of entrepreneurship is considered. Note that the authors here mean products produced by the enterprise based on world-class scientific developments. The authors write: “Our key finding is that changes in Federal research commitments to universities are uniquely linked to positively correlated changes in the quality-adjusted quantity of entrepreneurship”. The article also points to universities as drivers of entrepreneurship in regions: “Research funding to universities seems to play a unique role in promoting the acceleration of local entrepreneurial ecosystems”. The authors distinguish two channels for the influence of universities: “On the one hand, universities can directly influence their entrepreneurial ecosystem through the establishment of spin-offs and technology transfer activities to industrial partners ... Second, the impact on local entrepreneurial ecosystems may arise from the impact of students and other non-permanent research staff involved in the research itself”.

Statistical substantiation of such conclusions is also given in a recent article [19]. In particular, here "... identifies the extent to which knowledge from U.S. universities drives industry agglomeration. Establishment-level data indicate faster growth in employment, wages, and corporate innovation after the 1980 Bayh-Dole Act's shock to the spread of innovation from universities in industries more closely related to the nearby university's innovative strengths. Federal research funding amplified the effect. University knowledge spillovers strengthen with geographic proximity, density, and local skills". It is necessary to pay attention to the fact that this article analyzes the consequences of changing the conditions of the functioning of science in the USA, that is, at the level of the country. It is emphasized that these changes led to the rapid quantitative and qualitative growth of regional entrepreneurship.

Also, in the early 2000s, [20] analyzed the relationship between invention (patenting) and the motivation of scientists to conduct research and development and found that "...economic incentives affect the number and commercial value of inventions created at universities. Using panel data for 102 US universities over the period 1991–1999, we find that universities that pay a higher share of royalties to researchers generate more inventions and higher license revenues, controlling for other factors including university size, quality, research funding, and technology licensing. The stimulating effect is much stronger in private universities than in public ones". The authors [20] also showed that: "...academic research and inventive activity in universities respond to variations in inventors' royalty shares. ...we find that universities with higher royalty shares generate higher levels of license income. This finding is important because it implies that the design of intellectual property rights, and other forms of incentives, in academic institutions can have real effects. ...these incentive effects work primarily by increasing the quality, rather than the quantity, of inventions". In other words, better incentives for scientists will lead to more world-class inventions.

The main result of scientific activity of scientific institutions/institutions of higher education is their contribution to world science. This contribution consists of the fact that the world community of scientists applies the obtained scientific results. Such an application is formalized by citing sources where scientific results obtained by scientists are published. Today, such sources are articles in scientific journals. And an effective measure of the scientific impact on world science is the number of citations of articles describing relevant scientific results. But this indicator does not used in the evaluation methodology of scientific institutions (see, e.g., [1–6]).

There is no need to be interested in the number of employees, their qualifications, the number of scientific equipment, the number of publications, etc. When all this is effectively used, it is reflected in the quality of the obtained scientific results. Finally, the number of citations of articles depends on the quality of the obtained scientific results. The high number of article citations is proportional to the level of their contribution to world science.

Thus, the situation in Ukraine with the method of state attestation of scientific results of higher education institutions/universities to receive state funding is unsatisfactory. Therefore, this method should be optimized.

Thus, there is a need to study quantitative indicators that can be used as a basis for analyzing the effectiveness of activities in the field of scientific research and development for universities and scientific institutes.

The article aims to develop a method for optimizing competitive funding of scientific research and development at universities and scientific institutes (with a discussion of the specifics of its application for Ukraine).

2 Evaluation Methodology

To optimize of funding of HEI/SI, it is necessary to have a powerful expert system that allows for objective evaluation of these subjects of economic activity.

As a rule, today the population receives the main body of knowledge related to the implementation of activities while studying at the university [21]. Therefore, the presented results will be applied primarily to tertiary education.

We have proposed to choose the entire world scientific community as an expert system for evaluating the scientific activity of HEI/SI. The number of citations of articles published in international journals by employees of HEI/SI can be served as the objective indicator. All articles in such journals are reviewed, where the reviewers are highly qualified scientists. At the same time, reviewers are motivated to objectively assess the scientific level of articles.

Taking into account the presence in Ukraine of a high level of imitation of scientific activity and a direct falsification of their evaluation by domestic scientists, it is necessary to carry out “cleaning” of such an indicator as citations. This can be done by existing methods today with the help of information technologies. The process of cleaning data on citations will be described in detail later in the article (for example, Scopus provides the ability to clean the Hirsch index from self-citations).

To optimize funding for HEI, an additional indicator should be used that objectively characterizes the contribution of this higher education institution to the state's economy. As an expert system, the entire set of economically efficient enterprises is involved in this. This can be done by measuring the number of graduates that such enterprises attract to work in the specialty for which they were trained in higher education institutions. This is an objective analysis because it is unprofitable for companies to hire incompetent employees.

Another useful indicator can be the amount of money the university received from the sale of patents or licenses.

Let us pay attention to an important aspect of financing scientific structures that carry out R&D. From the point of view of financing, these structures should be beneficial for the economy. This means that the “best” will be those scientific structures that provide a greater “output of scientific output (citations, employed students in their specialty, profit from the sale of patents, licenses, etc.)” per “unit of financing”.

As a result, the following indicators (“the more the better”) will be important for analyzing the economic efficiency of producing scientific R&D by each university or scientific institute.

$$\begin{aligned} & \text{Cost of one cite} \\ &= \frac{\text{number of citations for a certain term}}{\text{quantity of received funds for this term}} \quad (\text{A}) \end{aligned}$$

Note. The “number of citations” can take into account the weight of the scientific publication (for example, a scientific journal) in which the article is published.

$$\begin{aligned} & \text{Effectiveness of R\&D} \\ & \quad (\text{amount offunds received for sales} \\ & \quad \quad \text{patents for a certain perid}) \\ &= \frac{\quad}{\text{amount of funds spend on R\&D for this period}} \quad (\text{B}) \end{aligned}$$

$$\begin{aligned} & \text{Educational costs for one specialist} \\ & \quad (\text{amount of funds allocated for all student education} \\ & \quad \quad \text{of the specialty for a certain period}) \\ &= \frac{\quad}{(\text{a quantity of working students in the specialty} \\ & \quad \quad \text{for this period})} \quad (\text{C}) \end{aligned}$$

It is these indicators that describe the effectiveness of managing funds provided to a scientific structure (university or scientific institute) so that the management system organizes the most effective use of these funds.

Note. The use of indicators (A), (B), (C) can ensure competition between universities and scientific institutes in the scientific field. Because only in this case will these scientific structures be forced to look for the most effective scientists in the field of scientific R&D.

3 Indicators for Evaluating the Activity of HEI/SI

3.1 *The First Criterion for Both HEI and SI: Is the Evaluation of the Impact on World Science*

The first criterion is the evaluation of the particular HEI/SI impact on world science.

The impact of HEI/SI on world science can be evaluated by the number of citations to scientific articles that teachers of this HEI/SI received in international scientific journals. For example, it can be Scopus or Web of Science (WoS) databases. Scopus data covers a larger number of scientific publications and therefore can be today more informative for the conditions of Ukraine.

Note. Why were selected the international scientific publications and the databases Scopus and WoS? Because in this case, the scientific level of Ukrainian scientists will be examined by the entire scientific community of the world. At the same time, it will

be an honest examination, because it will be carried out by scientists from developed countries, where the rules of academic integrity are strictly adhered to. In addition, it will also be an evaluation of the impact of Ukrainian scientists on world science. Professional journals of Ukraine cannot be used for such evaluation, because their citation in international databases is an extremely rare exception. Therefore, the evaluation of the impact of Ukrainian HEI/SI on world science cannot be representative when using Ukrainian specialized scientific journals.

To form an indicator that allows them to compare themselves HEI/SI and, at the same time, to evaluate their scientific impact at the state level as a whole, the following steps must be taken. Such a calculation must be carried out for the sets of higher education institutions and scientific institutions separately.

1. Ordering HEI/SI by the number of citations from the highest to the lowest value will be made. HEI/SI will then be numbered with natural numbers $j \in [1, N]$, where N is the total number of HEI/SI ($j = 1$ – HEI/SI with the largest number of citations, HEI/SI with $j = N$ has the least number of citations). The number of citations for the i th HEI/SI is denoted as C_j .

Note. The indicator C_j can to calculate using formula (A).

2. The total number of citations of all HEI/SI in Ukraine can find using the formula

$$C = \sum_{j=1}^{i=N} C_j \quad (1)$$

3. We determine the fraction of citations that falls on the j th HEI/SI according to the following formula:

$$p_j = \frac{C_j}{C} = \frac{C_j}{\sum_{j=1}^{i=N} C_j} \quad (2)$$

4. We build a cumulative fraction of citations Y_j for each j th HEI/SI, that is, the sum of all values p_k up to and including p_j . That is, for the j th HEI/SI, the corresponding value of the Y_j indicator should be calculated according to the following formula:

$$Y_j = \sum_{k=1}^{k=j} p_k \quad (3)$$

Thus, the value Y_j is the total fraction of citations of the j th HEI/SI and citations of all other HEI/SI that have more citations.

5. For the j th HEI/SI, we find the indicator F_j , which is its request for state funding. We also calculate the value of the accumulated funding of all HEI/SI that has a higher number of citations than it. The calculation is carried out according to the following formula:

Table 1 Database for calculating the optimal funding of the required number of HEI/SI

No.	Name of HEI/SI	Y_j	W_j
1	Name of HEI/SI with number 1	$Y_1 = p_1$	$W_1 = F_1$
2	Name of HEI/SI with number 2	$Y_2 = p_1 + p_2$	$W_2 = F_1 + F_2$
3	Name of HEI/SI with number 3	$Y_3 = p_1 + p_2 + p_3$	$W_3 = F_1 + F_2 + F_3$
...	...	...	...
j	Name of HEI/SI with number j	Y_j	W_j
...	...	...	...
N	Name of HEI/SI with number N	Y_N	W_N

$$W_j = \sum_{m=1}^{m=j} F_m \quad (4)$$

6. We form the resulting Table 1.

Thus, in Table 1, the universities are ordered by the number of citations (from the largest number to the smallest), and the amount of state funding is also indicated for each of them.

3.2 Formulation of the Problem of One-Factor Funding Optimization of the HEI/SI System

This one-factor funding optimization is carried out taking into account only the scientific impact.

The task of one-factor optimization of the state funding of the HEI/SI system can be set in two variants.

Version 1. Optimization based on scientific input.

In this case, only and only those HEI/SI are financed, the total scientific contribution of which, expressed through the number of citations in given international databases, is given by the critical value $0 < q_{opt} < 1$.

In other words, only those HEI/SI that jointly ensures the fulfillment of the ratio $Y_j = q_{opt}$ are financed. From here can find the j_{opt} .

The optimization problem is mathematically written by the following expression:

$$j_{opt} = \arg \min \{ (q_{opt} - Y_j) \geq 0 \} \quad (5)$$

The list of HEI/SI is taken from Table 1, and consists of all HEI/SI with numbers from 1 to j_{opt} .

The funding is then spent at $W_{j_{opt}}$.

Version 2. Limited funding.

In this case, which is more typical for the conditions of Ukraine, the maximum financing is given by the value of F_b . In this case, only those HEI/SI whose total funding does not exceed F_b will be financed.

The optimization problem for this case is written as follows:

$$j_{opt} = \arg \min \{ (F_b - W_j) \geq 0 \} \quad (6)$$

The list of HEI/SI is taken from Table 1, and consists of all HEI/SI with numbers from 1 to j_{opt} .

The number of citations that will be financed will be Y_{iopt} 's share of the total number of citations that were in HEI/SI before optimization. Note that it may well turn out that as a result of financial optimization, the result obtained for Y_{iopt} will completely satisfy the state.

3.3 Building a Database of Citations

The result of optimization problems depends on two databases. The first database formed from requests from HEI/SI for state funding. The second database is a database of citations of scientists currently working in HEI/SI. This database today allows a high level of inaccuracies and falsifications.

Therefore, below we will focus in detail on the analysis of the formation of such a database of citations of scientific publications of HEI/SI in international databases (primarily WoS and Scopus), which will most fully characterize the scientific level of their employees.

First of all, the database of citations should be formed exclusively only for those employees who currently work in this HEI/SI. Therefore, its source should not be data on the employee's affiliation, but data from his account in the international database. This will make it possible to exclude from the citation database those employees who no longer work in this HEI/SI. Today, the "Institution" filter in international databases takes into account employees who have transferred to work at other institutions (or, on the contrary, have just transferred to this institution), deceased, etc.

Next, the selection should be carried out according to the authors of the publications. For example, a publication (and, accordingly, the number of its citations) by several authors from the same organization should be counted only once. In addition, selection should be carried out according to the number of self-citations of authors and/or the author's team. Moreover, it is possible to falsify the number of citations due to the cross-citation of publications by authors who belong to the same HEI/SI. Falsification is also possible due to cross-quoting of author groups belonging to different HEI/SI. Such falsifications can be detected and eliminated with using of specialized software. Also, a large number of citations of a certain scientist's articles in scientific journals of a relatively low level (as a rule, the 4th or 3rd quartile) can serve as a filter for detecting "rapid growth" of the number of citations.

The time frame for which the citation count database is being compiled is important. As a rule, funding of scientific activities in the world is formed within the framework of competitive grant projects. The duration of such projects does not exceed, as a rule, 3–5 years. Therefore, the citation database should take into account the citations of publications in the last 5–7 years. The issue of determining the time interval requires a separate scientific study. But the term of 5–7 years seems quite sufficient to take into account only actively working scientists, which is important precisely for the optimization of state funding.

Another issue is that the co-authors of the publication do not make the same scientific contribution. Today, many scientific journals are more and more actively demanding that the personal contribution of each author be indicated in the article. Such data appear more and more frequently in each article. And, of course, this will affect the “level of citation of the article due to the contribution of this co-author”. The contribution of all co-authors related to this HEI/SI should be taken into account. However, this indicator still needs a more serious scientific analysis, and therefore it can be ignored for now.

As a result, it is possible to propose such a method for building a database of citations for a given HEI/SI.

1. For each HEI/SI, a database is created from the accounts of their employees in international databases (e.g., WoS and/or Scopus). For example, it can be done within the Ukrainian Joint State Electronic Database on Education for higher education institutions.
2. For each account, a list of scientific publications for the last 5 (or 7) years is generated from the database of HEI/SI. (The number of years to be taken into account may be changed as a result of the analysis of statistical data that will be obtained as a result of using this method).
3. For each publication, a list of its citations is created in the International Database. This list is cleared from self-citations and cross-citations both within this HEI/SI and for cross-citations in scientific groups of other HEI/SI.
4. The importance of the contribution of the scientists from the HEI/SI is taken into account (if possible) in each of the articles. For example, this can be taken into account by a certain coefficient to reduce the number of “credited” citations.
5. The total number of citations is formed for each HEI/SI.
6. The resulting database is created on the number of citations for each of the HEI/SI.
7. The resulting database is ordered by the number of citations from highest to lowest.

3.4 *Second Criteria. Improvement of the Financing Method by Taking into Account the Specific Conditions for Higher Education Institutions (HEI)*

The task of HEI is not only scientific activity. HEI trains students for their integration into the economic activity of society after graduation. The success of this task of HEI should be evaluated by the following indicator v_i : “The number of graduates who got a job in their specialty within 6 months after graduating from HEI”. The term can be specified after a separate study.

Note. In this case, we can use the indicator (C).

It is possible to obtain a quantitative expression for the rating of HEI, similar to formulas (1)–(3). Relevant data for Ukrainian enterprises of different forms of ownership can be obtained from the data of the pension fund and, possibly, with the involvement of data from fiscal organizations and/or employment centers.

1. Order HEIs by the number of employed graduates v_i from the largest to the value smallest. HEIs will then be numbered with natural numbers $i \in [1, N]$, where N is the total number of HEIs (HEI with $i = 1$ has the largest number of employed graduates, HEI with $i = N$ has the smallest number of employed graduates).
2. We find the total number of employed graduates for all HEIs in Ukraine using the formula

$$V = \sum_{i=1}^{i=N} v_i \quad (7)$$

3. Determine the share of employed graduates that fall on the i th HEI according to the following formula:

$$u_i = \frac{v_i}{V} = \frac{v_i}{\sum_{i=1}^{i=N} v_i} \quad (8)$$

4. Build the cumulative fraction of employed graduates R_i for each i th HEI. That is, for the i th HEI, the corresponding value of the indicator R_i should be calculated according to the following formula:

$$R_i = \sum_{k=1}^{k=i} u_k \quad (9)$$

Thus, the value of R_i is the total fraction of employed graduates of the i th HEI and employed graduates for all other HEIs that have a larger number of employed graduates.

For state funding, a formula similar to (4) will be used, only for ranking HEIs by the number of employed graduates.

According to formula (9) and similarly to formula (4), it is necessary to create a rating table, similar to Table 1.

Note. One can also enter an indicator of the percentage of employed graduates, which is found as the number of employed graduates divided by the total number of all graduates from all state-funded HEIs. This will make it possible to create another indicator, which can be called “the efficiency of spending public funds on the training of a new employee with a higher education”.

3.5 *Additional Variants for Optimization of HEI*

Optimization for state funding of HEIs can also be carried out according to two options, whose are similar formulas (5) for optimization according to the required number of employed graduates or according to formula (6).

Optimization according to the required number of employed graduates is carried out similarly to formula (5):

$$i_{opt} = \arg \min \{ (R_{opt} - R_i) \geq 0 \} \quad (10)$$

Optimization based on available finances is carried out according to formula (6), but according to the rating of higher education institutions according to employed graduates):

$$i_{opt} = \arg \min \{ (F_b - W_i) \geq 0 \} \quad (11)$$

3.6 *The Motivational Effect of the Proposed Optimization Methods*

The motivational effect of optimization conditions (5) or (6) is that HEI/SI will be forced to create the conditions for obtaining scientific results at an international level. Moreover, since the highest citations in international databases are received by scientific articles published in the best scientific journals in the world, HEI/SI will be forced to:

- (1) involve in scientific work only those scientists who will be able to create high-level scientific results;
- (2) to ensure that high-level managers of HEI do not use the “force of power” to include them in the team of authors of articles (because this will demotivate scientists to do scientific work);

- (3) spend money specifically on creating the conditions for the scientific work of highly effective scientists (including, for example, sending them, and not representatives of the administration, on business trips to foreign conferences);
- (4) to significantly reduce the apparatus of units, which do administrative support of research, while increasing the efficiency of their work;
- (5) spend grant funds on “indirect costs”, which contain in the grants, specifically for the purchase of scientific equipment and creation of conditions for more effective work of scientists;
- (6) both the HEI administration and scientists will be interested in the transition to grant funding of scientific activities;
- (7) both the administration of HEI and scientists will be motivated to engage senior students as support staff for conducting scientific research.

It changes the existing institutional conditions in HEI/SI. For the conditions of the proposed state funding, it is easy to build a game-theoretic model [22] that will describe the “scientist—administration” and “administration—Ministry of Education and Science” interaction. For such games, under the conditions of the new approach to financing HEI/SI, Nash equilibrium strategies will be the “carry out effective allocation of funds” both for the role of “scientist” in the game “scientist—administration” and for the role of “administration HEI” in the game “administration—Ministry of Education and Science”.

Note that for the conditions existing today in Ukraine, in [23] it is shown that for the role of “professor/scientist of HEI” the equilibrium strategy of both Nash and Stackelberg is the strategy of “imitating activity”. The role of “administration of HEI” in the game “Administration HEI—Ministry of Education and Science” is also the strategy of Nash and Stackelberg and the strategy of “imitating activity”.

4 Discussion and Outlook

For SI, it is enough to choose one of the optimization conditions (5) or (6), depending on the choice of public administration at this stage of the development of the economy and society.

Optimization conditions (5), (6), (10), and (11) should be the basis of the corresponding two-factor optimization for the HEI. These optimization conditions can be conflicting. Therefore, which of them to apply should be a decision based on the current conditions for the development of the state, economy, and society. It is also possible to apply several or even all criteria using certain weighting factors. The objective function for the optimization is always determined by the subjects of a higher level of the hierarchy.

To eliminate conflicts, it is possible to propose a division of higher education institutions into two separate classes: research HEIs and educational HEIs. The first is financed according to the criterion of the maximum scientific result (5), while the

second is financed according to the criterion of the maximum output of specialists needed by the state (10).

It is also possible to propose dividing the funding of HEIs into two separate parts: funding with criteria (5) for the training of highly qualified personnel (masters and PhDs, post-doctoral positions, invitations of world-leading scientists, etc.) and funding with criteria (10) for the training of specialists needed by the state/region (bachelor's and master's degrees, possibly PhDs in applied/regional directions). Note that in the last case for the process of regional-level personnel training, it will be possible to attract regional funding as well, using the optimization criterion (6).

It is also possible to propose separate funding for each HEI of science and training of specialists. For example, separate funds from the state budget can be allocated specifically for scientific activities in HEIs. In addition, a part of the funds can be allocated by the state for grant funding of scientific activities in Ukraine, within which each of the HEIs and SIs will fight to win their academic applications for grants. A fixed amount will be allocated to educational activities for all HEIs according to one of the optimization options (10) or (11).

The use of the results of this work makes it possible to obtain an information system for the formation of relevant databases. Such a set of databases is needed for further analysis of the existing constituent institutions of science and education and the development of a program for the development of scientific activity in Ukraine and such bodies of higher education, which are characteristic of developed countries. It will be possible to adjust the quantitative indicators as a result of the analysis of the application of the proposed methods of state financing and corresponding databases.

The use of the proposed method must allow to development of a system of criteria that can be used to manage the fields of science and education. A feature of the construction of relevant information systems is that the same ranking algorithms will be used both for the management of science and for the management of higher education.

The method is based on objective characteristics that allow for obtaining objective quantitative indicators. This makes it possible to reduce the probability of making wrong decisions.

The detailed development of the financing strategy for HEI/SI for today's Ukraine requires modern decisions and is beyond the scope of this article.

It should be emphasized that the optimization of state funding proposed in the article will be effective at the beginning of Ukraine's transition to the status of a developed country. After some time, when the developed institutions will be implemented in Ukraine, a large number of other powerful players, which will belong to other forms of ownership, will appear in the market of both higher education and the creation of scientific developments and research. Then, in the new conditions, the proposed optimization methods will need some adjustment.

5 Conclusion

Optimization of state financing of HEI/SI should be carried out based on the results of their activities. This will stimulate them to increase the efficiency of their activities due to the involvement of almost their entire team in decision-making.

The number of citations of scientists actively working at this institution was chosen as an objective indicator for the results of scientific activity. This indicator indicates the level of integration of the scientific results of the scientists of this institution into world science. For this, it is proposed to use the Scopus and WoS databases, which have a powerful analytical apparatus.

The number of graduates who are involved in participating in the economic activity of the state in the field of specialization within 6 months after graduation was selected as an indicator for evaluating the effectiveness of the functioning of higher education institutions. This indicator takes into account that the task of higher education is to prepare a new generation for active participation in the economic life of the country.

The proposed system of conditions for evaluating the optimality of state financing. The choice of specific optimization conditions and their ratio allows effective management of state funding of HEI/SI, creating powerful motivational conditions for the entire staff of their employees.

Application of the obtained results will allow the Government of Ukraine to make decisions based on a system of objective indicators for the contribution to the country's economy made by HEI/SI.

References

1. Cannings J, Knuth K, Booth J, Hagelberg Ph et al (2024) The economic, social, and cultural impact of the University of Birmingham. London Economics
2. Ajilore O, Smornov O (2015) Economic impact of the University of Toledo. University of Toledo
3. Arik M, Devi R, Lawrence A (2024) Alumni impact 2024. Middle Tennessee State University, Business and Economic Research Center at MTSU
4. Ubfal D, Maffioli A (2011) The impact of funding on research collaboration: evidence from a developing country. *Res Policy* 40(9):1269–1279
5. Jamil S (2009) The challenge of establishing world-class universities. The World Bank
6. Annual Report 2023: Illuminating the Unknown. Institute of Science and Technology Austria (ISTA) (2023). https://ista.ac.at/wp-content/uploads/2024/05/240430_ISTA-Annual-Report-2023-EN_Web.pdf. Accessed 17 Feb 2025
7. Pysarenko TV, Kuranda TK et al (2020) Scientific and scientific and technical activity in Ukraine in 2019: scientific-analytical report. Kyiv, UkrINTEI (in Ukrainian)
8. Pisarenko TV, Kuranda TK, Kvasha TK et al (2021) The state of scientific and innovative activity in Ukraine in 2020: scientific and analytical note. Kyiv, UkrINTEI (in Ukrainian)
9. Pisarenko TV, Kuranda TK et al (2022) Results of the implementation of the order of the Ministry of Education and Science of Ukraine for conducting scientific research and scientific and technical (experimental) developments, other areas of budget financing of the sphere of scientific and scientific and technical activity in 2021: scientific and analytical report. Kyiv, UkrINTEI (in Ukrainian)

10. Pysarenko TV, Kuranda TK et al (2024) Scientific, scientific-technical and innovative activity in Ukraine in 2023: scientific-analytical report. Kyiv, UkrINTEI (in Ukrainian)
11. On conducting a competitive selection of scientific and technical (experimental) developments under a state order, the implementation of which begins in 2024 at the expense of the state budget. Order of the Ministry of Education and Science № 808 from 06.06.2024. <https://mon.gov.ua/static-objects/mon/uploads/public/667/949/95d/66794995d4c08352545480.pdf>. Accessed 17 Feb 2025 (in Ukrainian)
12. On the announcement of the competitive selection of projects for fundamental scientific research, applied scientific research and scientific and technical (experimental) developments of young scientists in 2024. Order of the Ministry of Education and Science № 1160 from 16.08.2024 [online]. https://nauka.gov.ua/docs/62/%D0%9D%D0%B0%D0%BA%D0%B0%D0%B7_%D0%9C%D0%9E%D0%9D_%D0%B2%D1%96%D0%B4_18_08_2024_1160.PDF. Accessed 17 Feb 2025 (in Ukrainian)
13. On holding a competitive selection of fundamental scientific research and applied scientific research in 2024. Order of the Ministry of Education and Science № 1361 from 24.09.2024 [online]. <https://mon.gov.ua/static-objects/mon/uploads/public/66f/2b2/8b5/66f2b28b56375117186449.pdf>. Accessed 17 Feb 2025 (in Ukrainian)
14. Etzkowitz H, Leydesdorff L (1995) The triple helix—university-industry-government relations: a laboratory for knowledge based economic development. *EASST Rev* 14(1):14–19. Available at SSRN: <https://ssrn.com/abstract=2480085>. Accessed 17 Feb 2025
15. Haitham J, Maysara AD (2022) The role of scientific research on sustainable development into organizations. Available at SSRN: <https://ssrn.com/abstract=4233050> or <https://doi.org/10.2139/ssrn.4233050>. Accessed 17 Feb 2025
16. Mongeon P, Brodeur C, Beaudry C, Larivière V (2016) Concentration of research funding leads to decreasing marginal returns. *Res Eval* 25:396–404
17. Larivière V, Macaluso B, Archambault É, Gingras Y (2010) Which scientific elites? On the concentration of research funds, publications and citations. *Res Eval* 19(1):45–53
18. Tartari V, Stern S (2021) More than an ivory tower: the impact of research institutions on the quantity and quality of entrepreneurship. NBER Working Paper Series. Working Paper 28846. Available at: <http://www.nber.org/papers/w28846>. Accessed 17 Feb 2025
19. Hausman N (2022) University innovation and local economic growth. *Rev Econ Stat* 104(4):718–735
20. Lach S, Schankerman M (2003) Incentives and invention in universities. NBER Working Paper Series. Working Paper 9727. Available at: <http://www.nber.org/papers/w9727>. Accessed 17 Feb 2025
21. Rauer JN, Kroiss M, Kryvinska N, Engelhardt-Nowitzki C, Aburaia M (2021) Cross-university virtual teamwork as a means of internationalization at home. *Int J Manag Educ* 19(3):100512. <https://doi.org/10.1016/j.ijme.2021.100512>
22. Gibbons R (1992) *Game theory for applied economists*. Princeton University Press, Princeton
23. Shiyani AA, Zlepko SM, Korniyenko VO (2008) Theoretical and game modeling of the effectiveness of the interaction “teacher–university” in the context of taking into account the negotiating power of the parties. *Bull Vinnytsia Polytech Inst* 1:27–30 (in Ukrainian)

Mathematical Models of the Milking Processes Duration



Pavlo Kulakov , Volodymyr Kucheruk , Tetiana Neskoro dieva ,
Olena Semenova , Roman Lishchuk , Serhii Kontseba ,
Anna Kulakova , and Wiktor ia Mankovska

Abstract The research identified the need to improve mathematical models of the machinery milking duration, which will contribute to increasing the accuracy of assessing the productivity of milking plants at the stages of their development and modernization. Also, the development of these models will allow to further create the methodology for designing information systems to support and ensure the technological process of milk production in the conditions of various types of dairy farms. The study examines the basic algorithms of modern milking machines. Based on the experimental data obtained, it was found that the duration of animal preparation for milking (if milking robots are not used) has a chi-square distribution. The duration of milking an animal when using the milking machine without a milking process control function, and the duration of a random milking time interval when using the milking machine with a milking control function, have a gamma distribution. Mathematical models of the machinery milking duration when using the milking robot, the milking

P. Kulakov (✉) · V. Kucheruk · T. Neskoro dieva · R. Lishchuk · S. Kontseba · W. Mankovska
Uman National University, Uman, Ukraine
e-mail: p.kulakov@udau.edu.ua

V. Kucheruk
e-mail: kucheruk@udau.edu.ua

T. Neskoro dieva
e-mail: t.neskoro deva@udau.edu.ua

R. Lishchuk
e-mail: roman.lishchuk@udau.edu.ua

S. Kontseba
e-mail: kontseba@udau.edu.ua

W. Mankovska
e-mail: mankovska@udau.edu.ua

O. Semenova
Vinnytsia National Technical University, Vinnytsia, Ukraine
e-mail: semenova.o.o@vntu.edu.ua

A. Kulakova
Vasyl' Stus Donetsk National University, Vinnytsia, Ukraine

machine without a milking process control function, and the milking machine with a milking process control function have been improved, which take into account the minimum operating time of the milking machine and the time period during which the animal is identified.

Keywords Dairy farms · Milking plant · Mathematical model · Information systems · Machine milking · Milking duration

1 Introduction

Modern livestock farm cannot function effectively without automatic control systems and information systems to support and ensure the technological process of milk production [1–3]. The productivity of the milking plant, its type, measured zootechnical and technological parameters determine the structure and characteristics of the above systems [4–6]. When designing new and modernizing existing livestock farms, high requirements are imposed on determining their characteristics, including the productivity of the milking plant, which is very difficult to implement in practice [7–9]. This explains the fact that foreign companies do not indicate its numerical value in their advertising brochures, or indicate it with low accuracy [10–18].

An integral part of the machine milking process is the technological process of preparing the animal. In [19], this process is considered and it is proposed to consider its duration as constant when calculating the productivity of the milking plant. This statement can be considered objective only for the case of using milking robots, in which the duration of preparation is a deterministic value. On non-robotic milking plants, the duration of animal preparation is a random variable that depends on a number of objective and subjective factors, for example, on the operator's qualifications, his responsibility, and other random factors [20, 21].

There are three modern methods of milking: using milking machines with a milking process control function, using milking machines without a milking process control function, using milking robots. The duration of machine milking depends on the method used, the time of preparation of the animal and the type of animal identification.

Milking duration is an important parameter of a milking plant, which has a dominant value in determining its productivity [22]. The complexity of determining the productivity of the plant is largely due to the imperfection of existing mathematical models of the milking process duration. Known models of milking duration do not provide sufficient accuracy in determining the productivity of the plant and its other parameters and require improvement [23].

Thus, there is a need to create mathematical models based on the statistical characteristics of the machinery milking duration for the above milking methods, which will increase the accuracy of determining the productivity of the milking plant when designing or modernizing it. Mathematical models based on statistical characteristics of processes are widely used in genetics [24], electronics [25], medicine [26],

energy [27], and other fields. Improving existing models will improve the quality of designing information systems to support and ensure the technological process of milk production and automatic farm management systems.

In [28], the mathematical model of milking duration on a group milking plant was proposed, in [29] the mathematical model of milking duration on a milking plant with a milklane was proposed, in [30] the mathematical model of milking duration on a milking plant with abreast milking machines was proposed, and in [31] the mathematical model of milking duration on a conveyor milking plant was proposed. All of the above models do not take into account the time period during which animal identification occurs, which necessitates their improvement.

Based on this, further improvement of mathematical models of milking duration is an actual task. The research conducted relates to the theory and practice of designing, modernizing, informatizing, and automating livestock farms.

2 Research Results

The process of preparing the animal for milking consists of the following technological operations: washing the udder; manual milking of the first streams of milk; udder massage; anti-mastitis treatment of the udder; putting on the teat cups; removing the teat cups; post-milking treatment of the udder. If the milking plant is equipped with an automatic process control system that provides for animal identification, the following technological operations may be added: manual entry of the animal number in the absence of automatic identification; raising the reader by the milker to the animal's transponder (for cases of semi-automatic identification); manual entry of the animal number if the identification system does not work. In addition, there may be occasional delays in technological operations due to insufficient qualifications or dishonesty of the milker, accidental falling of milking cups, failures of machine milking equipment, and other similar reasons.

Based on the research conducted by the authors, it was found that the distribution law of time t_P of preparing an animal for milking approaches the chi-square distribution for both tethered and untethered animal keeping. The analytical expression for the chi-square distribution is determined by the expression [32, 33]

$$p_{PD}(t_P) = \frac{t_P^{\frac{k_{PD}}{2}-1} e^{-\frac{t_P}{2}}}{2^{\frac{k_{PD}}{2}} \Gamma(k_{PD}/2)}, \quad (1)$$

where k_{PD} —distribution function parameter; $\Gamma(z)$ —Euler's gamma function.

For the chi-square distribution, the expected value is given by the expression

$$M_{PD} = \int_0^{+\infty} \frac{t_P^{\frac{k_{PD}}{2}} e^{-\frac{t_P}{2}}}{2^{\frac{k_{PD}}{2}} \Gamma(k_{PD}/2)} dt_P = k_{PD}, \quad (2)$$

and the dispersion

$$D_{PD} = \int_0^{+\infty} \frac{(t_P - k_{PD})^2 t_P^{\frac{k_{PD}}{2}-1} e^{-\frac{t_P}{2}}}{2^{\frac{k_{PD}}{2}} \Gamma(k_{PD}/2)} dt_P = 2k_{PD}. \quad (3)$$

The milking time of an animal is a random variable that to some extent depends on the principle of dividing animals into groups, the algorithm of the milking machine, the type of milking machine, the type of milking plant, etc. In [21], it is assumed that the milking time interval has a normal distribution, and it is proposed to predict the milking duration using quadratic regression. In [19] it is proposed to assume that the time interval of milking duration on conveyor milking plants, group milking plants, and milking plants with a milkline has a normal distribution law. It should be noted that the normal distribution does not reflect an important feature of the milking duration time interval, namely that the milking duration cannot be less than a certain value or zero. In addition, due to the presence of a certain number of animals with non-standard milk yield parameters, the milking duration distribution function must have a certain asymmetry, which the normal law does not have. The possibility of an animal not in lactation entering the milking plant is also not taken into account when using the normal law. The paper [34] presents the results of studies based on which it is proposed to consider the duration of milking of an animal as distributed according to the lognormal law, which is widely used for statistical evaluation of the parameters of many biological processes. As a result of the conducted research, it was found that the distribution function of the milking time of an animal approaches lognormal if the animals are divided into a large number of groups according to the stage of their lactation period. In reality, this condition is very rarely met.

Figure 1 shows the cyclogram of the typical algorithm for working with the milking machine with the milking process control function.

After preparing the animal for a time t_P and putting on the teat cups, the milker starts the milking machine, which stimulates the udder for a deterministic time t_S . After the stimulation phase ends, there is a transition to the uncontrolled milking phase. During the uncontrolled milking phase, the milk yield intensity of the animal

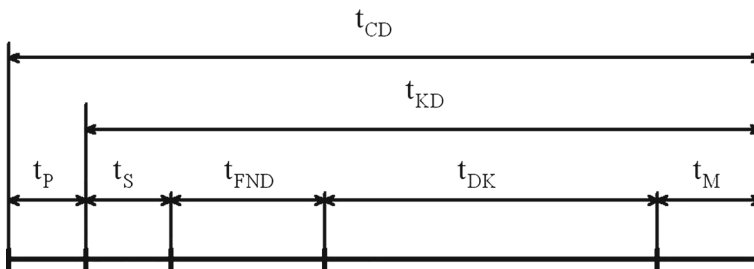


Fig. 1 Cyclogram of the typical algorithm for working with the milking machine with the milking process control function

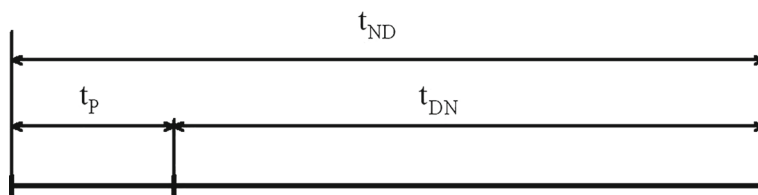


Fig. 2 Cyclogram of the typical algorithm for working with the milking machine without the milking process control function

is not measured for a deterministic time t_{FND} . After this, the transition to the controlled milking phase is carried out, during which the intensity of milk output is measured, and depending on its value, the optimal frequency of vacuum pulsations in the milking cups of the milking machine is set. The controlled milking phase has a random duration t_{DK} . After the milk yield intensity decreases below 0.8 kg/min, there is a transition to the finishing phase, during which periodic udder massage is performed for a deterministic time t_M . By increasing the intensity of milk output during the finishing phase, its duration can be prolonged. The controlled milking time t_{KD} when using a milking machine with the function of controlling the milking process is determined as the sum of the stimulation phase time, the uncontrolled milking phase time, the controlled milking phase time, and the udder massage phase time. The total interaction time with the animal when using a milking machine with a milking process control function is defined as the sum of the controlled milking time and the time to prepare the animal for milking.

Figure 2 shows a cyclogram of a typical algorithm for working with a milking machine without the milking process control function.

After preparing the animal for milking for a time t_p , the milker starts the milking machine, which milks the animal for a random uncontrolled milking time t_{DN} . After a certain period of time, the milker visually assesses the current state of the milking process and, based on certain signs, makes a subjective conclusion about the need to complete it. Thus, when using a milking machine without a milking process control function, the uncontrolled milking time t_{DN} , during which the animal is milked, is a random variable. The total interaction time with the animal when using a milking machine without a milking process control function is defined as the sum of the uncontrolled milking time and the milking preparation time. As a result of conducting research on the time of uncontrolled milking t_{DN} for milking machines without the function of controlling the milking process, it was found that the distribution of this interval approaches the gamma distribution and is determined by the expression

$$p_{DN}(t_{DN}) = \frac{t_{DN}^{l_{DN}} e^{-\frac{t_{DN}}{m_{DN}}}}{m_{DN}^{l_{DN}+1} \Gamma(l_{DN} + 1)}, \quad (4)$$

where l_{DN} , m_{DN} —distribution function parameters.

The expected value of the uncontrolled milking time t_{DN} when using milking machines without the function of controlling the milking process is determined by the expression

$$M_{DN} = \int_0^{+\infty} \frac{t_{DN}^{l_{DN}+1} e^{-\frac{t_{DN}}{m_{DN}}}}{m_{DN}^{l_{DN}+1} \Gamma(l_{DN} + 1)} dt_{DN} = m_{DN}(l_{DN} + 1), \quad (5)$$

and the dispersion of the uncontrolled milking time t_{DN} when using milking machines without the function of controlling the milking process is determined by the expression

$$\begin{aligned} D_{DN} &= \int_0^{+\infty} \frac{(t_{DN} - m_{DN}(l_{DN} + 1))^2 t_{DN}^{l_{DN}} e^{-\frac{t_{DN}}{m_{DN}}}}{m_{DN}^{l_{DN}+1} \Gamma(l_{DN} + 1)} dt_{DN} \\ &= m_{DN}^2(l_{DN} + 1). \end{aligned} \quad (6)$$

As a result of the study of the random time of the controlled milking phase t_{DK} of milking machines with the function of controlling the milking process, it was found that the distribution function of this interval also approaches the gamma distribution and is determined by the expression

$$p_{DK}(t_{DK}) = \frac{t_{DK}^{l_{DK}} e^{-\frac{t_{DK}}{m_{DK}}}}{m_{DK}^{l_{DK}+1} \Gamma(l_{DK} + 1)}, \quad (7)$$

where l_{DK} , m_{DK} —distribution function parameters.

Thus, the expected value of the controlled milking phase time is determined by the expression

$$M_{DK} = \int_0^{+\infty} \frac{t_{DK}^{l_{DK}+1} e^{-\frac{t_{DK}}{m_{DK}}}}{m_{DK}^{l_{DK}+1} \Gamma(l_{DK} + 1)} dt_{DK} = m_{DK}(l_{DK} + 1). \quad (8)$$

The dispersion of the controlled milking phase time is determined by the expression

$$\begin{aligned} D_{DK} &= \int_0^{+\infty} \frac{(t_{DK} - m_{DK}(l_{DK} + 1))^2 t_{DK}^{l_{DK}} e^{-\frac{t_{DK}}{m_{DK}}}}{m_{DK}^{l_{DK}+1} \Gamma(l_{DK} + 1)} dt_{DK} \\ &= m_{DK}^2(l_{DK} + 1). \end{aligned} \quad (9)$$

When using a milking machine without a milking process control function, the total interaction time with the animal is determined as the sum of two time intervals—the duration of the animal's preparation for milking and the time of uncontrolled milking.

$$t_{ND} = t_P + t_{DN}. \quad (10)$$

These two random variables are independent, so the distribution function of their sum $p_{ND}(t_{ND})$ is found as the convolution of their distribution functions

$$\begin{aligned} p_{ND}(t_{ND}) &= p_{PD}(t_{ND}) * p_{TD}(t_{ND}) \\ &= \int_0^{+\infty} \frac{\tau^{\frac{k_{PD}}{2}-1} (t_{ND} - \tau)^{l_{DN}} e^{-\frac{2t_{ND} + \tau(m_{DN}-2)}{2m_{DN}}}}{2^{\frac{k_{PD}}{2}} m_{DN}^{l_{DN}+1} \Gamma(l_{DN} + 1) \Gamma(k_{PD}/2)} d\tau. \end{aligned} \quad (11)$$

The given integral is not analytically defined and can only be calculated using numerical methods.

The expected value of the total interaction time with the animal for a milking machine without a milking process control function is determined by the expression

$$M_{ND} = m_{DN}(l_{DN} + 1) + k_{PD}, \quad (12)$$

and the dispersion

$$D_{ND} = m_{DN}^2(l_{DN} + 1) + 2k_{PD}. \quad (13)$$

The controlled milking time for a milking machine with the function of controlling the milking process consists of the stimulation phase time, the uncontrolled milking phase time, the controlled milking phase time, and the finishing phase time, and is determined by the expression

$$t_{KD} = t_S + t_{FND} + t_{DK} + t_M. \quad (14)$$

Expected value of controlled milking time

$$M_{KD} = t_S + t_{FND} + t_M + m_{DK}(l_{DK} + 1), \quad (15)$$

dispersion of controlled milking time

$$D_{KD} = m_{DK}^2(l_{DK} + 1). \quad (16)$$

The total interaction time with the animal when using a milking machine with the milking process control function is defined as the sum of the animal preparation time and the controlled milking time.

$$t_{CD} = t_P + t_{KD} = t_P + t_S + t_{FND} + t_{DK} + t_M. \quad (17)$$

The random variables in expression (17) are the animal preparation time t_P and the controlled milking phase time t_{DK} . All other variables are deterministic, the distribution function of the sum of the above random variables is defined by the expression

$$\begin{aligned} p_{CD}(t_{CD}) &= p_{PD}(t_{CD}) * p_{DK}(t_{CD}) \\ &= \int_0^{+\infty} \frac{\tau^{\frac{k_{PD}}{2}-1} (t_{CD} - \tau)^{l_{DK}} e^{-\frac{2t_{CD} + \tau(m_{DK}-2)}{2m_{DK}}}}{2^{\frac{k_{PD}}{2}} m_{DK}^{l_{DK}+1} \Gamma(l_{DK} + 1) \Gamma(k_{PD}/2)} d\tau. \end{aligned} \quad (18)$$

The given integral is similar to the integral (11), it is not analytically determined, it can be calculated only using numerical methods.

The expected value of the total interaction time with the animal when using milking machines with the function of controlling the milking process is determined by the expression

$$M_{CD} = t_S + t_{FND} + t_M + m_{DK}(l_{DK} + 1) + k_{PD}, \quad (19)$$

and the dispersion

$$D_{CD} = m_{DK}^2(l_{DK} + 1) + 2k_{PD}. \quad (20)$$

When using a milking robot, the time to prepare the animal for milking is a deterministic value t_{PR} . Based on this, the total time of working with the animal when using a milking robot is

$$t_{RD} = t_{PR} + t_S + t_{FND} + t_{DK} + t_M. \quad (21)$$

The expected value of the total interaction time with the animal when using a milking robot is given by the expression

$$M_{RD} = t_{PR} + t_S + t_{FND} + m_{DK}(l_{DK} + 1) + t_M, \quad (22)$$

and the dispersion

$$D_{RD} = m_{DK}^2(l_{DK} + 1). \quad (23)$$

For further statistical evaluation of the duration of milking machine operation, let t_{pi} denote the time of preparation of the i th animal for milking, and t_{TDi} denote the time of milking the i th animal using one of the types of milking machines.

When using a milking machine without a milking control function, the time interval t_{TDi} is equal to the uncontrolled milking time t_{DN} . Thus, in this case, the

expected value of the time interval t_{TDi} is determined by the expression

$$M_{TD} = M_{DN} = m_{DN}(l_{DN} + 1), \quad (24)$$

and the dispersion

$$D_{TD} = D_{DN} = m_{DN}^2(l_{DN} + 1). \quad (25)$$

When using a milking machine with a milking process control function, the time interval t_{TDi} is equal to the controlled milking time t_{KD} . In this case, the expected value of the time interval t_{TDi} is determined by the expression

$$M_{TD} = M_{KD} = t_S + t_{FND} + t_M + m_{DK}(l_{DK} + 1), \quad (26)$$

and the dispersion

$$D_{TD} = D_{KD} = m_{DK}^2(l_{DK} + 1). \quad (27)$$

Experimental studies of machine milking processes with different methods of keeping animals and different options for their identification were conducted at the agricultural enterprise “Radivske”, Radivka village, Kalynivskyi district, Vinnytsia region (Ukraine). When determining the distribution of the duration of the animal preparation for milking t_P , a sample of 1200 observations was used. As a result of the research, it was found that with untethered keeping with automatic animal identification $M_{PD} = 28$ s, $D_{PD} = 56$ s², with untethered keeping and automated identification (the milker brings the reader closer to the transponder manually) $M_{PD} = 35$ s, $D_{PD} = 70$ s², with tethered keeping and no animal identification $M_{PD} = 32$ s, $D_{PD} = 64$ s². Figure 3 shows graphs of the obtained experimental distribution functions of the time required to prepare an animal for milking under different methods of keeping and different options for animal identification.

Figure 4 shows a graph of the obtained time distribution function of uncontrolled milking for a milking machine without a milking process control function.

In the study of time t_{DN} , the manual milking machine without a milking process control function manufactured by the “Bratslav” company (Ukraine) was used. A sample of 700 observations was used to determine the distribution of this parameter. As a result of the experiment, it was found that for the studied herd with tethered keeping $M_{DN} = 306$ s, the variance $D_{DN} = 5750$ s².

The graph of the obtained time distribution function of the controlled milking phase when using a milking machine with the function of controlling the milking process when animals are kept without tethering is shown in Fig. 5.

When studying time t_{DK} , the milking process control unit “Bigmilk” manufactured by the company “Bratslav” (Ukraine) was used, which has the function of controlling the milking process and has $t_S = 15$ s, $t_{FND} = 30$ s, $t_M = 30$ s. The sample of 700 observations was used to determine the distribution function of this parameter. As a result of the experiment, it was found that for the studied herd with untethered

Fig. 3 Probability density functions of the animal preparation time for different methods of keeping and different options for animal identification

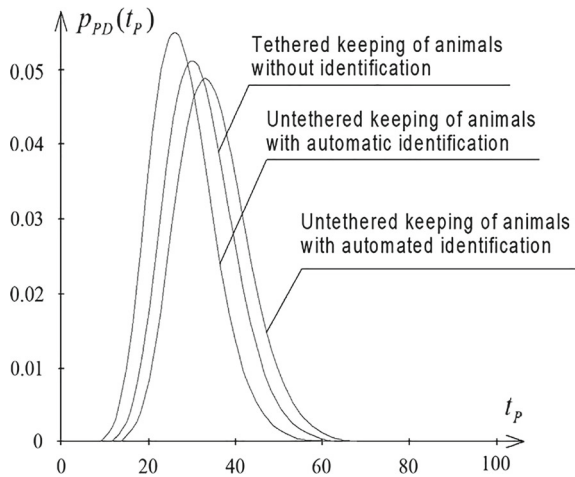
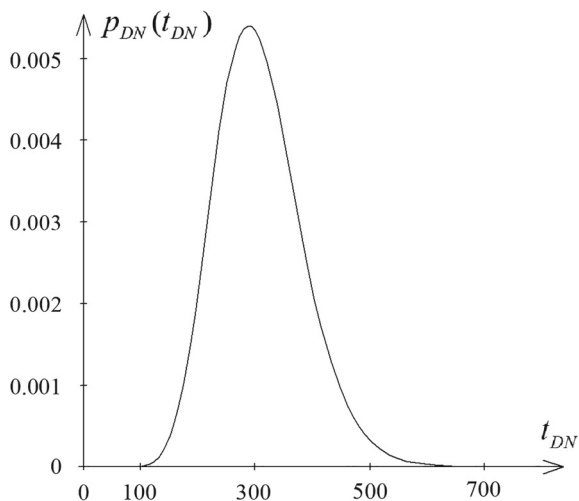


Fig. 4 Probability density function of time of uncontrolled milking using a milking machine without milking process control function



keeping, the expected value of the controlled milking phase time is $M_{DK} = 223$ s, the dispersion of the controlled milking phase time is $D_{DK} = 3745$ s².

Figure 6 shows the obtained graph of the distribution function of the total time of interaction with the animal when tethered keeping and using a milking machine without the function of controlling the milking process, for which the expected value $M_{ND} = 338$ s, dispersion $D_{ND} = 5814$ s².

Figure 7 shows the graph of the probability density function of the total interaction time with the animal when using the milking machine with the function of controlling the milking process and untethered keeping of the animals and automatic identification, for which the expected value $M_{CD} = 326$ s, dispersion $D_{CD} = 3801$ s².

Fig. 5 Probability density function of the controlled milking phase duration when using a milking machine with a milking process control function

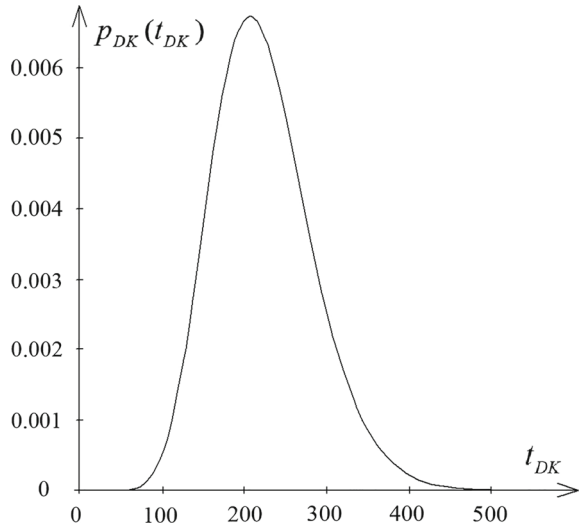
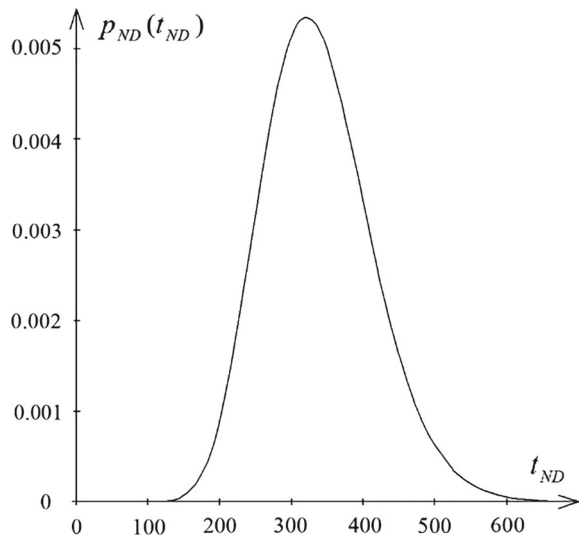
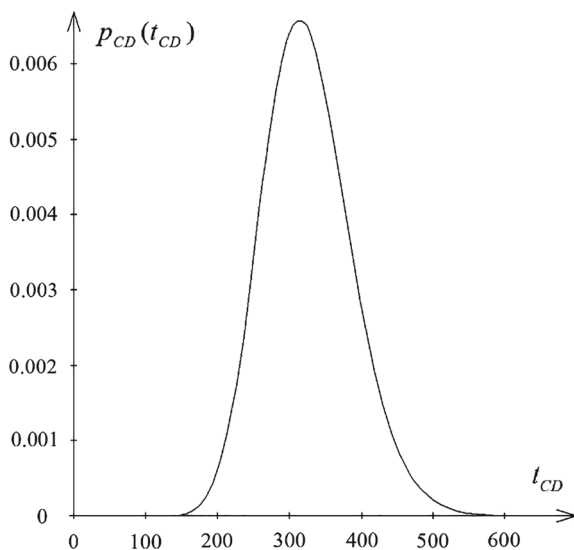


Fig. 6 Probability density function of the total interaction time with the animal when using the milking machine without the function of controlling the milking process and tethered keeping of the animals



To assess the closeness of the experimental data distribution to the adopted distribution function model, the Pearson agreement criterion was used in all cases.

Fig. 7 Probability density function of the total interaction time with the animal when using the milking machine with the function of controlling the milking process and untethered keeping of the animals and automatic identification



3 Conclusions

Based on the conducted research, it was found that the time interval of the technological process of preparing animals for milking has a chi-square distribution.

Based on the conducted research, it was found that the time interval of milking duration when using milking machines without a milking control function and the time interval of controlled milking when using milking machines with a milking control function have the gamma distribution.

Formulas were obtained that describe the mathematical model of the machine milking duration when using the milking robot, the milking machine without a milking control function, and the milking machine with a milking control function, which will allow to increase the accuracy of determining the productivity of a milking plant during its design or modernization, and will allow to develop the methodology for designing information systems to support and ensure the technological process of milk production and automatic farm management systems.

Existing mathematical models of the milking process duration have been developed by taking into account the time interval required for animal identification.

References

1. Ozella L, Brotto Rebuli K, Forte C, Giacobini M (2023) A literature review of modeling approaches applied to data collected in automatic milking systems. *Animals* 13(12):1916. <https://doi.org/10.3390/ani13121916>

2. Chikurtev D, Chikurteva A, Blagoeva E (2024) Technological analysis of types of milking systems and robots: a review. In: Pisla D, Carbone G, Condurache D, Vaida C (eds) *Advances in service and industrial robotics*. RAAD 2024. Mechanisms and machine science, vol 157. Springer, Cham. https://doi.org/10.1007/978-3-031-59257-7_57
3. Martin T, Gasselin P, Hostiou N et al (2022) Robots and transformations of work in farm: a systematic review of the literature and a research agenda. *Agron Sustain Dev* 42:66. <https://doi.org/10.1007/s13593-022-00796-2>
4. Nimbalkar V, Verma HK, Singh J (2021) Dairy farming innovations for productivity enhancement. New advances in the dairy industry. IntechOpen, London. <https://doi.org/10.5772/intechopen.101373>
5. Calcante A, Tangorra FM, Oberti R (2016) Analysis of electric energy consumption of automatic milking systems in different configurations and operative conditions. *J Dairy Sci* 99(5):4043–4047. <https://doi.org/10.3168/jds.2015-10490>
6. Gaworski M (2021) Implementation of technical and technological progress in dairy production. *Processes* 9:2103. <https://doi.org/10.3390/pr9122103>
7. Boguniewicz-Zablocka J, Klosok-Bazan I, Naddeo V (2019) Water quality and resource management in the dairy industry. *Environ Sci Pollut Res* 26:1208–1216. <https://doi.org/10.1007/s11356-017-0608-8>
8. Lage CFdA, Marques TC, Bruno DR, Endres MI, Ferreira F, Pires APA, Leão K, de Lima FS (2024) Farmers' perceptions on implementing automatic milking systems in large USA dairies: decision-making process, management practices, labor, and herd performance. *Animals* 14(2):218. <https://doi.org/10.3390/ani14020218>
9. Research and Markets (2023) Global milking robots market report 2023. Research and Markets, Dublin, Ireland
10. «DeLaval» company. Tetra Laval Group website. Accessed from: <https://www.delaval.com/>
11. GEA Group company. GEA Farm Technologies website. Accessed from: <https://www.gea-farmtechnologies.com/>
12. «Boumatic» company. «Boumatic» company website. Accessed from: <https://www.boumatic.com/>
13. «Panazoo Italiana Srl» company. «Panazoo Italiana Srl» company website. Accessed from: <https://www.panazoo.it/>
14. Impulsa AG company. Impulsa AG company website. Accessed from: <https://www.impulsa-ag.de/>
15. Afimilk company. Afimilk company website. Accessed from: <https://www.afimilk.com/>
16. Sac company. Aktieselskabet S.A.Christensen & Co. Accessed from: <https://www.sacmilking.com/>
17. «AGRO-PROMSERVIS» company. «AGRO-PROMSERVIS» company website. Accessed from: <https://agropromservis.net.ua/>
18. «Bratslav» Group company. Corporate website of the group of companies «Bratslav». Accessed from: <https://www.bratslav.com/>
19. Marques TC, Lage CFD, Bruno DR, Fausak ED, Endres MI, Ferreira FC, Lima FS (2023) Geographical trends for automatic milking systems research in non-pasture-based dairy farms: a scoping review. *J Dairy Sci* 106:7725–7736. <https://doi.org/10.3168/jds.2023-23313>
20. Kucheruk V, Palamarchuk Y, Kulakov P, Gnes T (2014) Statistical models of machinery milking duration. *Eastern Eur J Enterpr Technol* 1(3(67)):4–7. <https://doi.org/10.15587/1729-4061.2014.20080>
21. Wildridge AM, Thomson PC, Garcia SC, Jongman EC, Kerrisk KL (2020) Transitioning from conventional to automatic milking: effects on the human-animal relationship. *J Dairy Sci* 103:1608–1619. <https://doi.org/10.3168/jds.2019-16658>
22. Milanese S, Donina D, Chierici Guido V, Zaghen F, Sora VM, Zecconi A (2024) Comparing the performance of automatic milking systems through dynamic testing also helps to identify potential risk factors for mastitis. *Animals* 14(19):2789. <https://doi.org/10.3390/ani14192789>
23. Kucheruk V (2016) Measurement of the number servings of milk and control of water content in milk on stall milking machines. In: Kucheruk V, Kulakov P, Storozhuk N (eds) *Proceedings*

- of the international conference SCIT 2016, 20–21 May 2016, Warsaw, Poland. Recent advances in systems, control and information technology. Part V. Advances in intelligent systems and computing, vol 543, pp 435–447. https://doi.org/10.1007/978-3-319-48923-0_46
24. Su R, Lv J, Xue Y et al (2025) Genomic selection in pig breeding: comparative analysis of machine learning algorithms. *Genet Sel Evol* 57:13. <https://doi.org/10.1186/s12711-025-00957-3>
 25. Semenov AO, Savytskyi AY, Bisikalo OV, Kulakov PI (2018) Mathematical modeling of the two-stage chaotic colpitis oscillator. In: 2018 14th international conference on advanced trends in radioelectronics, telecommunications and computer engineering (TCSET), Lviv-Slavske, Ukraine, pp 835–839. <https://doi.org/10.1109/TCSET.2018.8336327>
 26. Sullivan B, Barker E, MacGregor L et al (2025) Comparing conventional and Bayesian workflows for clinical outcome prediction modelling with an exemplar cohort study of severe COVID-19 infection incorporating clinical biomarker test results. *BMC Med Inform Decis Mak* 25:123. <https://doi.org/10.1186/s12911-025-02955-3>
 27. Jin K, Banizaman H, Gharehveran SS et al (2025) Robust power management capabilities of integrated energy systems in the smart distribution network including linear and non-linear loads. *Sci Rep* 15:6615. <https://doi.org/10.1038/s41598-025-89817-0>
 28. Kucheruk V, Palamarchuk Y, Kulakov P (2014) The statistical models of machinery milking duration by group milking machines. *Eastern-Eur J Enterpr Technol* 4(4):13–17. <https://doi.org/10.15587/1729-4061.2014.26287>
 29. Kucheruk V, Palamarchuk Y, Kulakov P, Gnes T (2014) The statistical model of mechanical milking duration of farmyard milking installation. *Eastern-Eur J Enterpr Technol* 2(4):31–37. <https://doi.org/10.15587/1729-4061.2014.23120>
 30. Kucheruk V, Palamarchuk Y, Kulakov P (2014) Development of statistical models of milking duration on abreast milking machines. *Eastern-Eur J Enterpr Technol* 5(4(71)):4–9. <https://doi.org/10.15587/1729-4061.2014.28013>
 31. Kucheruk V, Palamarchuk Y, Kulakov P (2014) Devising statistic models of milking duration on the conveyor milking machines. *Eastern-Eur J Enterpr Technol* 6(4(72)):4–8. <https://doi.org/10.15587/1729-4061.2014.28951>
 32. Semenov AO, Baraban SV, Osadchuk OV, Semenova OO, Koval KO, Savytskyi AY (2019) Microelectronic pyroelectric measuring transducers. In: IFMBE proceedings. Springer, pp 393–397. https://doi.org/10.1007/978-3-030-31866-6_72
 33. Semenov A, Voznyak O, Osadchuk O, Baraban S, Semenova O, Rudyk A, Klimek J, Orazalieva S (2019) Development of a non-standard system of microwave quadrupoles parameters. In: Romaniuk RS, Linczuk M (eds) *Photonics applications in astronomy, communications, industry, and high-energy physics experiments*. SPIE, p 80. <https://doi.org/10.1117/12.2536704>
 34. Silva Boloña P, Upton J, Cabrera V, Erker T, Reinemann DJ (2022) Simulation model of quarter milk flowrates to estimate quarter and cow milking duration and automated milking system's box duration. *J Dairy Sci* 105(5):4156–4170. <https://doi.org/10.3168/jds.2021-20464>

Innovative Magnetosensitive Frequency Microsensors Based on Transistor Structures for Monitoring Systems in Biomedicine



Volodymyr Martyniuk , Oleksandr Osadchuk , Andrii Semenov ,
Iaroslav Osadchuk , and Olena Semenova

Abstract This study presents the development and analysis of innovative magnetosensitive frequency microsensors based on double-collector transistor structures, intended for use in biomedical monitoring systems. The research introduces mathematical models of microelectronic magnetic field frequency transducers that incorporate the influence of magnetic fields on nonlinear equivalent circuit elements. Analytical expressions for conversion functions and sensitivity equations have been derived to enable optimised engineering calculations and sensor design. These models use various transistor configurations, such as double-gate field-effect transistors and bipolar transistors with active inductive elements. This achieves enhanced sensitivity in the range of weak magnetic fields that are typical of biological processes. Experimental validation confirms that the proposed designs demonstrate high sensitivity ranging from 60 Hz/mT to 7.2 kHz/mT in the magnetic induction range of 0–60 mT, with close agreement between theoretical and experimental results ($\pm 5\%$). These findings lay the groundwork for the development of cost-effective, highly sensitive microsensors for use in medical diagnostics, functional monitoring and neurophysiological applications.

Keywords Magnetic field · Equivalent circuit · Double-collector transistor · Conversion function · Sensitivity equation

V. Martyniuk · O. Osadchuk · A. Semenov (✉) · I. Osadchuk · O. Semenova
Vinnitsia National Technical University, Vinnitsia, Ukraine
e-mail: semenov.a.o@vntu.edu.ua

O. Semenova
e-mail: semenova.o.o@vntu.edu.ua

1 Introduction

Modern biomedicine requires high-precision, non-invasive, sensitive equipment to monitor human physiological parameters. Magnetosensitive sensors that can detect weak magnetic fields generated by bioelectrical activity in the heart, brain and muscles are particularly interesting. However, existing magnetometric technologies, such as SQUID or optical sensors, are too expensive and complex for widespread use. There is therefore a need for miniature, energy-efficient, highly sensitive microsensors that can be integrated into portable or implantable medical devices.

The modern stage of scientific and technical progress can be characterized by rapid development of analytical instrument-making based on modern achievements in microelectronics [1–4]. This provides developing sensors for monitoring ambient parameters using group methods of technology [5–7]. One of the urgent today's problems is the design and production of microelectronic transducers.

Now accuracy requirements for measuring transducers, including magnetic induction transducers are increasing, and operating conditions are complicating, this forces the development of new measurement methods that would allow solving new emerging problems [8–10].

At the current stage of the measuring technology, various methods for converting magnetic induction into an electrical signal exist [11–15]. Presentation of measuring information in analog form demands using analog-to-digital converters, which require additional costs due to ensuring system immunity from interference, which creates a number of restrictions on the application of traditional measurement transducers compatible with digital means of information processing [16–20].

Applying a frequency signal as an informative parameter of primary transducers provides high interference resistance, simplicity and significant accuracy of conversion into a digital code, as well as the convenience of switching in multi-channel measuring systems [21–25].

Therefore, designing magnetic induction frequency transducers will contribute application of these advantages and provide full utilization of integral technology for the manufacture of devices [26–30].

2 Design of the Double-Collector Magnetotransistor and Its Mathematical Model

One of the most sensitive to the magnetic field elements is a double-collector magnetotransistor (DMT) with vertical collectors (Fig. 1) [31, 32]. Along with the effect of carrier redistribution between collectors, the magnetic field reduces the effective base thickness if the C2 collector current increases and, accordingly, increases the base thickness for the C1 collector. Thus, changing the effective thickness of base additionally increases the collector C2 current and also decreases the collector C1 current. This leads to an additional increase in the magnetic sensitivity of DMT.

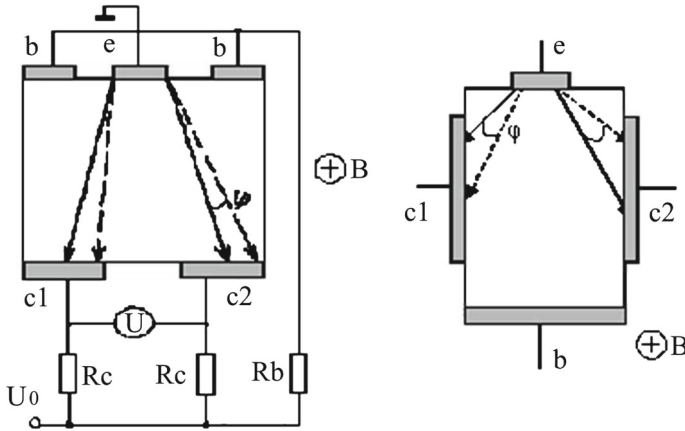


Fig. 1 Design of the double-collector magnetotransistors

Consider the magnetic field dependence of the elements of the magnetosensitive double-collector transistor equivalent circuit. Their dependence on the magnetic field is determined allowing for analytical dependences of the equivalent circuit parameters [33]. The ohmic resistance of the base region depends on the action of the magnetic field in this way

$$R_B = R_{B0}(1 + c\mu_p^2 B^2), \quad (1)$$

where R_{B0} is the base region resistance without the action of magnetic field, μ_p is the mobility of holes, c is the coefficient depending on holes scattering, B is the magnetic induction.

Magnetic field dependence of the emitter junction resistance is described by the expression

$$R_E = \frac{kT(1 + c\mu_p^2 B^2)}{qI_0 \cdot \exp(qU_{E0}/kT)}, \quad (2)$$

where k is the Boltzmann constant, q is the electron charge, I_0 is the reverse current of the emitter p-n junction, U_{E0} is the constant voltage at the emitter junction of the magnetosensitive transistor. The diffusion capacity of the emitter junction depends on the magnetic field according to the formula

$$C_E = \frac{qI_0\tau_p \cdot \exp(qU_{E0}/kT)}{kT(1 + c\mu_p^2 B^2)}, \quad (3)$$

where τ_p is the lifetime of holes. The capacity of the collector junction is described by the formula

$$C_K = S \left[\frac{\varepsilon \varepsilon_0}{2U_K \mu_p \rho_0 (1 + c \mu_p^2 B^2)} \right]^{1/2}, \quad (4)$$

where S is the collector junction area, ρ_0 is the resistivity of the base region, U_K is the reverse voltage of the magnetosensitive transistor collector junction, ε , ε_0 are the dielectric constants of the semiconductor material in the base region and in vacuum respectively.

Magnetic field dependence of the forward current gain in the common base circuit in the low-frequency region can be represented as

$$\alpha = 1 - \frac{1}{2} \left[\frac{W_0}{L_p (1 + c \mu_p^2 B^2 / 2)} \right]^2, \quad (5)$$

where L_p is the diffusion length of holes, W_0 is the base thickness without the influence of the magnetic field.

Thus, having determined the magnetic field dependence of the elements of the magnetosensitive transistor equivalent circuit, we proceed to determining the total resistance active and reactive components. This can be done allowing for the equivalent circuit of the magnetic field frequency transducer shown in general form in Fig. 2.

For this reason, we convert the equivalent circuit (Fig. 2) into a circuit for alternating current (Fig. 3).

Fig. 2 Equivalent circuit of the bipolar double-collector transistor

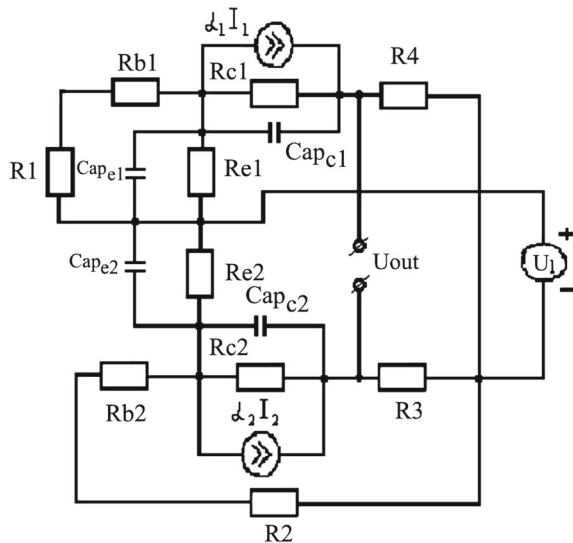
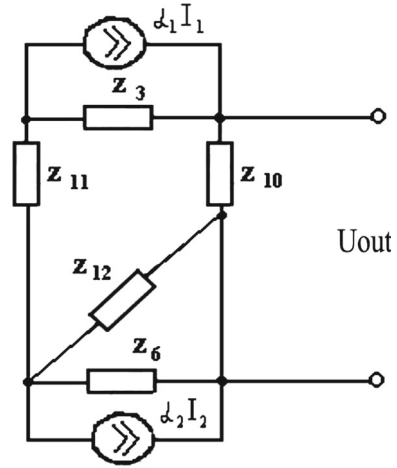


Fig. 3 Transformed equivalent circuit of the double-collector bipolar transistor for alternating current



The total resistances are described by expressions:

$$Z_3 = \frac{R_{C1}}{1 + \omega^2 R_{C1}^2 C_{C1}^2} - j \frac{R_{C1}^2 \omega C_{C1}}{1 + \omega^2 R_{C1}^2 C_{C1}^2}, \quad Z_6 = \frac{R_{C2}}{1 + \omega^2 R_{C2}^2 C_{C2}^2} - j \frac{R_{C2}^2 \omega C_{C2}}{1 + \omega^2 R_{C2}^2 C_{C2}^2},$$

$$Z_{12} = R_{B2} + R_3, \quad Z_{10} = R_4,$$

$$Z_{11} = \frac{(R_1 + R_{B1})Z_4}{R_1 + R_{B1} + Z_4} + \frac{R_{E2}}{1 + \omega^2 R_{E2}^2 C_{E2}^2} - j \frac{R_{E2}^2 \omega C_{E2}}{1 + \omega^2 R_{E2}^2 C_{E2}^2},$$

$$Z_4 = \frac{R_{E1}}{1 + \omega^2 R_{E1}^2 C_{E1}^2} - j \frac{R_{E1}^2 \omega C_{E1}}{1 + \omega^2 R_{E1}^2 C_{E1}^2}.$$

We regard this transistor as active one-port and apply the “star-delta” transformation to find the total resistance. It is described by the formula:

$$Z_0 = \frac{B_8 C_5 + B_7 C_6}{C_5^2 + C_6^2} - j \frac{B_7 C_5 - B_8 C_6}{C_5^2 + C_6^2}, \quad (6)$$

where

$$B_8 = (KB_3 + YB_5)P, \quad B_7 = (B_4K + B_6Y)P, \quad K = Z_{10} + Z_{12} + Z_{10}Z_{12}/Z_8,$$

$$Y = Z_8 + Z_{12} + Z_8Z_{12}/Z_{10}, \quad P = Z_{10} + Z_8 + Z_{10}Z_8/Z_{12},$$

$$B_3 = B_1(Y + W_6) - B_2V_6, \quad B_4 = B_2(Y + W_6) + B_1V_6, \quad B_5 = W_6(B_1 + K) - B_2V_6$$

$$B_6 = V_6(B_1 + K) + B_2W_6, \quad B_1 = W_3 + A_5, \quad B_2 = V_3 + A_6, \quad C_5 = B_8/P + C_3P.$$

$$C_6 = B_7/P + C_4P, \quad C_3 = C_1(Y + W_6) - C_2V_6, \quad C_4 = C_2(Y + W_6) + C_1V_6,$$

$$C_1 = W_3 + A_5 + K, \quad C_2 = V_3 + A_6, \quad A_5 = A(A_2/A_3) + W_5,$$

$$A_6 = A(A_4/A_3) + V_5, \quad A_1 = A + W_4, \quad A_2 = W_4A_1 + V_4^2, \quad A_3 = A_1^2 + V_4^2,$$

$$A_4 = A_1V_4 - V_4W_4, \quad A = Z_1 + Z_2, \quad W_3 = R_{C1}/(1 + \omega^2 R_{C1}^2 C_{C1}^2),$$

$$V_3 = (R_{C1}^2 \omega C_{C1})/(1 + \omega R_{C1}^2 C_{C1}^2), \quad W_4 = R_{E1}/(1 + \omega^2 R_{E1}^2 C_{E1}^2),$$

$$V_4 = (Re_1^2 \omega C_{E1})/(1 + \omega Re_1^2 C_{E1}^2), \quad W_5 = R_{E2}/(1 + \omega^2 R_{E2}^2 C_{E2}^2),$$

$$V_5 = (R_{E2}^2 \omega C_{E2})/(1 + \omega R_{E2}^2 C_{E2}^2), \quad W_6 = R_{C2}/(1 + \omega^2 R_{C2}^2 C_{C2}^2),$$

$$V_6 = (R_{C2}^2 \omega C_{C2})/(1 + \omega R_{C2}^2 C_{C2}^2).$$

Figure 4a and b presents magnetic induction dependences of the active (a) and reactive components allowing for (6). The absolute sensitivity of the reactive component to magnetic induction is 0.35 Ω/T , and it is 2 Ω/mT for the active component.

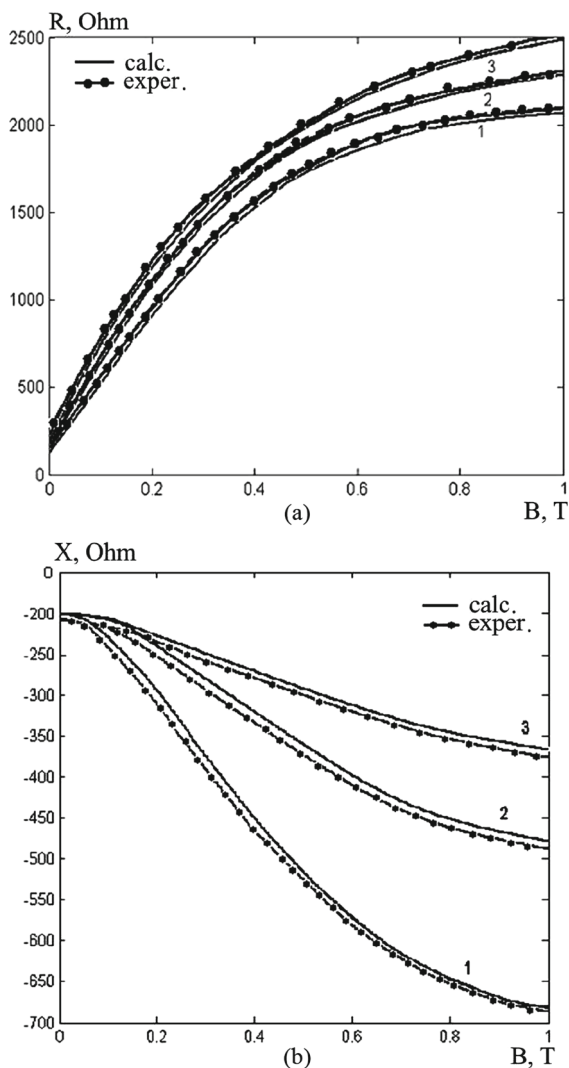
3 The Magnetic Field Transducer Based on a Magnetosensitive Bipolar Transistor and a Field Effect Transistor

An electric circuit of the magnetic field transducer is shown in Fig. 5. It consists of a double-collector magnetosensitive bipolar transistor and a two-gate field-effect transistor.

In order to determine behavior of the magnetic field transducer in dynamic mode, dependence of active and reactive components of the total resistance on the drain-collector electrodes, the oscillation frequency (conversion function) and sensitivity to the magnetic field action. Calculations were performed considering the equivalent circuit (Fig. 5).

Figure 6 illustrates an equivalent circuit of the radio measuring magnetic induction transducer. It takes into account nonlinear properties of the circuit, because

Fig. 4 Magnetic induction dependence of the total resistance active **a** and reactive **b** components for the bipolar magnetotransistor at frequencies of (1) $f = 1000$ Hz, (2) $f = 3000$ Hz, (3) $f = 5000$ Hz



the autogenerator can operate in both linear and nonlinear modes. To facilitate the calculations, the equivalent circuit is presented in Fig. 7.

Consider how the elements of the magnetosensitive double-collector transistor equivalent circuit depend on the magnetic field action. The dependence can be determined taking into account analytical dependences of the equivalent circuit parameters [34]. The base region ohmic resistance depends on magnetic field as follows

$$R_B = R_{B0} \left(1 + c \mu_p^2 B^2 \right), \quad (7)$$

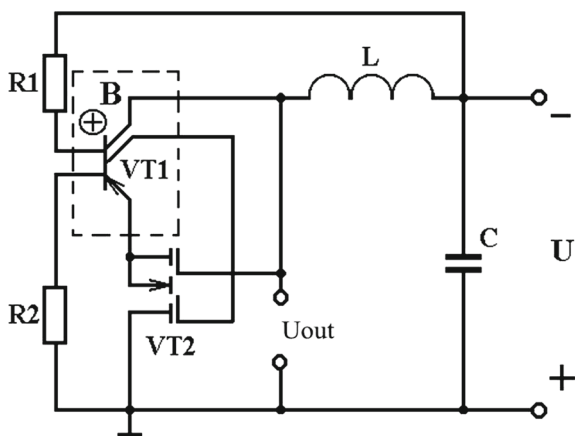


Fig. 5 Electrical circuit of the radio measuring magnetic induction transducer based on a magnetosensitive transistor

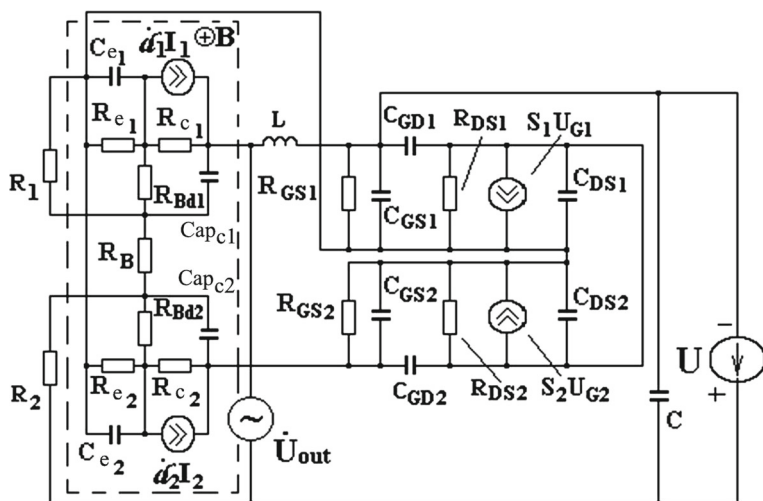


Fig. 6 Nonlinear equivalent circuit of the frequency transducer on a double-collector magnetotransistor

where R_{B0} is the base region resistance without action of magnetic field, μ_p is the mobility of holes, c is the coefficient that depends on holes scattering, B is the magnetic induction.

The magnetic field dependence of the emitter junction resistance is described by the expression

$$R_E = \frac{kT(1 + c\mu_p^2 B^2)}{qI_0 \cdot \exp(qU_{E0}/kT)}, \quad (8)$$

$$\alpha = 1 - \frac{1}{2} \left[\frac{W_0}{L_p(1 + c\mu_p^2 B^2/2)} \right]^2, \quad (11)$$

where L_p is the diffusion length of holes, W_0 is the base thickness without the magnetic field effect.

Thus, having determined the magnetic field dependence of elements of the magnetosensitive transistor equivalent circuit, we proceed to defining a conversion function and a sensitivity equation.

To define the conversion function, the magnetic induction dependence of the oscillation frequency must be determined. This can be done by solving the system of Kirchhoff equations, which was compiled for alternating current considering the equivalent circuit (Fig. 7). Solving the system of Eq. (12) yields the value of the total resistance at collector-collector electrodes of the transducer. By decomposing the total resistance into real and imaginary components, we determine the equivalent capacitance of the oscillating circuit, which depends on the magnetic induction.

Considering directions of the currents, the system of Kirchhoff equations is

$$\left\{ \begin{array}{l} 0 = (Z_1 + Z_2 + Z_{11})i_1 - Z_1i_3 - Z_2(i_5 + I_1) - Z_{11}i_9 + Z_1i_2, \\ 0 = (Z_1 + Z_3 + Z_5 + Z_6 + Z_8 + Z_7)i_2 + Z_1(i_1 - i_3) + Z_3(i_5 - i_3) \\ \quad + (Z_5 + Z_6)i_6 - Z_8i_4, \\ 0 = (Z_1 + Z_3 + Z_{R_1})i_3 - Z_1(i_1 + i_2) - Z_3(i_2 + i_5), \\ 0 = (Z_8 + Z_9 + Z_{10})i_4 - Z_8i_2 + Z_9I_2 + Z_{10}i_8, \\ 0 = (Z_3 + Z_2 + Z_4)i_5 + Z_3(i_2 - i_3) + Z_2(I_1 - i_1) - Z_4i_6, \\ U = (Z_4 + Z_5 + Z_6 + Z_{R_2})i_6 - Z_4i_5 + (Z_5 + Z_6)i_2 - Z_{R_2}i_8, \\ U = Z_Li_7 - Z_Li_8, \\ 0 = (Z_L + Z_{R_2} + Z_{10} + Z_{17} + Z_{12})i_8 - Z_Li_7 - Z_{R_2}i_6 + Z_{10}i_4 + Z_{17}i_{13} - Z_{12}i_9, \\ 0 = (Z_{11} + Z_{12} + Z_{13})i_9 - Z_{11}i_1 - Z_{12}i_8 + Z_8i_{10} + Z_{11}(I_3 + I_4) + Z_{12}(I_3 + I_4) \\ \quad - Z_{13}(I_3 + I_4), \\ 0 = (Z_{13} + Z_{14})i_{10} + Z_{13}i_9 - Z_{14}i_{11} - Z_{13}(I_3 + I_4) + Z_{14}(I_3 + I_4), \\ 0 = (Z_{14} + Z_{15})i_{11} - Z_{14}i_{10} - Z_{15}i_{12} - Z_{14}(I_3 + I_4) + Z_{15}(I_3 + I_4), \\ 0 = (Z_{15} + Z_{16})i_{12} - Z_{15}i_{11} - Z_{16}i_{13} - Z_{15}(I_3 + I_4) + Z_{16}(I_3 + I_4), \\ 0 = (Z_{16} + Z_{17} + Z_{18})i_{13} - Z_{16}i_{12} + Z_{17}i_8 - Z_{16}(I_3 + I_4) + Z_{18}(I_3 + I_4) \\ \quad + Z_{17}(I_3 + I_4), \end{array} \right. \quad (12)$$

where

$$Z_1 = \frac{R_{E1}}{1 + \omega^2 R_{E1}^2 C_{E1}^2} - j \frac{R_{E1}^2 \omega C_{E1}}{1 + \omega^2 R_{E1}^2 C_{E1}^2}, Z_2 = R_{k_1}, Z_3 = R_{Bd_1}, Z_4 = -\frac{j}{\omega C_{k_1}}, Z_5 = R_{B_1}, \\ Z_7 = \frac{R_{E2}}{1 + \omega^2 R_{E2}^2 C_{E2}^2} - j \frac{R_{E2}^2 \omega C_{E2}}{1 + \omega^2 R_{E2}^2 C_{E2}^2}, Z_6 = R_{B_2}, Z_8 = R_{Bd_2}, Z_9 = R_{k_2}, Z_{10} = -\frac{j}{\omega C_{k_2}},$$

$$\begin{aligned}
Z_{11} &= \frac{R_{GS1}}{1+\omega^2 R_{GS1}^2 C_{GS1}^2} - j \frac{R_{GS1}^2 \omega C_{GS1}}{1+\omega R_{GS1}^2 C_{GS1}^2}, Z_{12} = -\frac{j}{\omega C_{GD1}}, Z_{13} = R_{DS1}, Z_{14} = -\frac{j}{\omega C_{DS1}}, \\
Z_{15} &= -\frac{j}{\omega C_{DS2}}, Z_{16} = R_{DS2}, Z_{17} = -\frac{j}{\omega C_{GD2}}, Z_{18} = \frac{R_{GS2}}{1+\omega^2 R_{GS2}^2 C_{GS2}^2} - j \frac{R_{GS2}^2 \omega C_{GS2}}{1+\omega R_{GS2}^2 C_{GS2}^2}, \\
Z_L &= j\omega L_1.
\end{aligned}$$

To simplify the system of Kirchhoff equations, we introduce the following notation:

$$\begin{aligned}
M_1 &= Z_1 + Z_2 + Z_{11}, M_2 = Z_1 + Z_3 + Z_5 + Z_6 + Z_8 + Z_7, M_3 = Z_1 + Z_3 + Z_{R1}, \\
M_4 &= Z_8 + Z_9 + Z_{10}, M_5 = Z_3 + Z_2 + Z_4, M_6 = Z_4 + Z_5 + Z_6 + Z_{R2}, M_8 = \\
Z_L &+ Z_{R2} + Z_{10} + Z_{17} + Z_{12}, M_9 = Z_{11} + Z_{12} + Z_{13}, M_{10} = Z_{13} + Z_{14}, M_{11} = Z_{14} + Z_{15}, \\
M_{12} &= Z_{15} + Z_{16}, M_{13} = Z_{16} + Z_{17} + Z_{18}.
\end{aligned}$$

Then system (12) becomes as follows

$$\begin{cases}
0 = M_1 i_1 - Z_1 i_3 - Z_2 (i_5 + I_1) - Z_{11} i_9 + Z_1 i_2, \\
0 = M_2 i_2 + Z_1 (i_1 - i_3) + Z_3 (i_5 - i_3) + (Z_5 + Z_6) i_6 - Z_8 i_4, \\
0 = M_3 i_3 - Z_1 (i_1 + i_2) - Z_3 (i_2 + i_5), \\
0 = M_4 i_4 - Z_8 i_2 + Z_9 I_2 + Z_{10} i_8, \\
0 = M_5 i_5 + Z_3 (i_2 - i_3) + Z_2 (I_1 - i_1) - Z_4 i_6, \\
U = M_6 i_6 - Z_4 i_5 + (Z_5 + Z_6) i_2 - Z_{R2} i_8, \\
U = Z_L i_7 - Z_L i_8, \\
0 = M_8 i_8 - Z_L i_7 - Z_{R2} i_6 + Z_{10} i_4 + Z_{17} i_{13} - Z_{12} i_9, \\
0 = M_9 i_9 - Z_{11} i_1 - Z_{12} i_8 + (Z_{11} + Z_{12} - Z_{13}) (I_3 + I_4), \\
0 = M_{10} i_{10} + Z_{13} i_9 - Z_{14} i_{11} - (Z_{13} - Z_{14}) (I_3 + I_4), \\
0 = M_{11} i_{11} - Z_{14} i_{10} - Z_{15} i_{12} - (Z_{14} - Z_{15}) (I_3 + I_4), \\
0 = M_{12} i_{12} - Z_{15} i_{11} - Z_{16} i_{13} - (Z_{15} - Z_{16}) (I_3 + I_4), \\
0 = M_{13} i_{13} - Z_{16} i_{12} + Z_{17} i_8 - (Z_{16} - Z_{18} - Z_{17}) (I_3 + I_4).
\end{cases} \quad (13)$$

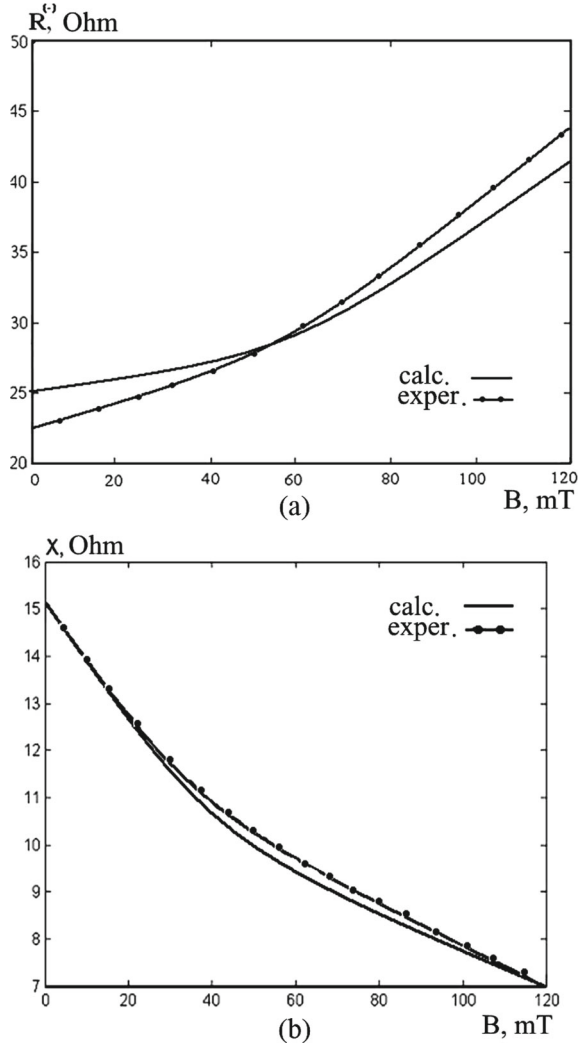
Having solved the system of equations, we obtained the theoretical magnetic induction dependence of active and reactive components of the total resistance, an oscillation frequency of and conversion functions.

Figure 8 shows theoretical and experimental magnetic induction dependences of the total resistance active and reactive components.

The conversion function of the device was determined by Lyapunov's method considering the circuit (13). This function describes the magnetic field induction dependence of the oscillation frequency. Analytical dependence of the conversion function is

$$F_0 = \frac{1}{2\pi} \sqrt{\frac{A_1 + \sqrt{A_1^2 + LC_{GD}(C_B(B)R_B(B))^2}}{2LC_{GD}(R_B(B)C_B(B))^2}}, \quad (14)$$

Fig. 8 Magnetic induction dependences of the total resistance active **a** and reactive **b** components



where

$$A_1 = LC_{GD} - (C_B(B)R_B(B))^2 - C_{GD}C_B(B)R_B^2(B),$$

L is the external inductance, C_B , R_B are the equivalent capacitance and resistance of the magnetosensitive transistor base, C_{GD} is the field-effect transistor gate-drain capacitance.

Figure 9 demonstrates graphical dependence of the conversion function. The sensitivity of the magnetic field sensor can be determined by expression (14) and described as equation

$$\begin{aligned}
S_B^{F_0} = & -0.0198 \left(-2C_B(B)R_B^3(B)C_{GD} \left(\frac{\partial C_B(B)}{\partial B} \right) \times \sqrt{A_1 + 2A_2} \right. \\
& - 2C_B^2(B)R_B^3(B) \left(\frac{\partial C_B(B)}{\partial B} \right) - 2C_B^3(B) \times R_B^2(B) \left(\frac{\partial R_B(B)}{\partial B} \right) \\
& - 3C_B(B)R_B^3(B)C_{GD} \left(\frac{\partial C_B(B)}{\partial B} \right) - 2C_{GD}C_B^2(B)R_B^2(B) \left(\frac{\partial R_B(B)}{\partial B} \right) \\
& + 8C_B^2(B)R_B^3(B)L \times C_{GD} \left(\frac{\partial C_B(B)}{\partial B} \right) + 8LC_{GD}C_B^2(B)R_B^2(B) \left(\frac{\partial R_B(B)}{\partial B} \right) \\
& + 4LC_{GD}R_B(B) \left(\frac{\partial C_B(B)}{\partial B} \right) \sqrt{A_1 + 2A_2} + 4R_B(B) \times \left(\frac{\partial C_B(B)}{\partial B} \right) LC_{GD} \\
& + 4C_B(B)LC_{GD} \left(\frac{\partial R_B(B)}{\partial B} \right) \times \sqrt{A_1 + 2A_2} + 4LC_{GD}C_B(B) \\
& \times \left(\frac{\partial R_B(B)}{\partial B} \right) \left. \right) / \left(\left(2\sqrt{A_1 + \sqrt{A_1 + 2A_2}} / A_2 \right) LC_{GD}C_B^3(B)R_B^3(B) \right. \\
& \times \sqrt{A_1 + 2A_2} \left. \right), \tag{15}
\end{aligned}$$

where

$$A_2 = 2LC_{GD}(C_B(B)R_B(B))^2.$$

Figure 10 presents the sensitivity graph. As can be seen from the graph, the highest sensitivity of the device lies in the range from 0 to 60 mT and is 200–60 Hz/mT.

To increase the sensitivity of the magnetic field transducer on a double-collector magnetosensitive bipolar transistor with active inductive element, we will consider

Fig. 9 Dependence of the oscillation frequency on the magnetic field induction of a magnetic field converter based on a magnetically sensitive bipolar transistor and a field-effect transistor

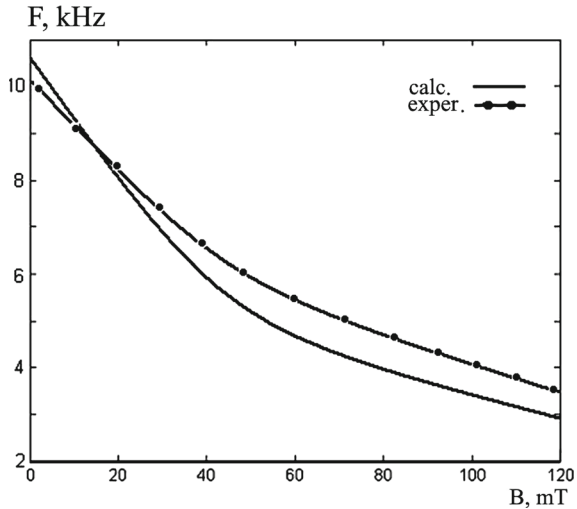
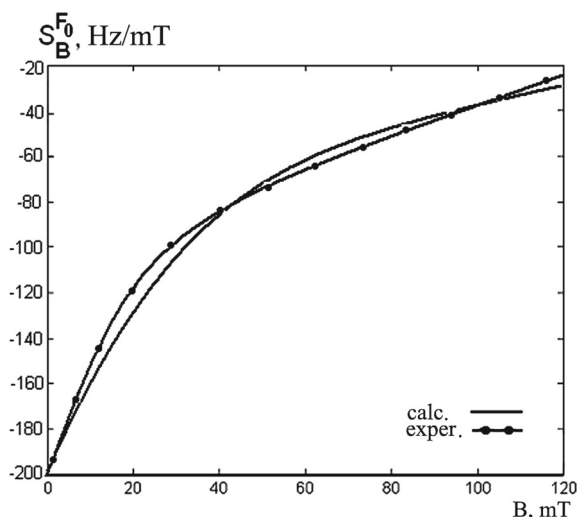


Fig. 10 Dependence of the sensitivity of a magnetic field transducer based on a magnetically sensitive bipolar transistor and a field-effect transistor on the magnetic field induction



and examine in the next section a circuit of a magnetic field transducer on a double-collector magnetosensitive bipolar transistor with active inductive element.

4 Magnetic Field Transducer on a Double-Collector Magnetosensitive Bipolar Transistor with Active Inductive Element

Figure 11 illustrates an electrical circuit of the magnetic field transducer. It is a hybrid integrated circuit that consists of a bipolar double-collector magnetosensitive transistor VT1 and a bipolar transistor VT2 with an RC circuit, which implements an active inductive element. This circuit forms an autogenerator device, its frequency depending on magnetic field. There is a total resistance at electrodes of the first collector and emitter of transistor VT1. Its active component has a negative value, and its reactive component is of capacitive nature. Connecting the active inductive element based on the bipolar transistor VT2 and an RC circuit to the first collector of VT1 and a common bus through the shorting capacity C2 creates an oscillating circuit in which energy losses are compensated by negative resistance.

Resistors R1-R4 and constant voltage sources U1 and U2 provide a power mode for the transducer circuit. When the magnetic field affects the transistor VT1, the equivalent capacitance of the autogenerator's oscillating circuit changes, which causes a change in its resonance frequency.

To determine the conversion function of the device, the inductance and Q-factor of the active inductive element on the bipolar transistor VT1 with an RC circuit must be specified. The inductance is determined by the next formula [35, 36]

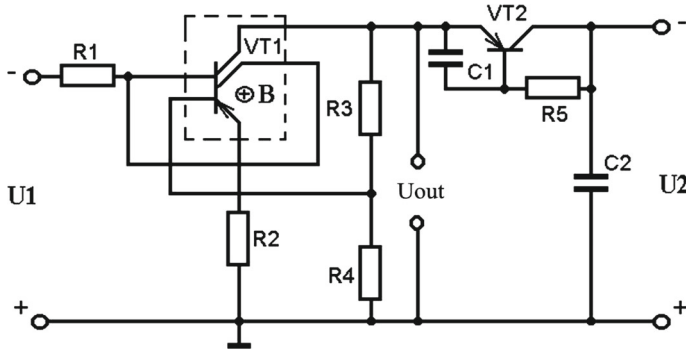


Fig. 11 Electric circuit for the magnetic field transducer

$$L = \frac{1}{\omega} \cdot \frac{A_4 A_1 - A_3 A_2}{A_1^2 + A_2^2}, \quad (16)$$

where

$$A_1 = \omega C \left[(r_B - R\alpha_1)^2 - (R\alpha_2)^2 + \frac{1}{(\omega C)^2} \right], \quad A_2 = \omega C [2R\alpha_2(r_B - R\alpha_1)],$$

$$A_3 = \left[R\alpha_1(r_B - R) + r_B \frac{R^2}{r_C} - 2 \frac{R^3}{r_C} \alpha_1 + \frac{R^2 \alpha_2^2}{(\omega C)^2} \right],$$

$$A_4 = \left[2R^2 \alpha_1 \alpha_2 - r_B R \alpha_2 + \frac{2R^3}{r_C} \alpha_2 \right].$$

$\alpha_1 = \frac{\alpha_0}{1+(f/f_\alpha)^2}$ is the real component of current transfer coefficient in the circuit with common base, $\alpha_2 = \frac{\alpha_0 f/f_\alpha}{1+(f/f_\alpha)^2}$ is the imaginary component of current transfer coefficient in the circuit with common base, r_B , r_C are the base and collector resistances of the transistor VT2 respectively, f_α is the limit frequency of the bipolar transistor in common base configuration, f is the operating frequency, $\omega = 2\pi f$ is the angular frequency.

The quality factor of the active inductive element is determined by the next formula [35, 36]

$$Q = \frac{(A_4 A_1 - A_3 A_2) \omega C}{\omega C (A_3 A_1 + A_4 A_2) - (A_1^2 + A_2^2)}. \quad (17)$$

Having determined how magnetic field affected the elements in the magnetosensitive transistor VT1 equivalent circuit, we proceed to defining of the conversion function and the sensitivity equation [37].

This can be done by considering an equivalent circuit of the magnetic field frequency transducer shown in general form in Fig. 12.

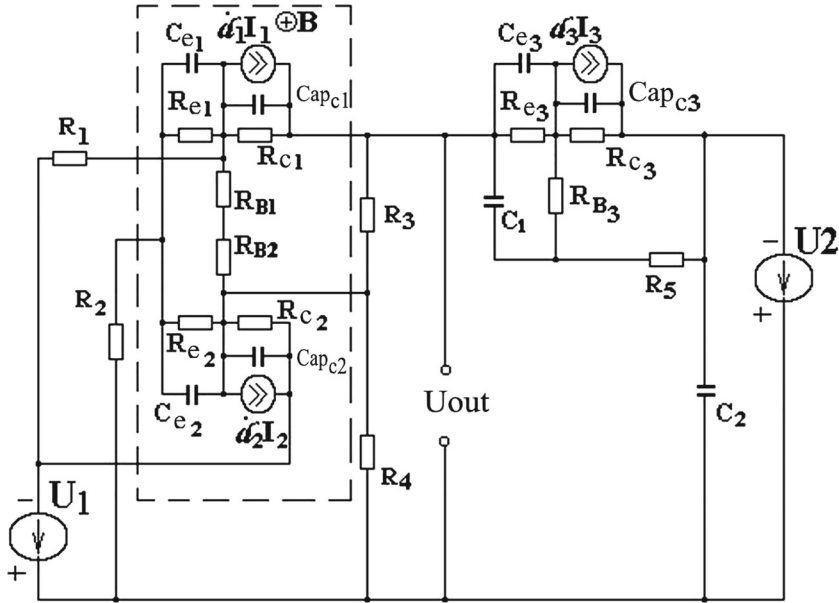


Fig. 12 General equivalent circuit of the magnetic field frequency transducer

We converted the equivalent circuit (Fig. 12) into a circuit for alternating current (Fig. 13), since we have to calculate the equivalent capacitance of the autogenerator's oscillating circuit determined by the reactive component of total resistance at the first collector electrode of the transistor VT1 with a common bus [38]. Let us write down the system of Kirchhoff equations, considering directions of the currents:

$$\begin{cases}
 0 = (Z_{R_1} + Z_{R_2} + Z_5 + Z_4 + Z_3)i_1 - Z_{R_2}i_3 - Z_5i_3 + Z_5i_2 + (Z_3 + Z_4)i_2 - (Z_3 + Z_4)i_5, \\
 0 = (Z_1 + Z_5 + Z_4 + Z_3)i_2 + Z_5(i_1 + i_3) + (Z_3 + Z_4)(i_1 - i_5), \\
 0 = (Z_{R_2} + Z_6 + Z_5)i_3 - Z_{R_2}i_1 + Z_6(I_2 - i_4) - Z_5(i_1 + i_2), \\
 0 = (Z_6 + Z_{R_4})i_4 - Z_6(I_2 + i_3) - Z_{R_4}i_6, \\
 0 = (Z_2 + Z_3 + Z_4 + Z_{R_3})i_5 + Z_2I_1 - (Z_3 + Z_4)(i_1 + i_2) - Z_{R_3}i_6, \\
 U = (Z_{R_3} + Z_{R_4})i_6 - Z_{R_3}i_5 - Z_{R_4}i_4, \\
 U = (Z_{C_1} + Z_{R_5})i_7 - Z_{C_1}i_8 - Z_{R_5}i_9, \\
 0 = (Z_7 + Z_8 + Z_{C_1})i_8 - Z_{C_1}i_7 - Z_8i_9, \\
 0 = (Z_9 + Z_8 + Z_{R_5})i_9 + Z_9I_3 - Z_8i_8 - Z_{R_5}i_7,
 \end{cases} \quad (18)$$

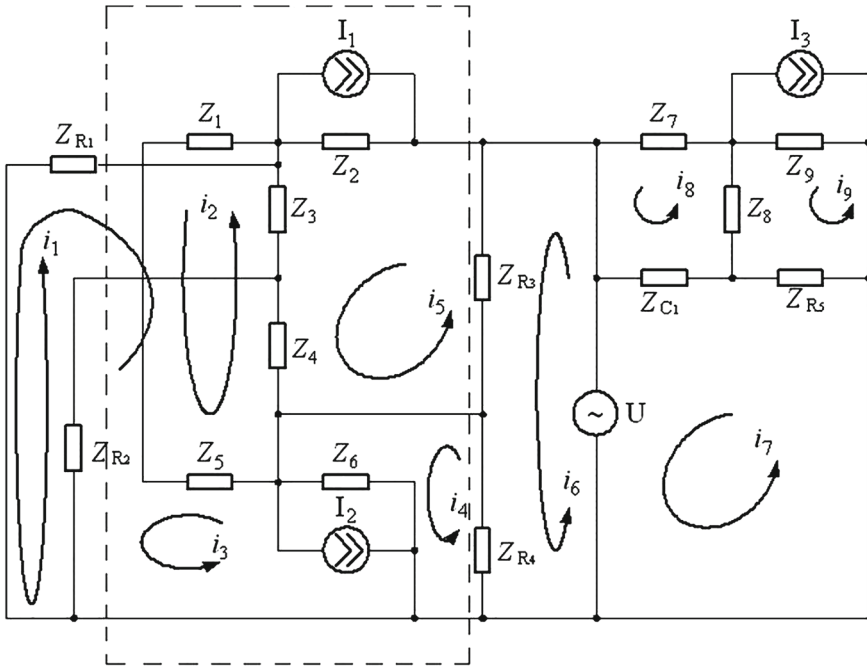


Fig. 13 Transformed equivalent circuit of an alternating current of a magnetic field frequency transducer based on a dual-collector magnetically sensitive bipolar transistor with an active inductive element

where

$$\begin{aligned}
 Z_1 &= \frac{R_{E1}}{1 + \omega^2 R_{E1}^2 C_{E1}^2} - j \frac{R_{E1}^2 \omega C_{E1}}{1 + \omega R_{E1}^2 C_{E1}^2}, & Z_2 &= \frac{R_{C1}}{1 + \omega^2 R_{C1}^2 C_{C1}^2} - j \frac{R_{C1}^2 \omega C_{C1}}{1 + \omega R_{C1}^2 C_{C1}^2}, \\
 Z_3 &= R_{B1}, & Z_4 &= R_{B2}, & Z_5 &= \frac{R_{E2}}{1 + \omega^2 R_{E2}^2 C_{E2}^2} - j \frac{R_{E2}^2 \omega C_{E2}}{1 + \omega R_{E2}^2 C_{E2}^2}, \\
 Z_7 &= \frac{R_{E3}}{1 + \omega^2 R_{E3}^2 C_{E3}^2} - j \frac{R_{E3}^2 \omega C_{E3}}{1 + \omega R_{E3}^2 C_{E3}^2}, & Z_8 &= R_{B3}, \\
 Z_9 &= \frac{R_{C3}}{1 + \omega^2 R_{C3}^2 C_{C3}^2} - j \frac{R_{C3}^2 \omega C_{C3}}{1 + \omega R_{C3}^2 C_{C3}^2}, & Z_{R1} &= R_1, & Z_{R2} &= R_2, & Z_{R3} &= R_3, \\
 Z_{R4} &= R_4, & Z_{R5} &= R_5, & Z_{C1} &= -j / (\omega C_1).
 \end{aligned}$$

To simplify the system of Kirchhoff equations, we introduce the following notation:

$$V_1 = Z_{R_1} + Z_{R_2} + Z_5 + Z_4 + Z_3, \quad V_2 = Z_1 + Z_5 + Z_4 + Z_3, \quad V_3 = Z_{R_2} + Z_6 + Z_5,$$

$$V_4 = Z_6 + Z_{R_4}, \quad V_5 = Z_2 + Z_3 + Z_4 + Z_{R_3}, \quad V_6 = Z_{R_3} + Z_{R_4}, \quad V_7 = Z_{C_1} + Z_{R_5}$$

$$V_8 = Z_7 + Z_8 + Z_{C_1}, \quad V_9 = Z_9 + Z_8 + Z_{R_5}.$$

Then system (18) becomes:

$$\begin{cases} 0 = V_1 i_1 - Z_{R_2} i_3 - Z_5 i_3 + Z_5 i_2 + (Z_3 + Z_4)(i_2 - i_5), \\ 0 = V_2 i_2 + Z_5(i_1 - i_3) + (Z_3 + Z_4)(i_1 - i_5), \\ 0 = V_3 i_3 - Z_{R_2} i_1 + Z_5(i_1 - i_3) + (Z_3 + Z_4)(i_1 - i_5), \\ 0 = V_4 i_4 - Z_6(i_2 + i_3) - Z_{R_4} i_6, \\ 0 = V_5 i_5 + Z_5 i_1 - (Z_3 + Z_4)(i_1 + i_2) - Z_{R_3} i_6, \\ U = V_6 i_6 - Z_{R_3} i_5 - Z_{R_4} i_4, \\ U = V_7 i_7 - Z_{C_1} i_8 - Z_{R_5} i_9, \\ 0 = V_8 i_8 - Z_{C_1} i_7 - Z_8 i_9, \\ 0 = V_9 i_9 + Z_9 i_3 - Z_8 i_8 + Z_{R_5} i_7. \end{cases} \quad (19)$$

Figure 14 illustrates theoretical and experimental magnetic induction dependences of the total resistance active and reactive components.

The conversion function of the device was determined by Lyapunov's method for stability [34] considering the equivalent circuit (Fig. 13). This function describes the magnetic field induction dependence of the oscillation frequency. Analytical dependence of the conversion function is

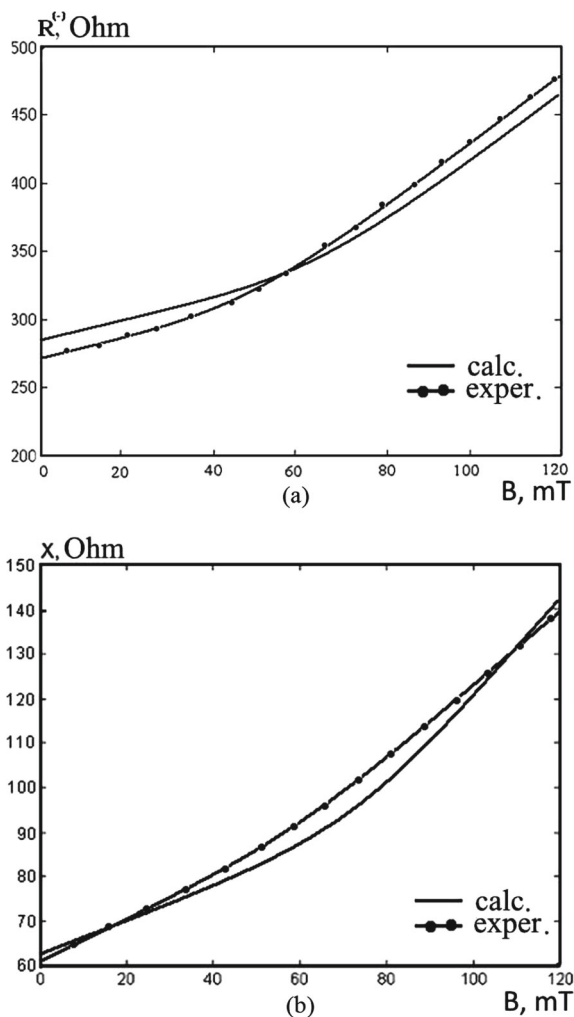
$$F_0 = \frac{1}{2\pi} \sqrt{\frac{R_B^2(B)C_B(B) - L}{LC_B^2(B)R_B^2(B)}}, \quad (20)$$

where R_B is the resistance of the magnetosensitive double-collector transistor base, C_B is the equivalent capacity of the magnetosensitive double-collector transistor, L is the active inductive element inductance.

The sensitivity equation is determined by expression (20)

$$\begin{aligned} S_B^{F_0} &= \frac{1}{4} \left(\left(C_B(B) \left(\frac{\partial R_B(B)}{\partial B} \right) + R_B^2(B) \left(\frac{\partial C_B(B)}{\partial B} \right) \right) / (LC_B^2(B)R_B^2(B)) \right. \\ &\quad \left. - \left(2(C_B(B)R_B^2(B) - L) \left(\frac{\partial C_B(B)}{\partial B} \right) \right) / (LC_B^3(B)R_B^2(B)) \right. \\ &\quad \left. - \left(2(R_B^2(B)C_B(B) - L) \left(\frac{\partial R_B(B)}{\partial B} \right) \right) / (LC_B^2(B)R_B^3(B)) \right) / \left(\pi \sqrt{\frac{C_B(B)R_B^2(B) - L}{LC_B^2(B)R_B^2(B)}} \right), \end{aligned} \quad (21)$$

Fig. 14 Magnetic induction dependencies of total resistance active (a) and reactive (b) components



The graphical dependence of the conversion function, at 5 V power supply, is shown in Fig. 15. The sensitivity of the magnetic field frequency transducer is shown in Fig. 16. As can be seen from the graph, the highest sensitivity of the device lies in the range from 0 to 40 mT and is 350–550 Hz/mT.

Theoretical and experimental investigations demonstrated that the sensitivity of the transducer is 450 Hz/mT.

To increase the sensitivity of the magnetic field transducer on a double-collector magnetosensitive bipolar transistor with active inductive element, we will consider and examine in the next section a circuit of a magnetic field transducer with a double-collector magnetosensitive bipolar transistor, a two-gate field-effect transistor, and a bipolar transistor.

Fig. 15 Dependence of the oscillation frequency on the magnetic field induction of a magnetic field converter based on a dual-collector magnetically sensitive bipolar transistor with an active inductive element

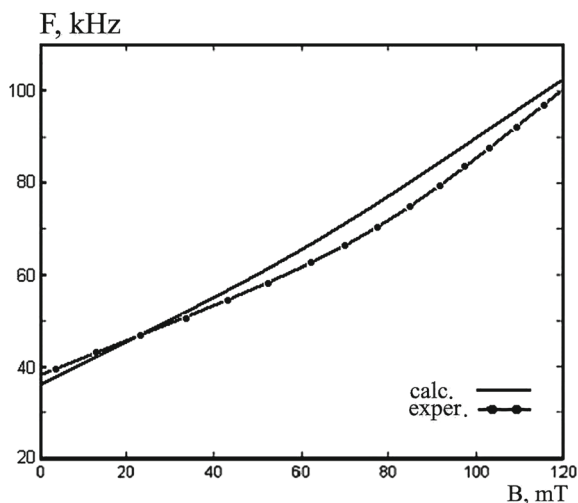
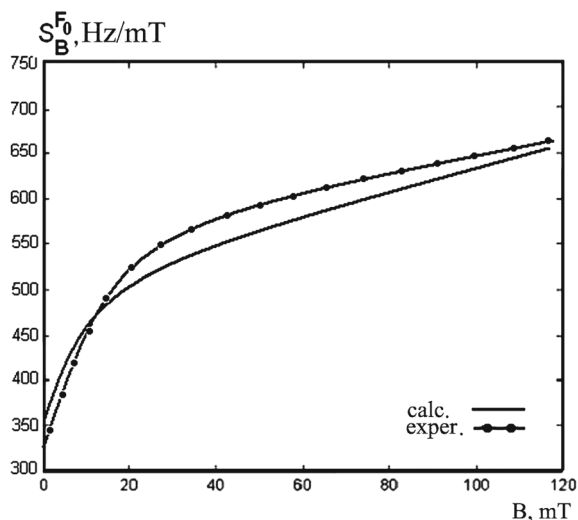


Fig. 16 Magnetic field induction dependence of the transducer sensitivity



5 Magnetic Field Transducer on a Double-Collector Magnetosensitive Bipolar Transistor, a Two-Gate Field-Effect Transistor and a Bipolar Transistor

The electrical circuit of the magnetic frequency sensor is shown in Fig. 17. It is a hybrid integrated circuit, which consist of a double-collector magnetosensitive transistor, a field-effect transistor and a bipolar transistor that create an autogenerator device and its oscillation frequency depends on magnetic field induction.

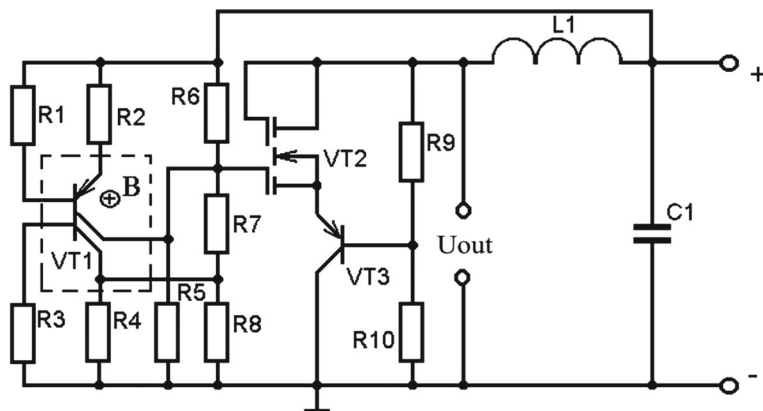


Fig. 17 Electric circuit of the magnetic frequency sensor

There is a total resistance at collector electrodes of the bipolar transistor VT3 and at the field-effect transistor VT2 drain. Its active component is negative and its reactive component is capacitive [39].

Connecting the inductance L1 to the double-gate field-effect transistor VT2 drain and to the common bus through the blocking capacitance C1 creates an oscillating circuit in which energy losses are compensated by negative resistance.

Resistors R9 and R10 provide the power mode by direct current. When the magnetic field affects the double-collector magnetosensitive transistor VT1, the equivalent capacitance of the oscillating circuit changes, this causes a change in the resonant frequency.

The equivalent circuit of the magnetic frequency sensor is presented in Fig. 18. It is a linear circuit, because the autogenerator operates in a linear operation mode at the low-frequency range [40].

To simplify calculations, we convert the equivalent circuit (Fig. 18) into a circuit for alternating current (Fig. 19).

Let us write a system of Kirchhoff equations, considering directions of the currents:

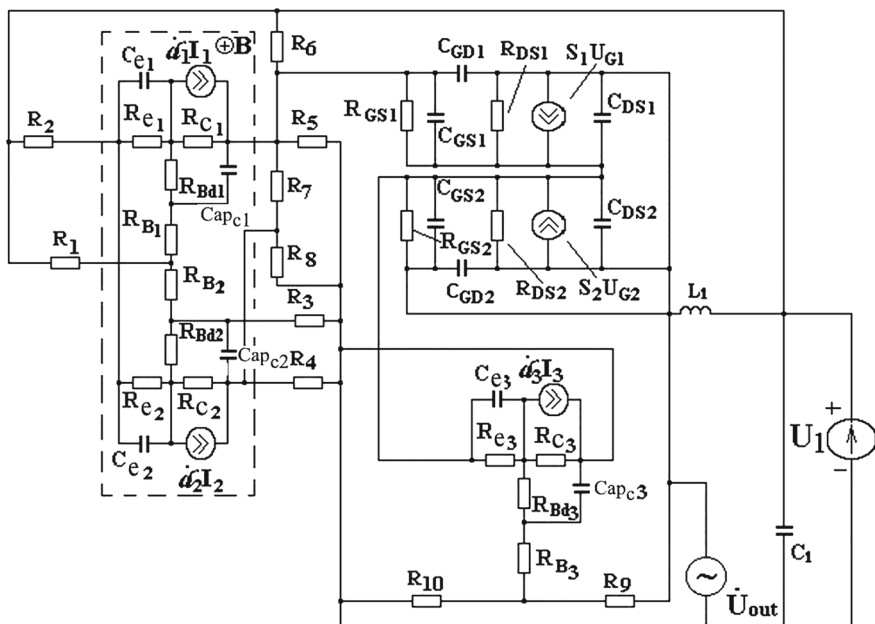


Fig. 18 Equivalent circuit of the magnetic frequency sensor

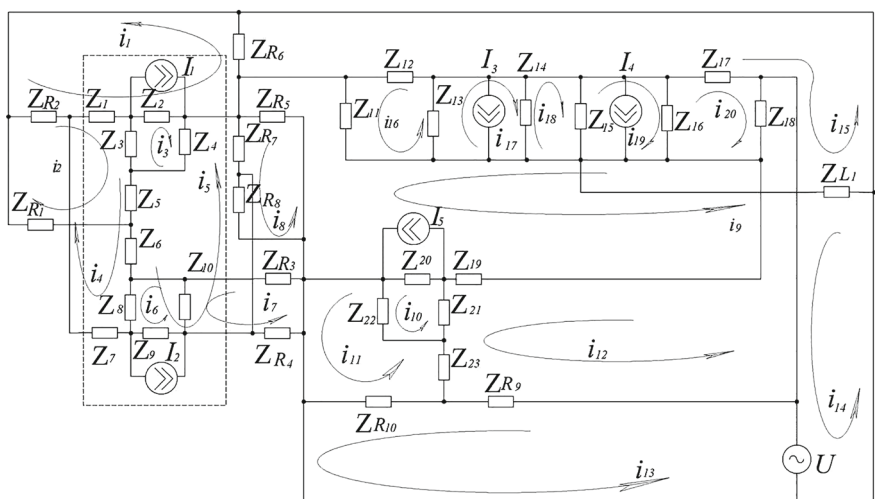


Fig. 19 Transformed equivalent circuit of an alternating current of a magnetic field frequency transducer based on a double-collector magnetosensitive bipolar transistor, a two-gate field-effect transistor and a bipolar transistor

$$\left\{ \begin{array}{l}
0 = (Z_{R_2} + Z_1 + Z_2 + Z_{R_6})i_1 + Z_{R_2}i_2 + Z_1i_2 + Z_2(i_3 - I_1) - Z_{R_6}i_{15} + Z_1i_4, \\
0 = (Z_{R_2} + Z_1 + Z_3 + Z_5 + Z_{R_1})i_2 + (Z_{R_2} + Z_1)i_1 - Z_3i_3 + Z_5(i_4 + i_5), \\
0 = (Z_2 + Z_3 + Z_4)i_3 + Z_2(i_1 - I_1) - Z_3i_2 + Z_4i_5, \\
0 = (Z_1 + Z_3 + Z_5 + Z_6 + Z_8 + Z_7)i_4 + Z_1(i_1 + i_2) - Z_3i_3 + (Z_5 + Z_6)i_5 \\
\quad + Z_8(i_5 + i_6), \\
0 = (Z_4 + Z_5 + Z_6 + Z_8 + Z_9 + Z_{R_7})i_5 + Z_4i_3 + (i_4 + i_5)Z_5 + Z_6i_4 \\
\quad + (i_4 + i_6)Z_8 + Z_9(i_6 + I_2) - Z_{R_7}i_8, \\
0 = (Z_8 + Z_9 + Z_{10})i_6 + Z_8(i_5 + i_4) + Z_9(i_5 + I_2) - Z_{10}i_7, \\
0 = (Z_{R_3} + Z_{R_4} + Z_{10})i_7 - Z_{10}i_6, \\
0 = (Z_{R_5} + Z_{R_7} + Z_{R_8})i_8 - Z_{R_7}i_5 - Z_{R_5}i_9, \\
0 = (Z_{R_5} + Z_{20} + Z_{19} + Z_{11})i_9 - Z_{R_5}i_8 + Z_{20}(I_5 - i_{10}) - Z_{19}i_{12} - Z_{11}i_{16}, \\
0 = (Z_{20} + Z_{21} + Z_{22})i_{10} - Z_{22}i_{11} - Z_{21}i_{12} - Z_{20}(I_5 + i_9), \\
0 = (Z_{22} + Z_{23} + Z_{R_{10}})i_{11} - Z_{22}i_{10} - Z_{23}i_{12} - Z_{R_{10}}i_{13}, \\
0 = (Z_{19} + Z_{21} + Z_{23} + Z_{R_9} + Z_{18})i_{12} - Z_{19}i_9 - Z_{21}i_{10} - Z_{23}i_{11} \\
\quad - Z_{R_9}i_{13} + Z_{18}i_{20}, \\
\dot{U} = (Z_{R_{10}} + Z_{R_9})i_{13} - Z_{R_{10}}i_{11} - Z_{R_9}i_{12}, \\
\dot{U} = Z_{L_1}i_{14} - Z_{L_1}i_{15}, \\
0 = (Z_{R_6} + Z_{12} + Z_{17} + Z_{L_1})i_{15} - Z_{R_6}i_1 - Z_{12}i_{16} + Z_{17}i_{20} - Z_{L_1}i_{14}, \\
0 = (Z_{11} + Z_{12} + Z_{13})i_{16} - Z_{12}i_{15} - Z_{11}i_9 + Z_{13}(I_3 + I_4) + Z_{13}i_{17} \\
\quad - Z_{11}(I_3 + I_4) - Z_{12}(I_3 + I_4), \\
0 = (Z_{13} + Z_{14})i_{17} + Z_{13}i_{16} - Z_{14}i_{18} + Z_{13}(I_3 + I_4) - Z_{14}(I_3 + I_4), \\
0 = (Z_{14} + Z_{15})i_{18} - Z_{14}i_{17} - Z_{15}i_{19} + Z_{14}(I_3 + I_4) - Z_{15}(I_3 + I_4), \\
0 = (Z_{15} + Z_{16})i_{19} - Z_{15}i_{18} - Z_{16}i_{20} + Z_{15}(I_3 + I_4) - Z_{16}(I_3 + I_4), \\
0 = (Z_{16} + Z_{17} + Z_{18})i_{20} - Z_{16}i_{19} + Z_{18}i_{12} + Z_{17}i_{15} + Z_{16}(I_3 + I_4) \\
\quad - Z_{17}(I_3 + I_4) - Z_{18}(I_3 + I_4),
\end{array} \right. \quad (22)$$

where

$$\begin{aligned}
Z_{R_2} &= R_2, Z_{R_1} = R_1, Z_{R_6} = R_6, Z_{R_5} = R_5, Z_{R_7} = R_7, Z_{R_8} = R_8, Z_{R_3} = R_3, \\
Z_{R_4} &= R_4, Z_{R_{10}} = R_{10}, Z_{R_9} = R_9, Z_1 = \frac{R_{E_1}}{1 + \omega^2 R_{E_1}^2 C_{E_1}^2} - j \frac{R_{E_1}^2 \omega C_{E_1}}{1 + \omega^2 R_{E_1}^2 C_{E_1}^2}, Z_2 = R_{C_1}, \\
Z_3 &= R_{Bd_1}, Z_4 = -\frac{j}{\omega C_{C_1}}, Z_5 = R_{B_1}, Z_6 = R_{B_2}, Z_7 = \frac{R_{E_2}}{1 + \omega^2 R_{E_2}^2 C_{E_2}^2} - j \frac{R_{E_2}^2 \omega C_{E_2}}{1 + \omega^2 R_{E_2}^2 C_{E_2}^2}, Z_8 = \\
R_{Bd_2}, Z_9 &= R_{C_2}, Z_{10} = -\frac{j}{\omega C_{C_2}}, Z_{11} = \frac{R_{GS_1}}{1 + \omega^2 R_{GS_1}^2 C_{GS_1}^2} - j \frac{R_{GS_1}^2 \omega C_{GS_1}}{1 + \omega^2 R_{GS_1}^2 C_{GS_1}^2}, Z_{12} = -\frac{j}{\omega C_{GD_1}}, \\
Z_{13} &= R_{DS_1}, Z_{14} = -\frac{j}{\omega C_{DS_1}}, Z_{15} = -\frac{j}{\omega C_{DS_2}}, Z_{16} = R_{DS_2}, Z_{17} = -\frac{j}{\omega C_{GD_2}}, Z_{18} = \\
\frac{R_{GS_2}}{1 + \omega^2 R_{GS_2}^2 C_{GS_2}^2} - j \frac{R_{GS_2}^2 \omega C_{GS_2}}{1 + \omega^2 R_{GS_2}^2 C_{GS_2}^2}, Z_{19} &= \frac{R_{E_3}}{1 + \omega^2 R_{E_3}^2 C_{E_3}^2} - j \frac{R_{E_3}^2 \omega C_{E_3}}{1 + \omega^2 R_{E_3}^2 C_{E_3}^2}, Z_{20} = R_{C_3}, Z_{21} = R_{Bd_3}, \\
Z_{22} &= -\frac{j}{\omega C_{C_3}}, Z_{23} = R_{B_3}, Z_{L_1} = j\omega L_1.
\end{aligned}$$

To simplify the system of Kirchhoff equations, we introduce the following notation:

$$\begin{aligned}
 K_1 &= Z_{R_2} + Z_1 + Z_2 + Z_{R_6}, K_2 = Z_{R_2} + Z_1 + Z_3 + Z_5 + Z_{R_1}, K_3 = Z_2 + Z_3 + Z_4, \\
 K_4 &= Z_1 + Z_3 + Z_5 + Z_6 + Z_8 + Z_7, K_5 = Z_4 + Z_5 + Z_6 + Z_8 + Z_9 + Z_{R_7}, \\
 K_6 &= Z_8 + Z_9 + Z_{10}, K_7 = Z_{R_3} + Z_{R_4} + Z_{10}, K_8 = Z_{R_5} + Z_{R_7} + Z_{R_8}, \\
 K_9 &= Z_{R_5} + Z_{20} + Z_{19} + Z_{11}, K_{10} = Z_{20} + Z_{21} + Z_{22}, K_{11} = Z_{22} + Z_{23} + Z_{R_{10}}, \\
 K_{12} &= Z_{19} + Z_{21} + Z_{23} + Z_{R_9} + Z_{18}, K_{13} = Z_{R_{10}} + Z_{R_9}, \\
 K_{15} &= Z_{R_6} + Z_{12} + Z_{17} + Z_{L_1}, K_{16} = Z_{11} + Z_{12} + Z_{13}, \\
 K_{17} &= Z_{13} + Z_{14}, K_{18} = Z_{14} + Z_{15}, K_{19} = Z_{15} + Z_{16}, K_{20} = Z_{16} + Z_{17} + Z_{18}.
 \end{aligned}$$

Then system (23) becomes

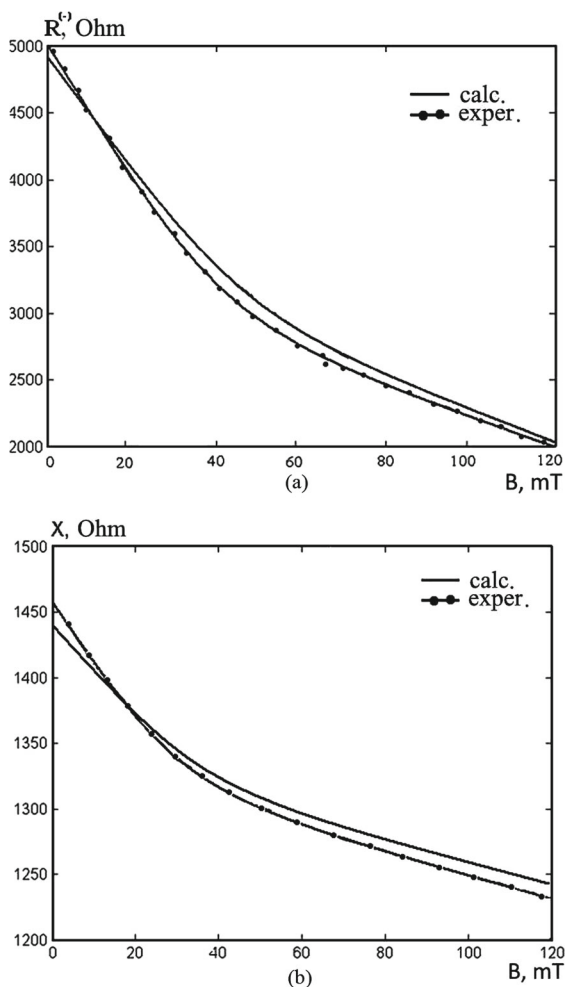
$$\left\{ \begin{aligned}
 0 &= K_1 i_1 + Z_{R_2} i_2 + Z_1 i_2 + Z_2 (i_3 - I_1) - Z_{R_6} i_{15} + Z_1 i_4, \\
 0 &= K_1 i_1 + Z_{R_2} i_2 + Z_1 i_2 + Z_2 (i_3 - I_1) - Z_{R_6} i_{15} + Z_1 i_4, \\
 0 &= K_2 i_2 + (Z_{R_2} + Z_1) i_1 - Z_3 i_3 + Z_5 (i_4 + i_5), \\
 0 &= K_3 i_3 + Z_2 (i_1 - I_1) - Z_3 i_2 + Z_4 i_5, \\
 0 &= K_4 i_4 + Z_1 (i_1 + i_2) - Z_3 i_3 + Z_5 i_5 + Z_6 i_5 + Z_8 (i_5 + i_6), \\
 0 &= K_5 i_5 + Z_4 i_3 + Z_5 i_4 + Z_5 i_2 + Z_6 i_4 + Z_8 i_4 + Z_8 i_6 + Z_9 (i_6 - I_2) - Z_{R_7} i_8, \\
 0 &= K_6 i_6 + Z_8 (i_5 + i_4) + Z_9 (i_5 - I_2) - Z_{10} i_7, \\
 0 &= K_7 i_7 - Z_{10} i_6, \\
 0 &= K_8 i_8 - Z_{R_7} i_5 - Z_{R_5} i_9, \\
 0 &= K_9 i_9 - Z_{R_5} i_8 + Z_{20} (I_5 - i_{10}) - Z_{19} i_{12} - Z_{11} i_{16}, \\
 0 &= K_{10} i_{10} - Z_{22} i_{11} - Z_{21} i_{12} - Z_{20} (I_5 + i_9), \\
 0 &= K_{11} i_{11} - Z_{22} i_{10} - Z_{23} i_{12} - Z_{R_{10}} i_{13}, \\
 0 &= K_{12} i_{12} - Z_{19} i_9 - Z_{21} i_{10} - Z_{23} i_{11} - Z_{R_9} i_{13} + Z_{18} i_{20}, \\
 \dot{U} &= K_{13} i_{13} - Z_{R_{10}} i_{11} - Z_{R_9} i_{12}, \\
 \dot{U} &= Z_{L_1} i_{14} - Z_{L_1} i_{15}, \\
 0 &= K_{15} i_{15} - Z_{R_6} i_1 - Z_{12} i_{16} + Z_{17} i_{20} - Z_{L_1} i_{14}, \\
 0 &= K_{16} i_{16} - Z_{12} i_{15} - Z_{11} i_9 + Z_{13} (I_3 + I_4) + Z_{13} i_{17} - Z_{11} (I_3 + I_4) \\
 &\quad - Z_{12} (I_3 + I_4), \\
 0 &= K_{17} i_{17} + Z_{13} i_{16} - Z_{14} i_{18} + Z_{13} (I_3 + I_4) - Z_{14} (I_3 + I_4), \\
 0 &= K_{18} i_{18} - Z_{14} i_{17} - Z_{15} i_{19} + Z_{14} (I_3 + I_4) - Z_{15} (I_3 + I_4), \\
 0 &= K_{19} i_{19} - Z_{15} i_{18} - Z_{16} i_{20} + Z_{15} (I_3 + I_4) - Z_{16} (I_3 + I_4), \\
 0 &= K_{20} i_{20} - Z_{16} i_{19} + Z_{18} i_{12} + Z_{17} i_{15} + Z_{16} (I_3 + I_4) - Z_{17} (I_3 + I_4) \\
 &\quad - Z_{18} (I_3 + I_4).
 \end{aligned} \right. \quad (23)$$

Figure 20 demonstrates theoretical and experimental magnetic induction dependencies of the total resistance active and reactive components.

Now, knowing how the equivalent circuit elements depend on the magnetic field effect, we may proceed to determining a conversion function and a sensitivity equation.

The conversion function of the device was determined by Lyapunov's method for stability [34] considering the equivalent circuit. This function is dependence of the oscillation frequency on the magnetic field induction. Analytical dependence of the conversion function is

Fig. 20 Magnetic induction dependencies of active (a) and reactive (b) components of the total resistance for 5 V supply voltage



$$F_0 = \frac{1}{2\pi} \sqrt{\frac{A_1 + \sqrt{A_1^2 + 4L_1 C_{GD}(C_B(B)R_B(B))^2}}{2L_1 C_{GD}(R_B(B)C_B(B))^2}}, \quad (24)$$

where

$$A_1 = LC_{GD} - (C_B(B)R_B(B))^2 - C_{GD}C_B(B)R_B^2(B),$$

L_1 is the external inductance, C_B , R_B are the equivalent capacitance and resistance of the magnetosensitive transistor base, C_{GD} is the gate-drain capacitance of the field-effect transistor.

Figure 21 presents a graph of the convention function. Sensitivity of the magnetic frequency sensor was determined by expression (25) and can be described by the next equation

$$\begin{aligned} S_B^{F_0} = & -0.0198 \left(-2C_B(B)R_B^3(B)C_{GD} \left(\frac{\partial C_B(B)}{\partial B} \right) \sqrt{A_1 + 2A_2} \right. \\ & - 2C_B^2(B)R_B^3(B) \left(\frac{\partial C_B(B)}{\partial B} \right) - 2C_B^3(B) \times R_B^2(B) \left(\frac{\partial R_B(B)}{\partial B} \right) \\ & - 3C_B(B)R_B^3(B)C_{GD} \left(\frac{\partial C_B(B)}{\partial B} \right) - 2C_{GD}C_B^2(B)R_B^2(B) \left(\frac{\partial R_B(B)}{\partial B} \right) \\ & + 8C_B^2(B)L \times R_B^3(B)C_{GD} \left(\frac{\partial C_B(B)}{\partial B} \right) + 8LC_{GD}C_B^2(B)R_B^2(B) \left(\frac{\partial R_B(B)}{\partial B} \right) \\ & + 4LC_{GD}R_B(B) \left(\frac{\partial C_B(B)}{\partial B} \right) \sqrt{A_1^2 + 2A_2} + 4R_B(B) \left(\frac{\partial C_B(B)}{\partial B} \right) LC_{GD} \\ & + 4C_B(B)LC_{GD} \left(\frac{\partial R_B(B)}{\partial B} \right) \sqrt{A_1^2 + 2A_2} + 4LC_{GD}C_B(B) \\ & \times \left(\frac{\partial R_B(B)}{\partial B} \right) \left. \right) / \left(\left(2\sqrt{A_1 + \sqrt{A_1^2 + 2A_2}} / A_2 \right) \right. \\ & \left. \times LC_{GD}C_B^3(B)R_B^3(B)\sqrt{A_1^2 + 2A_2} \right), \end{aligned} \quad (25)$$

where

$$A_2 = 2LC_{GD}(C_B(B)R_B(B))^2.$$

Figure 22 presents a sensitivity graph, where the greatest sensitivity of the device being 2.1–1.1 kHz/mT is in the range from 0 to 50 mT.

Theoretical and experimental investigations yielded that sensitivity of the transducer was 0.75–2.1 kHz/mT.

Fig. 21 Dependence of the oscillation frequency on the magnetic field induction of a magnetic field converter based on a double-collector magnetosensitive bipolar transistor, a double-gate field-effect transistor, and a bipolar transistor

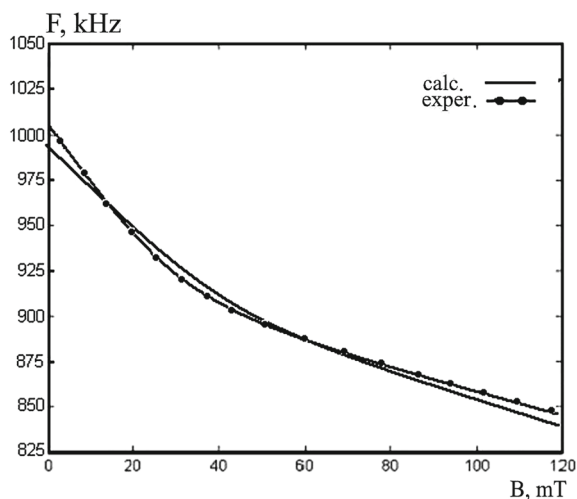
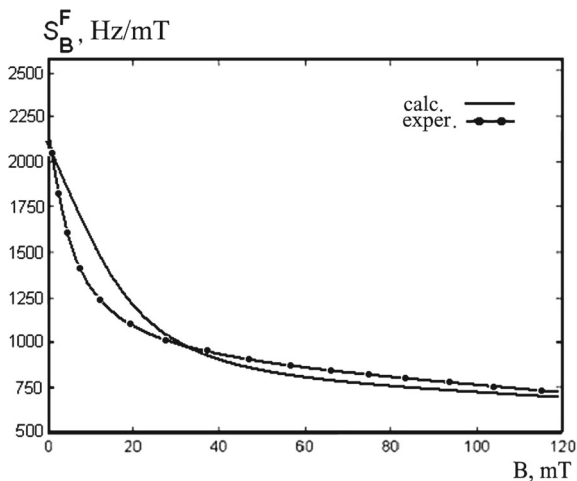


Fig. 22 Dependence of the sensitivity of a magnetic field transducer based on a dual-collector magnetically sensitive bipolar transistor, a dual-gate field-effect transistor, and a bipolar transistor on the magnetic field induction



6 Conclusion

This paper discusses innovative, frequency-sensitive magnetic sensors based on transistor structures. In particular, it focuses on a dual-collector magnetic-sensitive bipolar transistor. These sensors convert magnetic induction into frequency, simplifying their use in digital biomedical systems. The proposed models are characterised by increased sensitivity in the range of weak magnetic fields that are typical of biological processes. This research is relevant due to the need for new solutions to create affordable sensor components for medical monitoring, functional diagnostics, and neurophysiology systems.

The authors obtained the following main result in the article:

1. Mathematical models of microelectronic magnetic field frequency transducers based on a double-collector magnetosensitive bipolar transistor have been developed. They differ from the existing models by allowing for the magnetic field effect on the elements of the transducer nonlinear equivalent circuits. This provides obtaining sensitivity functions and an equation for converting the magnetic field induction into frequency.
2. Analytical expressions were obtained as a result of mathematical modeling, they can be employed for engineering calculation of the conversion function and sensitivity equation of microelectronic magnetic field frequency transducers thus ensuring selection of an optimal mode of their electric power supply.
3. The performed experimental studies proved that the highest sensitivity of the microelectronic magnetic field transducer on a double-collector magnetosensitive bipolar transistor and a double-gate field-effect transistor lay in the range from 0 to 60 mT and was 200–60 Hz/mT, the highest sensitivity microelectronic magnetic field transducer on a double-collector magnetosensitive bipolar transistor with active inductive element lay in the range from 0 to 40 mT and was 350–550 Hz/mT, the highest sensitivity microelectronic magnetic field transducer on a double-collector magnetosensitive bipolar transistor, a double-gate field-effect and bipolar transistor lay in the range from 0 to 50 mT and was 2.1–1.1 kHz/mT, the highest sensitivity microelectronic magnetic field transducer on a double-collector magnetosensitive bipolar transistor, double-gate field-effect and bipolar transistor with active inductive element lay in the range of 0 up to 60 mT and was 7.2–6.4 kHz/mT. The discrepancy between theoretical and experimental results is $\pm 5\%$.

References

1. Wang Z, Wang X, Li M, Gao Y, Hu Z, Nan T, Liang X, Chen H, Yang J, Cash S et al (2016) Highly sensitive flexible magnetic sensor based on anisotropic magnetoresistance effect. *Adv Mater* 28:9370–9377. <https://doi.org/10.1002/adma.201602910>
2. Pascal J, Hebrard L, Frick V, Blonde JP (2008) Three-dimensional Hall probe integrated in 0.35 μm CMOS technology for magnetic field pulses measurements. In: Proceedings of the 2008 joint 6th international IEEE northeast workshop on circuits and systems and TAISA conference (NEWCAS-TAISA 2008), Montreal, QC, Canada, 22–25 June 2008, pp 97–100. <https://doi.org/10.1109/NEWCAS.2008.4606330>
3. Elsayed MY, Cicek P, Nabki F, El-Gamal MN (2015) Surface micromachined combined magnetometer/accelerometer for above-IC integration. *J Microelectromech Syst* 24:1029–1037. <https://doi.org/10.1109/JMEMS.2014.2375574>
4. Long L, Wang M, Zhong S (2016) A torsion MEMS magnetic sensor with permanent magnet and fiber-optic detection. *J IEEE Sens J* 16:8426–8433. <https://doi.org/10.1109/JSEN.2016.2581922>
5. Cubells-Beltran MD, Reig C, Madrenas J, De Marcellis A, Santos S, Freitas PP (2016) Integration of GMR sensors with different technologies. *Sensors* 16:939. <https://doi.org/10.3390/s16060939>

6. Weiss R, Mattheis R, Reiss G (2013) Advanced giant magnetoresistance technology for measurement applications. *Meas Sci Technol* 24(8):082001. <https://doi.org/10.1088/0957-0233/24/8/082001>
7. Schell JB, Kammerer JB, Hébrard L, Breton E, Gounot D, Guvillon L, de Mathelin M (2012) CMOS 3D Hall probe for magnetic field measurement in MRI scanner. In: *Proceedings of the 2012 IEEE 10th international new circuits and systems conference (NEWCAS)*, Montreal, QC, Canada, 17–20 June 2012, pp 517–520. <https://doi.org/10.1109/NEWCAS.2012.6329070>
8. Osberger L, Frick V, Hébrard L (2016) High resolution shallow vertical hall sensor operated with four-phase bi-current spinning current. *Sens Actuators A Phys* 244:270–276. <https://doi.org/10.1016/j.sna.2016.04.036>
9. Sun J, Kosel J (2012) Finite-element modelling and analysis of hall effect and extraordinary magnetoresistance effect. *Mech Eng* 119. <https://doi.org/10.5772/47777>
10. Uchiyama T, Mohri K, Honkura Y, Panina LV (2012) Recent advances of pico-Tesla resolution magneto-impedance sensor based on amorphous wire CMOS IC MI sensor. *J IEEE Trans Magn* 48:3833–3839. <https://doi.org/10.1109/TMAG.2012.2198627>
11. Tikhonov RD (2008) Dual-collector lateral bipolar magnetotransistor: negative sensitivity and galvanomagnetic effects. *Open Electr Electron Eng J* (2):14–26. <https://www.benthamopen.com/contents/pdf/TOEEJ/TOEEJ-2-14.pdf>
12. Ramsden E (2011) Hall-effect sensor: theory and application. Elsevier, Philadelphia, PA, USA
13. Popovic RS (2003) Hall effect devices. CRC Press, Boca Raton, FL, USA
14. Hotra OZ, Samila AP, Rozorinov HM, Hres OV (2019) Current status and development prospects of nuclear quadrupole resonance pulsed spectroscopy methods: a review. *Telecom Rad Eng* 78(16):1483–1496. <https://doi.org/10.1615/TelecomRadEng.v78.i16.60>
15. Samila A (2015) Simulation of magnetic field topology in a saddle-shaped coil of nuclear quadrupole resonance spectrometer. *PIER Lett* 56:67–73. <https://doi.org/10.2528/PIERL15072306>
16. Chen YT, Kolhatkar AG, Zenasni O, Xu SJ, Lee TR (2017) Biosensing using magnetic particle detection techniques. *Sensors* 17:2300. <https://doi.org/10.3390/s17102300>
17. Hotra O, Samila A, Politsky L (2018) Synthesis of the configuration structure of digital receiver of NQR radiospectrometer. *Przeglad Elektrotechniczny* 94(7):58–61. <https://doi.org/10.15199/48.2018.07.14>
18. Samila A (2017) Peculiarities of using s-simulation for parametric identification of multiplet 115 In NQR spectra in InSe. *Measurement* 106:109–115. <https://doi.org/10.1016/j.measurement.2017.04.035>
19. Rudyk AV, Semenov AO, Kryvinska N et al (2020) Strapdown inertial navigation systems for positioning mobile robots—MEMS gyroscopes random errors analysis using Allan variance method. *Sensors* 20:4841. <https://doi.org/10.3390/s20174841>
20. Kucheruk V, Kulakov P, Storozhuk N (2016) Measurement of the number servings of milk and control of water content in milk on stall milking machines. *Advances in intelligent systems and computing*, pp 435–447. https://doi.org/10.1007/978-3-319-48923-0_46
21. Taranchuk A, Pidchenko S (2021) Quartz pulse wave sensor with a capacitive control for healthcare solutions. *IEEE Sensors J* 21(6):8613–8620. <https://doi.org/10.1109/JSEN.2020.3049065>
22. Yanenko A, Totsky A, Pidchenko SK, Taranchuk AA (2018) Experimental study of microwave radiation caused by the materials contacting with human body. *Telecom Rad Eng* 77(7):635–644. <https://doi.org/10.1615/TelecomRadEng.v77.i7.60>
23. Semenov A, Baraban S, Semenova O et al (2019) Statistical express control of the peak values of the differential-thermal analysis of solid materials. *SSP* 291:28–41. <https://doi.org/10.4028/www.scientific.net/SSP.291.28>
24. Semenov AO, Baraban SV, Osadchuk OV et al (2019) Microelectronic pyroelectric measuring transducers. *IFMBE Proc* 393–397. https://doi.org/10.1007/978-3-030-31866-6_72
25. Semenov AO, Savitskiy AY, Bisikalo OV, Kulakov PI (2018) Mathematical modeling of the two-stage chaotic colpitis oscillator. In: *2018 14th international conference on advanced trends in radioelectronics, telecommunications and computer engineering (TCSET)*, pp 835–839. <https://doi.org/10.1109/tcset.2018.8336327>

26. Riccobene C, Wachutka G, Burgler J, Baltes H (1994) Operating principle of dual collector magnetotransistors studied by two-dimensional simulation. *IEEE Trans Electron Devices* 41:32–43. <https://doi.org/10.1109/16.259617>
27. Castagnetti R, Leme CA, Baltes H (1993) Dual collector magnetotransistors with on-chip bias and signal conditioning circuitry. *Sens Actuators A* 37–38:698–702. [https://doi.org/10.1016/0924-4247\(93\)80118-Z](https://doi.org/10.1016/0924-4247(93)80118-Z)
28. Riccobene C, Wachutka G, Baltes H, Biirgler J (1991) Operation of vertical and lateral dual collector magnetotransistors studied by exact 2D-simulation. *Simul Semicond Devices Processes* 4:349–358
29. Tikhonov RD (2009) Dual-collector bipolar magnetotransistor: defining and determining its magnetic-field sensitivity. *Russ Microelectron* 38:239–250. <https://doi.org/10.1134/S1063739709040027>
30. Glauberma MA, Kozel VV, Nakhabin AV (2000) Charge-carrier transport in a double-collector magnetotransistor. *Semiconductors* 34:603–605. <https://doi.org/10.1134/1.1188036>
31. Osadchuk O, Koval K, Prytula M, Semenov A (2016) Comparative analysis of radiomeasuring frequency converters of the magnetic field. In: 2016 13th international conference on modern problems of radio engineering, telecommunications and computer science (TCSET). <https://doi.org/10.1109/TCSET.2016.7452034>
32. Vikulin IM, Vikulina LF, Stafeev VI (2001) Magnetotransistors. *Semiconductors* 35:1–9. <https://doi.org/10.1134/1.1340281>
33. Osadchuk A, Osadchuk I, Martyniuk V et al (2022) Mathematical simulation of a microelectronic transducer with frequency output for measuring the induction of the magnetic field. *IAPGOS* 12:43–49. <https://doi.org/10.35784/iapgos.2921>
34. Schubert TF Jr, Kim EM (2015) *Fundamentals of electronics*. Springer. <https://doi.org/10.1007/978-3-031-79873-3>
35. Osadchuk VS, Osadchuk AV (2011) The microelectronic radiomeasuring transducers of magnetic field with a frequency output. *Elektronika ir Elektrotechnika* 4:67–70
36. Osadchuk VS, Osadchuk AV (2011) The magneticreactive effect in transistors for construction transducers of magnetic field. *Elektronika ir Elektrotechnika* 3:119–122
37. Semenov A, Baraban S, Kovtun V, Baraban M, Arseniuk I, Rudyk A (2024) Development and validation of a mathematical model for pyroelectric temperature measurement sensors for application in mobile robotic systems. *Electronics* 13(16):3173. <https://doi.org/10.3390/electronics13163173>
38. Semenov A, Semenova O, Osadchuk O (2015) The UHF oscillators based on a HEMT structure with negative conductivity. In: 2015 international Siberian conference on control and communications (SIBCON). IEEE, pp 1–4. <https://doi.org/10.1109/sibcon.2015.7147215>
39. Semenov A, Voznyak O, Osadchuk O, Baraban S, Semenova O, Rudyk A, Klimek J, Orazaliyeva S (2019) Development of a non-standard system of microwave quadripoles parameters. In: Romaniuk RS, Linczuk M (eds) *Photonics applications in astronomy, communications, industry, and high-energy physics experiments 2019*. Photonics applications in astronomy, communications, industry, and high-energy physics experiments 2019. SPIE, p 80. <https://doi.org/10.1117/12.2536704>
40. Semenov A (2016) Reviewing the mathematical models and electrical circuits of deterministic chaos transistor oscillators. In: 2016 international Siberian conference on control and communications (SIBCON). IEEE, pp 1–6. <https://doi.org/10.1109/sibcon.2016.7491758>

Formation of the Concept of Investment Decision-Making in the European Tourism Sector Using Economic Indicators



Tetiana But , Daria Mamotenko , Olga Korniienko ,
Galyna Kozytska , and Nataliia Pavlenchuk

Abstract Europe remains a leading region in the global tourism market due to its economic influence, innovation, and environmental sustainability. The study aims to create an effective investment decision-making system in the European tourism sector by using key economic indicators to analyse market conditions, identify priority investment areas, and improve the effectiveness of management actions in a changing economic environment. The research methodology included correlation, statistical, graphical, abstract logical, systemic, and comprehensive analyses. The key economic indicators of the European market were selected for the study's analysis. The result is an implicational map of the industry cycle life stages based on the integration of indicators. This allowed a comprehensive assessment of the current state and potential of economic development in Europe, considering the regional division into Western, Central, and Eastern Europe. This division formed the decision-making system of flexible investment management. The originality of the study lies in the development of a concept of investment decision-making that takes into account such pan-European priorities as green transformation, digitalization, inclusive growth, and regional tourism features and allows integrating methods of modern economic analysis and ensures investment decision-making based on expected benefits and dynamics of system parameters. In general, the conclusions

T. But (✉)

Faculty of Entrepreneurship and Law, Department of Tourism, Pan-European University, Praha, Czech Republic

e-mail: tatyanabut1979@ukr.net

D. Mamotenko · O. Korniienko

Faculty of Humanities, Department of Tourism, Hotel and Restaurant Business, National University Zaporizhzhia Polytechnic, Zaporizhzhia, Ukraine

G. Kozytska

Faculty of Business Technologies and Economics, Department of Economics and Customs, National University Zaporizhzhia Polytechnic, Zaporizhzhia, Ukraine

N. Pavlenchuk

Faculty of Tourism, Department of Economics and Management, Ivan Boberskyi Lviv State University of Physical Culture, Lviv, Ukraine

drawn in this article are novel and have important policy implications for promoting the sustainable development of the European tourism sector.

Keywords Investment decision-making · Economic indicators · Tourism sector · Life stages · Europe

1 Introduction

Thanks to its rich cultural heritage, economic strength, innovation, and environmental sustainability, Europe remains one of the leading regions in the global tourism market. However, Europe must adapt to new challenges and implement modern development strategies to stay competitive.

The study aims to create an effective investment decision-making system in the European tourism sector by using key economic indicators to analyse market conditions, identify priority investment areas, and improve the effectiveness of management actions in a changing economic environment.

For several reasons, researching priority areas for investment in the European tourism sector is extremely important. At the end of 2024, the European tourism sector showed significant growth. The number of foreign tourists increased by 6.3% in 2019 and by 6.7% in 2023. The number of overnight stays also increased by 5.9% compared to 2019 and by 4.8% year-on-year. Despite economic difficulties and extreme weather conditions, demand for travel services remained strong. Inflation forced tourists to look for destinations with the best value for money. Tourism spending grew by 7.8%, reaching €705 billion, with Western Europe accounting for almost 75% of this amount [1].

The analysis of investment activity in sustainable tourism in Europe includes studying the dynamics of investment flows in the tourism sector, sources of financing for sustainable tourism projects, and the principal risks and barriers to investment. Understanding how investment affects tourism development and what factors facilitate or hinder effective investment is essential. Sustainable tourism plays a key role in ensuring the long-term sustainability of the European tourism sector [2, 3]. Investment and innovation strategies in tourism take into account the European economy's life cycle, which affects tourism's development through changing economic conditions [4], technological innovations, and environmental requirements.

Developing investment strategies [5–8] for sustainable tourism requires a close partnership between the state, business, and international organizations. Joint efforts in implementing innovative solutions, financing sustainable projects, and supporting environmental initiatives [3, 9, 10] will contribute to the growth and development of the European tourism sector, ensuring its competitiveness and sustainability in the long term.

2 Literature Review

The study analyses global economic trends and their impact on the European tourism market, taking into account investment policy. The impact of investment on overcoming uncertainty is an important issue, which indicates that in conditions of instability, it is more important to focus on concrete actions than on emotional reactions [11]. To do this, it is necessary to support entrepreneurs in turbulent times by developing strategies for investing in the European tourism sector, taking into account the importance of active problem-solving, effective teamwork, and adaptation to uncertainty.

Investment decisions in the tourism sector can be seen as an example of sustainable development based on incremental eco-labelling of hotels, which is more effective than categorical labelling in increasing the sensitivity of tourists to environmental value [12]. This approach indicates that investing in ecolabelling can meet regulatory requirements and initiatives to reduce environmental impacts, thus contributing to the growth and development of the tourism sector while ensuring sustainability and competitiveness.

Another approach points out that investment leads to poverty reduction for the population, improved health and well-being, affordable and clean energy, decent work and economic growth, reduced inequality, equity, and partnerships for achieving goals [2, 13]. All of this shows that unsustainable practices, on the contrary, exacerbate hunger and reduce biodiversity. Therefore, investing in quality education, gender equality, clean water, and sanitation improves sustainable development by favouring sustainable tourism.

Identifying suitable areas for different types of ecotourism can make these destinations more attractive to tourists [3]. As we can see, the emphasis on the balanced preservation of natural and cultural landscape values will contribute to the long-term sustainability of tourist destinations, which is important for investors seeking to promote sustainable development and growth of ecotourism in Europe.

Today, there is a heterogeneity in the digitization of rural areas, which has a negative impact on tourism development at the county level. The moderating effect of tourism's contribution to GDP and GDP per capita also mitigates this impact [4]. However, this approach emphasizes the importance of a geographical perspective in understanding the complex relationship between digital technologies and tourism development and requires recommendations for tourism policymakers on digitalization not only in rural areas but also in European cities.

The growth of outbound tourism in emerging markets contributes to an increase in foreign direct investment (FDI) by tourism firms [5]. Understanding that state-owned enterprises can dampen the effect of outbound tourism on FDI will help investors make more informed decisions about investing in private or state-owned companies. Therefore, strategic planning of investments is necessary, taking into account different institutional logics and their impact on FDI.

European studies [6, 14–16] should also be taken into account, which confirm that investment in tourism, in particular capital investment and the direct contribution of tourism to employment, effectively reduces income inequality and poverty, which underlines the importance of tourism as a tool for socio-economic development. Increased international tourist arrivals can boost tourism revenues, making investment in the sector more attractive.

The author of [7] found that in the short term, tourism expenditure is positively related to economic growth, while foreign direct investment is negatively correlated. This approach highlights the importance of considering both long-term and short-term effects when planning investments in the tourism sector.

How investments in infrastructure have had a positive impact on the development of international tourism by increasing the number of inbound tourist's [8] points to the key role of transport accessibility and the level of information technology in this process. Therefore, investing in transport accessibility and information technology can significantly increase the attractiveness of tourist destinations, increasing the number of international tourists and tourism revenues.

Another approach indicates that green technological innovations and tourism contribute to reducing environmental damage by reducing CO₂ emissions [17]. Despite the conflicting empirical evidence on the impact of technological innovation on carbon emissions, the study confirms the positive effect of green innovation and foreign investment on environmental quality.

Another approach identifies the impact of tourism's direct contribution to employment, income equality, and poverty reduction, which is explained by investments in the sector's enablers, which are the main catalysts for job creation [13]. In particular, increased investment in the tourism sector accelerates the growth of high-profit expectations and fuels greater tourism demand, which is stimulated by higher incomes.

It should be noted that investors are attracted to such innovative processes as the implementation of the Smart City concept, as they contribute to the comfort and safety of residents, making cities more attractive for living and working. In this way, investors demonstrate their readiness for investment decisions and development and seek to support sustainable development and social responsibility [16].

Thus, the analysis of global economic trends and their impact on the European tourism market highlights the importance of actively overcoming uncertainty and adapting to change. Supporting entrepreneurs in turbulent times and introducing green technologies will contribute to the sustainable development and competitiveness of the tourism sector. The key aspects for attracting investment are the importance of socio-economic development through investment in tourism, reducing income inequality and poverty, as well as improving infrastructure and digitalization.

3 Methodological Research

In the course of the study, the following methods of economic research were used: correlation analysis (to form an implicit map of the stages of the market life cycle), statistical (to analyse economic indicators according to world reports), graphical and tabular methods (to visually display the materials of statistical research); abstract and logical (to make theoretical generalizations and formulate conclusions); visualization of theoretical and practical provisions; systematic and comprehensive analysis (to determine trends in the development of the European.

The research was supported by the documentation of information and official reports of the European Union, European Commission, WEF, IMD, UNCTAD, Smart Specialization Strategy, World Smart Cities Outlook 2024, and European Innovation Report.

The creation of an effective system of investment decision-making in the European tourism sector was carried out according to the following stages: determination of the strategic positioning of Europe in the global tourism market; analysis of the market conditions in Europe as a basis for investment decisions in the tourism sector; study of investment policy in the European tourism sector in modern conditions.

The period of economic indicators for determining the number of international tourist arrivals worldwide was selected from 2014–2024; the data on the dynamics of the EU Innovation Index was selected for the period from 2017–2024; data for 2022–2023 were used to determine foreign direct investment in most regions of the world by economic groups; the period from 2009–2022 was chosen to determine the stages of the European market life cycle. To build an implication map of the life cycle stages of the European market, the following economic indicators were selected for the following periods: the overall rating of the Travel and Tourism Development Index 2024, the average rating positions in 2024 at the level of competitiveness factors by economic indicators, production efficiency, business efficiency, infrastructure; GDP 2017–2023; GDP per capita growth in 2009–2022, forecast of the number of 5G mobile subscribers 2019–2028; global CO₂ emission intensity in kilograms per US dollar for the period from 2000–2022; productivity per employee, 1990–2023; increase in productivity per hour for the period 2020–2023. salaries in selected countries of the world, 2013–2021; average regional ranking positions in overall competitiveness, 2020–2024; foreign direct investment, 2023/2022; participation of various stakeholders in the development of the «SMART CITY» Development Concept for 2024.

4 Research Results

4.1 *Strategic Positioning of Europe in the Global Tourism Market*

Europe remains an important player in the global tourism market, but its position requires constant improvement and adaptation to current trends. Innovations, affordability, infrastructure development and consideration of social changes are key aspects of maintaining the region's competitiveness.

The global economic growth is boosting tourist arrivals. However, the affordability of travel and price levels in Europe affect the region's competitiveness. High accommodation and transport costs can deter tourists, especially from lower-income countries.

The Asia-Pacific region and Africa are showing high growth rates in the tourism industry. These regions attract tourists with their unique nature, pleasant climate, and cultural attractions. However, Europe is facing the challenge of a conventional «aging» of the tourism product, which requires innovative approaches to maintain competitiveness.

In 2024, international tourism showed a significant increase in revenues in many countries, including Albania, Andorra, Malta, Serbia, and Moldova, indicating that the industry is recovering from the pandemic. These countries also saw double-digit revenue growth in 2024 compared to 2023.

European tourism has undergone a change in the structure of popularity in 2024, with a decrease in the number of tourists to temperate countries such as Denmark (+ 38%), Norway (+ 18%) and Sweden (+ 9%) [18].

Tourism spending in 2024 increased in the UK, Germany, Australia, Italy and France.

Tourism revenues in Europe recorded 747 million international arrivals in 2024 (+ 1% compared to 2019 and 5% compared to 2023), which contributed to the growth of gross domestic product (GDP) due to tourist spending on accommodation, food, entertainment and other services.

In total, international tourism revenues in Europe reached \$1.6 trillion, which is about 3% more than in 2023 and 4% more than in 2019 [19]. This is due to strong support for intra-regional demand.

Total tourism exports reached a record high of \$1.9 trillion, which is about 3% more than before the pandemic (in real terms) [19]. This indicates that the industry is recovering from the COVID-19 pandemic, which has had a significant negative impact on the tourism industry and new opportunities for the development of domestic tourism and ecotourism.

All sub-regions have exceeded pre-pandemic levels, except Central and Eastern Europe, which is still suffering from the lingering effects of Russia's aggression in Ukraine. Europe remains the world's largest tourist destination [20].

The dynamics of international tourist arrivals worldwide from 2014 to 2024 by region is shown in Fig. 1. In 2025, the number of international tourist arrivals is

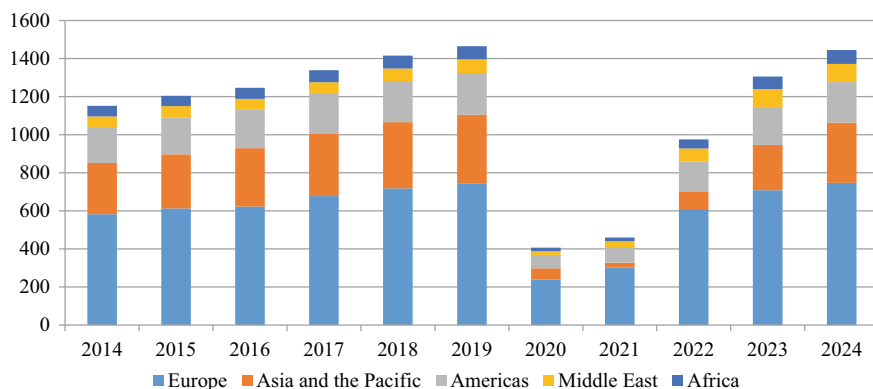


Fig. 1 Number of international tourist arrivals worldwide from 2014 to 2024 by region (in millions) [21]

expected to increase by 3–5% compared to 2024, provided that the economic recovery in Asia Pacific continues and growth in most other regions is stable.

According to Fig. 1, the highest growth rates in Europe in 2024 are observed in Albania (+80%), Andorra (+35%), Malta, and Serbia (both + 29%), where the number of arrivals significantly exceeded the figures for 2019.

As we can see, the development of international tourism in Europe stimulates investment in infrastructure, including the construction of new hotels, airports, roads, and other facilities. This helps to improve the overall infrastructure of the country. The development of international tourism in Europe also improves the quality of life of residents by creating new opportunities for recreation and entertainment; raises environmental awareness among tourists and locals alike, contributing to the preservation of natural resources and ecosystems; promotes cultural exchange that can lead to the enrichment of local culture and support the integration of local cultures into tourism products; and contributes to the preservation of historical and cultural monuments, as it can be used for their maintenance and restoration.

Tourism development in Europe accounts for more than 10% to the European economy, and according to WTTC forecasts, the sector's contribution to Europe's GDP is expected to grow to \$3 trillion by 2034, accounting for about 10% of the region's total GDP [21].

An analysis of the TTDI results for 2024 (Table 1) shows that Europe remains the leader in the global tourism market, with the highest scores in most components, including the share of international tourist arrivals and the contribution of the tourism industry to GDP. Asia-Pacific is in second place, showing significant development, while Sub-Saharan Africa has shown the greatest improvement since 2019.

These data show Europe's stable position and prospects for other regions in the global tourism environment.

Innovations improve the quality of tourism services, reduce costs, attract new segments of tourists, adapt to changes in the external environment, and maintain

Table 1 TTDI scores in 2024 [20, 22, 23]

Regions		Share of international tourist arrivals, %	Share of T&T industry GDP, %	Share of T&T industry employment, %
The Americas	3,86	16,9	38,9	0,5
Europe and Eurasia	4,26	64,9	32,6	0,4
Middle East and North Africa	3,86	9,5	6,7	0,9
Sub-Saharan Africa	3,33	1,8	1,3	2,1
Asia-Pacific	4,11	6,9	20,6	0,5

Note Arrivals, industry GDP and employment data are 2022 figures (or the latest available) and represent TTDI-ranked economies only

cultural heritage, contributing to the overall development and attractiveness of the tourism industry. That is why Fig. 2 shows an analysis of innovation activity in the EU countries over the period 2017–2024.

The results of Fig. 2 indicate a steady increase in innovation in the EU over the period 2017–2024. In particular, the innovation index increased from around 0.49 in 2017 to around 0.55 in 2024. This indicates a steady improvement in the innovation potential of EU countries, which may be the result of effective policies aimed at supporting scientific and technological development, investment in research and development, and the introduction of new technologies. In general, these dynamics indicate positive trends in the development of the EU’s innovation environment, which contributes to the region’s competitiveness in the global market.

Foreign direct investment (FDI) inflows by economic group and region are shown in Fig. 3.

As can be seen, the dynamics of foreign investment in 2023/2022 indicates that Europe faced a significant decline in FDI in 2023, which may affect its positioning

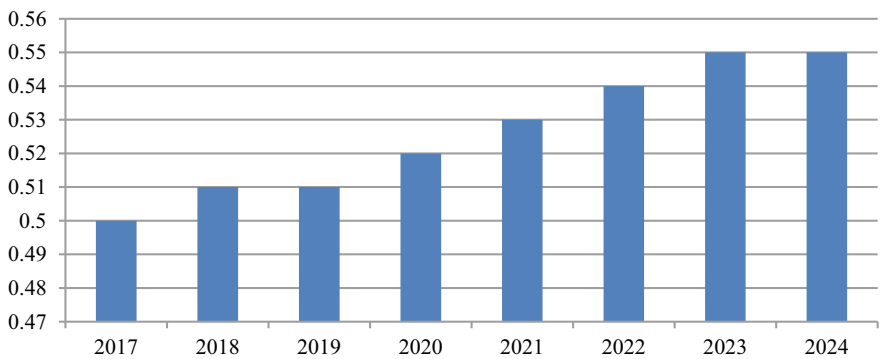


Fig. 2 Dynamics of the EU innovation index 2017–2024 [24]

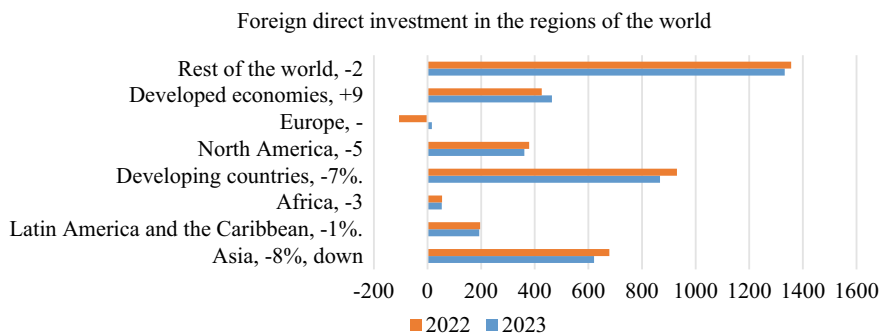


Fig. 3 Foreign direct investment in most regions of the world by economic group in 2023/2022 [25]

in the global tourism market. In particular, investments in Europe decreased by 44%, which may limit opportunities for infrastructure development and innovative projects necessary to maintain the region's competitiveness.

The decline in foreign direct investment may lead to insufficient funding for infrastructure projects and innovative initiatives, and Asia and Latin America are showing significant investment growth, which could increase competition in the global tourism market for Europe.

Development of domestic tourism:

However, a decline in foreign investment could stimulate the development of domestic tourism and ecotourism in Europe, which would help preserve natural resources and cultural heritage.

And, despite the decline in investment, Europe can focus on introducing innovative technologies and business models to improve the quality of tourism services and attract new segments of tourists.

The European region has significant potential for growth by investing in quality infrastructure, improving tourism services, and using technology to remain competitive and position itself as the most attractive destination for international tourists.

Strategic planning, consideration of regional specificities, and the introduction of innovative approaches will help ensure the long-term sustainability and growth of Europe's tourism sector, through tourism investments that reduce income inequality and poverty, contributing to the region's socio-economic development.

4.2 Analysis of the European Market Conditions as a Basis for Investment Decisions in the Tourism Sector

In modern economic science, the life cycle approach to analysing market dynamics is becoming increasingly relevant, allowing to identify patterns of industry development, assess investment potential and develop adaptive strategies for economic growth. The European market, as one of the most developed in the world, demonstrates a complex multi-level structure, where regional differences play a key role in shaping the overall market environment.

The tourism sector is an extremely dynamic, multi-sectoral system that is both an indicator and a catalyst for changes in the socio-economic environment. Its development is closely linked to general macroeconomic trends, the level of infrastructure provision, digital transformation, sustainable resource management practices, as well as the level of political stability and institutional capacity of countries and regions. That is why life cycle analysis in the European space through the prism of the tourism industry is an important tool for formulating an effective investment policy.

The method combines the classical theory of the market life cycle [26], adapted to the regional conditions of Europe. Each stage—emergence, growth, maturity, and decline—reflects a certain logic of market development that correlates with the dynamics of investment, productivity, infrastructure, institutional capacity, and market behaviour.

The implication mapping method of the market life cycle is a systematic framework for identifying industry development patterns regarding geographical, institutional, and infrastructural changes. It allows us to integrate statistical indicators such as GDP dynamics, labour productivity, investment intensity, infrastructure provision, digital transformation, and competitiveness indices with the life cycle stages (from initial development to decline), providing a qualitatively new level of management and investment decision-making.

To build an implicit map of the market life cycle, we used a comprehensive table (Table 2) that combines indicators from official reports [14, 22, 25, 27, 28], covering four key stages: initial, growth, stability, and decline. Each stage captures the level of institutional maturity, digital readiness, economic efficiency, and investment attractiveness.

The implication map of the market life cycle stages, built based on the integration of indicators, allows for a comprehensive assessment of Europe's current state and potential for economic development. The analysis is based on the regional division into Western, Central, and Eastern Europe, ensuring reliable verification of spatial differences.

Identification of life cycle phases by indicator groups:

At the initial stage (Emergence), indicators such as low GDP per capita in 2009–2022 (particularly in Central and Eastern Europe), initial 5G adoption, high CO₂ emissions per unit of GDP, low labour productivity, and relatively low innovation activity are

Table 2 Implication map of European market life cycle stages by key economic indicators [14, 22, 25, 27, 28]

Indicator	Data	The initial stage (emergence)	The growth stage	The stage of stability (maturity)	The stage of decline
Overall rating of the travel and tourism development index 2024	Europe and Eurasia			+	
Economic indicators	Western Europe			+	
	Eastern Europe		+		
Production efficiency	Western Europe			+	
	Eastern Europe		+		
Business efficiency	Western Europe			+	
	Eastern Europe		+		
Infrastructure	Western Europe			+	
	Eastern Europe		+		
GDP 2017–2023	Euro Area		+		
GDP growth per capita, 2009–2022	European Union	+			
	Central Europe and the Baltics		+		
Forecast of the number of 5G mobile subscribers 2019–2028	Western Europe			+	
	Central and Eastern Europe		+		
Global CO ₂ intensity in kilograms per US dollar 2000–2022	Europe	+			
Productivity per employee, 1990–2023 (2000 = 100)	Europe	+			
Increase in productivity per hour 2020–2023	Europe	+			
Salaries in selected countries of the world 2013–2021	(Germany, Switzerland)			+	
Average ranking positions by region in overall competitiveness 2020–2024	Western Europe			+	

(continued)

Table 2 (continued)

Indicator		Data	The initial stage (emergence)	The growth stage	The stage of stability (maturity)	The stage of decline
		Eastern Europe		+		
Foreign direct investment 2023/2022		Europe	+			
Participation of various stakeholders in the development of the SMART CITY development concept	Universities	Europe			+	
	local businesses				+	
	Private enterprises			+		

recorded. The data shows potential but realising investment opportunities have not reached critical mass. This is typical for countries with transformational economies.

The Growth stage is characterised by dynamic economic development, gradual competitiveness improvement, digital infrastructure intensification, labour productivity growth, and foreign investment. This phase is typical for Central Europe and the Baltics, where institutional reforms, digitalisation, and increased participation in trans-European networks can lead to a gradual transition to an innovative economy.

The Maturity stage is characteristic of Western European countries that demonstrate high but stabilised performance in all key indicators, including infrastructure, wages, competitiveness index, and digital coverage. There is an institutionally established environment for innovation here that requires a strategic shift in investment focus—from expansion to optimisation and efficiency.

The Decline stage covers mainly those countries or industry segments where economic and innovation growth is slowing or showing negative dynamics. This may be due to structural problems, demographic pressure, exhaustion of the resource model, or lack of new sources of growth. For example, some Eastern European countries are experiencing a decline in foreign direct investment, lack of participation in SMART initiatives, or lagging in digital convergence.

An important indicator of the maturity of the economic environment is the participation of various stakeholders in shaping conceptual development models, such as Smart City. The data shows that universities and large private companies are already well integrated into investment planning processes, while local businesses are involved selectively. This points to the need to intensify investment policies to support small and medium-sized enterprises.

The developed implication map of the European market life cycle by key economic indicators shows that Western Europe is at the stage of maturity, requiring strategies to support technological upgrades, inclusion and impact-oriented growth. Central Europe and the Baltics are at the growth stage, with the potential to become leaders if supported by digital transformation and ESG factors. Eastern Europe is mostly

between the initial and transitional stages, where institutional modernization and pilot innovation programs play an important role.

The results confirm the need to differentiate investment policy: countries at the growth stage need tools to accelerate digital and green technologies; countries at the maturity stage need to rethink strategies through the prism of sustainable tourism and impact investment; while countries in the initial phase, the key is to create a favourable institutional environment and infrastructure foundation.

Thus, the European tourism industry is not homogeneous but is being formed as a complex multi-level system, where the market life cycle requires constant monitoring, adaptation of strategies and flexible investment management. The presented map allows us to identify development vectors, laying the foundation for a new paradigm of investment policy in Europe.

4.3 Investment Policy in the European Tourism Sector in the Current Context

The current investment policy in the European tourism sector is characterized by transformational dynamics, driven by a combination of external challenges and internal structural changes. In the context of recovery from the pandemic, high inflation, rising energy costs, and environmental sustainability requirements, investment policy is undergoing a strategic rethink at both the government and corporate levels.

First, investment policy in the European tourism sector is focused on sustainable development, environmental transformation, and digitalization. There is a reorientation of financial flows to projects that meet the principles of green tourism: eco-hotels, energy-efficient infrastructure, biodiversity conservation, and sustainable transport development. For example, the InvestEU program, which runs from 2021 to 2027, aims to increase the competitiveness and sustainability of the tourism sector through innovative measures that reduce its environmental footprint. The InvestEU program aims to attract €372 billion of investment to the EU over the 2021–2027 period. It encourages investments to improve the competitiveness, sustainability, and value chains of the tourism sector. The portal provides a wealth of information, including tourism funding criteria, program descriptions, and examples of supported projects. The main goal of InvestEU is to support measures that are sustainable, innovative, and digital, which have the potential to reduce the sector's impact on climate and the environment [29]. InvestEU: Enhancing tourism's competitiveness and sustainability [30].

Under the umbrella of Smart Tourism, the European Commission has launched two competitions aimed at promoting smart and sustainable tourism in the EU, networking and strengthening destinations, and facilitating the exchange of best practices. The awards recognize outstanding achievements of European cities as tourist destinations in four categories: sustainability, accessibility, digitalization, and cultural heritage and creativity. The European Green Pioneer of Smart Tourism, the

successor to the European Destinations of Excellence (EDEN) competition, recognizes small destinations that have implemented successful strategies to promote sustainable tourism through green transition practices. Smart Tourism is an EU initiative currently funded under the Single Market Program (SMP), the former COSME program. It builds on the successful experience of the Preparatory Actions proposed by the European Parliament and implemented by the European Commission [31].

In November 2024, Greece announced the creation of a fund in partnership with the European Commission and the European Investment Bank to support the decarbonization of its islands, which are heavily dependent on tourism. The goal is to switch from expensive, oil-dependent energy sources to more sustainable ones, reducing dependence on fossil fuels and energy costs [32]. In 2022, the European Commission adopted the European Tourism Agenda 2030, which plans to make tourism greener, more digital, and more sustainable [33]. Secondly, the role of decentralization and support for local businesses, local and regional projects aimed at developing domestic tourism, supporting SMEs, and forming tourism clusters and smart destinations is growing.

The European Regional Development Fund (ERDF) plays a key role in enhancing the competitiveness, sustainable development, and quality of tourism at regional and local levels by providing funding for the development of natural, historical, and cultural resources, as well as innovative tourism products and services. These measures contribute to making cities and regions more attractive as places to live, work and play, and create the preconditions for long-term economic growth. The ERDF recommends that initiatives be aligned with one or more thematic objectives and investment priorities, be in line with the SWOT analysis of national, regional, or transnational operational programs, focus on the effective valuation of local cultural and tourism assets, promote innovation and diversification of products, processes, and services, and support specialization in market niches to overcome dependence on low value-added, seasonal employment and ensure stable economic activity over time.

Particular attention should be paid to tripling the capacity for excellence, innovation, and internationalization of SMEs and clusters, and (cross-)clustering activities [34], including strengthening internal and external linkages.

As one of the particular strengths of the tourism sector is that barriers to entry are particularly low, the focus should be on supporting entrepreneurship and the creation of new companies. Tourism will continue to play a prominent role in the planned ERDF investments, as well as related investments in the preservation, protection, promotion, and development of natural and cultural heritage, with a planned ERDF allocation of around EUR 8 billion [35].

The Destination SMEs program aims to help European territories develop effective destination management strategies. This creates a favourable environment for the growth of tourism SMEs and enhances their competitiveness. The program emphasizes the importance of supporting local businesses and developing tourism clusters. Destination SMEs aim to help European territories develop effective destination management strategies to provide tourism SMEs with a favourable environment for growth and competitiveness. The project brings together 7 partners from 7 European

countries (France, Italy, Malta, Ireland, Finland, Latvia, Slovenia), which consist of a combination of public authorities at the national, regional, and local levels and agencies that have delegated powers from their public authorities [36].

In April 2025, the European Bank for Reconstruction and Development (EBRD) launched the €377 million Go Digital in the Western Balkans program. The program, supported by the European Union, aims to promote digitalization, automation, and the introduction of green technologies in small and medium-sized enterprises in the region, including the tourism sector. Particular attention is paid to supporting women-led businesses [37]. Austria, Portugal, and Greece give preference to preferential loans, France, the United Kingdom, and Italy pay special attention to subsidies, and Spain pays special attention to low value-added tax. In Italy, Spain, and Poland, potential investors are exempt from land taxes for a certain period [38].

Third, investment priorities are increasingly integrated into strategic initiatives of the European Union, in particular within the framework of the Green Deal, Digital Europe, Smart Specialization strategy and cohesion policy [15].

The European Green Deal aims to transform the EU into a modern, resource-efficient, and competitive economy with zero greenhouse gas emissions by 2050. This involves reorienting financial flows to projects that meet the principles of green tourism, including eco-hotels, energy-efficient infrastructure, biodiversity conservation, and sustainable transport. [39].

Smart Specialization Strategies are an important tool for stimulating knowledge-based growth in the regions and countries of the European Union, with a particular focus on supporting projects aimed at developing domestic tourism [15], strengthening small and medium-sized enterprises, creating tourism clusters [34], and creating «smart destinations». In the new programming period 2021–2027, increased attention is paid to governance through a new thematic enabling condition—«Good governance of a national or regional smart specialization strategy», which includes seven key criteria: conducting an up-to-date analysis of the challenges to innovation and digitalization; having a competent body responsible for implementing the strategy; Implementation of performance monitoring and evaluation tools; active involvement of stakeholders in the entrepreneurial discovery; implementation of measures to improve the research and innovation system; support for industrial transition, if necessary; and strengthening cross-border cooperation in priority areas of smart specialization [40].

Investment priorities are increasingly integrated into the strategic initiatives of the European Union. The Digital Decade program sets specific goals and targets until 2030 to guide Europe's digital transformation, including the development of digital skills, infrastructure, businesses and governments [41].

International tourism in Europe is facing a number of challenges that are affecting its development, despite high growth rates. Economic, geopolitical and negative climatic factors also influence the decision of tourists to travel. Inflation and changes in oil prices lead to higher prices for transportation and hotel accommodation [35]. Against this background, tourists will continue to look for the best value for money.

In addition to the ongoing political conflicts, geopolitical risks are a growing concern, ranking third among the main factors after economic ones, according to

experts. In addition to extreme weather events, staff shortages are also a serious problem [19].

Environmental problems in the world have hurt the development of winter sports and tourism, which are facing an «existential crisis». Due to climate change, the International Ski and Snowboard Federation (FIS) and the World Meteorological Organization (WMO) are developing tools related to the development of sports tourism due to the threat of «extinction» of winter sports and winter tourism. According to the FIS, the 2023–2024 winter season is organized for 616 World Cup races on 166 tracks. Twenty-six of these competitions were cancelled for weather-related reasons. [35].

In 2025, Europe's tourism sector is facing complex challenges that are significantly changing the structure of demand, approaches to organizing services, and strategic investment directions. The analysis of current problems allows us to formulate a consistent link between the problems, their impact on the market, and the corresponding investment strategies (Table 3).

The challenges considered: economic, environmental, technological, social, and demographic, not only create serious barriers to the stable functioning of the tourism industry but also open up new strategic niches for attracting capital. Accordingly, there is a need to develop a holistic, flexible, and adaptive concept of investment decision-making in the tourism sector. Investing in tourism experiences, authentic products, cultural events, and personalized offers. An effective investment policy should not only respond to current risks but also proactively direct resources to the most promising areas of development, taking into account macro trends and regional specifics (Fig. 4).

In the context of transformations of the European tourism market caused by geopolitical, environmental, and socio-economic factors, there is an objective need to rethink the strategic guidelines of investment policy. Its optimization should be based on a multi-level approach that takes into account the territorial peculiarities of development, the dynamics of demand, and the need to form sustainable institutional foundations.

1. Differentiation of investment approaches according to the stage of development of the tourism sector. The implementation of investment policy should be based on the typology of regions according to the level of maturity of the tourism system. In regions with growing tourism potential, priority should be given to investments in basic infrastructure, transport accessibility, hotel development, and branding of territories. For regions with a high level of maturity, investment support for the modernization of existing tourism facilities, diversification of the product portfolio, and implementation of sustainable development principles is advisable. In depressed regions, comprehensive projects to restructure assets, upgrade human resources, and attract new consumer segments can be effective.
2. Focus on high-potential areas of tourist demand. Given the transformation of tourist behaviour, investment policy should be aimed at supporting sectors that demonstrate sustainable development dynamics. These include medical and wellness tourism, eco-tourism, and rural tourism as tools for regional diversification,

Table 3 Challenges for the development of international tourism in Europe in 2025 and their impact on investment policy [developed by the authors]

Negative issues	Consequences of the challenges	Implications for investment policy
Economic instability	Rising energy prices, inflation, and currency fluctuations reduce the purchasing power of tourists	The need to stimulate investment in budget and domestic tourism and the development of local infrastructure
Environmental problems	Increasing sustainability requirements: reducing emissions, preserving biodiversity	Reorientation of investments to green projects, eco-hotels, sustainable mobility, and certification of facilities
Technological changes	Intensification of digitalization: online platforms, big data, AR/VR in tourism; cybersecurity risks	Investments in IT infrastructure, process automation, digital security, and the development of new digital services
Post-event impacts	A prolonged return to full mobility, new security requirements, and changes in travel habits	The need to invest in flexible business models, digital control, contactless service, and medical standards
Social change	Demand is shifting to staycation, workation, slow tourism, local travel, and individual recreation	Support for small tourism enterprises, development of local tourism infrastructure and smart destinations
Demographic trends	The ageing of Europe's population, the growing number of tourists aged 60+, and new requirements for convenience and service	Investments in inclusive tourism, accessibility, medical tourism, and specialized services for older travellers
Regulatory barriers	Tightening security standards, visa policy, sustainability standards, ESG reporting	Directing investments to bring infrastructure in line with regulatory requirements, ESG reporting, and certification
Competition with alternatives	The growing popularity of cultural, online educational and hybrid leisure formats that partially replace travel	Investing in travel experiences, authentic products, cultural events, and personalized offers

off-season products that can smooth out seasonal imbalances, and event tourism that contributes to a multiplier effect in local economies.

3. Development of tourism clusters as a mechanism for enhancing synergies. The cluster approach to investment allows for the creation of conditions for concentrating resources, effective cooperation between stakeholders, and increasing the

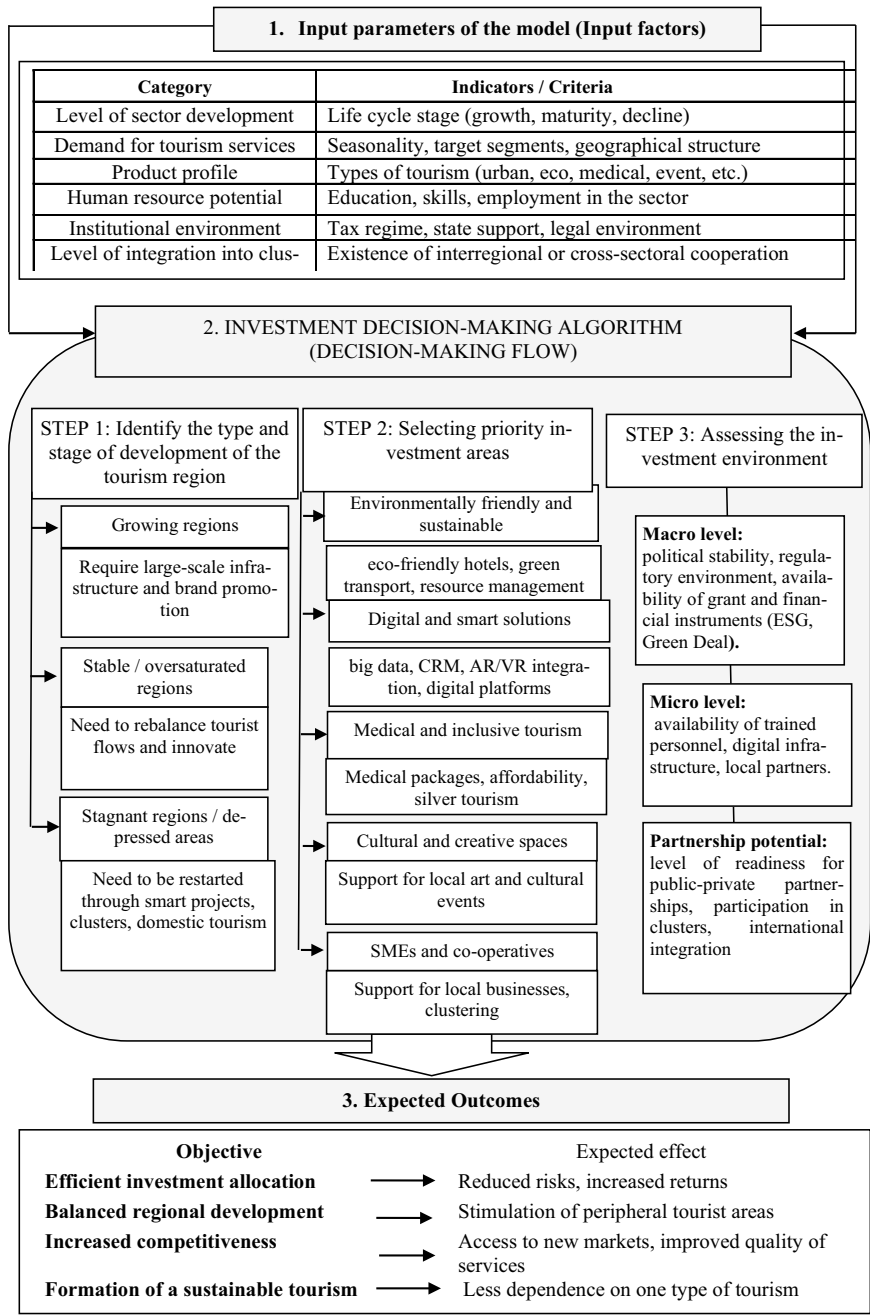


Fig. 4 The concept of investment decision-making in the tourism sector in Europe. [authors' development]

- innovation capacity of regions. This includes support for inter-municipal cooperation, financing of common facilities, and coordination of marketing initiatives within tourism clusters.
4. Investing in human capital as a strategic priority. One of the key vectors for optimizing investment policy is the formation of highly skilled human potential. Investments in vocational education and training for managers, guides, and travel organizers, as well as the creation of regional training centres and the development of digital and language competencies in the service sector, are of paramount importance.
 5. Improving the institutional environment for attracting investment. Effective implementation of investment policy requires a favourable institutional climate. This means ensuring transparency of procedures, predictability of the regulatory framework, expanding access to EU financial instruments, and active use of public–private partnership mechanisms to implement strategic tourism projects.

Thus, modern investment policy in the European tourism sector should evolve towards a comprehensive, adaptive, and inclusive approach [42]. Its effectiveness largely depends on the ability to combine regional development peculiarities with the European priorities of sustainable growth, digital transformation, and social cohesion.

The developed concept of investment decision-making in the European tourism sector is flexible and can be adapted to the conditions of individual countries, regions, or tourism clusters, taking into account their unique economic, social, and environmental characteristics. This approach allows for the integration of specific regional factors, such as the type of tourism services, seasonality of demand, and the level of infrastructure development, into the process of formulating investment strategies.

In the future, to develop a model of investment policy optimization based on the concept of investment decision-making, which can be formalized in the form of a multi-criteria evaluation system that will ensure informed decision-making based on modern analytical tools [43]. In particular, the use of multi-criteria analysis methods, such as AHP (Analytical Hierarchy Process) or TOPSIS (Technique for Order of Preference by Similarity to Ideal Solution), will allow the integration of numerous factors and critical parameters that affect investment efficiency. This, in turn, will contribute to more accurate forecasting of expected results, reduce risks, and increase the efficiency of investment decisions in the tourism sector.

5 Discussion

To remain competitive, Europe must continue to adapt to new challenges, such as changes in consumer preferences, economic fluctuations and extreme weather conditions. Implementing modern development strategies, such as investing in infrastructure, supporting innovation and environmental projects, are key to ensuring the long-term sustainability of the tourism sector.

The analysis of investment activity in sustainable tourism in Europe includes the study of the dynamics of investment flows in the tourism sector, sources of financing for sustainable tourism projects, as well as the main risks and barriers to investment. Therefore, it is important to understand how investment affects tourism development and what factors facilitate or impede effective investment [8, 13]. Investments in quality education, gender equality, clean water, and sanitation can enhance sustainable development by favouring sustainable types of tourism [3, 9, 10, 12, 17]. This helps to reduce poverty and inequality [2, 13]. Strategic investment planning should be taken into account [5–7].

As we can see, to remain competitive, it is important to focus on specific actions, support entrepreneurs, introduce green technologies and innovations, develop digitalization, invest in education, gender equality, and infrastructure, and implement the Smart City concept [16], which will contribute to sustainable development and job creation [15, 34].

The development of investment strategies for sustainable tourism requires close partnerships between the state, business, and international organizations [11]. Joint efforts in implementing innovative solutions, financing sustainable projects, and supporting environmental initiatives will contribute to the growth and development of the European tourism sector, ensuring its competitiveness and sustainability in the long term.

Further research will be aimed at understanding the relationships between tourism receipts, expenditures and foreign direct investment and the importance of investing in the European tourism market.

6 Conclusion

Thus, the results of the study indicate the relevance of transforming the investment decision-making system in the European tourism sector in the context of the growing complexity of the global economic environment. The European tourism space, despite the high level of development in several countries, demonstrates significant heterogeneity in the structure of investment potential and the dynamics of economic indicators, which requires the construction of adaptive mechanisms for managing investment flows.

The key conclusion of the analysis is the confirmation that investment policy in the tourism industry cannot be unified: it should reflect the phase of development of the industry in a particular country or region. For countries at the growth stage, it is advisable to introduce tools that help accelerate digital and environmental innovations. Countries that have reached the stage of maturity need to rethink their investment strategy in the direction of sustainable tourism, impact investing, and support for institutional maturity. In turn, emerging economies should focus on creating a favourable environment for investment through infrastructure modernization, human capital development, and regulatory stability.

The developed implication map of the European market life cycle by key economic indicators shows that Western Europe is at the stage of maturity, requiring strategies to support technological upgrades, inclusion and impact-oriented growth. Central Europe and the Baltics are at the growth stage, with the potential to become leaders if supported by digital transformation and ESG factors. Eastern Europe is largely between the early and transition stages, where institutional modernization and pilot innovation programs play an important role.

The results indicate that the life stages of the tourism market cycle are a complex analytical tool and form strategies for flexible investment management. The developed concept of investment decision-making takes into account pan-European priorities (green transformation, digitalization, inclusive growth) and regional peculiarities (type of tourism, intensity of seasonality, depth of infrastructure maturity), and opens up new opportunities for more accurate identification of areas with a high investment multiplier.

The institutional flexibility of the concept allows for the integration of modern economic analysis methods that ensure decision-making based on expected benefits and the dynamics of system parameters. This integration helps to reduce uncertainty, improve forecast accuracy and create a sustainable investment architecture that can adapt to changing external conditions.

Thus, the creation of an effective investment decision-making system in the European tourism sector has all the hallmarks of a new paradigm that combines systemic analytics, economic adaptability, and strategic focus on long-term sustainable development. In the future, further research efforts should be focused on testing this model in the context of specific EU member states, taking into account their institutional context, financial capacity, and innovation landscape. Overall, the findings in this paper are novel and have vital policy implications for promoting the sustainable development of Europe's tourism industry.

References

1. ETC (2025) European tourism: trends and prospects. 2024. <https://etc-corporate.org/reports/european-tourism-2024-trends-prospects-q3-2024/>. Accessed 12 Mar 2025
2. Çapar H, Arslanoğlu A (2025) The impact of sustainable development goals as a driving force in the development of medical tourism. *Eval Program Plann* 111(102587). <https://doi.org/10.1016/j.evalprogplan.2025.102587>
3. Gürbüz E, Pirselimoglu BZ (2025) Evaluation of ecologically based activities within the scope of sustainable tourism and recreation planning. *Sustain (Switz)* 17(5):2136. <https://doi.org/10.3390/su17052136>
4. Zhang Z, Zhang H, Lin MS, Jia W (2025) Is more rural digitization always better for tourism? An innovative multiscale spatial perspective. *Tour Econ* 31(2):285–308. <https://doi.org/10.1177/13548166241273607>
5. Peng X, Liang Y, Zhu T et al (2025) Unlocking global horizons through outbound tourism: an institutional logics approach to driving outward FDI by tourism firms from emerging countries. *Tourism Manage* 109:105126. <https://doi.org/10.1016/j.tourman.2024.105126>

6. Castilho D, Fuinhas JA (2025) Exploring the effects of tourism capital investment on income inequality and poverty in the European Union countries. *J Econ Struct* 14(1):6. <https://doi.org/10.1186/s40008-025-00349-2>
7. Sadekin MN (2025) Relationship among tourism, FDI, and economic growth in Bangladesh. *Discov Sustain* 6(1):98. <https://doi.org/10.1007/s43621-025-00874-x>
8. Yang Z, Lv P, Sun S (2025) The impact of new infrastructure investment on the international tourism industry: evidence from provincial-level panel data in China. *Sustain (Switz)* 17(6):2334. <https://doi.org/10.3390/su17062334>
9. Tsviliy S, Mykhailyk D, Ogloblina V et al (2024a) The mechanism of activation of investment potential in the PostConflict period of development of the tourism industry. *Qubahan Acad J* 4(3):175–194. <https://doi.org/10.48161/qaj.v4n3a213>
10. Tsviliy S, Mykhailyk D, Gurova D et al (2024b) Strategy for the development of the investment potential of the tourism industry of Ukraine in the international economic system. *Theor Pract Res Econ Fields* 15(3):601–619. [https://doi.org/10.14505/tpref.v15.3\(31\).08](https://doi.org/10.14505/tpref.v15.3(31).08)
11. Ribeiro MA, Adam I, Kimbu AN et al (2025) Resilience in uncertainty: the impact of coping strategies on tourism entrepreneurs' subjective well-being in sub-Saharan Africa. *Tour Manage* 110:105169. <https://doi.org/10.1016/j.tourman.2025.105169>
12. Ni X, Li H, Wang D (2025) How hotel eco-labels can be powerful nudges: an evaluability perspective. *Tour Manage* 110:105191. <https://doi.org/10.1016/j.tourman.2025.105191>
13. Nguyen CP, Schinckus C et al (2024) Tourism and contribution to employment: global evidence. *J Econ Dev*. <https://doi.org/10.1108/JED-07-2024-0269>
14. But T, Mamotenko D (2025) Increasing the economic development of the EU countries through the implementation of the «Smart City» concept. *Manage Entrepreneurship: Trends Dev* 1(31):27–37. <https://doi.org/10.26661/2522-1566/2025-1/31-03>
15. But T, Pulina T, Bielina O et al (2025) Assessment of the development potential of the tourism industry in Czechia on the basis of smart specialization. *E&M Econ Manage* 28(1):117–134. <https://doi.org/10.15240/tul/001/2025-1-008>
16. But T, Mamotenko D, Lnenicka L et al (2023) A conceptual model for creating smart cities in Czechia based on smart specialization. *Sci Papers Univ Pardubice* 31(2):1736. <https://doi.org/10.46585/sp31021736>
17. Yue X-G, Liao Y, Zheng S et al (2021) The role of green innovation and tourism towards carbon neutrality in Thailand: evidence from bootstrap ADRL approach. *J Environ Manage* 292:112778. <https://doi.org/10.1016/j.jenvman.2021.112778>
18. ONU Turismo (2025) European tourism experienced strong momentum in the first quarter of 2024. <https://www.ifema.es/en/fitur/news/european-tourism-first-quarter-2024>. Accessed 12 Mar 2025
19. Khudaverdieva V, Merchanskyi V (2022) International tourist exchange for business purposes as a type of foreign economic relations. The economic discourse. *Int Sci J* 3–4:92–105. <https://doi.org/10.36742/2410-0919-2022-2-9>
20. World Travel and Tourism Council (2025) World Travel and Tourism Council (WTTC) represents the travel and tourism sector globally. <https://wtcc.org/>. Accessed 12 Mar 2025
21. Statista.com (2025) Number of international tourist arrivals worldwide from 2005 to 2024, by region. <https://www.statista.com/statistics/186743/international-tourist-arrivals-worldwide-by-region-since-2010/#statisticContainer>. Accessed 12 Mar 2025
22. World Economic Forum (2024) Travel and tourism development index 2024. https://www3.weforum.org/docs/WEF_Travel_and_Tourism_Development_Index_2024.pdf. Accessed 12 Mar 2025
23. UN Tourism (2025) UN tourism world tourism barometer 2025-excerpt. <https://www.unwto.org/news/international-tourism-recovers-pre-pandemic-levels-in-2024>, Accessed 04 June 2025, 12 Mar 2025
24. European Commission: Directorate-General for Research and Innovation, European Innovation Scoreboard 2024, Publications Office of the European Union (2024) <https://data.europa.eu/doi/10.2777/779689>. Accessed 12 Mar 2025

25. Unctad. UN trade and development (2025) World investment report 2023. <https://unctad.org/publication/world-investment-report-2023>. Accessed 12 Mar 2025
26. Porter ME (1980) Competitive strategy: techniques for analysing industries and competitors. Free Press, New York. (Republished with a new introduction, 1998)
27. IMD (2024) IMD world competitiveness booklet 2024. https://imd.widen.net/content/rjlc6f12jl/pdf/booklet_wcy_2024.pdf. Accessed 12 Mar 2025
28. UN-Habitat (2024) World smart cities outlook 2024, p 106. Retrieved from: https://unhabitat.org/sites/default/files/2024/12/un_smart_city_outlook.pdf. Accessed 12 Mar 2025
29. EU (2025a) InvestEU: enhancing tourism's competitiveness and sustainability. <https://transition-pathways.europa.eu/support-resources/investeu-enhancing-tourisms-competitiveness-and-sustainability?>. Accessed 12 Mar 2025
30. EU (2025b) EU tourism platform. InvestEU: enhancing tourism's competitiveness and sustainability. <https://transition-pathways.europa.eu/support-resources/investeu-enhancing-tourisms-competitiveness-and-sustainability?>. Accessed 12 Mar 2025
31. EU (2025c) European capital and green pioneer of smart tourism. An initiative to reward smart and sustainable tourism practices in European cities. https://smart-tourism-capital.ec.europa.eu/index_en?. Accessed 12 Mar 2025
32. Reuters (2025a) Greece sets up fund to decarbonise its islands, reduce hefty fuel imports. <https://www.reuters.com/sustainability/sustainable-finance-reporting/greece-will-set-up-fund-decarbonise-its-islands-2024-11-21/?>. Accessed 12 Mar 2025
33. EU (2025d) European tourism agenda 2030: commission welcomes the commitment of EU countries to make tourism greener, more digital and resilient. https://single-market-economy.ec.europa.eu/news/european-tourism-agenda-2030-commission-welcomes-commitment-eu-countries-make-tourism-greener-more-2022-12-02_en?. Accessed 12 Mar 2025
34. Bezkhlibna A, But T, Nikonenko S (2018) Assessment of tourism industry clustering potential. *Sci Papers Univ Pordubice* 44:17–28. <https://hdl.handle.net/10195/71485>
35. EC (2025a) Tourism. https://ec.europa.eu/regional_policy/policy/themes/tourism_en?
36. Interreg Europe (Destination SMEs) (2025) Project summary. Destination strategies for the competitiveness of the local SMEs. <https://projects2014-2020.interregurope.eu/destinations/mes/?>. Accessed 12 Mar 2025
37. Reuters (2025b) EBRD launches \$417 million programme to help Western Balkans go digital. <https://www.reuters.com/markets/europe/ebrd-launches-417-million-programme-help-western-balkans-go-digital-2025-04-10/?>. Accessed 12 Mar 2025
38. Melnyk A, Chir N (2019) Modern experience of creating the country's tourist attractiveness and opportunities for its improvement. Proceedings of the international scientific and practical internet conference «tendencies and prospects for the development of science and education in the context of globalisation» 28.09.2019, 51, Pereyaslav-Khmelnyskyi, Ukraine. https://www.researchgate.net/publication/366016469_sucasnij_dosvid_stvorennia_turysticnoi_privablivosti_kraini_ta_mozlivosti_ii_pidvisenna
39. EC (2025b) The European green deal striving to be the first climate-neutral continent. https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/european-green-deal_en?. Accessed 12 Mar 2025
40. EC (2025c) About S3. Smart specialisation strategies foster knowledge-based growth in EU regions and countries. They are crucial to achieving a smarter, sustainable and inclusive Europe. https://ec.europa.eu/regional_policy/policy/communities-and-networks/s3-community-of-practice/about_en?. Accessed 12 Mar 2025
41. EC (2025d) Europe's digital decade: digital targets for 2030. https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/europe-fit-digital-age/europes-digital-decade-digital-targets-2030_en?. Accessed 12 Mar 2025
42. Voynarenko M, Dzhezdzhula V, Yepifanova I (2016) Modelling the process of making decisions on sources of financing of innovation activity. *Econ Ann-XXI* 160(7–8):126–129. <https://doi.org/10.21003/ea.v160-25>
43. Kryvinska N, Strauss C (2013) Conceptual model of business services availability versus interoperability on collaborative IoT-enabled eBusiness platforms. In studies in computational

intelligence. Springer Berlin Heidelberg, pp 167–187. https://doi.org/10.1007/978-3-642-34952-2_7

Optical Measurement Converter “Gas Concentration–Voltage” for Environmental Monitoring of Aggressive Atmospheric Environments



Igor Dudatiev , Andrii Semenov , Kostyantyn Ovchynnykov ,
Maksym Prytula , and Serhii Maltsev

Abstract The paper proposes an approach to monitoring the concentration of carbon dioxide (CO₂) in aggressive gas environments using the optical absorption method. The relevance of the study is due to the need for high-precision and stable means of monitoring the air composition in the face of increasing emissions of toxic and greenhouse gases that adversely affect the environment, human health, and climatic conditions. The main element of the measuring transducer is a photodetector based on a photodiode-operational amplifier pair, for which a mathematical model describing the relationship between gas concentration and the system output voltage has been developed. The model is based on the fundamental Bouguer–Lambert–Beer law and takes into account the parameters of infrared radiation, spectral characteristics of the photodiode, the effect of temperature, pressure, and the width of the spectral absorption band. Particular attention is paid to modeling the absorption process in the selected spectral range (central wavelength of 4.267 μm), where CO₂ has a pronounced intensity without significant overlap with other gases. The results of laboratory tests using calibrated gas mixtures confirmed the high accuracy and reproducibility of the readings. The error analysis demonstrated that the maximum absolute and relative errors do not exceed the established metrological tolerances, and the model is adequate over the entire operating range. The proposed approach allows for effective control of CO₂ content in complex environments, including emissions from industrial, energy, and transportation sources. The novelty of the work lies in the creation of a comprehensive optical measurement model that takes into account destabilizing factors and allows the system to be adapted to different operating conditions. The results obtained indicate the prospects of implementing the developed sensor in new generation environmental monitoring systems. The results of the study can be adapted to detect other harmful gases (NO_x, SO₂, CH₄), taking

I. Dudatiev · A. Semenov (✉) · K. Ovchynnykov · M. Prytula · S. Maltsev
Vinnytsia National Technical University, Vinnytsia, Ukraine
e-mail: semenov.a.o@vntu.edu.ua

M. Prytula
e-mail: pritulamo@ukr.net

into account their spectral characteristics and physicochemical properties, which opens up prospects for the further development of sensor systems for environmental monitoring in industry, energy, and agriculture.

Keywords Environmental monitoring · Optical sensor · Measuring transducer · Concentration

1 Introduction

In today's environment, the task of efficient and accurate monitoring of aggressive gas environments is key to maintaining ecological balance, protecting the health of living organisms, and complying with environmental regulations and standards. Aggressive gas environments are environments that contain gases or mixtures of gases that can have a negative impact on the environment and the health of living organisms. These may include gases that are highly corrosive, toxic, or contribute to air pollution and climate change. These gases include: carbon dioxide (CO_2) is a major greenhouse gas that plays a key role in global warming; nitrogen oxides (NO_x)—gases formed during fuel combustion, which can cause acid rain and air pollution sulfur dioxide (SO_2)—a toxic gas that contributes to acid rain and has a harmful effect on organisms, including vegetation and aquatic ecosystems; methane (CH_4)—a powerful greenhouse gas that has a significant effect on global warming; carbon monoxides (CO)—poisonous gases that, even in low concentrations, can be dangerous to human and animal health.

Aggressive gases can come from a variety of sources, including industrial emissions, transportation, fuel combustion [1], agriculture, and natural processes (e.g., volcanic activity). They can cause corrosion of metal structures, soil and water pollution, and change the composition of the atmosphere, which has global climate implications.

A review of current methods for measuring gas concentrations shows that optical methods are the most optimal and accurate [2]. These methods are based on measuring the absorption of light by gases in certain spectral ranges [3]. The use of laser technology ensures high sensitivity and accuracy of measurements even at low gas concentrations. In addition, optical methods allow for non-contact monitoring, which reduces the risk of damage to measuring systems and increases the reliability of the results [4].

The object of study in this paper is the environmental monitoring of CO_2 concentration in chemically aggressive atmospheres.

The purpose of the work is to improve the accuracy of carbon dioxide concentration measurement in aggressive gas environments by assessing and compensating for the influence of destabilizing factors based on the optical absorption method.

2 Materials and Methods

An overview of modern methods for measuring gas concentrations reveals a wide variety of technologies and approaches actively used across numerous fields, including industrial safety, environmental monitoring, medical diagnostics, and scientific research. Among the most commonly employed techniques are electrochemical, thermochemical, acoustic, mass spectrometric, laser-based, and optical methods. Each of these has its own advantages and limitations, typically associated with factors such as cost, accuracy, sensitivity, selectivity, response time, and environmental operating conditions.

Electrochemical methods, for instance, are widely applied for detecting toxic or explosive gases in both household and industrial environments. These sensors are relatively inexpensive, compact, and energy-efficient. However, they often suffer from limited operational lifespans, sensitivity to ambient temperature and humidity changes, and insufficient selectivity when multiple gas species are present. Similar constraints apply to thermochemical methods, which rely on heat-based reactions and can be significantly affected by fluctuations in environmental temperature, thus reducing measurement accuracy.

Mass spectrometric and gas chromatographic techniques offer extremely high precision and the ability to separate complex gas mixtures into individual components. Nevertheless, such methods typically require bulky and sophisticated instrumentation, often needing specialized operational environments. Furthermore, they are generally not suitable for real-time or continuous monitoring applications, which are crucial in safety-critical or dynamic environmental scenarios.

Unlike other approaches, optical techniques for measuring gas concentrations are notable for their versatility, precision, rapid response, and ability to operate reliably in challenging or remote conditions [2]. These techniques rely on fundamental physical principles, including the Beer–Lambert law, which explains how light intensity decreases when traveling through a gas. Leveraging these principles, optical techniques can detect even trace amounts of target gases using spectral absorption analysis.

Infrared (IR)-based optical sensors [3], in particular, exhibit high sensitivity to gases with distinct absorption features in specific wavelength regions. Gases such as carbon dioxide (CO_2), methane (CH_4), and nitric oxide (NO) have well-defined absorption bands in the mid-infrared range. This property enables the development of sensors that not only detect the presence of specific gases but also quantify their concentrations with impressive precision.

One of the significant advantages of optical methods lies in their high selectivity. Since each gas possesses a unique spectral fingerprint, the risk of cross-sensitivity—common in many electrochemical systems—is greatly minimized. When combined with modern digital signal processing techniques and accurate mathematical modeling, optical systems achieve high calibration fidelity, long-term stability, and strong resistance to environmental disturbances such as vibration, temperature variation, or contamination.

Recent advances in photonics and microelectronics have also enabled the development of compact and miniaturized optical sensors suitable for integration into portable devices. This technological evolution expands the applicability of optical gas sensors beyond traditional industrial use, reaching into areas such as home air quality monitoring, automotive diagnostics, and even space-based atmospheric research.

Another critical advantage of optical sensors is their rapid response time. Most optical systems are capable of detecting gas concentration changes within seconds—or even milliseconds—which is vital in safety and emergency scenarios where immediate detection is paramount. Moreover, optical devices do not rely on consumable materials, which significantly reduces maintenance requirements and extends operational lifetime, thus lowering the total cost of ownership.

In conclusion, a comprehensive analysis of available gas sensing technologies indicates that optical methods are currently the most optimal solution in terms of a well-balanced combination of accuracy, stability, speed, selectivity, and adaptability. As a result, they form the technological foundation of most state-of-the-art gas monitoring systems used in environmental science, industrial automation, healthcare, and space exploration.

The Bouguer–Lambert–Beer law, a well-established principle, explains how light intensity diminishes as it traverses a uniform absorbing medium. The physical basis of the Bouguer–Lambert–Beer law is that the intensity of light decreases as it passes through a substance depending on two main factors: the thickness of the layer of this substance and the concentration of particles that can absorb light (absorbers). If the layer of a substance becomes thicker, the light has to travel a greater distance through the medium, and thus the probability that photons will collide with molecules and be absorbed increases. Similarly, when the concentration of a substance increases, i.e. the number of molecules per unit volume, the density of absorbers increases, and this also increases the probability of light absorption. As a result, the more absorbers there are in the light's path or the longer this path is, the lower the intensity of the light coming out of the substance will be.

The molecules of a substance are capable of absorbing light. When a light beam passes through a solution, photons interact with the molecules of the substance and can be absorbed by them. The more molecules in the layer (i.e., the higher the concentration) and the thicker the layer, the more photons will be absorbed.

The decrease in light intensity in this process is not linear but exponential. This means that each subsequent part of the light path does not absorb a fixed amount, but a certain fraction of the light remaining after the previous layer. That is why the dependence of intensity on concentration and thickness is described by a logarithmic function.

It is also important that light absorption is an additive process, meaning that the total absorbed amount of light is the sum of the contributions from each layer of matter. However, the Bouguereau-Lambert–Beer law is only fulfilled under certain conditions: the substance must be homogeneous, the light must be monochromatic (of one wavelength), and there must be no scattering, fluorescence, or other side processes that can change the intensity of the light.

Let's represent mathematically the Bouguer–Lambert–Beer law, which formulates the dependence of transmitted irradiation on the studied gas concentration, the optical path length, and the absorption coefficient in the expression

$$I_l = I_0 \cdot e^{-(k_a + k_s) \cdot C \cdot l}, \quad (1)$$

where I_0 is the initial irradiation intensity, C is the corresponds to the concentration of the gas being investigated, l is the length of the absorption path, k_a is the absorption coefficient, k_s is the scattering coefficient.

An LED serves as the source of infrared radiation [5]. In optical gas sensors, the infrared (IR) radiation source plays a key role, as it enables the interaction of the light signal with gas molecules, which is essential for analyzing the concentration of the substance. Among the various possible radiation sources, the infrared light-emitting diode (IR-LED) is the most appropriate choice due to its combination of structural, energetic, and spectral advantages.

Firstly, IR-LEDs provide narrow-band emission in a specific region of the infrared spectrum. This is particularly important in the context of absorption-based optical spectroscopy, as most gases (such as CO_2 , CH_4 , NO , NH_3) exhibit distinct absorption peaks in the mid-infrared range. Therefore, the use of IR-LEDs allows for precise tuning of the sensor to a specific gas without the need for additional spectral filtering, which significantly simplifies the sensor's design.

Secondly, light-emitting diodes feature a fast response time and stable emission, ensuring high repeatability of measurements and accurate detection of rapid changes in gas concentration in real time. This is especially critical for sensors used in environmental monitoring applications.

Another important advantage of IR-LEDs in gas sensors is their energy efficiency and low heat emission. They consume significantly less energy than thermal sources (such as miniature incandescent or halogen lamps), which makes it possible to develop compact, autonomous devices powered by batteries. Due to their low thermal load, LEDs also do not introduce thermal disturbances into the measurement chamber, which could otherwise distort the results.

Moreover, IR-LEDs offer a long service life—tens of thousands of operating hours—and high mechanical reliability, as they contain no fragile elements and are not sensitive to vibrations or mechanical shocks. This ensures long-term stable operation of the gas sensor even under harsh operating conditions, such as in industrial zones, vehicles, or field stations.

Thanks to their low physical mass and compact size, LEDs also allow for significant reduction in the overall dimensions of the sensor module. This makes it possible to create miniature gas analyzers for personal use or integration into mobile or portable devices.

In conclusion, the use of infrared light-emitting diodes as the radiation source in optical gas sensors ensures precise, fast, stable, and energy-efficient operation of the device, while maintaining high selectivity and reliability. For these reasons, IR-LEDs are the standard choice in modern commercial and scientific optical sensing systems for gas analysis.

The LED emits infrared radiation with a line width of 40–1000 nm, which satisfies the requirements of the current task. The radiation lines were chosen to avoid strong absorption lines of other gases within the emission range [6].

Thanks to modern HL LEDs with a narrow spectrum, the scattering index in the Bouguereau-Lambert–Beer law disappears

$$I_l = I_0 \cdot e^{-k_s Cl}, \quad (2)$$

The monograph [7] describes in detail metametric transition from the general absorption coefficient k_s to the absorption coefficient $k_s(\lambda)$ in a certain spectral range (using the Clapeyron-Mendeleev law and adopting normal conditions in accordance with the recommendations of the International Union of Pure and Applied Chemistry)

$$k_s = k_s(\lambda) \cdot \frac{8333.(3) \cdot T}{P}, \quad (3)$$

where T is the temperature of the gas mixture, P is the pressure of the gas mixture.

$K_s(\lambda)$ is represented as the sum of the absorption coefficients of the individual spectral lines in the operating spectral range these are the volt–ampere characteristics. In the high-signal regime, the bipolar transistor model can be simplified to a two-segment, voltage-controlled, piecewise-linear resistor in series with a linear, current-controlled current source [7]:

$$k_s(\lambda) = \sum_i \frac{S(\lambda_i)}{\pi} \cdot \frac{\sigma}{\sigma^2 + \left(\frac{1}{\lambda} - \frac{1}{\lambda_i}\right)^2}, \quad (4)$$

where S is the measure of the absorption band intensity, $S(\lambda_i)$ is the intensity or strength of the absorption line at the resonant wavelength λ_i , $\Sigma S(\lambda_i)$ is the sum of all absorption lines i , λ and λ_i are the length and center (resonant length) of the absorption band, σ is the width of the absorption band (related to expansion effects: pressure, temperature, etc.).

Then

$$k_s = \sum_i \frac{S(\lambda_i)}{\pi} \cdot \frac{\sigma}{\sigma^2 + \left(\frac{1}{\lambda} - \frac{1}{\lambda_i}\right)^2} \cdot \frac{8333.(3) \cdot T}{P} \quad (5)$$

Paper [8] presents a mathematical model of radiation transfer in the flue gas environment of boiler plants, where the absolute pressure is defined as the difference between atmospheric and actual pressure. However, the object of monitoring in this work is the flue gases of boiler plants, where it is really important to take into account the vaporization pressure in the chimney. For the purposes of this study, absolute pressure will be presented in a more general form:

$$P_{abs} = P_{atm.} - \rho \cdot g \cdot h \quad (6)$$

where P_{abs} is the absolute pressure (Pa), P_{atm} is the atmospheric pressure, ρ is the density of gases (kg/m^3), g is the acceleration of free fall (approximately 9.81 m/s^2), h is the altitude above sea level. In turn

$$\rho = \frac{P_{abs} \cdot M}{R \cdot (T + 273.15)} \quad (7)$$

where M is the molar mass of the gases, kg/mol (approximately 0.029 kg/mol), R is the universal gas constant [$8.314 \text{ J/(mol}\cdot\text{K)}$], T is the temperature of the gases in $^{\circ}\text{C}$.

Then the absolute pressure on the object of control is expressed as

$$P_{abs} = \frac{P_{atm}}{1 + \frac{M \cdot g \cdot h}{R \cdot (T + 273.15)}}. \quad (8)$$

Thus, substituting formulas (7) and (8) into (1), the model will look like this:

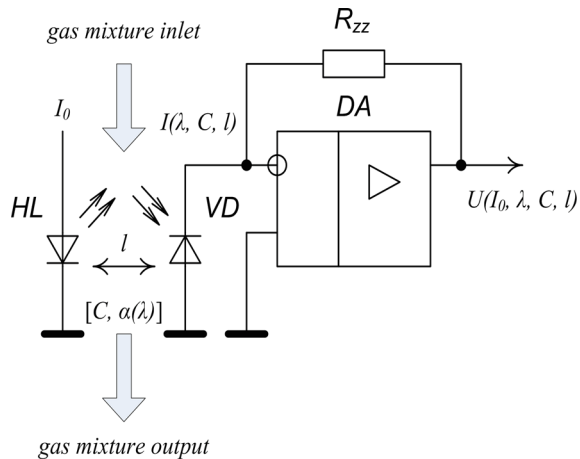
$$I_l = I_0 \cdot e^{-\left(\sum_i \frac{S(\lambda_i)}{\pi} \cdot \frac{\sigma}{\sigma^2 + (\frac{1}{\lambda} - \frac{1}{\lambda_i})^2}\right) \cdot \frac{8333 \cdot (3) \cdot T}{P_{atm}} \cdot \frac{C \cdot l}{1 + \frac{M \cdot g \cdot h}{R \cdot (T + 273.15)}}}. \quad (9)$$

The seminal work [9] reports theoretical studies on the photoelectric area-to-voltage transducer and derives the fundamental mathematical relationships governing its operation. In these transducers, the illuminated area of the photodetector's photo-sensitive layer varies in response to the measured physical quantity, making the output voltage a function of that quantity.

Figure 1 illustrates the structure of a measuring transducer based on the combination of a photodiode and an operational amplifier.

In the above configuration, the photodiode VD functions as a current source, while the feedback operational amplifier DA converts this current into a voltage. The output

Fig. 1 Model of measuring transducer substance concentration–voltage



voltage of the photodetector, based on the photodiode–operational amplifier pair and accounting for zero offset, input current differences, and noise voltage, is described by expression [9]

$$U_F = I_l \cdot S_{I0} \cdot S \cdot R_{ZZ}. \quad (10)$$

Expression (10) indicates that the output voltage of a photodetector, based on a photodiode–operational amplifier pair, is linearly proportional to the incident radiation intensity. The relation (10) describes the photoelectric concentration-voltage measurement transformation.

Then, substituting formulas (9) into (10), we obtain:

$$U_F = I_0 \cdot e^{-\left(\sum_i \frac{S(\lambda_i)}{\pi} \cdot \frac{\sigma}{\sigma^2 + \left(\frac{1}{\lambda} - \frac{1}{\lambda_i}\right)^2}\right)} \cdot \frac{8333 \cdot (3) \cdot T}{\frac{P_{appM}}{M \cdot g \cdot h}} \cdot C \cdot l \cdot S_{I0} \cdot S \cdot R_{ZZ}. \quad (11)$$

Consequently, a comprehensive mathematical model was established that characterizes the functional dependence of the output voltage of a photodetector on the gas concentration. This model is based on the photodiode–operational amplifier configuration and accounts for the underlying electrical and photoelectric interactions governing the transducer’s response.

3 Results and Discussion

In this study, CO₂ is chosen as a model object, the absorption of which is analyzed in the context of the impact of an aggressive gas environment in order to determine the effective absorption coefficient. The methodological scheme provides for flexibility in choosing the object of study: instead of carbon dioxide, it is possible to parameterize and model absorption for other gases, taking into account their physical and chemical properties.

The required information regarding the gas, including its absorption intensity and spectral half-width, is available in the HITRAN database [10, 11]. For accurate measurement of gas concentrations using optical methods, it is essential to have access to physical parameters that describe the interaction of gas molecules with electromagnetic radiation. Among the most critical of these parameters are the intensity of spectral absorption lines and their spectral linewidth (full width at half maximum), which determines how broad or narrow a given line appears in the frequency or wavelength domain [12]. These data enable researchers to model how effectively a specific molecule absorbs light within a given spectral range, forming the basis for calculations in systems based on infrared (IR) spectroscopy [13].

All of these important parameters are collected and systematized in one of the most renowned and reliable international scientific databases—HITRAN (High-resolution Transmission Molecular Absorption Database). This is a specialized database that

contains thoroughly verified and standardized information on the spectral properties of molecules, necessary for simulating the absorption and transmission of light through the atmosphere or any other gaseous medium.

The HITRAN database is continually updated and expanded, incorporating both experimental and theoretical data. The primary purpose of the database is to provide accurate and reliable input parameters for modeling the optical transparency of Earth’s atmosphere across a wide range of wavelengths—from ultraviolet to far-infrared.

The key parameters included in HITRAN are:

- the central frequency or wavelength of the spectral line,
- line intensity (under standard temperature and pressure conditions),
- pressure-broadened linewidth (full width at half maximum due to air collisions),
- pressure-induced line shift,
- molecular species and isotope identification,
- electronic, vibrational, and rotational transition data,
- temperature dependence of line intensity.

The intensity of a spectral line determines what fraction of radiation at a given wavelength will be absorbed by a unit concentration of gas molecules at standard temperature. The higher the intensity, the stronger the light absorption at that frequency, making the gas more readily detectable. The spectral linewidth, on the other hand, defines the range of frequencies over which absorption occurs and is influenced by factors such as temperature, pressure, and intermolecular interactions.

One of the distinctive features of HITRAN is its high spectral resolution. This means that the data in the database account for individual absorption lines, even if they are extremely narrow or weak. Such precision is crucial for modeling atmospheric absorption in applications like satellite remote sensing, laser absorption spectroscopy, and calculating the optical thickness of instruments.

HITRAN also serves as the foundation for numerous software tools and modeling platforms such as LBLRTM, MODTRAN, and HAPI (HITRAN Application Programming Interface). These tools allow users to simulate light transmission through the atmosphere or other gaseous media with high accuracy and reliability.

Access to HITRAN is typically provided through the official website (<https://hitran.org>), where the latest version of the database is available, along with documentation, usage examples, visualization tools for spectral lines, and an API for integrating HITRAN data into user-developed applications. For academic and research purposes, access to HITRAN is free of charge, which promotes the dissemination of high-quality and reproducible scientific modeling across the global research community.

In conclusion, HITRAN is a fundamental resource in the field of precise gas concentration measurements using optical methods. The availability of this database makes it possible to develop accurate mathematical models, calibrate sensors, analyze the spectral behavior of new substances, and perform measurements with high credibility under a wide range of conditions. Without access to such data, any

spectroscopy-based system would be significantly limited in terms of efficiency, accuracy, and reliability.

It is also necessary to take into account gases with an average expected concentration above 0.1% by volume (Figs. 2 and 3).

Analysis of the absorption spectra indicates that CO₂ exhibits its most suitable absorption band at a central wavelength of 4.267 μm. It should be noted that the strongest CO₂ absorption band occurs at 2.688 μm; however, at this wavelength, there is significant overlap with H₂O vapor absorption.

The modeling of the dependence of CO₂ concentration on voltage in the full range is shown in Fig. 4.

The simulation was performed at a specified set of parameters: input intensity of 3 mW, a feedback resistance of 9.88 MΩ (E192 series), an integrated current sensitivity

id	name	formula	bands	lines	WNmin, cm ⁻¹	WNmax, cm ⁻¹	Smin, cm/mol	Smax, cm/mol	Sv, cm/mol
1	Water vapour	H ₂ O	728	224515	0.007002	25710.8255	1.393e-37	2.651e-18	7.374e-17
2	Carbon dioxide	CO ₂	4064	471847	0.736369	12784.056441	3.934e-35	3.543e-18	1.132e-16
4	Nitrous oxide	N ₂ O	351	47843	0.838022	7796.63311	1.229e-28	1.004e-18	7.191e-17
5	Carbon monoxide	CO	49	4606	3.462498	8464.881965	1.378e-56	4.461e-19	1.020e-17
6	Methane	CH ₄	334	468013	0.001063	11501.8725	1e-37	2.114e-19	1.783e-17
8	Nitrogen oxide	NO	293	105079	1e-06	9273.21434	1.451e-95	2.322e-20	4.735e-18
9	Sulfur dioxide	SO ₂	20	95121	0.017394	4092.94822	3.101e-30	4.851e-20	4.010e-17
10	Nitrogen dioxide	NO ₂	11	104223	0.497999	3074.15265	4.24e-28	1.302e-19	6.232e-17

Fig. 2 Main components of aggressive gaseous media

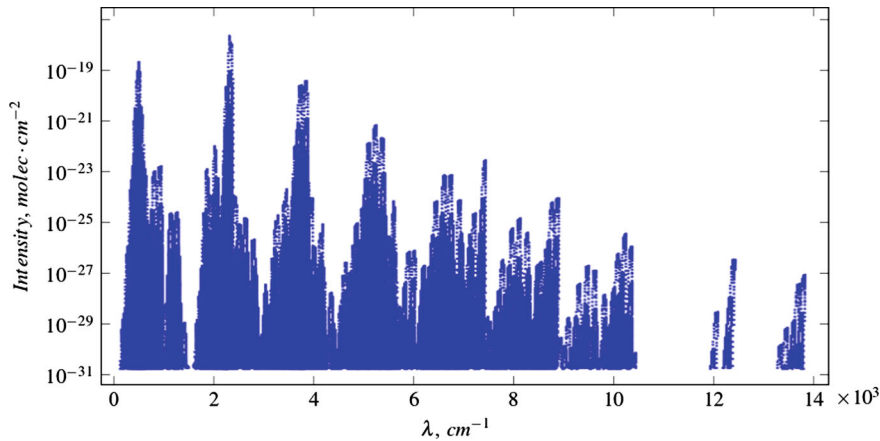


Fig. 3 CO₂ absorption spectrum in the range of 0.75–15 microns

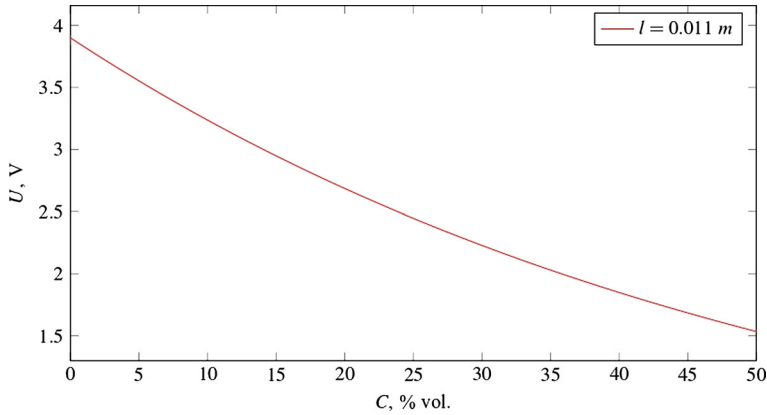


Fig. 4 Voltage–concentration relationship (full range)

of the photodiode [InSb (p-type)] of 5.06 A/m², and a photosensitive layer area of 26 mm².

The radiation power exhibits a clear dependence on the device temperature, decreasing as the temperature of the diode rises [13].

Investigation of the transmission coefficient and sensitivity of the optical sensor “gas concentration–voltage”. Transmission coefficient is a dimensionless physical quantity that reflects the ratio of the signal transmitted through a certain medium (in our case, optical) to the input signal under standard or controlled conditions [14]. It is an objective indicator of the system throughput under specified conditions. To ensure metrological reliability, it is important that the transmission coefficient measurement gives the same results when repeatedly measured under the same conditions (repeatability) [15, 16]. It depends on the sensitivity of the sensor, the linearity of the response, noise in the electronics, and the stability of the light source. For an accurate measurement of the transmittance, the instruments must be calibrated with reference media (e.g., clean air, vacuum, or a medium with a known absorber concentration) [17].

As the output signal of the optical sensor “gas concentration–voltage”, we present the transmittance of irradiation in a cuvette with gas

$$k_{trans}(C, L, \alpha(\lambda)) = \frac{U(I_0, \lambda, C, L)}{U_0(I_0, \lambda)}. \quad (12)$$

When using IRS and PD with spectral characteristics that overlap a certain frequency band $\lambda = \lambda_2 - \lambda_1$. Shifts in the spectral characteristics of both the radiation source and the detector, resulting from variations in temperature and pressure, are inherent to all infrared sources (IRS) and PD. To quantify the influence of these temperature and pressure variations on the optical sensor’s ‘gas concentration–voltage’ relationship—and consequently on the measured gas concentration—we introduce the parameters T and P into the spectral characteristic functions of the IR source and receiver. Since the photodetector responds to the integrated radiation

signal, the transfer function of the optical sensor, relating gas concentration to output voltage, can be represented by an integral expression

$$k_{trans}(C, L, P, T) = \frac{\int_{\lambda_1}^{\lambda_2} S_{PhD}(\lambda, T, P) \cdot I_{LED}(\lambda, T, P) \cdot \exp[-\alpha(\lambda) \cdot L \cdot C] d\lambda}{\int_{\lambda_1}^{\lambda_2} S_{PhD}(\lambda, T, P) \cdot I_{LED}(\lambda, T, P) d\lambda}. \quad (13)$$

where $S_{PhD}(\lambda, T)$ is the spectral sensitivity of the photodetector, $I_{LED}(\lambda, T)$ is the spectral power of the source, $\alpha(\lambda)$ is the spectrum of the gas absorption coefficient, L is the length of the optical path (the length of interaction of radiation with the gas).

For the optical sensor ‘gas concentration–voltage,’ the information parameter is defined as the variation in intensity of infrared radiation after it passes through the gas sample

$$D = \frac{\int_{\lambda_1}^{\lambda_2} S_{out}(\lambda, C\%, l) d\lambda}{\int_{\lambda_1}^{\lambda_2} S_{out}(\lambda, l) d\lambda}. \quad (14)$$

Using the developed mathematical model of the CO_2 absorption spectrum, the dependence of D on $\%_{\text{vol}}$. The relationship between the optical gas sensor’s information parameter and the carbon dioxide concentration is presented in Fig. 5.

For the developed system for monitoring carbon dioxide concentration, the total measurement error is considered to comprise two components: instrumental and methodological. A comprehensive metrological analysis of the monitoring device is carried out by theoretically evaluating each of these components.

Instrumental errors arise directly within the measurement channels—including optical, temperature, pressure, and humidity channels—at the points where they

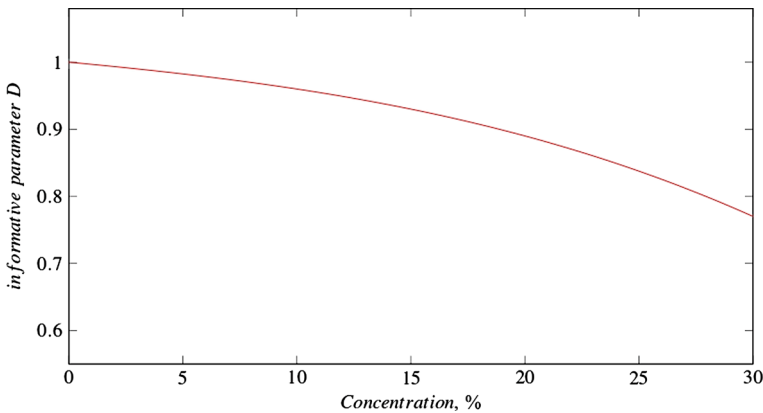


Fig. 5 Dependence of the optical sensor’s information parameter D (‘gas concentration–voltage’) on the level of carbon dioxide concentration

occur. For the developed carbon dioxide monitoring system, the principal components of instrumental error are as follows:

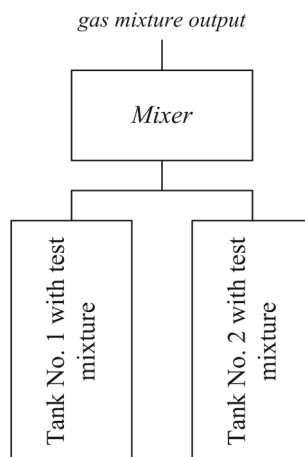
- Radiation source error. In the present system, a laser diode serves as the radiation source. Its emission wavelength is set with a finite deviation determined by manufacturing tolerances. The magnitude of this error is specified by the manufacturer and documented in the component’s datasheet;
- Radiation receiver conversion error. This error is intrinsic to spectral measurement instruments and represents a significant contribution to the overall instrumental error;
- Optical transducer errors. These arise within the optical measuring transducer and are primarily influenced by the quality of the inner surface coating. Additional contributions may result from variations in reflection angles or changes in the optical path length caused by surface roughness of the applied coating;
- Auxiliary measurement channel errors. Errors in the temperature and pressure measurement channels can directly influence the readings of the carbon dioxide concentration within the flue gases of boiler plants;
- Analog-to-digital conversion (ADC) error. The dominant component of ADC error is the quantization error, while other contributions are negligible in comparison.

By substituting approximate error values obtained from the datasheets of the system’s constituent modules, the total instrumental error for the developed monitoring and control system is estimated to be approximately 0.5%.

Methodological errors originate from inherent limitations of the measurement technique itself, independent of the quality or precision of the control device, as well as from approximations introduced during the design of the measurement system. The predominant component of methodological error in the present carbon dioxide monitoring system stems from the use of a relative measurement approach. The principal contributions to methodological error include:

- Error of measurement transformation in the mathematical model: Within the operational range, the maximum deviation associated with this component is approximately 0.2%;
- Additivity errors. This comprehensive error term encompasses multiple sources:
 - Characteristic wavelength selection error: Deviations arising from the choice of the spectral wavelength at which the measurement is conducted, estimated according to the absorption spectrum of the target gas;
 - Doppler broadening approximation error. Errors arising from neglecting Doppler broadening of the gas absorption spectral lines in the calculations. This can be quantified using a radiation absorption coefficient that incorporates the Doppler effect (as per the relevant formula);
 - Deviation from the Bouguer–Lambert–Beer law. Errors resulting from the inherent limitations of the law, which underpins the absorption spectroscopy methodology;

Fig. 6 Relationship between the information parameter of the optical sensor and carbon dioxide concentration in the ‘gas concentration–voltage’ characteristic



- Exclusion of radiation scattering effects. Errors associated with neglecting the radiation scattering index in calculations, which may occur due to fluorescence of gas molecules or the reduction in effective path length of the radiation beam within the primary optical transducer

By substituting the corresponding error contributions into the analysis, the total methodological error is estimated to be approximately 0.4%. Consequently, the overall measurement error—defined as the sum of instrumental (0.5%) and methodological (0.4%) errors—is 0.9%.

For a practical assessment of the main metrological characteristics and the adequacy of the developed mathematical model of the gas concentration-voltage measuring transducer, we use the following diagram (Fig. 6).

The cylinders with the test gas mixture (under a pressure of 85 atm.) are used as *No. 1* CO₂ [concentration of DSTU 4817:2007 (99.99%_{vol.})] and *No. 2* N₂ + SO₂ [concentration of DSTU EN 14211:2018 (99.99%_{vol.})]. The gas mixture is fed alternately from the required cylinder through a mixer with a built-in rotameter, which allows you to supply a gas mixture with a known concentration of components to the sensor (Table 1).

Table 1 Composition of cylinders with standardized gas mixture

Numbers of cylinders	Volume fraction of CO ₂ (%vol.)	Volume fraction of N ₂ + SO ₂ (vol.)
No. 1	0	100
No. 2	10	90
No. 3	12	88
No. 4	14	86
No. 5	16	84

Let's calculate the CO_2 concentration according to the mathematical model. Consequently, a set of carbon dioxide concentration measurements was obtained and is presented in Table 2.

Based on the calculations, the value exhibiting the greatest deviation from the true value is 10.092%vol. Thus, the maximum absolute, relative and total error of concentration determination using a gas concentration–voltage measuring transducer is

$$\Delta = X_{\text{measured}} - X_{\text{real}} = 0.092\% \text{vol.} \quad (15)$$

$$\delta = \frac{\Delta}{X_{\text{real}}} = \frac{10.092 - 10.000}{10.000} \cdot 100\% = 0.92\% \quad (16)$$

Based on this, the admissible boundaries of absolute error can be identified

$$\Delta_{\text{limit}} = \pm \frac{\gamma_{\text{limit}}}{100\%} \cdot X_N = \pm 0.1\% \text{vol.} \quad (17)$$

And also to determine the permissible limits of relative error

$$\delta_{\text{limit}} = \pm \gamma_{\text{limit}} \cdot \frac{X_N}{X_{\text{measured}}} = \pm 0.99\% \quad (18)$$

Consequently, the absolute error remains constant and independent of the measured quantity, whereas the relative error increases as the measured value decreases. This implies that the measurement accuracy of the monitoring system diminishes when the device readings are substantially lower than the upper measurement limit.

Having carried out measurements according to the reading of the device X_{reading} , we obtain an estimate of the measured value, the true value of which X lies within

$$\begin{aligned} X_{\text{reading}} - \Delta_{\text{limit}} < X < X_{\text{reading}} + \Delta_{\text{limit}} \\ \Rightarrow -9.9\% \text{vol.} < X < 10.1\% \text{vol.} \end{aligned} \quad (19)$$

The obtained theoretical calculations of the error of the optical sensor are consistent with the results of experimental studies, which allows us to draw a reasonable conclusion about the adequacy of the developed mathematical model.

4 Conclusion

In the paper we have developed a gas concentration-voltage sensor for monitoring the concentration of carbon dioxide in aggressive gas environments was developed using the optical absorption method, featuring enhanced metrological performance.

Table 2 CO₂ concentration according to the mathematical model (11)

No. of cycle	measurement	actual value	temp. °C	pressure Pa	1	2	3	4	5	6	7	8	9	10
0	24\	98,791			0.004	0.001	0.002	0.001	0.003	0.002	0.001	0.001	0.001	0.002
10	24\	98,780			10.035	10.035	10.044	10.092	10.092	10.092	10.092	10.092	10.035	10.035
12	25\	98,745			12.017	12.018	12.007	12.034	12.055	12.043	12.023	12.011	12.015	12.017
14	25\	98,701			14.029	14.027	14.021	14.022	14.029	14.021	14.023	14.028	14.021	14.022
16	25\	98,658			16.017	16.019	16.007	16.009	16.003	16.011	16.015	16.016	16.017	16.015

A mathematical model of radiation transfer to the gas medium with the value of the output voltage of the photodetector, which takes into account the characteristics of the monitoring object, was synthesized.

Laboratory tests using standard gas mixtures have confirmed the adequacy of the developed model and its ability to accurately monitor the CO₂ concentration within the permissible metrological errors. The results obtained indicate the prospects of using the developed measuring transducer in industrial and environmental monitoring systems.

References

1. Kucheruk V, Kurytnik I, Dudatiev I (2013) Automatic control of concentration of flue gases. *Pomiary avtovatyka kontrola* 59(7):621–623
2. Sigrist M (2017) Trace gas monitoring by laser photoacoustic spectroscopy and related techniques. *Rev Sci Instrum* 88(12)
3. Tittel F, Richter D, Fried A (2015). Mid-infrared laser applications in spectroscopy. In: *Solid-State mid-infrared laser sources* (c. 445–510)
4. Hodgkinson J, Tatam RP (2012) Optical gas sensing: A review. *Meas Sci Technol* 24(1):012004. <https://doi.org/10.1088/0957-0233/24/1/012004>
5. Amara S, Aljedaibi A, Alrashoudi A, Ben Mbarek S, Khan D, Massoud Y (2023) High-performance MTJ-based sensors for monitoring of atmospheric pollution. *AIP Adv* 13(3):035329. <https://doi.org/10.1063/9.0000496>
6. Engelbrecht R (2004) A compact NIR fiber-optic diode laser spectrometer for CO and CO₂. *Spectrochim Acta Part A Mol Biomol Spectrosc* 60(14):3291–3298. <https://doi.org/10.1016/j.saa.2003.11.033>
7. Petruk V, Ischenko V (2010) Висококочутливі засоби контролю малих концентрацій газів. VNTU, 2010. – 138 s
8. Vasilevskiy O, Dudatiev I, Ovchynnykov K (2019) Tool control the concentration of carbon dioxide in the flue gas boilers based on the optical absorption method. *Informatyka Automatyka, Pomiary w Gospodarce i Ochronie Środowiska* 9(3):30–34. <https://doi.org/10.35784/iapgos.232>
9. Podzharenko VA, Kulakov PI (2001) Photoelectric angle converter. In: *Svechnikov SV, Kojemiako VP, Kostyukevych SA (eds) SPIE proceedings, vol 4425, p 452. International Conference on Optoelectronic Information Technologies. SPIE.* <https://doi.org/10.1117/12.429768>
10. Giorgetta FR, Peischi J, Herman DI, Ycas G, Coddington I, Newbury NR, Cossel KC (2021) Open-path dual-comb spectroscopy for multispecies trace gas detection in the 4.5–5 μm spectral region. *Laser Photon Rev* 15(9):2000583. <https://doi.org/10.1002/lpor.202000583>
11. Rothman LS, Gordon IE, Barbe A, Benner DC, Bernath PF, Birk M, Boudon V, Brown LR, Campargue A, Champion JP, Chance K, Coudert LH, Dana V, Devi VM, Fally S, Flaud JM, Gamache RR, Goldman A, Jacquemart D et al (2009) The HITRAN 2008 molecular spectroscopic database. *J Quant Spectrosc Radiative Transf* 110(9–10):533–572. <https://doi.org/10.1016/j.jqsrt.2009.02.013>
12. Kucheruk V, Kulakov P, Storozhuk N (2016) Measurement of the number servings of milk and control of water content in milk on stall milking machines. *Adv Intell Syst Comput* 543:435–447. https://doi.org/10.1007/978-3-319-48923-0_46
13. Semenov AO, Baraban SV, Osadchuk OV, Semenova OO, Koval KO, Ayu, Savitskyi (2019) Microelectronic pyroelectric measuring transducers. *IFMBE Proc* 77:393–397. https://doi.org/10.1007/978-3-030-31866-6_72

14. Semenov AO, Savvitskiy AY, Bisikalo OV, Kulakov PI (2018) Mathematical modeling of the two-stage chaotic colpitis oscillator. In: 2018 14th international conference on advanced trends in radioelectronics, telecommunications and computer engineering (TCSET), pp 835–839. <https://doi.org/10.1109/tcset.2018.8336327>
15. Dzhedzhula VV, Yu Yepifanova I (2024) Increase in the energy efficiency of the biogas production process through vibration activation of heat and mass exchange processes in the bioreactor. IOP Conf Ser Earth Environ Sci 1415(1):012017. <https://doi.org/10.1088/1755-1315/1415/1/012017>
16. Semenov A, Voznyak O, Osadchuk O, Baraban S, Semenova O, Rudyk A, Klimek J, Orazaliyeva S (2019) Development of a non-standard system of microwave quadripoles parameters. In: Romaniuk RS, Linczuk M (eds) Photonics applications in astronomy, communications, industry, and high-energy physics experiments 2019, p 80. <https://doi.org/10.1117/12.2536704>
17. Dzhedzhula VV, Yepifanova IY (2023) Neural network model of investment process of biogas production. IOP Conf Ser Earth Environ Sci 1254(1):012103. <https://doi.org/10.1088/1755-1315/1254/1/012103>

Formation of Pharmaceutical Enterprises' Strategy Based on the Development of Operational Capabilities



Inna Kuznetsova , Olha Balabash , and Yuliia Karpenko 

Abstract The article proposes a methodological toolkit that enables a comprehensive examination of the operational capabilities of a pharmaceutical enterprise and the formulation of its development strategy. The study employs a set of methods that enabled the achievement of the stated objective. The EFAS analysis was used to examine the business landscape of the pharmacy segment. An interrelationship diagram was used to identify the root causes of decreased turnover of medicinal products. ABC analysis enabled the evaluation of medicine sales by trade names in terms of profitability and demand. The article presents an EFAS analysis of the business landscape within the pharmacy segment, which has revealed a range of external challenges exerting a considerable impact on the strategic position of enterprises operating in the pharmaceutical sector. The most critical risks were identified as factors related to the competitive environment and supplier activity, in particular marketing pressure, price volatility, and rigid contractual terms. The study also examines internal operational challenges, specifically the decline in pharmaceutical turnover resulting from the presence of low-demand items within the product assortment. Through the application of ABC analysis, an inventory management model was developed that integrates product prioritisation and enables cost optimisation. A logical framework for assortment management was developed that ensures the rational utilisation of resources. The practical value of the study lies in the justification of a business development strategy based on the integration of external threat analysis and internal capability evaluation.

Keywords Management · Enterprise · Strategic management · Enterprise strategy · Development strategy · Operational strategy · External environment of pharmaceutical business enterprises · Enterprise resources · Operational capabilities · Inventory management · Assortment management · Inventory management model · Pharmacy business landscape · Optimisation of operational business processes

I. Kuznetsova · O. Balabash (✉) · Y. Karpenko
Odesa National Economic University, Odesa, Ukraine
e-mail: balabash.olga@gmail.com

1 Introduction

The modern business environment of the pharmaceutical sector is characterised by a high level of dynamic change and competitive pressure. Under such conditions, the effectiveness of enterprises' strategic development largely depends on their ability to adapt to changes through the mobilisation of internal potential and the optimisation of operational processes. The development of an enterprise's operational capabilities—conceived as a combination of resources, competencies, and managerial technologies—plays a key role in ensuring its competitiveness and its ability to achieve strategic objectives. The implementation of operational capabilities enables effective adaptation to the external environment and facilitates the formulation of a development strategy. Given these considerations, the importance of strategic management focused on operational capabilities increases significantly, thereby highlighting the relevance of the chosen research direction.

The aim of this article is to improve the methodological toolkit that enables a comprehensive analysis of the operational capabilities of a pharmaceutical business enterprise and the formulation of its strategy.

Objectives:

- to identify the operational capabilities of pharmaceutical enterprises;
- to establish cause-and-effect relationships between existing operational problems and capabilities in pharmaceutical enterprises;
- to conduct an ABC analysis of operational capabilities and determine optimal approaches to their implementation depending on their group classification;
- to develop a logically formalised model for managing operational capabilities.

Methods: The time series method—the dynamics of the volumes of products sold by pharmacies were examined; EFAS analysis—to investigate the specific features of the pharmacy business landscape that must be taken into account when formulating the development strategy of a pharmaceutical business enterprise; an interrelationship diagram was used to investigate the underlying problems contributing to the decline in the turnover of medicines in pharmaceutical enterprises. Using ABC analysis, the volumes of medicine sales by trade names were examined in terms of their profitability and customer demand. To address the problem of declining product turnover, a methodology for managing the assortment of medicines in a pharmacy was formalised using a logic diagram.

The practical significance of the study lies in the development of a toolkit for the strategic management of the pharmacy business based on the development of the enterprise's operational capabilities.

2 Literature Review

The theoretical aspects of the formulation and implementation of enterprise strategies have been reflected in the scholarly contributions of researchers. The conceptual model of the actions required for strategy implementation, as well as the methods for measuring the degree of its execution, have been studied by researchers Oliveira et al. [1]. The interrelationship between strategy and external environmental factors is also the subject of scholarly debate. The contentious issues related to the implementation of specific types of strategies that enable adaptation to change or the maintenance of stability to sustain market position are raised by authors Anwar and Hasnu [2]. In particular, the sequence of firms in their strategic choices or their adaptation to environmental conditions, as well as the effectiveness of the selected type of strategy in relation to firm size and the industry in which they operate, is the subject of scholarly discussion. In the study by Ozkan [3], it is argued that the primary reason for firms exiting the market is, in most cases, the misalignment of the firm's strategy with the risk environment of the market. The distinctive aspects of strategy development for new enterprises operating in complex environments have been explored by Du and Kim [4]. The researchers propose using fuzzy qualitative comparative analysis to identify different strategic configurations of entrepreneurial and market orientation under complex conditions of hostility, competition, and a lack of institutional support. Studies of the mutual influence between enterprise strategy and market structure, as well as the behaviour of market participants (the so-called market stimulation strategies), have been conducted by Sthakopoulos et al. [5]. It is argued by researchers that the outcomes of strategy implementation are reflected in a firm's financial indicators, customer metrics, reputation, and sustainable competitive advantage, while strategy implementation is driven by the introduction of both radical and incremental product innovations.

The issue of strategy formulation for enterprises in specific sectors is actively discussed in the scholarly literature. In particular, the study by Yin J., Wei Sh., Chen X., and Wei J. examines the alignment between firms' competitive strategies and their strategic roles in the IT industry, as well as how such alignment influences firm performance [6]. The process of enterprise strategy formation, including its stages—in particular, the stage of external environment analysis in accordance with the industry life cycle concept—has been examined by Karpenko et al. [7]. The proposed developments have been applied with regard to enterprises in the agricultural sector, on the basis of which a development strategy for these enterprises has been recommended. A methodology for identifying and analysing the determinants of industry development, which enables the formulation of a set of applied recommendations for developing the strategy of cable industry enterprises, has been proposed by researchers Balabash et al. [8]. The formation of strategies aimed at enhancing the energy resilience of manufacturing enterprises, including issues of ensuring reliability, relevance, and rigour in addressing these challenges, is examined by researchers Lebepe and Mathaba [9]. There is a wide range of scholarly developments for tourism enterprises: Islam and Sadhukhan [10] have identified

key strategic directions for the development of creative tourism, while Chandra and Kumar [11] have proposed promising strategies for the development of sustainable tourism business.

Despite the existing scientific research and available scholarly developments, the problem of formulating a strategy for pharmaceutical enterprises—one that takes into account the specific features of industry development and firms' operational capabilities—remains unresolved.

3 Overview of Economic Development Indicators for the Ukrainian Pharmaceutical Sector

The pharmaceutical sector occupies a strategically significant place in the national economy. According to Fig. 1, pharmaceutical manufacturing accounts for 2.3% of share of GDP contributed by industry, which is lower than the share of the automotive sector (2.7%) but higher than that of the chemical industry (2.2%). In terms of capital investment, the pharmaceutical sector represents 1.7%, while its share of total output sold stands at 1.6% [12].

According to [13], the share of domestically produced medicines in total sales increased during 2023–2024, supported by Ukrainians' desire to support national manufacturers (Fig. 2).

An analysis of the leading players in Ukraine's pharmaceutical market indicates the dominance of domestic companies, which demonstrate not only stable market

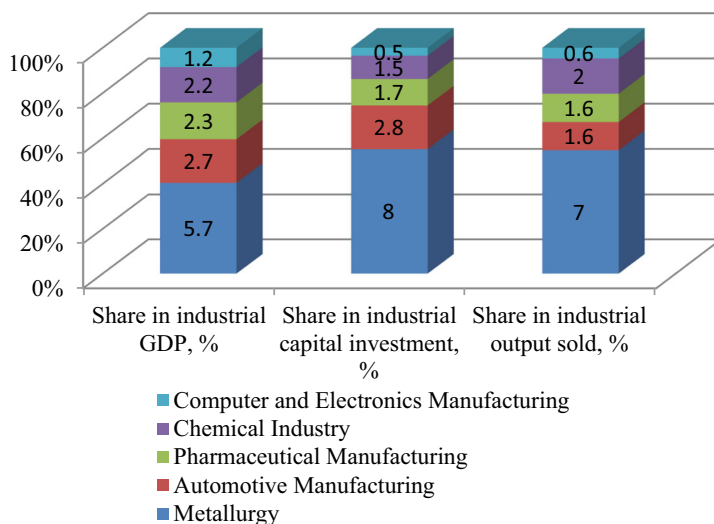


Fig. 1 Position of pharmaceuticals among Ukraine's other economic sectors

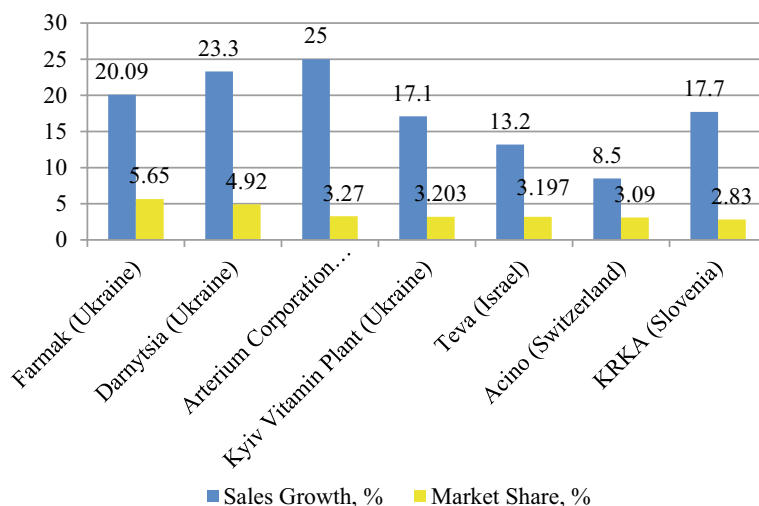


Fig. 2 Market leaders in Ukraine's pharmaceutical sector in 2024

presence but also dynamic growth. The largest market share is held by Farmak, accounting for 5.65% with a sales growth rate of 20.09%. In second place is Darnysia, with a share of 4.92% and an even higher growth rate of 23.3%. The third position in terms of growth is occupied by the Arterium Corporation, which, with a market share of 3.27%, recorded a sales growth of 25% [14].

This indicates active competition among domestic manufacturers, who successfully adapt to the challenging market conditions.

The pharmacy retail market in Ukraine developed dynamically during 2018–2021, with sales volumes increasing each year. Annual growth ranged from 8 to 20%. An exception was 2022 (the onset of active military operations), when sales volumes declined by 6%. However, in 2023, this decline was offset by a 25% increase (see Fig. 3) [14].

Experts forecast that the market will grow annually by 16–18%. The continuation of hostilities, the increase in blast and other injuries among the civilian population, and the rising overall psychological burden on society also contribute to the growth of the pharmacy retail market.

As for medicine purchasers, 88% of medicines are bought by patients, while 12% are purchased by the state under the reimbursement programme (Fig. 4) [14].

Over the past year, medicine prices—and consequently the average receipt—have increased by 16%, and by 11% to 18% for certain categories of medicines. The smallest price increase was recorded for dietary supplements (see Fig. 5) [15].

It should be noted that over the years, the number of legal entities as well as sole proprietors has been declining (the share of sole proprietors fluctuates at around 69–70% of the total number of pharmacies). However, pharmacy chains continue to expand (a single enterprise—a large pharmacy chain—can include from 1000 to

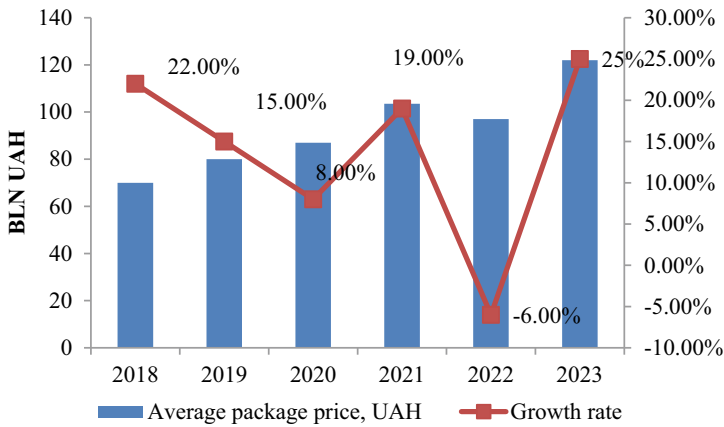


Fig. 3 Dynamics of pharmacy sales volumes, UAH billion

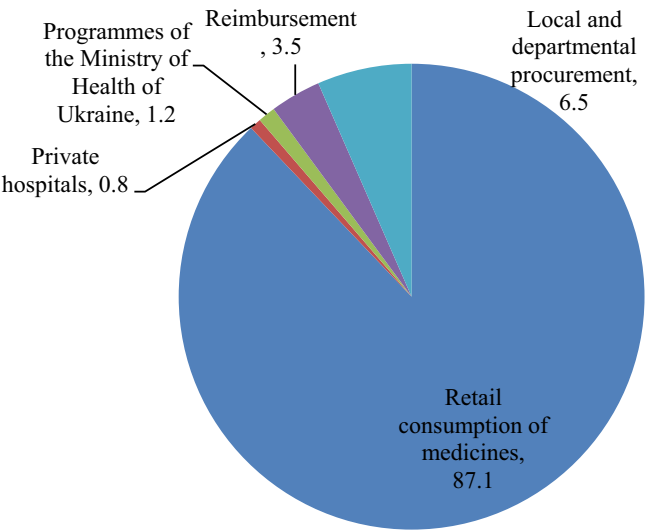


Fig. 4 Structure of medicine consumption by source of funding

2000 pharmacies). A growth trend for pharmacy chains can be observed: during the first ten months of 2024, 977 new pharmacies were opened, 377 of which belong to “Podorozhnyk” [16].

As the market continues to grow (see Fig. 3) against the backdrop of a declining number of pharmacies, these trends indicate market concentration and the crowding out of small pharmacies in favour of the expansion of national pharmacy chains. This is confirmed by the increase in annual sales volume per enterprise, expressed in thousand UAH (see Table 1) [12].

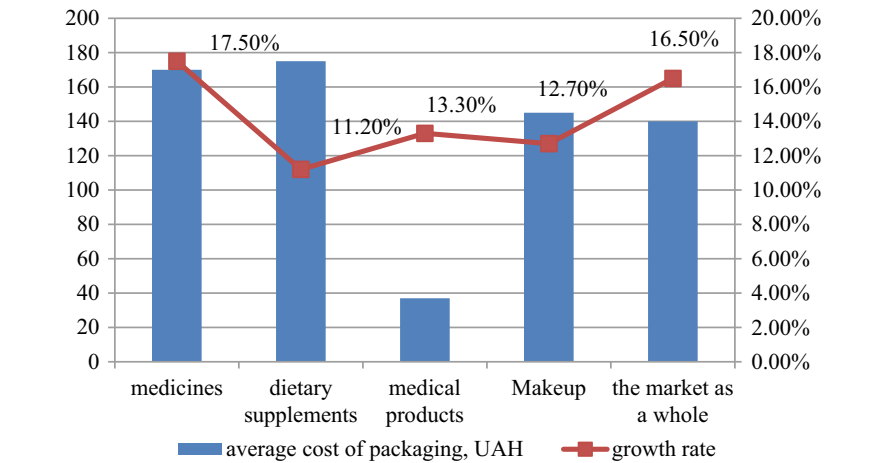


Fig. 5 Dynamics of the average price per package for different categories of the “pharmacy basket” as of the first nine months of 2024, indicating growth or decline rates compared to the previous year

Table 1 Dynamics of sales volume per pharmaceutical enterprise, thousand UAH

Pik	Sales volume per pharmaceutical enterprise, thousand UAH	Growth rate	
		Absolute, thousand UAH	Relative, %
2014	4814.46	–	–
2015	5990.51	1176.05	24.4
2016	8091.29	2100.77	35.1
2017	9930.20	1838.91	22.7
2018	12,441.28	2511.08	25.3
2019	14,894.85	2453.58	19.7
2020	18,517.00	3622.14	24.3
2021	24,687.84	6170.84	33.3
2022	27,466.87	2779.03	11.3
2023	33,890.73	6423.86	23.4

If we consider the largest pharmacy chains, they currently control more than 70% of the total turnover of medicines (see Table 2) [17].

As can be seen, the largest pharmacy chain is Podorozhnyk (2823 units), which currently accounts for 15% of the total number of pharmacies and 18% of the total turnover of medicines. The smallest among the listed chains is Dobroho Dnya (1123 units), which represents 6.24% of the total number of pharmacies and 7.4% of the total turnover of medicines, respectively. Thus, it can be concluded that competition for customers among pharmacy chains is fairly active.

Alongside the growth in sales volumes, domestic pharmacies are increasingly facing the problem of excessive inventory accumulation. One of the key factors

Table 2 Information on the largest pharmacy chains in Ukraine as of 2024

Name of chain	Number of pharmacies		Share in total turnover of medicines, %
	Units	%	
Podorozhnyk	2823	15.00	18.0
911	1813	10.07	13.7
ANC	1429	7.94	12.4
Bazhayemo zdorovya	1242	6.90	11.7
Dobroho Dnya	1123	6.24	7.4
Others	–	53.85	36.8

contributing to this is the marketing pressure exerted by distributors [18, 19]. Within their commercial policy, suppliers often employ aggressive sales promotion strategies. As a result, warehouse stocks become oversaturated, the share of low-turnover products increases, which leads to a decline in the liquidity of the pharmacy assortment, the freezing of working capital, a higher risk of write-offs for products approaching their expiry dates, and an imbalance between demand and availability. This issue is particularly critical for small and medium-sized pharmacies, which lack advanced analytical forecasting systems and are more vulnerable to the influence of the external marketing environment.

To summarise the results of the external environment analysis and to identify the factors to which the greatest degree of adaptation is required, an EFAS analysis of the pharmacy business landscape will be conducted.

The EFAS analysis will be carried out according to the methodology described in [20]. This approach makes it possible to take into account the specifics of the pharmacy business landscape and to identify the key strategic factors that need to be considered in operational activities.

In the EFAS analysis (External Strategic Factors Analysis Summary) of the business landscape, the analysis is carried out by segment: competitors, suppliers, and consumers.

A three-point scale was used for the evaluation. The results of the analysis are presented in Table 3.

The results of the EFAS analysis of the pharmacy business landscape indicate the presence of significant external challenges and potential. The greatest impact among external factors has been recorded in the segments of the competitive environment and suppliers, where threats prevail, including marketing pressure, price policy instability, strict contractual conditions, and risks of declining supply quality. These identified trends should be taken into account when formulating the development strategy of a pharmaceutical business enterprise, particularly in the context of implementing supply models aimed at cost reduction and greater flexibility [21].

Table 3 EFAS analysis of the pharmacy business landscape

Characteristics of the business landscape by segment	Threats “–” if a threat	Opportunities “+” if an opportunity	Score of impact
<i>1. Competitors segment</i>			
1.1 Growth rates and increasing market share of competitors	–		2.0
1.2 Ability to adapt quickly to market changes		+	0.5
1.3 Ability to respond quickly and effectively to competitors' actions	–		0.5
1.4 Active use of marketing measures to boost sales		+	2.0
1.5 Active use of price competition and ability to compete by reducing costs	–		2.0
1.6 Positive reputation among consumers	–		1.0
Total:			8.0 / 7 = 1.14
<i>2. Consumers segment</i>			
2.1 Growth rates of the consumer market		+	1.0
2.2 Changes in demand due to shifts in consumer preferences	–		0.5
2.3 Societal trends that shift consumers' basic needs	–		1.0
2.4 Increasing price sensitivity of buyers	–		1.5
2.5 Emergence of substitute products			0.5
Total:			4.5/5 = 0.9
<i>3. Suppliers segment</i>			
3.1 Marketing pressure from distributors	–		2.5
3.2 Changes in pricing policy	–		2.0
3.3 Changes in supply quality	–		1.5
3.4 Emergence of new suppliers		+	0.5

(continued)

Table 3 (continued)

Characteristics of the business landscape by segment	Threats “–” if a threat	Opportunities “+” if an opportunity	Score of impact
3.5 Strict supply conditions under contracts	–		1.5
Total:			8.0/5 = 1.6

4 Results

The development of a company’s strategy involves conducting an audit, which primarily consists of examining its ability to produce goods or provide services with characteristics that meet market requirements [22, 23]. As a result, two profiles are built—market requirements and the operational capabilities of the enterprise—which are compared to identify discrepancies (so-called strategic gaps). Based on the results of the profile comparison, management decisions should be made regarding the development of the operational capabilities of the business entity or the application of tools to influence the formation of market requirements [24, 25]. Since the analysis of the pharmaceutical market made it possible to draw conclusions about the significant and contradictory influence of drug suppliers on the development of market enterprises, attention should be focused on developing internal capabilities to make more effective use of available operational resources.

Most pharmaceutical companies today face the problem of reduced drug inventory turnover and product expiration. To investigate this issue, we propose using an interrelationship diagram. An interrelationship diagram is one of the seven quality management tools that allows analysts to identify logical or cause-and-effect relationships between problems and factors or events; it helps to structure the analysis of complex problems where many elements are interconnected; and it visualises the influence of some factors on others, which supports the identification of key causes or critical points for decision-making. An interrelationship diagram is used when a problem is complex or unclear and when it is necessary to clarify the relationships among numerous interconnected elements.

The interrelationship diagram developed to address the problem of low turnover of goods is presented in Fig. 6.

As can be seen from the diagram, the underlying issue behind the low turnover of goods is the presence of low-demand items within the medicines assortment. Therefore, it is advisable to implement measures for assortment optimisation by analysing the sales volumes of medicines under different trade names, grouped by international non-proprietary names, in terms of their profitability and demand among pharmacy customers.

Thus, Group A includes only five items, or 25% of the total number of Ibuprofen products, which account for 70.2% of total sales volume. Group C, in contrast,

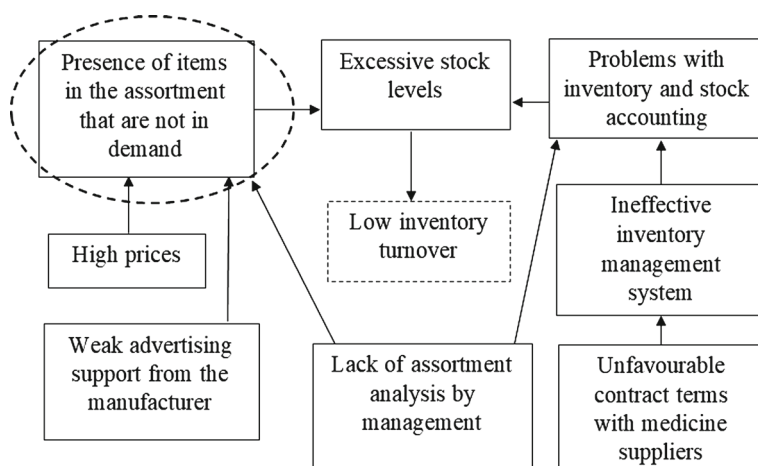


Fig. 6 Interrelationship diagram for the problem of low inventory turnover

includes nine items whose combined share in sales amounts to 8.1%. The approaches to inventory management for medicines in these groups should differ accordingly.

In particular, we conducted an ABC analysis of medicine sales within the pharmacy chain for the international non-proprietary name “Ibuprofen” (see Fig. 7).

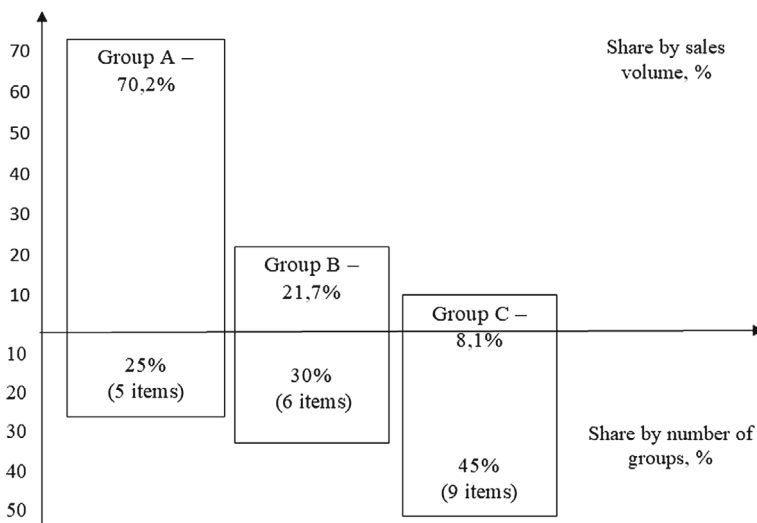


Fig. 7 Results of ABC analysis of sales volumes of medicines with the international non-proprietary name “Ibuprofen”

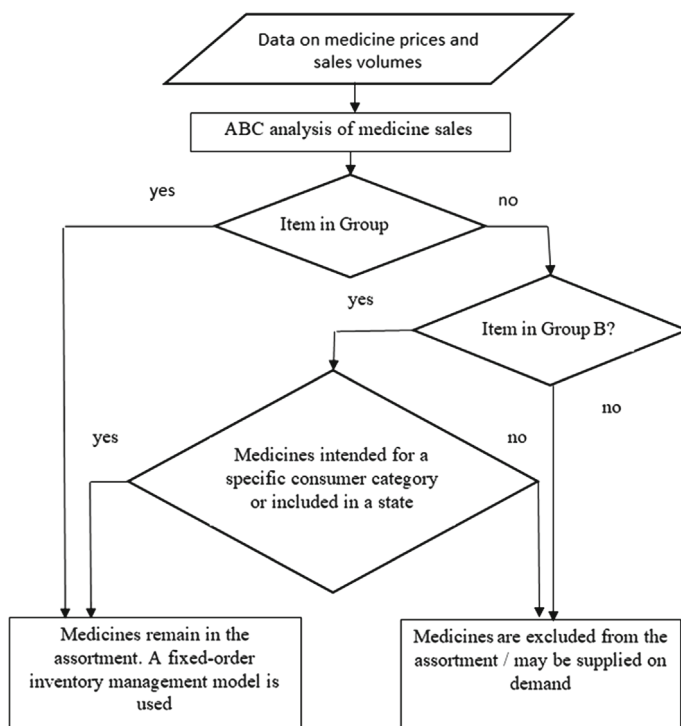


Fig. 8 Logical scheme for assortment management of medicines in a pharmaceutical enterprise

To optimise the management of medicine inventories and taking into account the demand for medicines by their trade names, we have proposed a logical scheme for assortment management (see Fig. 8).

For the ordering of medicines that fall into Group A, a fixed-order inventory management model should be used, as the availability of these medicines in the pharmacy is critical for the customer. The fixed-order inventory management model assumes that the pharmacy continuously monitors the stock levels of these medicines and places each subsequent order as soon as the stock decreases to a certain critical level. Medicines in Group B may remain in the assortment selectively: if they are intended for a specific consumer category—such as children, diabetics, and so on—or if they are included in state programmes (such as the “Available Medicines” programme). In such cases, they may also be ordered in accordance with the fixed-order inventory management model. Purchases of medicines in Group C should be reduced, as these products are not in high demand among consumers and can always be replaced with medicines containing the same active ingredient.

5 Conclusion

Based on the results of the EFAS analysis of the pharmacy business landscape, the presence of significant external challenges has been established. The greatest impact has been recorded in the segments of the competitive environment and suppliers, where threats prevail, in particular marketing pressure, price policy instability, strict contractual conditions, and the risks of declining supply quality. It has been substantiated that these identified trends must be critically taken into account when formulating the development strategy of a pharmaceutical business enterprise.

To make more effective use of the available operational resources and capabilities of pharmaceutical enterprises, the underlying problems associated with reduced medicine turnover were investigated. In particular, by means of an interrelationship diagram, the main cause of low turnover was identified as the presence in the assortment of medicines that are not in demand. The analysis of medicine sales volumes by trade names in terms of their profitability and customer demand, carried out using ABC analysis, made it possible to distinguish groups of medicines for which inventory formation requires different approaches. A methodology for managing the pharmacy's assortment of medicines based on ABC analysis and a fixed-order inventory management model has been proposed and formalised through a logical scheme. In particular, for all medicines in Group A and selectively for those in Group B, it is recommended to apply the fixed-order inventory management model; medicines in Group C should preferably be supplied on demand.

The conducted study has formed the basis for the development of the company's strategy, as it has enabled the identification of critical shortcomings in assortment management and the development of an effective model for inventory optimisation. This has ensured more rational use of resources, adaptation to actual customer needs, and cost reduction. The results obtained have enabled the company to formulate a development strategy based on objective analytics.

References

1. Oliveira CA, Carneiro J, Esteves F (2019) Conceptualizing and measuring the strategy execution construct. *J Bus Res* 105:333–344. <https://doi.org/10.1016/j.jbusres.2018.03.012>
2. Anwar J, Hasnu SAF (2017) Strategic patterns and firm performance: comparing consistent, flexible and reactor strategies. *J Organ Chang Manag* 30(7):1015–1029. <https://doi.org/10.1108/JOCM-03-2016-0053>
3. Ozkan KSL (2020) International market exit by firms: misalignment of strategy with the foreign market risk environment. *Int Bus Rev* 29(6). <https://doi.org/10.1016/j.ibusrev.2020.101741>
4. Du Y, Kim PH (2021) One size does not fit all: strategy configurations, complex environments, and new venture performance in emerging economies. *J Bus Res* 124:272–285. <https://doi.org/10.1016/j.jbusres.2020.11.059>
5. Stathakopoulos V, Kottikas KG, Painesis G, Theodorakis IG, Kottika E (2022) Why shape a market? Empirical evidence on the prominent firm-level and market-level outcomes of market-driving strategy. *J Bus Res* 139:1240–1254. <https://doi.org/10.1016/j.jbusres.2021.10.057>

6. Yin J, Wei S, Chen X, Wei J (2020) Does it pay to align a firm's competitive strategy with its industry IT strategic role? *Inf Manage* 57(8). <https://doi.org/10.1016/j.im.2020.103391>
7. Karpenko Y, Kuznetsova I, Chykurkova A, Matveyeva M, Hridin O, Nakonechna K (2021) Formation of the enterprise strategy based on the industry life cycle. *Independent J Manage Prod* 12(3):262–280. <https://doi.org/10.14807/ijmp.v12i3.1537>
8. Balabash O, Kuznetsova I, Levytska I, Zaika S, Skrypko T (2022) Strategic management of the enterprise: analysis of sectoral determinants. *Independent J Manage Prod* 13(3):196–214. <https://doi.org/10.14807/ijmp.v13i3.1992>
9. Lebepe P, Mathaba TND (2025) Enhancing energy resilience in manufacturing enterprises: a systematic mapping of challenges to strategies. *J Econ Technol* 3:82–94. <https://doi.org/10.1016/j.ject.2025.01.002>
10. Islam N, Sadhukhan S (2025) Relationship among creative tourism development strategies, creative industries, and activities: a case study of Lucknow, India. *J Destination Mark Manage* 36. <https://doi.org/10.1016/j.jdmm.2024.100988>
11. Chandra P, Kumar J (2021) Strategies for developing sustainable tourism business in the Indian Himalayan Region: insights from Uttarakhand, the Northern Himalayan State of India. *J Destination Mark Manage* 19. <https://doi.org/10.1016/j.jdmm.2020.100546>
12. Governmentportal of the State Statistics Committee of Ukraine: information on the development of pharmacies (2024). Available: <http://ukrstat.gov.ua>
13. Travina O (2024) Pharmaceutical market of Ukraine in the structure of the modern global pharmaceutical market: current state and development trends. *Investments: Pract Experience* 12:192–196. <https://doi.org/10.32702/2306-6814.2024.12.192>
14. Proxima Research—Ukraine: Ukrainian pharmaceutical market: pharmacy sales and forecasts. Available: <https://proximaresearch.com/ua/ua/novini/rynok-likarskyh-zasobiv-ukrayiny-aptechnyj-prodazh-ta-prognozy/>
15. Proxima Research—Ukraine: pharmacy sales for 9 months of 2024. Available: <https://proximaresearch.com/ua/ua/novini/aptechnyj-prodazh-za-pershi-9-mis-2024-roku/>
16. Business Center: pharmacy boom: since the beginning of the year, almost 1000 new pharmacies have opened in Ukraine. Available: <https://censor.net/n3524152>
17. Ovchinnikova O. Who controls the pharmacy market of Ukraine: the rating of the largest chains has been published. *Censor*. Available: <https://censor.net/ua/b3544331>
18. Khmelyarchuk M, Demko I, Kozmuk N, Balueva O, Balabash O (2019) The role of entrepreneurship in strategic management of the tourism industry. *J Entrepreneurship Educ* 22(2). Available: <https://www.abacademies.org/articles/The-roleof-entrepreneurship-in-strategic-management-1528-2651-22-2-320.pdf>
19. Balabash O, Kuznetsova I, Karpenko J, Dudziak O, Semenyshena N (2020) Management of biofuel production development on the basis of scenario planning. *J Environ Res, Eng Manage* 76(3):35–46. <https://doi.org/10.5755/j01.erem.76.3.25681>
20. Balabash O, Kuznetsova I (2020) Technologies of strategic management of sustainable development of modern organizations. In: Kuznetsova I (ed) *Disa Plyus*. Kharkiv. Available: <http://dspace.oneu.edu.ua/jspui/handle/123456789/11832>
21. Tkachenko R, Izonin I, Kryvinska N, Chopyak V, Lotoshynska N, Danylyuk D (2018) Piecewise-linear approach for medical insurance costs prediction using SGTM neural-like structure. In: 1st international workshop on informatics and data-driven medicine (IDDM 2018). CEUR workshop proceedings, vol 2255, paper 16, pp 170–179. [online] <https://ceur-ws.org/Vol-2255/paper16.pdf>
22. Stoshikj M, Kryvinska N, Strauss C (2016) Service systems and service innovation: two pillars of service science. *Procedia Comput Sci* 83:212–220. <https://doi.org/10.1016/j.procs.2016.04.118>
23. Kryvinska N, Kaczor S, Strauss C, Greguš M (2014) Servitization—its raise through information and communication technologies. In: *Lecture notes in business information processing*. Springer International Publishing, pp 72–81. https://doi.org/10.1007/978-3-319-04810-9_6
24. Yepifanova I, Dzhezdzhula V (2021) Modelling of potential level of industrial enterprises. *WSEAS Trans Environ Dev* 17:556–565. <https://doi.org/10.37394/232015.2021.17.53>

25. Voynarenko M, Dzhezdzhula V, Yepifanova I (2016) Modelling the process of making decisions on sources of financing of innovation activity. *Econ Ann-XXI* 160(7–8):126–129. <https://doi.org/10.21003/ea.v160-25>

Data-Centric Innovations for Psychological Support: Integrating LLMs and Vector Search in Professional Web Applications



Inna Baraban , Serhii Baraban , Olha Voitsekhovska ,
Mariia Baraban , and Nazarii Shportun

Abstract This paper describes the design and implementation of the Smart Psychologist Assistant, a data-centric web application that automates routine clinical workflows and augments the decision-making of mental-health professionals. The system leverages a modular client–server architecture in which large language models (LLMs)—GPT-4 (Turbo) from OpenAI and high-throughput Groq deployments of open-weight models—are orchestrated through function calling and tool-use pipelines. Text embeddings generated by the LLMs are stored in PostgreSQL with the PGVector extension, enabling sub-second semantic search across patient dialogues, therapist notes, and knowledge-base articles. Benchmarking on a synthetic workload that mirrors medium-size practices (10 therapists, 500 patients) demonstrates average end-to-end response times below 400 ms and horizontal scalability to thousands of concurrent sessions. The open, service-oriented design facilitates future extensions such as emotion-state inference, multimodal data ingestion, or the integration of

I. Baraban

Faculty of Information Electronic Systems, Vinnytsia National Technical University, Vinnytsia, Ukraine

e-mail: baraban.i.o@vntu.edu.ua

S. Baraban (✉)

Faculty of Computing and Telecommunications, Poznan University of Technology, Poznan, Poland

e-mail: serhii.baraban@put.poznan.pl

O. Voitsekhovska · M. Baraban

Faculty of Intelligent Information Technology and Automation, Vinnytsia National Technical University, Vinnytsia, Ukraine

e-mail: o_voytsekhovska@vntu.edu.ua

M. Baraban

e-mail: baraban@vntu.edu.ua

N. Shportun

Resolver Inc. Kroll, Toronto, ON, Canada

e-mail: nazarii.shportun@kroll.com

additional evidence-based therapeutic frameworks. The proposed approach illustrates how LLM-driven vector databases combined with modern web technologies can create practical, secure, and extensible tools that advance data-centric innovation in psychological support services.

Keywords Intelligent assistant · Psychology · Web application · Artificial intelligence · OpenAI · Groq · PostgreSQL · Vector search · Ethics

1 Introduction

In the modern world, psychological health is becoming increasingly important. The fast pace of life, constant stress, information overload, and social challenges are causing an increased demand for qualified psychological help. At the same time, opportunities for personal consultations with a psychologist remain limited due to a lack of specialists, time or geographical constraints, and financial factors. In this context, technological solutions that can support psychologists in their professional activities and provide a basic level of interaction with clients in a convenient format are coming to the fore [1].

Artificial intelligence (AI) is gradually being integrated into the field of mental health care. Smart assistants using modern natural language processing models are able to analyze user requests, offer psychologically sound answers, help formulate emotions, remind about self-help techniques, and provide initial contact before communicating with a specialist. Such tools do not replace psychologists, but act as their assistants, capable of automating routine tasks, structuring information, and improving the effectiveness of assistance [2].

This work involves the creation of a web application—a smart psychologist’s assistant that combines an interface for a specialist with the integration of an artificial intelligence language model. Such a tool allows a psychologist to get help in analyzing requests, formulate individualized advice for clients, and improve the quality of sessions by saving the history of requests, modeling dialogues, and context-based prompts.

Modern web technologies allow us to create user-friendly and secure interfaces that meet the requirements of privacy and personal data protection. Particular attention is paid to compliance with ethical standards, in particular, avoiding the replacement of a specialist with an algorithm, as well as providing users with a clear and transparent mechanism for interacting with artificial intelligence [3].

The main scientific and practical result of the work is the development of software for a smart psychologist’s assistant that can be integrated into the workflows of a specialist, facilitating preliminary analysis of client requests, generating answers and recommendations, and automating routine communication.

The aim of the work is to improve the efficiency of psychological practice by developing a web application with an integrated intelligent module that provides

automated support for psychologists in analyzing client requests, managing sessions, and creating therapeutic plans.

To achieve the goal, the following tasks need to be solved:

1. Develop the architecture of the web application taking into account security and functionality requirements.
2. Integrate an AI language model to support the processing of user requests.
3. Implement a user interface for a psychologist and provide convenient interaction management.
4. To test the system for correctness, efficiency, and convenience.

The object of research is the process of automated support for a psychologist using intelligent systems.

The subject of the study is the means and methods of developing web applications with built-in language models for use in psychological practice.

The research methods include analysis and synthesis, design and programming methods, the use of libraries for integrating artificial intelligence models [4], testing methods, and UX analysis.

The scientific and practical result is the creation of a web application that provides interactive assistance to a psychologist, has a user-friendly interface, uses artificial intelligence to generate answers, stores query history, and meets modern privacy requirements.

2 Material and Methods

2.1 *Justification of the Architectural Approach to Building the System*

The projected system should provide a convenient, flexible, and scalable platform for interactive interaction between a psychologist and users, with the ability to use modern AI services, save communication history, and save notes related to sessions. Taking these requirements into account, we chose a multi-tiered client–server architecture with a clear division into three main components: frontend, backend, and external AI services (Fig. 1).

The frontend is implemented using React, TypeScript, and the Material UI (MUI) library. Each of these technologies plays an important role in building the client part of the system, providing flexibility, scalability, and high quality of the user interface.

React [5] is a popular JavaScript library for building user interfaces created by Meta. The main idea of React is to build an interface using components—isolated, independent blocks that can be reused. In the context of this system, React allows you to divide the interface into logical modules: login form, chat with AI, note viewing, navigation bar, etc. This greatly simplifies code development, testing, and maintenance.

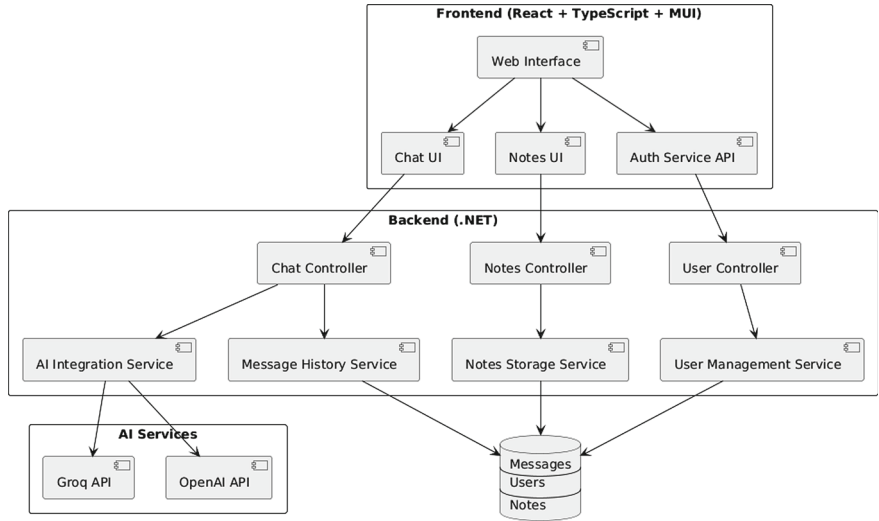


Fig. 1 Diagram of system components

TypeScript [6] is a JavaScript superset that adds static typing. Thanks to types, developers can describe the structure of objects, function parameters, variable types, etc. This improves development security, allows you to detect errors before the program is launched, facilitates autocomplete in the code editor, and simplifies working with APIs. Within a project, TypeScript is especially useful when interacting with the backend, where structured data is transmitted, such as user objects, chat messages, or notes.

Material UI (MUI) [7] is a set of interface components based on the principles of Google Material Design. MUI allows you to quickly create modern and consistent UI elements such as buttons, forms, dialogs, tabs, tables, and more. In the project, MUI is used to build all the main interfaces: an authentication form, a note editor, navigation elements, chat messages, etc. The use of MUI ensures usability, adaptability to different screens, and visual integrity of all parts of the system.

In general, this combination of technologies makes it possible to create a flexible, extensible, and intuitive user interface that works effectively in conjunction with server logic and AI models.

The backend, implemented on the .NET platform [8], acts as a central logical level of request processing, as well as an intermediary between the client side, AI services, and the database. The main controllers are responsible for authentication, message processing, and note management. All requests from the front-end are routed to the relevant services through these controllers. A separate AI integration service ensures the system's interaction with external language model providers—OpenAI and Groq. Both services provide access to modern artificial intelligence models through APIs, which allows you to integrate them into your own infrastructure without having to host the models yourself or spend resources on their maintenance.

OpenAI [9] offers powerful proprietary models, such as GPT-4.1, which demonstrate high quality text generation, contextual analysis, summarization, and other cognitive tasks. These models are well suited for implementing intelligent chats, assistants, and agents that work with large contexts, follow instructions, and analyze textual data.

Groq [10], in turn, provides access to open source models (e.g., LLaMA [11], Mistral [12], Gemma [13]) and features an innovative hardware architecture based on its own processors—LPUs [14] (Language Processing Units). In practice, this means a significantly faster response speed when executing queries to models, even with complex or large computational tasks. Thanks to this, Groq integration allows the system not to waste time waiting when generating large responses or processing large amounts of text, for example, when analyzing notes, summarizing chat history, or working with vector queries.

Thus, the system can dynamically choose between OpenAI and Groq depending on the type of query, performance priorities, or budget. The entire logic of routing and processing model queries is concentrated inside the AI service, which simplifies the overall architecture and allows scaling functionality without interfering with other components.

In addition, separate services for storing message history and working with notes have been implemented.

Special attention is paid to data storage. All data is centrally stored in the PostgreSQL database [15], which is logically divided into three parts: user storage, message storage (chat history), and psychologist's note storage. Such segmentation provides flexibility in accessing different types of information, allows scaling individual parts of the system as needed, and improves overall performance.

One of the key reasons for choosing PostgreSQL as the main DBMS was the availability of the PGVector extension [16], which provides native support for storing and searching vector data representations. This is especially relevant for tasks related to vector search in notes or dialog history. For example, each note or message can be represented as a feature vector (embedding) derived from a language model. These vectors are stored in a special vector column, which allows you to efficiently find similar records using approximate nearest neighbor search operations—without the need to create a separate storage or use external services.

In addition, PostgreSQL is great for containerization and can be easily deployed in a Docker environment [17], which allows for flexible configuration and storage management. If necessary, the database can even be mounted inside a desktop or mobile application as local storage—for example, in cases of offline or offline operation. This expands the possibilities for future scaling and adaptation of the system to different use cases.

This architectural approach allows for a high level of modularity and future extensibility. For example, it is easy to add new AI sources or functionality (such as a session scheduler or analytics) without changing the fundamental parts of the system. Moreover, a clear division of responsibilities between services greatly simplifies testing, debugging, and maintaining the system.

Thus, the chosen architecture meets the modern requirements for reliable and scalable web systems and ensures effective interaction of all system components in real time.

2.2 Rationale for Choosing a Language Model of Artificial Intelligence

When building a modern system that uses artificial intelligence to interact with a user and support a psychologist, it is important to clearly define what types of machine learning models should be involved. Each model in the system has its own role: one is responsible for finding related content, another for conducting a conversation, and yet another for following complex instructions and analyzing data. It is the division of responsibilities between the models that allows us to achieve high accuracy and speed without putting too much stress on individual components.

The proposed architecture provides for the use of three categories of models:

- Embedding models—for searching and contextualizing data.
- Language models—for chatting.
- AI agents to execute analytical or complex instructions.

An embedding model converts a text fragment into a vector—a set of numbers that mathematically reflects the meaning of this text. This allows you to search not by keywords, but by semantic proximity. For example, if a psychologist has created a note about “anxiety and restlessness,” the system will be able to find it even if you search for “tension and excitement,” although there may be no specific word matches.

Vector representations allow you to:

- Perform a semantic search on notes or chat history.
- Store vectors in a PostgreSQL database with the pgvector extension.
- Identify similar records using cosine distance or other similarity metrics.

For example, the text-embedding-3-small model from OpenAI is suitable for this purpose, which gives a vector of 1536 features for each text. Once generated, the vector can be saved and used for each search without the need to recalculate it.

The dimensionality indicates the number of values that can fit in one vector, for example, 1536, which is the standard, means that the vector will have the form: [1, 2, 3, ..., 1536]. Only for vectorization, not integers are used, but with a certain value after the decimal point, it can even go up to millionths. And it turns out that after processing certain content, it is stored in the form of vectors in a vector database and serves as a contextual unit for finding relevant results during the next query (Fig. 2).

To search for relevant vectors, the Euclidean distance formula [18] is usually used—it is a way to measure the distance between two points in space. It is defined as

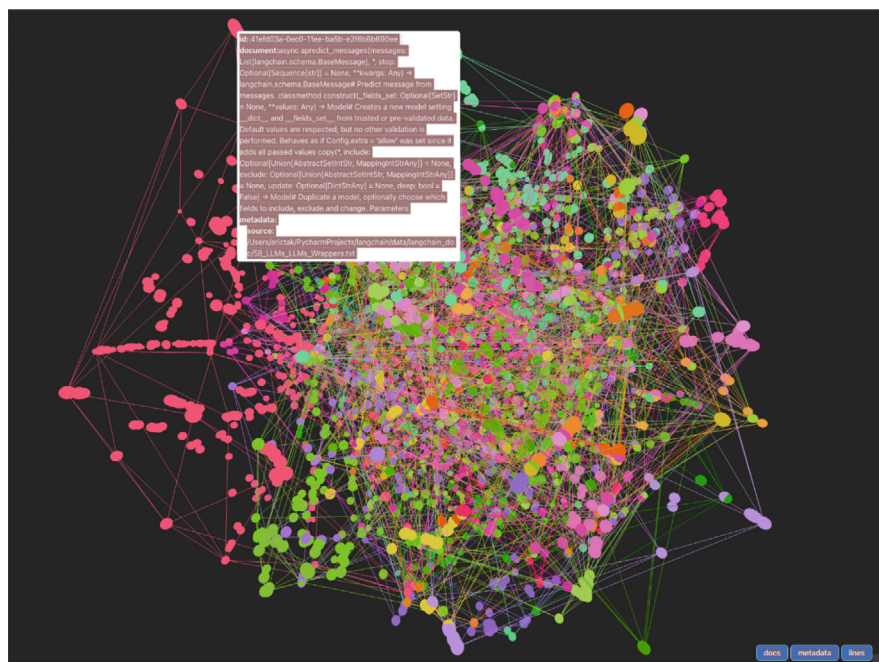


Fig. 2 Visualization of the vector storage

the length of the line connecting these points and is calculated using the Pythagorean theorem based on their coordinates.

For example, in two-dimensional space, the distance between the points $A(x_1, y_1)$ and $B(x_2, y_2)$ is calculated using the equation

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \tag{1}$$

In general, for an n -dimensional space, it is calculated by the equation

$$d = \sqrt{\sum_{i=1}^n (x_i - y_i)^2} \tag{2}$$

This metric is widely used in machine learning, computer vision, and clustering, as it allows you to estimate how close objects are in a multidimensional space.

The basis of the dialog interface is a large-scale language model (LLM—Large Language Model), which responds to user messages by understanding the context. Models such as OpenAI’s GPT-4-turbo or Meta’s LLaMA 3 are trained on billions of texts and are capable of supporting meaningful conversations.

Features of such models:

- token-by-token response generation, taking into account the entire history of the dialog;
- support for long contexts (from 8000 to 128,000 tokens depending on the model);
- controlling the behavior of the model using a system prompt (instruction);
- connection via API or integration through an intermediate service.

This allows users to receive not just responses to requests, but meaningful, logically constructed responses that take into account both the tone and purpose of the interaction [19].

For more complex tasks, such as analyzing message history, preparing reports, or generating interpretations, agents are used. These are not separate models, but rather customized LLM use cases in which the model performs a series of actions based on a goal. An agent can combine text generation, database work, third-party API calls, and even decision-making logic.

In our case, agents are used to:

- generating a psychological portrait of a client based on notes;
 - detecting changes in the user's state between sessions;
 - building session summaries and identifying key topics of conversation;
 - providing recommendations to the psychologist for further actions.
- Platforms that allow you to easily implement such agents are LangChain [20], Dify [21], OpenAI Functions, or your own script with “tool use” support.

Such agents allow you to build a behavior that is closer to a consultant or assistant—not just a request fulfillment, but an active situation analyst.

Building a flexible and scalable system is impossible without dividing tasks between specialized models [19]. Vector representations provide fast and accurate search, language models guarantee natural conversation, and agent scripts add depth to analysis and automation. The interaction between these components creates an efficient system that is able not only to answer queries but also to support the full-fledged work of a psychologist.

2.3 Rationale for Choosing the Platform and Means of Implementation (Web Application)

To implement the system, we chose the approach of building a web application that runs in a browser without the need to install additional software. This format is convenient for psychologists and clients, provides cross-platform compatibility, and allows the application to be scaled to mobile or desktop versions based on the same code in the future.

The client side is implemented using React, TypeScript, and Material UI (MUI), a powerful stack that allows you to quickly create interactive, responsive, and maintainable interfaces. React provides the component structure of the interface, TypeScript

guarantees safe typing and improves integration with the backend, and MUI provides a set of ready-to-use components that adhere to modern design standards.

On the backend, the .NET platform is used, which, thanks to its performance, modularity, and support for modern API standards, is suitable for implementing the logic of request processing, authorization, interaction with the database, and third-party services. The choice of .NET is also due to its extensive support, active community, convenient testing, and the ability to host both in the cloud and on-premises.

AI services, such as OpenAI and Groq, play a special role in the system, providing access to large language models via APIs. These platforms were chosen due to their different advantages: OpenAI stands for stability and quality of closed models, while Groq is characterized by high performance thanks to innovative LPU chips, which is especially important when processing large requests without delays. This combination allows the system to dynamically select the most suitable generation source in each case, balancing performance and cost.

For data storage, PostgreSQL was chosen as a reliable relational database with support for the PGVector extension, which is necessary for vector search. This allows you to store not only structured data but also vector representations of messages and notes, which ensures efficient search by content. PostgreSQL is also easy to containerize and can be adapted to various infrastructures—from cloud services to local environments.

Docker was used in the implementation to isolate services, facilitate deployment, and scale the system. This approach makes it easy to run individual components (database, backend, frontend, AI proxy) independently of each other, which is critical for development, testing, and updates.

Thus, the selected platforms and implementation tools allow us to build a reliable, productive, and scalable system that meets modern requirements for interactive web applications in the field of digital psychological assistance.

2.4 Development of a General Structural Scheme for the Work of a Smart Psychologist's Assistant

A smart psychologist's assistant functions as an intermediary between the user (patient or psychologist) and powerful external AI models, providing personalized support, data storage, and a convenient interaction interface. The focus is on the logic of query processing, context preservation, feedback, and analytics support.

At a general level, the system's workflow consists of the following logical stages:

1. Initiation of interaction—the user logs in and sends a message via the chat interface.
2. Request processing—the message is transmitted to the system that verifies the session, performs routing logic, and completes the context.

3. Calling AI—the generated request is sent to the selected AI provider (for example, OpenAI or Groq).
4. Response generation—AI processes the request, returns a response, which is then supplemented or filtered by the system.
5. Data storage—all messages and responses are stored in the database, along with emails, if the context saving mode is enabled.
6. Feedback—the response is sent to the user and can also be used for analytics or note-taking by the psychologist.

2.5 Software Implementation of the Application “Smart Psychologist’s Assistant”

At the initial stage of development, the user interface design of the Smart Psychologist Assistant web application was created. The main goal of designing the interface was to ensure intuitive user interaction with the system, minimize cognitive load, and meet modern UI/UX requirements.

Modern approaches to web application design were used to create interface layouts, in particular:

- minimalist style: simple shapes, calm colors, clear typography;
- responsive layout: the interface was designed to be displayed equally well on different screen sizes;
- convenient navigation: the placement of elements was based on the ease of access to the main functions of the application (for example, launching a session, viewing notes, viewing analytics).

During the development of the interfaces, several layout options were created, which were evaluated internally for compliance with functional and visual criteria. Based on the selected layout, the final prototype was formed, which became the basis for further implementation in the code.

To create sketches of the layouts, we partially used auxiliary tools with artificial intelligence support, which allowed us to speed up the generation of visual solutions. However, the final structuring of elements, layout logic, and choice of interface style were done manually, taking into account the specifics of the target audience and the functional purpose of the system.

The registration page contains only the necessary information, without any additional burden on the potential user, focusing on intuitive clarity. These are the fields for entering information from the account and the ability to log in using third-party services such as Google or Facebook (Fig. 3).

The main page of the web application has a convenient and logical structure that facilitates quick navigation and efficient access to the main functions. At the top of the interface is a header containing a search field and a notification icon. This allows you to quickly find the information you need in the system and respond to new events or notifications in a timely manner (Fig. 4).

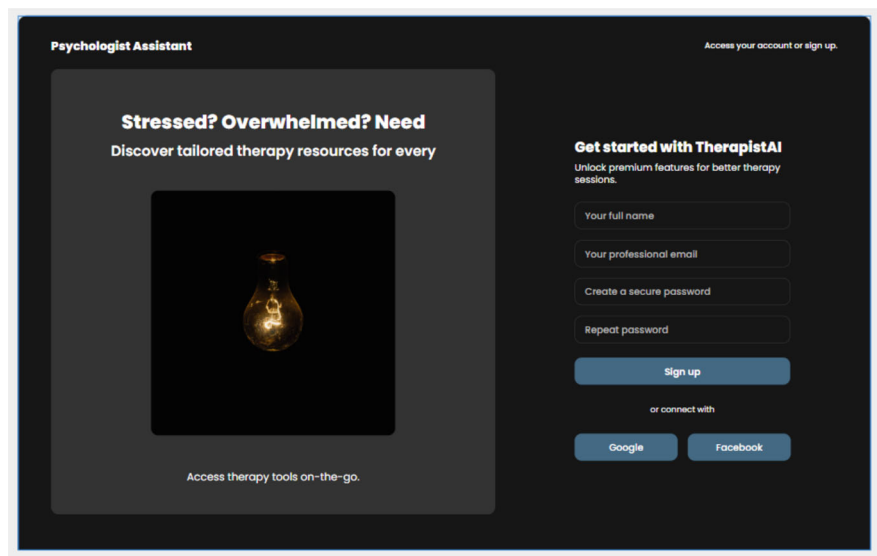


Fig. 3 Registration page

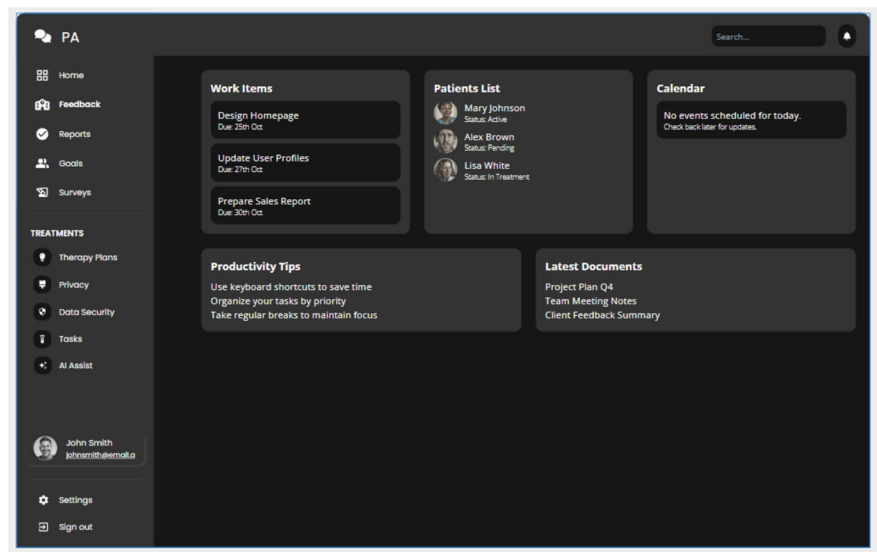


Fig. 4 The main page of the application

On the left is the sidebar, which serves as the main navigation menu. It contains the main sections, such as: home page, reviews, reports, goals, surveys, as well as functional blocks related to treatment, therapy plans, privacy, data security, tasks, and access to the AI assistant. The lower part of the sidebar contains user profile controls: name, email address, and buttons to go to the settings or log out.

The central part contains the key elements that provide quick access to important information and help to organize the work of a psychologist. The main element is a list of current tasks with deadlines for their completion—this allows the specialist to keep track of deadlines and not miss important tasks. Next to it is a list of active patients, which shows their names and statuses, allowing you to instantly assess the situation and make decisions without having to go to other sections.

There is also a calendar that displays scheduled classes, appointments, or events for the current day, allowing you to quickly navigate the schedule. In additional information blocks, you can find tips for improving productivity or view the latest work-related documents. This structure allows psychologists to have everything they need for their daily work at their fingertips and ensures convenient interaction with the system interface.

The AI Assist page is a key element of the intelligent functionality of the Smart Psychologist Assistant web application. It acts as a virtual assistant that helps the user (in our case, a psychologist) to quickly receive consultations, advice, and solutions on working with patients, using the system, and performing routine tasks (Fig. 5).

The central part of the interface is a chat where users interact directly with AI. Artificial intelligence is able to generate answers to user queries, analyze the documents provided, give practical advice on treatment plans, offer therapy options, or

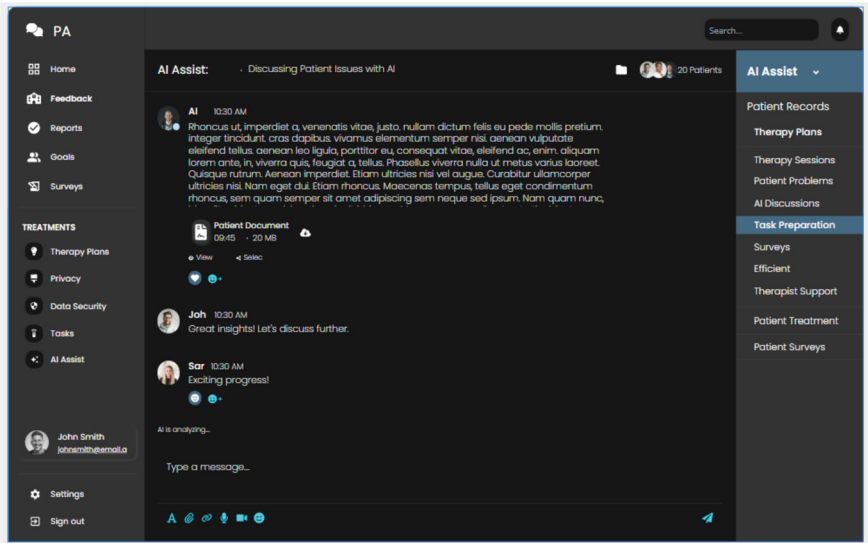


Fig. 5 AI assistant page

help with workflows. For example, a psychologist can ask a question like: “What is the best approach for a patient with anxiety?” and receive a recommendation based on generally accepted methods or the contents of the patient’s card.

The chat also has the ability to send files, such as patient documents, notes, or questionnaires. AI is able to automatically analyze these files, highlight key information, and even generate short summaries or suggestions for further processing. This significantly reduces the workload of a specialist, allowing them to focus on decision-making rather than mechanical data processing.

Other users can participate in the chat, such as colleagues or consultants. This makes it possible to create joint discussions of complex clinical cases or coordinate work between several specialists. At the bottom, there is a text field for entering messages, as well as a set of icons for adding attachments, reactions, or quick actions.

On the right, there is an additional navigation menu that allows you to switch between different sections of the assistant: “Patient Records, Therapy Plans, Patient Problems, AI Discussions, Task Preparation, etc. Each section corresponds to a specific type of consultation or set of actions that AI can perform. For example, in the Task Preparation section, you can discuss with AI how to better organize your work schedule for the week or create a list of tasks by priority.

Thus, the AI Assist page is not just an ordinary chat room—it is a full-fledged tool for integrated analytics, decision support, and navigation of the entire system’s functionality. It is aimed at improving the quality of psychological assistance, increasing the efficiency of a specialist, and creating the most comfortable environment for interaction with patients.

The questionnaire creation page is a key tool for a psychologist or specialist working with patients—it allows you to quickly and conveniently create individual or standard questionnaires aimed at diagnosing psycho-emotional state, assessing the dynamics of therapy, or identifying risks. The interface is designed in such a way that the creation process is intuitive and does not require technical knowledge. A psychologist can easily add questions of different types: single-choice, scale (for example, to assess mood or anxiety), and multiple-choice questions for situations where multiple answers are allowed (Fig. 6).

A special feature of the page is the integration of artificial intelligence, which helps to generate meaningful and professionally formulated questions. The AI Assistance panel on the right constantly suggests contextually appropriate questions that can be added with a single click. For example, the system itself suggests questions such as “How often do you feel stressed?” or “Please rate your mood on a scale from 1 to 10”—these are basic but important queries that are often used in daily surveys. If the user wants to add a unique or situational question, they can use the Generate from Prompt button and simply describe what they want to ask—the system will generate the question automatically.

The created survey is saved in the system and can be used in several ways. First, it can be exported in a document format (e.g., PDF or Word) to be printed and used offline—for example, for patients without access to a computer or in a clinic setting. Secondly, for each survey, you can generate a link that is sent to the patient via email, messenger, or personal account. The patient clicks on the link, completes the survey

Fig. 6 Survey creation page

in a convenient format, and the results are automatically saved in the psychologist's database.

Immediately after the patient completes the survey, the system can visualize the results in the form of graphs, tables, or just a list of answers. This allows the psychologist to respond quickly to critical changes or use the results as a basis for a consultation. In addition, the results can be linked to a specific date, session, or therapy plan, which allows you to build analytics in the dynamics—how the patient's condition changed during the month or the entire period of therapy.

Thus, this page is not just a tool for creating forms—it is a full-fledged module for interacting with patients, collecting data, analyzing and documenting progress. It integrates with other parts of the system, ensuring the continuity and integrity of the psychotherapeutic process.

The therapeutic plans page is the central element of the system, designed to organize, manage, and control all aspects of the psychotherapeutic process for each client individually. This section is designed to meet the real needs of psychologists, psychotherapists, coaches, and other mental health professionals who deal with a large number of patients, long periods of therapy, and the need to clearly track progress (Fig. 7).

On the left side of the page, the “Recent updates” block is displayed, which allows you to quickly see all the latest changes related to therapeutic plans, patient assessments, progress reports, etc. Each entry is accompanied by a brief description (for example, “TherapistAI created a new task” or “TherapistAI marked the treatment complete”), which allows the specialist to keep abreast of the therapeutic process.

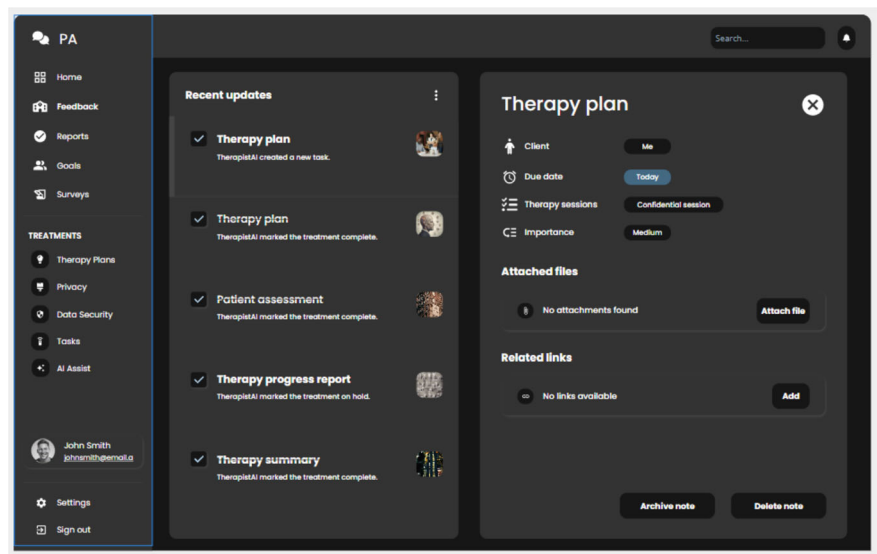


Fig. 7 Therapy plans page

This is especially useful in teamwork or when assisted by artificial intelligence, which can independently create or close tasks based on the collected patient data.

On the right side of the page, a detailed form of an individual therapeutic plan opens. It displays the basic information: for which client the plan is intended (for example, “Me” in the case of a personal plan or testing), the set date (with the ability to filter by urgency), the type of sessions (for example, “Confidential session”) and the importance (for example, “Medium”). This allows you not only to record basic information but also to prioritize tasks, working with the most critical cases first.

An important functionality is the ability to add attachments and related links—documents, reports, tables, medical reports, or even external resources related to a particular case. Thus, the therapeutic plan is not an abstract note—it is a full-fledged case card, where everything that a specialist may need in their work is stored in one place. The “Attach file” and “Add link” buttons allow you to quickly supplement the plan with the necessary information without switching between windows or modules.

Two important options are worth mentioning separately: Archive note and Delete note. The first option allows you to archive completed or temporarily inactive plans without completely deleting the information, which is useful for maintaining treatment history. Deletion is used only when the information is completely outdated or was created by mistake.

As a result, this page is not just a form of recording therapeutic tasks, but a functional tool for managing the psychotherapeutic process. It allows you to see the whole picture of therapy: what has already been done, what is planned, which tasks have the highest priority, what materials are attached, and what patient dynamics are

observed. All of this provides flexibility, reduces administrative burden, and allows you to pay more attention to working directly with the client.

The interface design for the psychotherapist support system combines modern visual aesthetics with high functionality and practicality. It is aimed at facilitating the daily work of a specialist, allowing them to quickly create and edit therapy plans, create surveys for patients, view progress and results, and interact with AI prompts.

Thanks to logically structured screens, well-thought-out placement of elements, history of changes, and the ability to attach files, the system makes managing the treatment process intuitive and efficient. Of particular value is the survey module, which allows you to collect data in a convenient format, export results, share them via links, and receive instant feedback after the patient has completed the survey.

As a result, the interface enhances not only the organizational but also the therapeutic capabilities of the specialist, contributing to a better and more personalized approach to each patient.

The server side of the “Smart Psychologist Assistant” web application was developed using the ASP.NET Core Web API platform, which provides high performance, modularity, and support for modern RESTful API development standards. This approach made it possible to create a reliable, scalable, and secure server infrastructure (Fig. 8) that processes requests from the client side, interacts with the database and external AI services, and provides the logic of business processes.

The development of the server side began with the creation of a basic project using the webapi template in ASP.NET Core. To do this, we used the command:

```
dotnet new webapi -n SmartAssistant.API
```

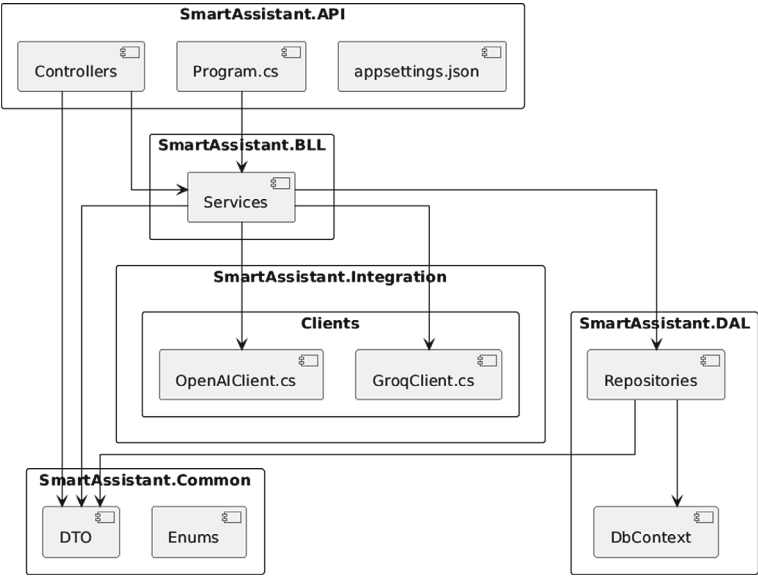


Fig. 8 Server application architecture

This template provides a basic configuration for the REST API, including a minimal set of controllers, middleware for handling requests and dependencies, and basic security settings. To ensure modularity and ease of code maintenance, a multi-layered architecture was created that includes the following projects within a single solution:

- SmartAssistant.API—entry point to the application, controllers, configuration.
- SmartAssistant.BLL—business logic.
- SmartAssistant.DAL—access to the database (Entity Framework Core).
- SmartAssistant.Common—common data structures, DTOs, enums.
- SmartAssistant.Integration—interaction with external services (OpenAI, Groq, etc.).

This structure complies with the principles of clean architecture, which provides a clear separation of duties, simplifies testing, and allows for easy scaling of the system in the future.

To store data such as user information, chat history, psychologist's notes, session settings, and results of interaction with AI services, we chose a PostgreSQL relational database with the PGVector extension to support vector search. PostgreSQL was chosen due to its reliability, support for complex queries, the ability to containerize via Docker, and native support for vector operations, which is critical for implementing semantic search on notes and messages.

The SmartAssistant.DAL project has created the AppDbContext database context, which inherits the DbContext class from the Entity Framework Core. This context defines data sets (DbSet) for the main entities of the system:

- Users—information about users (psychologists and clients).
- ChatSessions—data about interaction sessions (for example, chats between a client and AI).
- Messages—the history of messages in chats.
- Notes—psychologist's notes related to clients or sessions.
- AIResponses—saved responses from AI services for further analysis.
- VectorEmbeddings—vector representations of notes and messages for semantic search.

Data models are defined as classes in the Entities catalog. For example, the Message class includes fields for the message text, session ID, author (user or AI), creation time, and vector representation. Relationships between entities are configured using data annotations and the Fluent API, which ensures the correct mapping of models to database tables.

A repository template is used to isolate the data access logic. Each repository (e.g., UserRepository, MessageRepository) implements a corresponding interface (IUserRepository, IMessageRepository) that provides a single contract for interacting with data. This allows you to abstract business logic from the details of database implementation, simplifies testing, and provides flexibility for replacing the DBMS in the future.

A code snippet for the database context is shown in Fig. 9.



```

public class AppDbContext : DbContext
{
    public DbSet<User> Users { get; set; }
    public DbSet<ChatSession> ChatSessions { get; set; }
    public DbSet<Message> Messages { get; set; }
    public DbSet<Note> Notes { get; set; }
    public DbSet<AIResponse> AIResponses { get; set; }
    public DbSet<VectorEmbedding> VectorEmbeddings { get; set; }

    public AppDbContext(DbContextOptions<AppDbContext> options) : base(options) { }

    protected override void OnModelCreating(ModelBuilder modelBuilder)
    {
        modelBuilder.Entity<Message>()
            .HasOne(m => m.ChatSession)
            .WithMany(cs => cs.Messages)
            .HasForeignKey(m => m.ChatSessionId);

        modelBuilder.Entity<VectorEmbedding>()
            .HasIndex(ve => ve.Embedding)
            .HasMethod("vector_cosine_ops");
    }
}

```

Fig. 9 Partial code snippet for the database context

The business logic of the system is concentrated in the SmartAssistant.BLL project, where services are implemented to handle the main scenarios of the application. The main services include:

- **AuthService**—responsible for user authentication and authorization. It uses JWT (JSON Web Tokens) to generate tokens after a successful login. Tokens are stored in localStorage on the client side and added to request headers for secure endpoints.
- **ChatService**—manages chat logic, including session creation, message storage, and routing requests to AI services.
- **NoteService**—allows creating, editing, deleting, and searching for psychologist notes. It uses vector search to find relevant notes by content.
- **SurveyService**—responsible for creating surveys, saving answers, and generating analytical reports.
- **AIService**—coordinates interaction with external AI services (OpenAI, Groq), including the selection of a model depending on the type of query, processing of answers, and saving of results.

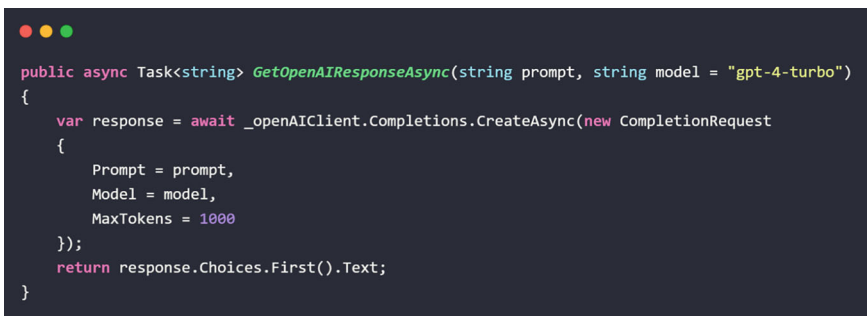
Each service is implemented as a separate class that depends on the respective repositories through interfaces. Dependencies are implemented through the built-in DI container of ASP.NET Core, which ensures modularity and testability of the code.

For example, the ChatService uses IMessageRepository and IChatSessionRepository to store messages and sessions, and calls methods from SmartAssistant.Integration to interact with AI.

To integrate with external AI services, we created the SmartAssistant.Integration project, which contains clients for working with OpenAI and Groq. The OpenAI client is based on the official OpenAI SDK library connected via NuGet. This library provides standardized access to the API, making it easier for to send requests and process responses. For example, a request to the GPT-4-turbo model looks like Fig. 10.

For Groq, which has an OpenAI-compatible API but uses its own models (e.g., LLaMA 3), a custom HTTP client based on HttpClient was created. This allowed us to fine-tune request parameters such as timeouts, headers, and error handling. An example of a Groq request is shown in Fig. 11.

Both clients encapsulate the logic of error handling, logging of requests and responses, and support dynamic model selection depending on the type of request (e.g., Groq for quick responses, OpenAI for complex analytical tasks). The logic of

A screenshot of a code editor showing a C# method named `GetOpenAIResponseAsync`. The method takes a `prompt` string and a `model` string (defaulting to "gpt-4-turbo") as parameters. It uses an `_openAIClient` to create a `CompletionRequest` with the prompt, model, and a `MaxTokens` of 1000. It then returns the text of the first choice from the response.

```
public async Task<string> GetOpenAIResponseAsync(string prompt, string model = "gpt-4-turbo")
{
    var response = await _openAIClient.Completions.CreateAsync(new CompletionRequest
    {
        Prompt = prompt,
        Model = model,
        MaxTokens = 1000
    });
    return response.Choices.First().Text;
}
```

Fig. 10 GPT model request

A screenshot of a code editor showing a C# method named `GetGroqResponseAsync`. The method takes a `prompt` string and a `model` string (defaulting to "llama3-70b") as parameters. It creates an `HttpRequestMessage` with a POST request to the "completions" endpoint, including an "Authorization" header with a Bearer token. The request body is a JSON object containing the prompt, model, and max_tokens. It uses an `_httpClient` to send the request and returns the text of the first choice from the response.

```
public async Task<string> GetGroqResponseAsync(string prompt, string model = "llama3-70b")
{
    var request = new HttpRequestMessage(HttpMethod.Post, "completions");
    request.Headers.Add("Authorization", $"Bearer {_groqApiKey}");
    request.Content = JsonContent.Create(new { prompt, model, max_tokens = 1000 });

    var response = await _httpClient.SendAsync(request);
    response.EnsureSuccessStatusCode();

    var responseData = await response.Content.ReadFromJsonAsync<GroqResponse>();
    return responseData.Choices.First().Text;
}
```

Fig. 11 Query to Groq provider

model selection is implemented in AIService, which analyzes the type of request (text generation, note analysis, survey creation) and directs it to the appropriate provider.

The server side configuration is stored in the appsettings.json file, which contains:

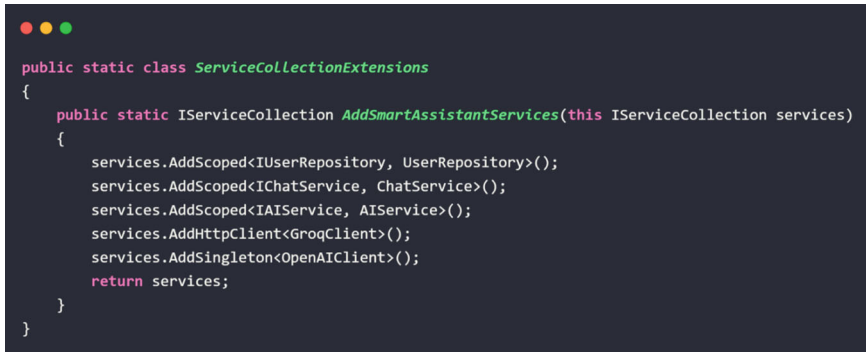
- PostgreSQL connection string.
- API keys for OpenAI and Groq.
- Logging settings (Serilog is used for structured logging).
- JWT parameters for authentication.
- CORS settings for secure interaction with the frontend.

All dependencies (repositories, services, clients) are registered in a DI container in Program.cs using the extensible methods template. For example, see Fig. 12.

This approach provides centralized management of dependencies and simplifies their substitution during testing.

To process HTTP requests from the frontend, a set of controllers has been created in the SmartAssistant.API project. Each controller is responsible for a separate functional module:

- AuthController—handles requests for registration, login, and token updates.
- ChatController—manages the creation of chat sessions, sending messages, and receiving responses from AI (Fig. 13).



```

public static class ServiceCollectionExtensions
{
    public static IServiceCollection AddSmartAssistantServices(this IServiceCollection services)
    {
        services.AddScoped<UserRepository, UserRepository>();
        services.AddScoped<IChatService, ChatService>();
        services.AddScoped<IAIService, AIService>();
        services.AddHttpClient<GroqClient>();
        services.AddSingleton<OpenAIClient>();
        return services;
    }
}

```

Fig. 12 Dependency registration



```

[HttpPost("send-message")]
public async Task<IActionResult> SendMessage([FromBody] SendMessageRequest request)
{
    var response = await _chatService.SendMessageAsync(request.SessionId, request.Message);
    return Ok(new { Response = response });
}

```

Fig. 13 Endpoint for sending a chat message

- **NoteController**—provides CRUD operations for notes and vector search.
- **SurveyController**—responsible for creating, editing, and processing survey results.

The controllers use services from SmartAssistant.BLL to execute business logic and return responses in JSON format.

To protect endpoints, the [Authorize] attribute is used, which checks for a valid JWT token in the request header.

The system security is ensured at several levels:

- **Authentication and authorization:** The use of JWT tokens with verification through the middleware.
- **Data encryption:** All sensitive data (e.g. passwords) is stored in hashed form using BCrypt.
- **CORS:** Configured to restrict access to the API only from authorized domains.
- **Error handling:** Unified exception handling through the middleware, returning structured JSON error messages.

For scalability, the system is deployed in Docker containers, which makes it easy to scale individual components (API, database, AI proxy). The use of PostgreSQL with PGVector ensures efficient processing of large amounts of data and vector queries.

The server side of the Smart Psychologist Assistant web application was successfully implemented using ASP.NET Core Web API, Entity Framework Core, and PostgreSQL. The modular architecture based on the principles of clean architecture provides a clear separation of duties between the levels of API, business logic, data access, and integration with AI services. Integration with OpenAI and Groq allows for flexible processing of queries to language models, and the use of PGVector ensures efficient vector search. The security, scalability, and configuration settings meet modern development standards, making the system ready for further development and integration of new features.

3 Results

Testing the Smart Psychologist Assistant web application is a key development stage aimed at ensuring the system's correctness, reliability, performance, and usability. This subsection describes the testing methodology, which includes unit tests, integration tests, and user interface testing, as well as the analysis of the results. Particular attention is paid to testing the server side (ASP.NET Core), interaction with the PostgreSQL database and external AI services (OpenAI, Groq), as well as assessing the system's compliance with security, ethics, and privacy requirements.

The testing was conducted using a multi-layered approach covering all major components of the system:

1. Unit tests: Checking individual modules of the server side (services, repositories) for the correct execution of their logic. We used the xUnit library together with the Moq mocking framework to isolate dependencies.
2. Integration tests: Checking the interaction between system components, including API controllers, business logic, database, and external AI services. For this purpose, we use built-in ASP.NET Core tools (TestServer) and a local PostgreSQL database in a Docker container.
3. Testing the user interface (UI): Checking the correct display of frontend components (React, TypeScript, MUI) and their interaction with the API. The React Testing Library was used to test the rendering of components and event handling.
4. Performance testing: Evaluation of the response speed of APIs and AI services, as well as the scalability of the system with an increase in the number of requests.
5. Security testing: Checking the protection against typical vulnerabilities (e.g. SQL injections, XSS) and the correctness of authentication/authorization via JWT.

The testing was carried out in an isolated environment using Docker containers for the database and API, which allowed us to simulate the real conditions of the system. A separate project `SmartAssistant.Tests` was created for unit tests, which contains tests for all major services and repositories.

Unit tests were developed to test key components of the business logic in the `SmartAssistant.BLL` project. The main goal is to make sure that each service performs its functions correctly regardless of external dependencies (database, AI services). To isolate dependencies, we used the Moq library, which allowed us to create mock objects for AI repositories and clients.

An example of a unit test for `ChatService`. The service is responsible for message processing and interaction with AI services.

This test checks:

- whether the method of saving the message to the repository is called;
- whether the AI service is called with the correct request text;
- whether the expected response is returned from the AI.

In the `SmartAssistant.Tests` project, 47 unit tests were created that cover the main methods of the `AuthService`, `ChatService`, `NoteService`, `SurveyService`, and `AIService` services (Fig. 14). Results of the tests:

- Successfully passed: 45 tests (95.7%).
- Failed tests: 2 tests related to the processing of incorrect input data in `NoteService` (fixed by adding validation).
- Code Coverage: ~ 85% for business logic, which meets the standards for projects of this scale.

The tests were performed using the command:

```
dotnet test SmartAssistant.Tests
```

```

public class ChatServiceTests
{
    private readonly Mock<IMessageRepository> _messageRepositoryMock;
    private readonly Mock<IAIService> _aiServiceMock;
    private readonly ChatService _chatService;

    public ChatServiceTests()
    {
        _messageRepositoryMock = new Mock<IMessageRepository>();
        _aiServiceMock = new Mock<IAIService>();
        _chatService = new ChatService(_messageRepositoryMock.Object, _aiServiceMock.Object);
    }

    [Fact]
    public async Task SendMessageAsync_ValidInput_SavesMessageAndReturnsAIResponse()
    {
        // Arrange
        var sessionId = Guid.NewGuid();
        var message = new Message { Content = "Як впоратися з тривожністю?", Author = "User", ChatSessionId = sessionId };
        var aiResponse = "Рекомендується спробувати техніки глибокого дихання.";
        _aiServiceMock.Setup(ai => ai.GetResponseAsync(message.Content)).ReturnsAsync(aiResponse);
        _messageRepositoryMock.Setup(r => r.AddAsync(It.IsAny<Message>())).Returns(Task.CompletedTask);

        // Act
        var result = await _chatService.SendMessageAsync(sessionId, message);

        // Assert
        _messageRepositoryMock.Verify(r => r.AddAsync(It.Is<Message>(m => m.Content == message.Content)), Times.Once());
        _aiServiceMock.Verify(ai => ai.GetResponseAsync(message.Content), Times.Once());
        Assert.Equal(aiResponse, result);
    }
}

```

Fig. 14 Unit test of saving messages

Integration tests checked the interaction between API controllers, services, repositories, and the database. To do this, we used a PostgreSQL test database deployed in a Docker container and a TestServer with ASP.NET Core to simulate HTTP requests.

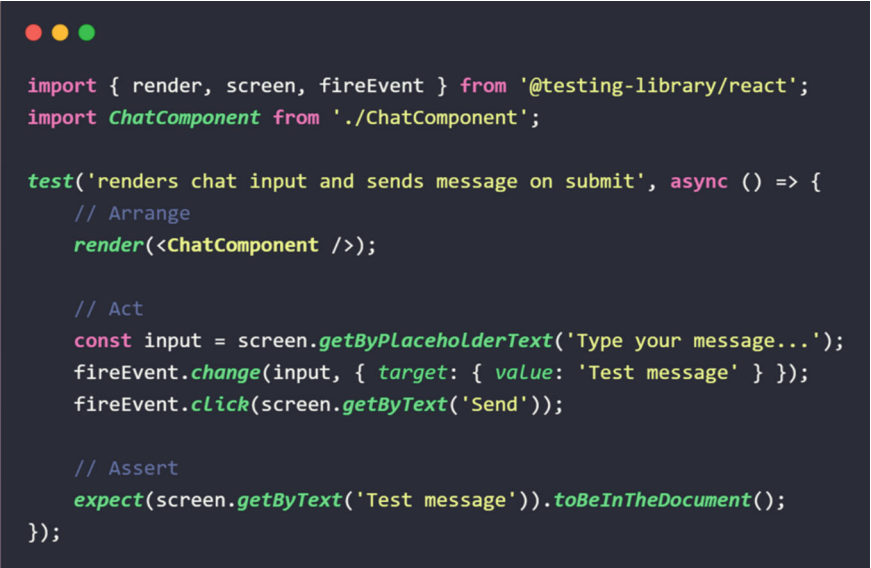
We performed 32 integration tests covering the main API endpoints (AuthController, ChatController, NoteController, SurveyController). Results:

- Successfully passed: 30 tests (93.75%).
- Failed tests: 2 tests related to CORS configuration errors and timeouts when requesting the Groq API (fixed by configuring CORS and increasing the timeout).
- API coverage: ~ 90% of endpoints have been tested.

React Testing Library was used to test the client side (React, TypeScript, MUI). The tests checked the rendering of components, event handling, and the correct display of data received from the API (Fig. 15).

This test checks:

- whether the message input field is rendered;
 - whether the sent message is displayed in the interface after the button is pressed.
- 25 tests were performed for the main components (LoginPage, MainPage, AssistantPage, SurveyPage, TherapyPage). Results:
- Passed: 24 tests (96%).



```
import { render, screen, fireEvent } from '@testing-library/react';
import ChatComponent from './ChatComponent';

test('renders chat input and sends message on submit', async () => {
  // Arrange
  render(<ChatComponent />);

  // Act
  const input = screen.getByPlaceholderText('Type your message...');
  fireEvent.change(input, { target: { value: 'Test message' } });
  fireEvent.click(screen.getByText('Send'));

  // Assert
  expect(screen.getByText('Test message')).toBeInTheDocument();
});
```

Fig. 15 Test for the chat component

- Failed tests: 1 test related to incorrect rendering of responsive design on mobile devices (fixed by updating MUI styles).
- Component coverage: ~ 80% of the main components have been tested.

Some tests of the client side were, of course, performed manually to check how the system works in different scenarios (Fig. 16).

Performance: Apache JMeter was used to evaluate the API response speed. Testing was conducted with 100 simultaneous users sending requests to the `/api/chat/send-message` and `/api/survey/create` endpoints. Average response time:

- OpenAI requests: ~ 1.2 s.
- Groq requests: ~ 0.8 s (thanks to LPU).
- Survey creation: ~ 0.3 s.

The system withstood the load of up to 500 requests per minute without a significant drop in performance, which confirms its scalability.

Security: Security testing included:

- Authentication testing (JWT): Successfully blocked unauthorized requests.
- SQL injection protection: Entity Framework Core automatically parameterizes queries.
- XSS protection: All incoming data is validated and screened on the frontend and backend.

The test results confirmed that the web application meets the basic requirements specified in the task:

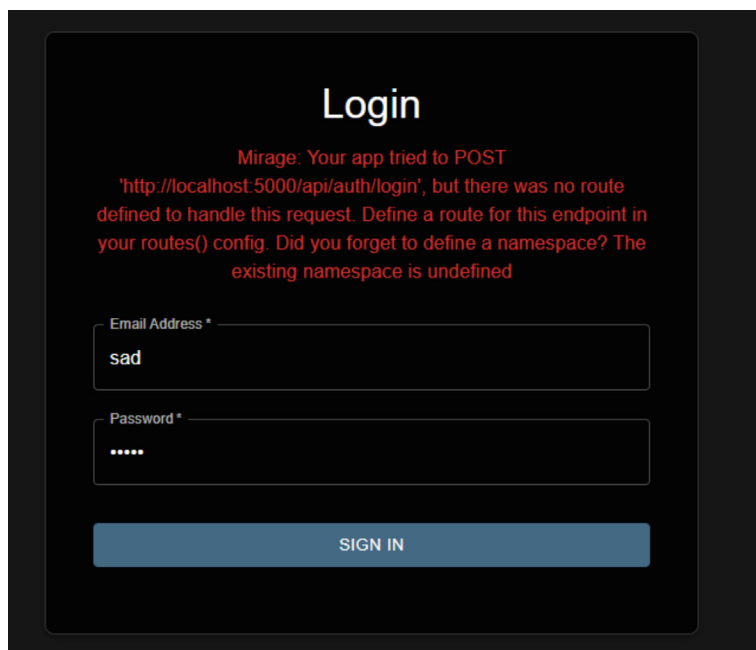


Fig. 16 Request submission error (internal)

- **Accuracy:** Unit and integration tests showed that the system correctly processes requests, stores data, and interacts with AI services. Code coverage (~ 85–90%) is sufficient to ensure reliability.
- **User-friendliness:** UI tests have confirmed the intuitiveness of the interface, and the responsive design ensures correct display on different devices.
- **Efficiency:** Performance tests have shown that the system processes requests quickly, especially when using Groq for AI answers.
- **Ethics and privacy:** Security checks confirmed the protection of sensitive data, and the modular architecture makes it easy to add new privacy policies.

A small number of failed tests (5 out of 104) were fixed by code modifications, which indicates the effectiveness of the testing process. The main problems were related to configuration (CORS, timeouts) and minor errors in the frontend that did not affect the overall functionality.

Testing of the Smart Psychologist Assistant web application confirmed its compliance with functional and non-functional requirements. Unit tests verified the isolated logic of the services, integration tests—the correctness of the interaction of components, and UI tests—the convenience and reliability of the interface. Performance and security tests showed that the system is scalable, fast, and secure. The results obtained allow us to recommend the application for use in psychological practice,

with the possibility of further improvement, for example, adding automated tests for vector search or expanding the coverage of tests.

4 Conclusions and Future Work

The development of the web application “Smart Psychologist’s Assistant” was a comprehensive study aimed at creating an innovative tool to improve the efficiency of psychological practice by integrating modern artificial intelligence technologies. In the process of performing the work, the goal was achieved—software was created that provides automated support for a psychologist in analyzing client requests, managing sessions, creating surveys, and maintaining therapeutic plans, while maintaining a high level of ethics and confidentiality.

The main results of the work are in line with the tasks set:

1. Development of the web application architecture: A client–server architecture with a clear division into frontend (React, TypeScript, MUI), backend (.NET Core), and integration module with AI services (OpenAI, Groq) was proposed and implemented. The use of PostgreSQL with the PGVector extension provided an efficient vector search for analyzing notes and messages, which increased the functionality of the system.
2. Integration of AI language models: A module has been implemented to process user requests using OpenAI (GPT-4-turbo) and Groq (LLaMA 3) models, which allows generating psychologically sound answers, analyzing the history of interactions, and creating recommendations. Flexible routing of requests between providers ensured a balance between the speed and quality of responses.
3. Implementation of the user interface: We created an intuitive interface with responsive design support, which includes modules for authorization, chat management, surveys, and therapy plans. The use of Material UI (MUI) ensured visual integrity and user experience.
4. System testing: Comprehensive testing was conducted, including unit tests (47 tests, 95.7% successful), integration tests (32 tests, 93.75% successful), and UI tests (25 tests, 96% successful). The results confirmed the correctness of the system’s operation, its performance (average API response time is ~ 0.3–1.2 s) and compliance with security and privacy requirements.

The scientific and practical result of the work is the creation of a web application that automates routine tasks of a psychologist, provides interactive support using AI, stores the history of interactions, and offers tools for analyzing client data. The application meets modern development standards, including modularity, scalability, and data protection, making it suitable for integration into real psychological practice.

The ethical aspects of using AI in psychology were taken into account by clearly delineating roles: the system acts as an assistant, not a substitute for a psychologist, ensuring transparency of interaction and protection of personal data through encryption and JWT authentication. The results of the work were tested at scientific and

practical conferences, where positive feedback was received on its relevance and practical value.

Prospects for further development include:

- Adding a module for automatic analysis of clients' emotional state based on text data.
- Integration of additional therapeutic techniques (e.g., ACT or DBT support).
- Expanding the functionality for teamwork of several psychologists.
- Optimization of vector search to process larger amounts of data.

Thus, the developed web application “Smart Psychologist Assistant” is a modern solution that increases the efficiency of mental health professionals, opens up new opportunities for automation, and meets the requirements of ethical use of AI.

References

1. Darcy AM, Louie AK, Roberts LW (2016) Machine learning and artificial intelligence: applications in mental health. *Acad Psychiatry* 40(2):263–268
2. Manning CD, Schütze H (1999) Foundations of statistical natural language processing. MIT Press, Cambridge, MA
3. Yarovyi A, Kudriavtsev D, Baraban S, Ozeranskyi V, Krylyk L, Smolarz A, Karnakova G (2019) Information technology in creating intelligent chatbots. In Romaniuk RS, Linczuk M (eds) *Photonics applications in astronomy, communications, industry, and high-energy physics experiments 2019*. Photonics applications in astronomy, communications, industry, and high-energy physics experiments 2019. SPIE, p 246. <https://doi.org/10.1117/12.2537415>
4. Touvron H, Lavril A, Izacard G et al (2023) LLaMA: open and efficient foundation language models
5. React (2024) React—JavaScript library for building user interfaces. <https://react.dev/>. Accessed 20 May 2025
6. TypeScript (2024) TypeScript—JavaScript with syntax for types. <https://www.typescriptlang.org/>. Accessed 20 May 2025
7. MUI (2024) MUI—component library for react. <https://mui.com/>. Accessed 20 May 2025
8. Freeman A (2022) *Pro ASP.NET Core 7*, 2nd edn. Apress, New York
9. OpenAI (2024) About OpenAI—creating advanced AI systems. <https://openai.com/about/>. Accessed 20 May 2025
10. Groq (2024) Hardware for accelerating AI computation. <https://groq.com/>. Accessed 20 May 2025
11. Touvron H, Lavril A, Izacard G et al (2023) LLaMA: open and efficient foundation language models. [arXiv:2302.13971](https://arxiv.org/abs/2302.13971)
12. Mistral AI (2024) Frontier AI LLMs, assistants, agents and services. <https://mistral.ai/>. Accessed 20 May 2025
13. Google DeepMind (2024) Gemma—open models from Google DeepMind. <https://deepmind.google/models/gemma/>. Accessed 20 May 2025
14. Groq (2024) The Groq LPU explained. <https://groq.com/the-groq-lpu-explained/>. Accessed 20 May 2025
15. PostgreSQL Global Development Group (2024) PostgreSQL—the world's most advanced open source database. <https://www.postgresql.org/>. Accessed 20 May 2025
16. pgvector—open-source vector search for PostgreSQL (2024) <https://github.com/pgvector/pgvector>. Accessed 20 May 2025

17. Docker (2024) Docker—the platform for building, running and shipping apps. <https://www.docker.com/>. Accessed 20 May 2025
18. Hastie T, Tibshirani R, Friedman J (2009) The elements of statistical learning. Springer, New York
19. Rauer JN, Kroiss M, Kryvinska N, Engelhardt-Nowitzki C, Aburaia M (2021) Cross-university virtual teamwork as a means of internationalization at home. *Int J Manag Educ* 19(3):100512. <https://doi.org/10.1016/j.ijme.2021.100512>
20. LangChain (2024) LangChain—framework for building AI applications. <https://www.langchain.com/>. Accessed 20 May 2025
21. Dify (2024) Dify—AI agent builder platform. <https://dify.ai/>. Accessed 20 May 2025

Cyber Security Manager After NIS2 Directive



Marián Mikolášik  and Daniela Pucicov

Abstract This study provides an in-depth analysis of the evolving role of the cybersecurity manager within organizations, focusing on the critical need for comprehensive cybersecurity education and the integration of the NIS2 Directive requirements. As digital transformation accelerates across industries, organizations face escalating cyber threats that exploit both technological vulnerabilities and human factors. The research draws on theoretical frameworks, empirical interviews with cybersecurity experts from multinational enterprises, and recent regulatory developments to identify persistent gaps in cybersecurity education, training, and organizational culture. Key findings reveal that while technological defenses are essential, the lack of structured, mandatory, and role-based cybersecurity training for all employees—combined with insufficient assessment, communication, and cultural support—remains a major vulnerability. The NIS2 Directive significantly raises the bar by mandating regular risk assessments, incident reporting, management accountability, and continuous professional development, making cybersecurity a strategic and legal imperative for senior leadership. The report recommends a holistic approach: implementing standardized assessments, mandatory and tailored training, continuous learning, effective communication, and a supportive culture that encourages incident reporting and proactive security behavior. The study also highlights the growing impact of AI-driven attacks, ransomware evolution, and Cybercrime-as-a-Service, underscoring the necessity of adaptive and forward-looking defense strategies. By embedding NIS2 requirements into all aspects of cybersecurity management and education, organizations can close existing gaps, enhance resilience, and ensure compliance in an increasingly complex digital landscape (European Union Agency for Cybersecurity in Cybersecurity roles and skills for NIS2 essential and important entities: mapping NIS2 obligations to ECSF. Publications Office, LU [1]). The research concludes that successful cybersecurity is not solely a technological challenge but a collective organizational responsibility, requiring sustained leadership commitment, cross-functional collaboration, and a culture of continuous improvement.

M. Mikolášik (✉) · D. Pucicov

Faculty of Management, Comenius University in Bratislava, Bratislava, Slovakia

e-mail: marian.mikolasik@fm.uniba.sk

Keywords Cybersecurity · Cybercrime · NIS2 directive · Cybersecurity education · Risk assessment

Foreword

In 2025, cyberattacks have surged significantly in both frequency and sophistication. According to Check Point Software's Q1 2025 report [2], organizations experienced an average of 1925 cyberattacks per week, marking a 47% increase compared to the same period in 2024. This translates to roughly 275 attacks per day per organization on average. Globally, cybersecurity statistics indicate that about 2200 cyberattacks occur daily, meaning a cyberattack happens approximately every 39 s [3]. These resources collectively underscore the necessity of implementing effective cybersecurity training programs to enhance employee awareness and mitigate risks associated with human error in organizations. The alarming rise in cyberattacks highlights the urgent need for organizations to prioritize cybersecurity awareness training, empowering employees to act as the first line of defense against these threats. This proactive approach not only helps in reducing incidents but also fosters a culture of security mindfulness within the organization.

1 Introduction

In the past several decades, humanity has experienced a drastic increase in the use of Information Technology in different areas of our lives and all sectors of society. The use of the internet, even solely its availability at any moment or in any circumstance, is no longer a matter of question or doubt, as the increased demand for digital connectivity has made it a “must” in our lives.

Technology is progressively becoming a more significant part of who we are as a species. Its pervasiveness has resulted in a profound transformation of our behavior, determining the way we communicate or interact, affecting our productivity and work culture, our health and security, our education and even our thinking. This is increasing to such an extent that we are progressively losing our autonomy in making decisions without relying (either partially or entirely) on technology.

Over the past few decades, humanity has witnessed a significant surge in the utilization of Information Technology throughout various aspects of life and all societal sectors. The ubiquity of the internet, regardless of time or circumstance, is no longer a subject of debate, as the heightened desire for digital connectivity has rendered it indispensable in our lives.

Technology is progressively becoming a more significant part of who we are as a species. Its pervasiveness has resulted in a profound transformation of our behavior, determining the way we communicate or interact, affecting our productivity and work culture, our health and security, our education and even our thinking. This is

increasing to such an extent that we are progressively losing our autonomy in taking decisions without relying (either partially or entirely) on technology.

However, not only is our decision-making impacted. In many cases, we have no other choice or path of getting things done, other than employing our electronic devices or engaging with some form of technology—whether it’s finding our way around a new city, purchasing a train ticket, booking accommodation, reading the latest news, getting a queue ticket when visiting a government office, paying bills, or even finding a job. Figure 1 illustrates the digital transformation benefits for businesses.

Table 1 provides the most costly incidents in the first four months of 2025, based on Verizon data.

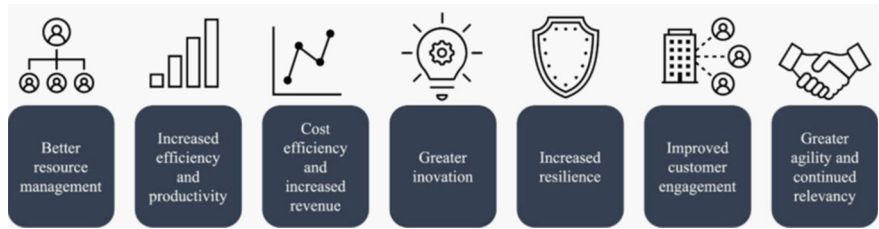


Fig. 1 The digital transformation benefits for businesses adapted from [4]

Table 1 Incidents and impact by various attack method based on [5]

Date	Incident/Target	Attack type/ Method	Impact/Details
Jan 2025	PowerSchool (US K-12 districts)	Data breach	Personal data of 62.4 million students and 9.5 million teachers exposed
Jan 2025	Green bay packers (NFL)	Retail data breach	Credit card data of 8500 + customers stolen from online store
Jan 2025	Phemex (Crypto exchange)	Crypto theft	\$85 million in cryptocurrency stolen from hot wallets
Feb 2025	ByBit (Crypto exchange, Dubai)	Crypto theft	\$1.5 billion in Ethereum stolen by North Korean hackers
Feb 2025	Meta/WhatsApp	Spyware attack	Sophisticated spyware targeted journalists and civil society via WhatsApp
Feb 2025	IoT data breach	Data exposure	2.7 billion records exposed in a major IoT platform breach
Mar 2025	Lazarus group (LinkedIn scam)	Social engineering/ malware	North Korea-linked group stole credentials and deployed malware via fake job links
Apr 2025	US bank (OCC)	Espionage	Hackers accessed 150,000 sensitive emails of 103 regulators over months
Apr 2025	North Korean cyber spies	Espionage/ extortion	Targeted European defense/government orgs, used extortion after infiltration

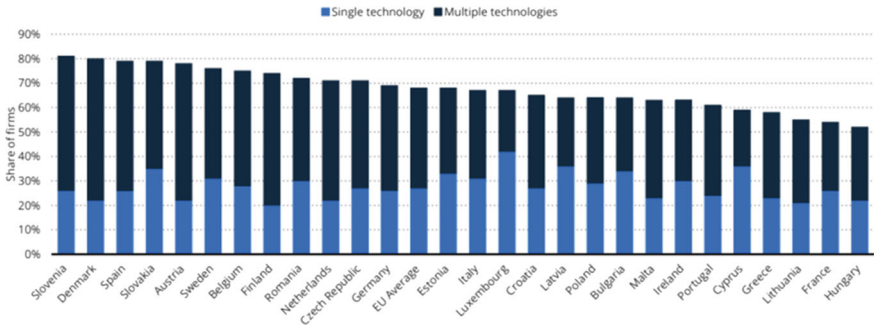


Fig. 2 Share of firms in the European Union that have implemented new digital technologies in their work processes in 2022, by country [7]

The digital transformation of society has revolutionized how organizations operate, communicate, and deliver value. However, this transformation has also exposed enterprises to an escalating array of cyber threats. The European Union’s NIS2 Directive marks a significant regulatory milestone, setting new standards for cybersecurity risk management, education, and organizational accountability. This report provides a comprehensive, detailed analysis of the intersection between cybersecurity management, the need for education, and the integration of NIS2 requirements, drawing on both theoretical research and empirical findings from multinational enterprises [1].

2 Digitalization and the Modern Threat Landscape

2.1 The Evolution of Digital Society

The past decades have witnessed a dramatic increase in the use of information technology across all sectors. Digitalization has become a defining feature of modern society, influencing communication, productivity, education, and even decision-making processes. The shift from digitization to digitalization and ultimately to digital transformation has fundamentally altered the core pillars of society, including institutions [6] (Fig. 2).

2.2 Digital Transformation of Enterprises

Businesses have embraced digital technologies to enhance efficiency, productivity, and competitiveness. The adoption of digital tools has created new job roles, transformed skill requirements, and expanded the reach of enterprises. According to recent

data, nearly 70% of businesses in the European Union have implemented new digital technologies in their work processes, with larger enterprises exhibiting higher degrees of digitalization [8].

Key Drivers of Digital Transformation

- Increased Efficiency and Productivity: Automation and digital workflows streamline operations
- Reduce manual labor and improve accuracy.
- Enhanced Customer Experience: Digital tools enable personalized, real-time interactions and services, increasing customer satisfaction and loyalty.
- Data-Driven Decision Making: Advanced analytics and big data allow organizations to make informed decisions and predict trends.
- Market Competitiveness: Embracing digital transformation helps companies stay ahead of competitors and respond quickly to market changes.
- Innovation: Digital platforms foster innovation in products, services, and business models.

Case Example: Key Technologies

Technology	Application Example
Cloud Computing	Flexible data storage and remote collaboration
Artificial Intelligence	Predictive analytics, chatbots, automation
Internet of Things	Smart factories, real-time asset tracking
Big Data Analytics	Customer insights, market trend analysis
Automation/Robotics	Manufacturing, logistics, and customer service

Digital transformation is essential for enterprises seeking to thrive in a digital-first world. It requires not only technological adoption but also a shift in mindset, culture, and organizational structure. While the journey involves challenges, the benefits—greater efficiency, innovation, and competitiveness—make it a strategic imperative for modern businesses [6].

2.3 Cybersecurity as a Core Business Concern

While digitalization offers significant benefits, it also introduces new vulnerabilities. Cyber-attacks now rank among the top global risks, threatening financial stability, reputation, legal compliance, and intellectual property. High-profile incidents such as the Equifax and Uber breaches have underscored the devastating consequences of inadequate cybersecurity measures, including regulatory fines, reputational damage, and loss of stakeholder trust [9].

2.3.1 The Equifax Data Breach (2017)

Incident Overview: In September 2017, Equifax, one of the largest credit reporting agencies in the United States, announced that its system had been hacked. This breach exposed the personal data of nearly 143 million customers, with approximately 209,000 also having their credit card numbers and other sensitive data compromised.

Vulnerability Exploited: The attackers exploited a known vulnerability in Equifax's web application software that had not been patched due to internal process failures.

Delayed Disclosure and Consequences: The breach began months before its discovery and went unnoticed for weeks. Once aware, Equifax controversially chose to keep the incident from investors, customers, and state regulators for over a month. This delay prevented stakeholders from taking preventive actions and hindered authorities from minimizing the consequences.

Impact: The incident caused widespread outrage among consumers and investors, leading to a significant loss of trust in the company's reliability. The sensitive nature of the information lost and the substantial financial costs—estimated at around 1.4 billion USD in fines, remediation, and settlements—made it one of the most significant cyberattacks in history [10].

2.3.2 The Uber Data Breach (2016)

Incident Overview: In October 2016, Uber faced a cybersecurity attack that exposed confidential data belonging to 57 million of its customers and employees (drivers).

Attack Method: The attackers gained access by using stolen login information to break into Uber's private source code storage. From there, they obtained a special access key that allowed them to copy a significant amount of user and driver information.

Attempted Cover-Up: The incident incurred substantial financial costs, reputational damage, and legal liability for Uber. However, the company initially attempted to conceal the breach for a year to avoid reputational harm. Uber's Chief Security Officer at the time orchestrated a scheme to pay the hackers 100,000 USD in Bitcoin and arranged for them to sign non-disclosure agreements with false information about the hack.

Discovery and Legal Ramifications: Uber's new management eventually discovered the truth, leading to public disclosure in November 2017. The Chief Security Officer was subsequently convicted of obstruction of justice and "misprision of felony," facing a potential prison sentence of up to seven years.

These cases underscore that cyberattacks can lead to significant financial losses, operational disruptions, reputational damage, and legal liabilities for organizations. The human factor, often exploited through methods like phishing, remains a critical vulnerability, as it is easier to exploit human negligence than to bypass sophisticated software security [11].

2.4 *The Evolving Concern for Security in the Digital Era*

The rapid adoption of digital technologies continues to deliver significant advantages for organizations, driving innovation, efficiency, and new business models. As a result, more companies than ever are embracing digital transformation to remain competitive. Yet, alongside these clear benefits, digitalization brings a host of complex and evolving risks that cannot be ignored.

Some of these risks are immediately apparent, such as increased dependency on technology, reduced face-to-face interaction, and workforce displacement due to automation. However, other dangers are subtler but potentially more damaging, most notably, the growing threats to privacy and cybersecurity. The consequences of privacy breaches or major security incidents can be catastrophic, affecting not only the financial health of organizations but also eroding public trust and causing widespread societal harm. The World Economic Forum's Global Risk Report [12] places cyber-attacks in the top ten greatest risks the world is expected to face based on both likelihood and impact. Figure 3 shows the position of cyber threats in comparison to other global risk factors.

Recent global analyses confirm the seriousness of these risks. According to the World Economic Forum's Global Risks Report 2024, cyber threats now rank among the top five global risks for both the public and private sectors, reflecting their frequency and potential for disruption. The report highlights that persistent cyber-attacks can undermine economic stability, disrupt critical infrastructure, and inflict long-term reputational and financial damage on organizations of all sizes [12].

This trend is intensifying as organizations become more reliant on interconnected digital systems and cloud-based services. Each new digital platform or technology expands the attack surface, creating fresh vulnerabilities and making it more challenging to safeguard sensitive data. The complexity of modern supply chains, increased use of artificial intelligence, and geopolitical tensions further amplify these risks, leading to a more unpredictable and hazardous cyber landscape [13].

As experts have noted, the very efficiency and accessibility provided by digital technologies also facilitate the proliferation of cybercrime and the emergence of new forms of digital misconduct. This underscores the urgent need for organizations to adopt a proactive, security-first approach, integrating robust cybersecurity measures into every stage of their digital transformation journey.

In summary, while digital transformation is essential for growth and innovation, it must be balanced with vigilant risk management and a deep commitment to cybersecurity. Only by recognizing and addressing these evolving threats can organizations fully realize the benefits of digitalization while safeguarding their assets, reputation, and stakeholders [14].

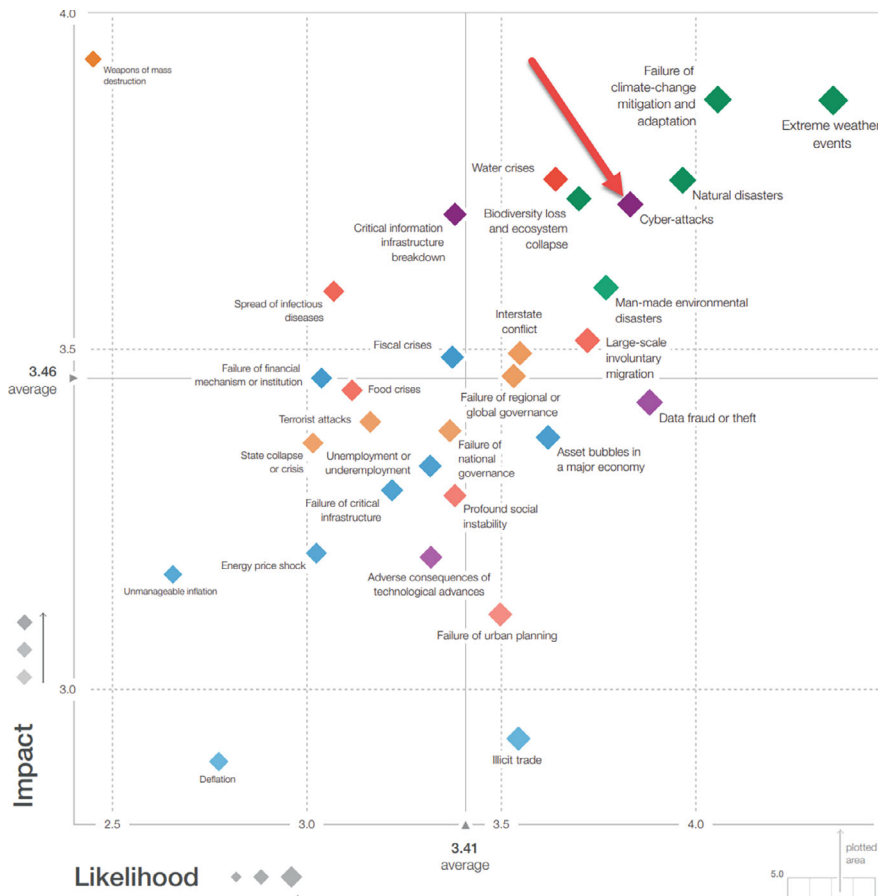


Fig. 3 Likelihood and impact of cyberattacks among other disaster events (own creation)

2.5 Recent Cybercrime Trends and Examples (2024–2025)

2.5.1 Key Cybercrime Trends in 2025

- **AI-Driven Attacks:** Cybercriminals are leveraging artificial intelligence and machine learning to automate and mutate malware, craft sophisticated phishing campaigns, and evade traditional security measures. AI-powered malware can adapt in real time, making detection and defense significantly more challenging.

AI-Driven Cyber Attacks: How Artificial Intelligence Is Transforming Cybercrime

Artificial intelligence (AI) is rapidly changing the landscape of cyber threats. While AI helps defenders detect and prevent attacks, cybercriminals are now leveraging

AI to automate, scale, and personalize their attacks, making them faster, more convincing, and harder to detect [15]. These AI-driven attacks are not entirely new in concept, but their speed, precision, and adaptability represent a significant escalation in cyber risk.

Key Characteristics of AI-Driven Attacks

- Automation: AI enables attackers to automate reconnaissance, vulnerability discovery, and attack execution, reducing manual effort and increasing attack speed.
- Personalization: AI can scrape data from public sources to craft highly targeted phishing and social engineering messages, increasing the likelihood of success.
- Adaptability: AI-powered malware and bots can learn from their environment, change tactics in real time, and evade detection by traditional security tools [16].
- Scalability: Attackers can launch thousands of unique, well-crafted attacks simultaneously, reaching more victims than ever before.
- Evasion: AI can help malware mutate its code (polymorphism) or mimic legitimate user behavior, making it hard to detect (Table 2).

Real-World Examples

Table 2 Cybersecurity attacks according main types (own creation)

Attack type	Description	Example/Technique
AI-powered phishing	AI generates convincing, personalized emails or messages to trick users into revealing data	LLMs (like WormGPT) craft flawless emails
Deepfake social engineering	AI creates realistic audio/video to impersonate executives or staff for fraud or extortion	Deepfake CEO voices for wire fraud
AI-generated malware	Malware that adapts its code and behavior to bypass detection and defenses	Polymorphic malware, AI-crafted ransomware
Automated reconnaissance	AI tools scan for vulnerabilities, exposed assets, or leaked credentials at scale	AutoGPT for reconnaissance
Password cracking	AI speeds up brute-force or dictionary attacks by analyzing user behavior and patterns	AI-optimized brute force
Botnet attacks	AI-powered bots mimic human behavior and adapt to evade detection in DDoS or credential stuffing	Adaptive botnets
Prompt injection	AI manipulates chatbots or LLMs to reveal sensitive data or perform unintended actions	Prompt-injection hijacks

- **Voice Deepfakes:** In 2019, criminals used AI to mimic a CEO's voice, convincing an executive to transfer \$243,000 to a fraudulent account.
- **Hyper-Personalized Phishing:** AI-generated phishing emails referencing real company events have tripled the success rate compared to traditional phishing.
- **Polymorphic Ransomware:** AI-driven ransomware adapts its code to each target, evading signature-based detection and encrypting data more efficiently.

Why AI-Driven Attacks Are So Dangerous

- **Speed:** Attacks can be launched and adapted in real time, leaving defenders little time to respond [17].
- **Believability:** AI-generated content (emails, audio, video) is often indistinguishable from legitimate communication [16].
- **Lower Barrier to Entry:** AI tools are increasingly accessible, enabling less-skilled attackers to launch sophisticated campaigns.
- **Evasion:** AI-powered attacks can bypass traditional security tools by continuously changing their tactics and appearance [17].

Defensive Strategies

- **Behavioral AI Detection:** Use security solutions that analyze behavior, not just signatures, to spot novel and adaptive threats.
- **User Education:** Train staff to recognize AI-enhanced phishing and deepfake scams.
- **Multi-Factor Authentication:** Reduces the risk of credential theft and brute-force attacks.
- **Continuous Monitoring:** Implement real-time monitoring and automated response to detect and contain threats quickly.
- **Regular Updates:** Patch systems and update defenses to address vulnerabilities that AI tools might exploit.

AI-driven cyber attacks are rapidly evolving, making cybercrime faster, smarter, and more difficult to defend against. Organizations must adapt by combining advanced AI-driven defenses, robust user education, and proactive security strategies to counter these emerging threats.

- **Ransomware Evolution:** Ransomware attacks have surged, with a 126% year-over-year increase in Q1 2025. Attackers are shifting from encrypting data to exfiltrating and threatening to leak it, making extortion more effective. Nearly half (44%) of all breaches now involve ransomware. Data theft, especially of credentials and sensitive information, often precedes ransom demands [18].

Ransomware has transformed from a crude form of digital extortion in the late 1980s to a sophisticated, multi-billion-dollar threat by 2025. Its evolution reflects advances in technology, changes in cybercriminal tactics, and the growing reliance of organizations on digital infrastructure.

Mainstream Emergence: 2005–2016

- 2005–2007: Encryption and Locker Ransomware
 - GPCoder and Archiveus used asymmetric encryption, making file recovery without payment difficult.
 - Locker ransomware emerged, locking users out of devices entirely.
- 2010: Cryptocurrencies Enable Anonymity
 - Bitcoin and other cryptocurrencies provided a new, untraceable payment method, fueling ransomware's growth.
- 2012: Ransomware-as-a-Service (RaaS)
 - Reveton introduced the RaaS model, allowing non-technical criminals to rent ransomware and share profits with developers.
- 2013: CryptoLocker
 - Used military-grade encryption and spread via email attachments and botnets.
 - Demonstrated the profitability of ransomware, with millions collected in ransoms.

Escalation and Innovation: 2016–2021

- 2016: Locky, Petya
 - Locky spread through phishing emails; Petya targeted the Master Boot Record, locking entire systems.
- 2017: WannaCry and NotPetya
 - WannaCry affected over 200,000 victims in 150 countries, exploiting a Windows vulnerability.
 - NotPetya introduced wiper functionality, destroying data rather than just encrypting it.
- 2019: Maze and Double Extortion
 - Maze pioneered double extortion, encrypting files and threatening to leak stolen data unless a ransom was paid [19].

Modern Era: 2021–2025

- Triple Extortion and Beyond
 - BlackCat and similar groups added DDoS attacks and threats to contact customers or partners, increasing pressure on victims [19].
- Stealth and Sophistication
 - Ransomware groups now use advanced loader chains (e.g., NETXLOADER, SmokeLoader) for stealthy, in-memory execution, making detection and analysis harder.

Table 3 Attack trends (own creation)

Trend	Description
Advanced loader chains	In-memory execution, sandbox evasion, reflective DLL injection
Abuse of legitimate software	Surveillance and credential theft via tools like Kickidler and SMOKEDHAM
Non-traditional delivery	Ransomware embedded in images or paired with decoy documents
Exploitation of enterprise Apps	Targeting vulnerabilities in software like SAP NetWeaver
RaaS ecosystem	Professionalization of ransomware operations, with affiliates and support infrastructure
Triple/Quadruple extortion	Encryption, data theft, DDoS, and threats to contact third parties

- Attacks increasingly exploit legitimate tools for reconnaissance and credential theft, and use non-traditional vectors like JPEG-embedded ransomware [20].
- Ransomware-as-a-Service (RaaS) Matures
 - The RaaS model dominates, with affiliate programs, customer support, and profit-sharing, making ransomware accessible to a wider range of criminals.
- Targeting and Impact
 - Critical infrastructure, healthcare, and manufacturing remain top targets.
 - The United States, Germany, and Canada are among the most targeted countries [20].
- Attack Volume and Trends
 - Ransomware attacks continue to rise: 331 incidents in May 2023, 520 in May 2024, and 545 in May 2025—a 15.95% year-over-year increase.
 - New groups (e.g., SafePay, SilentRansomGroup) and evolving tactics (e.g., JPEG delivery, SAP NetWeaver exploits) are reshaping the landscape [20].

Key Trends in 2025

Ransomware has evolved from simple file encryption and crude extortion to a highly professionalized, multi-faceted threat. In Table 3, we can see various types of attacks. Modern ransomware campaigns are stealthier, more targeted, and more damaging than ever, leveraging advanced technology and business models to maximize profit and impact. Organizations must adapt with layered defenses, rapid detection, and robust incident response to counter this ever-evolving threat [19] (Fig. 4).

- Cybercrime-as-a-Service (CaaS): The dark web is seeing a rise in marketplaces where criminals can buy or rent attack tools and services. This lowers the barrier to entry for cybercrime, enabling less-skilled actors to launch sophisticated attacks.

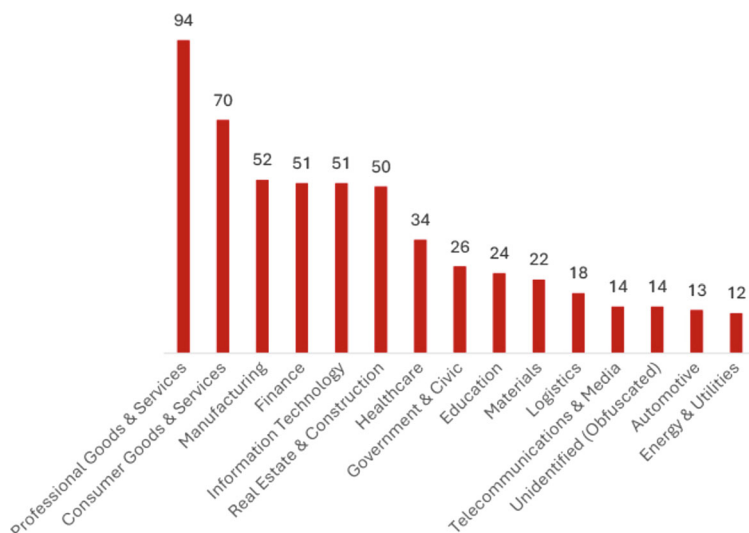


Fig. 4 Industries targeted in May 2025 based on [20]

Cybercrime-as-a-Service (CaaS): The New Era of Organized Digital Crime

What Is Cybercrime-as-a-Service?

Cybercrime-as-a-Service (CaaS) is a business model in which cybercriminals offer their tools, skills, and infrastructure to others, often through underground marketplaces, enabling anyone, regardless of technical expertise, to launch sophisticated cyberattacks for a fee. This model mirrors legitimate “as-a-service” platforms in the IT sector, such as Software-as-a-Service (SaaS), but is focused entirely on enabling and monetizing criminal activity.

How CaaS Works

- **Vendors:** Skilled hackers develop attack tools, malware, or exploit kits and offer them for rent or sale.
- **Clients:** Individuals or groups (often with little technical skill) purchase or subscribe to these services, typically paying in cryptocurrency for anonymity.
- **Marketplace:** Transactions and support often occur through dark web forums, invite-only chat groups, or encrypted messaging apps.
- **Support:** Many CaaS providers offer customer service, updates, and even profit-sharing affiliate programs.

Key Trends in 2025

- **Democratization of Cybercrime:** Anyone can launch complex attacks without technical skill, dramatically increasing the volume and diversity of cyber threats [21].

- **AI-Enabled Services:** CaaS platforms now offer AI-driven phishing campaigns, deepfake social engineering, and adaptive malware, making attacks more convincing and harder to detect see Table 4.
- **Subscription and Affiliate Models:** CaaS offerings are often sold on a subscription basis, with profit-sharing for successful attacks (especially in Ransomware-as-a-Service) [21].
- **Professionalization:** Many CaaS vendors provide customer support, regular updates, and even “guarantees” of service, mimicking legitimate business practices [21].
- **Rapid Proliferation:** As soon as one CaaS operation is disrupted by law enforcement, others quickly fill the void, learning from previous takedowns.

Real-World Impact

- **Lower Barriers to Entry:** Novice attackers can now launch sophisticated campaigns, increasing the overall threat volume.
- **Increased Attack Sophistication:** Even skilled cybercriminals use CaaS to access specialized tools or scale their operations [22].
- **Financial Scale:** The CaaS ecosystem generates billions in criminal revenue annually, with ransomware payments alone exceeding \$1.1 billion in 20,234.
- **Enterprise Threat:** Over half of all cyber threats to organizations in 2025 are attributed to Malware-as-a-Service and related CaaS models [22].

Defensive Strategies for Organizations

Table 4 Cybercrime as a service. (own creation)

Service type	Description	Example/Trend (2025)
Ransomware-as-a-service	Pre-packaged ransomware kits with management dashboards, payment portals, and support	Dominant in major breaches
Malware-as-a-service	Trojans, viruses, spyware, and other malware available for deployment	Accounts for 57% of threats
Phishing-as-a-service	Ready-made phishing kits, fake website templates, hosting, and automation	60–70% of phishing attacks
DDoS-as-a-service	Botnets-for-hire to launch distributed denial-of-service attacks	Attacks for as little as \$20/month
Exploit-as-a-service	Leasing or selling zero-day vulnerabilities and exploit kits	Growing, especially for zero-days
Access brokers	Sale of stolen credentials or direct network access	Increasingly common
AI-driven attack kits	AI-powered phishing, deepfakes, autonomous malware, and password cracking tools	Rapid growth in 2025

- **Behavioral Detection:** Rely on behavioral analytics and AI-driven security tools to detect novel and adaptive threats.
- **Employee Training:** Regularly educate staff about phishing, deepfakes, and social engineering tactics enabled by CaaS68.
- **Incident Response:** Develop and test robust incident response plans that account for rapid, large-scale attacks.
- **Threat Intelligence:** Monitor dark web and underground forums for emerging CaaS trends and tools.
- **Layered Security:** Combine endpoint protection, network monitoring, and strong authentication to reduce risk.

Cybercrime-as-a-Service has fundamentally transformed the threat landscape, making advanced cyberattacks accessible to anyone with money and intent. In 2025, the proliferation of AI-driven tools, subscription-based attack kits, and professionalized criminal services means organizations must adopt proactive, intelligence-driven defenses and foster a strong culture of cybersecurity awareness to stay ahead of this rapidly evolving threat [22].

- **Deepfakes and Social Engineering:** The use of AI to generate convincing deepfake videos, audio, and images is proliferating—deepfake content is expected to reach 8 million items in 2025. These are increasingly used for fraud, disinformation, and impersonation attacks [9].
- **Credential Theft and Account Takeover:** Attackers are focusing on stealing login credentials and session cookies, which are then used for further attacks or sold on underground markets. Infostealers and phishing remain primary vectors [23].
- **Targeting Critical Infrastructure and Supply Chains:** There is a marked increase in attacks against government, healthcare, education, and supply chain targets, often using ransomware or espionage tactics [23].
- **Quantum Computing Threats:** While not yet mainstream, intercepted encrypted data is being stockpiled by threat actors in anticipation of future quantum decryption capabilities.
- **Proliferation of Nation-State Attacks:** State-sponsored actors are conducting espionage, sabotage, and data theft, particularly targeting government, defense, and financial sectors.

In Table 5, we can see a sample of the latest documented attacks.

2.5.2 Emerging Threats and Defensive Strategies

- **Zero Trust Security:** Organizations are moving toward zero trust architectures, with continuous authentication and micro-segmentation to limit lateral movement after breaches [24].
- **AI-Driven Defense:** Security teams are adopting AI-powered anomaly detection and automated response to counter advanced, AI-driven threats [16].

Table 5 Latest documented incidents (own creation)

Date	Incident/Target	Attack type/ Method	Impact/Details
Jan 2025	PowerSchool (US K-12 districts)	Data breach	Personal data of 62.4 million students and 9.5 million teachers exposed
Jan 2025	Green bay packers (NFL)	Retail data breach	Credit card data of 8500 + customers stolen from online store
Jan 2025	Phemex (Crypto exchange)	Crypto theft	\$85 million in cryptocurrency stolen from hot wallets
Feb 2025	ByBit (Crypto exchange, Dubai)	Crypto theft	\$1.5 billion in Ethereum stolen by North Korean hackers
Feb 2025	Meta/WhatsApp	Spyware attack	Sophisticated spyware targeted journalists and civil society via WhatsApp
Feb 2025	IoT data breach	Data exposure	2.7 billion records exposed in a major IoT platform breach
Mar 2025	Lazarus group (LinkedIn scam)	Social engineering/ malware	North Korea-linked group stole credentials and deployed malware via fake job links
Apr 2025	US bank regulators (OCC)	Espionage	Hackers accessed 150,000 sensitive emails of 103 regulators for over a year
Apr 2025	North Korean cyber spies	Espionage/ extortion	Targeted European defense/government orgs, used extortion after infiltration
Dec 2024	Romanian election systems	DDoS/ credential leaks	85,000 + attacks and credential leaks before presidential vote

- **Focus on Identity Security:** With credential theft on the rise, organizations are prioritizing identity protection, rapid password changes, and session management.

These trends highlight the growing complexity and scale of cybercrime in 2025, with AI and data theft at the forefront, and both criminal and nation-state actors targeting a wide range of sectors with increasingly sophisticated tactics [9] (Fig. 5).

Most critical weak spots for cyberattacks are, among others, the following:

1. **Software Vulnerabilities:** Software can often contain security flaws or vulnerabilities that can be exploited by hackers to gain unauthorized access, steal data, or cause system malfunctions [25].
2. **Unpatched Systems:** Refers to the known vulnerabilities in a company’s systems that have not yet been addressed or “patched” to improve security. Failure to apply these security patches and updates on time can leave systems exposed to known exploits and attacks Due to the rising use of the Internet of Things (IoT), the connectivity of networks increases the risk of Distributed Denial of Servic attacks [26].

3 The Human Factor in Cybersecurity

When discussing cybersecurity, it is often perceived as a field reserved for highly skilled IT professionals, with knowledge that seems inaccessible or overly complex for the average computer user. This perception leads many people to dismiss cybersecurity as irrelevant to their daily lives, believing it is unlikely to impact them or assuming they will never become victims of cyberattacks.

However, safeguarding an organization’s cybersecurity cannot be the sole responsibility of experts. It must be recognized as a shared duty across all staff members, since many cyber threats specifically target the inattentiveness or lack of awareness of regular users rather than exploiting only technical system vulnerabilities.

As Aloul highlights, while organizations frequently invest in advanced security technologies and specialized training for their IT teams, far less attention is devoted to building security awareness among everyday users, making them the most vulnerable point in the organization’s defences [28]. Encouragingly, as Gutzwiller [29] and colleagues note, the human element in cyber defense is receiving growing recognition, with increasing efforts to address this crucial factor in organizational security [9].

3.1 Cognitive Biases and Risk Perception

Despite investments in advanced security technologies, human error remains a leading cause of security incidents. Cognitive biases—such as overconfidence, familiarity bias, and risk underestimation—can lead employees to overlook threats or engage in risky behaviors. Emotional factors like stress, distraction, and pressure further increase vulnerability to attacks such as phishing see Table 6 (Fig. 6).

Table 6 Impact of various biases (own creation)

Bias	Description	Impact on security
Overconfidence	Overestimating one’s own knowledge or ability	Ignoring security protocols
Familiarity bias	Trusting familiar sources without verification	Falling for spoofed emails
Confirmation bias	Seeking information that confirms existing beliefs	Overlooking warning signs
Anchoring	Relying too heavily on initial information	Ignoring new threat intelligence

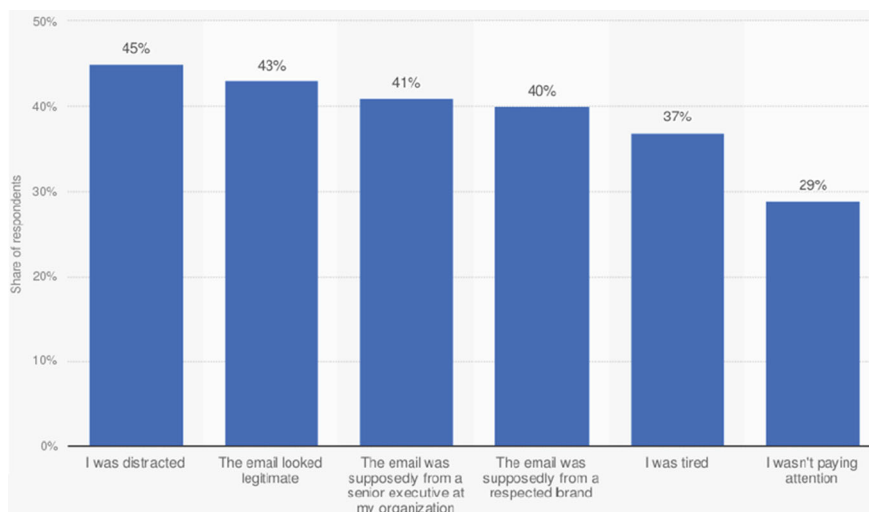


Fig. 6 Statistics of failures to protect from malicious email [30]

3.2 Work Environment and Attitudes

The organizational environment significantly shapes employee behavior. Stressful or unsupportive work cultures can increase the likelihood of mistakes, while positive, learning-oriented cultures foster safer practices. Attitudes towards security vary, ranging from compliance and indifference to proactive advocacy. Most employees fall into compliant or indifferent categories, highlighting the need for cultural change [9].

Rephrased Passage: Employee Attitudes Toward Cybersecurity.

A review of various studies on employee behavior within organizations reveals a consistent pattern in how staff attitudes toward cybersecurity are categorized. These attitudes can be positive, negative, or neutral, and are outlined as follows:

- **Compliant:** Employees with a compliance-oriented mindset view cybersecurity as a mandatory obligation. They tend to follow security policies and procedures closely, often without fully understanding the rationale behind them. While compliance is important, relying solely on this approach may result in a lack of genuine engagement or proactive behavior toward security.
- **Indifferent:** Those who are indifferent to cybersecurity often downplay the significance of security measures or fail to appreciate the potential impact of cyber threats. This lack of interest or concern can lead to neglecting or ignoring established security practices, creating substantial vulnerabilities within the organization.
- **Reactive:** Reactive employees only address cybersecurity when confronted with immediate issues or after a security incident has occurred. Their attention to

security is typically triggered by direct consequences or specific instructions from management, which can result in delayed or inconsistent responses to threats.

- **Proactive:** Proactive individuals demonstrate vigilance and initiative regarding cybersecurity. They actively seek out information, monitor emerging risks, and implement measures to safeguard organizational assets. Their forward-thinking approach helps the organization stay ahead of potential threats.
- **Security Advocate:** Security advocates go beyond personal responsibility, championing cybersecurity throughout the organization. They take on leadership roles, share expertise and best practices, and work to educate others about security awareness. These individuals play a key part in fostering a robust security culture.

It is important to recognize that these attitudes can coexist within a single organization, and individuals may display different behaviors depending on the situation.

To strengthen cybersecurity, organizations should strive to encourage positive and proactive attitudes by offering comprehensive training, awareness initiatives, regular communication, and ongoing evaluation of security practices. Nurturing the right mindset across the workforce is essential for building a resilient security culture and enhancing the organization's overall cybersecurity posture.

3.3 Training, Awareness, and Skills Gaps

Surveys reveal that nearly half of European employees feel inadequately informed about cyber risks. Inadequate or outdated training contributes to human error, especially in the face of sophisticated social engineering attacks. Regular, relevant, and engaging training is critical to building a resilient workforce¹. A survey conducted by the European Commission in 2019 on the European's attitudes towards cyber security (done in 28 countries, with a total of 27,609 respondents) 55 showed that 47% of people feel either not very well informed about the risks of cyber crime, or not informed at all (see Fig. 7).

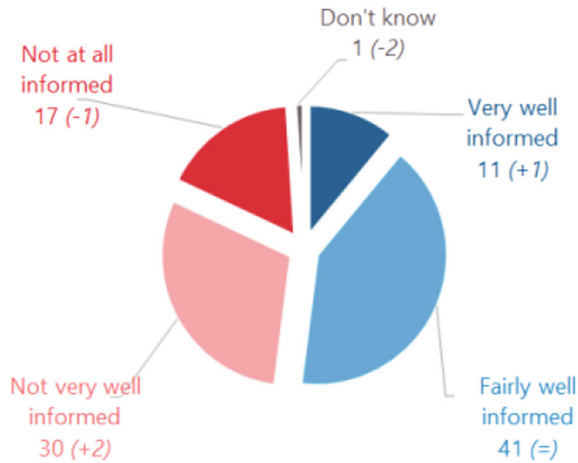
Lack of Awareness and Appropriate Training in Cybersecurity

A persistent lack of cybersecurity awareness and insufficient training among employees remains one of the most critical vulnerabilities for organizations. This gap is widely recognized as a leading cause of security incidents, with human error accounting for a significant proportion of breaches and cyberattacks. Despite investments in advanced security technologies, organizations often overlook the importance of equipping all staff—not just IT professionals—with the knowledge and skills necessary to recognize and respond to threats [9].

Key Issues

1. Insufficient Awareness

Fig. 7 Level of awareness about cybersecurity threats [30]



Low Perceived Relevance: Many employees perceive cybersecurity as a technical issue relevant only to IT staff, leading to disengagement from security training and guidelines.

Limited Understanding of Threats: Surveys indicate that nearly half of European employees feel poorly informed about cyber risks, making them more susceptible to phishing, social engineering, and other attacks [9].

Underestimation of Personal Impact: There is a common misconception that cybersecurity incidents are unlikely to affect individual users, resulting in complacency and risky behaviors (see Table 7).

2. Inadequate Training Programs

- **Lack of Mandatory Training:** In many organizations, cybersecurity training is optional or only required for certain roles, leaving large segments of the workforce unprepared [9].
- **One-Size-Fits-All Approach:** Training is often generic and fails to address the specific needs or risk profiles of different employee groups.

Table 7 Consequences of different risk areas

Risk area	Consequence
Human error	Increased likelihood of falling for phishing, malware, and social engineering
Data breaches	Unauthorized access and disclosure of sensitive information
Financial loss	Costs from incidents, downtime, and regulatory fines
Reputational damage	Loss of customer trust and market credibility
Regulatory non-compliance	Potential legal penalties for failing to meet training and awareness standards

- **Outdated or Irrelevant Content:** Training materials may not be regularly updated to reflect evolving threats, new attack vectors, or changes in organizational processes.
- **Limited Practical Application:** Most training focuses on theoretical knowledge or awareness rather than practical skills, such as incident response or secure data handling.

3. Ineffective Assessment and Reinforcement

- **Minimal Knowledge Testing:** Many organizations do not assess employee cybersecurity knowledge during onboarding or as part of ongoing development, allowing knowledge gaps to persist.
- **Superficial Progress Tracking:** Completion of training modules is often tracked, but there is little measurement of knowledge retention or behavioral change.
- **Lack of Continuous Learning:** Training is typically a one-time event rather than an ongoing process, failing to reinforce good practices or adapt to new threats.

Access control attacks occur when a legitimate access link is broken. The confidentiality of the entire system is jeopardized even if a single device or user is hacked, thus making the whole system vulnerable to various attacks. Considering this, there should be a secure access control management system to provide access to the system resources to only authorized users, processes, and applications [31].

Contributing Factors

- **Cognitive Biases:** Employees may underestimate risks due to overconfidence, familiarity bias, or lack of perceived immediacy [9].
- **Workplace Stress:** High-pressure environments can lead to mistakes, such as clicking on malicious links or mishandling sensitive data.
- **Cultural Barriers:** In some organizations, employees fear repercussions for reporting incidents, leading to underreporting and missed learning opportunities.

Recommendations

Make Training Mandatory: All employees, regardless of role, should participate in regular, role-appropriate cybersecurity training.

- **Tailor Content:** Develop training that addresses specific risks and responsibilities for different job functions.
- **Update Regularly:** Ensure training materials are current and reflect the latest threat landscape. Use reliable sources of information as CSIRT, ENISA or others.
- **Emphasize Practical Skills:** Incorporate hands-on exercises, simulations, and real-world scenarios.
- **Assess and Reinforce:** Use quizzes, phishing simulations, and feedback to measure knowledge retention and adapt training as needed.

- **Promote a Security Culture:** Encourage open communication, reward proactive behavior, and reduce stigma around reporting mistakes.

The lack of awareness and appropriate training in cybersecurity exposes organizations to significant risk. Addressing this gap requires a strategic, organization-wide commitment to continuous education, practical skill development, and a culture that values security as a shared responsibility [9].

4 The Role of the Cybersecurity Manager

The NIS2 Directive (Directive (EU) 2022/2555) significantly elevates the importance of the cybersecurity manager within organizations classified as “essential” or “important entities” across the EU. The cybersecurity manager is pivotal in ensuring compliance, managing risks, and embedding cybersecurity into the organization’s strategy and daily operations [1].

4.1 Key Responsibilities

The cybersecurity manager is pivotal in shaping an organization’s security posture. Core duties include:

- Assessing training needs and knowledge gaps
- Developing and implementing security awareness programs
- Ensuring accessibility and alignment of training materials
- Communicating policies and updates
- Monitoring and evaluating training effectiveness
- Staying abreast of emerging threats and best practices¹

4.2 Competencies and Skills

Effective cybersecurity managers combine technical expertise with strong interpersonal skills. They must understand human behavior, manage cross-functional teams, and drive organizational change. Key competencies include risk assessment, communication, and strategic decisions.

Core Responsibilities of the NIS2 Cybersecurity Manager (Fig. 8).

- **Policy Development & Implementation**
 - Develop and maintain the organization’s cybersecurity policies, procedures, and plans.

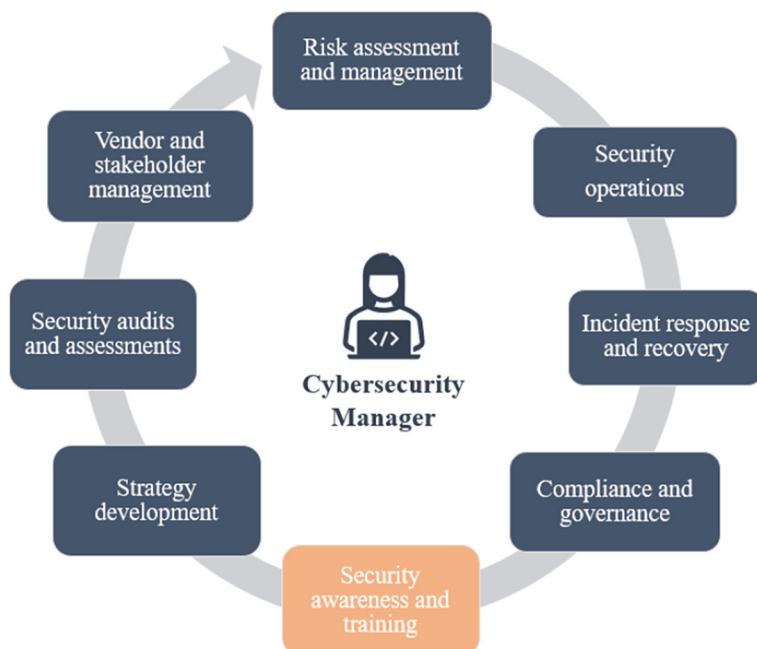


Fig. 8 Cybersecurity Manager competencies [1]

- Ensure alignment with NIS2 requirements, including risk management, incident response, and business continuity [32].
- Risk Management
 - Lead regular cybersecurity risk assessments.
 - Oversee the implementation of technical, operational, and organizational measures to mitigate risks.
 - Interpret and communicate risk management results to senior management [32].
- Incident Management & Reporting
 - Establish and maintain incident response procedures.
 - Ensure prompt reporting of significant incidents to competent authorities—within 24 h for early warning, 72 h for a full report, and a final report within one month [1].
- Compliance & Governance
 - Monitor the effectiveness of security controls and ensure continuous compliance with NIS2.
 - Prepare for and facilitate security audits, penetration testing, and regulatory reviews [32].

- **Training & Awareness**
 - Oversee regular cybersecurity training for all employees, including management, to foster a security-aware culture.
 - Ensure that training is up-to-date and tailored to the organization's risk profile.
- **Supply Chain** Ensure that supply chain partners meet NIS2 standards [1].
- **Resource Management**
 - Advocate for and manage the necessary budget, authority, and personnel to fulfill cybersecurity objectives [33].

Required Qualifications and Competencies

- **Professional Competence**
 - Demonstrated expertise in cybersecurity, with at least one year of relevant experience or a university degree in a related field [1].
 - Comprehensive understanding of both ICT and broader business operations to support business continuity and resilience.
- **Authority and Accountability**
 - Must have sufficient authority and independence within the organization to implement and enforce cybersecurity measures.
 - Cannot be responsible for the operation of regulated services or core IT systems, ensuring a separation of duties [33].
- **Continuous Learning**
 - Stay updated on evolving threats, regulatory changes, and best practices.
 - Participate in ongoing professional development and, if necessary, upskill in legal and compliance matters relevant to cybersecurity12.

NIS2 Compliance: Management's Role

- **Direct Accountability**
 - Senior management is directly accountable for cybersecurity compliance under NIS2, including approving and overseeing cybersecurity strategies and measures.
 - Non-compliance can result in substantial fines and even temporary bans from management roles.
- **Resource Provision**
 - Management must allocate adequate resources—budget, personnel, and technology—to support the cybersecurity manager's mandate.
- **Culture of Security**
 - Foster a culture of transparency, regular reporting, and continuous improvement in cybersecurity practices across the organization Security

Table 8 Key tasks of a cybersecurity manager in the main areas

Area	Key tasks
Policy and strategy	Develop, implement, and update cybersecurity policies and strategies
Risk management	Conduct regular risk assessments, oversee mitigation, report to management
Incident response	Establish, test, and update incident response and business continuity plans
Compliance	Monitor NIS2 compliance, prepare for audits, maintain documentation
Training and awareness	Organize regular, role-based training and awareness campaigns
Supply chain security	Evaluate and manage third-party and supply chain cybersecurity risks
Reporting	Ensure timely incident reporting as required by NIS2

- Assess and manage cybersecurity risks associated with third-party vendors and suppliers.

Key tasks of a cybersecurity manager in the main areas are visualised in Table 8.

The NIS2 cybersecurity manager is a critical role, responsible for the strategic and operational management of cybersecurity within an organization. This position requires a blend of technical expertise, risk management acumen, and organizational authority. The competencies are summarized in Fig. 9. The role is central to ensuring compliance with the NIS2 Directive, protecting the organization from cyber threats, and fostering a culture of security awareness and resilience [1, 34].

4.3 *Developing a Cybersecurity Culture*

Building a cybersecurity culture involves fostering shared values, attitudes, and norms that prioritize security. Managers must create environments that encourage openness, learning from mistakes, and proactive risk management. This cultural shift is essential for sustainable security improvement.

Fostering a cybersecurity culture is a continuous journey that demands sustained commitment and effort. The cybersecurity manager plays a pivotal role in cultivating an environment where positive beliefs, attitudes, knowledge, values, and norms related to security can thrive. This involves empowering individuals from diverse operational areas to operate securely and recognize that they are a crucial line of defense against cyber threats within the organization. The whole ecosystem is visualized in Fig. 10.

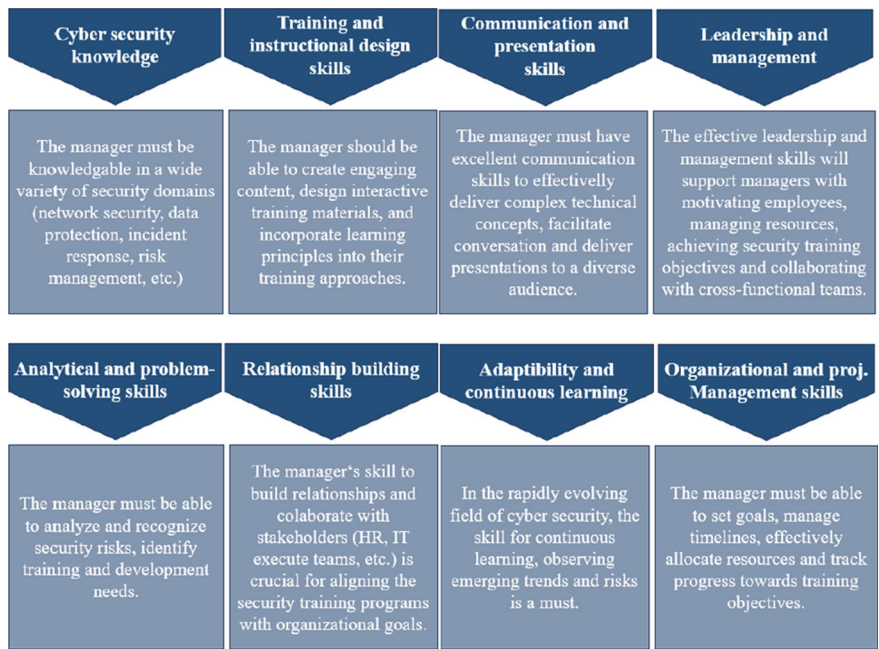


Fig. 9 Cybersecurity manager skills and competencies [1]

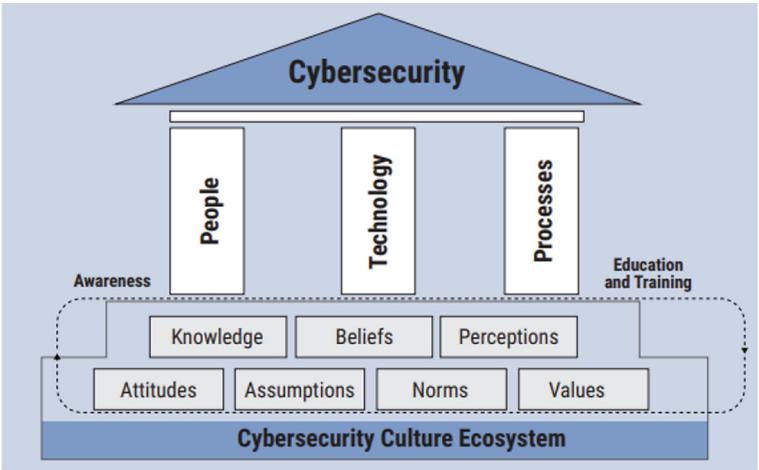


Fig. 10 Cybersecurity ecosystem [33]

5 Gaps in Cybersecurity Education and Management

Slovakia implemented NIS2 into its Cybersecurity law. The responsible authority is the NBU (National Security Office), which established the Cybercompetence office to train and certify cybersecurity managers and provide a cybersecurity audit of compliance with the NIS2 directive. The training schema in Fig. 11 consists of six main parts.

The following are bullet points for Fig. 11.

1. Certification at two levels: Manager and Auditor CS
2. Expert training for manager level 1 and level 2 level 2 can go for the certification exam
3. Specialized training and workshops
4. Information security according to ISO/IEC 27001:2022
5. Basic training (no certification, CS Basics, and CS overview)

5.1 Assessment and Onboarding

- Lack of Standardized Testing: None of the enterprises systematically assesses the cybersecurity knowledge of new hires. There is an overreliance on assumed “common sense” or prior experience.
- Minimal Entry-Level Testing: New employees are rarely tested on cybersecurity awareness at onboarding, leading to undetected knowledge gaps.

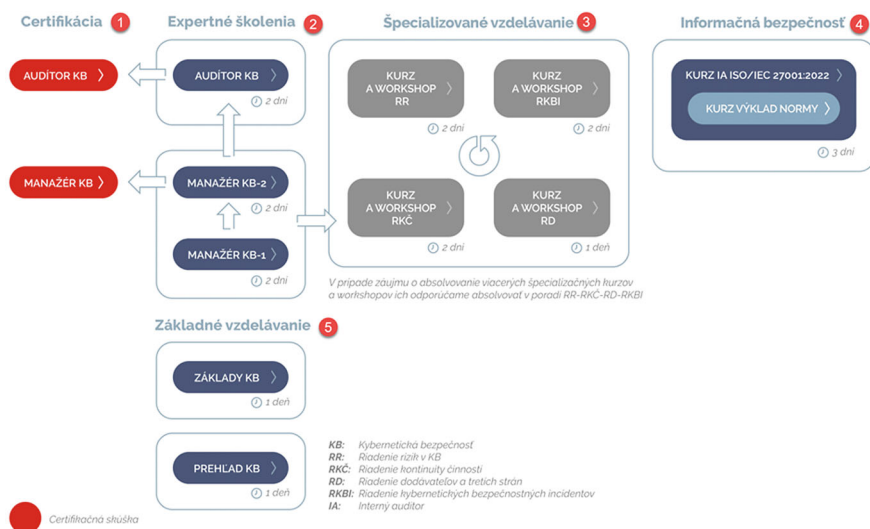


Fig. 11 Educational and certification schema of SlovakCybercompetence center (own creation)

Gaps in Cybersecurity Education and Management: Assessment and Onboarding

1. Many organizations lack standardized processes to assess the cybersecurity knowledge of new hires, relying instead on assumptions about “common sense” or prior experience
2. Entry-level testing for cybersecurity awareness is rarely conducted during onboarding, leaving knowledge gaps undetected at the most critical stage [9].
3. Most onboarding programs do not include scenario-based or practical cybersecurity assessments, which are essential for identifying vulnerabilities in user behaviour [1].
4. The absence of initial assessments means that employees with limited or no cybersecurity awareness may gain access to sensitive systems without adequate preparation [9].
5. Assessment methods, when present, are often limited to completion of generic e-learning modules rather than targeted evaluations of actual understanding or skills [2].
6. There is little differentiation in onboarding assessments between roles with varying levels of risk or access, resulting in a one-size-fits-all approach that fails to address specific needs [1].
7. Organizations frequently overlook the documentation and tracking of assessment results, making it difficult to identify trends or tailor follow-up training for individuals or groups [9].
8. The lack of early and structured assessment perpetuates the misconception that cybersecurity is solely an IT issue, rather than a shared responsibility across all roles.
9. Without effective onboarding assessments, organizations miss the opportunity to establish security expectations and reinforce a culture of vigilance from the outset.
10. Ultimately, these gaps in assessment and onboarding contribute to persistent vulnerabilities, as unprepared employees remain a primary target for cyber threats and social engineering attacks [9].

5.2 Training Programs and Accessibility

- **Non-Mandatory Training:** In some enterprises, cybersecurity training is optional or limited to specific roles. Mandatory courses often focus on compliance rather than security.
- **Limited Enforcement:** Even when training is available, completion is not always enforced or tracked [9].

Problems in Training Programs and Accessibility

Many organizations do not make cybersecurity training mandatory for all employees, limiting participation and leaving large segments of the workforce unprepared for current threats. When training is available, it is often generic and not tailored to the specific roles or risk profiles of different employee groups, reducing its practical relevance and effectiveness. Completion and enforcement of training are inconsistent, with some staff never required to complete essential modules or refresh their knowledge regularly. Training content is frequently outdated, failing to keep pace with the rapidly evolving threat landscape and new attack techniques. Finally, accessibility issues—such as poorly publicized resources or user-unfriendly platforms—further hinder employee engagement and limit the overall impact of security education efforts [9].

5.3 Tracking and Measuring Effectiveness

- **Superficial Tracking:** Progress is often measured by course completion rather than knowledge retention or behavioral change.
- **Reliance on Phishing Simulations:** Phishing campaigns are widely used, but there is limited diversity in assessment methods.

Problems with Tracking and Measuring Effectiveness in Cybersecurity Training

Many organizations primarily track cybersecurity training by monitoring course completion rates rather than assessing whether employees have truly understood or retained the material. This superficial approach fails to measure actual knowledge retention or behavioral change, leaving gaps in the organization's security posture. Overreliance on a single method—such as phishing simulations—limits the ability to capture a comprehensive view of employee readiness and can miss other critical vulnerabilities. Progress reports and assessment data are often not prioritized by line managers, reducing opportunities for targeted improvement and follow-up. As a result, organizations may believe their workforce is prepared, when, in reality, significant knowledge gaps and risky behaviors persist, undermining overall cybersecurity resilience [9].

5.4 Career-Long Learning and Development

- **No Career-Long Strategies:** Structured, ongoing development plans for cybersecurity education are lacking. Training is typically front-loaded or ad hoc.
- **Limited Integration with Career Progression:** Cybersecurity skills are rarely embedded in broader career development or promotion criteria.

Many organizations lack structured, ongoing development plans for cybersecurity education, resulting in training that is often limited to initial onboarding or delivered on an ad hoc basis. As a consequence, employees rarely receive regular updates or advanced learning opportunities to keep pace with rapidly evolving cyber threats and technologies. Cybersecurity skills are seldom integrated into broader career development or promotion criteria, which diminishes motivation for continuous learning among staff. Without a clear framework for career-long learning, knowledge gaps persist and employees may become complacent, leaving the organization more vulnerable to emerging risks. Ultimately, the absence of sustained professional development undermines both individual preparedness and the organization's overall security resilience [9].

5.5 Communication and Guidelines

- **Selective Communication:** Security updates are often sent only to IT/security staff or those who opt in, leaving many employees uninformed.
- **Poor Visibility and Usability:** Guidelines exist but are often not user-friendly or well-publicized, reducing engagement and compliance.

5.6 Organizational Culture and Incident Reporting

- **Fear of Repercussions:** In some companies, employees fear punishment for reporting incidents, leading to underreporting.
- **Lack of Proactive Culture:** There is insufficient emphasis on treating incidents as learning opportunities.

5.7 Training Focus and Early Education

- **Awareness Over Skills:** Training for general employees focuses on awareness rather than practical skills or incident response.
- **Late Start:** Cybersecurity education typically begins at the corporate level, with little integration into earlier educational stages.

Some of the preparedness gaps are summarized in Table 9.

Table 9 Preparedness gaps (own creation)

Area	Gap description
Assessment	No standardized or entry-level testing for all employees
Training	Cybersecurity courses often optional, not enforced
Effectiveness	Overreliance on phishing simulations, lack of diverse assessments
Career-long learning	No structured, ongoing cybersecurity development
Communication	Updates not consistently shared with all employees
Guidelines	Poor awareness/accessibility of policies
Culture	Fear of reporting incidents, lack of learning-oriented environment
Training focus	Emphasis on awareness, minimal practical skills training
Early education	Lack of integration into early education and habit formation

6 NIS2 Directive: Requirements and Implications

6.1 Overview of NIS2

The NIS2 Directive (Directive (EU) 2022/2555) is the EU’s latest legislative framework for cybersecurity, expanding the scope and depth of requirements for critical sectors and large organizations. Key aims include:

- Strengthening risk management and incident response
- Expanding coverage to more sectors and entities
- Mandating regular training for all employees
- Making senior management directly accountable
- Introducing strict incident reporting timelines

6.2 Key Provisions Relevant to Education and Management

- Risk Management: Regular risk assessments and implementation of security controls.
- Corporate Accountability: Senior management is responsible for compliance, with penalties for non-compliance.
- Mandatory Training: Regular, role-based training for all employees, including management.
- Incident Reporting: Early warning within 24 h, full report within 72 h, final report within one month.
- Business Continuity: Robust incident response and continuity plans, regularly tested and updated.
- Supply Chain Security: Assessment and management of third-party risks [1].

6.3 Practical Integration into Organizational Practice

Assessment of Employee Knowledge

- NIS2: Requires regular evaluation of employee knowledge through training records, quizzes, and scenario-based assessments.
- Practice: Standardize entry-level and ongoing assessments for all staff.

Mandatory Training Programs

- NIS2: Mandates role-based, regular training for all employees.
- Practice: Develop structured curricula, track completion, and require regular refreshers.

Measuring Training Effectiveness

- NIS2: Requires procedures for measuring effectiveness, including audits and feedback.
- Practice: Use tests, scenario challenges, and feedback sessions, maintaining records for audits.

Career-Long Learning

- NIS2: Ongoing training to address evolving threats.
- Practice: Integrate cybersecurity skills into career development and promotion criteria.

Communication of Security Updates

- NIS2: Timely communication to all relevant employees.
- Practice: Use targeted notifications, ensure critical information reaches all staff.

Accessibility of Guidelines

- NIS2: Policies must be accessible, updated, and user-friendly.
- Practice: Centralize documentation, ensure awareness and usability.

Organizational Culture

- NIS2: Encourage incident reporting without penalty.
- Practice: Provide clear guidelines and ensure psychological safety.

Training Content

- NIS2: Balance awareness with practical skills.
- Practice: Tailor content to roles, with advanced modules for technical and management staff.

Early Education

- NIS2: Recommends early education to build a culture of cyber hygiene.
- Practice: Advocate integration into school curricula and community programs [1].

7 Case Studies: Real-World Implementation and Challenges

Assume three hypothetical enterprises with different approaches to educating employees.

7.1 Enterprise 1

- Assessment: No mandatory assessment for general users; relies on assumed knowledge.
- Training: Optional courses available; mandatory training limited to compliance topics.
- Tracking: Progress tracked via course completion reports; phishing campaigns used for assessment.
- Culture: Emphasizes non-punitive response to incidents but lacks structured career-long learning.

7.2 Enterprise 2

- Assessment: Assumes no prior knowledge for new hires; provides basic training to all.
- Training: Mandatory onboarding and role-based courses; tailored for management.
- Tracking: Uses phishing simulations and course completion; developing expert career paths.
- Culture: Encourages risk identification and open reporting; building a proactive culture.

7.3 Enterprise 3

- Assessment: Relies on phishing campaign results; no entry-level testing.
- Training: Mandatory courses for all; interactive and gamified learning.
- Tracking: Monitors phishing results and feedback; no career-long strategy.
- Culture: Employees quick to report incidents; guidelines available but not widely accessed.

Methodology and research methods

As we've discussed the importance of the human factor in avoiding potential security breaches, I've decided to look into what is the current attitude of bigger companies operating in Slovakia towards educating and assessing the understanding of basic cyber security concepts among their employees, with the main goal of understand what is currently being practiced, determine the gaps and make an analysis of various companies' processes in regards to their employee's involvement in keeping the security of their company intact.

7.4 Method of Data Obtaining and Sources Used

The data used in this work were obtained by conducting several interviews with cybersecurity experts from world-renowned companies, where the main areas of discussion were:

- Current level of security literacy among employees
- Knowledge assessments
- Trainings
- Availability of resources
- Recommendations for improvement

7.5 Used Methods of Evaluation and Interpretation of Results

The interviews were not evaluated using typical statistical evaluation methods but were assessed through the means of a comparative analysis (see Table 10).

Similarities

It is evident that the three enterprises take distinctly different approaches regarding IT knowledge and the necessity for assessing cybersecurity awareness and training among their employees. Staff members working within cybersecurity departments typically undergo regular training, with their expertise evaluated either through their prior professional experience or through formal educational achievements in the field.

For the broader employee base, however, such assessments are far less routine. When evaluations do occur, they are generally broad in scope and often focus on observing how individuals respond to simulated phishing emails—a method that, as previously discussed, is effective for raising awareness, given the prevalence of phishing attacks among cyber threats today.

A notable contrast emerges between Enterprise 1 and Enterprise 2 in their assumptions about new employees' cybersecurity literacy. Enterprise 1's expectation that

Table 10 Three representative enterprises

Enterprise 1	Enterprise 2	Enterprise 3
• Assessment of CS staff is done through quizzes and certifications	• The assessment of CS staff is based on investigation of academic background and work experience	• The employee literacy is assessed by the means of the phishing email results and the CS course completeness
• No mandatory CS assessment exists for the simple users	• Assessment of CS knowledge for simple users is done by inspecting the CS courses completeness	
• “Common sense” in CS is expected to be present for each employee	• The CS team always assumes that at the time of joining the company, the employees’ knowledge is null, so basics are taught to everyone	
• No necessity is seen by the company to have the simple users well versed in CS, as the technical teams have it under their responsibility		
	CS knowledge and expertise are not being assessed at the time an employee joins the company in any of the cases, as it is deemed unnecessary	

all staff possess a certain level of “common sense” in cybersecurity can be problematic, as it risks leaving knowledge gaps unaddressed and may result in missed opportunities for targeted education and training. In contrast, Enterprise 2 adopts a more prudent approach by presuming that newcomers, regardless of their background or expertise, may lack sufficient cybersecurity knowledge. This perspective ensures that all employees receive foundational training, allowing those with limited knowledge to learn essential concepts while enabling more experienced individuals to strengthen or update their understanding where needed.

8 Recommendations for Bridging the Gap

8.1 Standardizing Assessment and Onboarding

- Implement standardized cybersecurity assessments for all new hires, regardless of role.
- Use scenario-based quizzes and practical exercises to identify knowledge gaps early.

Standardizing Assessment and Onboarding in Cybersecurity

1. Standardizing assessment and onboarding ensures that every new employee, regardless of their role, receives a consistent baseline evaluation of their cybersecurity knowledge and awareness [9].
2. Organizations should implement structured entry-level assessments—such as quizzes, scenario-based exercises, or practical demonstrations—to identify knowledge gaps before granting system access
3. These standardized assessments help eliminate assumptions about “common sense” security practices, which often lead to undetected vulnerabilities among new hires.
4. Onboarding programs must include tailored cybersecurity modules that introduce company-specific policies, procedures, and threat landscapes, ensuring immediate relevance.
5. Regular updates to onboarding materials are essential to reflect evolving threats, regulatory changes, and organizational priorities [9].
6. Assessment results should be documented and used to assign targeted follow-up training, ensuring that employees with lower scores receive additional support.
7. Integrating cybersecurity checks into onboarding reinforces the message that security is a shared responsibility from day one, not just an IT concern.
8. Automated onboarding platforms can streamline the assessment process, track completion rates, and generate compliance records for audit purposes.
9. Standardization also supports regulatory compliance, such as NIS2, by providing clear evidence that all staff have met minimum security awareness requirements [9].
10. Ultimately, a standardized assessment and onboarding process reduces risk, accelerates employee readiness, and helps build a more resilient organizational security culture.

8.2 *Mandatory, Role-Based, and Continuous Training*

- Make cybersecurity training mandatory for all employees, tailored to specific roles.
- Schedule regular refresher courses and ensure completion is tracked and enforced.
- Develop advanced modules for management and technical staff.

Mandatory, Role-Based, and Continuous Training in Cybersecurity

Mandatory cybersecurity training ensures that every employee, regardless of position, receives essential knowledge to recognize and respond to cyber threats. Role-based training tailors content to the specific risks, responsibilities, and access levels associated with different job functions, making the learning directly relevant and practical. For example, management and technical staff receive advanced modules focused on

leadership accountability, incident response, and technical controls, while general staff are trained on safe data handling and recognizing phishing attempts.

Continuous training is essential because the cyber threat landscape evolves rapidly, with new attack techniques and regulatory requirements emerging regularly. Regular refresher courses and ongoing learning opportunities help reinforce good security habits and ensure that employees remain vigilant against emerging risks. Tracking and enforcing completion of mandatory training modules is critical for regulatory compliance, such as with the NIS2 Directive, and for demonstrating due diligence during audits.

Assessment methods—such as scenario-based exercises, phishing simulations, and knowledge quizzes—should be integrated to measure retention and adapt training to address observed weaknesses. Providing accessible, up-to-date training resources encourages a culture of shared responsibility and empowers employees to act as the first line of defense. Ultimately, a structured, mandatory, role-based, and continuous training program strengthens organizational resilience and reduces the likelihood and impact of cyber incidents [9].

8.3 Effective Communication and Accessible Guidelines

- Centralize all security policies and guidelines in an easily accessible portal.
- Use targeted, relevant notifications to communicate updates and threats.
- Regularly update and user-test documentation for clarity and usability.

Effective Communication and Accessible Guidelines

Effective communication in cybersecurity requires that all employees receive timely, relevant, and clearly worded updates about threats, policies, and required actions. Organizations should centralize security policies and guidelines in an easily accessible portal, ensuring that employees know where to find the information they need. Regular updates and user-friendly formats—such as FAQs, interactive guides, or short videos—help increase engagement and comprehension across diverse roles. Targeted notifications can prevent information overload and ensure that critical alerts reach the right people without causing notification fatigue. User-testing documentation for clarity and usability is essential, as guidelines that are difficult to understand or locate are less likely to be followed. Ultimately, accessible and well-communicated guidelines foster higher compliance, empower employees to act securely, and support a resilient organizational security culture [9].

8.4 Fostering a Proactive Security Culture

- Encourage open reporting of incidents without fear of punishment.

- Recognize and reward proactive security behavior.
- Integrate cybersecurity into broader organizational values and performance metrics.

Fostering a Proactive Security Culture

Fostering a proactive security culture involves creating an environment where employees are encouraged and empowered to take initiative in identifying and addressing potential cyber risks before they escalate. Leadership must model transparency and openness, making it clear that reporting incidents or near-misses is valued and will not result in punitive action. Regular communication, recognition of positive security behaviors, and clear guidelines help reinforce the importance of vigilance and shared responsibility. Training programs should not only focus on compliance but also on developing critical thinking and situational awareness, enabling staff to anticipate and respond to emerging threats. Cross-functional collaboration and open dialogue between IT, management, and other departments further embed security into daily operations. Ultimately, a proactive culture transforms employees from passive rule-followers into active defenders of the organization's digital assets, strengthening overall resilience against cyber threats [9].

8.5 Early and Societal Education Initiatives

- Advocate for the integration of cybersecurity basics into school curricula.
- Partner with community organizations to promote cyber hygiene from an early age.
- Support parental education on digital safety for children.

Early and Societal Education Initiatives

Early and societal education initiatives in cybersecurity aim to instill safe digital habits and foundational knowledge from a young age, recognizing that prevention starts long before individuals enter the workforce. Integrating cybersecurity basics into school curricula helps children understand the risks of sharing information online, recognize common threats, and develop responsible digital behaviors. Community programs and parental education play a crucial role in reinforcing these lessons at home, ensuring that digital safety becomes a shared family value. Organizations and governments increasingly advocate for public awareness campaigns and partnerships with educational institutions to broaden the reach of cyber hygiene education. By fostering early awareness and societal engagement, these initiatives help build a more resilient and security-conscious population, reducing vulnerabilities across all age groups [9].

Responsible parties for different cybersecurity requirements are summarized in Table 11.

Table 11 Responsible parties for different cybersecurity requirements

Requirement	Description	Responsible parties
Risk assessments	Regular evaluation and mitigation of cyber risks	Management, IT, risk teams
Security policies	Documented, updated, and accessible security policies	Management, compliance
Mandatory training	Regular, role-based training for all employees, including management	HR, IT, management
Incident reporting	24-h notification, 72-h detailed report, 1-month final report	IT, security, management
Business continuity	Tested and documented business continuity and incident response plans	IT, operations, management
Supply chain security	Assessment and management of third-party cybersecurity risks	Procurement, IT
Effectiveness measurement	Security audits, penetration testing, training effectiveness evaluations	IT, audit, HR
Documentation and record-keeping	Maintain evidence of compliance, training records, incident logs	Compliance, HR, IT

The organisation’s management sought practical guidance to implement the cybersecurity risk management measures, the main steps are shown in Fig. 12.

They recognized that the European Cybersecurity Skills Framework (ECSF) could help them identify and address organisational capability gaps—whether through upskilling, reskilling, hiring, or procuring external services. In fact, the ECSF can serve as a foundational tool for the development of the roles, responsibilities and competences. With the roadmap illustrated in Table 12, the organisation now has a clear and actionable starting point (Fig. 13).



Fig. 12 Three steps of NIS2 implementation [33]

Table 12 Organisational tasks

Organisational tasks
<i>1. Define and implement cybersecurity plans and policies</i>
Develop strategies and policies to comply with cybersecurity regulations
Oversee the cybersecurity policy and assign responsibilities to implement measures and fulfil reporting obligations
Manage relationships with suppliers and ensure their security to protect the supply chain
Regularly update the management board on cybersecurity plans and strategies
Create plans to ensure the business can continue operating and recover quickly in case of a disaster
<i>2. Manage the cybersecurity risk</i>
Develop and apply policies for risk analysis, system security, supplier security, encryption, access control, asset management, and HR security
Teach management about cybersecurity risks, threats, and their impact on the organisation
Inform management about cybersecurity risk management measures
<i>3. Collaborate with authorities and other stakeholders</i>
Share information and provide reports to authorities and other interested parties
Ensure compliance with binding instructions or orders from authorities
<i>4. Ensure compliance through cybersecurity audits</i>
Conduct independent reviews of information and network security of the organisation
Conduct internal audits and provide reports to authorities when requested
<i>5. Oversee and assure compliance with the directive</i>
Make sure the organisation follows data protection laws (NIS2 and GDPR) when handling personal data
Submit required information to authorities to get listed as an essential entity and update them about any changes
Ensure proper handling and access to domain name registration data by relevant entities
<i>6. Inform stakeholders regarding compliance obligations</i>
Tell management they are responsible for approving cybersecurity measures and must complete cybersecurity training
Ensure top management knows if they need to appoint a representative for companies not based in the EU but offering services there
<i>7. Plan, design, and implement security solutions and controls Plan and build secure networks and ensure a secure environment</i>
Implement strong security features like multi-factor authentication (MFA), and secure communications for voice, video, text, and emergencies
Develop, set up, manage, and monitor cybersecurity measures such as secure network configurations and system integrations
Apply cybersecurity measures and safeguards based on the risk management results and planning and address issues identified during the audits
<i>8. Deliver cybersecurity training</i>
Create and deliver appropriate cybersecurity training programmes aimed at members of management bodies of entities and their employees

(continued)

Table 12 (continued)

Organisational tasks
Implement basic cyber hygiene practices and designs and deliver cybersecurity training
9. <i>Collect and analyse information to identify threats</i>
Collect and analyse cyber threats, then create reports and share them with relevant parties
Oversee the process of collecting, analysing, and sharing cybersecurity information, including threats, Near misses, vulnerabilities, and attack techniques
10. <i>Ensure effective response and reporting of cybersecurity incidents Develop and establish a plan for responding to cybersecurity incidents</i>
Evaluate and report any vulnerabilities to the Computer Security Incident Response Team (CSIRT). Manage cybersecurity incidents and disclose any vulnerabilities found
Identify, analyse, and report cybersecurity incidents to CSIRTs, authorities, and service recipients
11. <i>Conduct security assessment tests</i>
Conducts penetration tests and vulnerability assessments to assess the effectiveness of cybersecurity solutions and provides, when requested, the vulnerability assessment and/or penetration testing reports to the competent authorities
Plan and continuous perform security tests (including but not limited to penetration tests and vulnerability assessments) as per the results of the risk management process. Document the findings and apply appropriate mitigating actions in case of critical findings
Assists the competent authorities' personnel on the security scans conducted

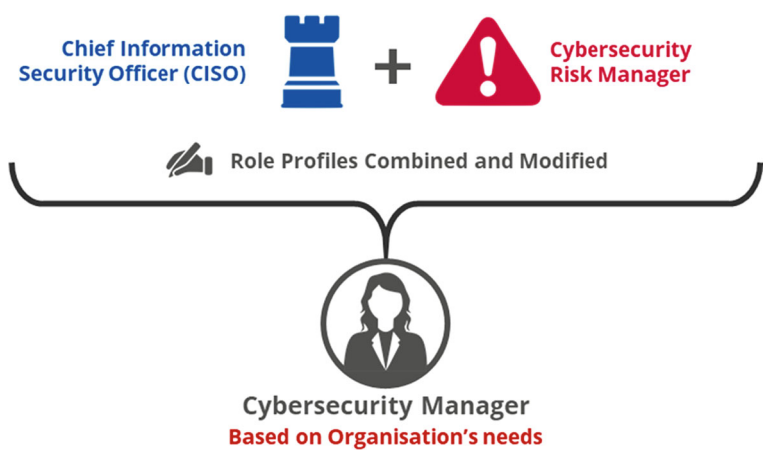


Fig. 13 Combination of roles to achieve compliance (own creation)

Methodology

Step 1: Identify Relevant NIS2 Articles

The first step was to identify the specific articles of the NIS2 Directive that outline the obligations for essential and important entities. To achieve this, we conducted a thorough review of each article, isolating only those provisions that directly pertain to these entities. References to other stakeholders, such as the European Commission, Member States' Competent Authorities, ENISA and CSIRTs, were deliberately excluded from this analysis.

Step 2: Specify Obligations and Tasks Per Article

In this step, we identified and extracted specific cybersecurity-related obligations from each article, paragraph and point of the NIS2 Directive. These obligations include tasks related to cybersecurity risk management measures, supervision, reporting, and communication.

For instance, Article 32(4)(f) states: "order the entities concerned to implement the recommendations provided as a result of a security audit within a reasonable deadline." Based on this, the corresponding task was defined as: "Implements cybersecurity solutions, within the time limit specified, to mitigate the findings of a security audit."

This approach ensures that each obligation is clearly formulated as an actionable task, making it easier for entities to understand and implement NIS2 requirements.

Step 3: Define Expected Outputs Per Task

In step 3, we defined the expected outputs or deliverables for each cybersecurity-related task identified in the previous phase. These outputs serve as concrete evidence of task completion. Clearly defining deliverables ensures that entities have measurable goals, can track progress, demonstrate accountability, and provide documentation for audits or supervisory reviews. For example, the task "Implements cybersecurity solutions, within the time limit specified, to mitigate the findings of a security audit" results in the deliverable "Implementation Report".

Step 4: Map Tasks With ECSF Role Profiles

In this step, we mapped each cybersecurity-related task identified in the previous phase to the relevant ECSF role profiles. This mapping was guided by the summary statements, mission tasks, main tasks, deliverables, key skills, key knowledge, and e-competences associated with each role.

To establish these connections, we identified relevant keywords and similar tasks. These elements helped align each NIS2 task with the appropriate ECSF role profiles, ensuring a structured and competency-driven approach.

Initially, this mapping was conducted broadly to capture all potential role associations. This approach ensured that no relevant role profiles were overlooked before refining the task-to-role connections in subsequent steps.



Fig. 14 Five steps to achieve compliance [1]

Step 5: Validate Role Mapping and Identify Overlaps

In the final step, we conducted a critical evaluation of each task-to-role association to enhance the accuracy and reliability of the mapping. This process involved a detailed review by cybersecurity experts to ensure that the selected ECSF skills and knowledge aligned with the practical execution of each task, rather than being influenced by predefined role assumptions.

Given the multidisciplinary nature of cybersecurity, we also identified task overlaps across multiple ECSF role profiles. For instance, while Article 23 primarily focuses on incident handling and reporting, its related tasks require expertise beyond the Cyber Incident Responder role. Instead of restricting the mapping to this profile alone, we carefully examined the full range of ECSF competencies that could realistically contribute to fulfilling these obligations. This methodology prevented role bias and ensured a comprehensive representation see Fig. 14.

Finally, we considered how collaboration between different ECSF role profiles may be necessary to deliver the outputs defined in Step 3. Many of the expected deliverables—such as cyber incident reports or implementation plans—require input or joint effort from multiple roles. By identifying where such collaboration is essential, the mapping can help entities structure their teams more effectively, allocate responsibilities clearly, and foster inter-role cooperation in meeting NIS2 obligations.

9 Conclusion

The integration of the NIS2 Directive into organizational practice represents a paradigm shift in how cybersecurity education and management are approached. The findings of this analysis reveal persistent gaps between best-practice recommendations and real-world implementation, particularly in assessment, training, communication, and culture.

To achieve compliance and enhance resilience, organizations must move beyond ad hoc or optional training and embrace a structured, mandatory, and continuously updated approach to cybersecurity awareness and skills development. Senior

management is now directly accountable, and compliance is not optional. By embedding NIS2 requirements into all aspects of cybersecurity education and management, organizations can close existing gaps, foster a proactive security culture, and ensure regulatory compliance in an increasingly digitalized world.

The comprehensive analysis of cybersecurity management and education, particularly in the context of the NIS2 Directive, underscores the urgent necessity for organizations to rethink and elevate their approach to cyber risk. The findings reveal that while technological defenses are indispensable, they are not sufficient on their own. The human factor remains the most unpredictable and vulnerable element in the cybersecurity landscape, and addressing this requires a fundamental shift in organizational priorities and culture.

Key Takeaways

Human-Centric Security: The majority of security incidents are rooted in human error, lack of awareness, or insufficient training. Organizations must move beyond viewing employees as the “weakest link” and instead empower them as active defenders through continuous education and engagement.

Leadership and Accountability: The NIS2 Directive places direct responsibility on senior management, making cybersecurity a board-level issue. This shift demands that leaders not only allocate resources but also model and enforce a culture of security throughout the organization.

Continuous Learning and Adaptation: Cyber threats are dynamic, evolving in sophistication and scale. Static or one-off training is inadequate; organizations need to implement ongoing, adaptive training programs that keep pace with emerging threats and changing technologies.

Integrated Approach: Effective cybersecurity requires collaboration across departments—IT, HR, compliance, procurement, and executive leadership. Policies, incident response plans, and training must be integrated into the fabric of everyday business operations.

Early and Broad Education: The consensus among experts is clear—cybersecurity education should begin early, both at home and in schools, to build lifelong habits and awareness. Organizations can play a role in advocating for and supporting broader societal education initiatives.

Strategic Imperatives

Mandatory, Role-Based Training: All employees, regardless of their position, must receive regular, tailored cybersecurity training. Specialized modules should be provided for those in high-risk or leadership roles.

Robust Assessment and Measurement: Organizations should implement standardized assessments to measure knowledge retention and behavioral change, not just training completion. Phishing simulations, scenario-based exercises, and feedback mechanisms are critical tools.

Accessible and Actionable Policies: Security guidelines and procedures must be easy to find, understand, and apply. Regular updates and user-friendly formats increase engagement and compliance.

Supportive Reporting Culture: Fear of punishment for mistakes must be replaced with a culture that encourages transparency, rapid reporting, and learning from incidents. This approach reduces underreporting and accelerates response and recovery.

Investment in Resources and Technology: Adequate funding, skilled personnel, and advanced security technologies are essential. However, these must be complemented by a strong focus on the human dimension of cybersecurity.

Looking Forward

The digital transformation of enterprises will only accelerate, bringing new opportunities and risks. As organizations become more interconnected and reliant on digital infrastructure, the attack surface expands, and the potential consequences of breaches grow more severe. Regulatory frameworks like NIS2 are raising the bar for compliance and resilience, but true security will require a proactive, holistic approach that blends technology, process, and, most importantly, people.

Organizations that succeed in embedding cybersecurity into their culture, leadership, and daily operations will not only achieve compliance but also gain a strategic advantage in trust, reputation, and long-term sustainability. The journey is ongoing, and the stakes are high, but with commitment and collaboration, it is possible to build a resilient digital future.

On Fig. 15 we can see schema of strengthening network DDOS attack detection in heterogeneous IoT environment [26].

The Internet of Things (IoT), considered an intriguing technology with substantial potential for tackling many societal concerns, has been developing into a significant component of the future. The foundation of IoT is the capacity to manipulate and track material objects over the Internet. The IoT network infrastructure is more vulnerable to attackers/hackers as additional features are accessible online. The complexity of cyberattacks has grown to pose a bigger threat to public and private sector organizations. They undermine Internet businesses, tarnish company branding, and restrict access to data and amenities. Enterprises and academics are contemplating using machine learning (ML) and deep learning (DL) for cyberattack avoidance because ML and DL show immense potential in several domains. A promising area is the use of AI automation and AI agents.

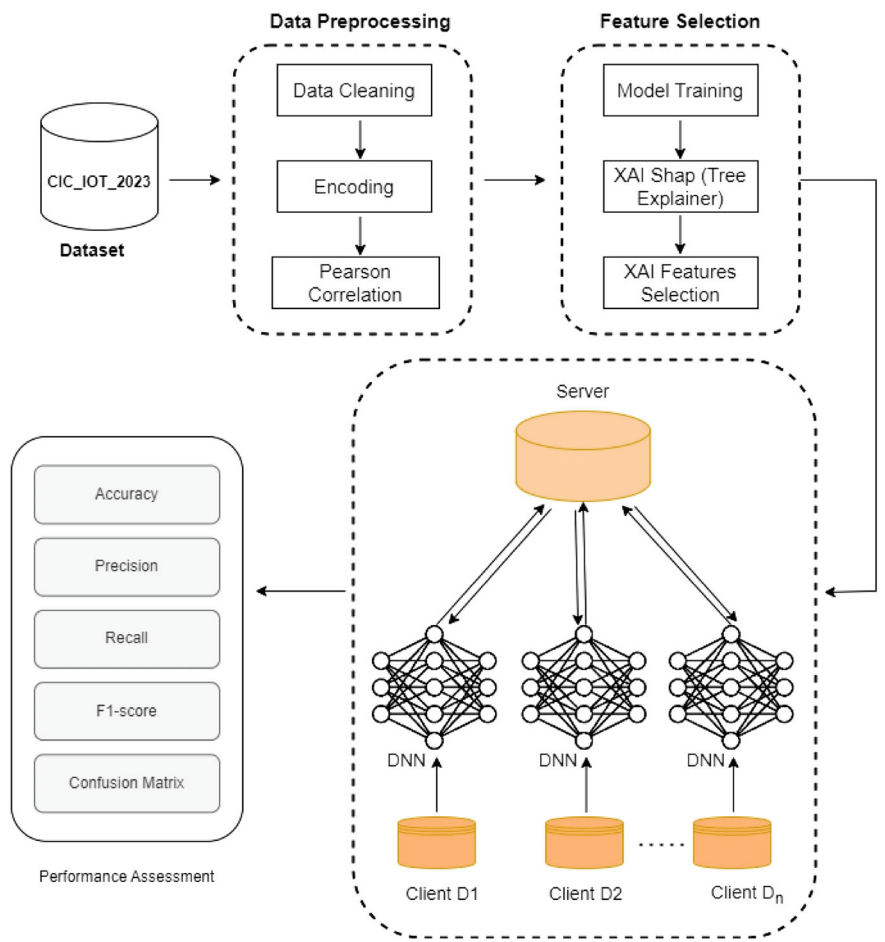


Fig. 15 DDOS attack detection [26]

References

1. European Union Agency for Cybersecurity (2025) Cybersecurity roles and skills for NIS2 essential and important entities: mapping NIS2 obligations to ECSF. Publications Office, LU
2. Gill M (2025) Check-point-software-first-quarter-2025-financial-results. Check Point, Tel Aviv, Israel
3. Petrosyan A (2023) Data records breached worldwide 2023. <https://www.statista.com/statistics/1307426/number-of-data-breaches-worldwide/>
4. Pratt MK (2021) 10 digital transformation benefits for businesses. <https://www.techtarget.com/searchcio/tip/Top-10-digital-transformation-benefits-for-business>
5. Verizon (2025) 2025 data breach investigations report (DBIR). Verizon
6. Bowersox DJ, Closs DJ, Drayer RW (2005) The digital transformation: technology and beyond. Supply Chain Manag Rev 9:22–29
7. Digitalisation in Europe—2025 edition. Interactive publications, Eurostat

8. European M & MEC on IS, Themistocleous M, Papadaki M, Kamal MM (2021) Information systems: 17th European, Mediterranean, and Middle Eastern conference, EMCIS 2020, Dubai, United Arab Emirates, November 25–26, 2020, proceedings. Springer, Cham
9. Pucicov BD (2023) Comenius University Bratislava faculty of management. Comenius University Bratislava Faculty of Management, Thesis
10. Equifax (2017) Equifax announces cybersecurity incident involving consumer information. <https://investor.equifax.com/news-and-events/news/2017/09-07-2017-213000628>
11. United States Department of Justice, U.S. Attorney's Office (2022) Former chief security officer of Uber convicted of federal charges for covering up data breach involving millions of Uber user records. <https://t.ly/ZzRWh>
12. World Economic Forum (2019) The global risks report 2019
13. Di Nardo C (2021) Digitization, digitalization and digital transformation. <https://deltalogix.blog/en/2021/03/09/digitisation-digitalisation-digital-transformation-whats-the-difference/>
14. Cyber insecurity and misinformation top WEF global risk list—infosecurity magazine. <https://www.infosecurity-magazine.com/news/top-wef-global-risk-list-2024-cyber/>. Accessed 13 Jul 2025
15. Yang X, Pan L, Zhao X et al (2023) A survey on detection of LLMs-generated content
16. CS B (2025) AI-powered cyber threats: how hackers are using AI for cyber attacks. In: Security Australia. <https://securitybrief.com.au/story/ai-powered-cyber-threats-how-hackers-are-using-ai-for-cyber-attacks>. Accessed 13 Jul 2025
17. Stanham L (2025) Most common AI-powered cyberattacks|CrowdStrike. <https://www.crowdstrike.com/en-us/cybersecurity-101/cyberattacks/ai-powered-cyberattacks/>. Accessed 13 Jul 2025
18. Abdo B, Schondorfer B, Hamdam K, Goody K (2019) Head fake: tackling disruptive ransomware attacks. In: FireEye. <https://www.fireeye.com/blog/threat-research/2019/10/head-fake-tackling-disruptive-ransomware-attacks.html>. Accessed 24 Nov 2020
19. Shea S (2025) The history and evolution of ransomware attacks. In: Search security. <https://www.techtarget.com/searchsecurity/feature/The-history-and-evolution-of-ransomware>. Accessed 13 Jul 2025
20. (2025) Tracking Ransomware: May 2025. In: CYFIRMA. <https://www.cyfirma.com/research/tracking-ransomware-may-2025/>. Accessed 13 Jul 2025
21. Field effect (2024) The rise of cybercrime-as-a-service. <https://fieldeffect.com/blog/cybercrime-as-a-service>. Accessed 13 Jul 2025
22. Matt High (2025) How cybercrime-as-a-service is a growing enterprise threat | cyber magazine. <https://cybermagazine.com/articles/how-cybercrime-as-a-service-is-a-growing-enterprise-threat>. Accessed 13 Jul 2025
23. Hill M (2025) Cyber attacks & data breaches in February 2025. In: Cyber security Hub. <https://www.cshub.com/attacks/articles/cyber-attacks-data-breaches-february-2025>. Accessed 13 Jul 2025
24. CLOUDFLARE zero trust security | what's a zero trust network? In: Cloudflare. <https://www.cloudflare.com/learning/security/glossary/what-is-zero-trust/>. Accessed 24 Nov 2020
25. Kuhn R, Raunak M, Kacker R (2017) It doesn't have to be like this: cybersecurity vulnerability trends. IT Prof 19:66–70. <https://doi.org/10.1109/MITP.2017.4241462>
26. Almadhor A, Altalbe A, Bouazzi I et al (2024) Strengthening network DDOS attack detection in heterogeneous IoT environment with federated XAI learning approach. Sci Rep 14:24322. <https://doi.org/10.1038/s41598-024-76016-6>
27. Abbas S, Bouazzi I, Ojo S et al (2024) Evaluating deep learning variants for cyber-attacks detection and multi-class classification in IoT networks. PeerJ Comput Sci 10:e1793. <https://doi.org/10.7717/peerj-cs.1793>
28. Aloul FA (2012) The need for effective information security awareness. J Adv Inf Technol 3:176–183
29. Gutzwiller RS, Fugate S, Sawyer BD, Hancock PA (2015) The human factors of cyber network defense. In: Proceedings of the human factors and ergonomics society annual meeting. SAGE Publications, Los Angeles, CA, pp 322–326

30. (2023) The top cybersecurity statistics of 2023. North America's trusted partner for penetration testing, Toronto | HQ 401 Bay Street, Suite 1600 Toronto, Ontario, Canada M5H 2Y4
31. Hashmi IF, Iqbal Z, Munir E et al (2024) SAAC: secure access control management framework for multi-user smart home systems. IEEE Access 12:133339–133355. <https://doi.org/10.1109/ACCESS.2024.3446180>
32. Kubíková K (2024) NIS2: security roles and their roll-out according to the new Cyber Security Act | Cybrela. <https://cybrela.com/en/nis2-security-roles-according-to-the-new-cyber-security-act/>. Accessed 13 Jul 2025
33. NIS2 requirements: a complete guide to compliance and implementation. <https://www.dataguard.com/nis2/requirements/>. Accessed 13 Jul 2025
34. Vakakis N, Nikolis O, Ioannidis D et al (2019) Cybersecurity in SMEs: the smart-home/office use case. 2019 IEEE 24th international workshop on computer aided modeling and design of communication links and networks (CAMAD). IEEE, Limassol, Cyprus, pp 1–7